

Engineering Activity Detail Reference

A 23-stage semiconductor program template — every engineering activity's purpose, work flow, inputs, relationship to the stage's key deliverables, risks, effort and criteria.

STAGES	ACTIVITIES	KEY DELIVERABLES	PROGRAM SPAN	TOTAL EFFORT
23	257	167	132 _w	3,653 _{M/M}

Reference basis: new-architecture 4nm-class AI accelerator, 2.5D packaging with 2xHBM3.
Generated from docs/activity-details.json · joins on activity id against docs/stage-template-v2.json.

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Every entry links to its page. On a tablet the PDF's bookmark panel carries the same tree.

DEFINE

01 Product Definition	DEF		w0–w8
DEF-01	Market and customer requirements consolidation	DEF-05	Memory and interface bandwidth requirements (HBM stacks, PCIe/CXL lanes, D2D)
DEF-02	Workload / use-case profiling and target KPI derivation (TOPS, TOPS/W, tokens/s)	DEF-06	Competitive benchmarking and product positioning
DEF-03	PPA target definition — max frequency, power, die area	DEF-07	Feasibility assessment against candidate technology nodes
DEF-04	Die size, yield and product cost model (wafer, mask, package, test, ASP)	DEF-08	Program schedule, resource and staffing plan drafting
		DEF-09	Business case preparation and funding approval package

02 Architecture	ARCH		w6–w24
ARCH-01	System-level performance modeling and workload simulation	ARCH-05	Power domain, clock domain and DVFS architecture; UPF intent draft
ARCH-02	Compute / memory / interconnect partitioning; monolithic vs chiplet decision	ARCH-06	PPA budget allocation per block
ARCH-03	Dataflow and memory hierarchy definition (on-die SRAM capacity, HBM channels)	ARCH-07	Security and safety architecture — secure boot, root of trust, fusing
ARCH-04	Interface and protocol selection (PCIe gen, CXL, UCIe, HBM3/3E)	ARCH-08	Chip-level floorplan intent, pin and bump budget
		ARCH-09	Architecture specification authoring and review
		ARCH-10	Block microarchitecture specification kickoff

TECHNOLOGY & IP ENABLEMENT

03 Technology & Foundry Selection	TECH		w0–w14
TECH-01	Foundry and node long-list to short-list evaluation	TECH-06	Foundry legal engagement — NDA, design agreement, IP licensing frame
TECH-02	Process option and flavor selection (HPC/HD, multi-Vt menu, backside PDN, RF/HV adders)	TECH-07	Foundry roadmap and risk-production timing alignment
TECH-03	Density / performance / leakage benchmarking against product targets (DTCO study)	TECH-08	Second-source and node-migration strategy assessment
TECH-04	Wafer price, mask and NRE quotation; MPW and volume pricing	TECH-09	OSAT and backend supply chain preliminary alignment
TECH-05	Fab capacity and tapeout slot reservation; hot-lot policy agreement		

04 PDK & Design Enablement **PDK**

w8–w44

PDK-01	PDK version roadmap tracking (0.1 → 0.5 → 1.0 → production) and delta impact analysis	PDK-08	EDA tool version qualification for the node — synthesis, P&R, STA, PV, EM/IR
PDK-02	Design rule manual review; restricted and recommended rule disposition	PDK-09	Foundry reference flow bring-up and internal methodology deck
PDK-03	Standard cell library selection and qualification (track height, multi-Vt, multi-bit flops)	PDK-10	Rule deck, QRC tech file and DRC/LVS deck version control
PDK-04	Memory compiler evaluation and instance generation (SRAM, register file)	PDK-11	Signoff corner and derate definition agreed with foundry (PVT, OCV/AOCV/POCV)
PDK-05	Compiler instance PPA characterization against block budgets — density, Vmin, access time, leakage	PDK-12	Compute farm, EDA license and storage capacity planning
PDK-06	Custom / pushed-rule memory instance decision, scope and schedule impact assessment	PDK-13	Golden design environment release and flow regression
PDK-07	IO, ESD and latch-up library qualification		

05 IP Strategy & Readiness **IPR**

w4–w26

IPR-01	Product requirement → IP requirement decomposition (IP bill of materials)	IPR-06	IP deliverable checklist review (GDS, LEF, multi-corner .lib, CDL, UPF, DV collateral, errata)
IPR-02	Reuse inventory audit — internal IP already available on the target node	IPR-07	Licence negotiation, PO issue and delivery-date commitment
IPR-03	Make / buy / reuse decision per IP block	IPR-08	Porting / hardening scope and schedule for IP not proven on the node
IPR-04	Vendor RFI / RFQ and technical evaluation	IPR-09	IP delivery schedule alignment to RTL and PD integration windows
IPR-05	Silicon-proven status check per IP on the selected node and process option	IPR-10	IP maturity risk rating and second-source contingency

06 Custom, AMS & Memory IP Development **AMS**

w18–w60

AMS-01	AMS IP specification and budget allocation (jitter, BER, PSRR, area)	AMS-09	Redundancy and repair scheme integration with the BISR architecture
AMS-02	PLL / clock generator design and closure	AMS-10	Custom memory view generation and compiler-equivalent characterization across corners
AMS-03	SerDes and PHY design or vendor hardening (PCIe/CXL, UCIe, HBM PHY)	AMS-11	Schematic design and pre-layout simulation
AMS-04	LDO, bandgap and power-management cell design	AMS-12	Custom layout with DRC / LVS / antenna and fill closure
AMS-05	Custom SRAM instance architecture — bitcell selection, array organization, sense amp and replica timing	AMS-13	Post-layout extracted simulation across PVT and Monte Carlo
AMS-06	Read / write assist circuit design for Vmin — WL boost, negative bitline, VDD collapse	AMS-14	Reliability checks — EM/IR, aging, ESD, latch-up
AMS-07	Custom array layout and pushed-rule DRC closure with foundry approval	AMS-15	Hard macro abstraction and view generation (LEF, .lib, CDL, GDS, behavioural model)
AMS-08	Statistical margin and yield analysis — importance sampling, sigma-Vmin, bitcell variation	AMS-16	AMS-digital co-simulation and integration support

07 Test Chip / MPW Shuttle **TC**

w22–w62

TC-01	Test chip objective definition and risk-item selection	TC-05	Shuttle tapeout and fabrication
TC-02	Test chip design and integration (IP under test, ring oscillators, PCM structures)	TC-06	Test chip board and lab setup preparation
TC-03	MPW shuttle slot booking and data submission scheduling	TC-07	Silicon characterization and correlation to simulation
TC-04	Test chip physical implementation and signoff	TC-08	Feedback into the production design — model correction and margin decisions

DESIGN & VERIFY

08 **RTL Design & Integration** **RTL**

w20–w52

RTL-01	Block microarchitecture specification	RTL-06	Lint, CDC and RDC closure
RTL-02	Block-level RTL implementation	RTL-07	Register map / RDL definition and header generation
RTL-03	Third-party and internal IP integration — wrappers, glue, configuration	RTL-08	Trial synthesis feedback loop on RTL PPA
RTL-04	Chip-level integration and top assembly	RTL-09	CI build, nightly regression and release management
RTL-05	Clock, reset and power intent (UPF) implementation	RTL-10	Specification change control and ECO board

09 **Verification** **DV**

w26–w66

DV-01	Verification plan and coverage model definition	DV-08	Emulation and FPGA prototype bring-up — capacity, partitioning, speed
DV-02	UVM environment and VIP bring-up	DV-09	Performance and bandwidth validation against the architecture model
DV-03	Block-level constrained-random and directed testing	DV-10	Coverage closure and regression stability management
DV-04	Chip-level and system-level scenario testing	DV-11	Gate-level simulation, functional and timing-annotated
DV-05	Formal property verification — control logic, connectivity, security	DV-12	Bug triage and disposition board
DV-06	Low-power (UPF) verification		
DV-07	AMS / mixed-signal co-simulation		

10 **DFT Architecture** **DFT**

w18–w58

DFT-01	DFT architecture definition — scan style, compression ratio, hierarchical DFT	DFT-07	ATPG pattern generation — stuck-at, transition, cell-aware — and coverage closure
DFT-02	Test coverage and test-time target negotiation with product and test engineering	DFT-08	Pattern validation by gate-level simulation and ATE format conversion (STIL/WGL)
DFT-03	MBIST / BIRA / BISR architecture for embedded memories	DFT-09	Scan compression and chain routing feasibility with physical design
DFT-04	Boundary scan and JTAG (1149.1 / 1687), TAP and debug access architecture	DFT-10	eFuse, chip ID and memory repair infrastructure
DFT-05	On-chip clock controller design for at-speed testing	DFT-11	Silicon debug hooks — trace, observability, design-for-debug
DFT-06	Scan insertion and DFT DRC during synthesis		

IMPLEMENT

11 **Logic Synthesis & Netlist Drops** **SYN**

w42–w66

SYN-01	SDC constraint development and validation	SYN-07	Low-power synthesis — isolation and level-shifter insertion against UPF
SYN-02	Technology mapping and optimization — multi-Vt, multi-bit, useful skew	SYN-08	N2 netlist drop — timing and congestion converged, late RTL changes absorbed
SYN-03	N0 flow-flush netlist release — functionally incomplete, structurally representative, issued so PD can build the flow	SYN-09	FFN (final full netlist) release — RTL frozen, ECOs closed, no functional change beyond this point
SYN-04	Physical-aware synthesis with congestion feedback from the PD flow setup	SYN-10	Formal equivalence checking per drop, RTL versus netlist
SYN-05	N1 netlist drop — first quality turn against PD placement and congestion feedback	SYN-11	Timing, area and power reporting against budget, per drop
SYN-06	Power optimization — clock gating, operand isolation, Vt mix	SYN-12	Netlist handoff and QoR delta review with physical design, per drop

12 **Physical Design** PD

w46–w76

PD-01	Flow setup and MMMC environment build on the N0 flow-flush netlist — scripts, decks, runtimes, capacity	PD-08	Detailed routing and DRC convergence
PD-02	Floorplan, macro placement and partition definition	PD-09	Multi-corner multi-mode timing closure
PD-03	Power delivery network design and early IR analysis	PD-10	Signal and power integrity iteration — crosstalk, EM
PD-04	Bump and RDL planning with chip-package co-design feedback	PD-11	Chip power model (CPM/GPS) extraction and handoff to package co-verification
PD-05	Turn 1 on the N1 drop — placement, CTS and first full route; QoR baseline established	PD-12	Scan chain reordering and DFT-aware routing
PD-06	Turn 2 on the N2 drop — congestion, timing and power convergence; closure risk quantified	PD-13	Final turn on the FFN — full closure, no functional change admitted
PD-07	Clock tree synthesis with skew and jitter budgeting, refined across turns	PD-14	Functional and timing ECO implementation
		PD-15	Chip finishing — seal ring, dummy and metal fill, alignment marks
		PD-16	Hierarchical block closure and top-level assembly

13 **Signoff** SO

w62–w78

SO-01	Signoff flow dry run on the Turn 2 database — decks, runtimes, capacity and violation triage rehearsed	SO-07	Reliability verification — ESD, latch-up, soft error / FIT
SO-02	Multi-corner multi-mode static timing analysis and closure on the final turn	SO-08	DFM and lithography checks, CMP and hotspot analysis
SO-03	Full-chip DRC, LVS, antenna and density verification	SO-09	Final formal equivalence and LVS netlist consistency
SO-04	EM and IR-drop signoff, static and dynamic	SO-10	Waiver review board and foundry waiver alignment
SO-05	Power and signal integrity signoff — crosstalk, noise	SO-11	Gate-level simulation with final SDF
SO-06	Chip-package-system co-analysis signoff review against SIPI results	SO-12	Signoff corner correlation against foundry decks

MANUFACTURE

14 **Tapeout & Mask Release (MTO)** T0

w78–w86

T0-01	Final GDSII / OASIS assembly on the FFN closure database, with layer map verification	T0-07	FEOL mask order confirmation and mask shop scheduling
T0-02	Final full-chip verification re-run on the released database	T0-08	BEOL design fix window — metal-layer ECOs and late fixes while FEOL masks are being cut
T0-03	Tapeout checklist completion and owner signoff	T0-09	BEOL layer re-verification — DRC, LVS, antenna and density on the fixed metal stack
T0-04	Open-issue risk assessment and waiver acceptance	T0-10	BEOL layer data preparation and MTO release
T0-05	Go / No-Go decision meeting	T0-11	BEOL mask order confirmation and full mask set completion tracking
T0-06	FEOL layer data preparation and MTO release to the mask shop		

15 **Fabrication** FAB

w79–w98

FAB-01	FEOL mask set fabrication, inspection and qualification	FAB-05	Back-end-of-line processing
FAB-02	Wafer start on FEOL mask availability	FAB-06	Inline metrology and defect inspection monitoring
FAB-03	BEOL mask set fabrication, inspection and qualification, in parallel with front-end processing	FAB-07	E-test / PCM data review
FAB-04	Front-end wafer processing	FAB-08	Wafer acceptance test and lot disposition
		FAB-09	Hot-lot management and WIP tracking
		FAB-10	Wafer shipment and logistics to sort and assembly

INTEGRATE

16 Package & Substrate Design

PKGD

w24–w76

- PKGD-01 Package architecture selection — 2.5D interposer type, organic substrate, layer count
- PKGD-02 Bump map, pitch and power-ground planning with physical design
- PKGD-03 Interposer / RDL routing design
- PKGD-04 Substrate stack-up, escape routing and package DRC
- PKGD-05 Package routing for signal integrity — length matching, reference plane and via design

- PKGD-06 Package PDN and decap footprint design (electrical closure owned by SIPI)
- PKGD-07 Thermal and mechanical simulation — warpage, co-planarity, TIM, lid
- PKGD-08 Test vehicle requirement definition and handoff to PTV
- PKGD-09 Substrate and interposer supplier selection and lead-time booking
- PKGD-10 OSAT selection and assembly process definition
- PKGD-11 Package design freeze, DRC and tooling release

17 Package Test Vehicle (CPI / TTV)

PTV

w36–w88

- PTV-01 Test vehicle strategy and risk item definition — CPI, warpage, thermal, ULK/BEOL stress
- PTV-02 Mechanical test vehicle (MTV) design — bump array, dummy die stack, substrate coupon
- PTV-03 Thermal test vehicle (TTV) design — embedded heater array and RTD sensor placement
- PTV-04 Daisy-chain electrical vehicle design — interconnect continuity and bump yield structures
- PTV-05 TV die fabrication and interposer / substrate vehicle build
- PTV-06 OSAT assembly of vehicles with process window DOE

- PTV-07 Warpage and co-planarity measurement across the reflow profile (shadow moiré)
- PTV-08 Chip-package interaction stress evaluation — low-k / ULK crack, bump and BEOL integrity
- PTV-09 Thermal characterization — Rjc / Rja, TIM performance, hotspot mapping against the thermal model
- PTV-10 Board-level reliability on vehicles — temperature cycle, drop, bend
- PTV-11 Daisy-chain continuity test and assembly yield learning
- PTV-12 Feedback into package design and assembly process window freeze

18 Chip-Package-System Co-Verification

SIPI

w50–w76

- SIPI-01 Chip power model (CPM/CPS) extraction per power domain and operating mode
- SIPI-02 Package and board model extraction — 3D EM, S-parameter and RLC network
- SIPI-03 Die-package-board PDN co-simulation — DC and AC impedance versus target
- SIPI-04 Dynamic voltage-drop analysis with package inductance and real switching profiles
- SIPI-05 Decoupling capacitor budget and placement optimization across die, package and board
- SIPI-06 SSN / SSO and simultaneous-switching noise analysis at the IO ring

- SIPI-07 High-speed channel simulation with extracted package models — insertion loss, crosstalk, return path
- SIPI-08 Eye diagram, jitter and BER budget closure per interface (PCIe/CXL, HBM, die-to-die)
- SIPI-09 Power-aware timing correlation — voltage drop back-annotated into STA
- SIPI-10 Electro-thermal co-analysis coupled with the package thermal model
- SIPI-11 Co-verification signoff review and criteria disposition with design and package teams

19 Assembly & Package Build

ASSY

w98–w106

- ASSY-01 Known-good-die sort and selection
- ASSY-02 HBM stack procurement and incoming inspection
- ASSY-03 Die attach, micro-bump and thermo-compression bonding
- ASSY-04 Interposer and substrate assembly, underfill, molding
- ASSY-05 Lid / TIM attach and ball attach

- ASSY-06 Assembly inline inspection — X-ray, CSAM, warpage metrology
- ASSY-07 Assembly yield analysis and process tuning
- ASSY-08 Package-level open / short and continuity test
- ASSY-09 Unit build for bring-up, qualification and customer samples

VALIDATE & RAMP

20 Validation Hardware / EVB **EVB**

w52–w90

EVB-01	Validation platform requirements and topology definition
EVB-02	EVB schematic design — power tree, clocking, host interface, connectors
EVB-03	PCB layout with high-speed channel SI/PI simulation
EVB-04	PCB fabrication and assembly
EVB-05	Power delivery, VRM and telemetry design and bring-up

EVB-06	Debug infrastructure — JTAG and trace pods, interposers, instrumentation headers
EVB-07	Board bring-up with socketed or dummy parts
EVB-08	Thermal solution and cooling for the lab platform
EVB-09	Lab instrumentation reservation and test rack build
EVB-10	Minimum host-side enablement for power-on (BSP handoff boundary)

21 Test Development **TEST**

w54–w96

TEST-01	Test plan and coverage strategy — wafer sort, final test, system-level test
TEST-02	ATE platform selection and tester time booking
TEST-03	Probe card design, fabrication and qualification
TEST-04	Load board / DUT board design, fabrication and bring-up
TEST-05	Wafer sort test program development
TEST-06	Final / package test program development

TEST-07	Characterization test content — shmoo, Vmin/Fmax, PVT sweeps
TEST-08	DFT pattern porting to ATE format and pattern debug
TEST-09	Test time and cost optimization — parallelism, site count
TEST-10	Correlation between ATE, bench and system results
TEST-11	Test data infrastructure — STDF collection and yield database

22 Silicon Bring-up **BU**

w102–w120

BU-01	Sample receipt, incoming inspection and board mounting
BU-02	Power-on, power sequencing and basic health check
BU-03	Reset, clocking and PLL lock validation
BU-04	Boot, firmware load and functional smoke test
BU-05	Interface bring-up — PCIe/CXL link training, HBM training, die-to-die
BU-06	Memory subsystem and bandwidth validation

BU-07	Shmoo across voltage, frequency and temperature
BU-08	Performance validation against the architecture model
BU-09	Silicon debug of anomalies and failure analysis requests
BU-10	Errata capture, workaround definition and documentation
BU-11	Respin versus metal-fix decision analysis — ECO scope, cost, schedule
BU-12	Customer sample readiness and release package

23 Qualification & Production **MP**

w106–w132

MP-01	Qualification plan definition against JEDEC / AEC standards
MP-02	Reliability stress — HTOL, HTS, temperature cycle, uHAST, THB
MP-03	ESD (HBM, CDM) and latch-up qualification
MP-04	Package qualification — MSL, drop, bend, board-level reliability
MP-05	Yield learning, failure pareto and defect analysis
MP-06	Production test program release and guard-band validation

MP-07	Test time reduction and multi-site conversion
MP-08	Process corner and split-lot validation
MP-09	Production readiness review and change control (PCN) setup
MP-10	Capacity, supply chain and ramp commitment — wafer starts, substrate, HBM
MP-11	Compliance and certification — PCIe/CXL compliance, RoHS/REACH, safety
MP-12	Documentation release — datasheet, user guide, errata, application notes

APPENDICES

A Glossary **TERMS**

184 entries

B Schedule tensions **SCHED**

49 late inputs

DEFINE

01 Product Definition

DEF

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH		
w0 – w8	8 _w	9	28 _{M/M}	5 _{of 9}		
ENGINEERING ACTIVITIES				KEY DELIVERABLES		
DEF-01	Market and customer requirements consolidation		4w	4 M/M	DEF-D1	Product requirements document (PRD)
DEF-02	Workload / use-case profiling and target KPI derivation (TOPS, TOPS/W, tokens/s)		5w	6 M/M	DEF-D2	Target specification — PPA and KPI table
DEF-03	PPA target definition — max frequency, power, die area		4w	3 M/M	DEF-D3	Product cost and margin model
DEF-04	Die size, yield and product cost model (wafer, mask, package, test, ASP)		4w	3 M/M	DEF-D4	Feasibility report
DEF-05	Memory and interface bandwidth requirements (HBM stacks, PCIe/CXL lanes, D2D)		3w	2 M/M	DEF-D5	Program charter, staffing and budget plan
DEF-06	Competitive benchmarking and product positioning		4w	2 M/M	DEF-D6	Kickoff Go / No-Go decision record
DEF-07	Feasibility assessment against candidate technology nodes		4w	3 M/M		
DEF-08	Program schedule, resource and staffing plan drafting		4w	3 M/M		
DEF-09	Business case preparation and funding approval package		3w	2 M/M		

01 DEF · PRODUCT DEFINITION

DEF-01

Market and customer requirements consolidation

TAKES	COSTS	OWNER	CRITICAL PATH
4w w0–w4 of the stage	4.0 M/M ~4.3 people while it runs	Requirements lead Stage lead approves	yes slipping moves the program

WHY IT EXISTS

Consolidate what customers and the market are asking for into a single, ranked and approved set of **product requirements** that the entire program is measured against.

Everything that follows is driven by these product requirements—from the PPA targets and cost model to the feasibility assessment and the architecture itself. Until the requirements are baselined and approved, all downstream decisions remain provisional. Changes introduced late in the program are significantly more expensive, often resulting in costly silicon redesigns.

WHAT IT DELIVERS

DEF-D1

Product requirements document (PRD)

The requirements list built here becomes the body of the PRD. This activity does not contribute to the PRD—it produces its content.

NEEDS FIRST

- Customer RFI / RFQ responses
- Field feedback and errata from the previous generation
- Analyst forecasts and segment sizing
- Competitor datasheets and public benchmarks
- Sales pipeline and committed-volume view

DONE WHEN

- Every requirement has an owner, a priority and an acceptance measure
- Conflicts resolved, or deferred with a named decision owner
- Register baselined and signed by product, engineering and sales

HOW IT GETS THERE

7 STEPS · 2 LANES

w0	w1	w2	w3	w4
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
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1	Stakeholder map and interview plan → Stakeholder map and interview schedule	0.5w
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#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Anchor-customer interviews — two accounts, six sessions → Interview notes, per account	1.5w
3	Market and analyst input, segment sizing → Market and analyst input with segment sizing	1w <i>runs in parallel</i>
4	Requirement extraction, de-duplication and requirements list build → Requirements list — ID, owner, priority, acceptance measure	1w
5	Conflict resolution and priority ranking — must-have, should-have, nice-to-have → Conflict and deferral log · Priority ranking record	0.5w
6	Traceability tagging to candidate KPIs and PPA targets → Requirement-to-KPI traceability tags	0.5w <i>runs in parallel</i>
7	Stakeholder review and baseline signoff → Baseline signoff sheet	0.5w

The seven steps total 5.5 weeks of work inside a 4-week window because steps 3 and 6 run in parallel with the main sequence. This is why the activity TAT does not equal the sum of its step durations. If either parallel step has to move onto the critical path, the activity takes 1.5 weeks longer.

WATCH OUT FOR

- ◆ **Requirements not signed off by all stakeholders.** Verbal agreement from sales is not a signature, and the Go / No-Go gate review checks for signatures. Missing approvals surface at the gate, when there is no time left to resolve them.
- ◆ **Late customer additions after baseline.** If no change-control process is agreed at signoff, requirements continue to change and no downstream activity can freeze its own inputs.
- ◆ **Requirements written as solutions.** "Use HBM3E" specifies an implementation; "819 GB/s sustained bandwidth" specifies a requirement. Writing the solution into the requirement removes the architecture team's options in ARCH-02.
- ◆ **Requirements without an acceptance measure.** A requirement that cannot be tested is never verified during the program and resurfaces at qualification, when the only remaining options are a respin or a customer waiver.
- ◆ **A single anchor account dominating the requirement set.** Requirements that serve one customer rather than a market segment produce a custom part that still carries a full product's development cost.

WHERE THE EFFORT GOES	4 M/M	WHO IS ON IT ~4.3 FTE	CONNECTIONS	TERMS HERE 10
Interviews and market input	1.5	Requirements lead <i>OWNS IT</i>	RUNS WITH DEF-06	TAT PPA KPI ASP
Register build and traceability	1.5	Customer engagement	FEEDS INTO DEF-02 DEF-04 DEF-05	PRD RFI RFQ GB/s
Workshops and review	0.5	Product cost analyst	DEF-07 DEF-08 ARCH-01	ID HBM3E
Documentation and signoff	0.5	Feasibility liaison	ARCH-07 TECH-01 TECH-09	
		Stage lead	IPR-01 PKGD-01 <i>Nothing upstream — this is the program's first activity. What it needs is bought, licensed or asked for, not produced by anything before it.</i>	

01 DEF · PRODUCT DEFINITION

DEF-02

Workload / use-case profiling and target KPI derivation (TOPS, TOPS/W, tokens/s)

TAKES	COSTS	OWNER	CRITICAL PATH
5w	6.0 M/M	Workload architect	yes
w1–w6 of the stage	~5.2 people while it runs	Architecture liaison approves	slipping moves the program

WHY IT EXISTS

The product requirements define what customers expect from the product. This activity defines **what the silicon must actually do**—the workloads it will run and the KPI targets derived from those workloads.

A KPI without a representative workload is just a marketing number. During ARCH-01, the architecture team spends fourteen weeks optimizing the design based on the workloads defined here. If those workloads do not reflect real customer usage, the silicon will be optimized for scenarios that customers never actually run. Verification will not catch this issue, because it only checks that the design functions correctly—not that it was optimized for the right workloads.

WHAT IT DELIVERS

DEF-D2

Target specification — PPA and KPI table

This activity derives the KPI half of the deliverable; DEF-03 derives the PPA half. The two are published as a single table.

NEEDS FIRST

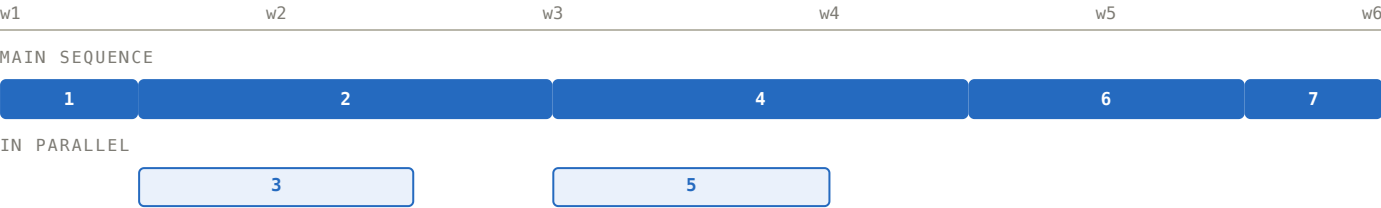
- Requirement register from DEF-01
- Customer-supplied models and traces, where the NDA allows
- Previous-generation silicon and its lab platform
- Public model zoo and framework benchmarks
- Deployment mix estimates from the segment sizing

DONE WHEN

- Every KPI traces to a named workload and stated conditions
- Workload suite runs end to end on the reference platform
- Architecture has accepted the handoff as modellable

HOW IT GETS THERE

7 STEPS · 2 LANES



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Use-case shortlist drawn from the requirement register → Use-case shortlist with deployment weights	0.5w
2	Workload trace capture — LLM inference, vision, recommendation → Captured workload traces per use case	1.5w
3	Reference runs on the previous generation for a baseline → Baseline measurements on the previous generation	1w <i>runs in parallel</i>
4	KPI derivation — TOPS, TOPS/W, tokens/s, time-to-first-token, batch scaling → KPI derivation worksheet, one row per requirement	1.5w
5	Precision and quantization policy — INT8, FP8, mixed → Precision and quantization policy note	1w <i>runs in parallel</i>
6	Sensitivity study — which KPI moves with which architectural knob → Sensitivity table — KPI against architectural knob	1w
7	Workload suite packaging and handoff to architecture → Workload suite — traces, models, and the harness to run them	0.5w

The main sequence fills the 5-week window exactly. Steps 3 and 5 run in parallel: the reference runs need lab time rather than analyst time, and the precision policy can be settled while traces are still being captured.

WATCH OUT FOR

- ◆ **Workload traces that flatter the design.** A workload suite assembled from benchmarks that already run well on the previous generation will not expose the cases that break the new architecture.
- ◆ **Customer models unavailable under NDA.** Substituting a public proxy workload is acceptable practice. Not recording that a proxy was used is not, because downstream teams will treat the results as customer data.
- ◆ **Deployment mix estimated rather than measured.** The architecture is tuned to the workload mix, so an incorrect mix allocates silicon area to the wrong functions. That area cannot be recovered after tapeout.
- ◆ **Precision policy left undecided.** The choice between INT8 and FP8 changes the datapath, the memory footprint and the KPI targets at the same time, and the cost of changing it increases every week it is deferred.
- ◆ **KPIs quoted without operating conditions.** A TOPS number with no batch size, sequence length or precision stated cannot be designed against and cannot be verified.

WHERE THE EFFORT GOES	6 M/M	WHO IS ON IT ~5.2 FTE	CONNECTIONS	TERMS HERE 10
Trace capture and reference runs	2.0	Workload architect <i>OWNS IT</i>	DEPENDS ON DEF-01	PPA KPI PRD NDA
KPI derivation	1.5	ML performance engineer	FEEDS INTO DEF-03 DEF-05 DEF-07	TOPS TOPS/W INT8
Sensitivity study	1.0	Systems analyst	ARCH-01 ARCH-03 DV-08	FP8 LLM ML
Precision and quantization policy	1.0	Customer engagement	DV-09	
Suite packaging and handoff	0.5	Architecture liaison		

01 DEF · PRODUCT DEFINITION

DEF-03

PPA target definition — max frequency, power, die area

TAKES	COSTS	OWNER	CRITICAL PATH
4w w3–w7 of the stage	3.0 M/M ~3.3 people while it runs	PPA lead Stage lead approves	yes slipping moves the program

WHY IT EXISTS

Convert the KPI set into the three numbers the design is actually held to—**max frequency, power and die area**—and allocate them so that every block owner inherits a specific budget rather than a program-level goal.

An unallocated target does not constrain anyone's design work. A board power target of 75 W does not bind any block owner until it is decomposed into a die-level budget, a per-domain budget and a per-block budget, each with a named owner. That decomposition is what makes a budget overrun visible in week 20 rather than in week 60, when the only remaining responses are reducing the speed bin or committing to a respin.

WHAT IT DELIVERS

DEF-D2

Target specification — PPA and KPI table

This activity produces the PPA half of the deliverable. DEF-02 produces the KPI half, and the two are published as a single table.

NEEDS FIRST

- KPI table from DEF-02
- Block list and reuse inventory from the architecture sketch
- Node PPA data from TECH-03
- Previous-generation power and area actuals
- Thermal and board constraints from the requirement register

DONE WHEN

- Frequency, power and area targets each carry a stated assumption
- Every block has an allocated budget and a named owner
- Architecture and physical design have signed the review

HOW IT GETS THERE

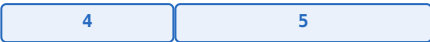
7 STEPS · 2 LANES

w3	w4	w5	w6	w7
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Frequency target from the KPI set and expected IPC → Frequency target with the IPC assumption behind it	0.75w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Power envelope decomposition — board, package, die, per-domain → Power budget tree — board to per-domain	1w
3	Die area budget from the block list and the reuse inventory → Die area budget by block	1w
4	Multi-Vt and operating voltage strategy, first pass → Voltage and multi-Vt strategy note	0.5w
5	PPA sensitivity against the candidate nodes → PPA sensitivity table per candidate node	0.75w
6	Per-block allocation with a named owner against each budget → Per-block allocation sheet with owners	0.75w
7	Target review with architecture and physical design → Target review record with architecture and physical design	0.5w

Allocation is the step that matters, and it sits late in the sequence because it cannot begin until the block list is stable. Steps 4 and 5 run in parallel with the budget work; both feed the review rather than the budgets themselves.

WATCH OUT FOR

- ◆ **Targets set from ambition rather than from process capability.** A max frequency target the standard cell library cannot reach leads to an extended timing closure effort that ends in either a respin or a reduced speed bin.
- ◆ **Power budget allocated with no margin.** A budget fully committed at definition leaves nothing for the ECO activity during physical design, where power consistently increases rather than decreases.
- ◆ **Area budget set without the IP reuse inventory.** Budgets set before the reuse decisions are known allocate area to blocks that will be reused and starve blocks that must be designed from scratch.
- ◆ **Budgets allocated without named owners.** A budget with no owner is not defended during implementation, and the available area or power is consumed by whichever block closes first.
- ◆ **Targets frozen before DEF-07 reports.** Feasibility can only be assessed against stated targets. Freezing targets that the selected node cannot support makes the feasibility verdict meaningless.

WHERE THE EFFORT GOES	3 M/M	WHO IS ON IT ~3.3 FTE	CONNECTIONS	TERMS HERE	5
Frequency and power budgets	1.0	PPA lead owns IT	DEPENDS ON DEF-02 DEF-05	PPA KPI IP ECO	
Die area budget	0.8	Architect	RUNS WITH TECH-03	IPC	
Node sensitivity	0.5	Power analyst	FEEDS INTO DEF-04 DEF-07 ARCH-01		
Per-block allocation	0.5	Physical design liaison	ARCH-05 ARCH-06 ARCH-08		
Review and arbitration	0.3	Stage lead	TECH-02 SYN-01 PD-09 PKG0D-07 BU-08 MP-01		

01 DEF · PRODUCT DEFINITION

DEF-04

Die size, yield and product cost model
(wafer, mask, package, test, ASP)

TAKES	COSTS	OWNER	CRITICAL PATH
4w w3–w7 of the stage	3.0 M/M ~3.3 people while it runs	Product cost analyst Finance partner approves	no has float

WHY IT EXISTS

Determine what the product will cost **before the program commits to building it**. Wafer, mask, package, test and yield are combined into a single model that states the cost per unit at volume and whether the resulting margin is acceptable.

The cost model also functions as a veto. A design that meets every KPI target but exceeds the cost ceiling is not a viable product, and the least expensive point to discover that is here, while the die size is still an estimate rather than a completed floorplan.

WHAT IT DELIVERS

DEF-D3

Product cost and margin model

This activity produces the deliverable. The workbook and its assumptions are handed over in full rather than summarized, because downstream teams need to re-run the model when inputs change.

NEEDS FIRST

- Die area budget from DEF-03
- Wafer, mask and NRE quotations from TECH-04
- Package architecture assumption from the requirement register
- HBM and substrate price indications
- Volume forecast and ASP from the segment sizing

DONE WHEN

- Cost model closes at the forecast volume with stated margin
- Every assumption is named, sourced and dated
- Sensitivity identifies the three exposures and their range

HOW IT GETS THERE

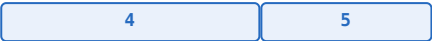
7 STEPS · 2 LANES

w3	w4	w5	w6	w7
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Die area estimate from the block list and the reuse inventory → Die area estimate with its tolerance	0.75w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Wafer cost and gross die per wafer, per node candidate → Gross die and yield curves per node candidate	0.5w
3	Defect density and yield model per node candidate → Defect density and yield curve per node	1w
4	Package and substrate cost — interposer, substrate layers, HBM stacks → Package and substrate cost breakdown	0.75w <i>runs in parallel</i>
5	Test cost model — sort and final, with a test-time assumption → Test cost model with its test-time assumption	0.5w <i>runs in parallel</i>
6	Volume, ASP and margin roll-up → Cost model workbook — wafer, package, test, yield · Margin roll-up at the forecast volume · Cost ceiling statement for the design	1w
7	Sensitivity — the three assumptions the model is most exposed to → Sensitivity analysis on the three dominant assumptions	0.75w

The sensitivity analysis in step 7 is not optional. On a 2.5D product, cost is dominated by yield, interposer supply and HBM pricing, and at this point in the program all three are assumptions rather than quoted prices.

WATCH OUT FOR

- ◆ **Yield assumed from a mature process node.** Defect density during early node ramp is significantly worse than the mature figure quoted by the foundry, and the entire difference is absorbed by product margin.
- ◆ **Package cost treated as a minor adder.** On a 2.5D product with two HBM stacks, package cost can approach die cost. Modeling it as a percentage adder hides the largest cost line the program can still influence.
- ◆ **Test cost estimated without a test time assumption.** Test cost is test time multiplied by the tester rate. Without a test time estimate from the DFT team, the test cost line carries no information.
- ◆ **A single-point ASP assumption.** An ASP with no downside scenario produces a margin number that cannot be committed to.
- ◆ **Die size estimate carried without a tolerance.** An area estimate that later grows by 15% reduces gross die per wafer non-linearly, but the model reports the change as a small input adjustment.

WHERE THE EFFORT GOES	3 M/M	WHO IS ON IT ~3.3 FTE	CONNECTIONS	TERMS HERE	7
Yield model per node	1.0	Product cost analyst owns IT	DEPENDS ON DEF-01 DEF-03 DEF-06 TECH-07	PPA KPI NRE ASP DFT HBM (ESD) HBM	
Cost roll-up and margin	0.8	Yield engineer	RUNS WITH TECH-04		
Package and test cost lines	0.8	Package cost liaison	FEEDS INTO DEF-07 DEF-09 ARCH-02 PDK-06 DFT-02 PKGD-01 TEST-01 TEST-07 MP-01 MP-05		
Sensitivity analysis	0.5	Test engineering liaison Finance partner	LATER INPUT AMS-08 +31w TC-08 +49w		

01 DEF · PRODUCT DEFINITION

DEF-05

Memory and interface bandwidth requirements
(HBM stacks, PCIe/CXL lanes, D2D)

TAKES	COSTS	OWNER	CRITICAL PATH
3w w2–w5 of the stage	2.0 M/M ~2.9 people while it runs	Memory systems architect Architecture liaison approves	no has float

WHY IT EXISTS

Determine the bandwidth the target workloads require at each level of the memory hierarchy, and convert that into **the interface count the package and floorplan must physically support**.

Bandwidth requirements are where the specification becomes a physical commitment. Selecting two HBM stacks rather than one is not a documentation change; it determines interposer size, substrate construction, power delivery and product cost. Making this decision during product definition is inexpensive. Revisiting it during architecture is not.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Workload traces and KPI table from DEF-02
- Requirement register — capacity and latency requirements
- HBM generation availability and roadmap
- Interface standard revisions in flight (PCIe, CXL, UCIe)
- Previous-generation bandwidth utilization data

DONE WHEN

- Bandwidth stated as sustained, per workload, with conditions
- Memory configuration and interface count recommended with a reason
- Bump budget accepted as buildable by package

HOW IT GETS THERE

6 STEPS · 2 LANES

w2

w3

w4

w5

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Bandwidth demand per workload, from the DEF-02 traces → Bandwidth demand table, per workload and per hierarchy level	0.75w
2	Memory hierarchy sizing — on-die SRAM, HBM capacity and channels → Memory configuration recommendation — capacity, stacks, channels	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Host interface sizing — PCIe / CXL generation and lane count → Host interface sizing — generation and lane count	0.5w runs in parallel
4	Die-to-die interface requirement, if a chiplet split is a candidate → Host and die-to-die interface requirement	0.5w runs in parallel
5	Translation into a pin and bump budget → Pin and bump budget estimate	0.75w
6	Requirement review with architecture and package → Review record with architecture and package	0.5w

Step 5 is the step that creates the commitment. A bandwidth figure stated in GB/s does not constrain any downstream team. The same figure expressed as a bump count constrains the floorplan, the package design and the cost model at the same time.

WATCH OUT FOR

- ◆ **Bandwidth sized on peak rather than sustained demand.** Peak figures size an interface the workload never reaches and add cost the product does not need.
- ◆ **Capacity and bandwidth requirements conflated.** HBM stack count is frequently set by capacity requirements while the discussion is conducted in GB/s. The two constraints do not scale together, and confusing them produces the wrong stack count.
- ◆ **Interface standard still under revision.** Committing to an interface generation whose specification is not yet ratified transfers the standards body's schedule risk into the program schedule.
- ◆ **Bump budget never derived.** A bandwidth requirement that is not translated into a bump count is discovered as a floorplan problem during PD-04, by which time the package has already been ordered.
- ◆ **Chiplet decision deferred.** A die-to-die interface is a major architectural commitment. Leaving the decision open past architecture freeze forces the program to adopt the monolithic option by default.

WHERE THE EFFORT GOES	2 M/M	WHO IS ON IT	~2.9 FTE	CONNECTIONS	TERMS HERE	11
Hierarchy sizing and bandwidth analysis	1.0	Memory systems architect OWNS IT		DEPENDS ON DEF-01 DEF-02	PPA KPI PRD NDA	
Interface sizing	0.5	Interface architect		FEEDS INTO DEF-03 ARCH-01 ARCH-03	SRAM HBM (ESD) HBM	
Bump budget translation and review	0.5	Workload analyst Package liaison Architecture liaison		ARCH-04 PKG-01 PKG-02 TEST-02 MP-10 MP-11	PCIe CXL UCIe GB/s	

01 DEF · PRODUCT DEFINITION

DEF-06

Competitive benchmarking and product positioning

TAKES	COSTS	OWNER	CRITICAL PATH
4w w0–w4 of the stage	2.0 M/M ~2.2 people while it runs	Product marketing lead Stage lead approves	no has float

WHY IT EXISTS

Establish where the product will land **against the parts that will be shipping when it ships**, rather than against the parts available today.

Benchmarking against the current generation produces a favorable comparison with no predictive value. The program's silicon reaches customers roughly thirty months from now, so the relevant comparison is against products launching in that window. Those figures have to be projected from public roadmaps and process node cadence rather than measured directly.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Competitor datasheets, disclosures and conference material
- Public benchmark submissions (MLPerf and equivalents)
- Foundry node cadence and process roadmaps
- Analyst launch forecasts
- Field intelligence from sales

DONE WHEN

- Every comparison is normalized to stated conditions
- Field projected to our launch window, not read from today
- Gap list handed to DEF-01 as candidate requirements

HOW IT GETS THERE

6 STEPS · 2 LANES

w0	w1	w2	w3	w4
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Competitor set definition and launch-window mapping → Competitor set with launch-window estimates	0.5w
2	Public benchmark, datasheet and disclosure collection → Collected competitor benchmarks and datasheets	1w
3	Normalization to comparable metrics and conditions → Normalized comparison table with stated conditions	1w

runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Projection to our own launch window → Projection of the competitive field at our launch	1w
5	Positioning statement and capability gap list → Positioning statement · Capability gap list, ranked	1w
6	Review with product and sales → Product and sales review record	0.5w

Normalization runs in parallel because it is time-consuming and independent of data collection. Vendors quote TOPS at different precisions, batch sizes and power points, and none of the published figures are comparable until they have been restated on a common basis.

WATCH OUT FOR

- ◆ **Benchmarking against currently shipping parts.** The comparison that determines the product's position is against parts launching thirty months out, and those numbers have to be projected rather than measured.
- ◆ **Vendor figures accepted without normalization.** A TOPS figure quoted at a precision, batch size and power point chosen by the vendor is not comparable until it has been restated at common conditions.
- ◆ **Capability gaps recorded without an owner.** A gap that does not become a requirement with a named owner remains an observation, and observations do not change the silicon.
- ◆ **Positioning based on a single metric.** A product positioned only on TOPS/W competes poorly against one positioned on total cost of ownership, and that comparison is not made until launch.
- ◆ **Competitor launch dates treated as fixed.** A schedule slip in the competitive field affects the product's position as much as a slip in this program, and neither can be predicted to within a quarter.

WHERE THE EFFORT GOES	2 M/M	WHO IS ON IT	~2.2 FTE	CONNECTIONS	TERMS HERE	4
Collection and competitor set	0.8	Product marketing lead OWNS IT		RUNS WITH DEF-01	ASP TOPS TOPS/W MLPerf	
Normalization and projection	0.8	Performance analyst		FEEDS INTO DEF-04 DEF-09 TEST-01		
Positioning and gap list	0.5	Technology analyst		MP-05		
		Sales liaison		LATER INPUT TEST-09 +58w		
		Stage lead				

01 DEF · PRODUCT DEFINITION

DEF-07

Feasibility assessment against candidate technology nodes

TAKES	COSTS	OWNER	CRITICAL PATH
4w w4–w8 of the stage	3.0 M/M ~3.3 people while it runs	Feasibility lead Stage lead approves	yes slipping moves the program

WHY IT EXISTS

Assess the requirement set, the PPA targets and the cost model against real process nodes, and return a verdict: **buildable as specified, buildable with named concessions, or not buildable**.

Feasibility is the only activity in this stage whose output can be "no", and its value depends entirely on that answer being available. A feasibility study run after the node has been selected and the budget committed documents a decision rather than informing one, and it cannot stop a program that should be stopped.

WHAT IT DELIVERS

DEF-D4

Feasibility report

This activity produces the deliverable. The verdict, the assessment matrix and the concession list are published together, because the verdict on its own cannot be acted on.

NEEDS FIRST

- Requirement register from DEF-01
- PPA targets from DEF-03
- Cost model from DEF-04
- Node characterization and DTCO study from TECH-03
- IP availability indications from vendors and internal inventory

DONE WHEN

- Every target has an achievability finding per candidate node
- Concessions named and costed, not described
- Verdict recorded with a recommendation and a dissent record

HOW IT GETS THERE

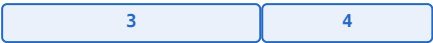
7 STEPS · 2 LANES

w4	w5	w6	w7	w8
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Candidate node shortlist agreed with TECH → Agreed candidate node shortlist	0.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	PPA achievability per node — density, frequency, leakage → Node-by-node achievability matrix	1.25w
3	IP availability check per node, ahead of IPR → IP availability finding per node	0.75w runs in parallel
4	Cost model re-run per node → Cost model results per node	0.5w runs in parallel
5	Package and thermal feasibility at the target power → Thermal and package feasibility note	0.75w
6	Concession list — what must be given up, per node → Concession list per node	0.75w
7	Verdict and recommendation → Feasibility verdict with a named recommendation	0.75w

The concession list is the part of the deliverable that actually gets used. A binary verdict gives the decision review nothing to act on. A list stating "this node, at 200 MHz below target, with the PHY licensed rather than developed internally" gives the review a decision it can make.

WATCH OUT FOR

- ◆ **Feasibility assessed after the node has been selected.** The study then documents an existing decision rather than informing one, and its independence is lost.
- ◆ **Optimistic assessment at the boundary.** A target described as reachable "with aggressive timing closure" is reachable in a presentation and not in a timing report.
- ◆ **IP availability taken from the vendor roadmap.** An IP block promised on the target node but not yet ported is a schedule risk, and recording it as available converts that risk into a hidden dependency.
- ◆ **Thermal feasibility checked at typical rather than maximum power.** Feasibility at average power says nothing about the part at its rated power, which is the condition the product is sold at.
- ◆ **No concession list produced.** A yes-or-no verdict gives the decision review no third option, and in practice the concession option is the one most often taken.

WHERE THE EFFORT GOES	3 M/M	WHO IS ON IT ~3.3 FTE	CONNECTIONS	TERMS HERE	4
PPA achievability per node	1.0	Feasibility lead owns it	DEPENDS ON DEF-01 DEF-02 DEF-03 DEF-04 TECH-03 TECH-04 TECH-07	PPA DTC0 IP PHY	
IP and cost checks	0.8	PPA analyst	RUNS WITH TECH-01		
Package and thermal feasibility	0.5	IP strategist	FEEDS INTO DEF-08 DEF-09 ARCH-01 IPR-03		
Concession list and verdict	0.8	Thermal and package liaison Stage lead			

DEF-08

Program schedule, resource and staffing plan drafting

TAKES	COSTS	OWNER	CRITICAL PATH
4w w4–w8 of the stage	3.0 M/M ~3.3 people while it runs	Program manager Finance partner approves	no has float

WHY IT EXISTS

Convert the milestone profile into a dated plan for **this** program, state what it requires in engineering headcount, and compare that requirement against the resources the organization actually has.

A schedule built on a headcount curve the organization cannot staff is not a plan; it is a slip that has already been committed to. The value of this activity is the comparison between required and available capacity, not the Gantt chart itself.

WHAT IT DELIVERS

DEF-D5

Program charter, staffing and budget plan

The schedule, the headcount curve and the effort roll-up are the operative content of the charter.

NEEDS FIRST

- Milestone profile and stage baselines
- Effort figures per activity from the template
- Organization capacity and current program commitments
- Supplier lead times — substrate, interposer, probe card, IP, shuttle
- Feasibility concessions from DEF-07

DONE WHEN

- Schedule dated from a committed kickoff
- Headcount curve reconciled against real capacity, gaps named
- Critical path derived and buffers placed against the fixed gates

HOW IT GETS THERE

7 STEPS · 2 LANES

w4	w5	w6	w7	w8
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Profile selection and kickoff date → Selected profile with a dated kickoff	0.5w
2	Stage baseline adjustment for this program's scope → Program schedule with dated stage boundaries and milestones	0.75w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Effort roll-up by stage and by discipline → Effort roll-up by stage and discipline	1w
4	Headcount curve set against the organization's real capacity → Headcount curve against available capacity	1w
5	External lead-time overlay — substrate, probe card, IP, shuttle → Lead-time overlay and order-by dates	0.75w <i>runs in parallel</i>
6	Risk buffer placement and critical path identification → Critical path and buffer placement record	0.75w
7	Schedule review and commitment → Schedule commitment record	0.5w <i>runs in parallel</i>

The supplier lead-time overlay runs in parallel and changes more dates than the internal plan does. A probe card and an advanced substrate each carry roughly twenty weeks of lead time, they are ordered against dates this activity sets, and once ordered they are outside the program's control.

WATCH OUT FOR

- ◆ **A headcount curve the organization cannot staff.** A peak requirement of 200 engineers in a 120-engineer organization is a schedule slip recorded as a plan.
- ◆ **Template baselines used without adjustment.** The profile describes a typical program. A derivative product or a first program on a new node is not typical, and the differences belong in the plan.
- ◆ **Supplier lead times discovered rather than planned.** Probe card and substrate lead times run to twenty weeks. Discovered late, they move the wafer-out date rather than the purchase order date.
- ◆ **Schedule buffer placed at the end of the program.** A buffer after the final stage protects nothing. Buffers belong immediately before the gates that cannot move.
- ◆ **Critical path assumed rather than derived.** On a 23-stage profile with parallel enablement work, the critical path frequently runs through technology or package enablement rather than through the design flow, and assuming otherwise directs management attention to the wrong stages.

WHERE THE EFFORT GOES	3 M/M	WHO IS ON IT	~3.3 FTE	CONNECTIONS	TERMS HERE	1
Effort roll-up by stage and discipline	1.0	Program manager <i>OWNS IT</i>		DEPENDS ON DEF-01 DEF-07	IP	
Headcount curve against capacity	0.8	Resource manager		FEEDS INTO DEF-09 TECH-05 PDK-12		
Lead-time overlay	0.5	Discipline leads		IPR-09 RTL-10 FAB-09		
Buffer and critical path	0.5	Procurement liaison		PKGD-09 TEST-02		
Review and commitment	0.3	Finance partner				

01 DEF · PRODUCT DEFINITION

DEF-09

Business case preparation and funding approval package

TAKES	COSTS	OWNER	CRITICAL PATH
3w w5–w8 of the stage	2.0 M/M ~2.9 people while it runs	Program manager Executive sponsor approves	yes slipping moves the program

WHY IT EXISTS

Assemble what the stage produced into **the business case the funding decision is made on**, and present a named recommendation to the people with the authority to decline it.

This activity produces no new analysis. Its work is selection and accuracy: the feasibility verdict, the cost model and the resourced schedule are carried into the review intact, with the downside scenario shown rather than described. A business case that only closes under base-case assumptions has not been adequately reviewed.

WHAT IT DELIVERS

DEF-D6

Kickoff Go / No-Go decision record

This activity produces the deliverable: the recorded decision, its conditions and any dissent, rather than the presentation that preceded it.

NEEDS FIRST

- Cost and margin model from DEF-04
- Feasibility verdict and concession list from DEF-07
- Resourced schedule and headcount curve from DEF-08
- Positioning and gap list from DEF-06
- Program risk register

DONE WHEN

- Case shown at base and downside, not base alone
- Decision recorded with named owners, conditions and dissent
- Approved budget written back into the program charter

HOW IT GETS THERE

6 STEPS · 2 LANES

w5	w6	w7	w8
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Financial model assembly — NRE, opex, revenue, breakeven → Financial model — NRE, opex, revenue, breakeven	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Scenario set — base, downside, upside → Scenario set with the downside made explicit	0.75w runs in parallel
3	Risk register summary written for the decision room → Risk summary for the decision room	0.5w
4	Recommendation and the ask, stated as a number → Recommendation with a stated ask	0.5w
5	Executive review pack → Executive review pack	0.75w
6	Decision meeting and record → Decision record with owners, conditions and dissent	0.25w

The decision meeting takes a quarter of a week and the preparation takes three weeks, which is the correct ratio. A gate decision argued in the room rather than resolved before it produces an outcome that depends on who attended.

WATCH OUT FOR

- ◆ **Base case presented alone.** A business case that closes at forecast volume and is never shown at downside volume transfers the risk to whoever inherits the program.
- ◆ **Conditions accepted verbally.** A Go granted subject to a condition that nobody records is an unconditional Go, and the condition is never revisited.
- ◆ **Dissent not recorded.** An objection that is raised but not minuted resurfaces at first silicon, with no record of who raised it or what was decided.
- ◆ **Feasibility concessions softened before the review.** A concession list edited on its way into the decision pack makes the entire feasibility activity worthless.
- ◆ **Decision authority unclear.** A gate review where no individual is identified as able to decline the program is a gate review that cannot decline it.

WHERE THE EFFORT GOES	2 M/M	WHO IS ON IT	~2.9 FTE	CONNECTIONS	TERMS HERE	1
Financial model assembly	0.8	Program manager owns it		DEPENDS ON DEF-04 DEF-06 DEF-07 DEF-08 TECH-04	NRE	
Scenario set	0.5	Finance partner				
Review pack	0.5	Product lead		FEEDS INTO ARCH-01 TECH-06 IPR-07		
Decision meeting and record	0.3	Stage lead Executive sponsor		LATER INPUT TECH-08 +1w IPR-10 +14w		

DEFINE

02 Architecture

ARCH

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w6 – w24	18 _w	10	111 _{M/M}	6 of 10

ENGINEERING ACTIVITIES

ARCH-01	System-level performance modeling and workload simulation	14w	28 _{M/M}
ARCH-02	Compute / memory / interconnect partitioning; monolithic vs chiplet decision	8w	14 _{M/M}
ARCH-03	Dataflow and memory hierarchy definition (on-die SRAM capacity, HBM channels)	8w	12 _{M/M}
ARCH-04	Interface and protocol selection (PCIe gen, CXL, UCle, HBM3/3E)	6w	6 _{M/M}
ARCH-05	Power domain, clock domain and DVFS architecture; UPF intent draft	7w	9 _{M/M}
ARCH-06	PPA budget allocation per block	4w	4 _{M/M}
ARCH-07	Security and safety architecture — secure boot, root of trust, fusing	6w	7 _{M/M}
ARCH-08	Chip-level floorplan intent, pin and bump budget	5w	5 _{M/M}
ARCH-09	Architecture specification authoring and review	9w	14 _{M/M}
ARCH-10	Block microarchitecture specification kickoff	6w	12 _{M/M}

KEY DELIVERABLES

ARCH-D1	Architecture specification
ARCH-D2	Performance model and workload analysis report
ARCH-D3	Block partitioning and PPA budget table
ARCH-D4	Interface and protocol definition document
ARCH-D5	Power / clock / reset architecture and UPF intent
ARCH-D6	Chip-level block diagram with pin and bump budget
ARCH-D7	Architecture Freeze review package

ARCH-01

System-level performance modeling and workload simulation

TAKES	COSTS	OWNER	CRITICAL PATH
14w w0–w14 of the stage	28.0 M/M ~8.7 people while it runs	Performance architect Silicon correlation engineer approves	yes slipping moves the program

WHY IT EXISTS

Build the model the architecture is **argued with**. Every partitioning, sizing and interface decision in this stage is settled by simulation against the DEF-02 workload suite rather than by opinion.

Fourteen weeks and 28 M/M make this the largest activity in the stage by a wide margin, and that is the right shape. Being wrong in a model costs a re-run; being wrong in RTL costs a quarter; being wrong in silicon costs a respin. The model's credibility is the stage's credibility—the moment nobody trusts it, every decision reverts to seniority.

WHAT IT DELIVERS

ARCH-D2

Performance model and workload analysis report

The model, its correlation and its confidence statement are the deliverable—the report without the model is an assertion.

NEEDS FIRST

- Workload suite and KPI table from DEF-02
- PPA targets from DEF-03
- Bandwidth requirements from DEF-05
- Previous-generation silicon and its measured performance
- Node library characterization from PDK-03

DONE WHEN

- Model correlated against silicon with a stated error band
- Every architectural decision in the stage traced to a sweep
- Regression harness handed over so later stages can re-run it

HOW IT GETS THERE

10 STEPS · 2 LANES

w0

w3

w6

w9

w12

w14

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Model scope and fidelity decision — analytical, trace-driven or cycle-approximate → Model scope and fidelity decision record	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Baseline model construction — compute, memory, interconnect → Baseline performance model	2.5w
3	Workload suite integration from DEF-02 → Workload suite integrated into the model	1w <i>runs in parallel</i>
4	Correlation against previous-generation silicon → Correlation report against previous-generation silicon	1.5w
5	Design-space exploration sweeps — cluster count, SRAM capacity, HBM channels → Design-space exploration results, per sweep	3w
6	Interconnect and NoC topology studies → Interconnect and NoC topology study results	2w <i>runs in parallel</i>
7	Power model overlay and TOPS/W projection → Power and TOPS/W projection	2w
8	Bottleneck analysis and roofline characterization → Bottleneck and roofline analysis	1.5w <i>runs in parallel</i>
9	Sensitivity analysis and a stated confidence interval → Sensitivity analysis with a confidence statement	3w
10	Model handoff and regression harness for later stages → Performance model and its source · Regression harness for downstream re-runs	1w

Correlation sits early on purpose. A model that has never been checked against silicon produces numbers, not answers, and every sweep run before correlation has to be re-run afterwards.

WATCH OUT FOR

- ◆ **Model never correlated against silicon.** An uncorrelated model has an unknown error bar, and every decision taken on it inherits that error silently.
- ◆ **Fidelity chosen for comfort.** An analytical model that runs in seconds will answer the wrong question quickly; a cycle-accurate model that takes a week per sweep will answer nothing in time to matter.
- ◆ **Sweeps run on an unrepresentative workload.** The model inherits whatever bias DEF-02 built into the suite, and amplifies it across every architectural decision.
- ◆ **Power model bolted on at the end.** TOPS/W is a KPI; a performance model with no power overlay cannot answer the question the product is sold on.
- ◆ **No confidence statement.** A projection quoted as a single number invites the organization to plan against a precision the model does not have.

WHERE THE EFFORT GOES	28 M/M	WHO IS ON IT ~8.7 FTE	CONNECTIONS	TERMS HERE 9
Model construction	7.0	Performance architect owns it	DEPENDS ON DEF-01 DEF-02 DEF-03 DEF-05 DEF-07 DEF-09 TECH-03 PDK-03	M/M PPA KPI SRAM NoC RTL TOPS/W HBM (ESD) HBM
Design-space exploration	6.0	Modeling engineers		
Power model overlay	4.0	Workload analyst		
NoC and interconnect studies	3.5	Power modeling engineer	FEEDS INTO ARCH-02 ARCH-03 ARCH-06 ARCH-09 DV-09	
Bottleneck and sensitivity analysis	3.5	Silicon correlation engineer	LATER INPUT DV-09 +28w	
Silicon correlation	3.0			
Handoff and regression harness	1.0			

ARCH-02

Compute / memory / interconnect partitioning; monolithic vs chiplet decision

TAKES	COSTS	OWNER	CRITICAL PATH
8w	14.0 M/M	Chief architect	yes
w2–w10 of the stage	~7.6 people while it runs	DFT liaison approves	slipping moves the program

WHY IT EXISTS

Decide what sits on which die and where the block boundaries fall—including **whether this is one die or several**.

The monolithic-versus-chiplet call is the most expensive decision in the stage. It sets the package, the die-to-die interface, the test strategy, the yield model and the cost model at once, and after Architecture Freeze it is effectively irreversible: unwinding it means redoing the floorplan, the package design and the test hardware together.

WHAT IT DELIVERS

ARCH-D3

Block partitioning and PPA budget table

The partitioning half of the deliverable; ARCH-06 supplies the budget half against these blocks.

NEEDS FIRST

- Design-space exploration results from ARCH-01
- Cost and yield model from DEF-04
- Reticle limit and node constraints from TECH-02
- Package architecture options from the requirement register
- Test strategy assumptions from DFT-01

DONE WHEN

- Partitioning decided and recorded with the alternatives it beat
- Every boundary carries a bandwidth and latency budget
- Test and yield implications priced, not deferred

HOW IT GETS THERE

8 STEPS · 2 LANES

w2	w4	w6	w8	w10
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Block decomposition from the functional requirements → Block decomposition and hierarchy	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Reticle limit and die size feasibility check → Reticle limit and die size feasibility finding	0.5w
3	Monolithic and chiplet options built as costed candidates → Costed partitioning candidates	2w
4	Yield and cost comparison per candidate, with the DEF-04 model runs in parallel → Yield and cost comparison per candidate	1.5w
5	Die-to-die interface implication study runs in parallel → Die-to-die interface implication study	1w
6	Partition boundary definition — what crosses, at what bandwidth → Partition boundary specification — signals, bandwidth, latency	3w
7	Known-good-die and test implication review runs in parallel → Known-good-die and test implication assessment	1w
8	Partitioning decision and record → Partitioning decision record with dissent	1.5w

Boundary definition takes three of the eight weeks because it is the part that gets skipped. Deciding "two dies" is quick; deciding exactly which signals cross at what bandwidth and latency is the work that makes the decision real.

WATCH OUT FOR

- ◆ **Chiplet chosen for the roadmap rather than for this product.** A die-to-die interface costs area, power and a test strategy; taking it on to be ready for the next program charges this one.
- ◆ **Boundaries drawn without bandwidth.** A partition that names blocks but not the traffic between them is redrawn at PD-02 when the floorplan cannot carry it.
- ◆ **Reticle limit checked against the estimate, not the growth.** Die area grows through implementation; a partition that fits at architecture and not at signoff is a partition that fails late.
- ◆ **Test implications deferred.** Known-good-die coverage on a chiplet part is a DFT and test-hardware program of its own, and it is priced after the decision is taken.
- ◆ **Decision recorded without the dissent.** The objection nobody minuted resurfaces at first silicon with nothing to point at.

WHERE THE EFFORT GOES	14 M/M	WHO IS ON IT	~7.6 FTE	CONNECTIONS	TERMS HERE	2
Candidate construction	4.0	Chief architect OWNS IT		DEPENDS ON DEF-04 ARCH-01 TECH-02	PPA DFT	
Boundary definition	3.5	Physical architect		FEEDS INTO ARCH-03 ARCH-04 ARCH-05		
Yield and cost comparison	2.5	Cost and yield analyst		ARCH-06 ARCH-07 ARCH-08		
Die-to-die interface study	2.0	Interconnect architect		ARCH-09 ARCH-10 IPR-01		
Test implication review	1.0	DFT liaison		DFT-01 PKG0-01 BU-08		
Decision record	1.0					

ARCH-03 Dataflow and memory hierarchy definition (on-die SRAM capacity, HBM channels)				
TAKES		COSTS	OWNER	CRITICAL PATH
8w		12.0 M/M	Memory systems architect	yes
w4–w12 of the stage		~6.5 people while it runs	Workload analyst approves	slipping moves the program
WHY IT EXISTS				
Define how data moves and where it lives— dataflow, on-die SRAM capacity and organization, HBM channel count —for each workload class the product serves.				
An inference accelerator is a memory system with arithmetic attached. A hierarchy that does not match the reuse pattern of the workload shows up as a utilization number nobody can lift afterwards: not by clock speed, not by more compute, and not by any amount of software.				
WHAT IT DELIVERS				
ARCH-D1 Architecture specification				
The memory hierarchy chapter is authored here and lands in the specification whole; ARCH-09 assembles rather than rewrites it.				
NEEDS FIRST				
- Workload traces and reuse analysis from DEF-02 - Bandwidth requirements from DEF-05 - Design-space results from ARCH-01 - Partition boundaries from ARCH-02 - Memory compiler capability from PDK-04				
DONE WHEN				
- Capacity and bandwidth justified by simulation, not by precedent - Dataflow chosen per workload class with the trade recorded - Memory instance requirements handed to PDK and AMS				
HOW IT GETS THERE			8 STEPS · 2 LANES	
w4	w6	w8	w10	w12
MAIN SEQUENCE				
12478				
IN PARALLEL				
356				
#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE			TAT
1	Dataflow selection per workload class — weight-stationary, output-stationary, hybrid → Dataflow specification per workload class			1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	On-die SRAM capacity sizing against measured reuse distance → SRAM capacity sizing with reuse-distance evidence	2w
3	SRAM banking, port and bandwidth organization → SRAM capacity, banking and port organization	1.5w runs in parallel
4	HBM channel count and address interleaving scheme → HBM channel count and interleaving scheme	1.5w
5	Cache and scratchpad policy → Cache and scratchpad policy note	1w runs in parallel
6	Prefetch and data-movement engine requirement → Prefetch and data-movement engine requirement	1w runs in parallel
7	Hierarchy simulation against the ARCH-01 model → Hierarchy simulation results	2w
8	Hierarchy specification and handoff → Memory hierarchy specification chapter	1w

Sizing precedes simulation and simulation revises sizing—the two weeks in step 7 include at least one full turn back through step 2. A hierarchy accepted on its first simulation has almost certainly not been swept.

WATCH OUT FOR

- ◆ **SRAM sized on average reuse.** Reuse distance has a tail; a capacity chosen at the median misses the workloads that define the product's worst case.
- ◆ **Banking decided after capacity.** Capacity and bandwidth are separate constraints, and a bank organization that cannot feed the compute wastes the area it took.
- ◆ **Memory compiler capability assumed.** The instance the hierarchy needs may not exist on the node—PDK-05 is where that is found, and it is found after this decision unless someone asks early.
- ◆ **Dataflow chosen once for all workloads.** A single dataflow optimized for one class starves another, and the deployment mix decides which mistake is cheaper.
- ◆ **Interleaving left to implementation.** Address interleaving decides whether HBM channels are actually parallel; deferred, it becomes a bandwidth shortfall nobody predicted.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT	~6.5 FTE	CONNECTIONS	TERMS HERE	8
SRAM capacity and organization	3.0	Memory systems architect OWNS IT		DEPENDS ON DEF-02 DEF-05 ARCH-01 ARCH-02	PPA PDK IP SRAM AMS RTL HBM (ESD) HBM	
Dataflow selection	2.5	Dataflow architect		FEEDS INTO ARCH-06 ARCH-08 ARCH-09 ARCH-10 PDK-04 PDK-05 PDK-06 AMS-05 RTL-01 PKGD-03 ASSY-02 EVB-01 EVB-02 TEST-06		
Hierarchy simulation	2.5	Performance modeling engineer				
HBM channels and interleaving	1.5	Memory IP liaison				
Cache, scratchpad and prefetch policy	1.5	Workload analyst				
Specification and handoff	1.0					

ARCH-04

Interface and protocol selection
(PCIe gen, CXL, UCIe, HBM3/3E)

TAKES	COSTS	OWNER	CRITICAL PATH
6w	6.0 M/M	Interface architect	no
w3–w9 of the stage	~4.3 people while it runs	Physical architect approves	has float

WHY IT EXISTS

Choose the external interfaces and **pin their revisions**—PCIe generation, CXL profile, HBM generation, die-to-die if the part is a chiplet—and turn each into a controller and PHY requirement IPR can go and buy.

Interfaces are where the program touches standards bodies and IP vendors, neither of which runs on the program's calendar. A generation selected before its specification is ratified imports someone else's schedule into this one, and the import is silent until the PHY is late.

WHAT IT DELIVERS

ARCH-D4

Interface and protocol definition document

This activity is the deliverable—the selections, their revisions and the requirements derived from them.

NEEDS FIRST

- Bandwidth and interface sizing from DEF-05
- Partitioning decision from ARCH-02
- Standards body revision status and ratification schedules
- Vendor PHY roadmaps under NDA
- Node PHY availability from TECH-03

DONE WHEN

- Every interface pinned to a named, ratified revision
- Controller and PHY requirements handed to IPR
- Compliance obligations listed with the hardware they need

HOW IT GETS THERE

8 STEPS · 2 LANES

w3	w5	w7	w9
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MAIN SEQUENCE

12378

IN PARALLEL

456

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Interface requirement extraction from DEF-05 → Consolidated interface requirement list	0.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Standard revision status and ratification timing review → Standard revision and ratification status record	1w
3	Host interface selection — PCIe generation and CXL profile → Host interface selection with generation and profile	1.5w
4	HBM generation selection and controller requirement → HBM generation selection and controller requirement	1w runs in parallel
5	Die-to-die interface selection, if a chiplet split survives ARCH-02 → Interface selection record with the revision pinned · Die-to-die interface requirement, where applicable	1w runs in parallel
6	Compliance and interoperability requirement definition → Compliance and interoperability requirement list	1w runs in parallel
7	Controller and PHY requirement handoff to IP strategy → Controller and PHY requirement specification	1w
8	Interface definition document and review → Interface definition document	2w

The handoff in step 7 is what makes this activity urgent rather than merely important. `IPR-04` cannot evaluate a PHY vendor against a generation nobody has chosen, and PHY license lead times run to a quarter.

WATCH OUT FOR

- ◆ **Specification not ratified at selection.** Designing to a draft revision transfers a standards-body schedule into the program, and the transfer is invisible until compliance testing.
- ◆ **PHY assumed available on the node.** A PHY proven on the previous node is not a PHY on this one; IPR-05 is where that is confirmed, and it is confirmed after this choice.
- ◆ **Compliance requirements written late.** Interoperability testing needs hardware, test suites and lab time booked against dates set here.
- ◆ **Lane count set by bandwidth alone.** Lanes cost bumps, area and power; a count chosen without ARCH-08 in the room is a count the floorplan will renegotiate.
- ◆ **Die-to-die selection deferred past ARCH-02.** The interface is a large architectural commitment, and deferring it forces the conservative default.

WHERE THE EFFORT GOES	6 M/M	WHO IS ON IT	~4.3 FTE	CONNECTIONS	TERMS HERE	8
Interface selection studies	2.0	Interface architect	OWNS IT	DEPENDS ON DEF-05 ARCH-02 TECH-03	NDA IP SerDes PHY	
Standards and ratification review	1.0	Standards liaison		FEEDS INTO ARCH-08 ARCH-09 PDK-07	HBM (ESD) HBM PCIe	
Compliance requirement definition	1.0	IP strategist		IPR-04 AMS-01 AMS-03	CXL	
IP requirement handoff	1.0	SerDes and PHY specialist		PKGD-02 PKGD-03 PKGD-05		
Definition document and review	1.0	Physical architect		SIPI-07 TEST-01 BU-06		
				FEEDS BACK INTO IPR-01		

ARCH-05

Power domain, clock domain and DVFS architecture; UPF intent draft

TAKES	COSTS	OWNER	CRITICAL PATH
7w w7–w14 of the stage	9.0 M/M ~5.6 people while it runs	Power architect Verification liaison approves	no has float

WHY IT EXISTS

Define the power and clock structure the whole implementation inherits—**domains, rails, DVFS operating points, clock crossings, reset topology**—and write the first UPF intent.

Power intent is one of the few architectural artifacts consumed literally by tools. A domain boundary drawn here becomes isolation and level-shifter cells at SYN-06 and a verification obligation at DV-06. Drawn late, or drawn differently by two teams, it forces both to be redone.

WHAT IT DELIVERS

ARCH-D5

Power / clock / reset architecture and UPF intent

This activity is the deliverable, and the UPF file is the part of it tools consume directly.

NEEDS FIRST

- Power envelope and per-domain budget from DEF-03
- Block list and partitioning from ARCH-02
- DVFS requirements from the workload analysis
- Library and IO capability from PDK-03
- Package power delivery constraints from PKGD-06

DONE WHEN

- Every clock crossing on the inventory with a named synchroniser strategy
- UPF intent parses in the target tools, not only on paper
- Physical design and verification have accepted the architecture

HOW IT GETS THERE

8 STEPS · 2 LANES

w7	w9	w11	w13	w14
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Power domain partitioning against the block list → Power domain definition and boundary map	1.5w
2	Voltage rail and regulator topology → Voltage rail and regulator topology	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	DVFS operating point definition and switching policy → DVFS operating point table and policy	1.5w runs in parallel
4	Clock domain architecture and crossing inventory → Clock domain architecture with a crossing inventory	1.5w
5	Reset topology and sequencing → Reset topology and sequence specification	1w runs in parallel
6	Power management controller requirement → Power management controller requirement	1w runs in parallel
7	UPF intent draft → UPF intent file, first draft	2w
8	Review with physical design and verification → Review record from physical design and verification	1w

The crossing inventory in step 4 is the artifact `RTL-06` closes CDC against. An inventory that is incomplete at Architecture Freeze becomes a CDC finding in week 40, when the fix is an RTL change under change control.

WATCH OUT FOR

- ◆ **Domains drawn for elegance rather than for control.** Every domain boundary costs isolation cells, level shifters and verification states; a domain that is never independently gated is pure overhead.
- ◆ **Crossing inventory incomplete.** A clock crossing missing from the inventory is a crossing nobody constrains and nobody verifies, and it fails intermittently in silicon.
- ◆ **DVFS policy without measured operating points.** Operating points chosen from the library rather than from silicon leave either performance or margin on the table.
- ◆ **UPF drafted after RTL starts.** Power intent that arrives mid-RTL forces retrofitting into blocks already written, and the retrofit is never as clean as the design.
- ◆ **Reset sequencing left to bring-up.** A reset order discovered in the lab is discovered on silicon that cannot be changed.

WHERE THE EFFORT GOES	9 M/M	WHO IS ON IT	~5.6 FTE	CONNECTIONS	TERMS HERE	7
Power domain partitioning	2.0	Power architect OWNS IT		DEPENDS ON DEF-03 ARCH-02 PDK-03	PPA RTL PDN IO DVFS UPF CDC	
UPF intent draft	2.0	Clocking architect		FEEDS INTO ARCH-06 ARCH-09 PDK-09		
Clock domain architecture	1.5	Low-power methodology engineer		PDK-11 AMS-01 AMS-02 AMS-04 AMS-06 RTL-05		
DVFS definition	1.5	Physical design liaison		DV-06 DFT-05 SYN-01 SYN-06 PD-03 PD-07		
Reset and power management controller	1.5	Verification liaison		SIPI-01 BU-03		
Review	0.5					

ARCH-06

PPA budget allocation per block

TAKES	COSTS	OWNER	CRITICAL PATH
4w w12–w16 of the stage	4.0 M/M ~4.3 people while it runs	PPA lead Program manager approves	yes slipping moves the program

WHY IT EXISTS

Divide the stage's PPA targets across the real block list, so **every block owner starts implementation with a number rather than an aspiration.**

This is DEF-03's allocation done again, and done properly. DEF-03 allocated against an estimated block list; this allocates against the partitioning ARCH-02 settled and the hierarchy ARCH-03 sized. It is the version RTL, synthesis and physical design are actually held to, and the version an overrun is measured against.

WHAT IT DELIVERS

ARCH-03

Block partitioning and PPA budget table

The budget half of the deliverable, published against the blocks ARCH-02 defined.

NEEDS FIRST

- PPA targets from DEF-03
- Block list and partitioning from ARCH-02
- Memory hierarchy sizing from ARCH-03
- Power domain definition from ARCH-05
- Floorplan intent from ARCH-08

DONE WHEN

- Every block carries an accepted area, power and timing budget
- Margin reserved and stated, not assumed
- Change-control rule published and agreed

HOW IT GETS THERE

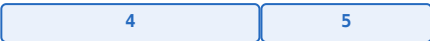
7 STEPS · 2 LANES

w12	w13	w14	w15	w16
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Reconcile the DEF-03 targets against the ARCH-02 block list → Reconciliation note against the DEF-03 targets	0.75w
2	Area allocation per block, from the floorplan intent → Per-block area allocation	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Power allocation per block and per domain → Per-block and per-domain power allocation	1w
4	Frequency and timing budget per clock domain → Per-block area, power and timing budget	0.75w runs in parallel
5	Margin reservation for ECO and late growth → Margin reservation record	0.5w runs in parallel
6	Owner assignment and explicit acceptance per block → Owner acceptance sheet per block	0.75w
7	Budget publication and change-control rule → Change-control rule for budget movement	0.5w

Step 6 is acceptance, not notification. A budget a block owner has not agreed to is a budget that will be exceeded and disputed, and the dispute happens at `PD-09` when there is no time for it.

WATCH OUT FOR

- ◆ **Budgets allocated to 100%.** A budget with no margin has nothing left for the ECO storm at physical design, where power and area always grow.
- ◆ **Allocation without acceptance.** An unaccepted budget is defended by nobody and consumed by whoever moves first.
- ◆ **No change-control rule.** Budgets that can be moved quietly stop being budgets; movement has to cost a conversation.
- ◆ **Reconciliation against DEF-03 skipped.** If the architecture's total quietly exceeds the definition target, the cost model is wrong and nobody has been told.
- ◆ **Timing budget by clock domain only.** Domains contain blocks with very different closure difficulty, and a flat budget hides which one is impossible.

WHERE THE EFFORT GOES	4 M/M	WHO IS ON IT ~4.3 FTE	CONNECTIONS	TERMS HERE 3
Area and power allocation	2.0	PPA lead OWNS IT	DEPENDS ON DEF-03 ARCH-01 ARCH-02 ARCH-03 ARCH-05 ARCH-08 IPR-02	PPA RTL ECO
Reconciliation against DEF-03	0.8	Chief architect	FEEDS INTO ARCH-09 ARCH-10 PDK-05 AMS-01 AMS-05 RTL-01 RTL-08 SYN-06 SYN-07 SYN-11 PD-02 PD-03 PD-16 EVB-10	
Timing budget	0.5	Block owners	LATER INPUT PDK-05 +0w PDK-06 +5w SYN-11 +22w	
Owner acceptance	0.5	Physical architect		
Publication and change control	0.3	Program manager		

ARCH-07

Security and safety architecture — secure boot, root of trust, fusing

TAKES	COSTS	OWNER	CRITICAL PATH
6w w6–w12 of the stage	7.0 M/M ~5.1 people while it runs	Security architect Compliance liaison approves	no has float

WHY IT EXISTS

Define how the part **boots trustworthily, protects its assets and identifies itself**—secure boot chain, root of trust, key storage, fuse map and lifecycle states—and the debug lockdown that goes with them.

Security is architectural or it is absent. A root of trust cannot be retrofitted onto a design with no immutable boot path and no fuse plan. The tension worth naming early is that the debug access DFT and bring-up want is the same access an attacker wants, and only an architecture can resolve it.

WHAT IT DELIVERS

ARCH-D1

Architecture specification

The security chapter is authored here in full—boot chain, lifecycle states and lockdown policy—and lands in the specification whole.

NEEDS FIRST

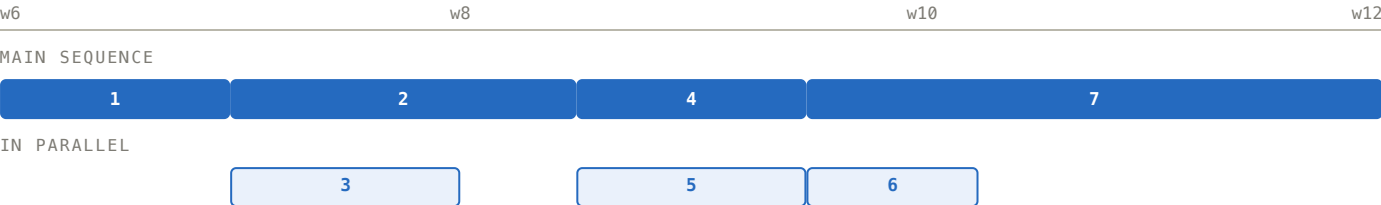
- Security requirements from the requirement register
- Customer and regulatory obligations
- Block list and boot architecture from ARCH-02
- eFuse and OTP capability from PDK-02
- DFT access architecture intent from DFT-04

DONE WHEN

- Threat model reviewed and every asset mapped to a control
- Lifecycle states defined with their transitions and their tests
- Debug lockdown agreed with DFT rather than imposed on it

HOW IT GETS THERE

7 STEPS · 2 LANES



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Threat model and asset inventory → Threat model and asset inventory	1w
2	Secure boot chain and immutable ROM definition → Secure boot chain specification	1.5w
3	Root of trust and key storage selection → Root of trust and key storage architecture	1w <i>runs in parallel</i>
4	Fuse map, chip identity and lifecycle state definition → Fuse map and lifecycle state definition	1w
5	Debug and test access lockdown policy, with DFT → Debug lockdown policy	1w <i>runs in parallel</i>
6	Cryptographic accelerator requirement → Cryptographic accelerator requirement	0.75w <i>runs in parallel</i>
7	Security architecture note and review → Security architecture note and review record	2.5w

Step 5 is a negotiation, not a decision taken alone. `DFT-04` needs JTAG and trace access for silicon debug, and every path it opens is a path this activity has to close in production lifecycle states.

WATCH OUT FOR

- ◆ **Security started after the block list is fixed.** A root of trust needs its own reset domain, its own ROM and its own fuse path; adding them later disturbs three architectures at once.
- ◆ **Debug access left open in production.** The lifecycle state that locks JTAG is the one nobody tests, because testing it makes the part undebuggable.
- ◆ **Fuse map without a lifecycle model.** Fuses are one-time; a map with no defined state transitions produces parts that cannot be returned to a testable state.
- ◆ **Key storage decided by IP availability.** Choosing the root of trust because a block was available is choosing a threat model by accident.
- ◆ **No named security owner past architecture.** Security decisions get eroded one waiver at a time through implementation unless someone owns them all the way to qualification.

WHERE THE EFFORT GOES	7 M/M	WHO IS ON IT ~5.1 FTE	CONNECTIONS
Boot chain and root of trust	2.5	Security architect <i>OWNS IT</i>	DEPENDS ON DEF-01 ARCH-02
Threat model and asset inventory	1.5	Boot and firmware architect	FEEDS INTO ARCH-09 ARCH-10 RTL-04
Fuse map and lifecycle states	1.0	DFT liaison	RTL-07 DV-05 DFT-03
Debug lockdown negotiation	1.0	Fuse and lifecycle engineer	DFT-04 DFT-10 DFT-11
Cryptographic accelerator requirement	0.5	Compliance liaison	MP-11
Review	0.5		TERMS HERE 5 IP ROM OTP DFT JTAG

ARCH-08

Chip-level floorplan intent, pin and bump budget

TAKES	COSTS	OWNER	CRITICAL PATH
5w w10–w15 of the stage	5.0 M/M ~4.3 people while it runs	Physical architect Chief architect approves	no has float

WHY IT EXISTS

Draw the chip as a **physical object** for the first time—block placement intent, IO ring, bump field—and publish the pin and bump budget the package will be designed against.

The bump map is the first artifact the die and the package share, and the first hard constraint on bandwidth. An interface needing more bumps than the field can supply is an interface that has to shrink, and having that argument here costs a meeting. Having it at PD-04, with the substrate already ordered, costs the substrate.

WHAT IT DELIVERS

ARCH-D6

Chip-level block diagram with pin and bump budget

This activity is the deliverable, and the bump map is the half of it the package team consumes directly.

NEEDS FIRST

- Partitioning and boundaries from ARCH-02
- Memory hierarchy and HBM channel count from ARCH-03
- Interface lane counts from ARCH-04
- Power envelope from DEF-03
- Package architecture options from PKGD-01

DONE WHEN

- Bump field defined with an explicit power-to-signal ratio
- Every interface has an allocated bump count package has accepted
- Placement intent reviewed by physical design as buildable

HOW IT GETS THERE

7 STEPS · 2 LANES

w10	w11	w12	w13	w14	w15
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Block placement intent from the partitioning and the hierarchy → Chip-level block diagram with placement intent	1w
2	IO ring and PHY placement plan → IO ring and PHY placement plan	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Bump field definition — pitch, count, power-to-signal ratio → Bump field definition and map	1.5w
4	Signal bump allocation per interface → Signal and power bump allocation	0.75w runs in parallel
5	Power delivery bump budget against the power envelope → Power delivery bump budget	1w runs in parallel
6	Package feasibility check with package design → Package feasibility finding	0.75w runs in parallel
7	Block diagram and bump budget publication → Published block diagram and bump budget	1.5w

The power-to-signal ratio in step 3 is the number that decides the argument. On a part at this power envelope most of the bump field is power delivery, and the signal budget is what is left over rather than what was asked for.

WATCH OUT FOR

- ◆ **Bump budget derived after interfaces are fixed.** Lane counts set without a bump budget produce a floorplan that cannot be built and an interface that has to be cut late.
- ◆ **Power delivery bumps underestimated.** Too few power bumps is an IR-drop problem discovered at SIPI-04, when the bump map is already in the package design.
- ◆ **Placement intent treated as decoration.** Physical design inherits this drawing; an intent that ignores routing feasibility is redrawn at PD-01 and the schedule pays for both.
- ◆ **Package feasibility checked against a generic assumption.** Bump pitch and field size depend on the actual interposer and substrate technology, which PKGD-01 may not have settled yet.
- ◆ **HBM PHY placement left open.** HBM PHYs must sit against the interposer edge facing their stacks; deciding this late constrains everything else on the die.

WHERE THE EFFORT GOES	5 M/M	WHO IS ON IT ~4.3 FTE	CONNECTIONS	TERMS HERE	5
Bump field definition and allocation	2.0	Physical architect owns it	DEPENDS ON DEF-03 ARCH-02 ARCH-03 ARCH-04	PPA PHY IO HBM (ESD) HBM	
Placement intent	1.3	IO and PHY architect	FEEDS INTO ARCH-06 ARCH-09 PDK-07 AMS-03 DFT-04 DFT-11 PD-01 PD-02 PD-04 PKGD-02 SIPI-02 EVB-01 EVB-08 TEST-04 BU-02		
Power bump budget	0.8	Power delivery engineer			
Package feasibility check	0.5	Package design liaison			
Publication	0.5	Chief architect			

ARCH-09

Architecture specification authoring and review

TAKES	COSTS	OWNER	CRITICAL PATH
9w w9–w18 of the stage	14.0 M/M ~6.8 people while it runs	Chief architect Program manager approves	yes slipping moves the program

WHY IT EXISTS

Write the document the rest of the program builds against, and run it through the review that ends in **Architecture Freeze**.

The specification is not a record of decisions already taken; writing it is what exposes the decisions nobody took. Every downstream stage reads this document, and every ambiguity left in it becomes an argument in RTL, a bug in verification, or a waiver at signoff. The consistency pass matters as much as the authoring.

WHAT IT DELIVERS

- ARCH-01

Architecture specification

This activity assembles and closes it. Chapters authored in ARCH-03, ARCH-05 and ARCH-07 arrive whole; the work here is coherence, not rewriting.
- ARCH-07

Architecture Freeze review package

The specification, the open-issue disposition and the downstream readiness findings, assembled for the gate.

NEEDS FIRST

- Every other ARCH activity's output
- Performance model results from ARCH-01
- PPA budget table from ARCH-06
- Feasibility concessions from DEF-07
- Open issues carried from the architecture risk register

DONE WHEN

- Every open issue dispositioned — closed, deferred with an owner, or accepted as risk
- Downstream readiness review completed before the gate, not after
- Freeze decision recorded with conditions and dissent

HOW IT GETS THERE

8 STEPS · 2 LANES

w9	w11	w13	w15	w17	w18
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Specification structure and section ownership assignment → Specification structure with section owners	0.5w
2	Section authoring across the architecture team → Architecture specification	3.5w
3	Consistency pass — every number traced to its source activity → Consistency and traceability record	1.5w runs in parallel
4	Open-issue register and disposition → Open-issue register with disposition per item	1w runs in parallel
5	Internal technical review, two rounds → Internal technical review findings, two rounds	2w
6	Downstream readiness review with RTL, DV, DFT and physical design → Downstream readiness review findings	1.5w
7	Freeze package assembly → Architecture Freeze package and decision record	1w runs in parallel
8	Architecture Freeze gate → Architecture Freeze gate decision	1.5w

The downstream readiness review in step 6 is the one that predicts the program. Architecture reviewing its own specification confirms it is coherent; RTL, DV and DFT reviewing it confirms it is implementable, and those are different questions.

WATCH OUT FOR

- ◆ **Freeze declared with open issues undispositioned.** An open issue carried past freeze is a decision deferred to whoever hits it first in RTL.
- ◆ **Consistency pass skipped under schedule pressure.** The same number appearing differently in two chapters is the defect that costs the most and is found the latest.
- ◆ **Downstream teams reviewing after freeze.** A readiness review that happens after the gate cannot change anything, and becomes a complaint rather than a review.
- ◆ **Specification written as a narrative.** Implementable specifications are tables, interfaces and constraints; prose is where ambiguity survives review.
- ◆ **Freeze treated as a date rather than a condition.** A gate that passes because it was scheduled to teaches the organization that gates do not mean anything.

WHERE THE EFFORT GOES	14 M/M	WHO IS ON IT	~6.8 FTE	CONNECTIONS	TERMS HERE	4
Section authoring	6.0	Chief architect OWNS IT		DEPENDS ON ARCH-01 ARCH-02 ARCH-03 ARCH-04 ARCH-05 ARCH-06 ARCH-07 ARCH-08	PPA RTL DV DFT	
Technical and readiness reviews	3.5	Section authors		FEEDS INTO ARCH-10 RTL-01 RTL-07 DV-01 DFT-01 SYN-01 PD-01		
Consistency and traceability pass	2.0	Technical editor				
Open-issue disposition	1.5	Downstream leads				
Freeze package and gate	1.0	Program manager				

ARCH-10

Block microarchitecture specification kickoff

TAKES	COSTS	OWNER	CRITICAL PATH
6w w12–w18 of the stage	12.0 M/M ~8.7 people while it runs	Block architects Chief architect approves	yes slipping moves the program

WHY IT EXISTS

Turn each architectural block into a **microarchitecture specification its RTL owner can implement from**—pipeline structure, interfaces, control, register map and the verification hooks.

The architecture specification says what each block does; this says how. Starting it inside the architecture stage rather than at RTL kickoff is what lets RTL-02 begin with something to write against instead of a paragraph—and it is the cheapest place to find out that two blocks disagree about the interface between them.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Architecture specification chapters from ARCH-09
- Partitioning and boundaries from ARCH-02
- Memory hierarchy specification from ARCH-03
- PPA budget per block from ARCH-06
- Verification strategy intent from DV-01

DONE WHEN

- Every block has a specification and a named owner
- Shared interfaces agreed by both sides, not by each
- RTL leads have accepted the handoff as implementable

HOW IT GETS THERE

8 STEPS · 2 LANES

w12	w14	w16	w18
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MAIN SEQUENCE

1	2	3	7	8
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IN PARALLEL

4	5	6
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#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Block specification template and review cadence → Microarchitecture specification per block	0.5w
2	Block owner assignment and kickoff → Block owner assignment and kickoff record	0.5w
3	Pipeline and datapath definition per block → Per-block pipeline and datapath definition	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Block interface and protocol definition against the NoC → Block interface and protocol definitions	1.5w runs in parallel
5	Control, configuration and register map, first pass → Register map first pass, per block	1.5w runs in parallel
6	Verification hook and observability requirement, with DV → Verification hook and observability requirements	1w runs in parallel
7	Microarchitecture review, first round → First-round microarchitecture review findings	1.5w
8	Handoff to RTL implementation → RTL implementation handoff package	1.5w

Step 4 runs in parallel and is where the interesting failures live. Two block owners each writing their side of a shared interface will disagree, and finding that here costs a meeting rather than an integration debug at `RTL-04`.

WATCH OUT FOR

- ◆ **Block owners not assigned.** A block whose specification has no author is a block whose RTL starts by inventing one, usually in week three of implementation.
- ◆ **Interfaces written from one side.** Each owner specifying their own side of a shared interface produces two consistent documents that disagree with each other.
- ◆ **Register map deferred to RTL.** The register map is the software contract; defining it late means firmware development starts late and the two never converge.
- ◆ **Verification hooks added after the design.** Observability designed in is cheap; observability retrofitted costs area, timing and an RTL change under control.
- ◆ **Microarchitecture that ignores the budget.** A pipeline designed without reference to ARCH-06's allocation is a budget overrun scheduled for synthesis.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT	~8.7 FTE	CONNECTIONS
Pipeline and datapath definition	4.0	Block architects OWNS IT		DEPENDS ON ARCH-02 ARCH-03 ARCH-06 ARCH-07 ARCH-09
Interface definition	2.5	Interconnect architect		FEEDS INTO RTL-01 RTL-02 RTL-07 DV-01 DFT-01
Register map first pass	2.0	RTL leads		TERMS HERE 4 PPA NoC RTL DV
Microarchitecture reviews	2.0	Verification liaison		
Verification hooks	1.0	Chief architect		
Handoff	0.5			

03 Technology & Foundry Selection

TECH

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w0 – w14	14 _w	9	26 _{M/M}	4 _{of 9}
ENGINEERING ACTIVITIES				KEY DELIVERABLES
TECH-01	Foundry and node long-list to short-list evaluation		5w 4 M/M	TECH-D1 Technology selection report and decision record
TECH-02	Process option and flavor selection (HPC/HD, multi-Vt menu, backside PDN, RF/HV adders)		5w 4 M/M	TECH-D2 Process option / flavor sheet agreed with foundry
TECH-03	Density / performance / leakage benchmarking against product targets (DTCO study)		7w 6 M/M	TECH-D3 Wafer, mask and NRE cost sheet
TECH-04	Wafer price, mask and NRE quotation; MPW and volume pricing		6w 3 M/M	TECH-D4 Executed foundry design agreement (DA) and NDA
TECH-05	Fab capacity and tapeout slot reservation; hot-lot policy agreement		5w 2 M/M	TECH-D5 Capacity and tapeout slot commitment
TECH-06	Foundry legal engagement — NDA, design agreement, IP licensing frame		8w 2 M/M	TECH-D6 Node risk assessment — maturity, defect density, yield learning curve
TECH-07	Foundry roadmap and risk-production timing alignment		4w 1.5 M/M	
TECH-08	Second-source and node-migration strategy assessment		4w 1.5 M/M	
TECH-09	OSAT and backend supply chain preliminary alignment		5w 2 M/M	

03 TECH · TECHNOLOGY & FOUNDRY SELECTION

TECH-01

Foundry and node long-list to short-list evaluation

TAKES	COSTS	OWNER	CRITICAL PATH
5w	4.0 M/M	Technology strategist	yes
w0–w5 of the stage	~3.5 people while it runs	Chief architect approves	slipping moves the program

WHY IT EXISTS

Narrow the field of foundries and nodes to a short-list the program can actually evaluate, and **record why the others were dropped**.

Every other decision in this stage is conditional on this one, and so is most of the enablement band: IP availability, PDK maturity, library qualification and capacity all become node-specific the moment the short-list closes. The screen has to be quick and it has to be written down—a selection nobody can reconstruct is one that gets reopened at every setback.

WHAT IT DELIVERS

TECH-01

Technology selection report and decision record

This activity owns the deliverable; TECH-03 supplies the technical evidence it rests on.

NEEDS FIRST

- Product requirements and PPA targets from DEF-01 and DEF-03
- Volume forecast and cost ceiling from the business case
- Foundry public roadmaps and node cadence
- Previous-program experience with each foundry
- Export control and supply-continuity guidance

DONE WHEN

- Short-list of at least two candidates, with the dropped options recorded
- Criteria weighted before the responses were read
- Recommendation carries availability windows, not only technical fit

HOW IT GETS THERE

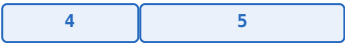
7 STEPS · 2 LANES

w0	w1	w2	w3	w4	w5
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Long-list construction — foundries, nodes and their availability windows → Foundry and node long-list with availability windows	0.75w
2	Screening criteria weighted against the product requirements → Weighted screening criteria	0.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Foundry engagement and formal information requests → Foundry information request responses	1.5w
4	Previous-program experience and relationship review → Previous-program experience review	0.5w
5	Geopolitical, export-control and supply-continuity screen → Screening record — what was dropped and why	0.75w
6	Short-list down to two or three candidates → Short-list of two or three candidates	1w
7	Selection recommendation and decision record → Selection recommendation and decision record	1.25w

Step 3 is the pacing item and almost none of it is our work—information requests sit with the foundry's technical marketing team, and the response time is theirs. Starting the engagement in week one is the only lever the program has over it.

WATCH OUT FOR

- ◆ **Short-list of one.** A screen that arrives at a single candidate has no negotiating position on price, capacity or terms, and the foundry knows it.
- ◆ **Selection driven by relationship.** The foundry the last program used is a legitimate input and a poor criterion; making it the criterion means the screen was theatre.
- ◆ **Availability window not checked.** A node that is right for the product but reaches risk production after the tapeout date is not a candidate, however good the numbers are.
- ◆ **Export control screened late.** A candidate that fails compliance after technical evaluation has consumed weeks of the stage for nothing.
- ◆ **Criteria written after the data arrives.** Weighting the screen once the responses are in produces the answer somebody already wanted.

WHERE THE EFFORT GOES	4 M/M	WHO IS ON IT ~3.5 FTE	CONNECTIONS	TERMS HERE	3
Foundry engagement	1.3	Technology strategist OWNS IT	DEPENDS ON DEF-01	PPA PDK IP	
Screening criteria and long-list	0.8	Foundry relationship manager	RUNS WITH DEF-07		
Short-listing	0.8	Program manager	FEEDS INTO TECH-02 TECH-03 TECH-04		
Continuity and compliance screen	0.5	Compliance liaison	TECH-05 TECH-06 TECH-07		
Decision record	0.8	Chief architect	TECH-08 TECH-09 PDK-01 IPR-02 IPR-05		

03 TECH · TECHNOLOGY & FOUNDRY SELECTION

TECH-02

Process option and flavor selection (HPC/HD, multi-Vt menu, backside PDN, RF/HV adders)

TAKES	COSTS	OWNER	CRITICAL PATH
5w w4–w9 of the stage	4.0 M/M ~3.5 people while it runs	Technology strategist Foundry technical account manager approves	yes slipping moves the program

WHY IT EXISTS

Choose the **flavor of the node**, not just the node—performance or density library, the multi-Vt menu, the metal stack, backside power delivery, and any analog or high-voltage adders the product needs.

A node name is not a specification. Two programs on the same node with different flavor sheets get different density, different leakage and different routing resource, and the difference is larger than most architectural decisions. This sheet is also what the foundry prices against, so it has to close before TECH-04 can finish.

WHAT IT DELIVERS

TECH-D2

Process option / flavor sheet agreed with foundry

This activity is the deliverable, and it is contractual rather than advisory.

NEEDS FIRST

- Short-list from TECH-01
- PPA targets from DEF-03
- DTCO findings as they emerge from TECH-03
- Reticle and die size direction from ARCH-02
- Analog and PHY requirements from ARCH-04

DONE WHEN

- Flavor sheet agreed and signed, not merely discussed
- Metal stack backed by a congestion study rather than by precedent
- Every analog and adder requirement captured before agreement

HOW IT GETS THERE

7 STEPS · 2 LANES

w4	w5	w6	w7	w8	w9
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Process flavor options against the PPA targets — HPC, HD, or mixed → Flavor option comparison against the PPA targets	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Multi-Vt menu selection and its leakage implication → Multi-Vt menu selection with leakage projection	1w
3	Backside power delivery evaluation, where the node offers it → Backside power delivery evaluation	0.75w runs in parallel
4	RF, high-voltage and analog adder requirements → Analog, RF and high-voltage adder list	0.5w runs in parallel
5	Metal stack selection and routing resource implication → Metal stack selection and routing resource note	1w
6	Option cost adders priced with TECH-04 → Priced option adders	0.5w runs in parallel
7	Flavor sheet agreement with the foundry → Process option / flavor sheet agreed with the foundry	2w

Two of the five weeks sit in step 7, which is signature time rather than engineering time. The sheet is a contractual artifact: once agreed it governs what the PDK contains and what the wafer costs, and changing it later reopens both.

WATCH OUT FOR

- ◆ **Flavor chosen for density alone.** A high-density library that cannot reach the frequency target trades an area saving for a performance failure, and the trade is discovered at trial synthesis.
- ◆ **Metal stack chosen without a congestion study.** Too few routing layers is the defect that surfaces at PD-06, when the only remaining fix is a floorplan change.
- ◆ **Multi-Vt menu wider than the flow supports.** Every extra Vt is a library to characterize, qualify and maintain; menus are chosen for leakage and paid for in methodology.
- ◆ **Backside power taken for the roadmap.** A first-generation backside PDN buys power integrity and costs schedule risk, and the trade belongs to this product rather than the next one.
- ◆ **Adders discovered after the sheet is agreed.** An analog or high-voltage requirement that arrives late is a sheet amendment, a price change and a PDK delta at once.

WHERE THE EFFORT GOES	4 M/M	WHO IS ON IT	~3.5 FTE	CONNECTIONS	TERMS HERE	10
Flavor and Vt studies	1.5	Technology strategist owns IT		DEPENDS ON DEF-03 TECH-01 TECH-03	PPA NRE PDK DTC0	
Metal stack selection	0.8	Library and CAD lead		FEEDS INTO ARCH-02 TECH-04 TECH-08	PHY RF PDN CAD	
Backside PDN and adder evaluation	0.8	Physical design liaison		PDK-01 PDK-02 PDK-03	HPC HD	
Sheet agreement	1.0	Analog lead		PDK-04 PDK-08 IPR-05		
		Foundry technical account manager		SYN-02 PD-06		

03 TECH · TECHNOLOGY & FOUNDRY SELECTION

TECH-03

Density / performance / leakage benchmarking
against product targets (DTCO study)

TAKES	COSTS	OWNER	CRITICAL PATH
7w w3–w10 of the stage	6.0 M/M ~3.7 people while it runs	DTCO lead Foundry technical contact approves	yes slipping moves the program

WHY IT EXISTS

Measure the candidate nodes against **this product's targets** rather than against the foundry's brochure—density, achievable frequency, leakage, SRAM Vmin and routing resource, on circuits that resemble the design.

This is the technical evidence the selection stands on. Published node comparisons are made on somebody else's circuits at somebody else's operating point; a DTCO study run on representative structures is the only way to know whether the PPA targets survive the node, and it is what DEF-07 tests feasibility against.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Short-list from TECH-01
- PPA targets from DEF-03
- Foundry-supplied preliminary libraries and models
- Previous-generation design as a benchmark reference
- Memory compiler capability indications

DONE WHEN

- Every target has a measured result per candidate node
- Findings state margin, not only pass or fail
- Benchmark set documented well enough to re-run on a later PDK

HOW IT GETS THERE

8 STEPS · 2 LANES

w3w5w7w9w10

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Benchmark circuit set definition, representative of the design → Benchmark circuit set and its rationale	0.75w
2	Standard cell density and area scaling study → Density and area scaling results per node	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Ring oscillator and critical-path frequency study → Ring oscillator and critical-path frequency study	1.5w
4	Leakage and dynamic power characterization across the Vt menu → Frequency and leakage characterization per Vt	1.25w runs in parallel
5	SRAM bitcell density and Vmin study → SRAM density and Vmin findings	1.25w runs in parallel
6	Routing congestion proxy study on the candidate metal stacks → Routing congestion proxy results	1w runs in parallel
7	Node comparison against the DEF-03 targets → Node comparison against the PPA targets	1.5w
8	DTCO findings and recommendation → DTCO findings and node recommendation	1.75w

Three of the eight steps run in parallel because they use different data and different people—power characterization, SRAM and congestion are separable studies. What cannot be parallelized is step 7: comparison needs all three to have landed.

WATCH OUT FOR

- ◆ **Benchmark circuits unrepresentative.** A study run on ring oscillators alone measures the node and not the design; the answer is precise and irrelevant.
- ◆ **Preliminary libraries treated as final.** Early PDK models move, sometimes by more than the difference between the candidates, and a comparison made on 0.5 data can invert.
- ◆ **SRAM left out.** On an accelerator the array is a large fraction of the die; a node comparison without a bitcell and Vmin study has skipped the dominant area and the dominant Vmin constraint.
- ◆ **Congestion never modelled.** Density that cannot be routed is not density, and the routing resource question is decided by the metal stack chosen in TECH-02.
- ◆ **Results reported without margin.** A target met exactly in a DTCO study will not be met in implementation, and reporting the number without the margin invites the wrong conclusion.

WHERE THE EFFORT GOES	6 M/M	WHO IS ON IT ~3.7 FTE	CONNECTIONS	TERMS HERE 6
Density and frequency studies	2.0	DTCO lead owns it	DEPENDS ON TECH-01	PPA NDA PDK DTCO SRAM Vmin
Leakage and SRAM characterization	1.5	Library characterization engineer	RUNS WITH DEF-03	
Node comparison	1.0	Memory design liaison	FEEDS INTO DEF-07 ARCH-01 ARCH-04	
Congestion proxy study	0.8	Physical design liaison	TECH-02 TECH-07 PDK-03	
Findings and recommendation	0.8	Foundry technical contact	PKG0-08 PTV-01 PTV-08	

03 TECH · TECHNOLOGY & FOUNDRY SELECTION

TECH-04

Wafer price, mask and NRE quotation; MPW and volume pricing

TAKES	COSTS	OWNER	CRITICAL PATH
6w w5–w11 of the stage	3.0 M/M ~2.2 people while it runs	Procurement lead Program manager approves	no has float

WHY IT EXISTS

Put a **price on the technology**—wafers at the forecast volume, the mask set, the NRE, the MPW shuttle, and every adder the flavor sheet carries—and hand it to the cost model as quotations rather than estimates.

The cost model in DEF-04 is only as good as this activity. A model built on published estimates produces a margin nobody can defend at the funding gate; a model built on quotations produces one that survives the first conversation with finance.

WHAT IT DELIVERS

TECH-D3

Wafer, mask and NRE cost sheet

This activity is the deliverable, and its value is that the numbers are quoted rather than estimated.

NEEDS FIRST

- Short-list and flavor sheet from TECH-01 and TECH-02
- Volume forecast and ramp profile from the business case
- Die size estimate from DEF-04
- Mask layer count implied by the metal stack
- MPW shuttle calendar from the foundry

DONE WHEN

- Every cost line quoted, with a validity period and currency term
- Tier structure reflects the actual ramp, not a notional volume
- Cost sheet loaded into the DEF-04 model, not summarized into it

HOW IT GETS THERE

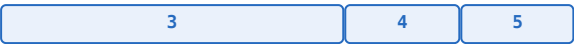
7 STEPS · 2 LANES

w5	w7	w9	w11
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Volume and wafer-start forecast from the business case → Volume and wafer-start forecast	0.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Wafer price request and negotiation → Wafer price quotations per foundry	2w
3	Mask set and NRE quotation per node and flavor → Mask set and NRE quotation per candidate	1.5w
4	MPW shuttle pricing and slot cost → MPW shuttle pricing and slot cost	0.5w
5	Option adder pricing against the TECH-02 sheet → Option adder pricing	0.5w
6	Volume tier and price-break structure → Wafer price quotation with tier structure	2.5w
7	Cost sheet publication into the DEF-04 model → Cost sheet published into the DEF-04 model	1w

The tier structure in step 6 is worth more than the headline price. A wafer price that only applies above a volume the product will not reach in its first year is a price the cost model should not be using.

WATCH OUT FOR

- ◆ **Price quoted at a volume the product will not reach.** Tier structures are where the real cost lives, and a headline price at ten times the forecast volume is a number the model should reject.
- ◆ **Mask cost estimated from layer count alone.** EUV layers dominate mask cost, and the count of them depends on the flavor sheet rather than on the total layer count.
- ◆ **Adders priced after the sheet is agreed.** Once TECH-02 closes, the adders are committed; pricing them afterwards removes the option of dropping one.
- ◆ **Currency and term exposure ignored.** A quotation with no validity period or currency term is not a quotation the cost model can carry to a funding gate.
- ◆ **Second-source pricing never requested.** Without a comparison the negotiation has no floor, and the first price is the last price.

WHERE THE EFFORT GOES	3 M/M	WHO IS ON IT	~2.2 FTE	CONNECTIONS	TERMS HERE	4
Price negotiation	1.3	Procurement lead OWNS IT		DEPENDS ON TECH-01 TECH-02	PPA NRE EUV MPW	
Mask, NRE and shuttle quotations	0.8	Product cost analyst		RUNS WITH DEF-04		
Tier structure	0.5	Technology strategist		FEEDS INTO DEF-07 DEF-09 TC-03		
Publication into the cost model	0.5	Finance partner Program manager		T0-07 MP-10		

TECH-05

Fab capacity and tapeout slot reservation; hot-lot policy agreement

TAKES	COSTS	OWNER	CRITICAL PATH
5w w8–w13 of the stage	2.0 M/M ~1.7 people while it runs	Foundry relationship manager Product engineering liaison approves	no has float

WHY IT EXISTS

Convert the program's schedule into **reserved fab capacity**—a wafer-start profile, a tapeout slot, hot-lot terms and an engineering-lot allocation—and get it committed in writing.

Capacity is not bought at tapeout; it is reserved a year before, against a date that will move. The commitment matters most in its escalation path: when the schedule slips or the market tightens, what the program has is whatever was written down, and a verbal understanding is worth exactly nothing at that moment.

WHAT IT DELIVERS

TECH-D5

Capacity and tapeout slot commitment

This activity is the deliverable, and it is worth only as much as the escalation path written into it.

NEEDS FIRST

- Program schedule and tapeout date from DEF-08
- Wafer volume forecast and ramp profile
- Foundry capacity availability at the node
- Hot-lot policy terms offered by the foundry
- Engineering and risk-lot requirements from the test plan

DONE WHEN

- Capacity committed in writing, with an escalation contact named
- Hot-lot terms explicit rather than understood
- Engineering and risk lots allocated separately from production

HOW IT GETS THERE

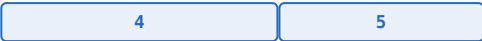
6 STEPS · 2 LANES

w8	w9	w10	w11	w12	w13
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Wafer-start profile derived from the program schedule → Wafer-start profile against the program schedule	0.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Capacity request and allocation discussion → Capacity allocation agreement	1.5w
3	Tapeout slot reservation against the DEF-08 date → Tapeout slot reservation	1w
4	Hot-lot policy and priority terms → Hot-lot policy and priority terms	1w
5	Engineering-lot and risk-wafer allocation → Engineering and risk-lot allocation	0.75w
6	Capacity commitment letter and escalation path → Capacity commitment letter with an escalation path	2w

Steps 4 and 5 look like details and behave like insurance. Hot-lot priority is what turns an 18-week fab cycle into an 11-week one at `FAB-04`, and it is only available to programs that negotiated it before they needed it.

WATCH OUT FOR

- ◆ **Slot reserved against a date that has not been committed.** A tapeout slot booked against an optimistic schedule is a slot that will be missed and re-queued behind whoever is ready.
- ◆ **Hot-lot terms left informal.** Priority granted by relationship evaporates when a larger customer needs the same capacity in the same week.
- ◆ **No escalation path.** A commitment with no named escalation contact is a commitment enforced by goodwill, which is not a mechanism.
- ◆ **Engineering lots forgotten.** Test-program development and characterization need silicon before the production ramp, and those wafers have to be allocated separately.
- ◆ **Capacity sized to the base case only.** An upside scenario the fab cannot supply is a revenue forecast the program cannot deliver on.

WHERE THE EFFORT GOES	2 M/M	WHO IS ON IT ~1.7 FTE	CONNECTIONS	TERMS HERE	1
Capacity allocation discussion	0.8	Foundry relationship manager OWNS IT	DEPENDS ON DEF-08 TECH-01 TECH-06	NRE	
Tapeout slot reservation	0.5	Program manager	FEEDS INTO TC-03 T0-06 T0-07		
Hot-lot and engineering-lot terms	0.5	Operations planner	FAB-02 FAB-09 MP-08		
Commitment letter	0.3	Procurement lead	MP-10		
		Product engineering liaison			

03 TECH · TECHNOLOGY & FOUNDRY SELECTION

TECH-06

Foundry legal engagement — NDA, design agreement, IP licensing frame

TAKES	COSTS	OWNER	CRITICAL PATH
8w w2–w10 of the stage	2.0 M/M ~1.1 people while it runs	Legal counsel Compliance officer approves	yes slipping moves the program

WHY IT EXISTS

Put the **legal frame** in place that everything technical depends on—NDA, design agreement, PDK and library licensing, the foundry IP frame, and liability terms—and execute it.

This is the longest activity in the stage at eight weeks and one of the smallest at 2 M/M, which is exactly the shape it should have: the elapsed time belongs to two legal departments and almost none of the work belongs to the program. It is on the critical path anyway, because production PDK access, memory compilers and foundry IP all sit behind the design agreement.

WHAT IT DELIVERS

TECH-D4

Executed foundry design agreement (DA) and NDA

This activity is the deliverable, and until it executes the enablement band cannot legally start.

NEEDS FIRST

- Short-list from TECH-01
- Corporate legal templates and negotiation positions
- Foundry standard agreement terms
- Export control and compliance guidance
- IP requirements from ARCH-04 and IPR-01

DONE WHEN

- NDA and design agreement executed, not merely agreed
- PDK, library and foundry IP access all covered by scope
- Access distributed to the teams that need it, and tracked

HOW IT GETS THERE

8 STEPS · 2 LANES

w2	w4	w6	w8	w10
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	NDA execution, to open technical discussion → Executed NDA	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Design agreement scope definition → Design agreement, scoped and executed	1w
3	PDK and library licensing terms → PDK and library license terms	1.5w
	runs in parallel	
4	Foundry IP licensing frame — memory compilers, IO, PHY → Foundry IP licensing frame	2.5w
5	Liability, indemnity and yield-responsibility terms → Agreed liability, indemnity and yield-responsibility terms	1.5w
	runs in parallel	
6	Export control and compliance review → Compliance and export-control record	1w
	runs in parallel	
7	Legal negotiation and redline cycles → Redline history with resolved positions	2.5w
8	Execution and distribution to the teams that need it → Distribution record — who is covered to access what	1w

Step 1 is deliberately first and deliberately short. Nothing technical can be discussed before the NDA executes, so every week it takes is a week subtracted from `TECH-03` — which is why the NDA is chased separately from the agreement it precedes.

WATCH OUT FOR

- ◆ **NDA treated as a formality and queued behind the agreement.** Technical evaluation is blocked until it executes, and every week lost here is taken out of the DTCO study.
- ◆ **Foundry IP frame left out of scope.** Memory compilers and IO libraries are licensed separately from the PDK; discovering that at PDK-04 stops library qualification.
- ◆ **Distribution not tracked.** An executed agreement nobody circulated leaves engineers unable to download the PDK they are waiting for, and the delay looks technical.
- ◆ **Yield responsibility left ambiguous.** Who owns a process excursion is a term, not a conversation to have after it happens.
- ◆ **Redline cycles unbudgeted.** Two legal departments will take longer than either forecasts, and the program's plan should assume the longer number.

WHERE THE EFFORT GOES	2 M/M	WHO IS ON IT	~1.1 FTE	CONNECTIONS	TERMS HERE	8
Negotiation support	0.8	Legal counsel	OWNS IT	DEPENDS ON DEF-09 TECH-01	M/M NDA PDK DTCO	
IP and licensing frame	0.5	Foundry relationship manager		FEEDS INTO TECH-05 PDK-01 PDK-04	IP PHY IO CAD	
Terms review	0.5			IPR-07 TC-03 FAB-10		
Execution and distribution	0.3	Procurement lead				
		CAD and methodology lead				
		Compliance officer				

TECH-07

Foundry roadmap and risk-production timing alignment

TAKES	COSTS	OWNER	CRITICAL PATH
4w w1–w5 of the stage	1.5 M/M ~1.6 people while it runs	Technology strategist Product cost analyst approves	no has float

WHY IT EXISTS

Establish where each candidate node **actually is on its maturity curve**—risk production timing, defect density trend, yield learning rate, capacity ramp—and turn that into a risk assessment the program plans against.

A node chosen at the right time is a node in volume production when the program taped out. A node chosen too early gives the program the foundry's learning curve as a yield problem, and the schedule to absorb it was never in the plan.

WHAT IT DELIVERS

TECH-D6

Node risk assessment — maturity, defect density, yield learning curve

This activity is the deliverable, and it is what the yield assumption in the cost model should be built on.

NEEDS FIRST

- Short-list from TECH-01
- Foundry roadmap and node cadence under NDA
- Defect density and yield learning data where available
- Program schedule and tapeout date from DEF-08
- DTCO findings as they emerge from TECH-03

DONE WHEN

- Yield curve stated at the program's own tapeout date, not in general
- Every maturity risk carries at least one mitigation option
- Assessment handed to DEF-04 as the model's yield input

HOW IT GETS THERE

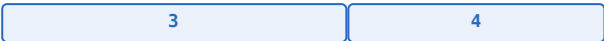
5 STEPS · 2 LANES

w1	w2	w3	w4	w5
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Node maturity and risk-production timing review → Node maturity and risk-production timing assessment	1w
2	Defect density trend and yield learning curve request → Defect density trend and yield learning curve	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Foundry roadmap alignment against the program schedule → Roadmap alignment note against the program schedule	1w runs in parallel
4	Capacity ramp profile at the node → Capacity ramp profile	0.75w runs in parallel
5	Node risk assessment and mitigation options → Node maturity assessment per candidate · Node risk assessment with mitigation options	2w

Step 2 asks for data the foundry treats as sensitive, and how much of it arrives is itself a signal. A foundry that will not share a defect density trend under NDA is telling the program something about the trend.

WATCH OUT FOR

- ◆ **Marketing defect density used as a planning number.** The published curve is a target; the yield the program gets is the curve at its own tapeout date, which is a different point.
- ◆ **Risk production date taken as the availability date.** Risk production is when the node exists, not when it is ready for a large die at volume, and the gap can be two quarters.
- ◆ **Roadmap alignment checked once.** Foundry roadmaps move; an alignment done at selection and never revisited is a stale assumption by tapeout.
- ◆ **No mitigation options developed.** Identifying a maturity risk without a mitigation—a later slot, a shuttle, a conservative design rule set—leaves the program with a finding and no action.
- ◆ **Large-die yield inferred from small-die data.** Defect density scales against area brutally, and a curve measured on small test die flatters a 600 mm² part.

WHERE THE EFFORT GOES	1.5 M/M	WHO IS ON IT	~1.6 FTE	CONNECTIONS	TERMS HERE	3
Maturity and roadmap review	0.5	Technology strategist	owns it	DEPENDS ON TECH-01 TECH-03	PPA NDA DTCO	
Defect density and yield data	0.5	Yield engineer		FEEDS INTO DEF-04 DEF-07 TECH-08		
Risk assessment and mitigations	0.5	Foundry technical contact		AMS-09 TC-01 DFT-02		
		Program manager		S0-08 FAB-06 MP-05		
		Product cost analyst				

03 TECH · TECHNOLOGY & FOUNDRY SELECTION

TECH-08

Second-source and node-migration strategy assessment

TAKES	COSTS	OWNER	CRITICAL PATH
4w w9–w13 of the stage	1.5 M/M ~1.6 people while it runs	Technology strategist Program manager approves	no has float

WHY IT EXISTS

Decide, deliberately, **whether the program accepts single-source risk**—and if it does not, what dual-sourcing or migration readiness would cost in area, schedule and IP.

Most leading-edge programs end up single-source, and that is a defensible answer. What is not defensible is arriving there by default. This activity prices the alternative so that the acceptance is a decision with a number attached rather than a shrug.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Selected node and flavor from TECH-01 and TECH-02
- IP bill of materials direction from IPR-01
- Design rule sets for alternative foundries
- Migration experience from previous programs
- Supply-continuity findings from TECH-01

DONE WHEN

- Migration cost estimated with IP portability included
- Recommendation recorded, including the option of accepting single source
- Accepted exposure entered on the program risk register with an owner

HOW IT GETS THERE

5 STEPS · 2 LANES

w9	w10	w11	w12	w13
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Second-source viability screen at the selected node class → Second-source viability finding	1w
2	Migration cost and schedule estimate → Migration cost and schedule estimate	1.25w
3	IP portability assessment across foundries → IP portability assessment	0.75w runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Design rule and library delta analysis → Design rule and library delta analysis	0.75w <i>runs in parallel</i>
5	Strategy recommendation — dual-source, migration-ready, or single-source accepted → Sourcing strategy recommendation and decision record	1.75w

Migration readiness is a design-time choice, not a later option: it constrains which IP can be used and how aggressively design rules can be pushed. Deciding after RTL starts means the decision has already been made by everyone who chose an IP.

WATCH OUT FOR

- ◆ **Single-source accepted by default.** A decision nobody made is a decision nobody owns, and it surfaces during a supply disruption as a surprise.
- ◆ **Migration cost estimated without IP portability.** Foundry-specific memory compilers, IO and PHY are the expensive part of a migration, and they are usually the part left out of the estimate.
- ◆ **Migration readiness assumed to be free.** Designing to the intersection of two rule sets costs density and frequency on the node actually used.
- ◆ **Decision taken after IP selection.** Once IPR-03 has committed to foundry-specific IP, migration readiness is no longer available at the price this study quotes.
- ◆ **Second source screened on capability alone.** Capability without allocated capacity is not a second source; it is a phone number.

WHERE THE EFFORT GOES	1.5 M/M	WHO IS ON IT	~1.6 FTE	CONNECTIONS	TERMS HERE	4
Migration cost and schedule estimate	0.5	Technology strategist owns it		DEPENDS ON TECH-01 TECH-02 TECH-07 IPR-01	IP PHY RTL IO	
Second-source viability screen	0.5	IP strategist		FEEDS INTO IPR-03 IPR-10 BU-11		
IP portability and rule delta	0.3	Physical design liaison		FEEDS BACK INTO DEF-09		
Strategy recommendation	0.3	Procurement lead Program manager				

03 TECH · TECHNOLOGY & FOUNDRY SELECTION

TECH-09

OSAT and backend supply chain preliminary alignment

TAKES	COSTS	OWNER	CRITICAL PATH
5w w9–w14 of the stage	2.0 M/M ~1.7 people while it runs	Backend supply chain lead Operations planner approves	no has float

WHY IT EXISTS

Open the **backend supply chain** early—OSAT capability and capacity, substrate and interposer suppliers, the foundry-to-OSAT logistics model, and who owns known-good die—so package design starts against real constraints.

On a 2.5D part the backend is not a downstream detail; it is a second supply chain with its own lead times, its own capacity constraints and its own qualification. Advanced packaging capacity in particular is allocated far ahead, and a program that starts asking at PKGD-10 is asking late.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Package architecture assumption from the requirement register
- Selected foundry and node from TECH-01
- Volume and ramp profile from the business case
- OSAT and substrate supplier capability data
- Interposer technology options from the foundry

DONE WHEN

- Advanced packaging capacity indicated, not assumed
- Known-good-die responsibility model agreed in principle
- Findings handed to PKGD before package architecture is chosen

HOW IT GETS THERE

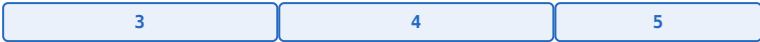
7 STEPS · 2 LANES

w9	w10	w11	w12	w13	w14
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	OSAT long-list against the package architecture assumption → OSAT long-list with capability assessment	0.75w
2	Advanced packaging capability and capacity screen → Advanced packaging capacity indication	1.5w
3	Foundry-to-OSAT logistics and interface model → Foundry-to-OSAT logistics model	1w

runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Substrate and interposer supplier landscape → Substrate and interposer supplier landscape	1w runs in parallel
5	Known-good-die handoff responsibility model → Known-good-die responsibility model	0.75w runs in parallel
6	Preliminary capacity indication and engagement → Preliminary OSAT capacity indication	1.5w
7	Backend supply chain finding and handoff to package design → Backend supply chain findings for package design	1.25w

Step 5 decides an argument before it happens. When a packaged unit fails, whether the die or the assembly is responsible is a contractual question, and settling it while nobody has lost money is far easier than settling it afterwards.

WATCH OUT FOR

- ◆ **Advanced packaging capacity assumed available.** CoWoS-class capacity is allocated quarters ahead and to a small number of customers; assuming access is the single most common backend planning error.
- ◆ **OSAT chosen after package design starts.** Assembly process capability constrains the package design; choosing the OSAT afterwards means redesigning to fit.
- ◆ **Known-good-die responsibility left open.** The first failed assembly lot is the wrong moment to discover that nobody agreed who owns the die.
- ◆ **Substrate lead time discovered rather than planned.** Twenty-week substrate lead times set the order date, and the order date is derived from a wafer-out date nobody has committed yet.
- ◆ **Logistics model ignored.** Wafers crossing borders between fab, sort and assembly carry customs, export control and transit time that the schedule rarely shows.

WHERE THE EFFORT GOES	2 M/M	WHO IS ON IT	~1.7 FTE	CONNECTIONS	TERMS HERE	2
Capability and capacity screen	0.8	Backend supply chain lead owns IT		DEPENDS ON DEF-01 TECH-01	NRE OSAT	
OSAT and supplier engagement	0.5	Package architect		FEEDS INTO FAB-10 PKGD-01 PKGD-09		
Logistics and supplier landscape	0.5	Procurement lead		PKGD-10 PTV-01 ASSY-01		
Handoff to package design	0.3	Quality liaison		ASSY-02		
		Operations planner				

04

PDK & Design Enablement

PDK

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH		
w8 – w44	36 _w	13	97 M/M	7 of 13		
ENGINEERING ACTIVITIES				KEY DELIVERABLES		
PDK-01	PDK version roadmap tracking (0.1 → 0.5 → 1.0 → production) and delta impact analysis		34w cont.	14 M/M	PDK-D1	PDK readiness dashboard — version, release date, open gap list
PDK-02	Design rule manual review; restricted and recommended rule disposition		6w	5 M/M	PDK-D2	Qualified library list with .lib / LEF / GDS views
PDK-03	Standard cell library selection and qualification (track height, multi-Vt, multi-bit flops)		8w	9 M/M	PDK-D3	Memory PPA gap analysis and custom-instance decision record
PDK-04	Memory compiler evaluation and instance generation (SRAM, register file)		9w	10 M/M	PDK-D4	EDA tool and version matrix (qualified and frozen)
PDK-05	Compiler instance PPA characterization against block budgets — density, Vmin, access time, leakage		6w	7 M/M	PDK-D5	Internal reference flow and methodology guide
PDK-06	Custom / pushed-rule memory instance decision, scope and schedule impact assessment		4w	3 M/M	PDK-D6	Signoff corner definition agreed with foundry
PDK-07	IO, ESD and latch-up library qualification		6w	5 M/M	PDK-D7	Compute and license capacity plan
PDK-08	EDA tool version qualification for the node — synthesis, P&R, STA, PV, EM/IR		10w	12 M/M	PDK-D8	Golden environment release notes
PDK-09	Foundry reference flow bring-up and internal methodology deck		10w	12 M/M		
PDK-10	Rule deck, QRC tech file and DRC/LVS deck version control		6w cont.	4 M/M		
PDK-11	Signoff corner and derate definition agreed with foundry (PVT, OCV/AOCV/POCV)		6w	5 M/M		
PDK-12	Compute farm, EDA license and storage capacity planning		6w	3 M/M		
PDK-13	Golden design environment release and flow regression		8w	8 M/M		

PDK-01

PDK version roadmap tracking (0.1 → 0.5 → 1.0 → production) and delta impact analysis

TAKES

34w cont.

w0–w34 of the stage

COSTS

14.0 M/M

~1.8 people while it runs

OWNER

PDK owner

Physical design liaison approves

CRITICAL PATH

yes

slipping moves the program

WHY IT EXISTS

Track what the foundry actually ships, version by version, and turn each release into **a delta the program can act on**—what changed, what it costs, and whether anything already done has to be redone.

A PDK is not a thing you receive; it is a sequence. Design starts on 0.5 because waiting for production would cost two quarters, and the 0.5-to-1.0 delta then arrives in the middle of physical design. Whether that delta is a two-week re-characterization or a reopened timing signoff depends entirely on whether anyone was watching.

WHAT IT DELIVERS

PDK-D1

PDK readiness dashboard — version, release date, open gap list

This activity is the deliverable, and it is a living document rather than a report—its value is entirely in being current.

NEEDS FIRST

- Foundry PDK release calendar and notes
- Selected node and flavor sheet from TECH-01 and TECH-02
- Design agreement access from TECH-06
- Library and deck versions in use across the teams
- Design status — what has been characterized against which version

DONE WHEN

- Every version transition assessed before adoption, not after
- All teams on one recorded version, with exceptions logged
- Production PDK adopted and frozen with an exception path defined

HOW IT GETS THERE

8 STEPS · 2 LANES

w0w7w14w21w28w34

MAIN SEQUENCE

12468

IN PARALLEL

357

#STEP, AND WHAT IT ADDS TO THE DELIVERABLETAT

1

Release calendar and version tracking framework

→ Release calendar and version tracking framework

1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	0.1 to 0.5 delta impact analysis → 0.1-to-0.5 delta impact analysis	4w
3	Gap list maintained against what the design actually needs → Open gap list against the design's requirements	3w runs in parallel
4	0.5 to 1.0 delta impact analysis → Delta impact analysis per version transition	5w
5	Library and deck re-qualification per version → Re-qualification record per library and deck	6w runs in parallel
6	Production PDK adoption and version freeze decision → Production PDK adoption and freeze decision	4w
7	Version freeze policy and exception handling → Version freeze decision and exception log	3w runs in parallel
8	Readiness dashboard maintenance across the stage → PDK readiness dashboard, maintained	20w

Step 8 runs for twenty of the thirty-four weeks and is the activity most likely to be dropped when the team is busy. It is also the only artifact that tells the program which of its assumptions have quietly expired.

WATCH OUT FOR

- ◆ **Design starts on a version the foundry calls preliminary and nobody re-checks.** The gap between 0.5 and production has reopened timing signoff on more than one program.
- ◆ **Delta analysis done on the release notes alone.** Foundry notes describe what changed; only re-running the design's own characterization says what it costs here.
- ◆ **Teams on different versions.** Synthesis on one library revision and physical design on another produces a mismatch that looks like a tool bug for weeks.
- ◆ **Freeze declared without an exception path.** A hard freeze with no route for a critical fix gets broken quietly, which is worse than breaking it openly.
- ◆ **Dashboard maintained until the team gets busy.** A stale readiness dashboard is more dangerous than none, because it is still trusted.

WHERE THE EFFORT GOES	14 M/M	WHO IS ON IT	~1.8 FTE	CONNECTIONS	TERMS HERE	2
Dashboard maintenance	4.0	PDK owner OWNS IT		DEPENDS ON TECH-01 TECH-02 TECH-06	PDK CAD	
Delta impact analyses	5.0	CAD methodology engineer		FEEDS INTO PDK-02 PDK-03 PDK-04		
Gap list against design needs	2.0	Library engineer		PDK-07 PDK-08 PDK-10		
Re-qualification coordination	2.0	Foundry technical contact		PDK-13 AMS-02 AMS-11		
Freeze policy and exceptions	1.0	Physical design liaison		SYN-01 PD-01 S0-10		

04 PDK · PDK & DESIGN ENABLEMENT

PDK-02

Design rule manual review; restricted and recommended rule disposition

TAKES 6w w1-w7 of the stage	COSTS 5.0 M/M ~3.6 people while it runs	OWNER CAD methodology lead Foundry technical contact approves	CRITICAL PATH no has float
WHY IT EXISTS Read the design rule manual as a constraint on this design rather than as a document—identify which restricted rules bite, dispose of every recommended rule, and get the exceptions requested early. Restricted and recommended rules are where a node's real density lives. A design that adopts every recommended rule is safe and large; one that ignores them is dense and fails DFM. The disposition is a series of deliberate trades, and making them here is far cheaper than discovering them as violations at physical verification.			
WHAT IT DELIVERS No key deliverable is owned here — this activity contributes to the ones below.			
NEEDS FIRST - Design rule manual for the selected node and flavor - Previous-node DRM for comparison - Floorplan intent from ARCH-08 - Custom and analog design intent from AMS-01 - Foundry exception request process			
DONE WHEN - Every recommended rule dispositioned with a reason, not defaulted - Exception requests submitted, with expected response dates - Disposition distributed to digital and custom teams alike			
HOW IT GETS THERE 6 STEPS · 2 LANES w1w3w5w7 1236 45			
#STEP, AND WHAT IT ADDS TO THE DELIVERABLE TAT 1DRM ingestion and change log against the previous node → Rule change log against the previous node1w 2Restricted rule identification and impact assessment → Restricted rule impact assessment1.5w			

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Recommended rule disposition — adopt, waive or defer → Recommended rule disposition record	1.5w
4	Density and spacing rule impact on the floorplan → Density and spacing rule impact on the floorplan	1w
5	Rule exception request preparation for the foundry → Exception requests submitted to the foundry	1w
6	DRM review record and distribution to the design teams → DRM review record for the design teams	2w

Exception requests in step 5 run in parallel because the foundry's response time is measured in weeks. Submitting them at the end of the review rather than as each one is identified adds a month to the answer.

WATCH OUT FOR

- ◆ **Recommended rules adopted wholesale.** Safe, and it costs density the product was sized against; the trade should be made per rule, not once.
- ◆ **Restricted rules discovered at physical verification.** A restricted rule that bites is a layout change, and finding it at S0-02 means finding it after routing closed.
- ◆ **Exception requests batched to the end.** Foundry turnaround runs to weeks per request; batching them converts a parallel wait into a serial one.
- ◆ **Analog rules read only by the digital team.** Pushed-rule and analog-specific sections matter to AMS-07, and they are the sections a digital-led review skips.
- ◆ **Disposition never distributed.** A decision the layout engineers never saw is a decision that does not exist.

WHERE THE EFFORT GOES	5 M/M	WHO IS ON IT	~3.6 FTE	CONNECTIONS	TERMS HERE	4
Rule impact assessment	2.0	CAD methodology lead owns it		DEPENDS ON TECH-02 PDK-01	PDK DRM DFM CAD	
Disposition decisions	1.5	Physical design representative		FEEDS INTO PDK-09 PDK-10 AMS-07 AMS-12 DFT-10 PD-06		
Exception preparation	0.8	Custom layout representative		PD-08 PD-15 S0-06		
Review record and distribution	0.8	DFM engineer		S0-07 S0-08		
		Foundry technical contact				

PDK-03

Standard cell library selection and qualification
(track height, multi-Vt, multi-bit flops)

TAKES	COSTS	OWNER	CRITICAL PATH
8w w4–w12 of the stage	9.0 M/M ~4.9 people while it runs	Library lead Foundry library contact approves	yes slipping moves the program

WHY IT EXISTS

Choose the standard cell libraries the design will be built from and **qualify them rather than assume them**—track height, Vt menu, multi-bit flops, and the timing models every downstream tool will trust.

Library selection is a PPA decision disguised as a procurement one. Track height sets density and drive strength; the Vt menu sets leakage; multi-bit flops set clock power. And a timing model that is wrong at one corner produces a design that closes in the tool and fails in silicon, which is the failure mode nobody catches until bring-up.

WHAT IT DELIVERS

PDK-D2

Qualified library list with .lib / LEF / GDS views

The standard cell half of the deliverable; PDK-04 adds memory and PDK-07 adds IO.

NEEDS FIRST

- Flavor sheet from TECH-02
- DTCO findings from TECH-03
- PDK release under tracking from PDK-01
- PPA targets and clock power budget from ARCH-06
- Foundry library release notes and characterization data

DONE WHEN

- Timing models checked at this design's corners, not only the foundry's
- Vt menu frozen with a maintenance owner per corner
- Abstract and layout views confirmed consistent

HOW IT GETS THERE

7 STEPS · 2 LANES

w4

w6

w8

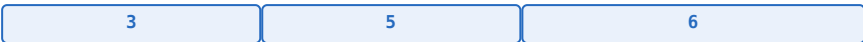
w10

w12

MAIN SEQUENCE



IN PARALLEL



#

STEP, AND WHAT IT ADDS TO THE DELIVERABLE

TAT

1	Track height and library family selection → Library family and track height selection record	1w
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#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Multi-Vt menu qualification against the flavor sheet → Qualified Vt menu with leakage projections	1.5w
3	Multi-bit flop and special cell evaluation → Multi-bit and special cell evaluation	1.5w runs in parallel
4	Timing model quality check across corners → Timing model quality report across corners	2w
5	Physical view and abstract consistency check → Physical view and abstract consistency findings	1.5w runs in parallel
6	Library re-characterization where the foundry data is thin → Re-characterization results where foundry data was thin	2w runs in parallel
7	Qualification record and library freeze → Library qualification record and freeze	3.5w

Step 4 is the step that earns the activity. Foundry timing models are qualified by the foundry against the foundry's test cases; checking them at this design's corners and slew ranges is what catches the model error before it becomes a silicon failure.

WATCH OUT FOR

- ◆ **Timing models trusted without a quality check.** A model error at one corner produces a design that closes in the tool and fails on silicon—the most expensive class of library defect.
- ◆ **Track height chosen from the previous program.** Density and drive strength trade differently at each node; inheriting the choice inherits an answer to a question nobody asked here.
- ◆ **Vt menu wider than the flow can maintain.** Every Vt is a characterization, a qualification and a maintenance burden across every version transition.
- ◆ **Multi-bit flops evaluated after synthesis starts.** Clock power is largely decided by flop banking, and retrofitting it means re-running synthesis and re-closing timing.
- ◆ **LEF and GDS inconsistency.** An abstract that disagrees with the layout produces DRC violations at the very end, after routing has been built on the abstract.

WHERE THE EFFORT GOES	9 M/M	WHO IS ON IT	~4.9 FTE	CONNECTIONS	TERMS HERE	8
Library re-characterization	3.0	Library lead owns it		DEPENDS ON TECH-02 TECH-03 PDK-01	PPA PDK DTC0 DRC I0 GDS LEF .lib	
Timing model quality checks	2.0	Characterization engineer		FEEDS INTO ARCH-01 ARCH-05 PDK-11 PDK-13 RTL-05 SYN-02		
Vt menu qualification	1.5	Synthesis liaison		SYN-06 SYN-07 PD-05		
Special cell evaluation	1.0	Physical design liaison		S0-12		
Qualification record and freeze	1.5	Foundry library contact		LATER INPUT TC-07 +30w TC-08 +36w		

PDK-04

Memory compiler evaluation and instance generation (SRAM, register file)

TAKES	COSTS	OWNER	CRITICAL PATH
9w w6–w15 of the stage	10.0 M/M ~4.8 people while it runs	Memory library engineer Foundry memory contact approves	yes slipping moves the program

WHY IT EXISTS

Get the memory compiler licensed, evaluate what it can actually produce, and **generate the instances the design asked for**—with every view a downstream tool needs.

On an inference accelerator the memory array is a large fraction of the die, and the compiler decides most of it. What the compiler can generate is a hard boundary on the architecture: a configuration outside its range is not a compiler setting, it is a custom design program in AMS-05 costing four months.

WHAT IT DELIVERS

PDK-D2

Qualified library list with .lib / LEF / GDS views

The memory half of the deliverable—instances with every view a downstream tool consumes.

NEEDS FIRST

- Memory hierarchy and capacity requirements from ARCH-03
- Compiler license terms from TECH-06
- PDK version under tracking from PDK-01
- BISR and repair architecture intent from DFT-03
- Block memory budgets from ARCH-06

DONE WHEN

- Every required configuration either generated or recorded as a gap
- All views present and checked, not merely produced
- Redundancy scheme aligned with the DFT architecture

HOW IT GETS THERE

7 STEPS · 2 LANES

w6	w8	w10	w12	w14	w15
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Memory requirement collection from the hierarchy definition → Memory requirement list per block	0.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Compiler license and access setup → Compiler license and access setup record	1w
3	Compiler capability screen against the requirement list → Compiler capability screen and gap list	1.5w
4	Instance generation across the required configurations → Generated instances across required configurations	2.5w
5	Redundancy and repair option evaluation → Redundancy and repair option evaluation	1.5w runs in parallel
6	Compiler view completeness check — .lib, LEF, GDS, CDL, models → Compiler view completeness findings	1.5w runs in parallel
7	Instance library assembly and release → Instance library with complete views	3.5w

Step 3 is the one with a schedule consequence. A capability gap found here opens the <code>PDK-06</code> decision with fifteen weeks of runway; found at instance generation, it opens the same decision with three.

WATCH OUT FOR

- ◆ **Compiler license assumed to be covered by the PDK agreement.** Memory compilers are licensed separately, and discovering that here stops the activity for weeks.
- ◆ **Capability screened after generation starts.** The gap is the same size either way; what differs is how much runway the custom path has.
- ◆ **Views generated but never checked.** A missing or inconsistent view surfaces at PD-02 when the macro cannot be placed, and the compiler run has to be repeated.
- ◆ **Redundancy chosen without DFT.** Repair schemes have to match the BISR architecture in DFT-03, and a mismatch is found at pattern generation.
- ◆ **Instance count explosion.** Generating every configuration a block owner asks for produces a library nobody can characterize or maintain; consolidation is cheaper here than later.

WHERE THE EFFORT GOES	10 M/M	WHO IS ON IT ~4.8 FTE	CONNECTIONS	TERMS HERE	9
Instance generation	3.5	Memory library engineer owns IT	DEPENDS ON ARCH-03 TECH-02 TECH-06 PDK-01 IPR-01	PPA PDK DFT BISR GDS LEF .lib CDL CAD	
Capability screen	2.0	Memory architect liaison	FEEDS INTO PDK-05 PDK-13 AMS-05 RTL-03 DFT-03 PD-02		
View completeness checks	1.5	DFT liaison			
Redundancy option evaluation	1.5	CAD methodology engineer			
Library assembly and release	1.5	Foundry memory contact			

PDK-05

Compiler instance PPA characterization against block budgets — density, Vmin, access time, leakage

TAKES	COSTS	OWNER	CRITICAL PATH
6w w14–w20 of the stage	7.0 M/M ~5.1 people while it runs	Memory characterization engineer Block owners approves	yes slipping moves the program

WHY IT EXISTS

Measure what the generated instances actually deliver—**density, Vmin, access time, leakage**—and set it against the budget each block was allocated.

This is the activity that decides whether the custom memory path opens. The compiler produces an instance that meets the specification; the question here is whether it meets the *budget*, and the answer is frequently no by a margin large enough to matter but small enough to be argued about.

WHAT IT DELIVERS

PDK-D3

Memory PPA gap analysis and custom-instance decision record

The gap analysis half—PDK-06 adds the decision that follows from it.

NEEDS FIRST

- Generated instances from PDK-04
- Block memory budgets from ARCH-06
- Vmin and operating voltage requirements from ARCH-05
- Access time requirements from the hierarchy specification
- DTCO SRAM findings from TECH-03

DONE WHEN

- Every instance measured against its block's budget
- Gaps quantified in area, power and Vmin, not described
- Results delivered with enough runway for PDK-06 to act

HOW IT GETS THERE

6 STEPS · 2 LANES

w14	w16	w18	w20
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Block memory budget extraction from ARCH-06 → Per-block memory budget extract	0.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Density measurement per generated instance → Density measurement per instance	1.5w
3	Vmin and access time characterization → Vmin and access time characterization results	1.5w
4	Leakage and dynamic power measurement → Leakage and dynamic power measurements	1w <i>runs in parallel</i>
5	Area efficiency against the block budget → Area efficiency comparison against budget	1w <i>runs in parallel</i>
6	Gap analysis and shortfall quantification → Quantified gap analysis per block	2.5w

The gap analysis is quantified rather than described on purpose. "The compiler misses the budget" starts an argument; "the L2 array misses its density budget by 14%, which is 3.1 mm² of die" starts a decision.

WATCH OUT FOR

- ◆ **Density measured on the datasheet rather than the instance.** Compiler datasheets quote best-case configurations; the instance the design needs is rarely one of them.
- ◆ **Vmin characterized at nominal only.** Vmin is a distribution, and the number that matters is at the corner and the sigma the product is sold at.
- ◆ **Gap described rather than quantified.** A shortfall without a number in square millimetres cannot be traded against schedule in PDK-06.
- ◆ **Measurement late enough to remove the option.** Characterization completing after the AMS window has closed makes the custom path unavailable regardless of the gap.
- ◆ **Access time measured without the wire load.** An instance that meets access time standalone and misses it in place has not been characterized usefully.

WHERE THE EFFORT GOES	7 M/M	WHO IS ON IT ~5.1 FTE	CONNECTIONS	TERMS HERE 5
Vmin and access time characterization	3.0	Memory characterization engineer <i>OWNS IT</i>	DEPENDS ON ARCH-03 ARCH-06 PDK-04	PPA DTC0 SRAM AMS
Density measurement	1.5	Memory library engineer	FEEDS INTO PDK-06 AMS-05 AMS-08	Vmin
Gap analysis	1.5	PPA lead	PD-02 MP-08	
Power measurement	1.0	Circuit design liaison	FEEDS BACK INTO ARCH-06	
		Block owners		

04 PDK · PDK & DESIGN ENABLEMENT

PDK-06

Custom / pushed-rule memory instance decision, scope and schedule impact assessment

TAKES	COSTS	OWNER	CRITICAL PATH
4w w19–w23 of the stage	3.0 M/M ~3.3 people while it runs	PDK owner Chief architect approves	yes slipping moves the program

WHY IT EXISTS

Decide what to do about the memory shortfall—**push rules, build a custom array, or change the architecture**—and hand the chosen scope to AMS with a schedule attached.

This is a gate, not a study. The compiler instance missed the budget; the options each cost something different and one of them has to be taken. Deferring the decision does not preserve the options—the custom array path needs its full window in AMS-05 through AMS-10, and every week of deferral removes one from the end.

WHAT IT DELIVERS

PDK-D3

Memory PPA gap analysis and custom-instance decision record

The decision half of the deliverable, and the record downstream stages will point at.

NEEDS FIRST

- Quantified gap analysis from PDK-05
- AMS capacity and window from the program schedule
- Foundry pushed-rule policy and approval process
- Architecture flexibility from ARCH-03
- Cost impact of the die area at stake from DEF-04

DONE WHEN

- Decision taken and recorded with the rejected options priced
- Foundry approval path confirmed if pushed rules are chosen
- Scope handed to AMS with a date, not only a requirement

HOW IT GETS THERE

6 STEPS · 2 LANES

w19	w20	w21	w22	w23
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Shortfall review against the block budgets → Shortfall review against the block budgets	0.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Option definition — pushed rules, custom array, or architecture change → Defined options — pushed rules, custom array, architecture change	1w
3	Schedule impact assessment against the AMS window → Schedule impact assessment against the AMS window	1w
4	Foundry appetite for pushed rules and the approval path → Foundry pushed-rule appetite finding	0.75w <i>runs in parallel</i>
5	Cost and risk comparison of the options → Option definition with cost, schedule and risk per path	0.75w <i>runs in parallel</i>
6	Decision and scope handoff to AMS → Decision record with the alternatives rejected · Scope handoff package to AMS	1.5w

Step 4 runs in parallel and can veto an option outright. A pushed-rule array the foundry will not approve is not an option, and finding that after the decision is taken costs the whole custom window.

WATCH OUT FOR

- ◆ **Decision deferred to preserve options.** Deferral removes the custom option rather than preserving it, because the window is fixed at the far end by physical design.
- ◆ **Pushed rules chosen without foundry commitment.** Approval is a process with its own duration, and an unapproved pushed-rule array is a layout nobody can tape out.
- ◆ **Architecture change dismissed too quickly.** Reducing the array is sometimes cheaper than building it; the option deserves the same costing as the other two.
- ◆ **Scope handed to AMS without a schedule.** A custom instance request with no date is a request that competes with the SerDes for the same engineers and loses.
- ◆ **Decision taken without the block owner.** The owner who accepted the budget in ARCH-06 is the one who has to accept the concession here.

WHERE THE EFFORT GOES	3 M/M	WHO IS ON IT ~3.3 FTE	CONNECTIONS	TERMS HERE	4
Option definition	1.0	PDK owner OWNS IT	DEPENDS ON DEF-04 ARCH-03 PDK-05	PPA PDK SerDes AMS	
Schedule impact assessment	0.8	Memory circuit lead	FEEDS INTO AMS-05 AMS-06 AMS-07		
Foundry approval path	0.5	Program manager	PD-02		
Decision record and handoff	0.8	Foundry technical contact Chief architect	FEEDS BACK INTO ARCH-06 AMS-01 TC-01		

04 PDK · PDK & DESIGN ENABLEMENT

PDK-07

IO, ESD and latch-up library qualification

TAKES	COSTS	OWNER	CRITICAL PATH
6w w8–w14 of the stage	5.0 M/M ~3.6 people while it runs	IO library engineer SI liaison approves	no has float

WHY IT EXISTS

Qualify the **IO, ESD and latch-up** collateral—the library that sits at the die edge and decides whether the part survives handling, and whether its interfaces meet their electrical specifications.

IO qualification is unglamorous and it is where parts die. An ESD strategy chosen late constrains the pad ring after the bump map is fixed; a latch-up rule missed produces a part that fails qualification for reasons no functional test would ever find.

WHAT IT DELIVERS

PDK-D2

Qualified library list with .lib / LEF / GDS views

The IO and ESD half of the deliverable—the cells at the die edge, qualified rather than assumed.

NEEDS FIRST

- Interface list from ARCH-04
- Bump and pad ring plan from ARCH-08
- Foundry IO library and ESD rule set
- Latch-up and reliability requirements from the qualification plan
- Package and board ESD requirements

DONE WHEN

- ESD strategy fixed before the pad ring is committed
- IO models validated for both timing and channel use
- Latch-up strategy reviewed against the qualification plan

HOW IT GETS THERE

6 STEPS · 2 LANES

w8	w10	w12	w14
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	IO requirement collection from the interface and bump plans → Consolidated IO requirement list	0.5w
2	IO library selection against the interface list → IO library selection per interface	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	ESD strategy and clamp selection → ESD strategy and clamp selection record	1.5w
4	Latch-up rule review and guard ring strategy → Latch-up and guard ring strategy	1w
5	IO timing and signal integrity model quality check → IO timing and SI model quality report	1w
6	Qualification record and release → IO qualification record	2.5w

The ESD strategy in step 3 has to precede pad ring layout, which in turn precedes the bump map freeze in `ARCH-08`. Late ESD is a pad ring change, and a pad ring change is a bump map change.

WATCH OUT FOR

- ◆ **ESD strategy chosen after the pad ring.** Clamp placement and rail topology constrain the ring; retrofitting means redoing the ring and the bump map with it.
- ◆ **IO models unchecked.** IO timing models feed both STA and the channel simulation in `SIPI-07`; an error propagates into two signoffs at once.
- ◆ **Latch-up rules read as a formality.** Guard ring strategy affects area at the die edge, and a late change competes with the interfaces for the same space.
- ◆ **Package and board ESD requirements ignored.** The part's ESD rating is a system property; qualifying only the die produces a rating the system cannot use.
- ◆ **Library selected per interface with no consolidation.** Several IO families in one ring multiplies the qualification and the pad ring complexity for little gain.

WHERE THE EFFORT GOES	5 M/M	WHO IS ON IT	~3.6 FTE	CONNECTIONS	TERMS HERE	4
IO library selection	1.5	IO library engineer	owns it	DEPENDS ON ARCH-04 ARCH-08 PDK-01	STA IO SI ESD	
ESD strategy	1.5	ESD engineer		FEEDS INTO PDK-11 PDK-13 AMS-14		
Model quality checks	1.0	Reliability engineer		PD-01 SO-07 SIPI-06		
Qualification record	1.0	Physical design liaison		SIPI-07 MP-03		
		SI liaison				

PDK-08

EDA tool version qualification for the node — synthesis, P&R, STA, PV, EM/IR

TAKES	COSTS	OWNER	CRITICAL PATH
10w w10–w20 of the stage	12.0 M/M ~5.2 people while it runs	CAD tools lead IT infrastructure approves	no has float

WHY IT EXISTS

Decide which tool at which version each flow step runs on, **qualify that combination on the node**, and freeze it—so that a result from one team means the same thing as a result from another.

Tool versions are a correctness question, not an IT one. A node this advanced needs certified versions for extraction, timing and physical verification, and mixing versions across teams produces disagreements that consume weeks before anyone suspects the tools rather than the design.

WHAT IT DELIVERS

PDK-D4

EDA tool and version matrix (qualified and frozen)

This activity is the deliverable, and the freeze is the part that matters.

NEEDS FIRST

- Selected node and flavor from TECH
- Vendor certification matrices for the node
- Foundry reference flow tool requirements
- Existing internal tool estate and licences
- Flow step definitions from PDK-09

DONE WHEN

- Every flow step has a qualified, named tool version
- Cross-tool interoperability checked, not assumed
- Freeze published with an exception route for critical fixes

HOW IT GETS THERE

8 STEPS · 2 LANES

w10

w12

w14

w16

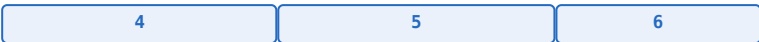
w18

w20

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Tool and version matrix definition per flow step → Tool and version matrix per flow step	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Vendor node certification status review → Vendor certification status record	1w
3	Synthesis and place-and-route version qualification → Synthesis and place-and-route version qualification results	2.5w
4	STA and signoff tool qualification → STA and signoff tool qualification results	2w
	runs in parallel	
5	Physical verification and EM/IR tool qualification → Qualification results per tool	2w
	runs in parallel	
6	Interoperability and database version check across tools → Interoperability check findings	1.5w
	runs in parallel	
7	Version freeze and upgrade exception policy → Version freeze and upgrade exception policy	4w
8	Tool matrix publication to every team → Published tool matrix	1.5w

Step 6 catches what per-tool qualification cannot. Each tool can be certified on the node and still disagree with the next one about a database format or a parasitic convention, and only a cross-tool check finds it.

WATCH OUT FOR

- ◆ **Teams on different versions.** Two teams with different STA versions will disagree about timing, and the disagreement is diagnosed as a design problem for weeks first.
- ◆ **Certification status assumed.** Vendor node certification lags node release; an uncertified version produces results the foundry will not accept at signoff.
- ◆ **Interoperability never checked.** Per-tool qualification passes and the handoff between them fails, usually on parasitic or database conventions.
- ◆ **Freeze with no upgrade path.** A critical tool fix will arrive mid-program, and a policy that has no route to accept it will be broken informally.
- ◆ **Matrix published once.** A matrix nobody re-reads after month three is a matrix that describes an estate nobody is running.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT	~5.2 FTE	CONNECTIONS	TERMS HERE	4
Synthesis and P&R qualification	3.0	CAD tools lead OWNS IT		DEPENDS ON TECH-02 PDK-01	STA EM/IR CAD EDA	
STA and signoff qualification	2.5	Flow engineers		FEEDS INTO PDK-09 PDK-12 PDK-13		
Physical verification and EM/IR qualification	2.5	Signoff methodology engineer		IPR-06 AMS-11 RTL-06		
Interoperability checks	2.0	EDA vendor account contacts		RTL-09 DV-05 SYN-01		
Matrix and freeze policy	2.0	IT infrastructure		SYN-10 PD-01 SO-01		

PDK-09

Foundry reference flow bring-up and internal methodology deck

TAKES	COSTS	OWNER	CRITICAL PATH
10w	12.0 M/M	Methodology lead	yes
w12–w22 of the stage	~5.2 people while it runs	Foundry flow contact approves	slipping moves the program

WHY IT EXISTS

Take the foundry's reference flow, **adapt it into a methodology this design can actually run**, and prove it on a representative block before the whole team depends on it.

A reference flow is a starting point written for a generic design. This one has hierarchical partitions, hard macros, multiple power domains and a memory subsystem the reference never contemplated. The pilot in the last step is what turns an adapted flow into a trusted one.

WHAT IT DELIVERS

PDK-D5

Internal reference flow and methodology guide

This activity is the deliverable, and the pilot is what makes it credible.

NEEDS FIRST

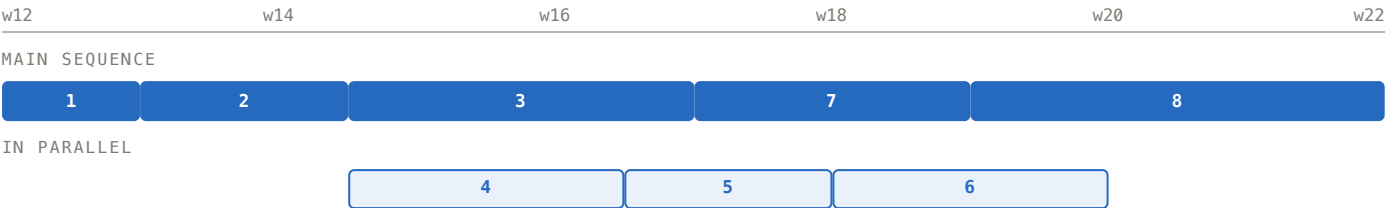
- Foundry reference flow and scripts
- Tool matrix from PDK-08
- Rule and extraction decks from PDK-10
- UPF intent from ARCH-05
- Hierarchical partitioning from ARCH-02

DONE WHEN

- Flow run end to end on a representative block, not a toy one
- Methodology deck matches the scripts as released
- Hierarchical and block flows both defined, not only the top

HOW IT GETS THERE

8 STEPS · 2 LANES



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Foundry reference flow ingestion → Ingested foundry reference flow	1w
2	Gap analysis against internal methodology and design needs → Gap analysis against internal methodology	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Synthesis and place-and-route flow adaptation → Adapted synthesis and place-and-route flow	2.5w
4	Signoff flow adaptation and deck integration → Adapted synthesis, P&R and signoff flows	2w
5	Low-power flow integration against the UPF intent → Low-power flow integration	1.5w
6	Hierarchical and block-level flow definition → Hierarchical and block-level flow definition	2w
7	Methodology deck authoring → Internal methodology deck	2w
8	Flow pilot on a representative block → Pilot results on a representative block	3w

Three of the ten weeks are the pilot, and it is the step under most pressure to be skipped. A flow that has never been run end to end on a real block is a flow whose first user is doing the debugging on the critical path.

WATCH OUT FOR

- ◆ **Pilot skipped under schedule pressure.** The flow's first real user then debugs it while on the critical path, which is the most expensive place to do it.
- ◆ **Reference flow adopted rather than adapted.** A generic flow on a hierarchical design with hard macros produces results nobody can close.
- ◆ **Low-power flow bolted on late.** UPF handling touches synthesis, placement and verification; adding it after the flow is built means rebuilding it.
- ◆ **Methodology deck written after the flow changes.** A guide that lags the scripts teaches the wrong flow to every engineer who joins.
- ◆ **Pilot block unrepresentative.** A small clean block pilots nothing; the pilot has to include a macro, a domain crossing and real congestion.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT	~5.2 FTE	CONNECTIONS	TERMS HERE	2
Flow adaptation	4.0	Methodology lead owns it		DEPENDS ON ARCH-05 PDK-02 PDK-08 PDK-10	PDK UPF	
Pilot on a representative block	3.0	Flow engineers		FEEDS INTO PDK-12 PDK-13 DFT-06 SYN-01 SYN-02 SYN-07 PD-01 PD-14 SO-01		
Signoff flow and deck integration	2.0	Low-power methodology engineer				
Methodology deck	1.5	Physical design pilot engineer				
Low-power flow integration	1.5	Foundry flow contact				

PDK-10

Rule deck, QRC tech file and DRC/LVS deck version control

TAKES	COSTS	OWNER	CRITICAL PATH
6w cont. w6–w12 of the stage	4.0 M/M ~2.9 people while it runs	CAD methodology engineer Foundry deck contact approves	no has float

WHY IT EXISTS

Put every rule deck, extraction file and verification runset **under version control**, validate them on known-good structures, and make deck changes a notified event rather than a surprise.

Decks are the quiet dependency. A DRC runset revision that arrives without notice changes what passes verification, and if two teams are running different revisions the disagreement looks like a layout problem. Version control here is what makes a signoff result reproducible six months later.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Foundry rule decks, runsets and tech files
- PDK version tracking from PDK-01
- Tool matrix from PDK-08
- Known-good test structures from previous programs
- Rule dispositions from PDK-02

DONE WHEN

- Every deck versioned and pinned, none used from a shared path
- Extraction and verification decks validated against known answers
- Change notification process running, with a log

HOW IT GETS THERE

6 STEPS · 2 LANES

w6	w8	w10	w12
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Deck inventory and version baseline → Deck inventory with version baseline	1w
2	Version control and release process definition → Version control and release process	1w
3	QRC tech file and extraction deck validation → Extraction deck validation results	1.5w runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	DRC and LVS deck validation on known-good structures → DRC and LVS validation results	1.5w
5	Deck change notification and impact process → Change notification process and log	1w
	runs in parallel	
6	Continuous deck version tracking across the stage → Deck version tracking log	2.5w

Validation on known-good structures in step 4 is what makes a deck trustworthy. A runset that has never been tested against a layout with a known answer will report both false positives and, far worse, false negatives.

WATCH OUT FOR

- ◆ **Decks used from a shared directory.** An unversioned deck makes every result unreproducible, and the problem is invisible until two results disagree.
- ◆ **Extraction deck unvalidated.** Parasitic extraction errors move timing by amounts that look like design problems, and they are found after weeks of closure work.
- ◆ **Deck updates applied silently.** A runset revision mid-closure changes what passes, and nobody can explain why yesterday's clean run is dirty today.
- ◆ **No known-good structures.** Without a layout whose answer is known, deck validation is an assertion.
- ◆ **Tracking stops when the stage ends.** Decks keep moving through signoff; a process that stops at PDK 1.0 leaves the most critical period unmanaged.

WHERE THE EFFORT GOES	4 M/M	WHO IS ON IT	~2.9 FTE	CONNECTIONS	TERMS HERE	5
Deck validation	1.5	CAD methodology engineer owns it		DEPENDS ON PDK-01 PDK-02	PDK DRC LVS QRC	
Version control process	1.0	Physical verification engineer		FEEDS INTO PDK-09 PDK-13 AMS-13	CAD	
Continuous tracking	1.0	Extraction engineer		PD-06 PD-08 S0-01		
Change notification process	0.5	PDK owner		S0-02 S0-03 S0-10		
		Foundry deck contact		S0-12 T0-02 T0-09		

04 PDK · PDK & DESIGN ENABLEMENT

PDK-11

Signoff corner and derate definition agreed with foundry (PVT, OCV/AOCV/POCV)

TAKES	COSTS	OWNER	CRITICAL PATH
6w w20–w26 of the stage	5.0 M/M ~3.6 people while it runs	Signoff methodology lead CAD infrastructure approves	no has float

WHY IT EXISTS

Agree with the foundry **exactly which corners and derates signoff will be run at**—PVT combinations, modes, and the OCV, AOCV or POCV methodology—and write it down before anyone closes timing against it.

A corner set is a contract. Sign off at corners the foundry does not recognize and the foundry will not accept the result; sign off at more corners than necessary and every closure iteration costs runtime and license capacity the program has not budgeted for.

WHAT IT DELIVERS

PDK-D6

Signoff corner definition agreed with foundry

This activity is the deliverable, and "agreed" is the operative word—an internal corner set has no standing.

NEEDS FIRST

- Operating conditions and DVFS points from ARCH-05
- Library characterization corners from PDK-03
- IO and ESD conditions from PDK-07
- Foundry signoff requirements and recommended derates
- Runtime and license capacity constraints

DONE WHEN

- Corner set agreed with the foundry in writing
- Every signoff corner covered by library characterization
- Runtime and license impact quantified and accepted

HOW IT GETS THERE

6 STEPS · 2 LANES

w20	w22	w24	w26
-----	-----	-----	-----

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Corner and mode enumeration from the operating conditions → Corner and mode enumeration	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	PVT corner set proposal → Proposed PVT corner set	1.5w
3	OCV, AOCV or POCV derate methodology selection → Derate methodology selection and rationale	1.5w <i>runs in parallel</i>
4	Foundry alignment on the corner set and derates → PVT corner set, agreed with the foundry	2w
5	Signoff condition documentation → Signoff condition document	1.5w
6	Corner count against runtime and license cost → Corner count impact on runtime and licences	1w <i>runs in parallel</i>

Step 6 exists because corner count multiplies. Nine corners across three modes is twenty-seven signoff runs per iteration, and the compute plan in `PDK-12` is sized against whatever number this activity settles on.

WATCH OUT FOR

- ◆ **Corner set agreed after closure starts.** Adding a corner mid-closure invalidates the work done, and the addition usually comes from the foundry.
- ◆ **Derate methodology chosen for margin.** OCV is conservative and expensive in area and power; POCV is tighter and demands better data. Choosing without the data is choosing margin blindly.
- ◆ **Corner count unbounded.** Every corner is a full signoff run per iteration; a set nobody trimmed becomes a schedule problem at S0-02.
- ◆ **Modes forgotten.** Corners multiply by modes, and a mode discovered late—a test mode, a low-power mode—multiplies the whole set.
- ◆ **Library corners narrower than the signoff set.** Signing off at a corner the library was never characterized at is signing off against extrapolation.

WHERE THE EFFORT GOES	5 M/M	WHO IS ON IT ~3.6 FTE	CONNECTIONS	TERMS HERE 9
Corner and mode definition	1.5	Signoff methodology lead <i>OWNS IT</i>	DEPENDS ON ARCH-05 PDK-03 PDK-07	STA PVT OCV AOCV POCV IO DVFS ESD
Foundry alignment	1.5	STA lead	FEEDS INTO PDK-12 AMS-10 AMS-13	CAD
Derate methodology	1.0	Library engineer	SYN-01 PD-01 PD-09	
Documentation	1.0	Foundry technical contact	S0-01 S0-02 S0-12	
		CAD infrastructure		

PDK-12 Compute farm, EDA license and storage capacity planning			
TAKES	COSTS	OWNER	CRITICAL PATH
6w	3.0 M/M	CAD infrastructure lead	no
w16–w22 of the stage	~2.2 people while it runs	IT operations approves	has float

WHY IT EXISTS

Model what the program will actually need in **compute, EDA licences and storage** across its peak, and procure it against a lead time rather than discovering it during timing closure.

Compute is a schedule input disguised as an IT line item. Timing closure at PD-09 runs multi-corner across a 600 mm² die; if the farm or the license pool caps concurrency, every iteration takes longer and the closure burn-down flattens for reasons nobody attributes to infrastructure.

WHAT IT DELIVERS

PDK-D7

Compute and license capacity plan

This activity is the deliverable, and its value is that it is dated against lead times.

NEEDS FIRST

- Flow step definitions and runtimes from PDK-09
- Corner and mode count from PDK-11
- Tool matrix from PDK-08
- Program schedule and stage overlap from DEF-08
- Existing farm and license estate

DONE WHEN

- Peak demand modelled per flow step, not per team
- Procurement dated against lead time, not against need date
- Plan published with a review cadence against actual usage

HOW IT GETS THERE

6 STEPS · 2 LANES

w16

w18

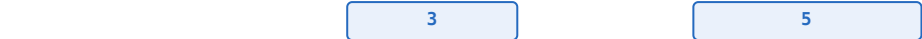
w20

w22

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Compute demand model per stage and per flow step → Compute demand model per stage and flow step	1.5w
2	EDA license demand model against peak concurrency → EDA license demand model at peak concurrency	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Storage and data retention requirement → Storage and retention requirement	0.75w runs in parallel
4	Farm capacity plan against procurement lead time → Farm capacity plan with procurement dates	1.5w
5	Licence negotiation support and pooling strategy → Licence pooling strategy	1w runs in parallel
6	Capacity plan publication and review → Published capacity plan and review record	1.5w

The demand model is built per flow step rather than per team because that is how contention actually happens: three teams running the same tool at the same point in the schedule contend for one license pool, whatever the org chart says.

WATCH OUT FOR

- ◆ **Demand modelled per team rather than per flow step.** Contention happens when several teams hit the same tool at the same time, which an org-chart model cannot see.
- ◆ **Peak sized to the average.** Signoff and closure peaks are several times the steady state, and a farm sized to the average flattens the burn-down exactly when it matters.
- ◆ **Licences procured without lead time.** Additional seats are a commercial negotiation with a duration; needing them next week is needing them next quarter.
- ◆ **Storage forgotten.** Multi-corner signoff databases are large, and a farm that runs out of storage stops as hard as one that runs out of cores.
- ◆ **Plan never reviewed against actual usage.** A capacity plan is a forecast; without a comparison to actuals nobody learns that it was wrong until closure.

WHERE THE EFFORT GOES	3 M/M	WHO IS ON IT	~2.2 FTE	CONNECTIONS	TERMS HERE	2
Demand modeling	1.5	CAD infrastructure lead OWNS IT		DEPENDS ON DEF-08PDK-08PDK-09PDK-11	CAD EDA	
Capacity plan	0.8	Flow engineers		FEEDS INTO RTL-09DV-10SYN-11PD-01PD-09S0-01		
Licence strategy	0.5	Procurement lead		LATER INPUT PD-01+16w		
Publication and review	0.3	Program manager				
		IT operations				

04 PDK · PDK & DESIGN ENABLEMENT

PDK-13

Golden design environment release and flow regression

TAKES	COSTS	OWNER	CRITICAL PATH
8w w24–w32 of the stage	8.0 M/M ~4.3 people while it runs	Methodology lead IT operations approves	yes slipping moves the program

WHY IT EXISTS

Package everything this stage produced into **one environment every team runs from**—tools, decks, libraries, scripts—prove it with a regression, and release it with known issues stated.

Enablement only becomes real when it is a single thing an engineer can load. Until then the program has a qualified library here, a validated deck there and a flow in someone's home directory, and every new engineer assembles a slightly different version of the design environment.

WHAT IT DELIVERS

PDK-D8

Golden environment release notes

This activity is the deliverable—the environment itself, with what is in it and what is known to be wrong with it.

NEEDS FIRST

- Qualified libraries from PDK-03, PDK-04 and PDK-07
- Validated decks from PDK-10
- Tool matrix from PDK-08
- Adapted flow and methodology deck from PDK-09
- Frozen PDK version from PDK-01

DONE WHEN

- Regression reproduces the pilot baseline, or the difference is explained
- Known issues published with the release, not after it
- Teams onboarded onto the environment rather than copying it

HOW IT GETS THERE

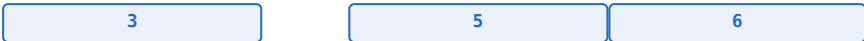
7 STEPS · 2 LANES

w24	w26	w28	w30	w32
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MAIN SEQUENCE



IN PARALLEL



#STEP, AND WHAT IT ADDS TO THE DELIVERABLE

TAT

1	Environment scope definition — tools, decks, libraries, scripts → Environment scope definition	1w
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#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Golden environment assembly → Golden design environment, versioned	2w
3	Flow regression suite definition → Flow regression suite	1.5w <i>runs in parallel</i>
4	Regression run on the reference block → Reference block regression results	2w
5	Result comparison against the pilot baseline → Regression results against the pilot baseline	1.5w <i>runs in parallel</i>
6	Release notes and known-issue list → Release notes and known-issue list	1.5w <i>runs in parallel</i>
7	Environment release and team onboarding → Onboarding material and team access	3w

Step 5 is what makes the release trustworthy. If the golden environment produces different results from the `PDK-09` pilot on the same block, something in the packaging changed the answer, and finding that after release means everyone's results are suspect.

WATCH OUT FOR

- ◆ **Environment released without a regression.** An untested package is a package whose first regression is run by whoever is closest to a deadline.
- ◆ **Results differ from the pilot and the difference is not chased.** A packaging change that moves results silently invalidates the pilot's credibility and everything built on it.
- ◆ **Known issues omitted.** An environment released as clean produces bug reports for problems the team already knew about, and the reports go to the wrong place.
- ◆ **No onboarding.** A golden environment nobody was shown how to use is a directory people copy from and modify, which is where divergence starts.
- ◆ **Released and then not maintained.** Decks and PDK versions keep moving; an environment frozen at release drifts away from what signoff will require.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT	~4.3 FTE	CONNECTIONS	TERMS HERE	2
Environment assembly	2.5	Methodology lead OWNS IT		DEPENDS ON PDK-01 PDK-03 PDK-04 PDK-07 PDK-08 PDK-09 PDK-10	PDK CAD	
Regression suite and runs	2.5	CAD engineers				
Onboarding	1.5	Regression owner		FEEDS INTO AMS-12 TC-02 TC-04 RTL-08 DFT-06 SYN-01 PD-01 SO-01		
Release notes	1.0	Physical design representative				
Baseline comparison	0.5	IT operations				

05 IP Strategy & Readiness

IPR

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w4 – w26	22 _w	10	47 M/M	6 of 10
ENGINEERING ACTIVITIES				KEY DELIVERABLES
IPR-01	Product requirement → IP requirement decomposition (IP bill of materials)		5w 5 M/M	IPR-D1 IP bill of materials with make / buy / reuse decision per block
IPR-02	Reuse inventory audit — internal IP already available on the target node		4w 3 M/M	IPR-D2 Vendor evaluation matrix and selection record
IPR-03	Make / buy / reuse decision per IP block		5w 4 M/M	IPR-D3 IP readiness report — silicon-proven status and maturity level per IP
IPR-04	Vendor RFI / RFQ and technical evaluation		9w 10 M/M	IPR-D4 IP deliverable acceptance checklist
IPR-05	Silicon-proven status check per IP on the selected node and process option		7w 6 M/M	IPR-D5 Executed licences and POs with committed delivery dates
IPR-06	IP deliverable checklist review (GDS, LEF, multi-corner .lib, CDL, UPF, DV collateral, errata)		7w 6 M/M	IPR-D6 IP delivery schedule folded into the program plan
IPR-07	Licence negotiation, PO issue and delivery-date commitment		12w 5 M/M	IPR-D7 IP risk register and contingency plan
IPR-08	Porting / hardening scope and schedule for IP not proven on the node		5w 4 M/M	
IPR-09	IP delivery schedule alignment to RTL and PD integration windows		4w 2 M/M	
IPR-10	IP maturity risk rating and second-source contingency		4w 2 M/M	

05 IPR · IP STRATEGY & READINESS

IPR-01

Product requirement → IP requirement
decomposition (IP bill of materials)

TAKES		COSTS	OWNER	CRITICAL PATH	
5w		5.0 M/M	IP strategist	yes	
w0-w5 of the stage		~4.3 people while it runs	Program manager approves	slipping moves the program	
WHY IT EXISTS					
Decompose the product into the IP it actually needs—every block, with its specification, its interfaces and the requirement it traces back to—and publish it as a bill of materials the program can be held to. An IP-BOM derived from the last program's list is the most common way to arrive at the wrong one. What this product needs follows from its requirements and its architecture, and the blocks that differ from last time are exactly the ones with the longest lead times.					
WHAT IT DELIVERS					
IPR-D1 IP bill of materials with make / buy / reuse decision per block The bill half of the deliverable; IPR-03 adds the decision against each line.					
NEEDS FIRST					
- Requirement register from DEF-01 - Architecture sketch and block list direction from ARCH-02 - Interface selections from ARCH-04 - Previous-program IP list as a starting reference - Foundry IP catalogue for the selected node					
DONE WHEN					
- Every IP traces to a requirement, and every requirement to an IP or a gap - Foundry IP included alongside third-party and internal blocks - Specification stubs detailed enough to evaluate a vendor against					
HOW IT GETS THERE				6 STEPS · 2 LANES	
w0	w1	w2	w3	w4	w5
MAIN SEQUENCE					
1		2		5	6
IN PARALLEL					
		3	4		
#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE				TAT
1	Requirement-to-IP decomposition against the requirement register → Requirement-to-IP decomposition				1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Block-level IP identification from the architecture sketch → Block-level IP list	1.5w
3	Interface and protocol IP identification → Interface and protocol IP list	0.75w runs in parallel
4	Foundry IP identification — memory compilers, IO, standard cells → Interface and foundry IP identification	0.75w runs in parallel
5	Per-IP specification stub — function, performance, interfaces → Per-IP specification stubs	1.5w
6	Bill of materials assembly and traceability check → Requirement-to-IP traceability matrix · IP bill of materials, block by block · Traceability gap list	1w

Step 6's traceability check is what separates a list from a bill of materials. An IP that traces to no requirement is an IP nobody needs; a requirement that traces to no IP is a gap nobody has staffed.

WATCH OUT FOR

- ◆ **Bill inherited from the previous program.** The blocks that differ are the ones with the longest lead times, and inheriting the list hides exactly those.
- ◆ **Foundry IP omitted.** Memory compilers, IO and standard cells are IP with licences and lead times, and a bill that lists only third-party blocks understates the procurement.
- ◆ **Specification stubs too thin to evaluate against.** A vendor cannot be assessed against "a PCIe PHY"; the stub has to carry performance, interface and process detail.
- ◆ **Traceability skipped.** Without it, blocks appear because someone assumed them and requirements go unserved because nobody noticed.
- ◆ **Bill frozen before architecture settles.** A bill written before ARCH-02 partitions will miss the die-to-die interface entirely if the part turns out to be a chiplet.

WHERE THE EFFORT GOES	5 M/M	WHO IS ON IT ~4.3 FTE	CONNECTIONS	TERMS HERE 5
Requirement decomposition	1.5	IP strategist owns IT	DEPENDS ON DEF-01 ARCH-02	IP PHY IO PCIe
Block and interface identification	1.5	Chief architect	FEEDS INTO TECH-08 PDK-04 IPR-02	CAD
Specification stubs	1.0	Interface architect	IPR-03 IPR-04	
Bill assembly and traceability	1.0	CAD and library lead	LATER INPUT ARCH-04 +0w	
		Program manager		

05 IPR · IP STRATEGY & READINESS

IPR-02

Reuse inventory audit — internal IP already available on the target node

TAKES

4w

w1–w5 of the stage

COSTS

3.0

M/M

~3.3 people while it runs

OWNER

IP strategist

Silicon validation liaison approves

CRITICAL PATH

no

has float

WHY IT EXISTS

Audit what the organization **already owns and can actually reuse on this node**—not what it owns in general, which is a different and much longer list.

Reuse is where the schedule is won, and it is also where optimism concentrates. A block proven on the previous node is not available on this one until it has been ported, re-characterized and re-verified, and that work is a project with a duration. The audit's job is to separate genuine reuse from hopeful reuse.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- IP bill of materials from IPR-01
- Internal IP catalogue and repositories
- Previous-program silicon results per block
- Node and flavor selection from TECH
- Team ownership records

DONE WHEN

- Every internal block classified as direct reuse, port, or rebuild
- Collateral gaps listed per block, not assumed away
- Blocks with no maintainer flagged as such

HOW IT GETS THERE

5 STEPS · 2 LANES

w1

w2

w3

w4

w5

MAIN SEQUENCE

1

2

5

IN PARALLEL

3

4

#

STEP, AND WHAT IT ADDS TO THE DELIVERABLE

TAT

1

Internal IP inventory against the bill of materials
→ Internal IP inventory mapped to the bill

0.75w

2

Node availability check per internal block
→ Node availability finding per block

1.25w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Collateral completeness audit — views, testbenches, documentation → Collateral completeness audit	1w runs in parallel
4	Maintenance and ownership status per block → Ownership and maintenance status	0.75w runs in parallel
5	Reuse classification — direct, port required, or rebuild → Reuse classification per block	2w

Step 4 asks who still maintains each block, and the answer is often nobody. An unmaintained block is technically reusable and practically a rebuild, because the first bug found has no owner.

WATCH OUT FOR

- ◆ **Reuse claimed on the previous node.** Availability is node-specific; a block not yet ported is a porting project, not a reuse.
- ◆ **Collateral assumed complete.** Missing multi-corner .lib files or an absent testbench turns reuse into re-characterization and re-verification.
- ◆ **Unmaintained blocks counted as reuse.** A block with no owner has no one to fix its first bug, and the fix lands on the integration team.
- ◆ **Audit run on the catalogue rather than the repository.** Catalogues describe intent; repositories contain what actually exists.
- ◆ **Silicon history ignored.** A block that shipped with an erratum is reusable with that erratum, and the erratum belongs in the audit.

WHERE THE EFFORT GOES	3 M/M	WHO IS ON IT ~3.3 FTE	CONNECTIONS	TERMS HERE 3
Node availability assessment	1.0	IP strategist OWNS IT	DEPENDS ON TECH-01 IPR-01	IP DV .lib
Inventory against the bill	0.8	IP librarian	FEEDS INTO ARCH-06 IPR-03 IPR-05	
Collateral audit	0.8	Design leads	IPR-08	
Reuse classification	0.5	Verification liaison		
		Silicon validation liaison		

05 IPR · IP STRATEGY & READINESS

IPR-03

Make / buy / reuse decision per IP block

TAKES	COSTS	OWNER	CRITICAL PATH
5w w5–w10 of the stage	4.0 M/M ~3.5 people while it runs	IP strategist Program manager approves	yes slipping moves the program

WHY IT EXISTS

Take each line of the bill and decide, with a reason, whether it is **made, bought or reused**—and record what the rejected options would have cost.

Make/buy is a schedule decision more than a cost one. Building a PHY internally is cheaper on paper and costs nine months in AMS-03; buying it costs a license and a vendor dependency. Neither is wrong, and the program only gets to argue about it once.

WHAT IT DELIVERS

IPR-D1

IP bill of materials with make / buy / reuse decision per block

The decision half of the deliverable, with the rejected options priced rather than forgotten.

NEEDS FIRST

- IP bill of materials from IPR-01
- Reuse classification from IPR-02
- AMS capacity and schedule from the program plan
- Sourcing strategy direction from TECH-08
- Cost model assumptions from DEF-04

DONE WHEN

- Every line decided with a criterion attached, not by default
- Make decisions backed by a named capacity commitment
- Rejected options priced and recorded

HOW IT GETS THERE

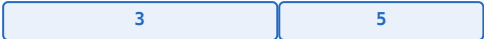
6 STEPS · 2 LANES

w5	w6	w7	w8	w9	w10
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Decision criteria definition — cost, schedule, differentiation, risk → Decision criteria and their weighting	0.75w
2	Differentiation screen — which blocks are the product's own value → Differentiation screen result	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Internal capacity check against the make candidates → Internal capacity commitment for make decisions	1w runs in parallel
4	Cost and schedule comparison per line → Cost and schedule comparison per line	1.5w
5	Foundry-specific versus portable IP consideration → Foundry-specific versus portable IP assessment	0.75w runs in parallel
6	Decision per line and record of rejected options → Make / buy / reuse decision per line · Rejected option record with cost and schedule	1.75w

Step 3 is the check that changes answers. A make decision assumes engineers who are also assigned to the custom SRAM and the PLL; capacity, not capability, is what usually decides.

WATCH OUT FOR

- ◆ **Make decided on capability rather than capacity.** The team can build it; whether it can build it alongside everything else it is already committed to is the real question.
- ◆ **Differentiating IP bought.** Buying the block that is the product's actual advantage buys a product the competitor can buy too.
- ◆ **Commodity IP built.** Building a standard interface consumes analog engineers who are the program's scarcest resource, for no differentiation.
- ◆ **Foundry-specific IP chosen without reference to sourcing strategy.** It forecloses the migration option TECH-08 was pricing, and usually after that study concludes.
- ◆ **Decisions recorded without the alternatives.** Every setback reopens a decision whose reasoning nobody can reconstruct.

WHERE THE EFFORT GOES	4 M/M	WHO IS ON IT	~3.5 FTE	CONNECTIONS	TERMS HERE	5
Cost and schedule comparison	1.5	IP strategist owns IT		DEPENDS ON DEF-07 TECH-08 IPR-01 IPR-02	IP SRAM PLL PHY AMS	
Differentiation screen	1.0	Chief architect		FEEDS INTO		
Capacity check	0.8	AMS lead		IPR-04 IPR-07 IPR-08 AMS-01 AMS-03		
Decision record	0.8	Procurement lead				
		Program manager				

05 IPR · IP STRATEGY & READINESS

IPR-04

Vendor RFI / RFQ and technical evaluation

TAKES	COSTS	OWNER	CRITICAL PATH
9w w6–w15 of the stage	10.0 M/M ~4.8 people while it runs	IP strategist Program manager approves	yes slipping moves the program

WHY IT EXISTS

Run a real evaluation of the vendors behind every buy decision—**technical capability, deliverable quality, support model and track record**—rather than choosing from a datasheet.

A PHY vendor is a nine-month dependency. What matters is less the block's peak specification than whether the deliverables arrive complete, whether the support engineer answers, and whether the last three customers taped out on schedule. None of that is on the datasheet.

WHAT IT DELIVERS

IPR-D2

Vendor evaluation matrix and selection record

This activity is the deliverable, and the matrix is what a later dispute is settled against.

NEEDS FIRST

- Buy decisions from IPR-03
- Specification stubs from IPR-01
- Node and process option from TECH-02
- Interface requirements from ARCH-04
- Vendor landscape and previous-program experience

DONE WHEN

- At least two candidates evaluated per significant line
- Reference customers contacted for the critical blocks
- Selection recorded against the matrix, with support model assessed

HOW IT GETS THERE

7 STEPS · 2 LANES

w6	w8	w10	w12	w14	w15
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Vendor long-list per IP line → Vendor long-list per IP line	0.75w
2	RFI issue and response collection → RFI responses per vendor	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Technical evaluation against the specification stubs → Technical evaluation against the stubs	2.5w
4	Reference customer checks and tapeout track record → Reference customer findings	1.5w <i>runs in parallel</i>
5	Support model and escalation path assessment → Support model and escalation path assessment	1w <i>runs in parallel</i>
6	RFQ issue and commercial comparison → RFI and RFQ responses	1.75w <i>runs in parallel</i>
7	Evaluation matrix and vendor selection → Vendor evaluation matrix and selection record	3.75w

Step 4 is the step most often skipped and the one that predicts the outcome best. A vendor's last three customers know whether the deliverables arrived complete and on time, and they will usually say.

WATCH OUT FOR

- ◆ **Evaluation from the datasheet.** Peak specifications are marketing; deliverable completeness and support responsiveness are what the integration actually depends on.
- ◆ **Reference checks skipped.** The vendor's previous customers know what the program is about to find out, and asking costs an hour.
- ◆ **Single vendor evaluated.** An evaluation with one candidate is a justification, and the commercial position that follows is weak.
- ◆ **Support model unexamined.** A block with no named support engineer becomes the integration team's problem the first time it misbehaves.
- ◆ **Node-specific capability assumed from a different node.** A PHY proven at the previous node is not the same product; IPR-05 is where that is confirmed.

WHERE THE EFFORT GOES	10 M/M	WHO IS ON IT ~4.8 FTE	CONNECTIONS
Technical evaluation	3.5	IP strategist <i>OWNS IT</i>	DEPENDS ON ARCH-04 IPR-01 IPR-03
RFI and RFQ handling	2.0	Technical evaluators	FEEDS INTO IPR-05 IPR-06 IPR-07
Reference checks	1.5	Procurement lead	AMS-03
Evaluation matrix and selection	2.0	Integration lead	TERMS HERE 4
Support model assessment	1.0	Program manager	RFI RFQ IP PHY

05 IPR · IP STRATEGY & READINESS

IPR-05

Silicon-proven status check per IP on the selected node and process option

TAKES	COSTS	OWNER	CRITICAL PATH
7w w8–w15 of the stage	6.0 M/M ~3.7 people while it runs	IP strategist Procurement lead approves	yes slipping moves the program

WHY IT EXISTS

Establish, for each bought and reused block, **whether it has actually run in silicon on this node and this process option**—and if not, what it would take to get it there.

"Silicon-proven" is the most abused phrase in IP procurement. It can mean proven on a different node, on a different process option, at a different speed grade, or in a test chip that was never characterized. This activity replaces the phrase with a fact, and the fact is what the integration schedule should be built on.

WHAT IT DELIVERS

IPR-D3

IP readiness report — silicon-proven status and maturity level per IP

This activity is the deliverable, and it is the one that replaces a claim with a fact.

NEEDS FIRST

- Vendor selections from IPR-04
- Reuse classification from IPR-02
- Node and process option from TECH-02
- Vendor silicon evidence and characterization data
- Errata from previous programs and vendor disclosures

DONE WHEN

- Every block carries a maturity level backed by evidence, not by claim
- Node and process option match verified per block
- Findings available before licences are signed in IPR-07

HOW IT GETS THERE

7 STEPS · 2 LANES

w8	w10	w12	w14	w15
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Proven-status definition — what the program will accept as evidence → Evidence standard the program accepts	0.75w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Evidence request per block — test chip, production part, characterization data → Collected silicon evidence per block	1.5w
3	Node and process option match verification → Silicon evidence per block, with node and option match	1.5w
4	Speed grade and operating condition match check → Speed grade and condition match findings	1w <i>runs in parallel</i>
5	Errata and known-issue review per block → Errata and known-issue summary	1w <i>runs in parallel</i>
6	Maturity level assignment per block → Maturity level per block and a gap list	1.25w
7	Readiness report and gap list → IP readiness report with its gap list	2w

Step 1 is not bureaucracy. Without an agreed evidence standard, every vendor's claim is accepted at its own definition, and the readiness report becomes a collection of adjectives.

WATCH OUT FOR

- ◆ **Vendor claim accepted at the vendor's definition.** Without an evidence standard, "silicon-proven" means whatever is most convenient.
- ◆ **Node matched, process option not.** The same node with a different flavor sheet is a different library, different rules and often a different PHY.
- ◆ **Errata not requested.** Every silicon-proven block has errata; a block presented with none has either not been characterized or is not being fully disclosed.
- ◆ **Assessment done after the license is signed.** The finding has no commercial consequence once the PO is placed, and the leverage is gone.
- ◆ **Speed grade ignored.** A PHY proven at a lower rate on the same node is an engineering project, not a proven block.

WHERE THE EFFORT GOES	6 M/M	WHO IS ON IT ~3.7 FTE	CONNECTIONS	TERMS HERE	4
Evidence assessment	2.5	IP strategist <i>OWNS IT</i>	DEPENDS ON TECH-01 TECH-02 IPR-02 IPR-04	PO IP PHY AMS	
Node and option verification	1.5	Technical evaluators			
Errata review	1.0	AMS lead	FEEDS INTO IPR-06 IPR-07 IPR-08		
Maturity assignment and report	1.0	Silicon validation liaison Procurement lead	IPR-10 TC-01 TC-02 RTL-03		

05 IPR · IP STRATEGY & READINESS

IPR-06

IP deliverable checklist review (GDS, LEF, multi-corner .lib, CDL, UPF, DV collateral, errata)

TAKES	COSTS	OWNER	CRITICAL PATH
7w w11–w18 of the stage	6.0 M/M ~3.7 people while it runs	IP integration lead Vendor technical contacts approves	no has float

WHY IT EXISTS

Define **exactly what a complete IP delivery contains**—every view, model, testbench and document—and get that list agreed before it becomes a contractual obligation.

Most IP delivery disputes are not about whether the block works; they are about whether a delivery counted as a delivery. A block that arrives without multi-corner timing views or without its verification collateral has arrived in a form the program cannot integrate, and the argument that follows is unwinnable if the checklist was never agreed.

WHAT IT DELIVERS

IPR-D4

IP deliverable acceptance checklist

This activity is the deliverable, and its value is entirely in being agreed before the PO rather than after the delivery.

NEEDS FIRST

- Vendor selections from IPR-04
- Tool matrix and view requirements from PDK-08
- Verification methodology needs from DV-01
- Integration requirements from RTL-03
- Previous-program IP delivery disputes

DONE WHEN

- Checklist agreed with each vendor before the PO is placed
- Acceptance procedure names who checks what, and when
- Views cover the signoff corners the program will actually run

HOW IT GETS THERE

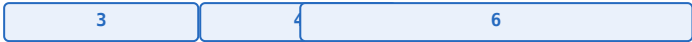
7 STEPS · 2 LANES

w11	w13	w15	w17	w18
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Deliverable checklist definition per IP class → Deliverable checklist per IP class	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	View and model requirements against the tool matrix → View and model requirement list	1.5w
3	Verification collateral requirements with DV → Verification collateral requirements	1w runs in parallel
4	Documentation, errata and integration guide requirements → Documentation, errata and integration guide requirements	1w runs in parallel
5	Acceptance criteria and incoming inspection procedure → Acceptance criteria and incoming inspection procedure	1.5w
6	Checklist agreement with each vendor → Vendor-agreed checklists	2w runs in parallel
7	Checklist publication and handoff to procurement → Published checklist with procurement handoff	3w

Step 5 turns the checklist into something enforceable. A list of expected files is a wish; an acceptance procedure that says what is checked, by whom, and what happens when it fails is a term.

WATCH OUT FOR

- ◆ **Checklist agreed after the PO.** Once the commercial terms are closed, a missing view is a favor rather than an obligation.
- ◆ **Views specified without the tool matrix.** A .lib set that does not cover the signoff corners in PDK-11 is incomplete in a way nobody notices until closure.
- ◆ **Verification collateral omitted.** A block delivered without its testbench has to be verified from scratch, and that effort was never in the DV plan.
- ◆ **No incoming inspection.** Deliverables accepted without a check are found incomplete weeks later, when the vendor's engineer has moved on.
- ◆ **One checklist for all IP classes.** A memory compiler, a PHY and a controller have genuinely different deliverable sets, and a single list serves none of them.

WHERE THE EFFORT GOES	6 M/M	WHO IS ON IT	~3.7 FTE	CONNECTIONS	TERMS HERE	6
Checklist and view requirements	2.0	IP integration lead owns IT		DEPENDS ON PDK-08 IPR-04 IPR-05	PO IP PHY DV .lib	
Acceptance criteria and inspection	1.5	CAD methodology engineer		FEEDS INTO IPR-07 IPR-09 RTL-03	CAD	
Verification collateral requirements	1.0	Verification lead		PD-02		
Vendor agreement and publication	1.5	Procurement lead Vendor technical contacts		LATER INPUT DV-01 +4w		

05 IPR · IP STRATEGY & READINESS

IPR-07

Licence negotiation, PO issue and delivery-date commitment

TAKES	COSTS	OWNER	CRITICAL PATH
12w w10–w22 of the stage	5.0 M/M ~1.8 people while it runs	Procurement lead Finance partner approves	yes slipping moves the program

WHY IT EXISTS

Turn selections into **executed licences and purchase orders with committed delivery dates**, including the deliverable checklist as a contractual schedule.

Twelve weeks for 5 M/M—this is the stage's longest activity and among its smallest in effort, because the elapsed time belongs to two legal departments and a procurement process. It is on the critical path regardless: nothing arrives until it executes, and RTL-03 has integration windows that assume it did.

WHAT IT DELIVERS

IPR-D5

Executed licences and POs with committed delivery dates

This activity is the deliverable, and the committed date is the part the schedule depends on.

NEEDS FIRST

- Vendor selections from IPR-04
- Readiness findings from IPR-05 as negotiating position
- Deliverable checklists from IPR-06
- Integration window requirements from the program schedule
- Corporate legal templates and positions

DONE WHEN

- Every license carries a committed delivery date, not an intention
- Deliverable checklist attached as a contractual schedule
- Access distributed and tracked to the teams that need it

HOW IT GETS THERE

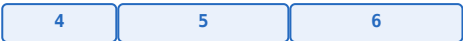
8 STEPS · 2 LANES

w10	w13	w16	w19	w22
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Commercial terms definition per IP line → Commercial terms per IP line	1.5w
2	Licence model negotiation — per-use, per-project, royalty → Agreed licence model per block	2.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Delivery date commitment and milestone schedule → Executed licences with delivery commitments	2w
4	Deliverable checklist attached as a contractual schedule → Deliverable checklist as a contractual schedule	1w
5	Support, maintenance and escalation terms → Support and escalation terms	1.5w
6	Liability, indemnity and IP warranty terms → Agreed liability, indemnity and warranty terms	1.5w
7	Legal redline cycles → Redline history with resolved positions	3.5w
8	Execution, PO issue and distribution → Purchase orders issued · Distribution record of what has been licensed to whom	2.5w

Step 3 is the one the program cares about and the one commercial negotiation treats as secondary. A license with excellent terms and no committed delivery date has bought a block that arrives when the vendor is ready.

WATCH OUT FOR

- ◆ **Licence executed without a committed delivery date.** The block then arrives when the vendor is ready, and the integration window absorbs the difference.
- ◆ **Readiness findings not used in negotiation.** A block that is not proven on the node is a commercial position; signing before IPR-05 reports discards it.
- ◆ **Checklist not attached.** An agreed checklist outside the contract is an understanding, and understandings do not survive a disputed delivery.
- ◆ **Redline duration underestimated.** Two legal departments take longer than either forecasts, and the program's plan should assume the longer figure.
- ◆ **Support terms omitted.** A block with no support commitment becomes unsupported precisely when integration needs help.

WHERE THE EFFORT GOES	5 M/M	WHO IS ON IT	~1.8 FTE	CONNECTIONS	TERMS HERE	3
		Procurement lead OWNS IT		DEPENDS ON DEF-09 TECH-06 IPR-03 IPR-04 IPR-05 IPR-06	M/M PO IP	
Negotiation support	2.0					
Terms and license model	1.5	Legal counsel		FEEDS INTO IPR-09 AMS-03 RTL-03 DV-02 PD-02		
Delivery commitment	0.8	IP strategist				
Execution and distribution	0.8	Program manager				
		Finance partner				

05 IPR · IP STRATEGY & READINESS

IPR-08

Porting / hardening scope and schedule for IP not proven on the node

TAKES

5w

w14–w19 of the stage

COSTS

4.0 M/M

~3.5 people while it runs

OWNER

IP strategist

Program manager approves

CRITICAL PATH

no

has float

WHY IT EXISTS

For every block that is **not proven on this node**, scope the work that gets it there—porting, hardening, re-characterization, re-verification—and put it in the schedule as a project with an owner.

Unproven IP does not become proven by being licensed. Someone ports it, re-characterizes it at the program's corners and re-verifies it, and that someone is either the vendor on a schedule the program does not control, or the program's own engineers who are already committed elsewhere.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Readiness report and gap list from IPR-05
- Reuse classification from IPR-02
- Make / buy decisions from IPR-03
- AMS capacity from the program plan
- Integration window requirements from RTL and PD

DONE WHEN

- Every unproven block has a scoped plan with an owner and a date
- Internal execution decisions backed by a capacity commitment
- Re-verification scope included, not assumed

HOW IT GETS THERE

6 STEPS · 2 LANES

w14w15w16w17w18w19

1246

35

#STEP, AND WHAT IT ADDS TO THE DELIVERABLETAT

1Unproven block list from the readiness report
→ Unproven block list0.5w

2Porting scope per block — layout, characterization, verification
→ Unproven block list with porting scope1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Vendor versus internal execution decision per block → Execution decision per block — vendor or internal	1w runs in parallel
4	Effort and duration estimate per block → Effort and duration estimates	1.25w
5	Dependency mapping against the integration windows → Dependency map against integration windows	0.75w runs in parallel
6	Hardening plan with owners and dates → Hardening plan with owners and dates	1.75w

Step 3 decides who absorbs the risk. Vendor-executed hardening is on a schedule the program cannot influence; internally executed hardening competes with <code>AMS-03</code> for the same analog engineers. Both are real answers and neither is free.

WATCH OUT FOR

- ◆ **Hardening treated as vendor overhead.** It is a project with a duration; leaving it off the plan means the block arrives on a date nobody derived.
- ◆ **Internal execution assumed available.** The engineers who would port a PHY are the engineers building the PLL and the custom array, and they cannot do all three.
- ◆ **Re-verification omitted from scope.** Porting changes the block; the verification collateral has to be re-run, and that effort belongs in the DV plan.
- ◆ **Characterization corners not matched to the program's.** A block re-characterized at the vendor's corners still cannot be signed off at the program's.
- ◆ **Plan without dates.** A hardening plan with owners and no dates does not integrate with the program schedule and is therefore not tracked.

WHERE THE EFFORT GOES	4 M/M	WHO IS ON IT	~3.5 FTE	CONNECTIONS	TERMS HERE	6
Porting scope definition	1.5	IP strategist OWNS IT		DEPENDS ON IPR-02 IPR-03 IPR-05	IP PLL PHY AMS RTL DV	
Effort and duration estimation	1.0	AMS lead		FEEDS INTO IPR-09 IPR-10 AMS-03		
Execution decision	0.8	Verification lead		DV-02		
Hardening plan	0.8	Vendor technical contacts Program manager				

05 IPR · IP STRATEGY & READINESS

IPR-09

IP delivery schedule alignment to RTL and PD integration windows

TAKES 4w w17–w21 of the stage	COSTS 2.0 M/M ~2.2 people while it runs	OWNER Program manager Procurement lead approves	CRITICAL PATH yes slipping moves the program
WHY IT EXISTS Line up every IP delivery date against the window that actually needs it—RTL integration, synthesis, floorplan—and surface the mismatches while they can still be fixed. An IP schedule is only meaningful against the design schedule. A PHY delivered in week 40 is early for RTL and late for the floorplan, because PD-02 needs the abstract long before RTL-03 needs the model. Alignment means matching each deliverable to the window that consumes it, not the block to a date.			
WHAT IT DELIVERS IPR-D6 IP delivery schedule folded into the program plan This activity is the deliverable, and "folded in" is the operative part—a separate IP schedule is one nobody tracks.			
NEEDS FIRST - Committed delivery dates from IPR-07 - Hardening plan from IPR-08 - RTL integration windows from the program schedule - Floorplan and macro placement needs from PD planning - Deliverable checklist from IPR-06			
DONE WHEN - Alignment done per deliverable, not per block - Every mismatch has an owner and an escalation path - IP schedule visible inside the program plan, not beside it			
HOW IT GETS THERE 6 STEPS · 2 LANES w17w18w19w20w21 1235 46			
MAIN SEQUENCE IN PARALLEL #STEP, AND WHAT IT ADDS TO THE DELIVERABLE TAT 1Integration window extraction — what each stage needs and when → Integration window map per consuming stage 0.75w			

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Committed delivery date collection per block and per deliverable → Committed delivery dates per block and deliverable	1w
3	Deliverable-level alignment — abstracts before models before collateral → Deliverable-level delivery schedule	1.25w
4	Hardening plan overlay from IPR-08 runs in parallel → Hardening overlay	0.75w
5	Mismatch identification and escalation list → Mismatch and escalation list	1w
6	Schedule integration into the program plan runs in parallel → IP schedule folded into the program plan	0.75w

Step 3 is where the useful detail lives. Vendors quote one date for a block; the design needs the LEF abstract for floorplan months before it needs the multi-corner .lib for signoff, and asking for the abstract early is often free.

WATCH OUT FOR

- ◆ **Block-level dates instead of deliverable-level.** One date per block hides that the abstract is needed months before the model, and the earlier need is the binding one.
- ◆ **IP schedule maintained separately.** A schedule outside the program plan is not reviewed at program reviews and drifts unnoticed.
- ◆ **Mismatches recorded without escalation.** A known gap with no owner and no escalation is a gap that is discovered again at integration.
- ◆ **Hardening overlay omitted.** Delivery dates for unproven blocks are meaningless without the porting window that follows them.
- ◆ **Floorplan needs underestimated.** Physical design needs abstracts and blockages very early; planning against RTL's needs alone produces a late floorplan.

WHERE THE EFFORT GOES	2 M/M	WHO IS ON IT ~2.2 FTE	CONNECTIONS	TERMS HERE	5
Deliverable-level alignment	0.8	Program manager OWNS IT	DEPENDS ON DEF-08 IPR-06 IPR-07 IPR-08	IP PHY RTL LEF .lib	
Window extraction and date collection	0.5	IP strategist	FEEDS INTO IPR-10 RTL-03 RTL-09 PD-02 PD-04		
Mismatch identification	0.5	RTL integration lead			
Schedule integration	0.3	Physical design lead			
		Procurement lead			

05 IPR · IP STRATEGY & READINESS

IPR-10

IP maturity risk rating and second-source contingency

TAKES	COSTS	OWNER	CRITICAL PATH
4w w18–w22 of the stage	2.0 M/M ~2.2 people while it runs	IP strategist Chief architect approves	no has float

WHY IT EXISTS

Rate every IP line for **maturity and supply risk**, and decide where a contingency is worth having—a second source, an internal fallback, or an accepted exposure with a trigger.

Some risk here is unavoidable and pretending otherwise is what makes it dangerous. The point is not to eliminate single-vendor dependencies but to know which ones would stop the program, and to have decided in advance what would be done about each.

WHAT IT DELIVERS

IPR-D7

IP risk register and contingency plan

This activity is the deliverable, and the triggers are what make it a plan rather than a list.

NEEDS FIRST

- Readiness report and maturity levels from IPR-05
- Hardening plan from IPR-08
- Delivery schedule and mismatches from IPR-09
- Vendor evaluation matrix from IPR-04
- Sourcing strategy from TECH-08

DONE WHEN

- Critical dependencies identified and separated from the rest
- Every critical block has a contingency with a trigger date
- Register has a review cadence, not just an author

HOW IT GETS THERE

5 STEPS · 2 LANES

w18	w19	w20	w21	w22
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Risk rating per IP line — maturity, vendor, schedule, supply → Risk rating per IP line	1w
2	Critical dependency identification — which failures stop the program → Critical dependency list	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Second-source viability per critical block → Second-source viability findings	0.75w runs in parallel
4	Internal fallback option assessment → Internal fallback assessment	0.75w runs in parallel
5	Contingency decision and trigger definition per critical block → Contingency plan with triggers and decision dates	2w

Step 5 defines triggers, not intentions. "We would build it internally if the vendor slipped" is worthless without the date at which that decision has to be taken to still be executable.

WATCH OUT FOR

- ◆ **Register without triggers.** A contingency with no decision date cannot be executed, because by the time the need is obvious the option has expired.
- ◆ **Every line rated as critical.** A register where everything is critical directs attention nowhere; the point is to identify the few that stop the program.
- ◆ **Second source assessed on capability alone.** An alternative vendor without a license, an evaluation and a delivery date is not a contingency.
- ◆ **Internal fallback assumed.** A fallback that depends on engineers already committed to the critical path is not a fallback.
- ◆ **Register written and not reviewed.** IP risk changes as deliveries land; a register reviewed once at IP Plan Freeze is stale by RTL integration.

WHERE THE EFFORT GOES	2 M/M	WHO IS ON IT	~2.2 FTE	CONNECTIONS	TERMS HERE	3
Risk rating	0.8	IP strategist owns it		DEPENDS ON TECH-08 IPR-05 IPR-08 IPR-09	IP AMS RTL	
Critical dependency analysis	0.5	Program manager		FEEDS INTO AMS-01 RTL-10		
Contingency options	0.5	Procurement lead		FEEDS BACK INTO DEF-09		
Trigger definition	0.3	AMS lead				
		Chief architect				

06

Custom, AMS & Memory IP Development

AMS

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w18 – w60	42 _w	16	296 _{M/M}	8 _{of 16}

ENGINEERING ACTIVITIES				KEY DELIVERABLES	
AMS-01	AMS IP specification and budget allocation (jitter, BER, PSRR, area)	6w	8	AMS-D1	AMS IP specifications and design review packages
AMS-02	PLL / clock generator design and closure	26w	32	AMS-D2	Characterization reports across PVT and Monte Carlo
AMS-03	SerDes and PHY design or vendor hardening (PCIe/CXL, UCIe, HBM PHY)	34w	60	AMS-D3	Custom SRAM instance specification with Vmin and sigma-yield report
AMS-04	LDO, bandgap and power-management cell design	20w	22	AMS-D4	Custom memory views characterized to compiler equivalence
AMS-05	Custom SRAM instance architecture — bitcell selection, array organization, sense amp and replica timing	10w	14	AMS-D5	Hard macro GDS with abstract views (LEF, .lib, CDL, UPF, wreal/Verilog model)
AMS-06	Read / write assist circuit design for Vmin — WL boost, negative bitline, VDD collapse	10w	12	AMS-D6	Per-macro DRC / LVS clean signoff, pushed rules approved by foundry
AMS-07	Custom array layout and pushed-rule DRC closure with foundry approval	16w	22	AMS-D7	Reliability report — EM/IR, ESD, latch-up, aging
AMS-08	Statistical margin and yield analysis — importance sampling, sigma-Vmin, bitcell variation	10w	12	AMS-D8	Integration guide with known limitations and errata
AMS-09	Redundancy and repair scheme integration with the BISR architecture	6w	6		
AMS-10	Custom memory view generation and compiler-equivalent characterization across corners	10w	12		
AMS-11	Schematic design and pre-layout simulation	16w	20		
AMS-12	Custom layout with DRC / LVS / antenna and fill closure	18w	26		
AMS-13	Post-layout extracted simulation across PVT and Monte Carlo	16w	20		
AMS-14	Reliability checks — EM/IR, aging, ESD, latch-up	10w	10		

AMS-15	Hard macro abstraction and view generation (LEF, .lib, CDL, GDS, behavioural model)	8w	8 M/M
AMS-16	AMS-digital co-simulation and integration support	16w cont.	12 M/M

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-01

AMS IP specification and budget allocation (jitter, BER, PSRR, area)

TAKES	COSTS	OWNER	CRITICAL PATH
6w w0–w6 of the stage	8.0 M/M ~5.8 people while it runs	AMS architect AMS lead approves	no has float

WHY IT EXISTS

Turn each custom block from a name on the make list into **a specification with budgets**—jitter, BER, PSRR, area, power—that a circuit designer can start against and a reviewer can hold them to.

Analog specifications are budgets before they are targets. Total jitter is divided between the PLL, the distribution network and the PHY; each owner gets a share, and a share nobody allocated becomes an argument at integration when the eye closes and no one can say whose fault it is.

WHAT IT DELIVERS

AMS-D1

AMS IP specifications and design review packages

This activity is the deliverable—the specifications every other AMS activity is executed and reviewed against.

NEEDS FIRST

- Make decisions from IPR-03
- Custom memory scope from PDK-06
- Interface specifications from ARCH-04
- Power and clock architecture from ARCH-05
- Block PPA budgets from ARCH-06

DONE WHEN

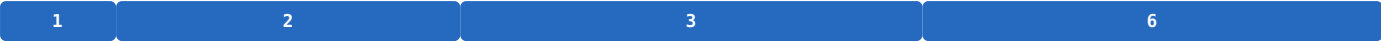
- Every system budget decomposed into per-block allocations
- Each specification accepted by its named owner
- Area and power allocations reconciled against ARCH-06

HOW IT GETS THERE

6 STEPS · 2 LANES

w0	w2	w4	w6
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Custom block list confirmation from the make decisions → Custom block list with owners	0.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	System-level budget decomposition — jitter, noise, BER → System-level budget decomposition · Per-block specifications with jitter, BER, PSRR and area budgets	1.5w
3	Per-block specification authoring → Per-block AMS specifications	2w
4	Area and power allocation per block → Area and power allocation per block	1w
5	PSRR and supply sensitivity requirements → PSRR and supply sensitivity requirements	1w
6	Specification review and owner acceptance → Specification review and acceptance record	2w

Step 2 is the one that prevents the integration argument. A jitter budget divided explicitly between PLL, distribution and PHY gives each owner a number; an unallocated total gives each owner a reason to believe the problem is somewhere else.

WATCH OUT FOR

- ◆ **Budgets stated as totals rather than allocations.** An unallocated jitter budget produces three designers each assuming the others left them room.
- ◆ **Specifications written without the PHY vendor.** Where a PHY is bought, its jitter tolerance bounds what the PLL must deliver, and the two have to be written together.
- ◆ **Area allocated before the architecture is understood.** Analog area is set by matching, isolation and routing, and an allocation made from a digital estimate is not usable.
- ◆ **PSRR requirements omitted.** Supply sensitivity is what turns a working block into a failing one once the digital neighbors switch, and it is rarely specified up front.
- ◆ **No owner acceptance.** A specification the designer has not accepted is one they will design around rather than to.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT	~5.8 FTE	CONNECTIONS	TERMS HERE	7
Per-block specification authoring	3.0	AMS architect OWNS IT		DEPENDS ON ARCH-04 ARCH-05 ARCH-06 IPR-03 IPR-10	PPA IP PLL PHY AMS BER PSRR	
Budget decomposition	2.0	Block circuit owners		FEEDS INTO AMS-02 AMS-03 AMS-04 AMS-05 AMS-11 TC-01		
Area and power allocation	1.5	Interface architect		LATER INPUT PDK-06 +3w		
Review and acceptance	1.5	Power architect				
		AMS lead				

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-02

PLL / clock generator design and closure

TAKES	COSTS	OWNER	CRITICAL PATH
26w w4–w30 of the stage	32.0 M/M ~5.3 people while it runs	PLL circuit designer AMS lead approves	yes slipping moves the program

WHY IT EXISTS

Design the **clock generation** the whole chip runs on—PLLs, their loop dynamics, their supply immunity and their jitter—and close them across process, voltage and temperature.

A PLL is small, slow to build and impossible to work around. Twenty-six weeks for a block of a few hundred microns square looks disproportionate until the first silicon that will not lock over temperature, at which point the entire program is waiting on a circuit nobody can patch.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Jitter and PSRR budgets from AMS-01
- Clock architecture and DVFS points from ARCH-05
- Reference clock specification from the system requirements
- Node device models and PDK from PDK-01
- Supply and rail topology from ARCH-05

DONE WHEN

- Jitter closed post-layout across PVT, not pre-layout
- Lock time and startup sequence characterized and documented
- Macro released with views and a behavioural model

HOW IT GETS THERE

10 STEPS · 2 LANES

w4	w10	w16	w22	w28	w30
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MAIN SEQUENCE

1	2	3	4	7	8	10
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IN PARALLEL

5	6	9
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#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Architecture selection — integer, fractional, ring or LC → PLL architecture selection record	2w
2	Loop dynamics and phase noise budget → Loop dynamics and phase noise budget	2w
3	VCO design and tuning range definition → VCO design with its tuning range	4w
4	Charge pump, loop filter and divider design → PLL architecture selection and loop design · VCO and loop component designs	4w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
5	Supply regulation and PSRR strategy → Supply regulation and calibration scheme	3w runs in parallel
6	Lock detection, calibration and startup sequencing → Lock detection, calibration and startup sequencing design	3w runs in parallel
7	Layout with matching, shielding and isolation → PLL layout with matching and shielding	6w
8	Post-layout jitter closure across PVT → Post-layout jitter results across PVT	5w
9	Clock distribution interface definition with physical design → Clock distribution interface definition	3w runs in parallel
10	Characterization and macro release → Characterized PLL macro with views	3w

Step 8 is where the schedule risk lives. Post-layout jitter is routinely worse than pre-layout by more than the margin, and the fix is a layout change that takes the six weeks of step 7 again.

WATCH OUT FOR

- ◆ **Post-layout jitter worse than budget.** The commonest analog schedule failure, and the fix is a layout iteration measured in weeks, not days.
- ◆ **PSRR ignored until integration.** A PLL that meets jitter on a clean supply and fails beside switching digital logic has met a specification nobody can use.
- ◆ **Tuning range too narrow across corners.** A VCO that cannot cover the DVFS range at the slow corner is a DVFS policy that has to be reduced in silicon.
- ◆ **Lock time and startup unspecified.** Boot sequencing depends on lock time, and discovering it at bring-up delays every subsequent test.
- ◆ **One PLL designed for several very different uses.** Sharing one design across HBM, SerDes and core clocking usually means it is optimal for none.

WHERE THE EFFORT GOES	32 M/M	WHO IS ON IT ~5.3 FTE	CONNECTIONS	TERMS HERE 12
VCO and loop design	9.0	PLL circuit designer OWNS IT	DEPENDS ON ARCH-05 PDK-01 AMS-01	PDK IP PLL VCO
Layout	7.0	Analog layout engineer	FEDS INTO AMS-11 AMS-12 AMS-13	SerDes AMS PVT DVFS
Post-layout jitter closure	6.0	Verification engineer, AMS	AMS-15 DFT-05 PD-05	HBM (ESD) HBM PSRR
Supply regulation and calibration	5.0	Clocking architect	PD-07 SIPI-08 BU-03	GDS
Characterization and release	3.0	AMS lead		
Distribution interface	2.0			

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-03

SerDes and PHY design or vendor hardening (PCIe/CXL, UCIe, HBM PHY)

TAKES	COSTS	OWNER	CRITICAL PATH
34w w4–w38 of the stage	60.0 M/M ~7.7 people while it runs	SerDes and PHY lead Vendor technical contact approves	yes slipping moves the program

WHY IT EXISTS

Deliver the **high-speed interfaces**—PCIe or CXL, die-to-die, HBM PHY—either designed from scratch or hardened from a vendor's deliverable onto this node and this flavor sheet.

At thirty-four weeks and sixty man-months this is the largest single activity in the entire template, and it is the one most likely to decide the tapeout date. Even a bought PHY is not free: hardening, integration, and closing its analog margins on this process option is a project, and the vendor's schedule for it is not the program's.

WHAT IT DELIVERS

AMS-D2

Characterization reports across PVT and Monte Carlo

The PHY's eye and BER characterization is the report's most consequential section.

NEEDS FIRST

- Interface selections and revisions from ARCH-04
- Vendor deliverables and licences from IPR-07
- Bump pitch and PHY placement from ARCH-08
- Channel budgets from SIPI-08
- Jitter budget from AMS-01

DONE WHEN

- BER and eye closed post-layout across PVT at the specified channel
- Compliance simulated against the interface specification, not only internally
- Macros released with a stated limitation list

HOW IT GETS THERE

10 STEPS · 2 LANES

w4	w11	w18	w25	w32	w38
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	PHY strategy per interface — build, buy hardened, or buy soft → PHY strategy per interface	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Vendor deliverable ingestion and gap assessment → Vendor deliverable gap assessment	3w
3	Equalization and link training architecture → Equalization and link training architecture	4w
	runs in parallel	
4	TX and RX front-end design or adaptation → TX/RX front-end and clock recovery designs	7w
5	Clock recovery and per-lane timing → Clock recovery and per-lane timing design	5w
	runs in parallel	
6	HBM PHY integration and training sequence → HBM PHY integration and training sequence	6w
7	Layout, matching and lane pitch closure → SerDes layout with lane pitch closure	8w
8	Post-layout BER and eye closure across PVT → Post-layout BER and eye results across PVT	6w
9	Compliance simulation against the interface specification → Compliance simulation results	4w
	runs in parallel	
10	Macro release and integration support → Released PHY macros with integration support	2w

Step 7 is eight weeks of layout because lane pitch is a floorplan constraint as much as a circuit one—the PHY has to fit the bump pitch <code>ARCH-08</code> defined, and a pitch mismatch found here forces a change at the die edge.

WATCH OUT FOR

- ◆

Bought PHY assumed to need no work. Hardening, integration and margin closure on this flavor sheet is a multi-quarter project regardless of who designed the circuit.
- ◆

Lane pitch incompatible with the bump map. Discovered at layout, it forces a change at the die edge where there is least room to make one.
- ◆

Compliance simulated late. A PHY that closes its own eye and fails the interface specification's compliance mask has to be reworked with no schedule left.
- ◆

Channel budget not shared with SI. The PHY closes against an assumed channel; if SIPI-07 extracts a worse one, the margin was never real.
- ◆

Vendor support model untested until integration. The first hard question is when a support relationship is discovered to be a ticketing system.

WHERE THE EFFORT GOES	60 M/M	WHO IS ON IT ~7.7 FTE	CONNECTIONS	TERMS HERE 13
TX and RX front-end design	14.0	SerDes and PHY lead owns it	DEPENDS ON ARCH-04 ARCH-08 IPR-03 IPR-04 IPR-07 IPR-08 AMS-01	SerDes PHY PVT SI HBM (ESD) HBM PCIe CXL TX/RX BER GDS TX RX
Layout and pitch closure	13.0	Analog circuit designers	RUNS WITH PD-04	
Post-layout BER and eye closure	11.0	Analog layout engineers	FEEDS INTO AMS-12 AMS-13 AMS-15 PKG0-05 SIPI-07 SIPI-08 BU-05	
HBM PHY integration	8.0	SI liaison	LATER INPUT SIPI-07 +2w	
Clock recovery and per-lane timing	7.0	Vendor technical contact		
Compliance simulation	5.0			
Macro release and support	2.0			

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-04

LDO, bandgap and power-management cell design

TAKES	COSTS	OWNER	CRITICAL PATH
20w w6–w26 of the stage	22.0 M/M ~4.8 people while it runs	Power management circuit designer Reliability engineer approves	no has float

WHY IT EXISTS

Build the **power management cells** everything else depends on—LDOs, bandgap references, bias generation—quietly and correctly, because every other block's specification assumes them.

These blocks are small, unglamorous and load-bearing. A bandgap that drifts moves every analog threshold on the die; an LDO with insufficient PSRR at the switching frequency turns the PLL's careful jitter budget into noise. They are specified against by everyone and reviewed by almost nobody.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Supply and rail topology from ARCH-05
- PSRR requirements from AMS-01
- Load profiles from the digital power estimate
- Node device models and precision element availability
- Test and trim capability from TEST-01

DONE WHEN

- PSRR verified at the digital switching frequency, not only at DC
- Stability held across the full load range post-layout
- Trim procedure agreed with test engineering

HOW IT GETS THERE

8 STEPS · 2 LANES

w6	w10	w14	w18	w22	w26
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MAIN SEQUENCE

1	2	3	6	7	8
---	---	---	---	---	---

IN PARALLEL

4	5
---	---

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Supply topology and regulation requirement per domain → Supply topology and regulation plan	1.5w
2	Bandgap reference design and trim strategy → Bandgap reference design with trim strategy	3.5w
3	LDO design — load range, stability, PSRR → LDO designs with stability and PSRR results	5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Bias generation and distribution → Bias generation and distribution scheme	3w runs in parallel
5	Startup, sequencing and brown-out behavior → Startup, sequencing and brown-out behavior definition	2.5w runs in parallel
6	Layout with matching and supply isolation → Regulator layout with supply isolation	4w
7	Post-layout stability and PSRR verification across PVT → Post-layout stability and PSRR results across PVT	4w
8	Trim and calibration procedure definition for test → Trim and calibration procedure for production test	2w

Step 8 is easy to leave until later and expensive to. A trim procedure that test engineering only sees at `TEST-05` becomes a test-time cost, and a bandgap with no defined trim is a part with a distribution nobody can tighten.

WATCH OUT FOR

- ◆ **PSRR specified at DC.** The frequency that matters is where the digital load switches, and a regulator flat at DC can be transparent at that frequency.
- ◆ **Stability verified only at nominal load.** LDO compensation has to hold across the full load range including the transitions, which is where it usually does not.
- ◆ **Bandgap trim undefined.** An untrimmed reference has a distribution wide enough to move every analog threshold, and there is no way to tighten it in production.
- ◆ **Brown-out and startup behavior unspecified.** The part's behavior at supply ramp is discovered at bring-up, on silicon that cannot be changed.
- ◆ **Reviewed by nobody.** Blocks everyone assumes and nobody reviews are the ones that fail in a way that looks like someone else's problem.

WHERE THE EFFORT GOES	22 M/M	WHO IS ON IT ~4.8 FTE	CONNECTIONS	TERMS HERE 8
LDO design	6.0	Power management circuit designer OWNS IT	DEPENDS ON ARCH-05 AMS-01	PLL LDO PHY DC
Bandgap and trim strategy	4.5	Analog layout engineer	FEEDS INTO AMS-12 AMS-13 AMS-14	PVT EM PSRR GDS
Layout	4.0	Power architect	AMS-15 SIPI-05 TEST-05	
Post-layout verification	3.5	Test engineering liaison	BU-03	
Bias generation	2.5	Reliability engineer		
Trim procedure for test	1.5			

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-05

Custom SRAM instance architecture — bitcell selection, array organization, sense amp and replica timing

TAKES	COSTS	OWNER	CRITICAL PATH
10w w2–w12 of the stage	14.0 M/M ~6.1 people while it runs	Memory circuit architect PDK owner approves	yes slipping moves the program

WHY IT EXISTS

Architect the **custom SRAM instance** the compiler could not deliver—bitcell choice, array organization, sense amplifier topology and replica timing—against the density and Vmin the block budget demands.

This activity exists only because PDK-06 opened it. That decision came with a quantified gap and a fixed window, and the architecture chosen here determines whether the rest of the custom memory chain can close inside it. A conservative architecture will not recover the density; an aggressive one may not close Vmin.

WHAT IT DELIVERS

AMS-D3

Custom SRAM instance specification with Vmin and sigma-yield report

The specification half of the deliverable; AMS-08 supplies the yield half.

NEEDS FIRST

- Custom instance decision and gap from PDK-06
- Density and Vmin measurements from PDK-05
- Memory hierarchy requirements from ARCH-03
- Block memory budget from ARCH-06
- Foundry bitcell options and pushed-rule policy

DONE WHEN

- Projected density closes the gap the decision was taken on
- Bitcell choice within the foundry's stated approval appetite
- Architecture frozen with an offset and Vmin budget attached

HOW IT GETS THERE

8 STEPS · 2 LANES

w2w4w6w8w10w12

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Scope confirmation against the PDK-06 decision and gap → Confirmed custom array scope	0.75w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Bitcell selection — foundry cell, pushed rules, or custom → Bitcell selection with rule implications	1.75w
3	Array organization — banking, segmentation, aspect ratio → Array organization and banking scheme	2w
4	Sense amplifier topology and offset budget → Sense amplifier topology and offset budget	2w
5	Replica timing and self-timed path design → Replica timing scheme	1.75w runs in parallel
6	Peripheral circuit architecture — decoders, drivers, IO → Peripheral circuit architecture	1.5w runs in parallel
7	Density projection against the block budget → Density projection against budget	3.5w
8	Architecture review and freeze → Architecture freeze record	1.5w runs in parallel

Step 7 closes the loop with `PDK-05`. If the projected density still misses the budget, the answer is to reopen the architecture now rather than to discover it after sixteen weeks of layout in `AMS-07`.

WATCH OUT FOR

- ◆ **Architecture that does not recover the gap.** A conservative organization produces a custom array with the compiler's density and none of the compiler's maintenance advantages.
- ◆ **Bitcell pushed beyond foundry appetite.** A cell the foundry will not approve is a layout that cannot tape out, and the approval path runs through PDK-06.
- ◆ **Sense amplifier offset budget optimistic.** Offset drives Vmin more than any other single choice, and an optimistic budget is discovered in AMS-08 when it is far too late to change topology.
- ◆ **Replica timing not proven across corners.** Self-timed paths that track at nominal and diverge at the corner produce read failures that look random.
- ◆ **Architecture frozen without the density projection.** Freezing before step 7 means committing sixteen weeks of layout to an organization that may not close the gap.

WHERE THE EFFORT GOES	14 M/M	WHO IS ON IT	~6.1 FTE	CONNECTIONS	TERMS HERE	6
Array organization	4.0	Memory circuit architect OWNS IT		DEPENDS ON ARCH-03 ARCH-06 PDK-04 PDK-05 PDK-06 AMS-01	PDK IP SRAM AMS IO Vmin	
Sense amplifier and offset budget	3.5	Bitcell and device engineer		FEEDS INTO AMS-06 AMS-07 AMS-08 AMS-09 AMS-10		
Bitcell selection	2.5	Sense amplifier designer				
Replica timing	2.0	Memory layout lead				
Density projection and review	2.0	PDK owner				

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-06

Read / write assist circuit design for Vmin
— WL boost, negative bitline, VDD collapse

TAKES

10w

w10–w20 of the stage

COSTS

12.0 M/M

~5.2 people while it runs

OWNER

Memory circuit designer

Memory architect approves

CRITICAL PATH

no

has float

WHY IT EXISTS

Design the **assist circuits that buy Vmin**—wordline boost, negative bitline, VDD collapse—and decide how much of each the array actually needs.

Assist is what makes an aggressive array workable, and it is not free: each technique costs area, power and complexity, and some of them interact badly. The design question is not which assist is best but which combination reaches the Vmin target at the least cost, verified statistically rather than at nominal.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Array architecture and Vmin budget from AMS-05
- Vmin requirements from ARCH-05 operating points
- Bitcell characterization data
- Node assist circuit precedents from the foundry
- Power budget for the array from ARCH-06

DONE WHEN

- Vmin target met statistically, not at nominal
- Assist interactions studied rather than assumed
- Boost levels within the reliability limits AMS-14 will check

HOW IT GETS THERE

7 STEPS · 2 LANES

w10w12w14w16w18w20

MAIN SEQUENCE

12567

IN PARALLEL

34

#STEP, AND WHAT IT ADDS TO THE DELIVERABLETAT

1Vmin gap analysis against the AMS-05 architecture
→ Vmin gap analysis with assist requirement1w

2Assist technique selection and interaction study
→ Assist technique selection and interaction findings2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Wordline boost circuit design → Wordline boost circuit design	2w runs in parallel
4	Negative bitline circuit design → Wordline boost, negative bitline and collapse circuits	2w runs in parallel
5	VDD collapse and read-assist timing → VDD collapse and read-assist timing design	2w
6	Assist control, timing and calibration → Assist control and calibration scheme	2w
7	Vmin verification with assist enabled, across corners → Vmin verification results with assist enabled	3w

Steps 3 and 4 run in parallel and step 2 decides whether both are needed. Write assist and read assist pull in opposite directions on the cell, and enabling both without studying the interaction can leave Vmin worse than either alone.

WATCH OUT FOR

- ◆ **Assist techniques combined without an interaction study.** Read and write assist push the cell in opposite directions; enabling both blindly can make Vmin worse.
- ◆ **Assist verified at nominal.** Vmin is a statistical quantity; verification at nominal proves nothing about the tail that decides yield.
- ◆ **Boost levels beyond reliability limits.** Wordline boost stresses the cell, and a level chosen for Vmin alone shortens life in a way AMS-14 will find late.
- ◆ **Assist timing not calibrated.** Fixed assist timing that works at one corner and misfires at another produces intermittent failures indistinguishable from a marginal cell.
- ◆ **Area cost of assist unbudgeted.** Assist circuits sit in the periphery and can consume the density the custom array was built to recover.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT ~5.2 FTE	CONNECTIONS	TERMS HERE	3
Assist circuit design	4.5	Memory circuit designer OWNS IT	DEPENDS ON ARCH-05 PDK-06 AMS-05	SRAM Vmin VDD	
Vmin verification	3.0	Bitcell and device engineer	FEEDS INTO AMS-07 AMS-08 AMS-10		
Assist control and calibration	2.5	Statistical analysis engineer	AMS-14		
Interaction study	2.0	Reliability engineer Memory architect			

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-07

Custom array layout and pushed-rule DRC closure with foundry approval

TAKES	COSTS	OWNER	CRITICAL PATH
16w w18–w34 of the stage	22.0 M/M ~6.0 people while it runs	Memory layout lead Foundry technical contact approves	yes slipping moves the program

WHY IT EXISTS

Lay out the custom array at **pushed rules, and get the foundry to approve it**—because a layout the foundry will not sign is a layout that cannot tape out.

Sixteen weeks of layout is what the density recovery actually costs. Pushed rules are the mechanism, and they carry an approval process with its own duration and its own possibility of refusal. Running layout and approval in sequence would not fit the window, so they run together, which means designing against rules that are not yet certain.

WHAT IT DELIVERS

AMS-D6

Per-macro DRC / LVS clean signoff, pushed rules approved by foundry

This activity is the deliverable, and the approval is the half that cannot be produced internally.

NEEDS FIRST

- Array architecture from AMS-05
- Assist circuits from AMS-06
- Pushed-rule policy and approval path from PDK-06
- Design rule manual and exceptions from PDK-02
- Density target from the block budget

DONE WHEN

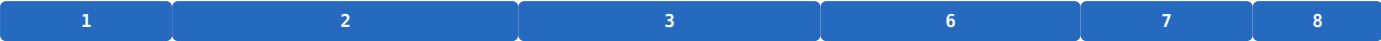
- Every pushed rule approved by the foundry in writing
- Achieved density meets the target the decision was taken on
- Layout DRC, LVS and antenna clean, and extractable

HOW IT GETS THERE

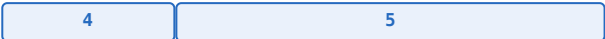
8 STEPS · 2 LANES

w18	w22	w26	w30	w34
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Pushed-rule set proposal and foundry pre-alignment → Pushed-rule set with foundry pre-alignment	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Bitcell array layout at pushed rules → Bitcell array and peripheral layout	4w
3	Peripheral and assist circuit layout → Peripheral and assist circuit layout	3.5w
4	Dummy, fill and density compliance → Density and fill compliance record	2w
5	Foundry approval submission and iteration → Foundry approval record and waivers	5w
6	DRC closure including pushed-rule waivers → DRC closure record with pushed-rule waivers	3w
7	LVS and antenna closure → DRC, LVS and antenna clean layout	2w
8	Layout freeze and extraction handoff → Frozen layout with extraction handoff	1.5w

Step 5 runs in parallel out of necessity rather than choice. Waiting for approval before laying out would not fit the window, so the array is laid out against rules that may still be refused—which is why step 1's pre-alignment matters more than its two weeks suggest.

WATCH OUT FOR

- ◆ **Foundry refuses a pushed rule after layout is built on it.** The mitigation is pre-alignment in step 1, and it is the single most valuable two weeks in the activity.
- ◆ **Density target missed despite the pushed rules.** The array then costs sixteen weeks and delivers the compiler's density, which is the worst available outcome.
- ◆ **Approval iterations underestimated.** Each foundry round trip runs to weeks, and the five-week allowance assumes no more than two.
- ◆ **Fill and density compliance left to the end.** Fill interacts with the array's matching and can undo careful layout if applied late.
- ◆ **Layout frozen before extraction is proven.** An array that extracts badly has to be re-laid out, and the extraction handoff is where that is found.

WHERE THE EFFORT GOES	22 M/M	WHO IS ON IT	~6.0 FTE	CONNECTIONS	TERMS HERE	4
Array layout	7.0	Memory layout lead OWNS IT		DEPENDS ON PDK-02 PDK-06 AMS-05 AMS-06	PDK SRAM DRC LVS	
Peripheral and assist layout	5.5	Layout engineers		FEEDS INTO AMS-08 AMS-10 AMS-12 AMS-15 S0-02		
Foundry approval iteration	3.5	PDK owner				
DRC and LVS closure	4.0	Physical verification engineer				
Fill and density compliance	2.0	Foundry technical contact				

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-08

Statistical margin and yield analysis — importance sampling, sigma-Vmin, bitcell variation

TAKES	COSTS	OWNER	CRITICAL PATH
10w w20–w30 of the stage	12.0 M/M ~5.2 people while it runs	Statistical analysis engineer AMS lead approves	no has float

WHY IT EXISTS

Prove the array works **at the tail of the distribution rather than at nominal**—sigma-Vmin, bitcell variation, sense amplifier offset—using importance sampling because brute-force Monte Carlo cannot reach six sigma.

A memory array of this size fails at the tail by definition: with hundreds of millions of cells, the six-sigma cell is not a hypothetical, it is present in every part. Nominal simulation says nothing about it, and the yield the cost model assumed depends entirely on where that tail sits.

WHAT IT DELIVERS

AMS-D3

Custom SRAM instance specification with Vmin and sigma-yield report

The yield half of the deliverable, and the half the cost model depends on.

NEEDS FIRST

- Array architecture from AMS-05
- Assist circuits and Vmin results from AMS-06
- Layout parasitics from AMS-07
- Foundry variation models and silicon data
- Redundancy scheme from AMS-09

DONE WHEN

- Sigma targets met for read, write and retention
- Variation model validated against silicon, not assumed
- Yield projection handed to the cost model with its methodology

HOW IT GETS THERE

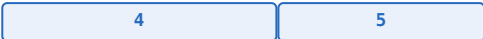
7 STEPS · 2 LANES

w20	w22	w24	w26	w28	w30
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Statistical methodology selection — importance sampling, scaled sigma → Statistical methodology and its validation	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Bitcell variation model validation against silicon data → Bitcell variation model validation record	2w
3	Sigma-Vmin analysis for read, write and retention → Sigma-Vmin results for read, write and retention	3w
4	Sense amplifier offset distribution analysis → Sense amplifier offset distribution analysis	2w
5	Peripheral and timing variation contribution → Offset and peripheral variation contributions	1.5w
6	Array-level yield projection with redundancy applied → Array yield projection with redundancy	2w
7	Statistical margin report and signoff → Statistical margin report	1.5w

Step 2 is the step that makes the rest meaningful. A variation model that has not been validated against silicon produces a six-sigma number with an unknown error, and the number is then used to justify a yield assumption in the cost model.

WATCH OUT FOR

- ◆ **Brute-force Monte Carlo used at the tail.** Reaching six sigma by direct sampling is computationally impossible; a run that terminates has not reached the tail it claims.
- ◆ **Variation model unvalidated.** The six-sigma answer inherits the model's error, and nobody knows its size.
- ◆ **Redundancy assumed to cover the tail.** Repair fixes a bounded number of failing bits; a distribution problem produces more than the repair scheme can absorb.
- ◆ **Retention overlooked.** Read and write Vmin get the attention; retention failures at low voltage and high temperature are found in qualification.
- ◆ **Result reported without the methodology.** A sigma number quoted without stating how it was obtained cannot be compared or challenged.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT ~5.2 FTE	CONNECTIONS	TERMS HERE
		Statistical analysis engineer owns it	DEPENDS ON PDK-05 AMS-05 AMS-06 AMS-07 AMS-09	SRAM AMS PVT Vmin
Sigma-Vmin analysis	4.0	Memory circuit designer		
Variation model validation	2.5	Device and bitcell engineer	FEEDS INTO AMS-10 AMS-14 MP-05	
Offset and peripheral analysis	2.5	Yield engineer	FEEDS BACK INTO DEF-04	
Yield projection	2.0	AMS lead		
Report and signoff	1.0			

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-09

Redundancy and repair scheme integration with the BISR architecture

TAKES	COSTS	OWNER	CRITICAL PATH	
6w w16–w22 of the stage	6.0 M/M ~4.3 people while it runs	Memory circuit designer Memory architect approves	no has float	
WHY IT EXISTS				
Build redundancy and repair into the array and make it match the BISR architecture DFT is designing—spare rows and columns, fuse allocation, repair algorithm, and the interface between them. Redundancy is worth several points of yield and only if the two halves agree. The array provides spares and a repair interface; DFT-03 provides the BIST engine and the repair analysis. A mismatch between them is found at pattern generation, when both are effectively frozen.				
WHAT IT DELIVERS				
No key deliverable is owned here — this activity contributes to the ones below.				
NEEDS FIRST				
● Array architecture from AMS-05 ● BISR architecture intent from DFT-03 ● Defect density projection from TECH-07 ● Fuse map and lifecycle model from ARCH-07 ● Yield target from the cost model				
DONE WHEN				
● Repair interface written jointly with DFT, not in parallel ● Spare allocation matched to the projected defect modes ● Fuse capacity sized against the worst-case repair set				
HOW IT GETS THERE			6 STEPS · 2 LANES	
w16	w18	w20	w22	
MAIN SEQUENCE				
1 2 3 6				
IN PARALLEL				
4 5				
#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE			TAT
1	Redundancy scheme selection — row, column, or both → Redundancy scheme selection record			1w
2	Spare allocation against the projected defect distribution → Redundancy scheme with spare allocation			1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Repair interface definition with the BISR architecture → Repair interface definition	1.25w
4	Fuse and repair storage allocation → Fuse and repair storage allocation	1w
5	Repair algorithm agreement with DFT → Repair interface specification, agreed with DFT · Repair algorithm agreement	1w
6	Redundancy verification and yield contribution analysis → Redundancy yield contribution analysis	2.25w

Steps 3 and 5 are joint work with `DFT-03` rather than handoffs. Both sides can specify a repair scheme that is internally consistent and mutually incompatible, and only writing them together prevents it.

WATCH OUT FOR

- ◆ **Repair interface specified independently by both sides.** Two consistent specifications that disagree with each other, discovered at ATPG when both are frozen.
- ◆ **Spare allocation not matched to the defect distribution.** Row spares against a column-clustered defect mode repair nothing.
- ◆ **Fuse capacity insufficient.** Repair data has to be stored; an allocation made without counting the worst-case repair set produces parts that cannot be repaired.
- ◆ **Redundancy credited twice.** Yield projections that apply repair in both the array analysis and the product model double-count the benefit.
- ◆ **Repair algorithm assumed rather than agreed.** The BIST engine implements one algorithm; the array assumes another, and the mismatch is silent until silicon.

WHERE THE EFFORT GOES	6 M/M	WHO IS ON IT ~4.3 FTE	CONNECTIONS	TERMS HERE	5
Redundancy scheme and spare allocation	2.0	Memory circuit designer owns it	DEPENDS ON TECH-07 AMS-05 DFT-03	SRAM DFT ATPG BIST	
Repair interface with DFT	1.5	DFT engineer	FEEDS INTO AMS-08 AMS-10 DFT-10	BISR	
Yield contribution analysis	1.5	Yield engineer	TEST-05 MP-03		
Fuse allocation and algorithm agreement	1.0	Fuse and lifecycle engineer	FEEDS BACK INTO DFT-03		
		Memory architect			

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-10

Custom memory view generation and compiler-equivalent characterization across corners

TAKES	COSTS	OWNER	CRITICAL PATH
10w w28–w38 of the stage	12.0 M/M ~5.2 people while it runs	Memory characterization engineer Signoff methodology engineer approves	yes slipping moves the program

WHY IT EXISTS

Generate the **views the custom array needs to behave like a compiler instance**—timing, power, physical, functional—characterized across the same corners, so nothing downstream has to treat it as a special case.

A custom array that cannot be dropped into the flow like a compiler macro has not finished. Synthesis, place-and-route, timing signoff and verification all expect a standard view set; anything missing becomes a manual exception that has to be remembered at every iteration by everyone.

WHAT IT DELIVERS

AMS-D4

Custom memory views characterized to compiler equivalence

This activity is the deliverable, and equivalence is the standard it is measured against.

NEEDS FIRST

- Layout and extraction from AMS-07
- Vmin and assist results from AMS-06
- Statistical results from AMS-08
- Signoff corner definition from PDK-11
- Tool view requirements from PDK-08

DONE WHEN

- Characterized at every corner signoff will run
- Equivalence proven by substitution, not by inspection
- Views released only after the layout freeze

HOW IT GETS THERE

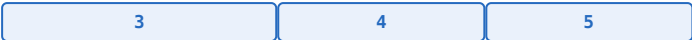
7 STEPS · 2 LANES

w28	w30	w32	w34	w36	w38
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	View requirement definition against the tool matrix → View requirement definition against the tool matrix	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Timing characterization across the signoff corners → Timing views across the signoff corners	3w
3	Power and leakage characterization → Power and leakage views	2w
4	Physical abstract, blockage and pin view generation → Physical abstract and blockage views	1.5w
5	Functional and behavioural model generation → Functional and behavioural models	1.5w
6	Compiler-equivalence check against a reference instance → Compiler-equivalence check results	2w
7	View release and integration verification → Released view set	4w

Step 6 is the test that matters. Dropping the custom instance into the flow in place of a compiler instance and confirming the tools behave identically is what proves the view set is complete—reading the file list does not.

WATCH OUT FOR

- ◆ **Characterized at fewer corners than signoff runs.** A missing corner turns into an extrapolation at S0-02, which the foundry will not accept.
- ◆ **Equivalence assumed from the file list.** A complete-looking view set can still behave differently in the tools, and only a substitution test finds it.
- ◆ **Behavioural model diverging from the circuit.** Verification passes against a model that does not match the array, and the mismatch reaches silicon.
- ◆ **Abstract blockages wrong.** Over-blocking wastes routing resource; under-blocking produces shorts found at S0-02.
- ◆ **Views released before the layout freezes.** Characterizing against a layout that then changes means doing the whole activity twice.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT	~5.2 FTE	CONNECTIONS	TERMS HERE	3
Timing characterization	4.0	Memory characterization engineer OWNS IT		DEPENDS ON PDK-11AMS-05AMS-06AMS-07AMS-08AMS-09	VminGDSCAD	
Power and leakage characterization	2.5					
Equivalence checking	2.0	CAD methodology engineer		FEEDS INTO AMS-15DV-11SYN-02PD-02S0-02		
Physical and functional views	2.0	Memory circuit designer				
Release and integration verification	1.5	Physical design liaison				
		Signoff methodology engineer				

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-11

Schematic design and pre-layout simulation

TAKES	COSTS	OWNER	CRITICAL PATH
16w w6–w22 of the stage	20.0 M/M ~5.4 people while it runs	Analog design leads PDK owner approves	no has float

WHY IT EXISTS

Run the **schematic-level design and pre-layout simulation** that every analog block passes through—the common engineering discipline underneath the PLL, the PHY, the regulators and the array periphery.

This is the shared substrate rather than a block: topology exploration, device sizing, pre-layout corner simulation and the design reviews that gate a block into layout. Separating it makes the schedule honest—a block that has not passed pre-layout review is not ready for the sixteen weeks of layout that follow.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Block specifications from AMS-01
- Node device models from the tracked PDK
- Design environment and simulator versions from PDK-08
- Reference designs from previous programs
- Reliability rules and limits from the DRM

DONE WHEN

- Every block passes a pre-layout gate before entering layout
- Corner simulation includes mismatch, not only process corners
- Regression suite exists per block and runs automatically

HOW IT GETS THERE

7 STEPS · 2 LANES

w6	w10	w14	w18	w22
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MAIN SEQUENCE

1	2	3	4	7
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IN PARALLEL

5	6
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#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Design environment and testbench framework setup → Design environment and testbench framework	1.5w
2	Topology exploration per block → Topology selection per block	3w
3	Device sizing and operating point definition → Sized schematics with operating points	3.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Pre-layout corner simulation → Pre-layout corner simulation results	3w
5	Pre-layout Monte Carlo and mismatch analysis → Pre-layout corner and Monte Carlo results	2.5w runs in parallel
6	Testbench and regression suite per block → Per-block regression suites	2.5w runs in parallel
7	Pre-layout design review and layout entry gate → Layout entry gate records	5w

Step 7 is a gate rather than a meeting. Layout is the longest and least reversible part of an analog block, and letting a schematic that has not closed at the corners enter layout guarantees that the corner failure is found after the layout is built.

WATCH OUT FOR

- ◆ **Blocks entering layout without passing the gate.** Sixteen weeks of layout on a schematic that fails at the corner is the most expensive way to discover a topology problem.
- ◆ **Corner simulation run at typical mismatch.** Mismatch dominates analog corner behavior, and a corner run without it is a nominal run in disguise.
- ◆ **Testbenches written per block by different people.** Inconsistent testbenches make results incomparable across blocks and unusable as a regression.
- ◆ **Device models used from an unfrozen PDK.** Analog is far more model-sensitive than digital, and a model update mid-design can invalidate the sizing.
- ◆ **No regression suite.** Every subsequent change is then verified by hand, and the coverage of that verification degrades under schedule pressure.

WHERE THE EFFORT GOES	20 M/M	WHO IS ON IT	~5.4 FTE	CONNECTIONS
Device sizing and operating points	5.5	Analog design leads OWNS IT		DEPENDS ON PDK-01 PDK-08 AMS-01 AMS-02
Topology exploration	4.5	AMS methodology engineer		FEEDS INTO AMS-12 AMS-13 AMS-14 AMS-16
Pre-layout corner simulation	4.0	Design reviewers		
Testbench and regression suites	3.0	AMS architect		
Monte Carlo and mismatch	2.0	PDK owner		TERMS HERE 6 PDK DRM IP PLL PHY AMS
Design reviews	1.0			

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-12

Custom layout with DRC / LVS / antenna and fill closure

TAKES	COSTS	OWNER	CRITICAL PATH
18w w20–w38 of the stage	26.0 M/M ~6.3 people while it runs	Analog layout lead Physical design liaison approves	yes slipping moves the program

WHY IT EXISTS

Lay out every analog and custom block to **DRC, LVS and antenna clean**, with the matching, shielding and fill discipline analog requires—and get the layouts extractable.

Eighteen weeks is the largest single block of time in the stage after the PHY, and it is largely irreducible. Analog layout is where matching, isolation and parasitic control are actually implemented, and none of it can be automated or parallelized the way digital place-and-route can.

WHAT IT DELIVERS

AMS-D6

Per-macro DRC / LVS clean signoff, pushed rules approved by foundry

Every analog macro's physical signoff, alongside the pushed-rule approvals from AMS-07.

NEEDS FIRST

- Sized schematics from AMS-11
- PLL, PHY and regulator designs from AMS-02 to AMS-04
- Design rules and exceptions from PDK-02
- Macro placement intent from ARCH-08 and PD-01
- Golden environment and decks from PDK-13

DONE WHEN

- Every macro DRC, LVS and antenna clean
- Fill applied without disturbing matching structures
- Pin placement agreed with physical design before freeze

HOW IT GETS THERE

8 STEPS · 2 LANES

w20	w24	w28	w32	w36	w38
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Floorplan and pin placement per macro → Macro floorplans and pin placement	2w
2	Matching-critical device layout → Matching-critical device layouts	4w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Shielding, guarding and isolation structures → Shielding and isolation structures	3w
4	Supply and bias routing with EM compliance → EM-compliant supply and bias routing	3w
	runs in parallel	
5	Fill and density compliance without disturbing matching → Fill and density compliance record	2.5w
	runs in parallel	
6	DRC closure per macro → Per-macro DRC closure	4w
7	LVS and antenna closure → DRC, LVS and antenna clean layouts	3w
8	Extraction setup and layout freeze → Extraction-ready frozen layouts	2w

Step 5 is where careful layout is most often undone. Fill applied without regard to matching structures changes the parasitics the whole design was tuned against, and the effect is only seen in post-layout simulation.

WATCH OUT FOR

- ◆ **Fill applied late and without care.** It disturbs the parasitics the design was tuned against, and the damage appears only in post-layout results.
- ◆ **Matching implemented by intent rather than by construction.** Common-centroid and dummy structures are a layout discipline; describing them in a review is not implementing them.
- ◆ **EM checked only at signoff.** Analog supply routing carries high current density in small geometries, and an EM failure found at S0-03 is a layout rebuild.
- ◆ **Pin placement not agreed with physical design.** A macro whose pins face the wrong way costs routing resource at the top level for the life of the design.
- ◆ **Layout frozen without an extraction check.** A macro that extracts slowly or badly stalls every post-layout simulation that follows.

WHERE THE EFFORT GOES	26 M/M	WHO IS ON IT	~6.3 FTE	CONNECTIONS	TERMS HERE	6
Matching-critical layout	7.0	Analog layout lead OWNS IT		DEPENDS ON PDK-02 PDK-13 AMS-02 AMS-03 AMS-04 AMS-07 AMS-11	PLL PHY DRC LVS EM GDS	
DRC closure	5.0	Layout engineers				
Shielding and isolation	4.5	Analog designers				
LVS and antenna closure	4.0	Physical verification engineer		FEEDS INTO AMS-13 AMS-14 AMS-15 PD-02 PD-15 S0-02		
Supply and bias routing	3.5					
Fill and extraction setup	2.0	Physical design liaison				

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-13

Post-layout extracted simulation across PVT and Monte Carlo

TAKES	COSTS	OWNER	CRITICAL PATH
16w w22–w38 of the stage	20.0 M/M ~5.4 people while it runs	Analog design leads AMS lead approves	yes slipping moves the program

WHY IT EXISTS

Simulate every block **with its layout parasitics extracted**, across process, voltage and temperature and across the Monte Carlo distribution—and close whatever the layout took away.

Post-layout is where analog designs are actually judged. Parasitics reduce bandwidth, add mismatch, degrade PSRR and move every margin in the wrong direction. The gap between pre- and post-layout is the real measure of layout quality, and closing it is what the sixteen weeks are for.

WHAT IT DELIVERS

AMS-D2

Characterization reports across PVT and Monte Carlo

This activity is the deliverable—the numbers every downstream consumer of these macros relies on.

NEEDS FIRST

- Frozen layouts and extraction from AMS-12
- Pre-layout baselines from AMS-11
- Signoff corner definition from PDK-11
- Extraction decks from PDK-10
- Block specifications and budgets from AMS-01

DONE WHEN

- Every block closed post-layout at every signoff corner
- Degradation from pre-layout quantified per block
- Characterization data complete enough to release the macros

HOW IT GETS THERE

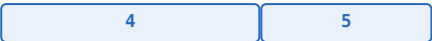
7 STEPS · 2 LANES

w22	w26	w30	w34	w38
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Extraction flow setup and parasitic model selection → Extraction flow and parasitic model selection	1.5w
2	Post-layout functional simulation per block → Post-layout functional and corner results per block	3w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	PVT corner closure post-layout → Post-layout PVT corner closure results	4w
4	Monte Carlo and mismatch analysis post-layout → Monte Carlo and mismatch results	3w
5	Pre- to post-layout degradation analysis → Degradation analysis pre- to post-layout	2w
6	Layout iteration requests and re-simulation → Layout iteration requests	4.5w
7	Characterization data assembly → Assembled characterization data	3w

Step 6 is planned iteration rather than contingency. Analog blocks essentially always need at least one layout turn after post-layout simulation, and a schedule with no allowance for it is a schedule that will slip on the first block.

WATCH OUT FOR

- ◆ **No layout iteration in the plan.** At least one turn per block is normal; a schedule with none slips on the first block and never recovers.
- ◆ **Corner closure on a subset of blocks.** The block skipped for time is the one that fails in silicon, and it is usually skipped because it looked simple.
- ◆ **Parasitic model too coarse.** A model that misses coupling produces post-layout results that are optimistic in exactly the cases that matter.
- ◆ **Degradation not analyzed.** Knowing a block passed post-layout is less useful than knowing how much margin the layout consumed, which is what predicts the next block.
- ◆ **Monte Carlo run at reduced sample count.** Cutting samples to save runtime narrows the distribution and flatters the tail.

WHERE THE EFFORT GOES	20 M/M	WHO IS ON IT	~5.4 FTE	CONNECTIONS	TERMS HERE	4
PVT corner closure	5.5	Analog design leads owns it		DEPENDS ON PDK-10 PDK-11 AMS-02 AMS-03 AMS-04 AMS-11 AMS-12 TC-07	IP AMS PVT PSRR	
Layout iteration and re-simulation	4.5	Extraction engineer		FEEDS INTO AMS-14 AMS-15 AMS-16 S0-02 BU-07		
Post-layout functional simulation	3.5	Characterization engineer		LATER INPUT TC-08 +0w		
Monte Carlo and mismatch	3.0	Analog layout lead				
Characterization assembly	2.5	AMS lead				
Extraction setup	1.0					

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-14

Reliability checks — EM/IR, aging, ESD, latch-up

TAKES	COSTS	OWNER	CRITICAL PATH
10w w28–w38 of the stage	10.0 M/M ~4.3 people while it runs	Reliability engineer Quality liaison approves	no has float

WHY IT EXISTS

Check every analog macro for the failure modes that **appear over time rather than at test**—electromigration, IR, aging, ESD and latch-up—because none of them show up in a functional simulation.

Analog blocks carry high current densities in small geometries and run continuously for the product's life. A design that passes every corner and fails EM has passed everything the design flow checks and none of what qualification will.

WHAT IT DELIVERS

AMS-D7

Reliability report — EM/IR, ESD, latch-up, aging

This activity is the deliverable, and it is what qualification will be checked against.

NEEDS FIRST

- Post-layout results and parasitics from AMS-13
- Reliability rules from PDK-02 and the DRM
- ESD strategy from PDK-07
- Mission profile and lifetime requirements from the qualification plan
- Assist boost levels from AMS-06

DONE WHEN

- Aging simulated against the product's real mission profile
- EM checked on signal routing as well as supplies
- Every waiver carries a quantified lifetime impact

HOW IT GETS THERE

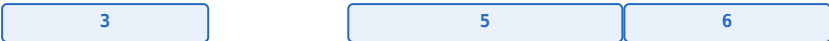
7 STEPS · 2 LANES

w28	w30	w32	w34	w36	w38
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Reliability requirement definition per macro → Reliability requirements per macro	1w
2	Electromigration analysis on supply and signal routing → EM analysis results on supply and signal routing	2.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	IR drop analysis within macros → In-macro IR drop results	1.5w runs in parallel
4	Aging and lifetime simulation at the rated conditions → Aging and lifetime simulation results	2.5w
5	ESD path verification through the macro → ESD path verification and latch-up review	2w runs in parallel
6	Latch-up and guard structure review → Latch-up and guard structure review findings	1.5w runs in parallel
7	Reliability report and waiver disposition → Reliability report with waiver dispositions	4w

Step 4 is the one that needs the product's actual mission profile. Aging simulated at nominal conditions for a nominal lifetime says nothing about a part that runs at the top of its DVFS range continuously, which is exactly how an accelerator is used.

WATCH OUT FOR

- ◆ **Aging simulated at nominal conditions.** An accelerator runs near its limit continuously; a nominal mission profile understates the stress by a wide margin.
- ◆ **EM checked on supplies only.** High-speed signal routing in a PHY carries current densities that matter, and signal EM is routinely omitted.
- ◆ **ESD path assumed from the strategy.** The strategy defines the intent; only tracing the path through the actual layout confirms it exists.
- ◆ **Assist boost stress not fed back.** The boost levels AMS-06 chose for Vmin have an aging cost, and the two activities have to reconcile.
- ◆ **Waivers granted without a lifetime calculation.** A reliability waiver without a quantified life impact is a decision nobody can defend at qualification.

WHERE THE EFFORT GOES	10 M/M	WHO IS ON IT ~4.3 FTE	CONNECTIONS	TERMS HERE 8
Aging and lifetime simulation	2.5	Reliability engineer owns it	DEPENDS ON PDK-07 AMS-04 AMS-06 AMS-08 AMS-11 AMS-12 AMS-13	DRM PHY EM IR DVFS EM/IR Vmin ESD
Electromigration analysis	2.5	Analog designers		
ESD path verification	2.0	ESD engineer	FEEDS INTO AMS-15 S0-07 MP-02 MP-03	
IR and latch-up review	1.5	Layout engineers		
Report and waiver disposition	1.5	Quality liaison		

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-15

Hard macro abstraction and view generation (LEF, .lib, CDL, GDS, behavioural model)

TAKES	COSTS	OWNER	CRITICAL PATH
8w w32–w40 of the stage	8.0 M/M ~4.3 people while it runs	AMS integration lead Verification liaison approves	yes slipping moves the program

WHY IT EXISTS

Wrap every finished block into a **hard macro physical design can place and verification can simulate**—GDS, abstract, timing views, netlist, power intent and a behavioural model.

This is the handoff the whole stage exists to make. The macro set has to arrive before PD-02 places it, and the abstracts specifically have to arrive well before the models do, because floorplan needs the shape long before signoff needs the timing.

WHAT IT DELIVERS

AMS-D5

Hard macro GDS with abstract views (LEF, .lib, CDL, UPF, wreal/Verilog model)

This activity is the deliverable, and it is the stage's milestone—AMS Macro Handoff.

NEEDS FIRST

- Frozen layouts from AMS-12
- Characterization data from AMS-13
- Custom memory views from AMS-10
- Reliability constraints from AMS-14
- View and tool requirements from PDK-08 and PD-01

DONE WHEN

- Abstracts released ahead of models, on the schedule PD needs
- Trial placement run and clean before general release
- Every macro version-controlled and its views mutually consistent

HOW IT GETS THERE

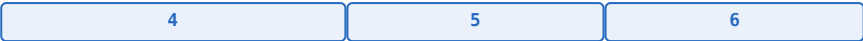
8 STEPS · 2 LANES

w32	w34	w36	w38	w40
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	View requirement confirmation with physical design and verification → Confirmed view requirement list	1w
2	GDS finalization and macro boundary definition → Finalized GDS per macro	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Abstract generation — blockages, pins, routing layers → Physical abstracts with blockages and pins	1.5w
4	Timing view generation from characterization data → Timing and power views	2w <i>runs in parallel</i>
5	CDL netlist and UPF power intent generation → CDL netlists and UPF power intent	1.5w <i>runs in parallel</i>
6	Behavioural and wreal model release → Behavioural and wreal models	1.5w <i>runs in parallel</i>
7	Integration verification on a trial placement → Trial placement integration verification results	2w
8	Macro release and version control → Released, version-controlled macro set	2w

Step 7 is worth the two weeks. Placing every macro in a trial floorplan and running the flow over it finds abstract errors, pin conflicts and missing views before the whole physical design team is depending on them.

WATCH OUT FOR

- ◆ **Abstracts delivered with the models.** Floorplan needs the shape months before signoff needs the timing; a single delivery date serves the later need and misses the earlier one.
- ◆ **Trial placement skipped.** Abstract and pin errors are then found by the physical design team, at the point where the whole floorplan depends on them.
- ◆ **Views inconsistent with the characterization.** A timing view generated from stale data produces a design closed against numbers the silicon does not have.
- ◆ **Macro released without version control.** Two teams on different macro revisions produce a mismatch that is diagnosed as a tool problem.
- ◆ **Power intent omitted.** A macro with no UPF view breaks low-power verification and forces a manual exception at every run.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT	~4.3 FTE	CONNECTIONS	TERMS HERE	5
Timing and power view generation	2.5	AMS integration lead <i>OWNS IT</i>		DEPENDS ON AMS-02 AMS-03 AMS-04 AMS-07 AMS-10 AMS-12 AMS-13 AMS-14	AMS UPF DV GDS CDL	
Abstract generation	1.5	Characterization engineer				
Trial placement verification	1.5	Layout engineer		FEEDS INTO AMS-16 DV-07 SYN-02 PD-01 PD-02 S0-02 T0-01		
GDS finalization	1.0	Physical design liaison				
Netlist, UPF and models	1.0	Verification liaison		FEEDS BACK INTO RTL-03 SYN-03		
Release and version control	0.5					

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-16

AMS—digital co-simulation and integration support

TAKES	COSTS	OWNER	CRITICAL PATH
16w cont. w26–w42 of the stage	12.0 M/M ~3.3 people while it runs	AMS verification engineer Validation liaison approves	no has float

WHY IT EXISTS

Keep the analog and digital halves talking—**co-simulation, integration support and the errata that come out of it**—through the whole period when the macros are being integrated.

Analog blocks fail at their boundaries: a reset that arrives before a rail is stable, a calibration sequence nobody scheduled, a control interface each side interpreted differently. Co-simulation is where those are found, and the integration guide is where they stop being rediscovered.

WHAT IT DELIVERS

AMS-D8

Integration guide with known limitations and errata

This activity is the deliverable, and it is what stops the same integration question being asked four times.

NEEDS FIRST

- Behavioural models from AMS-15
- Circuit simulation results from AMS-13
- RTL integration environment from RTL-03
- Verification methodology from DV-07
- Startup and sequencing requirements from ARCH-05

DONE WHEN

- Every behavioural model correlated against circuit simulation
- Boundary and sequencing verified, not assumed
- Integration guide published and maintained, not held in email

HOW IT GETS THERE

7 STEPS · 2 LANES

w26	w30	w34	w38	w42
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	AMS co-simulation environment setup → AMS co-simulation environment	2w
2	Behavioural model correlation against circuit simulation → Model correlation results	2.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Boundary and control interface co-verification → Boundary and control interface verification results	3w
4	Startup, calibration and sequencing co-simulation → Startup and calibration co-simulation results	2.5w
5	Integration support to RTL and physical design → Integration support record for RTL and physical design	5w
6	Errata capture and integration guide authoring → Integration guide with known limitations	3w
7	Bring-up support material and debug hooks → Errata list and bring-up debug material	5.5w

Step 5 runs across the whole window and is the reason this activity is marked continuous. Integration questions do not arrive in a block; they arrive one at a time for four months, and the engineer who can answer them has to still be assigned.

WATCH OUT FOR

- ◆ **Behavioural models never correlated.** Digital verification then passes against a model that does not match the circuit, and the mismatch reaches silicon undetected.
- ◆ **Boundary conditions unverified.** Reset ordering, rail sequencing and enable timing are where analog blocks actually fail, and none of them are internal to the block.
- ◆ **Support engineer unassigned.** Integration questions arrive for months; without a named engineer they are answered by guesswork or not at all.
- ◆ **Errata captured in email.** A limitation known to one integrator and nobody else is rediscovered by every other integrator in turn.
- ◆ **Calibration sequence not co-simulated.** A calibration that works in the circuit and is never exercised from the digital side is first tried at bring-up.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT	~3.3 FTE	CONNECTIONS	TERMS HERE	3
Integration support	4.0	AMS verification engineer	OWNS IT	DEPENDS ON AMS-11 AMS-13 AMS-15 RTL-03	IP AMS RTL	
Boundary and interface co-verification	2.5	AMS integration engineer		FEEDS INTO DV-07 PD-02 BU-03		
Model correlation	2.0	Analog designers		BU-05		
Integration guide and errata	1.5	RTL integration lead		FEEDS BACK INTO RTL-03		
Bring-up support material	1.0	Validation liaison				
Environment setup	1.0					

TECHNOLOGY & IP ENABLEMENT

07

Test Chip / MPW Shuttle

TC

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w22 – w62	40 _w	8	83 _{M/M}	6 _{of 8}

ENGINEERING ACTIVITIES				KEY DELIVERABLES	
TC-01	Test chip objective definition and risk-item selection		5w	4 M/M	TC-D1 Test chip specification and risk coverage matrix
TC-02	Test chip design and integration (IP under test, ring oscillators, PCM structures)		12w	18 M/M	TC-D2 Test chip GDS and shuttle submission record
TC-03	MPW shuttle slot booking and data submission scheduling		4w	1 M/M	TC-D3 Test chip silicon and characterization report
TC-04	Test chip physical implementation and signoff		12w	20 M/M	TC-D4 Silicon-to-model correlation report
TC-05	Shuttle tapeout and fabrication		16w	4 M/M	TC-D5 Design guidance and margin decisions for the production chip
TC-06	Test chip board and lab setup preparation		9w	8 M/M	
TC-07	Silicon characterization and correlation to simulation		12w	20 M/M	
TC-08	Feedback into the production design — model correction and margin decisions		6w	8 M/M	

07 TC · TEST CHIP / MPW SHUTTLE

TC-01 Test chip objective definition and risk-item selection			
TAKES	COSTS	OWNER	CRITICAL PATH
5w	4.0 M/M	Test chip lead	yes
w0–w5 of the stage	~3.5 people while it runs	Device and technology engineer approves	slipping moves the program

WHY IT EXISTS

Decide what the test chip is **for**—which specific risks it retires—and select the structures that will actually answer those questions rather than the ones that are easy to include.

A test chip with no objective becomes a collection of everything anyone wanted to try, and it answers nothing in time to matter. The discipline is subtractive: name the two or three uncertainties that would change a production decision, and carry only what measures them.

WHAT IT DELIVERS

TC-D1

Test chip specification and risk coverage matrix

This activity is the deliverable, and the matrix is what every later result is reported against.

NEEDS FIRST

- Node risk assessment from TECH-07
- IP readiness gaps from IPR-05
- Custom memory decision from PDK-06
- AMS block risk list from AMS-01
- Production design schedule from DEF-08

DONE WHEN

- Every carried structure traces to a named risk and a decision rule
- Schedule feasibility confirmed against the decision dates
- Area and cost budget fixed before structures are finalized

HOW IT GETS THERE

6 STEPS · 2 LANES

w0	w1	w2	w3	w4	w5
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Objective definition — which production decisions this chip informs → Test chip objective statement	1w
2	Risk item selection and prioritization → Prioritized risk item list	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Schedule feasibility — does the answer arrive before the decision → Schedule feasibility finding	1w runs in parallel
4	Structure selection per risk item → Structure selection per risk	1w
5	Cost and area budget for the shuttle die → Shuttle die cost and area budget	0.75w runs in parallel
6	Specification and risk coverage matrix → Test chip specification and risk coverage matrix	1.5w

Step 3 can cancel the activity, and should be allowed to. A test chip whose results land after the production design has committed is an expense with no decision attached, and finding that out in week three is far better than in week forty.

WATCH OUT FOR

- ◆ **Scope grows until the shuttle is missed.** Every team has something worth measuring, and a chip that carries all of them misses the slot that made it useful.
- ◆ **Results arrive after the decision.** A test chip is only worth its cost if the answer lands while the production design can still act on it.
- ◆ **Structures chosen for measurability rather than for risk.** Ring oscillators are easy and tell you about the node; the risk is usually in a specific circuit nobody has built before.
- ◆ **No decision rule attached to a result.** Measuring something without agreeing in advance what each outcome would change produces data and no decisions.
- ◆ **Area budget set after structure selection.** Shuttle die area is expensive and fixed; selecting first and budgeting second guarantees a painful subtraction later.

WHERE THE EFFORT GOES	4 M/M	WHO IS ON IT	~3.5 FTE	CONNECTIONS	TERMS HERE	3
Risk item selection	1.3	Test chip lead owns IT		DEPENDS ON TECH-07 IPR-05 AMS-01	PCM IP AMS	
Structure selection	1.0	Chief architect		FEEDS INTO TC-02 TC-03 TC-04		
Specification and coverage matrix	1.0	AMS lead		TC-06 TC-07		
Objective definition	0.8	Program manager		LATER INPUT PDK-06 +0w		
		Device and technology engineer				

07 TC · TEST CHIP / MPW SHUTTLE

TC-02

Test chip design and integration (IP under test, ring oscillators, PCM structures)

TAKES 12w w2–w14 of the stage	COSTS 18.0 M/M ~6.5 people while it runs	OWNER Test chip design lead Layout lead approves	CRITICAL PATH yes slipping moves the program
WHY IT EXISTS Design and integrate the vehicle itself—the IP under test, its access wrappers, the process monitors and the measurement structures—as a small chip that has to work first time. A test chip gets one shuttle. There is no bring-up iteration and no respin, so the access paths, the monitors and the pad ring have to be right on submission. That makes it a small design with an unusually high verification burden relative to its size.			
WHAT IT DELIVERS No key deliverable is owned here — this activity contributes to the ones below.			
NEEDS FIRST - Test chip specification and coverage matrix from TC-01 - IP-under-test deliverables from AMS and IPR - Process monitor structure definitions from the foundry - Shuttle die area and pad count limits from TC-03 - PDK and libraries from PDK-13			
DONE WHEN - Every structure reachable through a verified access path - Monitors distributed to see across-die variation - Pad budget met without unverified multiplexing			
HOW IT GETS THERE 8 STEPS · 2 LANES w2 w5 w8 w11 w14 MAIN SEQUENCE 1 2 4 6 8 IN PARALLEL 3 5 7			
# STEP, AND WHAT IT ADDS TO THE DELIVERABLE TAT			
1 IP-under-test integration plan → IP-under-test integration plan 1w			
2 Device-under-test wrapper and access path design → IP-under-test integration with access wrappers 3w			

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Ring oscillator and process monitor array → Ring oscillator and process monitor arrays	2w runs in parallel
4	Measurement and PCM structure design → Measurement and PCM structures	3w
5	Test structure library assembly → Assembled test structure library	2w runs in parallel
6	Top-level integration and pad assignment → Top-level netlist with pad assignment	2.5w
7	Pad ring and ESD for the shuttle die → Pad ring and ESD design for the shuttle	1.5w runs in parallel
8	Verification of access paths and measurement modes → Access path and measurement mode verification results	2.5w

Step 8 gets two and a half weeks for a chip that is mostly structures, because an access path that does not work makes the structure behind it unmeasurable. Every mode has to be exercised in simulation, since there is no second chance in silicon.

WATCH OUT FOR

- ◆ **Access paths unverified.** A structure that cannot be reached is a structure that was not measured, and the failure is only discovered in the lab.
- ◆ **Pad count exceeded.** Shuttle die have few pads; a measurement scheme needing more forces multiplexing that has to be designed and verified.
- ◆ **IP under test modified for the vehicle.** A block wrapped or altered to fit the test chip is no longer the block the production design will use.
- ◆ **Monitors placed without regard to position.** Process variation across the die is part of what is being measured, and monitors clustered in one corner cannot see it.
- ◆ **No redundancy in critical structures.** A single instance of the structure that matters most means one defect loses the answer entirely.

WHERE THE EFFORT GOES	18 M/M	WHO IS ON IT ~6.5 FTE	CONNECTIONS
DUT wrappers and access paths	4.5	Test chip design lead owns IT	DEPENDS ON PDK-13 IPR-05 TC-01 TC-03
Measurement and PCM structures	4.0	Analog designers	FEEDS INTO TC-04 TC-06 TC-07
Top-level integration	3.5	Device engineer	TERMS HERE 7
Verification	3.0	Verification engineer	PDK PCM IP AMS
Ring oscillator arrays	2.0	Layout lead	DUT ESD GDS
Pad ring and ESD	1.0		

07 TC · TEST CHIP / MPW SHUTTLE

TC-03

MPW shuttle slot booking and data submission scheduling

TAKES	COSTS	OWNER	CRITICAL PATH
4w w5–w9 of the stage	1.0 M/M ~1.1 people while it runs	Program manager Physical design lead approves	no has float

WHY IT EXISTS

Book the shuttle slot and **work backwards from its submission date**—because the shuttle calendar is fixed, and missing a slot costs a quarter rather than a week.

One man-month across four weeks: almost all of this is scheduling and paperwork. Its importance is out of proportion to its effort, because the submission date it fixes is what every other test chip activity is planned against.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Test chip objective and schedule feasibility from TC-01
- Shuttle calendars from the foundry or broker
- Capacity and pricing terms from TECH-05
- Production decision dates from DEF-08
- Submission format requirements

DONE WHEN

- Slot confirmed in writing with a submission date
- Submission requirements collected and understood
- Backward schedule agreed with physical design

HOW IT GETS THERE

5 STEPS · 2 LANES

w5	w6	w7	w8	w9
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Shuttle calendar review and slot options → Shuttle slot options and pricing	1w
2	Foundry or broker engagement and pricing → Foundry or broker engagement with pricing	0.75w <i>runs in parallel</i>
3	Slot booking against the production decision dates → Booked slot with confirmation	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Data format and submission requirement collection → Data format and submission requirements	0.75w runs in parallel
5	Submission schedule and checklist working backwards from the slot → Backward-planned submission schedule · Submission checklist	2w

Step 5 is the useful output. A shuttle slot is a date on which a compliant database must exist; scheduling backwards from it is what tells `TC-04` when physical verification actually has to finish.

WATCH OUT FOR

- ◆ **Slot booked without checking the production decision date.** An available slot that returns silicon too late buys nothing.
- ◆ **Submission requirements collected late.** Shuttle submissions have their own format and layer rules, and discovering them near the date forces rushed rework.
- ◆ **Backward schedule not enforced.** A submission date that is not translated into a physical verification deadline gets missed by exactly the amount by which signoff runs long.
- ◆ **No contingency slot identified.** A missed shuttle costs a quarter, and knowing the next available date in advance changes how the schedule is managed.
- ◆ **Cost approved after booking.** Shuttle die area is charged by area and layer count; booking before the budget is agreed produces an invoice nobody planned.

WHERE THE EFFORT GOES	1 M/M	WHO IS ON IT ~1.1 FTE	CONNECTIONS	TERMS HERE	1
Slot booking	0.4	Program manager owns it	DEPENDS ON TECH-04TECH-05TECH-06 TC-01	GDS	
Calendar review	0.3	Foundry relationship manager	FEEDS INTO TC-02TC-04TC-05		
Submission schedule	0.3	Test chip lead Procurement Physical design lead			

07 TC · TEST CHIP / MPW SHUTTLE

TC-04

Test chip physical implementation and signoff

TAKES	COSTS	OWNER	CRITICAL PATH
12w w12–w24 of the stage	20.0 M/M ~7.2 people while it runs	Test chip physical lead Program manager approves	yes slipping moves the program

WHY IT EXISTS

Implement the test chip physically and **sign it off to shuttle submission standard**—the same DRC, LVS and antenna closure a production die needs, on a fraction of the schedule.

The vehicle is small and the standard is not reduced. A shuttle rejects a non-compliant database, and the rejection costs the slot. What compresses the schedule is that timing closure is easy here; what does not compress is physical verification, which has to be as clean as production.

WHAT IT DELIVERS

TC-D2

Test chip GDS and shuttle submission record

This activity is the deliverable, and the submission record is what proves the slot was met.

NEEDS FIRST

- Test chip netlist and structures from TC-02
- Submission requirements and date from TC-03
- PDK, libraries and decks from PDK-13
- Shuttle frame and seal ring rules from the foundry
- Measurement accuracy requirements from TC-01

DONE WHEN

- Database clean against production physical verification standards
- Shuttle frame and format compliance confirmed before submission
- Submission made on the booked date

HOW IT GETS THERE

8 STEPS · 2 LANES

w12w15w18w21w24

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Floorplan and structure placement → Floorplan and routed database	1.5w
2	Place and route → Placed and routed shuttle database	2.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Power delivery network for measurement accuracy → Measurement-grade power delivery network	2w runs in parallel
4	Timing closure on access and control paths → Timing closure results on access and control paths	2.5w
5	Fill, seal ring and shuttle frame compliance → Fill, seal ring and frame compliance record	1.5w runs in parallel
6	Physical verification — DRC, antenna, density → Physical verification clean reports	3w
7	LVS closure → LVS closure record	1.5w runs in parallel
8	GDS assembly and shuttle submission → Shuttle frame compliant GDS · Submission package and record	2.5w

Step 3 exists because a test chip measures small signals. A power delivery network adequate for function but noisy in the measurement bands turns a characterization result into an argument about the vehicle rather than the device.

WATCH OUT FOR

- ◆ **Submission rejected on frame or format compliance.** Shuttle frames have their own rules, and a rejection costs the slot rather than a re-run.
- ◆ **PDN adequate for function and inadequate for measurement.** The results then carry vehicle noise that cannot be separated from device behavior.
- ◆ **Physical verification compressed because the chip is small.** Size does not reduce the standard, and a DRC violation is as fatal here as on a production die.
- ◆ **Structures placed without regard to the measurement plan.** Placement affects what across-die variation can be seen, and it is decided here rather than in the lab.
- ◆ **No margin against the submission date.** Everything before this activity can slip and this one cannot, so its schedule needs the buffer.

WHERE THE EFFORT GOES	20 M/M	WHO IS ON IT	~7.2 FTE	CONNECTIONS	TERMS HERE	5
Place and route	5.0	Test chip physical lead		DEPENDS ON	PDK DRC LVS PDN	
Physical verification	4.5	Physical design engineers		PDK-13 TC-01 TC-02	GDS	
Timing closure	4.0	Physical verification engineer		TC-03		
GDS assembly and submission	2.5	Analog layout engineer		FEEDS INTO		
Power delivery network	2.5	Program manager		TC-05 TC-07		
Fill and seal ring	1.5					

07 TC · TEST CHIP / MPW SHUTTLE

TC-05

Shuttle tapeout and fabrication

TAKES	COSTS	OWNER	CRITICAL PATH
16w w22–w38 of the stage	4.0 M/M ~1.1 people while it runs	Program manager Test chip lead approves	yes slipping moves the program

WHY IT EXISTS

Wait, visibly—**mask, fabricate, dice and deliver**—while keeping the program informed about a schedule it has almost no ability to influence.

Sixteen weeks and four man-months. The work here is tracking and logistics rather than engineering, and the value is that the rest of the program knows what date to plan against. A shuttle lot moves for reasons that have nothing to do with this program, and being told late is the avoidable part.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Submitted GDS from TC-04
- Shuttle slot confirmation from TC-03
- Foundry WIP reporting
- Packaging requirements from TC-06
- Delivery logistics arrangements

DONE WHEN

- Parts delivered in the quantity the characterization plan needs
- PCM and e-test data reviewed before parts arrive
- Schedule communicated at every status change, not only at delivery

HOW IT GETS THERE

6 STEPS · 2 LANES

w22	w26	w30	w34	w38
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Data submission and acceptance confirmation → Submission acceptance confirmation	1w
2	Shuttle mask fabrication → Fabricated shuttle mask set	4w
3	WIP tracking and schedule communication → WIP status reporting against the program schedule	2w

runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Wafer processing → Processed wafers	8w
5	E-test and PCM data review as it becomes available → E-test and PCM data	1.5w runs in parallel
6	Dicing, packaging and delivery → Packaged test chip parts · Delivery record	3w

Step 5 matters more than its 1.5 weeks suggest. PCM and e-test data arrives before packaged parts do, and it answers some of the process questions the chip was built for weeks earlier than the lab can.

WATCH OUT FOR

- ◆ **Shuttle lot slips and the program hears late.** The slip cannot be prevented; the surprise can, and the surprise is what damages downstream plans.
- ◆ **Packaging not arranged in advance.** Diced die with no package and no socket cannot be measured, and arranging it after delivery adds weeks.
- ◆ **PCM data ignored until parts arrive.** Some process questions are answered by e-test alone, weeks earlier and at no additional cost.
- ◆ **Insufficient parts requested.** Shuttle quantities are small; a characterization plan needing more parts than were ordered cannot be run.
- ◆ **Logistics and customs not planned.** Cross-border shipment of engineering samples carries paperwork that stops parts at a border for weeks.

WHERE THE EFFORT GOES	4 M/M	WHO IS ON IT ~1.1 FTE	CONNECTIONS
WIP tracking and communication	1.5	Program manager OWNS IT	DEPENDS ON TC-03 TC-04
PCM and e-test review	1.0	Foundry relationship manager	FEEDS INTO TC-07
Submission and acceptance	0.8	Operations planner	TERMS HERE 3
Logistics and delivery	0.8	Device engineer	WIP PCM GDS
		Test chip lead	

07 TC · TEST CHIP / MPW SHUTTLE

TC-06

Test chip board and lab setup preparation

TAKES	COSTS	OWNER	CRITICAL PATH
9w w24–w33 of the stage	8.0 M/M ~3.9 people while it runs	Characterization engineer Analog designer approves	no has float

WHY IT EXISTS

Have the **board, sockets, instruments and automation ready before the parts arrive**, so that characterization starts on the day of delivery rather than six weeks later.

This is the small-scale version of the same mistake the production program makes with EVB. Parts arriving into a lab with no board is a delay that consumes the schedule advantage the test chip was supposed to buy, and it is entirely avoidable by starting during fabrication.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Coverage matrix and measurement plan from TC-01
- Pad assignment and access modes from TC-02
- Package and die dimensions from TC-05
- Lab instrument availability
- Measurement accuracy requirements

DONE WHEN

- Board bring-up complete before parts are delivered
- Sockets in hand, not on order
- Automation able to run the matrix without manual steps

HOW IT GETS THERE

6 STEPS · 2 LANES

w24	w26	w28	w30	w32	w33
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Measurement requirement definition from the coverage matrix → Measurement requirement specification	1w
2	Board design — supplies, bias, measurement paths → Test chip board design and assembled boards	2.5w
3	Socket or probe card selection and lead time → Socket or probe card	1.5w

runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Board fabrication and assembly → Fabricated and assembled test boards	2.5w
5	Measurement automation and data capture scripting → Measurement automation and capture scripts	1.5w runs in parallel
6	Board bring-up and instrument setup → Bring-up record and instrument setup	3w

Step 3 has a lead time that can exceed the whole activity. Sockets for an unusual package are made to order, and a socket ordered when the parts ship arrives after they do.

WATCH OUT FOR

- ◆ **Board started when parts ship.** The lab then waits, and the schedule advantage the test chip existed to buy is spent waiting for a PCB.
- ◆ **Socket lead time discovered late.** Custom sockets are made to order and can take longer than the board itself.
- ◆ **Measurement automation deferred.** Manual measurement of a full characterization matrix is slow enough to consume the window and inconsistent enough to weaken the data.
- ◆ **Board noise floor above the signal of interest.** A board adequate for function cannot measure a low-jitter clock or a small offset, and the limitation is found with the parts in hand.
- ◆ **Instruments not reserved.** A shared high-end instrument booked by another program stops characterization as effectively as a missing board.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT ~3.9 FTE	CONNECTIONS
Board design	2.5	Characterization engineer OWNS IT	DEPENDS ON TC-01 TC-02
Bring-up and instrument setup	2.0	Board designer	FEEDS INTO TC-07 EVB-01
Measurement automation	1.5	Lab manager	TERMS HERE 2
Board fabrication	1.0	Test chip lead	PCB EVB
Socket procurement	1.0	Analog designer	

07 TC · TEST CHIP / MPW SHUTTLE

TC-07

Silicon characterization and correlation to simulation

TAKES	COSTS	OWNER	CRITICAL PATH
12w w28–w40 of the stage	20.0 M/M ~7.2 people while it runs	Characterization engineer Failure analysis engineer approves	yes slipping moves the program

WHY IT EXISTS

Measure the silicon and **correlate it against the models the production design is being built on**—which is the only reason the chip exists.

Characterization without correlation produces a data set. Correlation produces a statement of the form "the model is optimistic by 8% on this parameter at this corner", and that statement is what TC-08 converts into margin decisions the production design can act on.

WHAT IT DELIVERS

- TC-D3

Test chip silicon and characterization report

The measurement half of the deliverable.
- TC-D4

Silicon-to-model correlation report

The half that makes the measurements actionable—a delta against what was predicted, per parameter and corner.

NEEDS FIRST

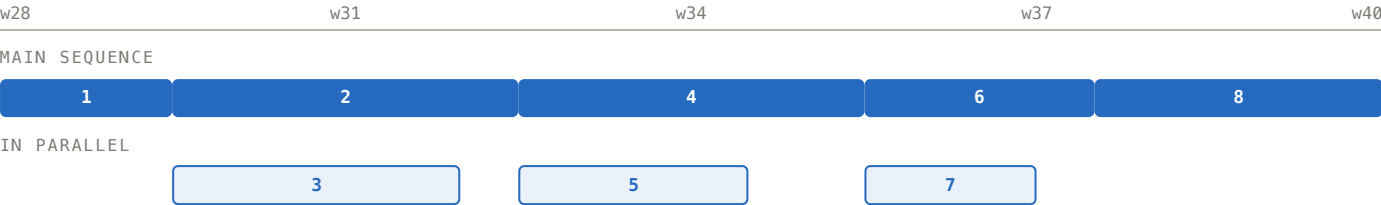
- Packaged parts and PCM data from TC-05
- Measurement setup from TC-06
- Coverage matrix from TC-01
- Simulation results the models predicted
- Foundry PCM reference data

DONE WHEN

- Every measured parameter compared against its prediction
- Measurement uncertainty stated alongside each correlation claim
- Results delivered with time for TC-08 to act

HOW IT GETS THERE

8 STEPS · 2 LANES



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Incoming inspection and basic functionality check → Incoming inspection and functionality results	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	IP-under-test characterization across the matrix → Functional and characterization results per structure	3w
3	Ring oscillator frequency and leakage measurement → Ring oscillator and leakage measurements	2.5w runs in parallel
4	PCM and device parameter extraction → Extracted device parameters	3w
5	Corner correlation setup against simulation → Corner correlation setup	2w runs in parallel
6	Across-wafer and lot-to-lot variability analysis → Variability analysis across wafer and lot	2w
7	Failure analysis on outliers → Failure analysis findings	1.5w runs in parallel
8	Correlation results assembly → Silicon-to-model correlation results	2.5w

This window overlaps the tail of fabrication because the stage as templated does not allow a clean sequence—PCM and e-test data are worked first, and device characterization follows once packaged parts arrive. It is worth noting that the stage's forty weeks are tight for its own chain.

WATCH OUT FOR

- ◆ **Characterization without correlation.** A data set with no comparison to prediction cannot change a design decision, and changing decisions was the point.
- ◆ **Sample size too small for variability.** Across-wafer and lot-to-lot variation is often the question, and a handful of parts cannot answer it.
- ◆ **Measurement uncertainty larger than the effect.** A correlation claim that sits inside the measurement error is not a finding.
- ◆ **Outliers discarded.** The parts that behave unexpectedly are usually the informative ones, and discarding them removes the tail the yield model depends on.
- ◆ **Results delivered too late for TC-08.** Correlation completed after the production design has committed is a report nobody can act on.

WHERE THE EFFORT GOES	20 M/M	WHO IS ON IT ~7.2 FTE	CONNECTIONS	TERMS HERE	1
IP-under-test characterization	6.0	Characterization engineer OWNS IT	DEPENDS ON TC-01 TC-02 TC-04 TC-05 TC-06	PCM	
PCM and device extraction	4.0	Device engineer	FEEDS INTO AMS-13 TC-08 S0-10		
Ring oscillator measurement	3.0	Analog designers	FAB-07		
Variability analysis	3.0	Modeling engineer	FEEDS BACK INTO PDK-03		
Correlation assembly	2.0	Failure analysis engineer			
Failure analysis	2.0				

07 TC · TEST CHIP / MPW SHUTTLE

TC-08

Feedback into the production design — model correction and margin decisions

TAKES	COSTS	OWNER	CRITICAL PATH
6w w34–w40 of the stage	8.0 M/M ~5.8 people while it runs	Test chip lead Program manager approves	yes slipping moves the program

WHY IT EXISTS

Convert correlation findings into **changes the production design actually makes**—corrected models, revised margins, methodology updates—and track that they were adopted.

This is where a test chip earns or wastes its cost. A correlation report that circulates and changes nothing has bought an expensive confirmation of what everyone assumed. The output has to be specific: this model corrected by this factor, this margin increased by this amount, owned by this person, adopted by this date.

WHAT IT DELIVERS

TC-05

Design guidance and margin decisions for the production chip

This activity is the deliverable, and adoption tracking is what makes it more than a report.

NEEDS FIRST

- Correlation results from TC-07
- Production design status and open margins
- Model and library data from PDK-03
- AMS post-layout results from AMS-13
- Production design change control process from RTL-10

DONE WHEN

- Every finding converted into a specific, owned action
- Margin released where silicon beat the model, not only tightened where it did not
- Adoption tracked to completion, not just published

HOW IT GETS THERE

6 STEPS · 2 LANES

w34	w36	w38	w40
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Correlation gap analysis — where model and silicon disagree → Correlation gap analysis	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Model correction and re-characterization requests → Model correction requests and re-characterization scope	1.5w
3	Design rule and methodology update proposals → Methodology and rule update proposals	1w <i>runs in parallel</i>
4	Margin decisions for the production design → Margin decisions for the production design	1.5w
5	Production design change requests → Production design change requests	1w <i>runs in parallel</i>
6	Guidance publication and adoption tracking → Adoption tracking record	2w

Step 6 is the one that distinguishes a useful test chip from an expensive one. Publishing guidance is easy; tracking that physical design actually changed its margin and that the model was actually re-characterized is what turns the finding into a change.

WATCH OUT FOR

- ◆ **Guidance published and not adopted.** The most common way a test chip wastes its cost: everyone reads the report and nobody changes a margin.
- ◆ **Findings arrive after the design has committed.** A margin decision for a block that has already closed is a decision that cannot be taken.
- ◆ **Model corrections requested but not scheduled.** Re-characterization is work; a correction with no owner and no date does not happen.
- ◆ **Only bad news acted on.** Where silicon is better than the model, the margin released is real and usually left on the table.
- ◆ **Guidance stated as observation.** "Leakage is higher than modelled" is not actionable; "increase the leakage budget for the compute cluster by 12%" is.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT	~5.8 FTE	CONNECTIONS	TERMS HERE	3
Model correction	2.5	Test chip lead <i>OWNS IT</i>		DEPENDS ON TC-07	PPA AMS ECO	
Margin decisions	2.0	Modeling and library engineers		FEEDS INTO PD-09 S0-10		
Gap analysis	1.5	PPA lead		FEEDS BACK INTO DEF-04 PDK-03 AMS-13		
Guidance and adoption tracking	1.0	Physical design lead				
Change requests	1.0	Program manager				

DESIGN & VERIFY

08

RTL Design & Integration

RTL

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w20 – w52	32 _w	10	410 _{M/M}	6 _{of 10}

ENGINEERING ACTIVITIES				KEY DELIVERABLES	
RTL-01	Block microarchitecture specification		10w 40 M/M	RTL-D1	Block and top-level RTL release, tagged
RTL-02	Block-level RTL implementation		24w 180 M/M	RTL-D2	Register map / RDL and generated headers
RTL-03	Third-party and internal IP integration — wrappers, glue, configuration		16w 45 M/M	RTL-D3	Integration testbench and build system
RTL-04	Chip-level integration and top assembly		12w 35 M/M	RTL-D4	Lint / CDC / RDC clean reports with waiver list
RTL-05	Clock, reset and power intent (UPF) implementation		10w 18 M/M	RTL-D5	UPF power intent file
RTL-06	Lint, CDC and RDC closure		16w 30 M/M	RTL-D6	IP integration report and version manifest
RTL-07	Register map / RDL definition and header generation		8w 10 M/M	RTL-D7	RTL Freeze package
RTL-08	Trial synthesis feedback loop on RTL PPA		12w 20 M/M		
RTL-09	CI build, nightly regression and release management		30w cont. 20 M/M		
RTL-10	Specification change control and ECO board		30w cont. 12 M/M		

RTL-01

Block microarchitecture specification

TAKES	COSTS	OWNER	CRITICAL PATH
10w w0–w10 of the stage	40.0 M/M ~17.4 people while it runs	Block owners Chief architect approves	yes slipping moves the program

WHY IT EXISTS

Complete the **microarchitecture specification for every block**—pipeline, interfaces, control, register map—to the level of detail an implementer can write code against without inventing anything.

ARCH-10 started these inside architecture; this finishes them. The distinction matters: a block whose specification is still being written while its RTL is being written produces code that documents itself, and the specification then follows the code instead of leading it.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Architecture specification and block chapters from ARCH-09
- Microarchitecture drafts from ARCH-10
- PPA budgets per block from ARCH-06
- Verification strategy from DV-01
- Interface definitions from ARCH-04

DONE WHEN

- Every block passes an entry gate before implementation starts
- Corner-case behavior defined, not left to the implementer
- Shared interfaces agreed by both sides

HOW IT GETS THERE

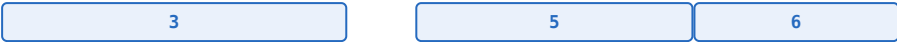
7 STEPS · 2 LANES

w0	w2	w4	w6	w8	w10
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Specification completion plan and block-owner assignment → Block microarchitecture specifications, complete	1w
2	Pipeline and datapath detail per block → Per-block pipeline and datapath detail	3w
3	Control and state machine specification → Control and state machine definitions	2.5w <i>runs in parallel</i>

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Block interface and protocol detail against the NoC → Interface and protocol detail per block	2.5w
5	Exception, error and corner-case behavior → Exception and corner-case behavior definitions	2w <i>runs in parallel</i>
6	Verification hook and observability requirements with DV → Verification hook and observability requirements	1.5w <i>runs in parallel</i>
7	Specification review and implementation entry gate → Implementation entry gate records	3.5w

Step 5 is the section most often left thin and the one verification spends the most time on. Corner-case behavior that the specification does not define is behavior the implementer chooses and the verification engineer has no basis to check.

WATCH OUT FOR

- ◆ **Specification written alongside the code.** The document then records what was built rather than defining it, and verification loses its independent reference.
- ◆ **Corner cases left undefined.** The implementer decides, verification cannot check, and the disagreement surfaces at chip-level integration.
- ◆ **Interfaces specified from one side.** Two blocks each documenting their own view of a shared protocol produces two consistent documents that disagree.
- ◆ **Entry gate skipped for schedule.** Coding against an incomplete specification is faster for two weeks and slower for two months.
- ◆ **Observability requirements added later.** Debug and coverage hooks designed in cost little; retrofitted they cost timing and an RTL change under control.

WHERE THE EFFORT GOES	40 M/M	WHO IS ON IT	~17.4 FTE	CONNECTIONS	TERMS HERE	4
Pipeline and datapath detail	14.0	Block owners <i>OWNS IT</i>		DEPENDS ON ARCH-03 ARCH-06 ARCH-09 ARCH-10	PPA NoC RTL DV	
Control and state machines	9.0	RTL lead		FEEDS INTO RTL-02 RTL-04 RTL-05		
Interface detail	7.0	Interconnect architect		RTL-07 DV-01 DV-02		
Exception and corner-case behavior	5.0	Verification leads		DV-03		
Reviews and entry gate	5.0	Chief architect				

08 RTL · RTL DESIGN & INTEGRATION

RTL-02

Block-level RTL implementation

TAKES	COSTS	OWNER	CRITICAL PATH
24w w4–w28 of the stage	180.0 M/M ~32.6 people while it runs	Block designers Configuration manager approves	yes slipping moves the program

WHY IT EXISTS

Write the **synthesisable logic for every block**, to a coding standard that survives lint, CDC, synthesis and gate-level simulation without rework.

At 180 man-months this is the largest single activity in the entire template—nearly 5% of the program's effort in one line. It is also the one most exposed to specification churn: every change upstream lands here, multiplied by the number of blocks that inherit it.

WHAT IT DELIVERS

RTL-D1

Block and top-level RTL release, tagged

The block half of the deliverable; RTL-04 assembles the top.

NEEDS FIRST

- Block specifications from RTL-01
- Coding and lint standards from the methodology
- IP integration requirements from RTL-03
- Verification feedback from DV-03
- Trial synthesis results from RTL-08

DONE WHEN

- Every block reviewed against its specification, not only linted
- Blocks frozen only after their verification loop has closed
- Releases tagged with known issues listed

HOW IT GETS THERE

8 STEPS · 2 LANES

w4	w9	w14	w19	w24	w28
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Coding standard, directory structure and naming conventions → Coding standard and repository structure	1.5w
2	Block RTL implementation, first pass → First-pass block RTL	9w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Block-level self-checking and assertion insertion → Embedded assertions and self-checking	4w runs in parallel
4	Block RTL review against specification → Block RTL review findings	3.5w
5	Parameterization and configurability implementation → Parameterization and configurability implementation	3w runs in parallel
6	Bug fix and refinement against block verification → Fixed and refined block RTL	7w
7	Block-level PPA refinement against trial synthesis → Block-level PPA refinement record	4w runs in parallel
8	Block freeze and release tagging → Block RTL source, tagged · Parameterized and configurable blocks · Block review records · Block release tags with known issues	3w

Step 6 is seven weeks of fixing rather than writing, and it is the honest half of RTL. First-pass code is written in step 2; what makes it correct is the loop against block-level verification, and shortening that loop is where schedule pressure does the most damage.

WATCH OUT FOR

- ◆ **Specification churn arriving mid-implementation.** Every upstream change lands here multiplied by the blocks affected, which is why RTL-10 exists.
- ◆ **Coding standard imposed after code exists.** Retrofitting a standard across 180 man-months of code is a project nobody schedules and everyone half-does.
- ◆ **Assertions treated as verification's job.** Assertions written by the implementer capture intent; written by someone else they capture observed behavior.
- ◆ **Blocks declared complete without verification feedback.** First-pass code is not correct code, and a block frozen before its verification loop closes freezes its bugs.
- ◆ **Configurability added everywhere.** Parameterization nobody uses multiplies the verification space at no benefit, and the cost lands on DV-03.

WHERE THE EFFORT GOES	180 M/M	WHO IS ON IT ~32.6 FTE	CONNECTIONS	TERMS HERE	5
Block implementation	70.0	Block designers owns it	DEPENDS ON ARCH-10 RTL-01 RTL-07 RTL-09	PPA IP RTL CDC RDC	
Bug fix and refinement	45.0	RTL lead	RUNS WITH RTL-08 DV-03		
Assertions and self-checking	25.0	Verification engineers	FEEDS INTO RTL-04 RTL-06 DV-05 SYN-02 SYN-03 SYN-05		
Block review	20.0	Synthesis liaison			
PPA refinement	12.0	Configuration manager			
Parameterization	8.0				

RTL-03

Third-party and internal IP integration
— wrappers, glue, configuration

TAKES 16w w8–w24 of the stage	COSTS 45.0 M/M ~12.2 people while it runs	OWNER IP integration lead Procurement liaison approves	CRITICAL PATH yes slipping moves the program
WHY IT EXISTS Bring every third-party, foundry and internal IP into the design—wrappers, glue, configuration and version control—and prove each one behaves as its documentation claims. IP integration fails at the boundary rather than in the block. A PHY that meets its specification and a controller that meets its specification can still disagree about reset ordering, and the disagreement is found here or it is found in the lab.			
WHAT IT DELIVERS RTL-D6 IP integration report and version manifest This activity is the deliverable, and the manifest is what makes an IP-related bug traceable to a version.			
NEEDS FIRST - IP deliveries against the schedule from IPR-09 - Acceptance checklists from IPR-06 - AMS macros and models from AMS-15 and AMS-16 - Block RTL from RTL-02 - Integration requirements from the architecture specification			
DONE WHEN - Every delivery inspected against its checklist on receipt - Boundary and reset behavior verified, not assumed from documentation - Version manifest complete and maintained through updates			
HOW IT GETS THERE 8 STEPS · 2 LANES w8w12w16w20w24 12358 467			
#STEP, AND WHAT IT ADDS TO THE DELIVERABLE TAT 1IP receipt, incoming inspection against the acceptance checklist → Incoming inspection records per delivery1.5w			

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Integration environment and wrapper framework → Integration environment and wrapper framework	2w
3	Per-IP wrapper and glue implementation → IP wrappers and glue logic	5w
4	Configuration, tie-off and parameter setting per IP runs in parallel → Per-IP configuration and tie-off	3w
5	Boundary and reset sequencing verification per IP → Boundary and reset sequencing verification results	4w
6	AMS macro integration with behavioural models runs in parallel → AMS macro integration with behavioural models	3.5w
7	IP version manifest and update handling runs in parallel → IP version manifest	2.5w
8	Integration report and issue escalation to vendors → Integration report and vendor issue log	3.5w

Step 1 is the acceptance check `IPR-06` defined, and it is the last point at which an incomplete delivery is the vendor's problem. Accepting a delivery without inspecting it transfers the missing views to the integration team.

WATCH OUT FOR

- ◆ **Deliveries accepted without inspection.** Missing views and collateral then become the integration team's problem rather than the vendor's.
- ◆ **Boundary behavior assumed from documentation.** Reset ordering, enable timing and calibration sequences are where IP integration actually fails.
- ◆ **Version manifest not maintained.** An IP bug that cannot be traced to a version cannot be reported usefully or fixed reproducibly.
- ◆ **IP updates absorbed silently.** A vendor drop applied without impact assessment can change behavior verification has already signed off.
- ◆ **Late IP compressing integration.** The integration window is fixed at its far end by RTL-04; a late delivery does not move it, it shortens it.

WHERE THE EFFORT GOES	45 M/M	WHO IS ON IT ~12.2 FTE	CONNECTIONS	TERMS HERE	4
Wrapper and glue implementation	14.0	IP integration lead OWNS IT	DEPENDS ON PDK-04 IPR-05 IPR-06 IPR-07 IPR-09	IP PHY AMS RTL	
Boundary and sequencing verification	11.0	Integration engineers	FEEDS INTO AMS-16 RTL-04 RTL-06		
AMS macro integration	7.0	Verification engineers	DV-03 DV-07 PD-02		
Configuration and tie-off	5.0	AMS integration engineer	LATER INPUT AMS-15 +6w AMS-16 +0w		
Incoming inspection	4.0	Procurement liaison			
Version manifest and reporting	4.0				

08 RTL · RTL DESIGN & INTEGRATION

RTL-04

Chip-level integration and top assembly

TAKES	COSTS	OWNER	CRITICAL PATH
12w w16–w28 of the stage	35.0 M/M ~12.7 people while it runs	Chip integration lead Software liaison approves	yes slipping moves the program

WHY IT EXISTS

Assemble the blocks and the IP into **one top-level design that elaborates, connects and boots**—the first point at which the chip exists as a single object.

Chip-level integration is where every optimistic assumption made in isolation meets every other one. Connectivity, clock and reset distribution, address maps and configuration all become concrete here, and the defects found are the ones no block-level test could have seen.

WHAT IT DELIVERS

RTL-D1

Block and top-level RTL release, tagged

The top half of the deliverable—the object every downstream stage actually consumes.

NEEDS FIRST

- Block RTL releases from RTL-02
- Integrated IP from RTL-03
- Power and clock intent from RTL-05
- Register map from RTL-07
- Top-level architecture from ARCH-09

DONE WHEN

- Connectivity verified automatically, not by inspection
- Chip boots in simulation before freeze is contemplated
- Top-level release tagged with its full IP manifest

HOW IT GETS THERE

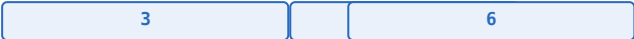
8 STEPS · 2 LANES

w16	w19	w22	w25	w28
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Top-level hierarchy and connectivity definition → Top-level hierarchy and connectivity	1.5w
2	NoC and interconnect integration → Integrated NoC and interconnect	3w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Clock and reset distribution implementation → Clock and reset distribution	2.5w runs in parallel
4	Address map and configuration space assembly → Assembled address map and configuration space	2w runs in parallel
5	Top-level elaboration and connectivity checking → Elaborated top level with connectivity check results	2.5w
6	Boot and bring-up path integration → Integrated boot and bring-up path	2.5w runs in parallel
7	Chip-level smoke tests and first boot → Chip-level smoke test results	2.5w
8	Integration issue closure and top release → Top-level RTL release	2.5w

Step 7—the first chip-level boot in simulation—is the moment the design becomes real. It is also slow, because the first boot exercises paths that no block test touched, and each failure takes a full-chip simulation to reproduce.

WATCH OUT FOR

- ◆ **Connectivity checked by inspection.** A chip this size has too many connections to review; automated connectivity checking is the only reliable method.
- ◆ **Integration started before blocks are stable.** Integrating moving blocks produces defects that are re-found every time a block updates.
- ◆ **Address map assembled from block documentation.** Generated maps agree with the code; hand-assembled maps agree with the documentation, which is a different thing.
- ◆ **First boot deferred.** The boot path exercises reset, clocking, configuration and the fabric together, and delaying it delays every defect that only appears in combination.
- ◆ **Clock and reset distribution treated as a wiring task.** It carries the crossing inventory from ARCH-05 and the CDC obligations that follow from it.

WHERE THE EFFORT GOES	35 M/M	WHO IS ON IT ~12.7 FTE	CONNECTIONS	TERMS HERE	4
NoC and interconnect integration	9.0	Chip integration lead owns IT	DEPENDS ON ARCH-07 RTL-01 RTL-02 RTL-03 RTL-05 RTL-07 RTL-09	IP NoC RTL CDC	
Chip-level smoke tests and boot	7.0	Integration engineers	FEEDS INTO RTL-06 RTL-08 DV-04 DV-08 DFT-06 SYN-01 SYN-02 BU-03		
Connectivity checking	6.0	Clock and reset owner			
Clock and reset distribution	5.0	Verification engineers			
Address map assembly	4.0	Software liaison			
Issue closure and release	4.0				

RTL-05

Clock, reset and power intent (UPF) implementation

TAKES	COSTS	OWNER	CRITICAL PATH
10w w6–w16 of the stage	18.0 M/M ~7.8 people while it runs	Low-power implementation lead Physical design liaison approves	no has float

WHY IT EXISTS

Implement the **clock, reset and power intent** the architecture defined—domain boundaries, isolation, retention, crossing synchronisers—and produce a UPF file the tools accept.

Power intent is where architecture becomes an obligation on every downstream tool. Synthesis inserts isolation cells from it, verification checks against it, and physical design implements it. An intent file that is wrong or incomplete propagates the same error into all three.

WHAT IT DELIVERS

RTL-D5

UPF power intent file

This activity is the deliverable, and "tool-accepted" is the standard rather than "reviewed".

NEEDS FIRST

- Power and clock architecture from ARCH-05
- Crossing inventory from ARCH-05
- Block list and hierarchy from RTL-01
- UPF methodology from PDK-09
- Library low-power cell availability from PDK-03

DONE WHEN

- UPF elaborates in synthesis, not only reviews well
- Every crossing on the inventory has an implemented synchroniser
- Intent accepted by synthesis, verification and physical design

HOW IT GETS THERE

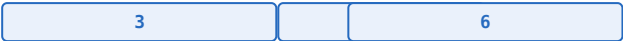
7 STEPS · 2 LANES

w6	w8	w10	w12	w14	w16
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Domain boundary implementation against the ARCH-05 definition → Implemented power domain boundaries	2w
2	Isolation, level shifter and retention strategy implementation → Isolation, level shifter and retention implementation	2.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Clock domain crossing synchroniser implementation → Implemented domain boundaries · CDC synchroniser implementation	2w runs in parallel
4	Reset sequencing and synchronization implementation → Reset sequencing logic	1.5w runs in parallel
5	UPF authoring and tool acceptance check → UPF power intent file, tool-accepted	2.5w
6	Power management controller RTL → Power management controller RTL	2w runs in parallel
7	Intent review with synthesis, verification and physical design → Power intent review record	3w

Step 5 includes a tool acceptance check because UPF that reads correctly to a human and fails to elaborate in synthesis is common. The file has to be exercised in the actual flow, not reviewed as a document.

WATCH OUT FOR

- ◆ **UPF written as a document.** Intent that has never been elaborated in synthesis is intent whose errors are found by the first person who runs the flow.
- ◆ **Crossings implemented without reference to the inventory.** A synchroniser strategy chosen per crossing by each designer produces inconsistency CDC will flag hundreds of times.
- ◆ **Retention strategy decided late.** Retention cells cost area and change the physical implementation; adding them after floorplan is expensive.
- ◆ **Reset sequencing implemented from the block view.** Reset order is a chip-level property, and blocks each releasing in their own order produces a boot that works in simulation and not in silicon.
- ◆ **Intent reviewed only by its author.** Synthesis, verification and physical design each consume it differently, and each has to confirm it works for them.

WHERE THE EFFORT GOES	18 M/M	WHO IS ON IT	~7.8 FTE	CONNECTIONS
Isolation and retention implementation	5.0	Low-power implementation lead OWNS IT		DEPENDS ON ARCH-05 PDK-03 RTL-01
Domain boundary implementation	4.0	Clock and reset owner		FEEDS INTO RTL-04 RTL-06 DV-06
UPF authoring and acceptance	3.5	Synthesis liaison		DV-11 SYN-06 SYN-07
CDC synchroniser implementation	2.5	Verification liaison		PD-03 BU-06
Power management controller	2.0	Physical design liaison		TERMS HERE 4
Intent review	1.0			RTL UPF CDC RDC

08 RTL · RTL DESIGN & INTEGRATION

RTL-06

Lint, CDC and RDC closure

TAKES	COSTS	OWNER	CRITICAL PATH
16w w12–w28 of the stage	30.0 M/M ~8.1 people while it runs	RTL quality lead CAD engineer approves	yes slipping moves the program

WHY IT EXISTS

Drive **lint, clock domain crossing and reset domain crossing to closure**—every violation either fixed or waived with a reason someone signed.

These checks find the class of defect that simulation cannot: a crossing without a synchroniser works in every simulation and fails intermittently in silicon. The discipline is in the waivers—an unexamined waiver list is where the surviving bugs live.

WHAT IT DELIVERS

RTL-D4

Lint / CDC / RDC clean reports with waiver list

This activity is the deliverable, and the waiver list is the part that needs review rather than filing.

NEEDS FIRST

- Block and top RTL from RTL-02 and RTL-04
- Crossing inventory from ARCH-05
- Synchroniser implementation from RTL-05
- Lint and CDC tool setup from PDK-08
- Waiver policy from the methodology

DONE WHEN

- Every violation fixed or waived with a signed reason
- Chip-level CDC clean, not only block-level
- Closure regressed in CI so it stays clean

HOW IT GETS THERE

7 STEPS · 2 LANES

w12	w16	w20	w24	w28
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Lint, CDC and RDC methodology and rule set definition → Lint, CDC and RDC rule sets	1.5w
2	Block-level lint closure → Block-level lint closure record	3.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Block-level CDC closure → Block-level CDC closure record	3.5w runs in parallel
4	Chip-level CDC closure against the crossing inventory → Block and chip-level closure results · Crossing inventory reconciliation	4w
5	RDC closure and reset domain analysis → RDC closure and reset domain analysis	3w runs in parallel
6	Waiver review and disposition → Waiver list with dispositions and signatures	3w
7	Closure reporting and regression setup → Closure regression in the CI flow	4w

Step 4 is chip-level rather than block-level because most real crossings are between blocks. Block-level CDC finds the easy ones; the crossing that fails in silicon is usually one neither block owner considered theirs.

WATCH OUT FOR

- ◆ **Waivers accumulated rather than reviewed.** A waiver list nobody re-reads is where the surviving crossings hide, and they are the hardest silicon bugs to diagnose.
- ◆ **CDC run only at block level.** The crossings that matter are usually between blocks, and neither owner runs the check that would find them.
- ◆ **Closure not regressed.** Clean at one release and dirty at the next, with nobody noticing until the freeze package is assembled.
- ◆ **Rule set weakened to reduce noise.** Turning off the rule that produces the most violations is the fastest way to a clean report and the slowest way to a working chip.
- ◆ **Crossing inventory not reconciled.** Crossings found by the tool but absent from ARCH-05's inventory mean the architecture and the implementation disagree.

WHERE THE EFFORT GOES	30 M/M	WHO IS ON IT ~8.1 FTE	CONNECTIONS	TERMS HERE 5
Chip-level CDC closure	8.0	RTL quality lead owns it	DEPENDS ON PDK-08 RTL-02 RTL-03 RTL-04 RTL-05	CI RTL CDC RDC CAD
Block-level lint closure	6.0	CDC specialist		
Block-level CDC closure	6.0	Block owners	FEEDS INTO RTL-10 DV-11 SYN-01 S0-09 EVB-10 BU-04	
RDC closure	4.0	Clock and reset owner		
Waiver review	3.0	CAD engineer		
Reporting and regression	3.0			

RTL-07

Register map / RDL definition and header generation

TAKES	COSTS	OWNER	CRITICAL PATH
8w w4–w12 of the stage	10.0 M/M ~5.4 people while it runs	Register map owner Verification liaison approves	no has float

WHY IT EXISTS

Define the **register map—the software contract**—and generate the RTL, headers and documentation from one source so they cannot disagree.

The register map is where hardware and firmware meet, and it is the interface most often maintained in three places. Generating the RTL, the C headers and the documentation from a single description is what makes them consistent; hand-maintaining them is what produces a driver that writes to the wrong offset.

WHAT IT DELIVERS

RTL-D2

Register map / RDL and generated headers

This activity is the deliverable, generated from one source rather than maintained in several.

NEEDS FIRST

- Block specifications from RTL-01
- Address map intent from ARCH-09
- Software and firmware requirements
- Security and lifecycle access rules from ARCH-07
- Verification model requirements from DV-01

DONE WHEN

- RTL, headers and documentation generated from one source
- Map reviewed by firmware before it is frozen
- Side effects and reset values specified for every register

HOW IT GETS THERE

7 STEPS · 2 LANES

w4	w6	w8	w10	w12
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Register description language and tooling selection → Register description source	1w
2	Address map allocation across blocks → Address map allocation	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Per-block register definition → Per-block register definitions	2.5w
4	Access policy, reset value and side-effect specification → Access policy and side-effect specification	1.5w runs in parallel
5	Generation flow — RTL, headers, documentation, UVM model → Generated RTL, headers, documentation and UVM model	2w runs in parallel
6	Software review of the map → Software review record	1.5w runs in parallel
7	Map release and change control → Released register map with change control	3w

Step 6 is a review by the people who will use the map rather than the people who wrote it. A register set that is natural to the hardware and awkward to drive is a set the driver will work around, and the workaround becomes the real interface.

WATCH OUT FOR

- ◆ **Map maintained in more than one place.** RTL, headers and documentation drift apart, and the driver eventually writes to an offset that moved.
- ◆ **Software reviews the map after freeze.** An awkward interface then gets worked around in the driver, and the workaround becomes the contract nobody documented.
- ◆ **Side effects undocumented.** A read that clears a status bit is a behavior firmware has to know about, and discovering it in the lab costs days per instance.
- ◆ **Security access rules omitted.** Registers that should be locked in production lifecycle states are locked late or not at all.
- ◆ **Reset values unspecified.** Registers whose reset value is whatever the implementer chose produce a boot sequence that depends on undocumented behavior.

WHERE THE EFFORT GOES	10 M/M	WHO IS ON IT ~5.4 FTE	CONNECTIONS	TERMS HERE	3
Per-block register definition	3.5	Register map owner owns IT	DEPENDS ON ARCH-07 ARCH-09 ARCH-10 RTL-01	RDL RTL UVM	
Generation flow	2.0	Block owners	FEEDS INTO RTL-02 RTL-04 DV-01 DV-02 DV-03 BU-04		
Address map allocation	1.5	Firmware lead			
Access policy and side effects	1.5	Security architect			
Release and change control	1.5	Verification liaison			

08 RTL · RTL DESIGN & INTEGRATION

RTL-08

Trial synthesis feedback loop on RTL PPA

TAKES	COSTS	OWNER	CRITICAL PATH
12w w10–w22 of the stage	20.0 M/M ~7.2 people while it runs	Synthesis engineer RTL lead approves	no has float

WHY IT EXISTS

Run **trial synthesis continuously against the evolving RTL** and feed timing, area and power back to block owners while the code can still change cheaply.

RTL that has never been synthesized is RTL whose PPA is unknown. A block that misses its area budget by 30% is a redesign if found at SYN-02 and an afternoon if found here, and the difference is only whether anyone was running synthesis during implementation.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Block RTL from RTL-02
- Constraints bootstrap from SYN-01
- Library and PDK from PDK-13
- PPA budgets from ARCH-06
- Floorplan intent from ARCH-08

DONE WHEN

- Every block has a current QoR result against its budget
- Feedback delivered on a cadence, not on request
- Overruns escalated rather than noted

HOW IT GETS THERE

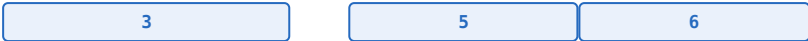
7 STEPS · 2 LANES

w10	w13	w16	w19	w22
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Trial synthesis flow and constraint bootstrap → Trial synthesis flow and constraints	1.5w
2	Per-block trial synthesis and QoR baseline → Per-block QoR baselines	3w
3	Timing feedback to block owners → Timing feedback to block owners	2.5w

runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Area and budget tracking against ARCH-06 → Budget tracking against ARCH-06	2w
5	Power and clock gating effectiveness feedback → Timing, area and power feedback reports	2w
6	Congestion and routability early indication → Congestion and routability indications	2w
7	Recurring trial synthesis cadence and reporting → Trial synthesis cadence and reporting record	5.5w

Step 7 is the activity's real shape—a weekly cadence rather than a one-off study. A single trial synthesis is a snapshot; a weekly one is a feedback loop, and only the loop changes how blocks are written.

WATCH OUT FOR

- ◆ **Trial synthesis run once.** A snapshot tells a block owner where they were; only a cadence tells them whether they are improving.
- ◆ **Feedback delivered without constraints.** Timing results against bootstrap constraints that do not resemble the real ones send block owners chasing paths that do not matter.
- ◆ **Area overrun tolerated because it is early.** Blocks rarely shrink after implementation, and an overrun accepted at trial synthesis is an overrun accepted permanently.
- ◆ **Congestion ignored.** Routability problems are cheaper to fix in RTL structure than in placement, and the early indication is the only chance to do so.
- ◆ **Results not tied to the budget.** QoR reported in absolute terms rather than against ARCH-06 tells nobody whether the block is acceptable.

WHERE THE EFFORT GOES	20 M/M	WHO IS ON IT	~7.2 FTE	CONNECTIONS
Recurring trial synthesis and reporting	7.0	Synthesis engineer owns it		DEPENDS ON ARCH-06 PDK-13 RTL-04
Per-block synthesis and baselines	4.5	Block owners		RUNS WITH RTL-02
Timing feedback	3.0	PPA lead		FEEDS INTO RTL-10 SYN-02 SYN-07
Area and budget tracking	2.5	Physical design liaison		PD-01
Power feedback	1.5	RTL lead		TERMS HERE 4
Congestion indication	1.5			PPA PDK RTL QoR

08 RTL · RTL DESIGN & INTEGRATION

RTL-09

CI build, nightly regression and release management

TAKES	COSTS	OWNER	CRITICAL PATH
30w cont. w2–w32 of the stage	20.0 M/M ~2.9 people while it runs	Configuration and release manager RTL lead approves	no has float

WHY IT EXISTS

Run the **build, regression and release machinery** the whole stage depends on—continuous integration, nightly regressions, tagged releases and a reproducible environment.

This is infrastructure, and its value shows only in its absence. Without a nightly build, a break introduced on Tuesday is found on Friday by whoever needed the design; without release tagging, a bug cannot be reproduced against the version that had it.

WHAT IT DELIVERS

RTL-D3

Integration testbench and build system

This activity is the deliverable, and its measure is that every release is reproducible from a tag.

NEEDS FIRST

- RTL source from RTL-02, RTL-03 and RTL-04
- Verification regressions from DV-10
- Compute and license capacity from PDK-12
- Tool versions from PDK-08
- Configuration management policy

DONE WHEN

- Every release reproducible from a tag
- Nightly regression completing within the night
- Failures triaged within a working day, with an owner

HOW IT GETS THERE

7 STEPS · 2 LANES

w2	w8	w14	w20	w26	w32
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Build system and dependency management setup → Build system and dependency management	2.5w
2	Continuous integration and per-commit checking → Continuous integration with per-commit checks	3w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Nightly regression definition and scheduling → Nightly regression suite and schedule runs in parallel	2.5w
4	Release tagging and version manifest process → Release tagging and version manifests	2w
5	Compute and license capacity management for regressions → Compute and license capacity management record runs in parallel	2w
6	Failure triage and notification workflow → Failure triage and notification workflow runs in parallel	2.5w
7	Continuous operation and maintenance across the stage → Regression operation and maintenance log	22.5w

Twenty-two of the thirty weeks are step 7, which is maintenance rather than construction. A CI system that is not maintained degrades into one everyone ignores, and an ignored red build is worse than no build at all.

WATCH OUT FOR

- ◆ **Red builds tolerated.** Once a broken build is normal, the signal is gone and every subsequent break is invisible.
- ◆ **Regression runtime exceeding the night.** A nightly that takes thirty hours is not nightly, and the feedback loop silently doubles.
- ◆ **Releases produced by hand.** An untagged release cannot be reproduced, and an IP bug found against it cannot be reported precisely.
- ◆ **Capacity not managed.** Regressions competing with verification for the same licences slow both, usually at the point in the schedule where both matter most.
- ◆ **Triage unowned.** Failures nobody is assigned to investigate accumulate until the regression result is meaningless.

WHERE THE EFFORT GOES	20 M/M	WHO IS ON IT ~2.9 FTE	CONNECTIONS
Continuous operation and maintenance	8.0	Configuration and release manager OWNS IT	DEPENDS ON PDK-08 PDK-12 IPR-09
CI and per-commit checking	4.0	CI engineer	FEEDS INTO RTL-02 RTL-04 DV-10 SYN-01
Build system setup	3.0	Verification liaison	TERMS HERE 4
Nightly regression setup	2.0	CAD infrastructure	CI IP RTL CAD
Triage workflow	1.5	RTL lead	
Release tagging process	1.5		

08 RTL · RTL DESIGN & INTEGRATION

RTL-10

Specification change control and ECO board

TAKES	COSTS	OWNER	CRITICAL PATH
30w cont. w2–w32 of the stage	12.0 M/M ~1.7 people while it runs	Program manager Configuration manager approves	yes slipping moves the program

WHY IT EXISTS

Control what changes and when—**specification change requests, ECO board decisions and the freeze itself**—so the design converges rather than oscillates.

Every stage downstream assumes RTL stops moving at some point. Change control is how that point is created: not by refusing changes, but by making each one visible, costed and decided by people who can see what it affects. The freeze is this activity's output, and its credibility is the whole program's.

WHAT IT DELIVERS

RTL-D7

RTL Freeze package

This activity is the deliverable—the freeze, its criteria, its exceptions and the change history behind it.

NEEDS FIRST

- Change requests from every source — architecture, verification, synthesis, IP
- Verification status from DV-12
- Trial synthesis and PPA status from RTL-08
- Lint and CDC closure status from RTL-06
- Freeze criteria from the program plan

DONE WHEN

- Freeze declared against criteria, not against a date
- Every change carries an impact assessment before approval
- Post-freeze exception policy published and used

HOW IT GETS THERE

7 STEPS · 2 LANES

w2	w8	w14	w20	w26	w32
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Change control process and ECO board definition → Change control process and ECO board charter	1.5w
2	Change request intake and impact assessment workflow → Change request log with impact assessments	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	ECO board operation and decision records → ECO board decision records	20w
4	Change tracking against blocks, verification and synthesis → Change tracking record across blocks, verification and synthesis	3w runs in parallel
5	Freeze criteria definition and readiness tracking → Freeze criteria and readiness tracking	2.5w runs in parallel
6	RTL Freeze package assembly → RTL Freeze package and declaration	3w
7	Freeze declaration and post-freeze exception policy → Freeze declaration with its exception policy	3.5w

Step 3 runs for twenty weeks and is a standing meeting rather than a task. Its discipline is impact assessment: a change that looks like an afternoon in one block can invalidate a month of verification, and the board exists so that cost is visible before the change is approved.

WATCH OUT FOR

- ❖ **Freeze declared by date rather than by criteria.** A gate that passes because it was scheduled teaches everyone that gates mean nothing.
- ❖ **Impact assessment skipped for small changes.** The cost of a change is rarely in the block that changes; it is in the verification that has to be re-run.
- ❖ **Post-freeze exceptions ungoverned.** Changes will be needed after freeze; without a policy they happen informally and nobody knows what the design contains.
- ❖ **Board without downstream representation.** Verification and synthesis are the ones who pay for most changes, and a board without them approves costs it cannot see.
- ❖ **Change log not reconciled with the release.** A freeze package that does not match what was actually built is a package nobody can rely on.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT	~1.7 FTE	CONNECTIONS
ECO board operation	5.0	Program manager owns it		DEPENDS ON DEF-08 IPR-10 RTL-06 RTL-08
Impact assessment workflow	2.5	RTL lead		FEEDS INTO DV-12 SYN-09 PD-14 T0-03
Freeze package assembly	1.5	Verification lead		TERMS HERE PPA IP RTL ECO CDC
Change tracking	1.5	Synthesis and PD representatives		
Process definition	1.0	Configuration manager		
Freeze declaration and policy	0.5			

DESIGN & VERIFY

09 Verification

DV

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w26 – w66	40 _w	12	650 _{M/M}	7 _{of 12}
ENGINEERING ACTIVITIES				KEY DELIVERABLES
DV-01	Verification plan and coverage model definition		8w 25 _{M/M}	DV-D1 Verification plan (vPlan) and coverage model
DV-02	UVM environment and VIP bring-up		14w 60 _{M/M}	DV-D2 UVM testbenches and integrated VIP
DV-03	Block-level constrained-random and directed testing		30w 200 _{M/M}	DV-D3 Regression and coverage dashboards
DV-04	Chip-level and system-level scenario testing		22w 110 _{M/M}	DV-D4 Formal proof reports with assumption list
DV-05	Formal property verification — control logic, connectivity, security		16w 40 _{M/M}	DV-D5 Low-power verification report
DV-06	Low-power (UPF) verification		10w 20 _{M/M}	DV-D6 Emulation platform and system test suite
DV-07	AMS / mixed-signal co-simulation		12w 20 _{M/M}	DV-D7 Gate-level simulation report
DV-08	Emulation and FPGA prototype bring-up — capacity, partitioning, speed		20w 55 _{M/M}	DV-D8 DV closure signoff package
DV-09	Performance and bandwidth validation against the architecture model		12w 30 _{M/M}	
DV-10	Coverage closure and regression stability management		26w cont. 45 _{M/M}	
DV-11	Gate-level simulation, functional and timing-annotated		10w 20 _{M/M}	
DV-12	Bug triage and disposition board		34w cont. 25 _{M/M}	

09 DV · VERIFICATION

DV-01

Verification plan and coverage model definition

TAKES	COSTS	OWNER	CRITICAL PATH
8w w0–w8 of the stage	25.0 M/M ~13.6 people while it runs	Verification lead Program manager approves	yes slipping moves the program

WHY IT EXISTS

Write the plan that says **what will be verified, how, and how anyone will know it is done**—feature by feature, with a coverage model behind each claim.

Verification is the largest line of effort in the program, and without a plan it is 650 man-months of activity with no definition of completion. The coverage model is what turns "we tested it" into a measurable statement, and the features nobody writes into the plan are the features nobody verifies.

WHAT IT DELIVERS

DV-D1

Verification plan (vPlan) and coverage model

This activity is the deliverable, and the coverage model is what every later closure claim is measured against.

NEEDS FIRST

- Architecture specification from ARCH-09
- Block specifications from RTL-01
- Register map from RTL-07
- IP verification collateral requirements from IPR-06
- Compute capacity model from PDK-12

DONE WHEN

- Every specified feature has a verification strategy and a coverage point
- Closure criteria written and agreed before verification starts
- Plan reviewed by architecture and design, not only by verification

HOW IT GETS THERE

6 STEPS · 2 LANES

w0	w2	w4	w6	w8
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Feature extraction from the architecture and block specifications → Feature list extracted from the specifications	1.5w
2	Verification strategy per feature — simulation, formal, emulation → Verification strategy per feature	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Coverage model definition — functional, code, assertion → Coverage model — functional, code and assertion	2w
4	Closure criteria and signoff definition per block → Closure criteria per block and at chip level	1w
5	Resource, compute and schedule modeling → Resource and schedule model	1w
6	Verification plan review with design and architecture → Reviewed verification plan	3w

Step 6 is a review by the people who wrote the specification, and its purpose is to find features the plan missed. Verification reviewing its own plan confirms it is coherent; architecture reviewing it confirms it is complete, and incompleteness is the failure mode that matters.

WATCH OUT FOR

- ◆ **Plan written from the RTL rather than the specification.** It then verifies what was built rather than what was specified, and the two differ exactly where the bugs are.
- ◆ **Coverage model that measures activity rather than intent.** Code coverage says lines were executed; functional coverage says features were exercised, and only the second answers the question.
- ◆ **Closure criteria left vague.** "High coverage" is not a criterion anyone can sign, and it becomes whatever the schedule allows.
- ◆ **Architecture not in the review.** The features the plan missed are exactly the ones only the specification's author would notice.
- ◆ **Resource model omitted.** A plan that needs more compute or more engineers than exist is a plan whose closure date is fiction.

WHERE THE EFFORT GOES	25 M/M	WHO IS ON IT ~13.6 FTE	CONNECTIONS	TERMS HERE	3
Coverage model definition	8.0	Verification lead owns it	DEPENDS ON ARCH-09 ARCH-10 RTL-01 RTL-07	IP RTL DV	
Feature extraction	5.0	Block verification owners			
Strategy per feature	5.0	Chief architect	FEEDS INTO DV-02 DV-03 DV-04		
Plan review	4.0	RTL lead	DV-05 DV-10 DV-12		
Closure criteria	2.0	Program manager	FEEDS BACK INTO IPR-06		
Resource modeling	1.0				

09 DV · VERIFICATION

DV-02

UVM environment and VIP bring-up

TAKES	COSTS	OWNER	CRITICAL PATH
14w w4–w18 of the stage	60.0 M/M ~18.6 people while it runs	Verification architect Register map owner approves	yes slipping moves the program

WHY IT EXISTS

Build the **verification environment**—UVM agents, scoreboards, sequences, VIP integration—that every test in the program will run inside.

The environment is infrastructure with a long lead time and no visible output until it works. Sixty man-months buys agents for every interface, scoreboards that actually check, and reusable sequences. Testbench bring-up running late is the single most common reason verification starts late, and it starts late invisibly.

WHAT IT DELIVERS

DV-D2

UVM testbenches and integrated VIP

This activity is the deliverable—the environment every subsequent verification activity runs inside.

NEEDS FIRST

- Verification plan from DV-01
- Block and interface specifications from RTL-01
- Register description source from RTL-07
- VIP licences from IPR-07
- Methodology standards and reuse libraries

DONE WHEN

- Environment running on a real block, not on a stub
- Scoreboards checking against reference models
- Environment documented and reused rather than rebuilt per block

HOW IT GETS THERE

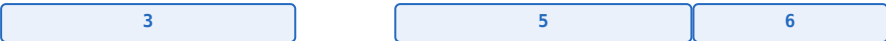
8 STEPS · 2 LANES

w4	w7	w10	w13	w16	w18
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Environment architecture and reuse strategy → Environment architecture and reuse strategy	1.5w
2	Agent development per interface → UVM agents per interface	4w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Third-party VIP integration and configuration → Integrated third-party VIP	3w runs in parallel
4	Scoreboard and reference model development → Scoreboards and reference models	4w
5	Sequence library and stimulus infrastructure → Sequence library	3w runs in parallel
6	Register model integration from the RDL source → Register model integration	2w runs in parallel
7	Environment bring-up on the first block → Environment bring-up results on the first block	2.5w
8	Environment release and documentation → Released environment with documentation	2w

Step 4 is where verification quality is decided. A scoreboard that compares against a reference model finds bugs; one that only checks protocol legality confirms the design does something rather than the right thing.

WATCH OUT FOR

- ◆ **Bring-up slower than planned and invisible.** Environment work has no test-pass metric, so slippage is not visible until testing should have started.
- ◆ **Scoreboards that check protocol rather than function.** A legal transaction carrying wrong data passes, and the bug survives to chip level.
- ◆ **No reuse strategy.** Every block building its own environment multiplies effort and makes results incomparable across blocks.
- ◆ **Register model hand-written.** A model that diverges from the RDL source produces tests that pass against the wrong offsets.
- ◆ **VIP licences short.** Verification concurrency is capped by VIP seats, and the cap is discovered when regressions start queuing.

WHERE THE EFFORT GOES	60 M/M	WHO IS ON IT ~18.6 FTE	CONNECTIONS	TERMS HERE	3
Agent development	16.0	Verification architect owns IT	DEPENDS ON IPR-07 IPR-08 RTL-01 RTL-07 DV-01	RDL UVM VIP	
Scoreboard and reference models	16.0	Agent and VIP engineers	FEEDS INTO DV-03 DV-04 DV-06 DV-07 DV-08 DV-10 DV-11		
VIP integration	10.0	Reference model engineers			
Sequence library	8.0	Methodology engineer			
Environment bring-up	5.0	Register map owner			
Register model and documentation	5.0				

09 DV · VERIFICATION

DV-03

Block-level constrained-random and directed testing

TAKES	COSTS	OWNER	CRITICAL PATH
30w w8–w38 of the stage	200.0 M/M ~29.0 people while it runs	Block verification owners Methodology engineer approves	yes slipping moves the program

WHY IT EXISTS

Verify every block against its specification with **constrained-random and directed testing**, and drive each one to its coverage closure criteria.

Two hundred man-months—the largest activity in the template—spread across thirty weeks and every block in the design. This is where most bugs are found and most verification time is spent, and its efficiency is set almost entirely by decisions taken in DV-01 and DV-02.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Verification environment from DV-02
- Block RTL releases from RTL-02
- Coverage model and closure criteria from DV-01
- Block specifications from RTL-01
- Regression capacity from DV-10

DONE WHEN

- Every block closed against the DV-01 criteria, not its own
- Coverage holes closed by tests, not by model changes
- Signoff records complete per block

HOW IT GETS THERE

8 STEPS · 2 LANES

w8	w14	w20	w26	w32	w38
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Per-block testbench instantiation from the shared environment → Per-block testbenches and test suites	2.5w
2	Directed test development for specified behavior → Directed test suite for specified behavior	5w
3	Constrained-random stimulus and constraint development → Directed and constrained-random tests	6w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Assertion development and binding → Bound assertion set	4w runs in parallel
5	Corner-case and error-injection testing → Corner-case and error-injection results	5w
6	Coverage analysis and hole closure per block → Coverage analysis and hole closure records	6w
7	Debug and bug reporting loop with block owners → Bug reports and debug log with block owners	8w runs in parallel
8	Block-level closure and signoff → Assertions bound to block interfaces · Block-level signoff records	5.5w

Step 7 runs in parallel across the whole window because debugging is continuous rather than a phase. It is also where the largest single share of the effort goes—finding a bug is usually quicker than proving what caused it.

WATCH OUT FOR

- ◆ **Coverage closed by adding coverage points rather than tests.** Redefining the model to match what was tested produces a closure number that measures nothing.
- ◆ **Constrained-random left unconstrained.** Random stimulus without well-designed constraints spends compute exploring illegal or uninteresting space.
- ◆ **Corner cases skipped because the specification is silent.** The gap belongs back in RTL-01, not quietly into the untested set.
- ◆ **Debug loop bottlenecked on block owners.** Verification finds faster than design fixes, and the queue is where the schedule quietly goes.
- ◆ **Blocks signed off against different standards.** Without enforced criteria from DV-01, each block closes to whatever its owner considered enough.

WHERE THE EFFORT GOES	200 M/M	WHO IS ON IT ~29.0 FTE	CONNECTIONS
Debug and bug reporting	55.0	Block verification owners OWNS IT	DEPENDS ON RTL-01 RTL-03 RTL-07 DV-01 DV-02
Constrained-random development	40.0	Verification engineers	RUNS WITH RTL-02
Coverage analysis and closure	35.0	Block designers	FEEDS INTO DV-04 DV-10 DV-12
Directed test development	30.0	Verification lead	TERMS HERE 4
Corner-case and error injection	22.0	Methodology engineer	RTL DV UVM VIP
Assertions and signoff	18.0		

09 DV · VERIFICATION

DV-04

Chip-level and system-level scenario testing

TAKES	COSTS	OWNER	CRITICAL PATH
22w w14–w36 of the stage	110.0 M/M ~21.7 people while it runs	Chip verification lead Software liaison approves	yes slipping moves the program

WHY IT EXISTS

Verify the chip as **one system running realistic scenarios**—boot, configuration, traffic, power transitions, error recovery—rather than as a collection of verified blocks.

Blocks that each pass in isolation routinely fail together. Chip-level verification is where arbitration, ordering, deadlock, reset sequencing and end-to-end data integrity are actually tested, and none of those are visible from inside a block.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Top-level RTL from RTL-04
- Block signoffs from DV-03
- Workload suite from DEF-02
- Power state definitions from ARCH-05
- Chip-level coverage model from DV-01

DONE WHEN

- Realistic workload scenarios run, not only synthetic ones
- Deadlock and arbitration tested under contention
- Chip-level coverage closed against the DV-01 model

HOW IT GETS THERE

8 STEPS · 2 LANES

w14	w19	w24	w29	w34	w36
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Chip-level environment assembly → Chip-level verification environment	2.5w
2	Boot and configuration scenario development → Boot and configuration scenarios	3w
3	End-to-end data path and integrity testing → End-to-end data integrity results	4w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Interconnect stress, arbitration and deadlock testing → Interconnect stress and deadlock findings	3.5w runs in parallel
5	Power state transition and DVFS scenario testing → Power transition scenario results	3w runs in parallel
6	Error injection and recovery at system level → System-level error injection and recovery results	3.5w
7	Realistic workload scenarios from the DEF-02 suite → Realistic workload scenarios in the testbench	3.5w runs in parallel
8	Chip-level coverage closure and signoff → Chip-level coverage closure and signoff	9w

Step 7 uses the same workloads `DEF-02` assembled and `ARCH-01` modelled. Running them in simulation is slow and worth it: it is the only place the design, the model and the product's actual intent are compared against each other before silicon.

WATCH OUT FOR

- ◆ **Started too late because blocks are not ready.** Chip-level defects take longest to find and fix, and compressing the window pushes them into silicon.
- ◆ **Scenarios written by verification alone.** Realistic use comes from the workload suite and from software; invented scenarios test what verification imagined.
- ◆ **Deadlock and arbitration untested.** These fail only under contention, which random block-level stimulus does not create.
- ◆ **Power transitions verified functionally only.** Retention, isolation and sequencing across a real power state change are where low-power designs actually fail.
- ◆ **Simulation too slow to run meaningful scenarios.** If a boot takes a day, the scenario set shrinks to what fits, and emulation in DV-08 is the answer.

WHERE THE EFFORT GOES	110 M/M	WHO IS ON IT ~21.7 FTE	CONNECTIONS
End-to-end and integrity testing	26.0	Chip verification lead OWNS IT	DEPENDS ON RTL-04 DV-01 DV-02 DV-03
Chip-level coverage closure	20.0	System verification engineers	RUNS WITH DV-08
Interconnect stress and deadlock	16.0	Interconnect verification specialist	FEEDS INTO DV-09 DV-10 DV-12 SYN-06 PD-11 SIPI-01 BU-04
Boot and configuration scenarios	16.0	Low-power verification engineer	TERMS HERE 3
Error injection and recovery	16.0	Software liaison	RTL DVFS DV
Workload scenarios and environment	16.0		

09 DV · VERIFICATION

DV-05

Formal property verification —
control logic, connectivity, security

TAKES	COSTS	OWNER	CRITICAL PATH
16w w10–w26 of the stage	40.0 M/M ~10.9 people while it runs	Formal verification lead Verification lead approves	no has float

WHY IT EXISTS

Use **formal proof where simulation is weak**—connectivity, control logic, arbiters, security properties, register access—and prove exhaustively what testing can only sample.

Formal answers a different question. Simulation shows that the cases tried worked; a proof shows that no case exists that does not. For arbiters, protocol state machines and security isolation, the space is too large to sample and the failure modes are exactly the ones random stimulus misses.

WHAT IT DELIVERS

DV-D4

Formal proof reports with assumption list

This activity is the deliverable, and the assumption list is what makes the proofs auditable.

NEEDS FIRST

- Verification plan target selection from DV-01
- Block RTL from RTL-02
- Security architecture properties from ARCH-07
- Interface protocol definitions from ARCH-04
- Formal tool setup from PDK-08

DONE WHEN

- Every proof reported with its assumptions and its bound
- Assumptions reviewed for over-constraint, not just listed
- Security properties proved rather than sampled

HOW IT GETS THERE

7 STEPS · 2 LANES

w10	w14	w18	w22	w26
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Target selection — where formal beats simulation → Formal target list with rationale	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Property specification per target → Property specifications per target	3w
3	Connectivity and structural proof at chip level → Connectivity and structural proof results	2.5w runs in parallel
4	Control logic and arbiter proofs → Control and arbiter proof results	3.5w
5	Security and isolation property proofs → Security property proof results	2.5w runs in parallel
6	Assumption review and over-constraint checking → Assumption review and over-constraint findings	2.5w
7	Proof closure, bounded results and reporting → Assumption list and proof closure report	5.5w

Step 6 is what separates a proof from a false claim. A property that passes because assumptions excluded the failing case has proved nothing, and reviewing assumptions is the only way to know which of the two happened.

WATCH OUT FOR

- ◆ **Over-constrained proofs.** Assumptions that exclude the failing case produce a green result and no information, and only assumption review catches it.
- ◆ **Formal applied where simulation is better.** Datapath and workload behavior are poor formal targets; applying it there consumes specialist time for little return.
- ◆ **Bounded results reported as proofs.** A property proved to depth twenty is not proved; the bound has to be stated with the result.
- ◆ **Properties written from the RTL.** A property derived from the implementation proves the implementation matches itself.
- ◆ **Security properties omitted.** Isolation and access control are exactly the properties simulation samples poorly and formal proves well.

WHERE THE EFFORT GOES	40 M/M	WHO IS ON IT ~10.9 FTE	CONNECTIONS
Property specification	10.0	Formal verification lead OWNS IT	DEPENDS ON ARCH-07 PDK-08 RTL-02 DV-01
Control and arbiter proofs	9.0	Formal engineers	FEEDS INTO DV-10 DV-12 S0-09
Proof closure and reporting	8.0	Security architect	TERMS HERE 2 RTL DV
Assumption review	6.0	Block designers	
Security property proofs	4.0	Verification lead	
Connectivity proofs	3.0		

09 DV · VERIFICATION

DV-06

Low-power (UPF) verification

TAKES	COSTS	OWNER	CRITICAL PATH
10w w18–w28 of the stage	20.0 M/M ~8.7 people while it runs	Low-power verification engineer Block designers approves	no has float

WHY IT EXISTS

Verify the design **against its power intent**—isolation, retention, level shifting, power sequencing—because none of it is visible in a normal functional simulation.

A design that passes every functional test can still corrupt state on a power-down, drive an isolated domain, or fail to retain what it needed. Low-power verification simulates the domains actually powering down, and it is the only place these failures appear before silicon.

WHAT IT DELIVERS

DV-D5

Low-power verification report

This activity is the deliverable, and it is the only evidence that the power architecture works as intended.

NEEDS FIRST

- UPF power intent from RTL-05
- Power state definitions from ARCH-05
- Verification environment from DV-02
- Power management controller RTL from RTL-05
- Low-power tool support from PDK-08

DONE WHEN

- Domains simulated powering down, not only statically checked
- Retention verified for both what is kept and what is not
- Power-aware tests running in the regular regression

HOW IT GETS THERE

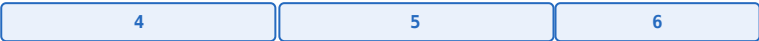
7 STEPS · 2 LANES

w18	w20	w22	w24	w26	w28
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	UPF-aware simulation environment setup → UPF-aware simulation environment	1.5w
2	Static UPF structural checking → Static UPF structural check results	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Isolation and level shifter verification → Isolation and level shifter verification results	2w
4	Retention and state restoration verification → Retention verification results	2w
5	Power sequencing and controller verification → Power sequencing verification results	2w
6	Power-aware regression integration → Power-aware regression integration	1.5w
7	Low-power coverage closure and reporting → Low-power coverage report	5w

Step 4 is the one that finds real bugs. Retention verification requires actually powering a domain down in simulation and checking what survives; a design that has never been simulated through a power cycle has never been tested for the failure mode.

WATCH OUT FOR

- ◆ **Only static checking performed.** Structural UPF checks find missing cells; only simulation finds a design that corrupts state when the domain actually goes down.
- ◆ **Retention verified for the intended state only.** What is not retained matters as much as what is, and a design that retains too little fails on resume.
- ◆ **Power sequences tested in isolation.** The failure is usually a power transition during traffic, not a clean idle transition.
- ◆ **Power-aware simulation excluded from regression.** It is slow, so it gets run once; a later RTL change then breaks it silently.
- ◆ **UPF used for verification differing from synthesis.** Two versions of the intent means verifying one design and building another.

WHERE THE EFFORT GOES	20 M/M	WHO IS ON IT ~8.7 FTE	CONNECTIONS
Isolation and retention verification	7.0	Low-power verification engineer OWNS IT	DEPENDS ON ARCH-05 RTL-05 DV-02 SYN-07
Power sequencing verification	4.0	Low-power implementation lead	FEEDS INTO DV-10 DV-12 S0-09 BU-02
Coverage closure and reporting	3.5	Verification methodology engineer	TERMS HERE 2
Environment setup	2.5	Power architect	
Static checking	1.5	Block designers	RTL UPF
Regression integration	1.5		

09 DV · VERIFICATION

DV-07

AMS / mixed-signal co-simulation

TAKES	COSTS	OWNER	CRITICAL PATH
12w w16–w28 of the stage	20.0 M/M ~7.2 people while it runs	AMS verification engineer Verification lead approves	no has float

WHY IT EXISTS

Verify the **boundary between the analog macros and the digital design**—control interfaces, calibration sequences, startup ordering—with the analog represented by models correlated to its circuits.

Analog blocks fail where they meet digital. A PLL that locks perfectly and a controller that sequences perfectly can still disagree about when lock is valid. Co-simulation is where those disagreements appear, and the alternative is finding them at bring-up with an oscilloscope.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Behavioural models from AMS-15
- Correlation results from AMS-16
- Integrated AMS macros from RTL-03
- Startup and sequencing requirements from ARCH-05
- Verification environment from DV-02

DONE WHEN

- Every model correlated against its circuit before being relied on
- Calibration sequences exercised from the digital side
- Degraded and fault behavior verified, not assumed

HOW IT GETS THERE

7 STEPS · 2 LANES

w16	w19	w22	w25	w28
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MAIN SEQUENCE

1	2	3	7
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IN PARALLEL

4	5	6
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#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Co-simulation environment and model integration → AMS co-simulation environment	2w
2	Model fidelity review against AMS characterization → Model fidelity review findings	2w
3	Control and configuration interface verification → Control and configuration interface results	2.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Calibration and training sequence verification → Calibration and training sequence results	2.5w runs in parallel
5	Startup, lock and ready-signaling verification → Startup and ready-signaling results	2w runs in parallel
6	Analog fault and degraded-mode behavior → Analog fault and degraded-mode results	1.5w runs in parallel
7	Co-simulation closure and findings report → Co-simulation findings report	5.5w

Step 2 decides whether any of this is meaningful. Verification against a behavioural model that does not match the circuit proves the digital side works with a fiction, and the correlation in `AMS-16` is what makes the model a proxy rather than a guess.

WATCH OUT FOR

- ◆ **Models never correlated.** Digital passes against a fiction, and the mismatch reaches silicon as a bring-up failure nobody predicted.
- ◆ **Calibration sequences untested from the digital side.** A calibration that works when the analog engineer drives it manually is first exercised by firmware at bring-up.
- ◆ **Ready and lock signaling assumed.** When a lock signal becomes valid is a contract; each side assuming a different answer produces intermittent boot failures.
- ◆ **Degraded modes unverified.** What the digital side does when a PLL fails to lock is behavior that only matters in the field, where it matters a great deal.
- ◆ **Co-simulation too slow to regress.** Run once and never again, it cannot catch a later change on either side.

WHERE THE EFFORT GOES	20 M/M	WHO IS ON IT	~7.2 FTE	CONNECTIONS
Control and calibration verification	7.0	AMS verification engineer	owns it	DEPENDS ON AMS-15 AMS-16 RTL-03 DV-02
Model fidelity review	4.0	Analog designers		FEEDS INTO DV-10 DV-12 BU-03 BU-05
Startup and signaling verification	3.5	Digital verification engineers		
Environment and model integration	3.0	Firmware liaison		TERMS HERE 6
Findings report	1.5	Verification lead		PLL AMS RTL DV
Degraded-mode testing	1.0			UVM VIP

09 DV · VERIFICATION

DV-08

Emulation and FPGA prototype bring-up — capacity, partitioning, speed

TAKES	COSTS	OWNER	CRITICAL PATH
20w w6–w26 of the stage	55.0 M/M ~11.9 people while it runs	Emulation lead Program manager approves	no has float

WHY IT EXISTS

Stand up **emulation and FPGA prototyping**—partitioning, capacity, speed, debug—so that scenarios too long for simulation can actually be run.

A boot that takes a day in simulation takes minutes in emulation, and workload scenarios that are impossible in one are routine in the other. Emulation is also the platform firmware development runs on, which makes its availability date a dependency for a team outside this stage entirely.

WHAT IT DELIVERS

DV-D6

Emulation platform and system test suite

This activity is the deliverable, and it serves firmware and validation as much as verification.

NEEDS FIRST

- Top-level RTL from RTL-04
- Emulator or FPGA capacity and licences
- Memory and interface models for the platform
- Workload suite from DEF-02
- Debug requirements from validation and firmware

DONE WHEN

- Design boots on the platform at usable speed
- Debug visibility sufficient to diagnose a failure
- Platform handed over to firmware on a scheduled date

HOW IT GETS THERE

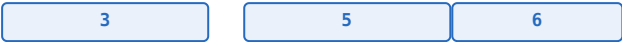
8 STEPS · 2 LANES

w6	w10	w14	w18	w22	w26
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Platform selection and capacity sizing → Platform selection and capacity plan	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Design partitioning across the platform → Design partitioning for the platform	3.5w
3	Memory and interface modeling on the platform → Memory and interface models	3w <i>runs in parallel</i>
4	Bring-up and first boot on emulation → Working emulation platform with boot	4w
5	Debug and visibility infrastructure → Debug and visibility infrastructure	3w <i>runs in parallel</i>
6	Performance and throughput tuning → Performance and throughput tuning record	2.5w <i>runs in parallel</i>
7	Long-scenario and workload execution → Long-scenario and workload results	8w
8	Platform handover to firmware and validation → Platform handover package to firmware and validation	2.5w

Step 8 makes this activity a dependency for people outside verification. Firmware cannot start on real workloads until the platform runs, and the handover date belongs in the program schedule rather than in the verification plan alone.

WATCH OUT FOR

- ◆ **Partitioning consuming more time than expected.** Splitting a large design across emulator capacity is difficult and iterative, and it is routinely underestimated.
- ◆ **Capacity contended with another program.** Emulator capacity is expensive and shared; a program without a booked allocation runs when the other one is not.
- ◆ **Debug visibility inadequate.** A platform that runs fast and cannot be probed produces failures that cannot be diagnosed.
- ◆ **Handover date not in the program schedule.** Firmware plans against it, and treating it as an internal verification milestone means nobody outside knows when it slips.
- ◆ **Models diverging from the design.** Memory and interface models built for the platform can drift from the RTL, and results then reflect the model.

WHERE THE EFFORT GOES	55 M/M	WHO IS ON IT ~11.9 FTE	CONNECTIONS	TERMS HERE	2
Design partitioning	14.0	Emulation lead <i>OWNS IT</i>	DEPENDS ON DEF-02 RTL-04 DV-02	RTL FPGA	
Bring-up and first boot	12.0	Emulation engineers	RUNS WITH DV-04		
Memory and interface modeling	10.0	Debug infrastructure engineer	FEEDS INTO DV-09 DV-10 BU-04		
Debug infrastructure	8.0	Firmware liaison	BU-09		
Long-scenario execution	7.0	Program manager			
Handover and tuning	4.0				

09 DV · VERIFICATION

DV-09

Performance and bandwidth validation
against the architecture model

TAKES 12w w22–w34 of the stage	COSTS 30.0 M/M ~10.9 people while it runs	OWNER Performance verification lead Chief architect approves	CRITICAL PATH no has float
WHY IT EXISTS Check that the design actually delivers the performance the architecture promised—bandwidth, latency, utilization—against the same workloads ARCH-01 modelled. Functional correctness and performance are separate questions, and passing every test says nothing about the second. A design that computes correctly at 60% of modelled throughput is a product that misses its headline KPI, and finding that in silicon leaves no options.			
WHAT IT DELIVERS No key deliverable is owned here — this activity contributes to the ones below.			
NEEDS FIRST - Emulation platform from DV-08 - Workload suite from DEF-02 - Performance model and predictions from ARCH-01 - Chip-level environment from DV-04 - KPI targets from DEF-02			
DONE WHEN - Real workloads measured, not only synthetic traffic - Every model disagreement resolved as design or model error - Findings delivered with time to act on them			
HOW IT GETS THERE 7 STEPS · 2 LANES w22w25w28w31w34 12367 45			
MAIN SEQUENCE 12367			
IN PARALLEL 45			
#STEP, AND WHAT IT ADDS TO THE DELIVERABLE TAT			
1Performance measurement infrastructure in the testbench → Performance measurement infrastructure 1.5w			
2Workload execution on emulation and simulation → Workload execution results 3w			
3Bandwidth and utilization measurement against the model → Bandwidth, utilization and latency measurements 2.5w			

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Latency and queueing analysis → Latency and queueing analysis	2w runs in parallel
5	Bottleneck identification and root cause → Bottleneck analysis with root causes	2.5w runs in parallel
6	Correlation against the ARCH-01 performance model → Model correlation results	2.5w
7	Findings, recommendations and model reconciliation → Findings and recommendations	2.5w

Step 6 closes a loop opened at architecture. Where the RTL and the model disagree, one of them is wrong—and finding out which is what tells the program whether it has a design problem or a modeling problem.

WATCH OUT FOR

- ◆ **Performance treated as a bring-up activity.** Discovered in silicon, a throughput shortfall has no remaining fix except software and marketing.
- ◆ **Measured only on synthetic traffic.** Synthetic patterns exercise the interconnect; real workloads exercise the design, and only the second predicts the product.
- ◆ **Model and RTL disagreement left unresolved.** Whichever is wrong, not knowing which means every downstream projection is unreliable.
- ◆ **Bottlenecks found without root cause.** Knowing throughput is low is not actionable; knowing which queue saturates is.
- ◆ **No time left to act.** Findings that arrive after RTL freeze can only be documented, not fixed.

WHERE THE EFFORT GOES	30 M/M	WHO IS ON IT ~10.9 FTE	CONNECTIONS	TERMS HERE	3
Bandwidth and utilization measurement	8.0	Performance verification lead OWNS IT	DEPENDS ON DEF-02 ARCH-01 DV-04 DV-08	KPI RTL DV	
Workload execution	6.0	Performance architect	FEEDS INTO DV-10 BU-08		
Bottleneck analysis	5.0	Emulation engineers	FEEDS BACK INTO ARCH-01		
Model correlation	5.0	Interconnect designers			
Latency analysis	3.0	Chief architect			
Findings and reconciliation	3.0				

09 DV · VERIFICATION

DV-10

Coverage closure and regression stability management

TAKES	COSTS	OWNER	CRITICAL PATH
26w cont. w12–w38 of the stage	45.0 M/M ~7.5 people while it runs	Regression owner CAD infrastructure approves	yes slipping moves the program

WHY IT EXISTS

Keep the regressions running and the **coverage moving toward closure**—triage failures, manage capacity, publish where every block stands, and close the holes that remain.

Coverage closure is not a phase at the end; it is a curve that either bends or does not. This activity owns the curve: which holes are real, which tests would close them, and whether the regression is stable enough for the numbers to mean anything.

WHAT IT DELIVERS

- DV-D3

Regression and coverage dashboards

This activity is the deliverable, and its value is that the numbers are current and trusted.
- DV-D8

DV closure signoff package

Closure is assembled here from the evidence every other DV activity produced.

NEEDS FIRST

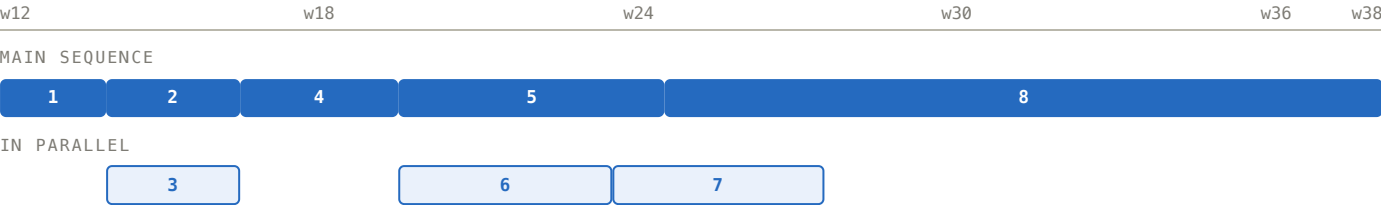
- Coverage model and criteria from DV-01
- Test suites from DV-03 through DV-09
- Regression capacity from PDK-12 and RTL-09
- Build and CI infrastructure from RTL-09
- Closure criteria per block

DONE WHEN

- Regression stable enough that a red result is believed
- Every hole analyzed before it is closed or waived
- Coverage curve flat at signoff, not still climbing

HOW IT GETS THERE

8 STEPS · 2 LANES



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Regression suite definition and tiering → Tiered regression suites	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Regression infrastructure and capacity management → Regression infrastructure and capacity plan	2.5w
3	Failure triage process and ownership → Failure triage process with ownership	2.5w runs in parallel
4	Coverage collection, merging and reporting → Merged coverage reporting	3w
5	Coverage hole analysis and test targeting → Hole analysis and targeted test list	5w
6	Regression stability and flake elimination → Regression stability and flake elimination record	4w runs in parallel
7	Dashboard publication and closure tracking → Published coverage dashboard	4w runs in parallel
8	Coverage closure and signoff assembly → Coverage dashboards and closure signoff	13.5w

Step 6 is unglamorous and load-bearing. A regression with intermittent failures trains everyone to ignore red results, and once that habit forms the regression has stopped being a signal.

WATCH OUT FOR

- ◆ **Flaky regressions tolerated.** Once red results are routinely ignored, the regression has stopped providing information and nobody notices when it matters.
- ◆ **Coverage merged without stability.** Numbers from an unstable regression are noise presented as a trend.
- ◆ **Holes closed by waiving rather than testing.** A hole waived without analysis is untested behavior recorded as covered.
- ◆ **Capacity insufficient at peak.** Regression queues at the point of closure directly extend the closure date.
- ◆ **Closure signed with the curve still climbing.** Coverage still rising at signoff means the plan's completion criteria were not what stopped the work.

WHERE THE EFFORT GOES	45 M/M	WHO IS ON IT	~7.5 FTE	CONNECTIONS	TERMS HERE	3
Coverage hole analysis and targeting	12.0	Regression owner OWNS IT		DEPENDS ON PDK-12 RTL-09 DV-01 DV-02 DV-03 DV-04 DV-05 DV-06 DV-07 DV-08 DV-09	CI DV CAD	
Closure and signoff assembly	10.0	Coverage lead				
Regression stability	8.0	Verification lead				
Coverage collection and reporting	6.0	Block verification owners		FEEDS INTO DV-12 SYN-05 S0-09 T0-03		
Triage process	5.0	CAD infrastructure				
Infrastructure and dashboards	4.0					

09 DV · VERIFICATION

DV-11

Gate-level simulation, functional and timing-annotated

TAKES	COSTS	OWNER	CRITICAL PATH
10w w28–w38 of the stage	20.0 M/M ~8.7 people while it runs	Gate-level simulation lead Physical design liaison approves	yes slipping moves the program

WHY IT EXISTS

Simulate the **gate-level netlist**, first functionally and then with real timing, to catch what RTL simulation structurally cannot—X-propagation, reset behavior, timing-dependent failures and synthesis or DFT-induced differences.

RTL simulation is optimistic about X and blind to timing. Gate-level simulation is slow, painful and the last check before the design becomes masks—it is also where scan insertion, clock gating and synthesis transformations are proved not to have changed behavior.

WHAT IT DELIVERS

DV-D7

Gate-level simulation report

This activity is the deliverable, and it is a tapeout checklist item rather than an optional confirmation.

NEEDS FIRST

- Gate-level netlist from SYN-01
- SDF timing from PD-09
- Scan-inserted netlist from DFT-06
- Verification environment from DV-02
- Power intent from RTL-05

DONE WHEN

- X-propagation issues fixed rather than suppressed
- Scan and test modes simulated, not only functional
- Timing simulation run at the corners that matter

HOW IT GETS THERE

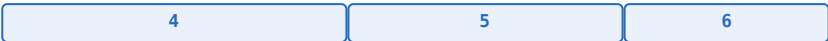
7 STEPS · 2 LANES

w28	w30	w32	w34	w36	w38
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Gate-level environment and netlist bring-up → Gate-level environment with the netlist brought up	1.5w
2	Zero-delay functional gate-level simulation → Functional gate-level results	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	X-propagation and reset initialization analysis → X-propagation and reset initialization findings	2w
4	SDF back-annotated timing simulation → SDF timing-annotated simulation results	2.5w
5	DFT and scan mode simulation → Scan and DFT mode simulation results	2w
6	Power-aware gate-level simulation → Gate-level simulation environment · Gate-level simulation report	1.5w
7	Failure debug and closure → Failure debug and closure record	4.5w

Step 3 finds the class of bug RTL simulation is structurally incapable of seeing. RTL X-optimism hides uninitialized state; at gate level the X propagates, and a design that boots in RTL can fail to come out of reset in gates.

WATCH OUT FOR

- ◆ **Skipped because it is slow.** The failures it finds—X propagation, reset, timing-dependent behavior—have no other pre-silicon detector.
- ◆ **Run only in functional mode.** Scan and test modes are how the part is tested at ATE, and a scan mode that does not work makes the part untestable.
- ◆ **Started after netlist churn has settled.** Waiting for a stable netlist means starting too late; running early on an imperfect netlist finds environment problems while there is time.
- ◆ **X failures suppressed rather than fixed.** Forcing initial values to make the simulation run hides exactly the uninitialized state the check exists to find.
- ◆ **Timing simulation on a subset of corners.** Timing-dependent failures appear at specific corners, and the corner skipped is the one that fails.

WHERE THE EFFORT GOES	20 M/M	WHO IS ON IT ~8.7 FTE	CONNECTIONS	TERMS HERE	5
Failure debug and closure	7.0	Gate-level simulation lead OWNS IT	DEPENDS ON AMS-10 RTL-05 RTL-06 DV-02 DFT-06 SYN-01 PD-09	RTL DV DFT SDF ATE	
Functional gate-level simulation	4.0	Verification engineers			
X-propagation analysis	3.5	Synthesis liaison			
SDF timing simulation	3.0	DFT engineer	FEEDS INTO DV-12 DFT-08 S0-09 S0-11 T0-02 BU-09		
Scan and DFT mode simulation	1.5	Physical design liaison			
Power-aware simulation	1.0				

09 DV · VERIFICATION

DV-12

Bug triage and disposition board

TAKES	COSTS	OWNER	CRITICAL PATH
34w cont. w6–w40 of the stage	25.0 M/M ~3.2 people while it runs	Verification lead Validation liaison approves	yes slipping moves the program

WHY IT EXISTS

Run the **bug disposition board**—triage every defect, decide what gets fixed, and make the risk of what does not explicit and owned.

Every program reaches tapeout with open bugs. The question is whether they were assessed and accepted or simply ran out of time, and the difference is entirely this activity. Its output at the end is the errata list bring-up will work against and the risk statement the Go / No-Go decision reads.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Defects from every DV activity
- Coverage and closure status from DV-10
- Change control decisions from RTL-10
- Silicon debug capability from DFT-11
- Tapeout criteria from the program plan

DONE WHEN

- Every deferred bug carries an assessed, owned risk
- Bug find rate turned over before closure is recommended
- Errata list ready for bring-up on day one

HOW IT GETS THERE

8 STEPS · 2 LANES

w6	w13	w20	w27	w34	w40
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Bug tracking, severity and triage process definition → Bug tracking and triage process	1.5w
2	Disposition board charter and membership → Disposition board decisions	1w
3	Continuous triage and disposition → Triage and disposition log	22w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Bug rate and closure trend analysis → Bug rate and closure trend analysis	4w runs in parallel
5	Deferred bug risk assessment → Deferred bug risk assessments	4w runs in parallel
6	Escape analysis on late-found defects → Escape analysis findings	3w runs in parallel
7	Open bug list and risk statement for tapeout → Open bug list and risk statement	5w
8	DV closure recommendation → DV closure recommendation	4.5w

Step 6 is the loop that improves the process rather than the design. Asking why a bug found in week 34 was not found in week 12 usually points at a missing coverage point or an untested mode, and fixing that finds the next one earlier.

WATCH OUT FOR

- ◆ **Bugs deferred without risk assessment.** An unassessed open bug at tapeout is a risk nobody quantified being accepted by nobody in particular.
- ◆ **Severity assigned by the finder.** Consistent severity requires a board; without one, the same defect is critical in one block and minor in another.
- ◆ **Bug rate still rising at closure.** A find rate that has not turned over means the design has more defects than have been found, whatever the coverage says.
- ◆ **No escape analysis.** The process that let a bug survive to week 34 will let the next one survive too.
- ◆ **Errata list assembled after tapeout.** Bring-up needs the known-issue list on day one, and assembling it later means debugging problems that were already known.

WHERE THE EFFORT GOES	25 M/M	WHO IS ON IT	~3.2 FTE	CONNECTIONS
Continuous triage and disposition	10.0	Verification lead OWNS IT		DEPENDS ON RTL-10 DV-01 DV-03 DV-04 DV-05 DV-06 DV-07 DV-10 DV-11
Open bug list and risk statement	4.0	RTL lead		FEEDS INTO S0-09 T0-03 T0-04 T0-05 BU-10
Deferred bug risk assessment	4.0	Chief architect		TERMS HERE 2 RTL DV
Trend analysis	3.0	Program manager		
Escape analysis	2.0	Validation liaison		
Closure recommendation	2.0			

DESIGN & VERIFY

10 DFT Architecture

DFT

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w18 – w58	40 _w	11	130 _{M/M}	5 _{of 11}
ENGINEERING ACTIVITIES				KEY DELIVERABLES
DFT-01	DFT architecture definition — scan style, compression ratio, hierarchical DFT		8w 12 _{M/M}	DFT-D1 DFT architecture specification and coverage plan
DFT-02	Test coverage and test-time target negotiation with product and test engineering		5w 4 _{M/M}	DFT-D2 MBIST / BISR insertion and test collateral
DFT-03	MBIST / BIRA / BISR architecture for embedded memories		10w 14 _{M/M}	DFT-D3 JTAG / IJTAG description files (BSDL, ICL, PDL)
DFT-04	Boundary scan and IJTAG (1149.1 / 1687), TAP and debug access architecture		8w 10 _{M/M}	DFT-D4 ATPG pattern sets with coverage report
DFT-05	On-chip clock controller design for at-speed testing		7w 8 _{M/M}	DFT-D5 DFT DRC clean report
DFT-06	Scan insertion and DFT DRC during synthesis		10w 14 _{M/M}	DFT-D6 Pattern validation (GLS) report and ATE-ready pattern files
DFT-07	ATPG pattern generation — stuck-at, transition, cell-aware — and coverage closure		16w 28 _{M/M}	DFT-D7 DFT signoff entry for the tapeout checklist
DFT-08	Pattern validation by gate-level simulation and ATE format conversion (STIL/WGL)		10w 14 _{M/M}	
DFT-09	Scan compression and chain routing feasibility with physical design		10w 10 _{M/M}	
DFT-10	eFuse, chip ID and memory repair infrastructure		6w 6 _{M/M}	
DFT-11	Silicon debug hooks — trace, observability, design-for-debug		8w 10 _{M/M}	

10 DFT · DFT ARCHITECTURE

DFT-01

DFT architecture definition — scan style, compression ratio, hierarchical DFT

TAKES		COSTS	OWNER	CRITICAL PATH
8w		12.0 M/M	DFT architect	yes
w0–w8 of the stage		~6.5 people while it runs	Physical design liaison approves	slipping moves the program
WHY IT EXISTS				
Decide how the chip will be tested before it is built to be testable—scan style, compression ratio, hierarchical partitioning, and the test modes the design will support. Every one of these choices constrains the design rather than following it. Compression ratio decides pattern volume and therefore test cost; hierarchical DFT decides whether a block can be tested independently; and both have to be settled before SYN-01 writes constraints against them.				
WHAT IT DELIVERS				
DFT-D1 DFT architecture specification and coverage plan This activity is the deliverable, and everything else in the stage is executed against it.				
NEEDS FIRST				
- Design hierarchy and partitioning from ARCH-02 - Coverage and test-time targets from DFT-02 - Memory list and BISR intent from ARCH-03 - Test platform constraints from TEST-02 - Debug access requirements from ARCH-07				
DONE WHEN				
- Architecture fixed before synthesis writes constraints against it - Compression ratio chosen against a test cost model - Specification reviewed by test and product engineering				
HOW IT GETS THERE				6 STEPS · 2 LANES
w0	w2	w4	w6	w8
MAIN SEQUENCE				
1 2 3 6				
IN PARALLEL				
4 5				
#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE			TAT
1	Test strategy definition — what is tested where and how → Test strategy statement			1.5w
2	Scan architecture and chain organization → Scan architecture and chain organization			2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Compression architecture and ratio selection → Compression architecture and ratio	1.5w
4	Hierarchical DFT partitioning against the design hierarchy → Hierarchical DFT partitioning	1.5w runs in parallel
5	Test mode definition and mode control → Test mode definition	1.5w runs in parallel
6	Architecture specification and review → DFT architecture specification	3w

Step 3 is a cost decision disguised as an architecture one. Compression ratio sets pattern volume, pattern volume sets tester memory and test time, and test time is a per-unit cost for the life of the product.

WATCH OUT FOR

- ◆ **Architecture decided after synthesis starts.** Scan insertion and constraints depend on it; deciding late means re-running both.
- ◆ **Compression ratio chosen without the test cost model.** Ratio trades pattern volume against silicon area, and only TEST-09's cost model says which side to favor.
- ◆ **Hierarchical DFT ignored on a large die.** Flat scan on a design this size produces pattern volumes and runtimes that become their own schedule problem.
- ◆ **Test modes defined without the security architecture.** Every test mode is an access path, and ARCH-07 has to lock the ones that should not exist in production.
- ◆ **Architecture written and not reviewed by test engineering.** A scheme that is elegant in design and unrunnable on the chosen ATE is discovered at TEST-08.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT	~6.5 FTE	CONNECTIONS	TERMS HERE	3
Scan and compression architecture	5.0	DFT architect OWNS IT		DEPENDS ON ARCH-02 ARCH-09 ARCH-10 DFT-02	DFT BISR ATE	
Architecture specification and review	3.0	Test engineering liaison		FEEDS INTO DFT-03 DFT-04 DFT-05 DFT-06 DFT-09 DFT-11 SYN-01 TEST-01		
Test strategy definition	2.0	Product engineering liaison				
Hierarchical partitioning	1.0	Security architect				
Test mode definition	1.0	Physical design liaison				

10 DFT · DFT ARCHITECTURE

DFT-02

Test coverage and test-time target negotiation with product and test engineering

TAKES	COSTS	OWNER	CRITICAL PATH		
5w w2–w7 of the stage	4.0 M/M ~3.5 people while it runs	DFT architect Program manager approves	no has float		
WHY IT EXISTS					
Agree the coverage and test-time targets with the people who pay for them—product engineering, who own quality, and test engineering, who own the tester seconds. Coverage and test time pull against each other, and neither belongs to DFT alone. A defect-per-million target sets required coverage; a cost-per-unit target sets allowable test time; and the architecture in DFT-01 is chosen to sit between them. Negotiating this first is what makes the architecture decidable.					
WHAT IT DELIVERS					
No key deliverable is owned here — this activity contributes to the ones below.					
NEEDS FIRST					
- Quality and DPPM targets from the product requirements - Test cost budget from DEF-04 - Defect density projection from TECH-07 - Test platform capability from TEST-02 - Previous-generation coverage and escape data					
DONE WHEN					
- Coverage target derived from DPPM, not inherited - Test time budget traced to a per-unit cost - Targets signed by product and test engineering					
HOW IT GETS THERE					
5 STEPS · 2 LANES					
w2	w3	w4	w5	w6	w7
MAIN SEQUENCE					
1 2 5					
IN PARALLEL					
3 4					
#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE				TAT
1	Quality target translation — DPPM to required coverage → Coverage target derived from the DPPM requirement				1w
2	Test time budget from the product cost model → Test time budget				1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Fault model selection — stuck-at, transition, cell-aware → Fault model selection	1w runs in parallel
4	Coverage target per fault model and per block class → Per-block-class coverage targets	1w runs in parallel
5	Target agreement and record → Agreed target record	3w

Step 3 has a large downstream cost. Cell-aware patterns catch defects the other models miss and multiply pattern count doing it, so the decision belongs here with the cost model in the room rather than at `DFT-07` when the patterns are being generated.

WATCH OUT FOR

- ◆ **Coverage target set by convention.** "99% stuck-at" carried from the last program may be far from what this product's DPPM requirement needs.
- ◆ **Test time budget not derived from cost.** A test time nobody costed becomes whatever the patterns need, and the per-unit cost is discovered at production.
- ◆ **Fault model chosen at pattern generation.** Adding cell-aware late multiplies pattern volume against a tester memory budget already fixed.
- ◆ **Targets agreed without product engineering.** The organization that owns field quality has to own the coverage number, or it will be renegotiated when patterns run long.
- ◆ **No per-block differentiation.** Memory-dominated blocks and control logic need different approaches, and a single flat target over-tests one and under-tests the other.

WHERE THE EFFORT GOES	4 M/M	WHO IS ON IT ~3.5 FTE	CONNECTIONS	TERMS HERE	3
Quality target translation	1.5	DFT architect owns it	DEPENDS ON DEF-04 TECH-07	DFT ATPG DPPM	
Test time budget	1.0	Product engineering	FEEDS INTO DFT-01 DFT-07 TEST-01		
Fault model selection	1.0	Test engineering	TEST-09		
Target agreement	0.5	Yield engineer			
		Program manager			

10 DFT · DFT ARCHITECTURE

DFT-03

MBIST / BIRA / BISR architecture for embedded memories

TAKES	COSTS	OWNER	CRITICAL PATH
10w w6–w16 of the stage	14.0 M/M ~6.1 people while it runs	DFT memory lead Verification engineer approves	yes slipping moves the program

WHY IT EXISTS

Architect **memory test and repair**—MBIST engines, BIRA analysis, BISR repair and the fuse path—for a design where memory is most of the die.

On an accelerator the arrays dominate both area and defect exposure, and they cannot be tested by scan. MBIST is how they are tested and repair is how the yield is recovered; both have to be designed jointly with AMS-09, because the array provides the spares and this activity provides the engine that uses them.

WHAT IT DELIVERS

DFT-D2

MBIST / BISR insertion and test collateral

This activity is the deliverable, including the repair collateral test engineering will run.

NEEDS FIRST

- Memory instances from PDK-04 and AMS-10
- Redundancy scheme from AMS-09
- Fuse map and lifecycle from ARCH-07
- Coverage targets from DFT-02
- Repair yield expectations from the yield model

DONE WHEN

- Repair interface written jointly with the memory designers
- Repair path verified end to end, not in pieces
- MBIST runtime inside the test time budget

HOW IT GETS THERE

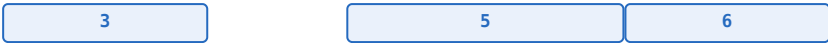
8 STEPS · 2 LANES

w6	w8	w10	w12	w14	w16
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Memory inventory and test requirement per instance → Memory inventory with test requirements	1.5w
2	MBIST architecture and engine allocation → MBIST architecture and engine allocation	2.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Algorithm selection per memory type → Algorithm selection per memory type	1.5w runs in parallel
4	BIRA and repair analysis architecture → BIRA and BISR architecture	2w
5	BISR and fuse path integration with ARCH-07 → Fuse path integration	2w runs in parallel
6	Repair interface agreement with the memory designers → Repair interface agreement with the memory designers	1.5w runs in parallel
7	MBIST insertion and integration → MBIST insertion and verification results	2.5w
8	Architecture verification and specification → Verified MBIST architecture specification	1.5w

Step 6 is the joint decision with <code>AMS-09</code>. Both sides can specify an internally consistent repair scheme that the other cannot implement, and writing the interface together is the only reliable way to avoid it.

WATCH OUT FOR

- ◆ **Repair interface specified separately from the array.** Two consistent specifications that disagree, found at pattern generation when both are frozen.
- ◆ **Algorithms chosen without the failure modes.** March algorithms differ in what they catch; a default choice misses the defect modes this array actually exhibits.
- ◆ **Fuse capacity insufficient.** Repair data has to be stored, and an allocation made without counting the worst-case repair set produces unrepairable parts.
- ◆ **MBIST runtime unbudgeted.** Memory test on an array-dominated die can dominate test time, and it is often costed after the architecture is fixed.
- ◆ **Repair not verified end to end.** A repair path proven in pieces and never exercised from failure through analysis to fuse blow fails at production test.

WHERE THE EFFORT GOES	14 M/M	WHO IS ON IT	~6.1 FTE	CONNECTIONS	TERMS HERE	5
MBIST architecture and insertion	5.0	DFT memory lead OWNS IT		DEPENDS ON ARCH-07 PDK-04 DFT-01	DFT MBIST BIRA BISR ATE	
BIRA and BISR architecture	3.5	Memory circuit designers		FEEDS INTO AMS-09 DFT-06 DFT-10		
Repair interface and fuse path	2.5	Fuse and lifecycle engineer		TEST-01 TEST-05 TEST-08		
Algorithm selection	1.5	Test engineering liaison		MP-05		
Architecture verification	1.5	Verification engineer		LATER INPUT AMS-09 +0w		

10 DFT · DFT ARCHITECTURE

DFT-04

Boundary scan and IJTAG (1149.1 / 1687),
TAP and debug access architecture

TAKES	COSTS	OWNER	CRITICAL PATH
8w w4–w12 of the stage	10.0 M/M ~5.4 people while it runs	DFT access lead Physical design liaison approves	no has float

WHY IT EXISTS

Design the **access network**—boundary scan, IJTAG, TAP controller and the debug paths—that every test, every repair and every silicon debug session runs through.

The TAP is the door into the chip. It is used at wafer sort, at package test, during bring-up and in the field, and it is also the door ARCH-07 needs to lock in production. Designing it is a negotiation between access and security rather than a purely technical exercise.

WHAT IT DELIVERS

DFT-D3

JTAG / IJTAG description files (BSDL, ICL, PDL)

This activity is the deliverable, and these files are what board test and ATE consume.

NEEDS FIRST

- DFT architecture from DFT-01
- Security lockdown policy from ARCH-07
- Debug requirements from validation and BU-09
- Package pin availability from ARCH-08
- IJTAG tooling from PDK-08

DONE WHEN

- Lockdown transitions tested, not only implemented
- Description files generated and validated before tapeout
- Debug access preserved for the lifecycle states that need it

HOW IT GETS THERE

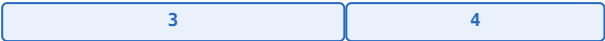
6 STEPS · 2 LANES

w4w6w8w10w12

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Access architecture — TAP, IJTAG network topology → TAP and IJTAG network architecture	1.5w
2	Boundary scan implementation per IEEE 1149.1 → Boundary scan implementation	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	IJTAG instrument network and description files → Instrument network with ICL and PDL descriptions · BSDL and description files	2w runs in parallel
4	Debug and trace access path definition → Debug and trace access paths	1.5w runs in parallel
5	Security lockdown integration per lifecycle state → Lifecycle-state lockdown integration	1.5w
6	Access network verification and description file generation → Access network verification results and description files	3.5w

Step 5 is where the negotiation with `ARCH-07` lands. Every path this activity opens for debug is a path security has to close in production lifecycle states, and the lock has to be testable without making the part permanently undebuggable.

WATCH OUT FOR

- ◆ **Debug access designed out for security.** A part that cannot be debugged in the field is a part whose first field failure has no diagnostic path.
- ◆ **Lockdown untested.** The lifecycle transition that locks the TAP is the one nobody exercises, because exercising it makes the part undebuggable.
- ◆ **Description files generated late.** BSDL and ICL are consumed by board test and ATE; generating them near tapeout leaves no time to validate them.
- ◆ **Access network verified only in functional mode.** The network exists to work when the design does not, and that is the mode that needs proving.
- ◆ **Pin budget for debug not reserved.** Trace ports need pins, and a bump map fixed without them means debug through a narrow channel.

WHERE THE EFFORT GOES	10 M/M	WHO IS ON IT	~5.4 FTE	CONNECTIONS
Access network verification and files	3.0	DFT access lead OWNS IT		DEPENDS ON ARCH-07 ARCH-08 DFT-01
TAP and IJTAG architecture	2.5	Security architect		FEEDS INTO DFT-08 DFT-10 DFT-11 TEST-08 BU-03
Boundary scan implementation	1.5	Validation liaison		
Debug and trace paths	1.5	Verification engineer		TERMS HERE 8
Security lockdown integration	1.5	Physical design liaison		DFT JTAG IJTAG ICL PDL BSDL TAP ATE

10 DFT · DFT ARCHITECTURE

DFT-05

On-chip clock controller design for at-speed testing

TAKES	COSTS	OWNER	CRITICAL PATH
7w w8–w15 of the stage	8.0 M/M ~5.0 people while it runs	DFT engineer, at-speed Physical design liaison approves	no has float

WHY IT EXISTS

Design the **on-chip clock controllers** that generate at-speed test clocks—because transition-fault testing needs clock pulses the tester cannot supply at frequency.

Stuck-at testing runs slowly and catches static defects. The defects that matter for a high-frequency part are timing-related, and catching them needs launch-capture pulses at functional speed. The tester cannot produce them, so the chip has to produce them for itself.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Clock architecture and domains from ARCH-05
- DFT architecture from DFT-01
- PLL interface from AMS-02
- Test platform clocking capability from TEST-02
- Transition fault coverage targets from DFT-02

DONE WHEN

- At-speed clocking available on every domain that needs coverage
- Cross-domain at-speed strategy defined and verified
- OCC inserted before clock tree synthesis, not after

HOW IT GETS THERE

6 STEPS · 2 LANES

w8	w10	w12	w14	w15
----	-----	-----	-----	-----

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	At-speed test requirement per clock domain → At-speed test requirements per domain	1w
2	OCC architecture and PLL interface → OCC architecture and PLL interface	1.5w
3	Launch-capture sequencing and pulse control → Launch-capture sequencing control	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Multi-domain and cross-domain at-speed strategy → Cross-domain at-speed strategy	1.5w runs in parallel
5	OCC insertion and integration → Inserted and verified OCC logic	1.5w runs in parallel
6	At-speed clocking verification → At-speed clocking verification results	3w

Step 4 is the difficult part. Single-domain at-speed test is well understood; testing paths that cross domains at speed requires the domains to have a defined phase relationship in test mode, which is a constraint on the clocking architecture rather than on DFT.

WATCH OUT FOR

- ◆ **At-speed test not supported on every domain.** Coverage then has a hole exactly where the fastest logic is, which is where transition defects concentrate.
- ◆ **Cross-domain paths untested at speed.** These are among the most timing-marginal paths in the design and the hardest to cover.
- ◆ **OCC interaction with the PLL unverified.** A controller that cannot get a clean clock from the PLL in test mode produces patterns that fail on good silicon.
- ◆ **Power during at-speed test ignored.** Scan shift followed by at-speed capture creates switching activity far above functional, and the IR drop can cause failures on good parts.
- ◆ **OCC inserted late.** It sits in the clock path, so adding it after clock tree synthesis means redoing the tree.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT ~5.0 FTE	CONNECTIONS	TERMS HERE	5
At-speed clocking verification	2.5	DFT engineer, at-speed owns it	DEPENDS ON ARCH-05 AMS-02 DFT-01	PLL IR DFT ATPG	
OCC architecture and interface	2.0	Clocking architect	FEEDS INTO DFT-06 DFT-07 SYN-02	OCC	
Launch-capture sequencing	1.5	PLL designer	PD-05 PD-07 TEST-01		
Cross-domain strategy	1.0	Test engineering liaison			
Insertion and integration	1.0	Physical design liaison			

10 DFT · DFT ARCHITECTURE

DFT-06

Scan insertion and DFT DRC during synthesis

TAKES	COSTS	OWNER	CRITICAL PATH
10w w14–w24 of the stage	14.0 M/M ~6.1 people while it runs	DFT implementation engineer DFT architect approves	yes slipping moves the program

WHY IT EXISTS

Insert the **scan chains, compression and test logic** into the design during synthesis, and drive DFT design-rule checking to clean.

This is where the architecture becomes logic. DFT DRC violations—uncontrollable clocks, unresettable flops, blocked scan paths—reduce coverage silently, and each one is a set of faults ATPG will never be able to target.

WHAT IT DELIVERS

DFT-D5

DFT DRC clean report

This activity is the deliverable, and clean means every violation fixed or explicitly accepted with its coverage cost.

NEEDS FIRST

- DFT architecture from DFT-01
- MBIST insertion from DFT-03
- OCC logic from DFT-05
- Synthesis flow from SYN-01
- Netlist from SYN-02

DONE WHEN

- Every DRC violation fixed or accepted with a measured coverage cost
- Chains balanced against the shift time budget
- Chain integrity verified in simulation before ATPG

HOW IT GETS THERE

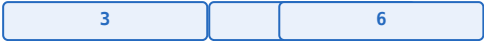
7 STEPS · 2 LANES

w14w16w18w20w22w24

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Scan insertion setup and configuration → Scan insertion configuration	1.5w
2	Scan chain insertion and stitching → Stitched scan chains	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Compression logic insertion → Inserted compression logic	1.5w runs in parallel
4	Test point insertion for coverage improvement → Test points for coverage improvement	1.5w runs in parallel
5	DFT DRC analysis and violation closure → DFT DRC clean results	3w
6	Scan chain integrity verification → Scan chain integrity verification results	1.5w runs in parallel
7	Insertion signoff and netlist handoff → Scan-inserted netlist	3.5w

Step 5 is the substance. Every DFT DRC violation is a region of logic ATPG cannot control or observe, and the coverage loss is silent—the tool simply reports a lower number without saying which violation caused it.

WATCH OUT FOR

- ◆ **DRC violations accepted without measuring their coverage cost.** Each one silently removes faults from the targetable set, and the total is invisible until ATPG reports a low number.
- ◆ **Test points inserted without area budget.** They improve coverage and cost area and timing, and adding them late competes with closure.
- ◆ **Chain balance ignored.** Unbalanced chains set shift time by the longest one, and shift time is most of the test time.
- ◆ **Insertion done once and not re-run.** Netlist changes after insertion leave chains stale, and the mismatch appears at pattern validation.
- ◆ **Chain integrity unverified.** A broken chain makes everything behind it untestable, and finding that at ATE rather than in simulation costs a silicon turn.

WHERE THE EFFORT GOES	14 M/M	WHO IS ON IT	~6.1 FTE	CONNECTIONS	TERMS HERE	6
DFT DRC closure	5.0	DFT implementation engineer OWNS IT		DEPENDS ON PDK-09 PDK-13 RTL-04 DFT-01 DFT-03 DFT-05	DRC DFT ATPG MBIST OCC ATE	
Scan insertion and stitching	3.5	Synthesis engineer		FEEDS INTO DV-11 DFT-07 DFT-09 SYN-05 SYN-10 PD-12 TEST-05		
Signoff and handoff	2.5	Block designers		LATER INPUT SYN-01 +0w		
Compression insertion	1.5	Physical design liaison				
Test point insertion	1.0	DFT architect				
Chain integrity verification	0.5					

10 DFT · DFT ARCHITECTURE

DFT-07

ATPG pattern generation — stuck-at, transition, cell-aware — and coverage closure

TAKES	COSTS	OWNER	CRITICAL PATH
16w w22–w38 of the stage	28.0 M/M ~7.6 people while it runs	ATPG engineer Product engineering approves	yes slipping moves the program

WHY IT EXISTS

Generate the **ATPG patterns**—stuck-at, transition and cell-aware—and drive coverage to the targets DFT-02 agreed, inside the pattern volume the tester can hold.

Two constraints pull against each other here: coverage has to reach the quality target and pattern volume has to fit tester memory and test time. Every technique that raises one raises the other, and closing both is the work.

WHAT IT DELIVERS

DFT-D4

ATPG pattern sets with coverage report

This activity is the deliverable, and the coverage report is what the quality target is measured against.

NEEDS FIRST

- Scan-inserted netlist from DFT-06
- Coverage targets and fault models from DFT-02
- Test time and memory budget from TEST-09
- Library fault models from PDK-03
- OCC and at-speed setup from DFT-05

DONE WHEN

- Coverage meets the DFT-02 targets per fault model
- Pattern volume inside tester memory without splitting
- Untestable faults analyzed rather than accepted

HOW IT GETS THERE

8 STEPS · 2 LANES

w22	w26	w30	w34	w38
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	ATPG setup, constraints and test mode configuration → ATPG setup and constraints	2w
2	Stuck-at pattern generation and coverage analysis → Stuck-at pattern set with coverage analysis	3w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Transition and at-speed pattern generation → Transition and at-speed pattern set	3.5w
4	Cell-aware pattern generation → Stuck-at, transition and cell-aware pattern sets	2.5w runs in parallel
5	Untestable fault analysis and coverage justification → Untestable fault analysis	2.5w runs in parallel
6	Pattern volume reduction and compaction → Compacted pattern volumes	3w
7	Coverage closure against the DFT-02 targets → Coverage report against targets	2w runs in parallel
8	Pattern set assembly and release → Released pattern sets	4.5w

Step 5 is what makes a coverage number honest. Faults reported as untestable are either genuinely untestable—tied off, redundant—or untestable because of a DRC violation, and only analysis distinguishes a real limit from a fixable one.

WATCH OUT FOR

- ◆ **Coverage target missed and accepted late.** The remedies—test points, DRC fixes, architecture changes—all need an RTL or netlist change, and none are available near tapeout.
- ◆ **Pattern volume exceeding tester memory.** Patterns that do not fit have to be split across loads, which multiplies test time and cost.
- ◆ **Untestable faults accepted without analysis.** A fixable DRC violation reported as an untestable fault is coverage given away.
- ◆ **Cell-aware added late.** It multiplies pattern count against a budget already set, forcing a choice between coverage and test time with no room to negotiate.
- ◆ **Patterns generated on a netlist that then changes.** Any netlist change after generation invalidates the set, and regeneration takes weeks.

WHERE THE EFFORT GOES	28 M/M	WHO IS ON IT	~7.6 FTE	CONNECTIONS	TERMS HERE	5
Pattern generation	10.0	ATPG engineer owns it		DEPENDS ON DFT-02 DFT-05 DFT-06 DFT-09	RTL DRC DFT ATPG OCC	
Pattern volume reduction	6.0	DFT architect		FEEDS INTO DFT-08 T0-03 T0-04 TEST-08 MP-06		
Untestable fault analysis	4.5	Test engineering liaison		LATER INPUT PD-12 +6w TEST-09 +6w		
Pattern set assembly and release	4.0	Block designers				
Coverage closure	3.5	Product engineering				

10 DFT · DFT ARCHITECTURE

DFT-08

Pattern validation by gate-level simulation and ATE format conversion (STIL/WGL)

TAKES	COSTS	OWNER	CRITICAL PATH
10w w28–w38 of the stage	14.0 M/M ~6.1 people while it runs	DFT validation engineer DFT architect approves	yes slipping moves the program

WHY IT EXISTS

Prove the patterns **actually run**—simulate them against the netlist with timing, convert them to tester format, and hand ATE a set that works.

A pattern set that passes in ATPG and fails in gate-level simulation is a pattern set that would have failed on silicon. This activity is the last place a pattern problem is cheap, and it is also where the format conversion for the tester is proved rather than assumed.

WHAT IT DELIVERS

- DFT-D6

Pattern validation (GLS) report and ATE-ready pattern files

This activity is the deliverable—patterns proved to run rather than assumed to.
- DFT-D7

DFT signoff entry for the tapeout checklist

DFT signs the checklist on validated patterns, not on generated ones.

NEEDS FIRST

- Pattern sets from DFT-07
- Gate-level netlist and SDF from PD-09
- Simulation environment from DV-11
- Tester format requirements from TEST-08
- Timing sets and tester capability from TEST-02

DONE WHEN

- Patterns validated with timing, not only zero-delay
- Format conversion validated against the actual tester capability
- DFT tapeout checklist entry signed on validated patterns

HOW IT GETS THERE

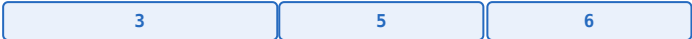
7 STEPS · 2 LANES

w28	w30	w32	w34	w36	w38
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Pattern simulation environment setup → Pattern simulation environment	1.5w
2	Zero-delay pattern validation by gate-level simulation → Zero-delay and timing-annotated validation results	2w
3	Timing-annotated pattern simulation runs in parallel → Timing-annotated pattern simulation results	2w
4	Pattern failure debug and regeneration → Regenerated patterns for failures	2.5w
5	STIL and WGL format conversion runs in parallel → STIL and WGL format files	1.5w
6	Tester format and timing set validation runs in parallel → Tester format validation results	1.5w
7	Pattern release and DFT tapeout signoff → Released ATE-ready pattern set	4w

Step 3 catches what zero-delay cannot: patterns that violate timing at the capture edge. On an at-speed pattern set this is not a formality—a launch-capture sequence that closes in ATPG and fails in timing simulation would have produced yield loss on good die.

WATCH OUT FOR

- ◆ **Validation skipped for schedule.** A pattern failure then appears at ATE bring-up, where the diagnosis competes with everything else happening at first silicon.
- ◆ **Only zero-delay simulation run.** Timing-related pattern failures are exactly what at-speed patterns are meant to detect, and they are invisible without SDF.
- ◆ **Format conversion assumed.** STIL and WGL translation has tool-specific quirks, and a set that converts cleanly can still time incorrectly on the tester.
- ◆ **Regeneration cost underestimated.** A pattern failure requiring regeneration puts the whole set back through DFT-07, which takes weeks not days.
- ◆ **Timing sets not validated against the actual tester.** A pattern that assumes an edge placement the ATE cannot produce fails on hardware for a reason no simulation shows.

WHERE THE EFFORT GOES	14 M/M	WHO IS ON IT	~6.1 FTE	CONNECTIONS	TERMS HERE	7
Pattern release and signoff	4.0	DFT validation engineer OWNS IT		DEPENDS ON DV-11 DFT-04 DFT-07 PD-09	GLS DFT ATPG STIL WGL SDF ATE	
Failure debug and regeneration	3.5	Gate-level simulation engineer		FEEDS INTO S0-11 T0-03 EVB-06 TEST-08 BU-02 BU-09 MP-06		
Gate-level validation	2.5	ATPG engineer				
Timing-annotated simulation	2.0	Test engineering				
Format conversion and validation	2.0	DFT architect				

10 DFT · DFT ARCHITECTURE

DFT-09

Scan compression and chain routing feasibility with physical design

TAKES	COSTS	OWNER	CRITICAL PATH		
10w w16–w26 of the stage	10.0 M/M ~4.3 people while it runs	DFT physical lead Clock architect approves	no has float		
WHY IT EXISTS					
Work with physical design on whether the scan chains and compression network can actually be routed—and fix the plan before routing rather than after. Compression networks connect every chain to a shared decompressor and compactor, which produces long routes across the die competing with functional signals for the same resource. A compression ratio that is optimal on paper and unroutable in practice is discovered at PD-06, when the remedies are expensive.					
WHAT IT DELIVERS					
No key deliverable is owned here — this activity contributes to the ones below.					
NEEDS FIRST					
- DFT architecture and compression from DFT-01 - Scan insertion from DFT-06 - Floorplan and partitioning from PD-01 and PD-02 - Routing resource from the metal stack in TECH-02 - Congestion analysis from PD-06					
DONE WHEN					
- Compression network confirmed routable at the chosen ratio - Chain reordering strategy preserves diagnosis capability - Physical DFT plan agreed before detailed routing starts					
HOW IT GETS THERE					
6 STEPS · 2 LANES					
w16	w18	w20	w22	w24	w26
MAIN SEQUENCE					
126					
IN PARALLEL					
345					
#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE				TAT
1	Chain routing feasibility analysis with the floorplan → Chain routing feasibility findings				1.5w
2	Compression network topology against physical partitioning → Compression topology aligned to physical partitions				2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Chain ordering and reordering strategy → Chain ordering and reordering strategy	1.5w runs in parallel
4	Congestion impact assessment → Congestion impact assessment	1.5w runs in parallel
5	Test clock distribution feasibility → Test clock distribution plan	1.5w runs in parallel
6	Physical DFT plan agreement with physical design → Agreed physical DFT plan	6.5w

Step 3 is where the two disciplines usually collide. DFT wants chains ordered for balance and diagnosis; physical design wants them ordered for routing. Reordering after placement is the standard compromise, and agreeing it here avoids arguing about it during closure.

WATCH OUT FOR

- ◆ **Compression ratio unroutable.** A ratio chosen for pattern volume alone produces a network that consumes routing resource the functional design needs.
- ◆ **Chain reordering breaking diagnosis.** Physical reordering improves routing and can destroy the chain-to-block mapping diagnosis depends on.
- ◆ **Test clock distribution treated as functional.** Scan shift clocks reach every flop and have their own skew and power requirements.
- ◆ **Feasibility assessed after routing starts.** The findings then arrive when the only remaining remedy is to reduce coverage.
- ◆ **Hierarchical partition boundaries ignored.** Chains crossing physical partitions carry a routing and timing cost that a logical view does not show.

WHERE THE EFFORT GOES	10 M/M	WHO IS ON IT ~4.3 FTE	CONNECTIONS	TERMS HERE	2
Compression topology and feasibility	3.5	DFT physical lead owns it	DEPENDS ON DFT-01 DFT-06	DRC DFT	
Physical DFT plan agreement	2.5	Physical design engineers	FEEDS INTO DFT-07 PD-06 PD-12		
Chain ordering strategy	1.5	DFT architect	LATER INPUT PD-01 +2w PD-02 +7w		
Congestion assessment	1.5	Diagnosis engineer			
Test clock distribution	1.0	Clock architect			

10 DFT · DFT ARCHITECTURE

DFT-10

eFuse, chip ID and memory repair infrastructure

TAKES	COSTS	OWNER	CRITICAL PATH
6w w12–w18 of the stage	6.0 M/M ~4.3 people while it runs	Fuse and infrastructure engineer Verification engineer approves	no has float

WHY IT EXISTS

Build the **fuse and identity infrastructure**—eFuse arrays, chip ID, repair storage, lifecycle state—that repair, traceability and security all depend on.

Fuses are written once and read forever. They store memory repair, trim values, chip identity and lifecycle state, and every one of those consumers has its own requirements. Getting the capacity or the read path wrong produces parts that cannot be repaired, identified or secured.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Repair storage requirement from DFT-03 and AMS-09
- Trim requirements from AMS-04
- Lifecycle and security states from ARCH-07
- Traceability requirements from MP-09
- eFuse capability from PDK-02

DONE WHEN

- Capacity sized against worst-case, not average, repair need
- Programming and read paths verified before tapeout
- Lifecycle transitions exercisable in test

HOW IT GETS THERE

7 STEPS · 2 LANES

w12	w14	w16	w18
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Fuse consumer inventory — repair, trim, ID, lifecycle, security → Fuse consumer inventory	1w
2	Capacity allocation against worst-case need → Capacity allocation with worst-case margin	1w
3	eFuse array integration and programming path → eFuse array integration and programming path	1.5w runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Fuse read, shadow register and distribution design → Fuse read and distribution logic	1.5w
5	Chip identity and traceability scheme → Chip identity scheme	1w
6	Lifecycle state encoding with ARCH-07 → Lifecycle state encoding agreed with the security architecture	1w
7	Fuse infrastructure verification → Verified fuse infrastructure	2.5w

Step 2 is the step that fails silently. Every consumer asks for a modest allocation, the total is set once, and the worst-case repair set—which is a distribution rather than an average—exceeds it on the parts that most needed repairing.

WATCH OUT FOR

- ◆ **Capacity sized to the average repair need.** Repair need is a distribution, and the parts needing most repair are exactly the ones the allocation fails.
- ◆ **Programming path untested.** Fuses are one-time; a programming path that does not work correctly produces parts that are permanently wrong.
- ◆ **Fuse read timing in the boot path unconsidered.** Repair and trim values have to be distributed before the logic that uses them is released from reset.
- ◆ **Traceability scheme designed late.** Chip identity is needed for yield analysis and field returns, and retrofitting it means parts that cannot be traced.
- ◆ **Lifecycle encoding without a test path.** A lifecycle transition that cannot be exercised in test is a transition whose correctness is unknown until the field.

WHERE THE EFFORT GOES	6 M/M	WHO IS ON IT	~4.3 FTE	CONNECTIONS	TERMS HERE	6
Fuse infrastructure verification	2.0	Fuse and infrastructure engineer owns it		DEPENDS ON ARCH-07 PDK-02 AMS-09 DFT-03 DFT-04	DFT MBIST BISR JTAG IJTAG ID	
Read and distribution design	1.5	DFT memory lead		FEEDS INTO TEST-05 BU-02 MP-06 MP-09		
Array integration and programming	1.0	Security architect				
Capacity allocation	0.8	Product engineering				
Identity and lifecycle encoding	0.8	Verification engineer				

10 DFT · DFT ARCHITECTURE

DFT-11

Silicon debug hooks — trace, observability, design-for-debug

TAKES	COSTS	OWNER	CRITICAL PATH
8w w6–w14 of the stage	10.0 M/M ~5.4 people while it runs	DFT debug lead Verification engineer approves	no has float

WHY IT EXISTS

Design the **observability the lab will need**—trace, triggers, internal visibility, design-for-debug—because at bring-up the only view into the chip is the one designed in months earlier.

Silicon debug is constrained entirely by what was built. A failure that cannot be observed cannot be diagnosed, and the difference between a two-day debug and a two-month one is usually whether someone put a trigger and a trace port where it was needed.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Debug requirements from validation and BU-09
- Access network from DFT-04
- Scan infrastructure from DFT-06
- Pin and bump budget from ARCH-08
- Security lockdown policy from ARCH-07

DONE WHEN

- Debug paths verified before tapeout, not first used at bring-up
- Trace bandwidth sufficient for the events validation named
- Capabilities documented for the people who will use them

HOW IT GETS THERE

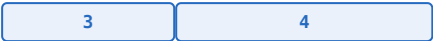
6 STEPS · 2 LANES

w6	w8	w10	w12	w14
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Debug requirement collection from validation and design → Debug requirement list	1w
2	Observability architecture — trace, triggers, snapshot → Observability architecture	2w
3	Trace port and bandwidth budget → Trace port and bandwidth budget	1w

runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Trigger and cross-trigger network → Trigger and cross-trigger network runs in parallel	1.5w
5	Internal state snapshot and scan dump capability → Scan dump and snapshot capability	1.5w
6	Debug infrastructure verification and documentation → Debug infrastructure documentation	3.5w

Step 5 is the cheapest debug capability available. Scan chains already reach every flop; adding the ability to halt and dump their contents turns the test infrastructure into a full internal state view for almost no extra area.

WATCH OUT FOR

- ◆ **Debug traded away for area.** The saving is small and the cost is paid at bring-up, in weeks, by a different team.
- ◆ **Requirements collected from design rather than validation.** The people who will debug the silicon know what they need to see; the people who built it usually assume it works.
- ◆ **Trace bandwidth inadequate.** A trace port too narrow to capture the event of interest produces a debug capability that cannot answer the question.
- ◆ **Debug infrastructure unverified.** It is used first when the chip is misbehaving, which is the worst moment to discover the debug path is broken.
- ◆ **No documentation.** A capability the lab does not know exists is a capability that was not built.

WHERE THE EFFORT GOES	10 M/M	WHO IS ON IT ~5.4 FTE	CONNECTIONS
Debug infrastructure verification and documentation	3.5	DFT debug lead owns it	DEPENDS ON ARCH-07 ARCH-08 DFT-01 DFT-04
Observability architecture	2.5	Validation liaison	FEEDS INTO TEST-10 BU-05 BU-09
Trigger network	1.5	Design leads	MP-05
Snapshot and scan dump	1.5	Security architect	TERMS HERE 3
Trace port budget	1.0	Verification engineer	DFT JTAG IJTAG

IMPLEMENT

11 Logic Synthesis & Netlist Drops

SYN

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w42 – w66	24_w	12	136_{M/M}	6_{of 12}

ENGINEERING ACTIVITIES

SYN-01	SDC constraint development and validation	8w	18 M/M
SYN-02	Technology mapping and optimization — multi-Vt, multi-bit, useful skew	8w	16 M/M
SYN-03	N0 flow-flush netlist release — functionally incomplete, structurally representative, issued so PD can build the flow	4w	6 M/M
SYN-04	Physical-aware synthesis with congestion feedback from the PD flow setup	7w	14 M/M
SYN-05	N1 netlist drop — first quality turn against PD placement and congestion feedback	5w	12 M/M
SYN-06	Power optimization — clock gating, operand isolation, Vt mix	7w	12 M/M
SYN-07	Low-power synthesis — isolation and level-shifter insertion against UPF	5w	8 M/M
SYN-08	N2 netlist drop — timing and congestion converged, late RTL changes absorbed	5w	12 M/M
SYN-09	FFN (final full netlist) release — RTL frozen, ECOs closed, no functional change beyond this point	4w	10 M/M
SYN-10	Formal equivalence checking per drop, RTL versus netlist	20w cont.	12 M/M
SYN-11	Timing, area and power reporting against budget, per drop	22w cont.	10 M/M
SYN-12	Netlist handoff and QoR delta review with physical design, per drop	22w cont.	6 M/M

KEY DELIVERABLES

SYN-D1	N0 flow-flush netlist for PD flow setup
SYN-D2	N1 and N2 netlist drops with QoR delta reports
SYN-D3	FFN — final full netlist, release-tagged
SYN-D4	Validated SDC constraint set per mode and corner
SYN-D5	Synthesis QoR report per drop against PPA targets
SYN-D6	Formal equivalence clean report per drop
SYN-D7	Power intent implementation report
SYN-D8	Physical design handoff package per drop

11 SYN · LOGIC SYNTHESIS & NETLIST DROPS

SYN-01

SDC constraint development and validation

TAKES	COSTS	OWNER	CRITICAL PATH
8w w0–w8 of the stage	18.0 M/M ~9.8 people while it runs	Constraints lead Signoff methodology engineer approves	yes slipping moves the program

WHY IT EXISTS

Write the **constraints the whole implementation is optimized and signed off against**—clocks, generated clocks, IO timing, exceptions, test modes—and validate them rather than assume them.

Constraints are the specification synthesis and physical design actually read. A missing clock definition produces a design optimized for a path nobody cares about; a wrong false path produces one that closes in the tool and fails in silicon. Every hour spent validating them here saves days at PD-09.

WHAT IT DELIVERS

SYN-D4

Validated SDC constraint set per mode and corner

This activity is the deliverable, and "validated" is what distinguishes it from a file of assertions.

NEEDS FIRST

- Clock architecture and DVFS points from ARCH-05
- Interface timing requirements from ARCH-04
- Corner and mode definition from PDK-11
- DFT test modes from DFT-01
- RTL hierarchy from RTL-04

DONE WHEN

- No unconstrained paths in the design
- Every exception justified by a design reason, not by a violation
- One constraint set used by both synthesis and signoff

HOW IT GETS THERE

6 STEPS · 2 LANES

w0	w2	w4	w6	w8
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
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1	Mode and corner enumeration against the operating conditions → Mode and corner enumeration	1.5w
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#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Clock and generated clock definition across the design → Clock and generated clock definitions	2w
3	Exception, false path and multicycle specification → Exception and multicycle specification	1.5w runs in parallel
4	IO and interface timing constraints → IO and interface constraints	2w
5	Test mode and scan constraints → Test mode constraints	1.5w runs in parallel
6	Constraint validation and quality checking → Constraint validation reports	2.5w

Step 6 is not review, it is tooling. Constraint quality checkers find unconstrained paths, conflicting exceptions and clocks that reach nothing, and they find them in minutes—which is why every drop should be re-checked rather than checked once.

WATCH OUT FOR

- ◆ **Unconstrained paths reported as clean.** A path with no constraint is a path the tool ignores, and it appears in the report as no violation at all.
- ◆ **Exceptions written to close timing.** A false path added because the path would not close is a silicon failure being converted into a green report.
- ◆ **Constraints diverging between synthesis and signoff.** Two constraint sets means optimizing against one and being judged against the other.
- ◆ **Test mode constraints written late.** Scan shift and at-speed modes have their own timing, and a design closed only in functional mode fails at ATE.
- ◆ **IO constraints assumed from the interface specification.** Actual budgets come from the package and channel analysis in SIPI-07, not from the protocol document.

WHERE THE EFFORT GOES	18 M/M	WHO IS ON IT	~9.8 FTE	CONNECTIONS	TERMS HERE	7
Clock and generated clock definitions	5.0	Constraints lead OWNS IT		DEPENDS ON DEF-03 ARCH-05 ARCH-09 PDK-01 PDK-08 PDK-09 PDK-11 PDK-13 RTL-04 RTL-06 RTL-09 DFT-01	RTL SDC QoR IO DVFS DFT ATE	
Constraint validation	4.5	Timing engineers		FEEDS INTO DV-11 SYN-02 SYN-03 SYN-05 SYN-11 PD-01 PD-09 S0-02		
IO and interface constraints	3.5	DFT engineer		FEEDS BACK INTO DFT-06		
Exception specification	3.0	Physical design liaison				
Test mode constraints	2.0	Signoff methodology engineer				

11 SYN · LOGIC SYNTHESIS & NETLIST DROPS

SYN-02

Technology mapping and optimization
— multi-Vt, multi-bit, useful skew

TAKES 8w w2–w10 of the stage	COSTS 16.0 M/M ~8.7 people while it runs	OWNER Synthesis lead PPA lead approves	CRITICAL PATH yes slipping moves the program
WHY IT EXISTS Map the RTL onto the qualified library and optimize it for the PPA budget—Vt mix, multi-bit banking, useful skew, datapath structure. Technology mapping is where architecture becomes gates and where most of the achievable PPA is decided. The same RTL mapped two ways can differ by 20% in area and a hundred millivolts in achievable frequency, and the difference is entirely in how this activity is configured.			
WHAT IT DELIVERS No key deliverable is owned here — this activity contributes to the ones below.			
NEEDS FIRST - Qualified libraries from PDK-03 - Constraints from SYN-01 - RTL from RTL-02 and RTL-04 - PPA budgets from ARCH-06 - Flow and methodology from PDK-09			
DONE WHEN - Area recovered on non-critical paths, not only speed optimized - Vt menu used across its range, not defaulted - QoR reported normalized so drops are comparable			
HOW IT GETS THERE 6 STEPS · 2 LANES w2 w4 w6 w8 w10 MAIN SEQUENCE 1 2 4 6 IN PARALLEL 3 5			
# STEP, AND WHAT IT ADDS TO THE DELIVERABLE TAT			
1 Library and mapping strategy selection → Mapping strategy and library configuration 1.5w			
2 Baseline mapping and optimization → Baseline mapped netlist 2.5w			
3 Datapath and arithmetic optimization → Datapath optimization results 1.5w runs in parallel			

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Multi-bit banking and useful skew configuration → Multi-bit and useful skew configuration	2w
5	Area recovery on non-critical paths → Area recovery results	1.5w runs in parallel
6	QoR analysis and optimization iteration → QoR analysis per iteration	2w

Step 5 is where area is won without cost. Most of a design is not on the critical path, and recovering area and leakage there is free performance-wise—but it only happens if someone configures the flow to do it rather than optimizing everything for speed.

WATCH OUT FOR

- ◆ **Everything optimized for speed.** Area and leakage on non-critical paths are given away for performance nobody asked for, and the die grows.
- ◆ **Multi-bit banking configured late.** Clock power depends heavily on it, and adding it after closure means re-timing the design.
- ◆ **Useful skew used to hide a timing problem.** It buys real margin and also makes the clock tree harder to build; using it to rescue a path that should be restructured moves the problem to PD-07.
- ◆ **Library subset used without knowing it.** Synthesis using only part of the Vt menu produces avoidable leakage, and nobody notices because the design closes.
- ◆ **QoR compared drop to drop without normalizing.** A netlist with more RTL in it is not directly comparable to the last one, and the trend becomes noise.

WHERE THE EFFORT GOES	16 M/M	WHO IS ON IT	~8.7 FTE	CONNECTIONS	TERMS HERE	3
Baseline mapping and optimization	5.0	Synthesis lead OWNS IT		DEPENDS ON TECH-02 PDK-03 PDK-09 AMS-10 AMS-15 RTL-02 RTL-04 RTL-08 DFT-05 SYN-01	PPA RTL QoR	
QoR analysis and iteration	4.0	Synthesis engineers				
Multi-bit and useful skew	3.0	Library engineer				
Datapath optimization	2.0	Block owners		FEEDS INTO SYN-03 SYN-04 SYN-05 SYN-06 SYN-07 SYN-10 SYN-11 PD-05		
Area recovery	2.0	PPA lead				

11 SYN · LOGIC SYNTHESIS & NETLIST DROPS

SYN-03

N0 flow-flush netlist release — functionally incomplete, structurally representative, issued so PD can build the flow

TAKES 4w w3–w7 of the stage	COSTS 6.0 M/M ~6.5 people while it runs	OWNER Synthesis lead Configuration manager approves	CRITICAL PATH yes slipping moves the program
--	--	--	---

WHY IT EXISTS

Release the **No flow-flush netlist**—functionally incomplete, structurally representative—so physical design can build and debug its flow against a real database months before the design is ready.

No exists to be wrong in the right way. It has the right hierarchy, the right macros and roughly the right size, and it does not need to work. Its entire purpose is to let PD-01 find its flow problems on a throwaway netlist rather than on the one the schedule depends on.

WHAT IT DELIVERS

SYN-D1

N0 flow-flush netlist for PD flow setup

This activity is the deliverable, and its value is entirely in arriving early rather than in being correct.

NEEDS FIRST

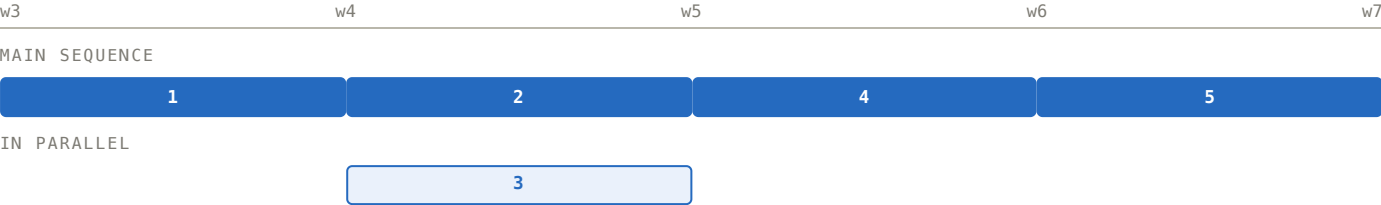
- RTL in whatever state exists from RTL-02
- Macro list and abstracts from AMS-15
- Constraints from SYN-01
- Flow setup requirements from PD-01
- Hierarchy from ARCH-02

DONE WHEN

- Representative in hierarchy, instance count and macro content
- Limitations stated explicitly with the release
- Delivered on the date PD's flow setup needs it, not when it is good

HOW IT GETS THERE

5 STEPS · 2 LANES



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	N0 scope definition — what must be representative, what may be stubbed → N0 scope definition — representative versus stubbed	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Black-box and stub strategy for incomplete blocks → N0 scope definition and stub strategy · Black-boxed and stubbed netlist	1w
3	Constraint bootstrap for N0 → Bootstrap constraint set	1w runs in parallel
4	Elaboration and structural sanity checking → Structural sanity check results	1w
5	Release and physical design handover → Released N0 netlist with a stated limitation list	1w

Step 1 is the whole activity. N0 has to be representative in the dimensions PD's flow cares about—hierarchy, macro count, instance count, pin count—and can be arbitrarily wrong in everything else. Getting that distinction right is what makes it cheap.

WATCH OUT FOR

- ◆ **N0 delayed until it is good.** The point is to be early; a correct N0 delivered late has become N1 and left PD's flow undebugged.
- ◆ **Not representative in the dimensions that matter.** A netlist with a tenth of the instances exercises none of the runtime and capacity problems PD needs to find.
- ◆ **Limitations not stated.** Physical design chasing timing on a stubbed netlist wastes a week before someone explains it was never going to close.
- ◆ **Stub strategy that changes the hierarchy.** A hierarchy that differs from the real one means the flow is set up for a structure that will not arrive.
- ◆ **Treated as a milestone rather than a tool.** N0 is scaffolding; celebrating it as progress distracts from the fact that no real design has been synthesized.

WHERE THE EFFORT GOES	6 M/M	WHO IS ON IT ~6.5 FTE	CONNECTIONS	TERMS HERE	1
		Synthesis lead OWNS IT	DEPENDS ON RTL-02 SYN-01 SYN-02	RTL	
Scope definition and stubs	2.0				
Elaboration and structural checks	1.5	Physical design lead	FEEDS INTO SYN-12 PD-01 PD-02		
Release and handover	1.5	Integration engineer			
Bootstrap constraints	1.0	Constraints engineer	LATER INPUT AMS-15 +1w		
		Configuration manager			

11 SYN · LOGIC SYNTHESIS & NETLIST DROPS

SYN-04

Physical-aware synthesis with congestion feedback from the PD flow setup

TAKES 7w w6–w13 of the stage	COSTS 14.0 M/M ~8.7 people while it runs	OWNER Synthesis engineers CAD engineer approves	CRITICAL PATH no has float
WHY IT EXISTS Synthesize with the floorplan in the loop—physical-aware mapping, congestion feedback, placement-driven restructuring—so the netlist that reaches physical design is already routable. Logic synthesis without physical information optimizes for a wire model that does not exist. On a large die the difference between estimated and actual wire delay is large enough to invalidate the optimization, and the correction happens either here or as weeks of congestion work at PD-06.			
WHAT IT DELIVERS No key deliverable is owned here — this activity contributes to the ones below.			
NEEDS FIRST - Floorplan from PD-02 - Mapped netlist from SYN-02 - Congestion analysis from PD-06 - Macro placement and blockages from PD-02 - Constraints from SYN-01			
DONE WHEN - Netlist synthesized against a floorplan PD is actually using - Congestion feedback flowing in both directions - Runtime cost justified by a measured QoR improvement			
HOW IT GETS THERE 6 STEPS · 2 LANES w6w8w10w12w13 1246 35			
#STEP, AND WHAT IT ADDS TO THE DELIVERABLETAT 1Physical-aware flow setup with the floorplan → Physical-aware synthesis flow1.5w 2Floorplan-aware mapping and optimization → Floorplan-aware mapped netlist2w			

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Macro and blockage awareness in placement → Macro and blockage aware placement	1.5w runs in parallel
4	Congestion feedback loop with physical design → Congestion feedback findings	2w
5	Placement-driven logic restructuring → Restructured logic in congested regions	1.5w runs in parallel
6	QoR comparison against non-physical synthesis → QoR comparison against non-physical synthesis	1.5w

Step 6 answers whether the added runtime is worth it. Physical-aware synthesis is slower; comparing its QoR against the wire-load-model result is what justifies the cost, and occasionally shows the floorplan is the problem rather than the logic.

WATCH OUT FOR

- ◆ **Floorplan too immature to synthesize against.** Physical-aware synthesis against a floorplan that then changes produces optimization for a layout that never exists.
- ◆ **Congestion feedback one-directional.** Synthesis restructures and physical design re-places; without both, each optimizes against the other's stale result.
- ◆ **Runtime cost not justified.** Physical-aware flows are slow, and without the QoR comparison nobody knows whether the extra hours bought anything.
- ◆ **Restructuring breaking hierarchy.** Aggressive cross-boundary optimization improves QoR and makes the netlist harder to close hierarchically and harder to equivalence-check.
- ◆ **Blockages not modelled.** Macros and their keep-out regions dominate placement on this die, and synthesis blind to them produces logic that cannot be placed where it assumed.

WHERE THE EFFORT GOES	14 M/M	WHO IS ON IT	~8.7 FTE	CONNECTIONS
Congestion feedback loop	4.0	Synthesis engineers	owns it	DEPENDS ON SYN-02 PD-02
Floorplan-aware mapping	3.5	Physical design engineers		FEEDS INTO SYN-05 SYN-08 PD-05
Flow setup	2.5	Synthesis lead		PD-06
Logic restructuring	2.0	Floorplan owner		TERMS HERE 2
QoR comparison	2.0	CAD engineer		QoR CAD

11 SYN · LOGIC SYNTHESIS & NETLIST DROPS

SYN-05

N1 netlist drop — first quality turn against PD placement and congestion feedback

TAKES	COSTS	OWNER	CRITICAL PATH
5w w8–w13 of the stage	12.0 M/M ~10.4 people while it runs	Synthesis lead PPA lead approves	yes slipping moves the program

WHY IT EXISTS

Release the **N1 drop—the first netlist meant to be closed**—against PD's feedback from its flow setup, and establish the QoR baseline every later drop is measured against.

No proved the flow ran. N1 is where the design's actual QoR first becomes visible: real timing, real area, real congestion. The number it establishes is the one PD-05 will work from and the one everyone will argue about for the next three months.

WHAT IT DELIVERS

SYN-D2

N1 and N2 netlist drops with QoR delta reports

The first of the two quality drops, with the baseline it establishes.

NEEDS FIRST

- RTL release from RTL-02
- Physical-aware results from SYN-04
- PD flow setup feedback from PD-01
- Constraints from SYN-01
- PPA budgets from ARCH-06

DONE WHEN

- QoR baseline established against a substantially complete design
- Delta against N0 explained, not only reported
- Issue list handed forward to N2

HOW IT GETS THERE

6 STEPS · 2 LANES

w8	w9	w10	w11	w12	w13
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	RTL drop intake and delta review against N0 → RTL drop intake and delta review	1w
2	Synthesis run and block-level closure → Block-level synthesis closure	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Physical design feedback incorporation → Physical design feedback incorporation record	1w runs in parallel
4	QoR delta analysis against the N0 baseline → QoR delta analysis against N0	1.5w
5	Issue list for the next drop → Issue list carried to N2	1w runs in parallel
6	Release and handoff to physical design → N1 netlist · Handoff package for PD turn 1	1w

Step 5 is what makes the drop cadence a process rather than a sequence of releases. Each drop should hand forward a named list of what it could not fix and why, so the next one starts from a position rather than from scratch.

WATCH OUT FOR

- ◆ **N1 released before RTL is worth synthesizing.** A drop taken from unstable RTL produces a baseline that means nothing and a PD turn that chases moving targets.
- ◆ **QoR delta not analyzed.** Knowing the numbers without knowing what changed them leaves the next drop guessing about what to fix.
- ◆ **Issue list not carried forward.** N2 then rediscovers the same problems, and the drop cadence loses its cumulative benefit.
- ◆ **PD feedback arriving after the drop.** Feedback from flow setup has to reach synthesis before N1 is run, not after it is released.
- ◆ **Baseline set on a partial design.** A QoR baseline missing significant blocks flatters the trend and hides the growth that follows.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT ~10.4 FTE	CONNECTIONS	TERMS HERE	3
Synthesis run and closure	4.0	Synthesis lead owns it	DEPENDS ON RTL-02 DV-10 DFT-06 SYN-01 SYN-02 SYN-04 PD-01	PPA RTL QoR	
QoR delta analysis	3.0	Synthesis engineers	RUNS WITH SYN-11		
Feedback incorporation	2.5	Physical design lead	FEEDS INTO SYN-08 SYN-12 PD-02 PD-05		
Intake and release	2.5	RTL lead PPA lead			

11 SYN · LOGIC SYNTHESIS & NETLIST DROPS

SYN-06

Power optimization — clock gating, operand isolation, Vt mix

TAKES	COSTS	OWNER	CRITICAL PATH
7w w9–w16 of the stage	12.0 M/M ~7.4 people while it runs	Power optimization engineer Library engineer approves	no has float

WHY IT EXISTS

Reduce power where it is actually spent—**clock gating, operand isolation, Vt mix**—driven by real switching activity rather than by defaults.

Power optimization without activity data is guesswork. Clock gating inserted everywhere costs area and saves little; inserted where the data says the logic is idle it saves a meaningful fraction of dynamic power. The activity data comes from verification, which makes this a cross-stage dependency more than a synthesis setting.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Switching activity from DV-04 workload simulations
- Power budgets from ARCH-06
- Mapped netlist from SYN-02
- Library Vt menu from PDK-03
- Clock architecture from ARCH-05

DONE WHEN

- Optimization driven by real activity data, not defaults
- Clock gating efficiency measured, not assumed
- Power reported at the operating point the product uses

HOW IT GETS THERE

6 STEPS · 2 LANES

w9	w11	w13	w15	w16
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Power analysis baseline and hot-spot identification → Power analysis baseline and hot spots	1.5w
2	Clock gating insertion and efficiency analysis → Clock gating insertion and efficiency report	2w
3	Vt mix optimization for leakage → Vt mix optimization results	1.5w

runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Operand isolation and datapath gating → Operand isolation implementation	2w
5	Activity-driven optimization with real switching data → Activity-driven optimization results	1.5w runs in parallel
6	Power QoR and comparison against the budget → Power QoR against budget	1.5w

Step 5 is what separates power optimization from power decoration. Switching activity from the `DV-04` workload simulations tells the tool which logic is actually idle; without it, gating is inserted by structural heuristic and saves a fraction of what it could.

WATCH OUT FOR

- ◆ **Optimization driven by default activity.** Uniform switching assumptions produce gating where it saves nothing and none where it would.
- ◆ **Clock gating efficiency unmeasured.** Gates inserted without measuring how often they actually gate cost area and leakage for no benefit.
- ◆ **Vt mix skewed to low-Vt for timing.** Leakage grows non-linearly, and a design closed on low-Vt everywhere blows its static power budget.
- ◆ **Power measured at the wrong operating point.** The number that matters is at the DVFS point the product runs at, not at nominal.
- ◆ **Optimization deferred to the final turn.** Power changes timing; a design closed on timing and then power-optimized has to be re-closed.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT	~7.4 FTE	CONNECTIONS	TERMS HERE	4
Clock gating	3.5	Power optimization engineer OWNS IT		DEPENDS ON ARCH-05 ARCH-06 PDK-03 RTL-05 DV-04 SYN-02	PPA QoR DVFS UPF	
Vt mix optimization	2.5	Synthesis engineers		FEEDS INTO SYN-08 SYN-11 PD-09 PD-10 S0-04		
Operand isolation	2.5	Verification liaison				
Power analysis	2.0	PPA lead				
QoR comparison	1.5	Library engineer				

11 SYN · LOGIC SYNTHESIS & NETLIST DROPS

SYN-07

Low-power synthesis — isolation and level-shifter insertion against UPF

TAKES	COSTS	OWNER	CRITICAL PATH
5w w10–w15 of the stage	8.0 M/M ~7.0 people while it runs	Low-power synthesis engineer Library engineer approves	no has float

WHY IT EXISTS

Implement the **power intent in the netlist**—isolation cells, level shifters, retention flops, always-on paths—from the UPF, and prove the netlist and the intent still agree.

UPF describes what should happen; synthesis is where it becomes cells. An isolation cell missing at a domain boundary is a design that corrupts data on power-down, and it is invisible in any functional check that does not model power. The consistency check between netlist and intent is the only reliable detector.

WHAT IT DELIVERS

SYN-D7

Power intent implementation report

This activity is the deliverable—what was inferred, what was inserted, and where intent and netlist agree.

NEEDS FIRST

- UPF from RTL-05
- Low-power cells from PDK-03
- Power domain definition from ARCH-05
- Mapped netlist from SYN-02
- Low-power flow from PDK-09

DONE WHEN

- Every domain boundary has verified isolation
- UPF and netlist consistent, checked rather than assumed
- Retention and always-on requirements handed to physical design

HOW IT GETS THERE

6 STEPS · 2 LANES

w10	w11	w12	w13	w14	w15
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	UPF ingestion and elaboration in the synthesis flow → Elaborated UPF in the synthesis flow	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Isolation and level shifter inference and insertion → Inserted isolation and level shifter cells	1.5w
3	Always-on and feed-through path handling → Always-on path implementation	1w runs in parallel
4	Retention cell insertion and control → Retention cell insertion and control logic	1.5w
5	UPF and netlist consistency checking → UPF-netlist consistency results	1w runs in parallel
6	Power intent implementation report → Power intent implementation report	1w

Step 5 runs in parallel and is the one that must not be skipped. The netlist and the UPF are two descriptions of the same intent, and any divergence between them propagates into physical design and verification as a defect nobody can attribute.

WATCH OUT FOR

- ◆ **Isolation inferred but not verified.** A missing cell at one boundary corrupts state on power-down and appears in no functional test.
- ◆ **Retention cells inserted without physical planning.** They have a different footprint and power connectivity, and adding them late disturbs the floorplan.
- ◆ **Always-on paths not identified.** Logic that must survive a domain shutting down and is not marked always-on is powered down with it.
- ◆ **UPF and netlist diverging.** Two descriptions of the same intent, and downstream tools reading different ones.
- ◆ **Level shifters omitted between voltage domains.** The design functions in simulation and fails at the voltage extremes in silicon.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT	~7.0 FTE	CONNECTIONS	TERMS HERE	1
Isolation and level shifter insertion	2.5	Low-power synthesis engineer OWNS IT		DEPENDS ON ARCH-06 PDK-03 PDK-09 RTL-05 RTL-08 SYN-02	UPF	
Retention cell insertion	2.0					
UPF elaboration	1.5	Low-power implementation lead		FEEDS INTO DV-06 SYN-08 SYN-10 PD-03 S0-09		
Consistency checking	1.0					
Implementation report	1.0	Physical design liaison				
		Verification liaison				
		Library engineer				

SYN-08

N2 netlist drop — timing and congestion converged, late RTL changes absorbed

TAKES	COSTS	OWNER	CRITICAL PATH
5w w14–w19 of the stage	12.0 M/M ~10.4 people while it runs	Synthesis lead Program manager approves	yes slipping moves the program

WHY IT EXISTS

Release the **N2 drop with everything enabled**—full optimization, power, low-power intent, physical awareness—and state honestly how much closure risk remains.

N2 is the last drop before the FFN and the last chance to change anything structurally. Its QoR is the best prediction available of what the final turn will achieve, and the closure risk statement it carries is what tells the program whether the tapeout date is credible.

WHAT IT DELIVERS

SYN-D2

N1 and N2 netlist drops with QoR delta reports

The second quality drop, and the last structurally changeable one.

NEEDS FIRST

- RTL updates and ECOs since N1
- Turn 1 results and feedback from PD-05
- Power optimization from SYN-06
- Low-power implementation from SYN-07
- Physical-aware results from SYN-04

DONE WHEN

- Closure risk stated specifically, with paths, blocks and options
- Turn-1 feedback incorporated rather than deferred
- Late changes assessed before inclusion

HOW IT GETS THERE

6 STEPS · 2 LANES

w14	w15	w16	w17	w18	w19
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	RTL and ECO intake since N1 → RTL and ECO intake record since N1	1w
2	Full synthesis with all optimizations enabled → N2 netlist with full optimization	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Late RTL change impact assessment → Late change impact assessment	1w runs in parallel
4	Congestion and timing convergence check → Congestion and timing convergence analysis	1.5w
5	Physical design turn-1 feedback incorporation → Turn-1 feedback incorporation record	1w runs in parallel
6	Release with a closure risk statement → Closure risk statement	1w

Step 6 is the honest part. A drop released with "timing is converging" tells nobody anything; one released with "forty paths in two blocks remain, all in the compute cluster, requiring a restructure or a frequency concession" gives the program a decision.

WATCH OUT FOR

- ◆ **Closure risk understated.** An optimistic risk statement at N2 removes the program's last opportunity to take a frequency or feature concession while it is still cheap.
- ◆ **Late RTL changes absorbed without assessment.** A functional change arriving between N1 and N2 can invalidate the turn-1 closure work entirely.
- ◆ **All optimizations enabled for the first time at N2.** Enabling power, low-power and physical awareness together for the first time produces a QoR nobody can attribute.
- ◆ **Turn-1 feedback not incorporated.** N2 then repeats N1's physical problems, and turn 2 spends its time on the same issues.
- ◆ **Treated as the final netlist.** N2 is not the FFN, and freezing behavior around it removes the RTL freeze discipline that SYN-09 depends on.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT ~10.4 FTE	CONNECTIONS	TERMS HERE	4
Full synthesis run	4.0	Synthesis lead owns it	DEPENDS ON SYN-04 SYN-05 SYN-06 SYN-07 PD-05	RTL ECO QoR FFN	
Convergence check	3.0	Synthesis engineers	RUNS WITH SYN-11		
Change impact assessment	2.5	Physical design lead	FEEDS INTO SYN-09 PD-06 S0-01		
Release and risk statement	2.5	RTL lead Program manager			

SYN-09

FFN (final full netlist) release — RTL frozen, ECOs closed, no functional change beyond this point

TAKES	COSTS	OWNER	CRITICAL PATH
4w w19–w23 of the stage	10.0 M/M ~10.9 people while it runs	Synthesis lead Physical design lead approves	yes slipping moves the program

WHY IT EXISTS

Release the **FFN—the final full netlist**—against frozen RTL, with constraints frozen and equivalence proved, and declare that nothing functional changes after it.

The FFN is a commitment more than a deliverable. Everything after it is physical: PD-13 closes on this netlist and S0-02 signs off on what PD produces. A functional change after the FFN restarts both, which is why the freeze around it has to be real.

WHAT IT DELIVERS

SYN-D3

FFN — final full netlist, release-tagged

This activity is the deliverable, and the milestone the stage ends on.

NEEDS FIRST

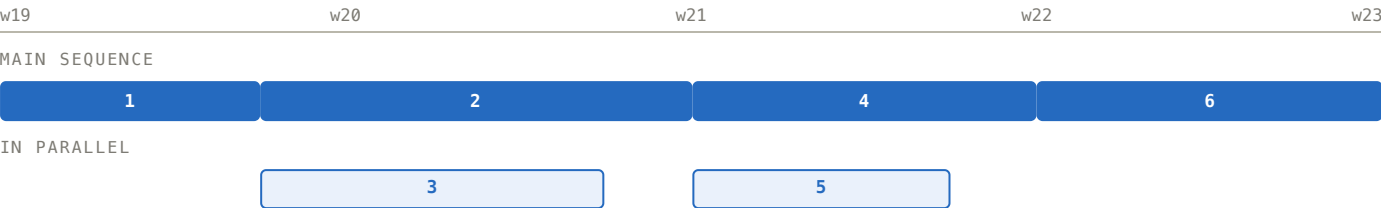
- Frozen RTL from RTL-10
- Constraints from SYN-01
- Optimization configuration from SYN-06 and SYN-07
- N2 results and closure risk from SYN-08
- Equivalence methodology from SYN-10

DONE WHEN

- Equivalence proved on the actual FFN, not on a proxy
- Constraints frozen with the netlist
- Functional freeze declared and its exception process published

HOW IT GETS THERE

6 STEPS · 2 LANES



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	RTL freeze confirmation and final intake → RTL freeze confirmation	0.75w
2	Final synthesis run → Final synthesis run and netlist	1.25w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Final constraint freeze → Frozen constraint set	1w runs in parallel
4	Final equivalence and UPF consistency verification → Final equivalence and UPF consistency results	1w
5	Handoff package assembly → Handoff package for the final turn	0.75w runs in parallel
6	FFN release and functional freeze declaration → FFN release with functional freeze declaration	1w

Step 1 is a confirmation rather than a formality. If `RTL-10` has not actually declared freeze, the FFN is a snapshot of a moving design, and the final turn in `PD-13` will be closing against something that changes underneath it.

WATCH OUT FOR

- ◆ **FFN released against RTL that is not actually frozen.** The final turn then closes against a moving netlist, and every change restarts it.
- ◆ **Constraints still changing after the FFN.** A design closed against one constraint set and signed off against another has not been signed off.
- ◆ **Equivalence not proved on the final netlist.** The last synthesis run is the one most likely to contain a transformation error, and the least likely to be checked under schedule pressure.
- ◆ **Functional freeze declared and not enforced.** Post-FFN functional ECOs are sometimes unavoidable and must be exceptional, visible and costed.
- ◆ **Handoff package incomplete.** The final turn needs constraints, UPF, DFT collateral and abstracts together; a missing item stops closure at its start.

WHERE THE EFFORT GOES	10 M/M	WHO IS ON IT	~10.9 FTE	CONNECTIONS	TERMS HERE	6
Final synthesis run	3.5	Synthesis lead	OWNS IT	DEPENDS ON RTL-10 SYN-08 SYN-10 PD-06	RTL SDC UPF LEC FFN DFT	
Equivalence and consistency	2.5	Constraints lead		FEEDS INTO PD-13 S0-02 S0-09 T0-01		
Release and freeze declaration	2.0	Equivalence engineer				
Constraint freeze	1.0	RTL lead				
Handoff package	1.0	Physical design lead				

11 SYN · LOGIC SYNTHESIS & NETLIST DROPS

SYN-10 Formal equivalence checking per drop, RTL versus netlist			
TAKES	COSTS	OWNER	CRITICAL PATH
20w cont. w4–w24 of the stage	12.0 M/M ~2.6 people while it runs	Equivalence engineer Synthesis lead approves	no has float

WHY IT EXISTS

Prove that **every netlist still means what the RTL meant**—formal equivalence per drop, including through DFT insertion, clock gating and low-power transformations.

Synthesis rewrites the design, and the tools that do it are not infallible. Equivalence checking is the only evidence that the netlist implements the RTL that verification actually verified; without it, a transformation error is discovered in silicon as behavior nobody can trace to a bug.

WHAT IT DELIVERS

SYN-D6 Formal equivalence clean report per drop

This activity is the deliverable, and "per drop" is what makes it a control rather than a final check.

NEEDS FIRST

- RTL from RTL-02 and RTL-04
- Netlists from every synthesis drop
- Scan-inserted netlist from DFT-06
- UPF and low-power implementation from SYN-07
- Equivalence tools from PDK-08

DONE WHEN

- Every drop checked, not only the final one
- Non-equivalences resolved rather than waived
- Chip-level equivalence clean on the FFN

HOW IT GETS THERE

8 STEPS · 2 LANES

w4	w8	w12	w16	w20	w24
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Equivalence methodology and tool setup → Equivalence methodology and setup	1.5w
2	Block-level equivalence per drop → Per-drop block-level equivalence results	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Black-box and constraint handling for incomplete drops → Black-box and constraint handling record	2w runs in parallel
4	Chip-level equivalence per drop → Block and chip-level equivalence results per drop	3w
5	DFT and clock gating equivalence handling → DFT and clock gating handling configuration	2w runs in parallel
6	Non-equivalence debug and resolution → Non-equivalence debug records	2.5w
7	Per-drop equivalence reporting → Per-drop equivalence reports	2w runs in parallel
8	Continuous checking across the drop sequence → Equivalence checking log across the drop sequence	11w

Step 5 is where most of the difficulty lives. Scan insertion and clock gating change the netlist in ways that are correct and not structurally equivalent, and setting up the checker to accept those and reject everything else is the skilled part of the activity.

WATCH OUT FOR

- ◆ **Equivalence run only on the final netlist.** A transformation error introduced at N1 is then found at the FFN, when the debug competes with tapeout.
- ◆ **Non-equivalences waived to make progress.** Each waiver is a place where the netlist may not implement the verified RTL, and they accumulate quietly.
- ◆ **DFT transformations not modelled.** The checker reports thousands of differences and everyone stops reading the report.
- ◆ **Black-box handling too permissive.** Boxing out a block to get a clean result excludes exactly the logic that was not checked.
- ◆ **Debug capacity underestimated.** A single real non-equivalence on a design this size can take a week to localize, and there is rarely a week available.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT	~2.6 FTE	CONNECTIONS	TERMS HERE	4
Continuous checking	4.0	Equivalence engineer	owns it	DEPENDS ON PDK-08 DFT-06 SYN-02 SYN-07	RTL UPF FFN DFT	
Non-equivalence debug	3.0	Synthesis engineers		FEEDS INTO SYN-09 S0-09 T0-02		
Chip-level equivalence	2.0	DFT engineer				
Methodology and setup	1.5	Low-power engineer				
DFT and low-power handling	1.5	Synthesis lead				

11 SYN · LOGIC SYNTHESIS & NETLIST DROPS

SYN-11

Timing, area and power reporting against budget, per drop

TAKES	COSTS	OWNER	CRITICAL PATH
22w cont. w2–w24 of the stage	10.0 M/M ~2.0 people while it runs	PPA reporting engineer Program manager approves	no has float

WHY IT EXISTS

Report **timing, area and power against the budget after every drop**, and escalate when a block is diverging rather than when it has already blown its allocation.

QoR reported in absolute terms tells nobody anything. Reported against the ARCH-06 allocation and trended across drops, it says which blocks are converging, which are not, and how much of the gap is still recoverable—which is the only version a program manager can act on.

WHAT IT DELIVERS

SYN-D5

Synthesis QoR report per drop against PPA targets

This activity is the deliverable, and "against targets" is what makes it usable.

NEEDS FIRST

- Synthesis results from every drop
- PPA budgets from ARCH-06
- Constraints from SYN-01
- Power results from SYN-06
- Physical-aware results from SYN-04

DONE WHEN

- Every report references the budget, not only the absolute number
- Divergence escalated on a rule, not on judgment
- Trends normalized so drops are comparable

HOW IT GETS THERE

6 STEPS · 2 LANES

w2	w7	w12	w17	w22	w24
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Reporting framework and budget ingestion → Reporting framework with budget ingestion	1.5w
2	Per-drop timing report and analysis → Per-drop timing, area and power report	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Budget comparison and escalation rules → Budget comparison per block · Escalation records runs in parallel	2w
4	Per-drop area and power report → Per-drop timing, area and power reports	2w
5	Trend analysis across the drop sequence → Trend analysis across drops runs in parallel	2w
6	Continuous reporting across the stage → Reporting log across the stage	16.5w

Step 5 is the part that predicts. A block 10% over budget and improving is a different problem from one 5% over and flat, and only a trend across drops distinguishes them.

WATCH OUT FOR

- ◆ **QoR reported in absolute terms.** Numbers with no budget reference cannot be acted on, and everyone forms their own opinion of whether they are acceptable.
- ◆ **No escalation rule.** Divergence noticed and not escalated becomes a surprise at the final turn, when there is nothing left to do about it.
- ◆ **Reports not normalized across drops.** More RTL in a later drop makes the trend meaningless unless the comparison accounts for it.
- ◆ **Power reported at nominal only.** The number that matters is at the product's operating point, and reporting at nominal understates it consistently.
- ◆ **Reporting stopping between drops.** The interesting question is whether a block is improving, and that needs data between releases as well as at them.

WHERE THE EFFORT GOES	10 M/M	WHO IS ON IT	~2.0 FTE	CONNECTIONS	TERMS HERE	3
Continuous reporting	4.0	PPA reporting engineer	owns it	DEPENDS ON ARCH-06 PDK-12 SYN-01 SYN-02 SYN-06	PPA RTL QoR	
Per-drop reports	2.5	Synthesis engineers		RUNS WITH SYN-05 SYN-08		
Budget comparison	2.0	PPA lead		FEEDS INTO SYN-12 PD-09		
Trend analysis	1.5	Block owners		FEEDS BACK INTO ARCH-06		
		Program manager				

11 SYN · LOGIC SYNTHESIS & NETLIST DROPS

SYN-12

Netlist handoff and QoR delta review with physical design, per drop

TAKES	COSTS	OWNER	CRITICAL PATH
22w cont. w2–w24 of the stage	6.0 M/M ~1.2 people while it runs	Synthesis lead PPA lead approves	no has float

WHY IT EXISTS

Hand each netlist to physical design as a **complete, accepted package**—netlist, constraints, UPF, DFT collateral, abstracts—and review the QoR delta together rather than throwing results over a wall.

The handoff is where two teams either share a picture of the design or maintain two. Six man-months across the stage buys a structured package and a joint review per drop, and it prevents the pattern where physical design spends a week discovering what synthesis already knew.

WHAT IT DELIVERS

SYN-D8

Physical design handoff package per drop

This activity is the deliverable, and acceptance rather than delivery is its standard.

NEEDS FIRST

- Netlists from every drop
- Constraints from SYN-01
- UPF and low-power implementation from SYN-07
- DFT collateral from DFT-06
- QoR reports from SYN-11

DONE WHEN

- Every package accepted rather than merely delivered
- QoR reviewed jointly per drop
- Issues tracked across drops rather than re-raised

HOW IT GETS THERE

6 STEPS · 2 LANES

w2	w7	w12	w17	w22	w24
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Handoff package definition — contents and format → Handoff package definition and format	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Per-drop package assembly → Per-drop handoff package	1.5w
3	Acceptance criteria agreed with physical design → Acceptance criteria agreed with PD	1.5w runs in parallel
4	QoR delta review with physical design per drop → QoR delta review records	1.5w
5	Issue tracking between drops → Issue tracking between drops	1.5w runs in parallel
6	Continuous handoff across the drop sequence → Assembled packages per drop	18w

Step 3 is what makes a handoff a transaction rather than a delivery. Acceptance criteria mean physical design can reject a package that is incomplete, and rejection at handoff is far cheaper than discovering the gap three days into a turn.

WATCH OUT FOR

- ◆ **Packages delivered without acceptance.** Physical design discovers the gap mid-turn, and the turn absorbs the delay.
- ◆ **QoR reviewed asynchronously.** A report read separately produces two interpretations, and the disagreement surfaces at the next escalation.
- ◆ **Issues not tracked between drops.** Each handoff then starts fresh, and the same problems are raised at every drop.
- ◆ **Package contents changing between drops.** Physical design automates against the format, and an inconsistent package breaks their flow.
- ◆ **Handoff treated as a synthesis milestone.** It is a shared event; a handoff that PD has not accepted has not happened.

WHERE THE EFFORT GOES	6 M/M	WHO IS ON IT ~1.2 FTE	CONNECTIONS	TERMS HERE 4
Continuous handoff	2.5	Synthesis lead OWNS IT	DEPENDS ON SYN-03 SYN-05 SYN-11	PPA QoR UPF DFT
QoR delta review	1.5	Physical design lead	FEEDS INTO PD-01 PD-05 PD-06	
Package assembly	1.0	Configuration manager	PD-13	
Acceptance criteria	1.0	DFT engineer		
		PPA lead		

IMPLEMENT

12 Physical Design

PD

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w46 – w76	30 _w	16	449 _{M/M}	8 _{of 16}
ENGINEERING ACTIVITIES				KEY DELIVERABLES
PD-01	Flow setup and MMMC environment build on the N0 flow-flush netlist — scripts, decks, runtimes, capacity		7w 22 M/M	PD-D1 Flow setup release — MMMC environment, scripts and runtime baseline
PD-02	Floorplan, macro placement and partition definition		9w 32 M/M	PD-D2 Turn 1 and Turn 2 databases with QoR delta reports
PD-03	Power delivery network design and early IR analysis		8w 20 M/M	PD-D3 Final-turn routed database on the FFN, per block and top
PD-04	Bump and RDL planning with chip-package co-design feedback		8w 14 M/M	PD-D4 Floorplan and PDN specification
PD-05	Turn 1 on the N1 drop — placement, CTS and first full route; QoR baseline established		8w 45 M/M	PD-D5 MCMM timing closure reports with violation burn-down across turns
PD-06	Turn 2 on the N2 drop — congestion, timing and power convergence; closure risk quantified		8w 45 M/M	PD-D6 Interim physical DRC / LVS clean
PD-07	Clock tree synthesis with skew and jitter budgeting, refined across turns		12w cont. 26 M/M	PD-D7 Bump map, RDL and package interface files
PD-08	Detailed routing and DRC convergence		16w cont. 40 M/M	PD-D8 ECO log and change control record
PD-09	Multi-corner multi-mode timing closure		22w cont. 60 M/M	PD-D9 Signoff-ready database handoff
PD-10	Signal and power integrity iteration — crosstalk, EM		10w 18 M/M	
PD-11	Chip power model (CPM/CPS) extraction and handoff to package co-verification		5w 6 M/M	
PD-12	Scan chain reordering and DFT-aware routing		6w 8 M/M	
PD-13	Final turn on the FFN — full closure, no functional change admitted		11w 55 M/M	
PD-14	Functional and timing ECO implementation		18w cont. 22 M/M	
PD-15	Chip finishing — seal ring, dummy and metal fill, alignment marks		5w 6 M/M	

PD-16	Hierarchical block closure and top-level assembly	14w	30 M/M
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12 PD · PHYSICAL DESIGN

PD-01

Flow setup and MMMC environment build on the N0 flow-flush netlist — scripts, decks, runtimes, capacity

TAKES	COSTS	OWNER	CRITICAL PATH
7w w0–w7 of the stage	22.0 M/M ~13.7 people while it runs	Physical design flow lead Synthesis liaison approves	yes slipping moves the program

WHY IT EXISTS

Build the flow—**scripts, MMMC environment, decks, runtimes, capacity**—and debug it on the No flow-flush netlist rather than on the netlist the schedule depends on.

This is why SYN-03 releases a netlist nobody expects to close. Every flow problem—a deck that does not load, a scenario that does not converge, a runtime that exceeds the night—is found here on a throwaway database, weeks before the same problem would have cost a turn.

WHAT IT DELIVERS

PD-D1

Flow setup release — MMMC environment, scripts and runtime baseline

This activity is the deliverable, and the runtime baseline is the part the schedule depends on.

NEEDS FIRST

- N0 netlist from SYN-03
- Golden environment and flow from PDK-13
- Constraints from SYN-01
- Corner and mode definition from PDK-11
- Compute and license capacity from PDK-12

DONE WHEN

- Flow runs end to end on N0, not on a fragment
- Runtime baseline measured and fed to capacity planning
- Hierarchical and block-level flows both defined

HOW IT GETS THERE

6 STEPS · 2 LANES

w0	w2	w4	w6	w7
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Flow scope definition and script framework → Physical design script framework	1.5w
2	MMMC environment build across corners and modes → MMMC environment across corners and modes	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Script, deck and library integration → Integrated decks and libraries	1.5w runs in parallel
4	N0 flow-flush run, end to end → N0 end-to-end run results	2w
5	Hierarchical and block-level flow definition → Hierarchical flow definition	1.5w runs in parallel
6	Runtime baseline and capacity assessment → Runtime baseline and capacity assessment	1.5w

Step 6 is what makes the schedule credible. A flow that runs correctly and takes forty hours per iteration sets a closure cadence nobody planned for, and knowing that number now is what lets `PDK-12` buy capacity before it is needed.

WATCH OUT FOR

- ◆ **Flow setup on the first real netlist.** Every flow problem then costs a turn, and the turns are the schedule.
- ◆ **MMMC scenarios incomplete.** A corner or mode missing from the setup is one the design is never closed at, and it is found at S0-02.
- ◆ **Runtime not measured.** A flow that works and takes two days per iteration produces a closure schedule nobody planned, discovered when closure starts.
- ◆ **Hierarchical flow deferred.** A design this size cannot be closed flat; deferring the block-level flow means building it under pressure.
- ◆ **Capacity assumed rather than booked.** The flow's peak concurrency is known here, and licences take weeks to procure.

WHERE THE EFFORT GOES	22 M/M	WHO IS ON IT ~13.7 FTE	CONNECTIONS	TERMS HERE	3
MMMC environment build	6.0	Physical design flow lead owns it	DEPENDS ON ARCH-08 ARCH-09 PDK-01 PDK-07 PDK-08 PDK-09 PDK-11 PDK-12 PDK-13 AMS-15 RTL-08 SYN-01 SYN-03 SYN-12	MMMC MCMM CAD	
N0 flow-flush run	5.0	CAD methodology engineer	FEEDS INTO SYN-05 PD-02 PD-05		
Flow scope and scripts	4.0	Timing methodology engineer	FEEDS BACK INTO PDK-12 DFT-09		
Deck and library integration	3.5	CAD infrastructure			
Hierarchical flow definition	2.0	Synthesis liaison			
Runtime baseline	1.5				

12 PD · PHYSICAL DESIGN

PD-02 Floorplan, macro placement and partition definition			
TAKES	COSTS	OWNER	CRITICAL PATH
9w	32.0 M/M	Floorplan owner	yes
w5–w14 of the stage	~15.4 people while it runs	Synthesis liaison approves	slipping moves the program

WHY IT EXISTS

Decide **where everything physically goes**—partitions, macros, block boundaries, pins, utilization—and freeze it, because everything downstream is built on it.

The floorplan is the most consequential and least reversible decision in the stage. Macro placement sets routing congestion, block boundaries set what can be closed independently, and pin placement sets the top-level routing burden. A floorplan changed at turn 2 invalidates the clock trees, the power grid and the routing built on it.

WHAT IT DELIVERS

PD-D4

Floorplan and PDN specification

The floorplan half; PD-03 supplies the power grid it is built against.

NEEDS FIRST

- Floorplan intent and bump map from ARCH-08
- Macro abstracts from AMS-15 and PDK-04
- Partitioning from ARCH-02
- Netlist from SYN-05
- Area budgets from ARCH-06

DONE WHEN

- Floorplan frozen before turn 1 starts
- Macro placement validated against congestion, not only access
- Pin placement reviewed for top-level routing cost

HOW IT GETS THERE

6 STEPS · 2 LANES

w5w7w9w11w13w14

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Partition definition against the design hierarchy → Partition definition	2w
2	Macro placement — memories, PHYs, analog → Macro placement	2.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Standard cell area planning and utilization targets → Area and utilization plan	2w runs in parallel
4	Block boundary and pin planning → Block boundaries and pin placement	2w
5	Keep-out, blockage and halo definition → Keep-out and blockage definition	1.5w runs in parallel
6	Floorplan review and freeze → Frozen floorplan	2.5w

Step 2 is where a memory-dominated die is won or lost. Hundreds of macro instances have to be placed so their access paths are short, their power can be delivered and the logic between them can be routed—and moving one late moves everything around it.

WATCH OUT FOR

- ◆ **Macro placement not congestion-aware.** Macros placed for access and not for the logic between them produce routing congestion that cannot be fixed at PD-08.
- ◆ **Floorplan changed after turn 1.** Clock trees, power grid and routing are all built on it, and all of them are invalidated.
- ◆ **Utilization set too high.** A dense floorplan closes area and cannot be routed or timing-closed, and the discovery comes after weeks of work.
- ◆ **Pin placement ignoring top-level routing.** Block pins facing the wrong way make top-level routing expensive for the whole rest of the program.
- ◆ **Partitions that do not match the design hierarchy.** Closing blocks that cut across logical boundaries makes timing budgets and equivalence checking harder for no gain.

WHERE THE EFFORT GOES	32 M/M	WHO IS ON IT	~15.4 FTE	CONNECTIONS	TERMS HERE	1
Macro placement	9.0	Floorplan owner owns IT		DEPENDS ON ARCH-06 ARCH-08 PDK-04 PDK-05 PDK-06 IPR-06 IPR-07 IPR-09 AMS-10 AMS-12 AMS-15 AMS-16 RTL-03 SYN-03 SYN-05 PD-01	PDN	
Partition definition	7.0	Physical design engineers		FEEDS INTO SYN-04 PD-03 PD-04 PD-05 PD-07 PD-14 PD-16		
Block boundary and pin planning	6.0	Architecture liaison		FEEDS BACK INTO DFT-09		
Area and utilization planning	5.0	Package liaison				
Review and freeze	3.0	Synthesis liaison				
Blockages and halos	2.0					

12 PD · PHYSICAL DESIGN

PD-03 Power delivery network design and early IR analysis			
TAKES	COSTS	OWNER	CRITICAL PATH
8w	20.0 M/M	Power delivery engineer	no
w6–w14 of the stage	~10.9 people while it runs	Floorplan owner approves	has float

WHY IT EXISTS

Build the **power delivery network**—grid topology, straps, switches, decap—and prove by early IR analysis that it can actually feed the die at its power envelope.

The PDN is built early because everything routes around it. Grid density trades directly against routing resource: too sparse and IR drop eats timing margin, too dense and there is nothing left to route with. Getting it wrong is discovered at S0-04, when the fix is a metal change across the whole die.

WHAT IT DELIVERS

PD-D4 Floorplan and PDN specification

The power grid half of the deliverable, frozen alongside the floorplan it sits in.

NEEDS FIRST

- Power budget per domain from ARCH-06
- Power domain definition from ARCH-05 and SYN-07
- Floorplan from PD-02
- Bump and rail assignment from PD-04
- Power estimates from SYN-06

DONE WHEN

- Static IR analyzed on a placed design with real power numbers
- Grid density justified against routing resource
- PDN frozen with the floorplan, not after it

HOW IT GETS THERE

6 STEPS · 2 LANES

w6w8w10w12w14

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	PDN topology and grid definition per domain → PDN topology and grid definition	1.5w
2	Grid implementation and strap insertion → Implemented grid and straps	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Power switch and always-on region planning → Power switch and always-on plan	1.5w runs in parallel
4	Early static IR analysis → Early static IR analysis results	2w
5	Decap planning and placement → Decap plan	1.5w runs in parallel
6	PDN refinement and freeze → Frozen PDN specification	2.5w

Step 4 has to happen while the grid can still change. Static IR on a placed design with real power numbers tells you whether the grid works; running it after routing means any fix competes with a routed design for the same metal.

WATCH OUT FOR

- ◆ **Grid density traded against routing without analysis.** Both constraints are real, and choosing by feel produces either IR problems or congestion.
- ◆ **IR analyzed only statically.** Dynamic IR under real switching is worse, and SIPI-04 with the package model is worse again.
- ◆ **Decap placed where there is room.** Decoupling works where the current is drawn, not where the floorplan left space.
- ◆ **Power switches planned late.** They need area, always-on routing and their own control, and retrofitting them disturbs the floorplan.
- ◆ **PDN frozen before power numbers are real.** A grid designed against estimates and fed a design that draws more is a grid that fails at signoff.

WHERE THE EFFORT GOES	20 M/M	WHO IS ON IT ~10.9 FTE	CONNECTIONS	TERMS HERE
Grid implementation	6.0	Power delivery engineer owns it	DEPENDS ON ARCH-05 ARCH-06 RTL-05 SYN-07 PD-02	MCMM IR PDN SI/PI
Early IR analysis	5.0	Physical design engineers	FEEDS INTO PD-04 PD-05 PD-08 PD-10 PD-11 S0-04 PKGD-03 PKGD-06 PKGD-07 SIPI-03 SIPI-05 SIPI-06 SIPI-10	
Topology definition	3.5	Power architect	LATER INPUT SIPI-05 +4w	
Decap planning	2.5	SI/PI liaison	FEEDS BACK INTO PKGD-02	
Power switch planning	2.0	Floorplan owner		
Refinement and freeze	1.0			

12 PD · PHYSICAL DESIGN

PD-04

Bump and RDL planning with chip-package co-design feedback

TAKES	COSTS	OWNER	CRITICAL PATH
8w w6–w14 of the stage	14.0 M/M ~7.6 people while it runs	Bump and RDL owner SI liaison approves	no has float

WHY IT EXISTS

Turn the architecture's bump budget into **an actual bump map and RDL**, iterating with package design until both sides can build what the other needs.

The bump map is the contract between the die and the package, and it is negotiated rather than declared. Signal escape on the package constrains bump position; power delivery constrains bump count; and the die's floorplan constrains where either can go. All three have to close at once.

WHAT IT DELIVERS

PD-D7

Bump map, RDL and package interface files

This activity is the deliverable, and it is consumed directly by package design.

NEEDS FIRST

- Bump budget and map intent from ARCH-08
- Package architecture from PKGD-01
- Floorplan and IO ring from PD-02
- Power delivery requirements from PD-03
- Substrate escape constraints from PKGD-04

DONE WHEN

- Bump map frozen on the date package design needs it
- Escape routing confirmed feasible on the substrate
- Power bump count sufficient for the delivery requirement

HOW IT GETS THERE

6 STEPS · 2 LANES

w6	w8	w10	w12	w14
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Bump map ingestion from the architecture budget → Ingested bump map	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	RDL and redistribution planning → RDL and redistribution plan	2w
3	Power bump and rail assignment → Bump map with signal and power assignment · Power bump and rail assignment	1.5w runs in parallel
4	Package co-design iteration → Package co-design iteration record	2w
5	Signal escape planning with the substrate → Signal escape plan with the substrate	1.5w runs in parallel
6	Bump map freeze and interface file release → Frozen bump map and package interface files	2.5w

Step 6 is a hard date rather than a milestone. `PKGD-11` freezes package design against this map, and the substrate order that follows has a twenty-week lead time—a late bump map moves wafer-out by moving the substrate.

WATCH OUT FOR

- ◆ **Bump map frozen late.** Package design and the substrate order both wait on it, and the substrate lead time makes the delay a wafer-out delay.
- ◆ **Escape routing not checked on the package side.** A map that is optimal on the die and unroutable on the substrate has to be redone.
- ◆ **Power bump count reduced to fit signals.** The signals that fit are then fed by a PDN that cannot deliver, and SIPI-04 finds it.
- ◆ **RDL treated as free.** Redistribution has its own rules and resistance, and a plan that ignores them produces IR and EM problems in the RDL itself.
- ◆ **Co-design conducted by file exchange.** Two teams iterating asynchronously on the same interface converge slowly or not at all.

WHERE THE EFFORT GOES	14 M/M	WHO IS ON IT	~7.6 FTE	CONNECTIONS	TERMS HERE	6
Package co-design iteration	4.0	Bump and RDL owner OWNS IT		DEPENDS ON ARCH-08 IPR-09 PD-02 PD-03 PKGD-01 PKGD-02	RDL EM IR PDN IO SI	
RDL planning	3.5	Package designer		RUNS WITH AMS-03		
Bump map ingestion and assignment	2.5	Power delivery engineer		FEEDS INTO PKGD-11 SIPI-02 SIPI-04		
Power bump assignment	2.0	Floorplan owner		FEEDS BACK INTO PKGD-02		
Escape planning	2.0	SI liaison				

PD-05

Turn 1 on the N1 drop — placement, CTS and first full route; QoR baseline established

TAKES	COSTS	OWNER	CRITICAL PATH
8w w8–w16 of the stage	45.0 M/M ~24.4 people while it runs	Physical design turn lead Synthesis liaison approves	yes slipping moves the program

WHY IT EXISTS

Run the **first real turn on the N1 netlist**—place, build clocks, route—and establish the QoR baseline that says how far the design is from closing.

Turn 1 is the first honest measurement of the design's physical difficulty. Everything before it was estimate; this produces real congestion, real timing and real power on a real floorplan, and the gap it reveals is what the next two turns exist to close.

WHAT IT DELIVERS

PD-D2

Turn 1 and Turn 2 databases with QoR delta reports

The first turn, and the baseline the second is measured against.

NEEDS FIRST

- N1 netlist and handoff package from SYN-05 and SYN-12
- Frozen floorplan from PD-02
- PDN from PD-03
- Flow and MMMC from PD-01
- Constraints from SYN-01

DONE WHEN

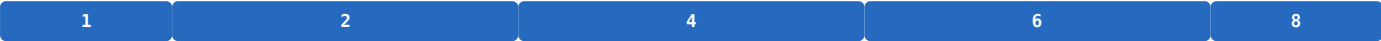
- QoR baseline measured on a substantially complete netlist
- Feedback to synthesis specific enough to act on
- Congestion characterized as structural or local

HOW IT GETS THERE

8 STEPS · 2 LANES

w8	w10	w12	w14	w16
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	N1 intake and turn setup → N1 intake and turn setup	1w
2	Placement and optimization → Placed and optimized N1 database	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Congestion analysis and mitigation → Congestion analysis and mitigation results	1.5w runs in parallel
4	Clock tree synthesis, first pass → First-pass clock trees	2w
5	Timing analysis and first ECO round → First-round timing analysis and ECOs	1.5w runs in parallel
6	First full route → First full route	2w
7	Power analysis on the placed and routed design → Turn 1 placed and routed database · Timing and power analysis results	1.5w runs in parallel
8	QoR baseline and feedback to synthesis → QoR baseline · Feedback list to synthesis and RTL	1w

Step 8 is what makes turn 1 worth its cost. The turn's purpose is not to close—it will not—but to produce a measured baseline and a specific list of what synthesis and RTL must change before turn 2.

WATCH OUT FOR

- ◆ **Turn 1 expected to close.** It will not, and treating the gap as a crisis rather than as data produces panic instead of a plan.
- ◆ **Feedback not specific.** "Congestion is high" changes nothing; "the compute cluster is over 90% utilization in three regions and needs restructuring" changes the next drop.
- ◆ **Turn run on an incomplete netlist.** A baseline missing blocks understates the difficulty and misleads the closure plan.
- ◆ **Congestion mitigated locally.** Fixing congestion by rip-up and reroute hides a structural problem that only synthesis or floorplan can solve.
- ◆ **Clock tree built before the floorplan is stable.** A CTS on a floorplan that then changes is work thrown away.

WHERE THE EFFORT GOES	45 M/M	WHO IS ON IT ~24.4 FTE	CONNECTIONS	TERMS HERE 7
Placement and optimization	11.0	Physical design turn lead OWNS IT	DEPENDS ON PDK-03 AMS-02 DFT-05 SYN-02 SYN-04 SYN-05 SYN-12 PD-01 PD-02 PD-03	RTL MMMC MCMM CTS ECO QoR PDN
First full route	10.0	Placement and routing engineers		
Clock tree synthesis	8.0	Clock tree engineer		
Timing analysis and ECO	7.0	Timing engineer	FEEDS INTO SYN-08 PD-06 PD-07 PD-08 PD-09 PD-11 SIPI-01	
Congestion analysis	5.0	Synthesis liaison		
QoR baseline and feedback	4.0			

12 PD · PHYSICAL DESIGN

PD-06

Turn 2 on the N2 drop — congestion, timing and power convergence; closure risk quantified

TAKES	COSTS	OWNER	CRITICAL PATH
8w	45.0 M/M	Physical design turn lead	yes
w14–w22 of the stage	~24.4 people while it runs	Program manager approves	slipping moves the program

WHY IT EXISTS

Run the **second turn on N2 and converge**—congestion, timing and power—then quantify honestly how much closure risk remains for the final turn.

Turn 2 is where the design either starts converging or reveals that it will not. Its output is a number: how many violations remain, in which blocks, and whether the trend from turn 1 projects to closure. That number is what the program uses to decide whether the tapeout date holds.

WHAT IT DELIVERS

PD-D2

Turn 1 and Turn 2 databases with QoR delta reports

The second turn, and the last one before the netlist is frozen.

NEEDS FIRST

- N2 netlist and handoff from SYN-08 and SYN-12
- Turn 1 database and QoR from PD-05
- Congestion feedback loop with SYN-04
- Clock trees from PD-07
- Timing closure status from PD-09

DONE WHEN

- Convergence trend measured against turn 1, not only reported
- Structural congestion addressed structurally
- Closure risk stated with paths, blocks and options

HOW IT GETS THERE

8 STEPS · 2 LANES

w14	w16	w18	w20	w22
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	N2 intake and delta assessment against turn 1 → N2 intake and delta assessment	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Placement with congestion mitigation → Placement with congestion mitigation	2w
3	Congestion hot-spot restructuring with synthesis → Congestion mitigation and restructuring results	1.5w runs in parallel
4	Clock tree refinement → Refined clock trees	2w
5	Timing convergence and ECO rounds → Timing convergence and ECO record	1.5w runs in parallel
6	Route convergence → Converged route	2w
7	Power and IR convergence → Timing and power convergence results	1.5w runs in parallel
8	Closure risk quantification for the final turn → Turn 2 placed and routed database · Closure risk quantification	1w

Step 3 is joint work with `SYN-04` rather than a physical design task. Congestion that survives placement optimization is structural, and the only remedies are logic restructuring or a floorplan change—both of which belong to someone else.

WATCH OUT FOR

- ◆ **Closure risk understated at turn 2.** This is the last point at which a frequency or feature concession is cheap, and an optimistic number removes the option.
- ◆ **Congestion fixed locally rather than structurally.** Rip-up and reroute buys a turn and leaves the problem for the final one, when there is no time.
- ◆ **Turn 2 not compared against turn 1.** The absolute number matters less than the trend, and only the delta says whether the design is converging.
- ◆ **Power convergence deferred.** Power fixes change timing; leaving them to the final turn means re-closing timing after every one.
- ◆ **Late RTL changes absorbed into turn 2.** A functional change mid-turn invalidates the comparison the turn exists to produce.

WHERE THE EFFORT GOES	45 M/M	WHO IS ON IT ~24.4 FTE	CONNECTIONS	TERMS HERE	5
Placement and congestion mitigation	11.0	Physical design turn lead owns it	DEPENDS ON TECH-02 PDK-02 PDK-10 DFT-09 SYN-04 SYN-08 SYN-12 PD-05 PD-07	RTL MCMM ECO QoR IR	
Route convergence	10.0	Placement and routing engineers			
Clock tree refinement	8.0	Synthesis liaison	FEEDS INTO SYN-09 PD-09 PD-13		
Timing convergence	8.0	Timing engineer	S0-01		
Power and IR convergence	4.0	Program manager			
Risk quantification	4.0				

12 PD · PHYSICAL DESIGN

PD-07

Clock tree synthesis with skew and jitter budgeting, refined across turns

TAKES	COSTS	OWNER	CRITICAL PATH	
12w cont. w10–w22 of the stage	26.0 M/M ~9.4 people while it runs	Clock tree engineer DFT engineer approves	no has float	
WHY IT EXISTS				
Build and refine the clock trees across turns—skew, insertion delay, jitter budget, power—for a design with dozens of domains and crossings between them. Clock tree synthesis is rebuilt every turn and improved each time. Skew directly buys or costs timing margin; insertion delay costs power and OCV; and cross-domain paths need a defined phase relationship that only a deliberately built tree provides. It is one of the few areas where a good result is worth several percent of frequency.				
WHAT IT DELIVERS				
No key deliverable is owned here — this activity contributes to the ones below.				
NEEDS FIRST				
● Clock architecture from ARCH-05 ● OCC insertion from DFT-05 ● Floorplan from PD-02 ● Placement from PD-05 and PD-06 ● PLL characteristics from AMS-02				
DONE WHEN				
● Clock power inside the ARCH-06 budget ● Cross-domain relationships defined and closed ● Clock DRC clean, not only timing-clean				
HOW IT GETS THERE			8 STEPS · 2 LANES	
w10	w13	w16	w19	w22
MAIN SEQUENCE				
1 2 4 6 8				
IN PARALLEL				
3 5 7				
#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE			TAT
1	Clock architecture implementation plan → Clock architecture implementation plan			1.5w
2	Clock tree synthesis per domain → Clock tree implementation plan · Synthesized clock trees per domain			2.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Clock power optimization → Clock power optimization results	2w runs in parallel
4	Skew and insertion delay balancing → Skew and insertion delay results	2.5w
5	Jitter and OCV budget management → Jitter and OCV budget records	2w runs in parallel
6	Cross-domain and useful skew handling → Cross-domain and useful skew handling	2.5w
7	Clock DRC and quality checking → Clock DRC and quality reports	2w runs in parallel
8	Continuous refinement across turns → Clock tree refinement record across turns	3w

Step 6 is the hardest part and the one that pays. Useful skew moves margin from paths that have it to paths that need it, and cross-domain handling decides whether those paths can be closed at all—but both make the tree harder to build and harder to verify.

WATCH OUT FOR

- ◆ **Clock power ignored.** The clock network is a large fraction of dynamic power on a design this size, and a tree optimized only for skew is expensive to run.
- ◆ **Useful skew applied without a plan.** It buys margin and complicates every subsequent ECO, because moving one path's timing moves its neighbors'.
- ◆ **Cross-domain phase relationship undefined.** Paths between domains cannot be closed reliably, and the failure appears as intermittent silicon behavior.
- ◆ **Insertion delay allowed to grow.** Deep trees increase OCV derate and jitter accumulation, consuming the margin the skew balancing bought.
- ◆ **Tree quality not checked.** Clock DRC violations—transition, capacitance, pulse width—degrade the tree in ways timing analysis reports as margin loss without explaining.

WHERE THE EFFORT GOES	26 M/M	WHO IS ON IT	~9.4 FTE	CONNECTIONS
Clock tree synthesis per domain	7.0	Clock tree engineer owns it		DEPENDS ON ARCH-05 AMS-02 DFT-05 PD-02 PD-05
Skew and insertion delay balancing	6.0	Clocking architect		FEEDS INTO PD-06 PD-08 PD-09
Refinement across turns	5.0	Timing engineer		PD-13 S0-02
Clock power optimization	3.0	Power engineer		TERMS HERE 8 PLL MCMM OCV CTS ECO DRC DFT OCC
Jitter and OCV budget	2.5	DFT engineer		
Clock DRC and quality	2.5			

12 PD · PHYSICAL DESIGN

PD-08

Detailed routing and DRC convergence

TAKES	COSTS	OWNER	CRITICAL PATH
16w cont. w12–w28 of the stage	40.0 M/M ~10.9 people while it runs	Routing lead Floorplan owner approves	yes slipping moves the program

WHY IT EXISTS

Route the design and drive it to **DRC clean**—global, detailed, antenna, via optimization—across every turn, on a die where routing resource is the scarcest commodity.

Routing is where the floorplan, the synthesis and the power grid are all judged. Sixteen weeks and forty man-months buys convergence on a design where congestion is structural rather than local, and where every metal layer is contested between signals, clocks and the power grid.

WHAT IT DELIVERS

PD-D6

Interim physical DRC / LVS clean

This activity is the deliverable at each turn, and clean here is what makes S0-03 tractable.

NEEDS FIRST

- Placement and clock trees from PD-05, PD-06 and PD-07
- Metal stack and routing rules from TECH-02 and PDK-02
- PDN from PD-03
- Scan routing requirements from PD-12
- DFM rules from PDK-02

DONE WHEN

- DRC clean at each turn, not only at the final one
- Congestion resolved structurally where it is structural
- Manufacturability addressed during routing, not after

HOW IT GETS THERE

8 STEPS · 2 LANES

w12	w16	w20	w24	w28
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Routing strategy and layer planning → Routing strategy and layer plan	1.5w
2	Global routing and congestion assessment → Global routing and congestion assessment	3w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Congestion-driven rip-up and reroute → Congestion resolution records	2.5w runs in parallel
4	Detailed routing → Globally and detail-routed database	3.5w
5	Antenna and via optimization → Antenna and via optimization results	2.5w runs in parallel
6	DRC convergence → DRC convergence results	4w
7	Routing quality and manufacturability → Routing quality and manufacturability findings	2.5w runs in parallel
8	Continuous routing across turns → Routing record across turns	4w

Step 6 takes four weeks because DRC convergence on an advanced node is iterative and asymptotic. The last few hundred violations take as long as the first hundred thousand, and they are the ones that need manual attention.

WATCH OUT FOR

- ◆ **Congestion treated as a routing problem.** Structural congestion cannot be routed around, and rip-up cycles spent on it consume time without converging.
- ◆ **Antenna violations left late.** Fixes need diodes or layer jumps, both of which need space that a fully routed design does not have.
- ◆ **DRC convergence assumed linear.** The last violations are the hardest, and planning as if the rate is constant produces a schedule that misses.
- ◆ **Manufacturability ignored until signoff.** DFM-clean routing is cheaper to produce than to retrofit, and S0-08 finds what was not.
- ◆ **Layer plan not respecting the PDN.** Signals and the power grid contend for the same upper layers, and a plan that ignores the grid produces conflicts everywhere.

WHERE THE EFFORT GOES	40 M/M	WHO IS ON IT	~10.9 FTE	CONNECTIONS	TERMS HERE	5
Detailed routing	11.0	Routing lead owns it		DEPENDS ON PDK-02 PDK-10 PD-03 PD-05 PD-07	DRC LVS DFM PDN FFN	
DRC convergence	9.0	Routing engineers				
Global routing	7.0	DFM engineer		FEEDS INTO PD-10 PD-12 PD-13 PD-15 S0-03 S0-08 EVB-02 EVB-05 BU-02		
Rip-up and reroute	6.0	Physical verification engineer				
Antenna and via optimization	4.0					
Routing quality	3.0	Floorplan owner				

12 PD · PHYSICAL DESIGN

PD-09

Multi-corner multi-mode timing closure

TAKES	COSTS	OWNER	CRITICAL PATH
22w cont. w8–w30 of the stage	60.0 M/M ~11.8 people while it runs	Timing closure lead Program manager approves	yes slipping moves the program

WHY IT EXISTS

Close **timing across every corner and every mode**—setup, hold, cross-corner, cross-mode—and drive the violation burn-down that decides the tapeout date.

Twenty-two weeks and sixty man-months, the largest activity in the stage. Multi-corner multi-mode closure on a large die is a long grind of analysis, ECO and re-analysis, and its burn-down curve is the single best predictor the program has of whether it will tape out on time.

WHAT IT DELIVERS

PD-D5

MCMM timing closure reports with violation burn-down across turns

This activity is the deliverable, and the burn-down is the program's best schedule predictor.

NEEDS FIRST

- Placed and routed databases from each turn
- Constraints from SYN-01
- Corner and derate definition from PDK-11
- Clock trees from PD-07
- Signoff correlation from SO-12

DONE WHEN

- Closed at every corner and mode, not a subset
- Implementation and signoff timing correlated within a stated tolerance
- Burn-down trend projecting closure before the tapeout date

HOW IT GETS THERE

8 STEPS · 2 LANES

w8	w13	w18	w23	w28	w30
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Timing setup and MMMC scenario definition → MMMC timing scenarios	1.5w
2	Setup timing closure → Setup timing closure results	3w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Violation burn-down tracking and reporting → Violation burn-down tracking	3w runs in parallel
4	Hold timing closure → Setup and hold closure results	3w
5	Timing ECO generation and application → Timing ECOs	3w runs in parallel
6	Cross-corner and cross-mode convergence → Cross-corner convergence results	3.5w
7	Signoff timing correlation with the SO flow → Signoff correlation findings	3w runs in parallel
8	Continuous closure across turns → Closure record across turns	11w

Step 7 prevents the worst surprise available. A design closed in the implementation tool and re-analyzed in the signoff tool will show differences; discovering their size early lets the team close with the right margin rather than re-closing after `SO-02` reports.

WATCH OUT FOR

- ◆ **Closing at some corners and deferring others.** The deferred corner is the one that fails, and it fails when there is no time to fix it.
- ◆ **Implementation and signoff timing diverging.** A design closed in one tool and judged in another can be weeks from closure without anyone knowing.
- ◆ **Hold fixed after setup.** Hold fixes add buffers that change setup, and doing them sequentially means iterating both.
- ◆ **Burn-down not tracked.** Without a trend nobody can distinguish a design that is converging slowly from one that is not converging at all.
- ◆ **ECO volume exceeding what the database can absorb.** Thousands of ECOs degrade the physical result, and at some point a re-run is cheaper than another ECO round.

WHERE THE EFFORT GOES	60 M/M	WHO IS ON IT ~11.8 FTE	CONNECTIONS	TERMS HERE
Setup timing closure	15.0	Timing closure lead owns it	DEPENDS ON DEF-03 PDK-11 PDK-12 TC-08 SYN-01 SYN-06 SYN-11 PD-05 PD-06 PD-07 S0-01	STA MMC MCMM ECO
Hold timing closure	12.0	STA engineers	RUNS WITH S0-12	
Continuous closure across turns	12.0	Physical design engineers	FEEDS INTO DV-11 DFT-08 PD-10 PD-13 PD-14 PD-16 S0-02 S0-11 SIPI-09	
Timing ECO generation	9.0	Signoff methodology engineer		
Cross-corner convergence	7.0	Program manager		
Signoff correlation	5.0			

12 PD · PHYSICAL DESIGN

PD-10

Signal and power integrity iteration — crosstalk, EM

TAKES	COSTS	OWNER	CRITICAL PATH
10w w16–w26 of the stage	18.0 M/M ~7.8 people while it runs	SI/PI engineer SIPI liaison approves	no has float

WHY IT EXISTS

Fix what the routing did to the electricals—**crosstalk, noise, electromigration, dynamic IR**—before signoff has to reject the design for them.

Signal and power integrity problems are created by routing and found by analysis. Crosstalk between adjacent nets, EM in high-current wires and dynamic IR under real switching are all consequences of physical decisions, and fixing them means changing routing that was closed for timing.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Routed database from PD-08
- PDN from PD-03
- Switching activity from SYN-06 and DV-04
- Signoff SI/PI methodology from SO-05
- Package co-analysis findings from SIPI-04

DONE WHEN

- Crosstalk included in timing closure, not analyzed after it
- EM checked on signal nets as well as power
- Dynamic IR analyzed with realistic, localized activity

HOW IT GETS THERE

6 STEPS · 2 LANES

w16	w18	w20	w22	w24	w26
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	SI setup and crosstalk analysis → SI setup and crosstalk analysis	1.5w
2	Crosstalk fixing — spacing, shielding, buffer sizing → Crosstalk fixes — spacing, shielding, sizing	2.5w
3	Noise and glitch analysis → Noise and glitch analysis results	2w runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Electromigration analysis and fixing → Crosstalk analysis and fixes · EM analysis and routing fixes	2.5w
5	Dynamic IR iteration with the PDN → Dynamic IR results and PDN adjustments	2w <i>runs in parallel</i>
6	SI and PI iteration and closure → SI and PI closure records	3.5w

Step 6 is an iteration rather than a step, because every fix perturbs something else. Widening a wire for EM changes its coupling; adding shielding consumes routing resource; and both change timing, which sends the design back to `PD-09`.

WATCH OUT FOR

- ◆ **SI analyzed after timing closes.** Crosstalk delay is timing, and closing without it means closing against numbers that will move.
- ◆ **EM checked only on power.** High-toggle signal nets on advanced nodes carry EM exposure too, and they are usually the ones nobody checks.
- ◆ **Dynamic IR analyzed with uniform activity.** Real switching is bursty and localized, and uniform assumptions understate the worst case badly.
- ◆ **Fixes applied without re-timing.** Shielding and spacing change delay, and a fix that breaks timing has moved the problem rather than solved it.
- ◆ **Package effects excluded.** Die-only dynamic IR is optimistic; SIPI-04 with package inductance is the number signoff will use.

WHERE THE EFFORT GOES	18 M/M	WHO IS ON IT	~7.8 FTE	CONNECTIONS	TERMS HERE	10
Crosstalk fixing	5.0	SI/PI engineer <i>OWNS IT</i> Routing engineers Power delivery engineer Timing engineer SIPI liaison		DEPENDS ON SYN-06 PD-03 PD-08 PD-09 FEEDS INTO PD-13 S0-04 S0-05 SIPI-04	MCMM DRC LVS EM IR PDN SI PI SI/PI SIPI	
EM analysis and fixing	4.5					
SI and PI iteration	3.5					
Noise and glitch analysis	2.5					
Dynamic IR iteration	2.5					

12 PD · PHYSICAL DESIGN

PD-11

Chip power model (CPM/CPS) extraction and handoff to package co-verification

TAKES	COSTS	OWNER	CRITICAL PATH		
5w w14–w19 of the stage	6.0 M/M ~5.2 people while it runs	Power modeling engineer Configuration manager approves	no has float		
WHY IT EXISTS					
Extract the chip power model the package co-verification needs—per domain, per mode, with realistic switching—and hand it over as a versioned artifact. SIPI-03 cannot analyze the die, package and board as one power delivery network without a model of the die's current draw. Producing that model is a small activity with an outsized dependency: without it the co-verification cannot start, and the co-verification gates mask release.					
WHAT IT DELIVERS					
No key deliverable is owned here — this activity contributes to the ones below.					
NEEDS FIRST					
- Placed and routed database from PD-05 or PD-06 - PDN from PD-03 - Switching activity from DV-04 and SYN-06 - Model format requirements from SIPI-01 - Power domain definition from ARCH-05					
DONE WHEN					
- Model extracted under realistic worst-case switching - Validated against the die's own IR analysis - Delivered versioned, on the date SIPI needs it					
HOW IT GETS THERE					
5 STEPS · 2 LANES					
w14	w15	w16	w17	w18	w19
MAIN SEQUENCE					
1 2 4 5					
IN PARALLEL					
3					
#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE				TAT
1	Power model requirement definition with SIPI → Power model requirement definition				1w
2	CPM extraction per domain and operating mode → Chip power model per domain and mode				1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Switching scenario selection for the extraction → Switching scenario selection record runs in parallel	1w
4	Model validation against internal IR analysis → Model validation against internal IR	1.5w
5	Handoff and version control → Versioned model handoff	1w

Step 3 determines whether the model is useful. A CPM extracted under average activity understates the transient the package has to supply; extracting under the worst realistic burst is what makes `SIPI-04`'s answer meaningful.

WATCH OUT FOR

- ◆ **Model extracted under average activity.** The package has to supply the transient, and an averaged model hides exactly the event that matters.
- ◆ **Delivered late.** Co-verification is on the critical path to mask release, and it cannot start without this model.
- ◆ **Model not validated.** A CPM that disagrees with the die's own IR analysis is wrong in one of the two places, and nobody knows which.
- ◆ **Single model for all modes.** Different DVFS points and workloads draw very differently, and one model cannot represent them.
- ◆ **No version control.** The package team analyzing against a stale model produces conclusions about a die that no longer exists.

WHERE THE EFFORT GOES	6 M/M	WHO IS ON IT ~5.2 FTE	CONNECTIONS	TERMS HERE 6
CPM extraction	2.5	Power modeling engineer owns it	DEPENDS ON DV-04 PD-03 PD-05 SIPI-01	RDL IR PDN DVFS CPM SIPI
Model validation	1.5	Power delivery engineer		
Requirement definition	1.0	SIPI liaison	FEEDS INTO S0-06 SIPI-03 SIPI-04	
Handoff and version control	1.0	Verification liaison	MP-03	
		Configuration manager	FEEDS BACK INTO SIPI-01	

12 PD · PHYSICAL DESIGN

PD-12

Scan chain reordering and DFT-aware routing

TAKES	COSTS	OWNER	CRITICAL PATH
6w w16–w22 of the stage	8.0 M/M ~5.8 people while it runs	DFT physical engineer Clock engineer approves	no has float

WHY IT EXISTS

Reorder and route the **scan chains physically**—shorter routes, less congestion—without destroying the chain-to-block mapping diagnosis depends on.

Scan chains as stitched by synthesis follow logical order and route terribly. Physical reordering fixes the routing and can scramble the relationship between chain position and physical location, which is exactly what failure diagnosis uses. Both constraints are real.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Scan-inserted netlist from DFT-06
- Physical DFT plan from DFT-09
- Placement from the current turn
- Routing resource from PD-08
- Test clock architecture from DFT-05

DONE WHEN

- Chains verified after every reorder
- Diagnosis mapping preserved or explicitly traded
- Chain order frozen before ATPG generates patterns

HOW IT GETS THERE

6 STEPS · 2 LANES

w16

w18

w20

w22

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Scan chain physical analysis and routing cost → Chain routing cost analysis	1w
2	Chain reordering for routing efficiency → Reordered scan chains	1.5w
3	Diagnosis mapping preservation → Diagnosis mapping record	1w

runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	DFT-aware routing and shielding of test paths → DFT-aware routing and shielding	1.5w
5	Test clock routing and skew → Test clock routing	1w runs in parallel
6	Chain verification after reorder → Post-reorder chain verification results	2w

Step 6 is not optional. Reordering rewrites the netlist's scan connectivity, and a chain broken by the reorder makes everything behind it untestable—a failure that would otherwise be discovered at ATE bring-up.

WATCH OUT FOR

- ◆ **Reordering breaking diagnosis.** Failure analysis relies on mapping a failing bit to a physical location, and a scrambled chain removes that.
- ◆ **Chains not verified after reorder.** A break makes everything behind it untestable, and ATE bring-up is a costly place to find that.
- ◆ **Patterns generated before the final reorder.** Any chain change after DFT-07 invalidates the pattern set and costs weeks of regeneration.
- ◆ **Test clock skew ignored.** Scan shift reaches every flop, and skew on the shift clock produces shift failures that look like design bugs.
- ◆ **Chain imbalance after reorder.** Physical reordering can lengthen one chain and shorten another, and shift time follows the longest.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT ~5.8 FTE	CONNECTIONS	TERMS HERE 5
Chain reordering	2.5	DFT physical engineer OWNS IT	DEPENDS ON DFT-06 DFT-09 PD-08	DRC LVS DFT ATPG
DFT-aware routing	2.0	Routing engineers	FEEDS INTO PD-13 S0-03	ATE
Post-reorder verification	1.5	Diagnosis engineer	FEEDS BACK INTO DFT-07	
Diagnosis mapping	1.0	ATPG engineer		
Test clock routing	1.0	Clock engineer		

12 PD · PHYSICAL DESIGN

PD-13

Final turn on the FFN — full closure, no functional change admitted

TAKES	COSTS	OWNER	CRITICAL PATH
11w w19–w30 of the stage	55.0 M/M ~21.7 people while it runs	Physical design lead Signoff lead approves	yes slipping moves the program

WHY IT EXISTS

Run the **final turn on the FFN and close it completely**—timing, power, DRC, LVS—with no functional change admitted and no next turn to fall back on.

Everything the stage has done is preparation for this. Fifty-five man-months over eleven weeks, on a frozen netlist, closing every corner and every mode to signoff standard. The discipline that matters is refusing functional change: an ECO that alters behavior restarts verification, and there is no schedule for that.

WHAT IT DELIVERS

PD-D3

Final-turn routed database on the FFN, per block and top

This activity is the deliverable, and it is what signoff and tapeout consume.

NEEDS FIRST

- FFN and handoff package from SYN-09 and SYN-12
- Turn 2 database and risk from PD-06
- Clock trees from PD-07
- SI/PI fixes from PD-10
- ECO infrastructure from PD-14

DONE WHEN

- Closed at every corner and mode to signoff standard
- No functional change admitted, or each one escalated and recorded
- Database frozen LVS-clean and accepted by signoff

HOW IT GETS THERE

10 STEPS · 2 LANES

w19	w22	w25	w28	w30
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	FFN intake and delta assessment against N2 → FFN intake and delta assessment	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Placement on the final netlist → Placement on the final netlist	2w
3	ECO-only change discipline and gatekeeping → ECO discipline records <i>runs in parallel</i>	1.5w
4	Final clock tree synthesis → Final clock tree	2w
5	Power and IR final convergence → Final power and IR convergence results <i>runs in parallel</i>	2w
6	Final routing → Final placed, clocked and routed database	2.5w
7	DRC and LVS convergence → DRC and LVS clean results <i>runs in parallel</i>	2w
8	Final timing closure across all corners and modes → Final timing closure across corners and modes	2w
9	Closure reporting and remaining-violation disposition → Closure report with remaining-violation disposition <i>runs in parallel</i>	1.5w
10	Database freeze and signoff handoff → Frozen signoff-ready database	1.5w

Step 3 runs across the whole turn and is a gate rather than a task. Every change request during the final turn has to be classified—timing ECO, functional change, or bug fix—and only the first is admitted without escalation.

WATCH OUT FOR

- ◆ **Functional change admitted during the final turn.** It restarts verification and equivalence, and no schedule allows for that at this point.
- ◆ **Closure gap larger than turn 2 predicted.** If the prediction was optimistic, the discovery comes with no remaining turns and only concessions available.
- ◆ **ECO volume degrading the database.** Thousands of ECOs on a final database produce a physical result worse than a clean re-run, but a re-run costs weeks.
- ◆ **Remaining violations dispositioned informally.** Every violation that survives to signoff must be a named, accepted waiver rather than an oversight.
- ◆ **Database frozen before LVS is clean.** A handoff that fails LVS at S0-03 sends everything back with no time to spare.

WHERE THE EFFORT GOES	55 M/M	WHO IS ON IT ~21.7 FTE	CONNECTIONS	TERMS HERE 9
Final timing closure	14.0	Physical design lead owns it	DEPENDS ON SYN-09 SYN-12 PD-06 PD-07 PD-08 PD-09 PD-10 PD-12 PD-14	RTL MCMC CTS ECO DRC LVS IR FFN SI/PI
Final routing	11.0	Placement, CTS and routing engineers		
Placement	9.0	Timing closure lead		
Final clock tree synthesis	7.0	Program manager	FEEDS INTO PD-15 PD-16 S0-02 S0-03 S0-04 S0-05 S0-07 S0-08 S0-09 S0-11 T0-01	
Power and IR convergence	6.0	Signoff lead		
DRC and LVS convergence	5.0			
Freeze and handoff	3.0			

PD-14

Functional and timing ECO implementation

TAKES	COSTS	OWNER	CRITICAL PATH
18w cont. w12–w30 of the stage	22.0 M/M ~5.3 people while it runs	ECO lead Configuration manager approves	no has float

WHY IT EXISTS

Run the **ECO machinery**—flow, spare cells, metal-only strategy, impact tracking—that every turn depends on and that the final turn depends on absolutely.

ECOs are how a closed design absorbs change without a full re-run. The infrastructure matters more than any individual ECO: a spare cell strategy decided at floorplan time determines whether a late fix is metal-only or needs a base layer change, and that distinction is worth several weeks and a mask set.

WHAT IT DELIVERS

PD-D8

ECO log and change control record

This activity is the deliverable, and the log is what makes a late database traceable to the design that was verified.

NEEDS FIRST

- Change requests from RTL-10, DV-12 and timing closure
- Placed and routed databases from each turn
- Spare cell placement from PD-02
- ECO flow support from PDK-09
- Timing ECOs from PD-09

DONE WHEN

- Metal-only fix path available at tapeout
- Every ECO tracked and reconcilable against the netlist
- Functional and timing ECOs distinguished in the log

HOW IT GETS THERE

6 STEPS · 2 LANES

w12	w16	w20	w24	w28	w30
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	ECO flow and methodology setup → ECO flow and methodology	1.5w
2	Functional ECO implementation capability → Functional ECO implementation capability	2.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	ECO impact assessment and tracking → Impact assessment and tracking records	2w runs in parallel
4	Timing ECO implementation → Timing ECO implementation	2.5w
5	Spare cell and metal-only ECO strategy → Spare cell and metal-only strategy	2w runs in parallel
6	Continuous ECO operation across turns → ECO log across turns	11.5w

Step 5 is decided early and matters late. Spare cells distributed through the floorplan cost area now and make a metal-only fix possible at tapeout; without them, a late functional fix needs base layers and a full mask set.

WATCH OUT FOR

- ◆ **No spare cell strategy.** A late functional fix then requires base layers and a full mask set rather than a metal-only change.
- ◆ **ECOs applied without impact tracking.** A database with untracked changes cannot be reconciled against the netlist that was verified.
- ◆ **ECO volume unbounded.** Beyond some number, the physical result degrades enough that a clean re-run would be better—and nobody notices the crossing point.
- ◆ **Functional and timing ECOs not distinguished.** They have completely different verification consequences, and conflating them lets functional change in unnoticed.
- ◆ **Spare cells consumed early.** Using them for convenience during turn 1 leaves none for the fix that actually needs them at tapeout.

WHERE THE EFFORT GOES	22 M/M	WHO IS ON IT	~5.3 FTE	CONNECTIONS
Continuous ECO operation	8.0	ECO lead OWNS IT		DEPENDS ON PDK-09 RTL-10 PD-02 PD-09
Timing ECO implementation	5.0	Physical design engineers		FEEDS INTO PD-13 S0-09 T0-03 T0-08
Functional ECO implementation	4.0	Timing engineers		
Impact assessment and tracking	3.0	Verification liaison		TERMS HERE ECO FFN
Spare cell strategy	2.0	Configuration manager		2

12 PD · PHYSICAL DESIGN

PD-15

Chip finishing — seal ring, dummy and metal fill, alignment marks

TAKES	COSTS	OWNER	CRITICAL PATH		
5w w24–w29 of the stage	6.0 M/M ~5.2 people while it runs	Chip finishing engineer Product engineering approves	no has float		
WHY IT EXISTS					
Finish the die as a manufacturable object—seal ring, metal fill, dummy, alignment marks, identification—and verify that none of it broke what was closed. Chip finishing is mechanical and it is not free. Metal fill changes coupling capacitance and therefore timing; the seal ring consumes die edge; and dummy fill interacts with the analog matching that AMS-12 spent weeks building. Every one has to be applied and then re-verified.					
WHAT IT DELIVERS					
<i>No key deliverable is owned here — this activity contributes to the ones below.</i>					
NEEDS FIRST					
● Routed database from PD-08 and PD-13 ● Seal ring and frame rules from PDK-02 ● Fill and density rules from the DRM ● Analog macro fill exclusions from AMS-12 ● Die identification requirements from MP-09					
DONE WHEN					
● Timing verified after fill, not before it ● Analog matching structures protected from fill ● Die identification present and readable					
HOW IT GETS THERE					
6 STEPS · 2 LANES					
w24	w25	w26	w27	w28	w29
MAIN SEQUENCE					
1 2 4 6					
IN PARALLEL					
3 5					
#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE				TAT
1	Seal ring and die edge implementation → Seal ring and die edge implementation				1w
2	Metal fill and density compliance → Metal fill meeting density rules				1w
3	Fill impact on timing and extraction → Fill timing impact analysis				1w
				runs in parallel	

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Alignment marks, logos and die identification → Alignment marks and die identification	1.5w
5	Dummy and CMP compliance → Dummy and CMP compliance	1w
	runs in parallel	
6	Finishing verification against the closed database → Finishing verification results	1.5w

Step 3 is why finishing is not a formality. Fill adds capacitance to every net near it, and a design closed before fill and taped out after it has been signed off against a database that no longer exists.

WATCH OUT FOR

- ◆ **Timing signed off before fill.** The taped-out database has fill and the signed-off one did not, which is a signoff against a different design.
- ◆ **Fill applied over analog matching structures.** It disturbs the parasitics AMS-12 tuned, and the damage is invisible until silicon characterization.
- ◆ **Seal ring area not reserved.** Discovered late, it forces the die edge inward and moves the bump map.
- ◆ **Die identification omitted.** Traceability at yield analysis and field return depends on being able to identify the die, and it cannot be added later.
- ◆ **Finishing treated as a checkbox.** Each step changes the physical database, and each needs verification against what was closed.

WHERE THE EFFORT GOES	6 M/M	WHO IS ON IT ~5.2 FTE	CONNECTIONS	TERMS HERE 4
Metal fill and density	2.0	Chip finishing engineer owns it	DEPENDS ON PDK-02 AMS-12 PD-08 PD-13	DRM CMP DRC LVS
Seal ring and die edge	1.5	Physical verification engineer	FEEDS INTO S0-03 S0-08 T0-01 TEST-03 MP-09	
Finishing verification	1.0	Timing engineer		
Fill timing impact	0.8	Analog layout liaison		
Marks and identification	0.8	Product engineering		

12 PD · PHYSICAL DESIGN

PD-16

Hierarchical block closure and top-level assembly

TAKES	COSTS	OWNER	CRITICAL PATH
14w w14–w28 of the stage	30.0 M/M ~9.3 people while it runs	Hierarchical closure lead Physical verification engineer approves	yes slipping moves the program

WHY IT EXISTS

Close the **blocks independently and assemble them into a top level** that meets timing across their boundaries—the hierarchical discipline a die this size requires.

A design of this size cannot be closed flat: runtime and capacity make it impossible. Hierarchical closure lets blocks converge in parallel, at the cost of interface budgets that have to be right. A block that closes internally and misses its boundary budget has not helped.

WHAT IT DELIVERS

PD-D9

Signoff-ready database handoff

This activity is the deliverable—the assembled top level signoff and tapeout actually consume.

NEEDS FIRST

- Partition definition from PD-02
- Block databases from each turn
- Interface budgets from ARCH-06
- Timing closure status from PD-09
- Final database from PD-13

DONE WHEN

- Block models correlated against flat analysis
- Every block closed to one shared criterion
- Top level assembled and verified before the final turn ends

HOW IT GETS THERE

8 STEPS · 2 LANES

w14	w17	w20	w23	w26	w28
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Block closure criteria and handoff definition → Block closure criteria	1.5w
2	Block-level closure → Block-level closure results	3w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Inter-block timing budget and interface closure → Inter-block timing budgets runs in parallel	2.5w
4	Block abstract and timing model generation → Block abstracts and timing models	3w
5	Top-level routing and feedthrough management → Top-level routing and feedthroughs runs in parallel	2.5w
6	Top-level assembly → Assembled top level	3w
7	Hierarchical timing correlation, block model against flat → Closed block databases runs in parallel	2w
8	Top-level integration verification → Assembled and verified top level	3.5w

Step 7 is the check that keeps hierarchy honest. A block's timing model is an abstraction, and if it disagrees with a flat analysis of the same block then the top level has been closed against a fiction.

WATCH OUT FOR

- ◆ **Block models disagreeing with flat analysis.** The top level is then closed against an abstraction that does not represent the block.
- ◆ **Interface budgets set late.** Blocks close internally and fail at their boundaries, and the fix requires reopening blocks that were declared done.
- ◆ **Feedthroughs not planned.** Signals crossing a block have to be routed through it, and a block closed without feedthrough provision blocks the top level.
- ◆ **Blocks closed to different criteria.** Without an enforced standard, each block closes to what its owner considered sufficient, and the top level inherits the weakest.
- ◆ **Top-level assembly left to the end.** Assembly finds interface problems, and finding them after the final turn leaves no time to fix them.

WHERE THE EFFORT GOES	30 M/M	WHO IS ON IT ~9.3 FTE	CONNECTIONS	TERMS HERE	1
Block-level closure	8.0	Hierarchical closure lead OWNS IT	DEPENDS ON ARCH-06 PD-02 PD-09 PD-13	FFN	
Top-level assembly	6.0	Block physical owners			
Integration verification	5.0	Top-level integration engineer	FEEDS INTO S0-02 S0-03 S0-09 T0-01		
Abstract and model generation	4.0	Timing engineer			
Interface budgets	4.0	Physical verification engineer			
Top-level routing	3.0				

IMPLEMENT

13 Signoff

S0

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w62 – w78	16 _w	12	159 _{M/M}	6 _{of 12}
ENGINEERING ACTIVITIES				KEY DELIVERABLES
S0-01	Signoff flow dry run on the Turn 2 database — decks, runtimes, capacity and violation triage rehearsed		6w14 _{M/M}	S0-D1STA signoff reports across all corners and modes, with waiver list
S0-02	Multi-corner multi-mode static timing analysis and closure on the final turn		10w34 _{M/M}	S0-D2Clean DRC / LVS / antenna / density reports
S0-03	Full-chip DRC, LVS, antenna and density verification		9w26 _{M/M}	S0-D3EM/IR and SI/PI signoff reports
S0-04	EM and IR-drop signoff, static and dynamic		8w20 _{M/M}	S0-D4Reliability reports — ESD, latch-up, FIT
S0-05	Power and signal integrity signoff — crosstalk, noise		6w12 _{M/M}	S0-D5DFM and lithography hotspot report
S0-06	Chip-package-system co-analysis signoff review against SIPI results		4w5 _{M/M}	S0-D6Final formal equivalence report
S0-07	Reliability verification — ESD, latch-up, soft error / FIT		6w10 _{M/M}	S0-D7Signoff summary and Design Freeze package
S0-08	DFM and lithography checks, CMP and hotspot analysis		6w10 _{M/M}	
S0-09	Final formal equivalence and LVS netlist consistency		5w8 _{M/M}	
S0-10	Waiver review board and foundry waiver alignment		6w6 _{M/M}	
S0-11	Gate-level simulation with final SDF		5w8 _{M/M}	
S0-12	Signoff corner correlation against foundry decks		5w6 _{M/M}	

13 S0 · SIGNOFF

S0-01

Signoff flow dry run on the Turn 2 database — decks, runtimes, capacity and violation triage rehearsed

TAKES	COSTS	OWNER	CRITICAL PATH
6w	14.0 M/M	Signoff lead	yes
w0-w6 of the stage	~10.1 people while it runs	Timing engineer approves	slipping moves the program
WHY IT EXISTS			
Rehearse the entire signoff flow on the turn 2 database—decks, runtimes, capacity, violation triage—so the final turn is not the first time it runs at full chip. Signoff is a large, slow, deck-heavy flow, and the first run always finds problems that have nothing to do with the design. Finding them on turn 2 costs a week of a database nobody is waiting for; finding them on the final turn costs a week of the tapeout schedule.			
WHAT IT DELIVERS			
No key deliverable is owned here — this activity contributes to the ones below.			
NEEDS FIRST			
- Turn 2 database from PD-06 - Decks and runsets from PDK-10 - Corner definition from PDK-11 - Tool matrix from PDK-08 - Compute and license capacity from PDK-12			
DONE WHEN			
- Flow runs end to end at full chip before the final turn - Runtime measured and the campaign scheduled against it - Rehearsal findings fixed, not only listed			
HOW IT GETS THERE		5 STEPS · 2 LANES	
w0	w2	w4	w6
MAIN SEQUENCE			
1 2 5			
IN PARALLEL			
3 4			
#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE		TAT
1	Signoff flow assembly and deck integration → Assembled signoff flow with integrated decks		1.5w
2	Dry run on the turn 2 database → Dry run results on turn 2		2w
3	Runtime and capacity measurement → Runtime and capacity measurements		1w
		runs in parallel	

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Violation triage process and tooling → Violation triage process and tooling	1w runs in parallel
5	Flow issue resolution and rehearsal findings → Rehearsal findings and flow fixes	2.5w

Step 3 sizes the signoff campaign. Signoff runtime on a die this size is measured in days per corner, and knowing the number before the final turn is what lets the schedule be built rather than discovered.

WATCH OUT FOR

- ◆ **Dry run skipped for schedule.** The flow's first full-chip run then happens on the critical path, and every problem it finds costs tapeout days.
- ◆ **Runtime not measured.** A signoff campaign planned without knowing its runtime is planned against a guess.
- ◆ **Dry run on an unrepresentative database.** Turn 2 has to be close enough in size and structure that the runtimes and violation counts mean something.
- ◆ **Triage tooling built during the real campaign.** Thousands of violations arrive at once, and sorting them by hand is how a week disappears.
- ◆ **Findings not fixed before the final turn.** A rehearsal that identifies problems and does not resolve them has only produced a list.

WHERE THE EFFORT GOES	14 M/M	WHO IS ON IT ~10.1 FTE	CONNECTIONS	TERMS HERE	2
Flow assembly and deck integration	4.5	Signoff lead OWNS IT	DEPENDS ON PDK-08 PDK-09 PDK-10 PDK-11 PDK-12 PDK-13 SYN-08 PD-06	STA CAD	
Dry run execution	3.5	Signoff methodology engineers	FEEDS INTO PD-09 S0-02 S0-03 S0-04 S0-05 S0-12		
Issue resolution	3.0	CAD infrastructure			
Triage process and tooling	1.5	Physical design liaison			
Runtime measurement	1.5	Timing engineer			

13 S0 · SIGNOFF

S0-02

Multi-corner multi-mode static timing analysis and closure on the final turn

TAKES	COSTS	OWNER	CRITICAL PATH
10w w5–w15 of the stage	34.0 M/M ~14.8 people while it runs	STA signoff lead Signoff methodology engineer approves	yes slipping moves the program

WHY IT EXISTS

Prove the final database meets timing **at every corner and every mode, in the signoff tool**, and dispose of every remaining violation as a fix or a signed waiver.

This is the largest activity in the stage and the one Design Freeze most depends on. It differs from PD-09 in that it is judgment rather than optimization: the signoff tool, the foundry-agreed derates and the full corner set, with no ECO loop to fall back on.

WHAT IT DELIVERS

S0-D1

STA signoff reports across all corners and modes, with waiver list

This activity is the deliverable, and the waiver list is the part that carries risk into silicon.

NEEDS FIRST

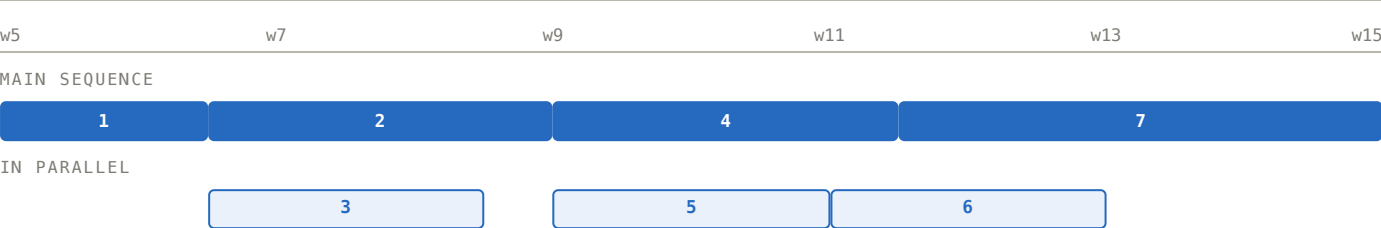
- Final database from PD-13 and PD-16
- Constraints from SYN-01
- Corner and derate definition from PDK-11
- Correlation findings from SO-12
- Timing closure status from PD-09

DONE WHEN

- Every corner and mode analyzed, none deferred
- Every waiver carries a root cause and a signature
- Test modes signed off alongside functional

HOW IT GETS THERE

7 STEPS · 2 LANES



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Signoff STA setup on the final database → Signoff STA setup	1.5w
2	Setup timing analysis across all corners and modes → Setup timing analysis across corners and modes	2.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Hold timing analysis across all corners and modes → Setup and hold analysis across all corners and modes	2w runs in parallel
4	Violation triage and root-cause classification → Violation triage with root causes	2.5w
5	ECO iteration with physical design → ECO requests to physical design	2w runs in parallel
6	Cross-mode and test-mode timing signoff → Test-mode timing signoff	2w runs in parallel
7	Timing signoff closure and waiver disposition → Timing signoff reports and waiver list	3.5w

Step 6 covers the modes that are easy to forget and expensive to miss. Scan shift, at-speed capture and any low-power mode each have their own timing, and a design signed off functionally and not in test mode produces parts that fail at ATE.

WATCH OUT FOR

- ◆ **Corners deferred to fit the schedule.** The deferred corner is the one that fails, and it fails in silicon where there is no fix.
- ◆ **Waivers granted without root cause.** A waived violation with no understood cause is a silicon risk being accepted blind.
- ◆ **Test modes not signed off.** Scan shift and at-speed timing failures produce parts that work functionally and fail at ATE.
- ◆ **Signoff and implementation disagreeing late.** If S0-12 has not correlated the two, the first signoff run can show a design weeks from closure.
- ◆ **ECO loop reopening after signoff starts.** Each ECO invalidates the analysis it was based on, and uncontrolled iteration never converges.

WHERE THE EFFORT GOES	34 M/M	WHO IS ON IT	~14.8 FTE	CONNECTIONS	TERMS HERE	4
Setup timing analysis	10.0	STA signoff lead owns it STA engineers Physical design liaison DFT engineer Signoff methodology engineer		DEPENDS ON PDK-10 PDK-11 AMS-07 AMS-10 AMS-12 AMS-13 AMS-15 SYN-01 SYN-09 PD-07 PD-09 PD-13 PD-16 S0-01 S0-04 S0-05 S0-12	STA ECO DFT ATE	
Hold timing analysis	7.0					
Violation triage and root cause	6.0					
Closure and waiver disposition	5.0					
Test-mode signoff	3.5			RUNS WITH SIPI-09		
ECO iteration	2.5			FEEDS INTO S0-06 S0-10 S0-11 T0-02 T0-03 T0-09		

13 S0 · SIGNOFF

S0-03

Full-chip DRC, LVS, antenna and density verification

TAKES	COSTS	OWNER	CRITICAL PATH
9w w6–w15 of the stage	26.0 M/M ~12.6 people while it runs	Physical verification lead Foundry liaison approves	yes slipping moves the program

WHY IT EXISTS

Run **full-chip physical verification**—DRC, LVS, antenna, density—to clean, on the database that will become masks.

Physical verification is binary and unforgiving: the foundry will not accept a database with violations it has not waived. On a die this size the runs take days, the violation counts start in the tens of thousands, and the last hundred are the ones that need a human.

WHAT IT DELIVERS

S0-D2

Clean DRC / LVS / antenna / density reports

This activity is the deliverable, and it is a precondition for mask release rather than an opinion.

NEEDS FIRST

- Final database from PD-13 and PD-16
- Verification decks from PDK-10
- Macro layouts and abstracts from AMS-12 and AMS-15
- Chip finishing from PD-15
- Foundry rule set from PDK-02

DONE WHEN

- DRC, LVS, antenna and density all clean or formally waived
- LVS mismatches resolved, not waived
- No database change after the last clean run

HOW IT GETS THERE

6 STEPS · 2 LANES

w6	w8	w10	w12	w14	w15
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Full-chip DRC setup and hierarchical run strategy → DRC run strategy and results	1.5w
2	Full-chip DRC execution and violation triage → Full-chip DRC results with triaged violations	3w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Antenna verification and fixing → Antenna verification and fixes	2w runs in parallel
4	LVS execution and mismatch debug → LVS results and mismatch resolution	2.5w
5	Density and fill verification → Density and fill verification	1.5w runs in parallel
6	Physical verification closure and waiver preparation → Physical verification clean reports and waiver candidates	2w

Step 4 is where the hardest problems hide. An LVS mismatch on a full chip can take days to localize, and the causes are usually elsewhere—a macro abstract that disagrees with its layout, or an ECO applied to one view and not the other.

WATCH OUT FOR

- ◆ **LVS mismatch traced late.** Full-chip mismatches take days to localize, and they are frequently caused by macro views rather than by the design.
- ◆ **Runs started too late.** Each full-chip run takes days, and the iteration count is what sets the schedule rather than any single run.
- ◆ **Antenna fixes needing space that is gone.** Diodes and layer jumps need room, and a fully routed design has none.
- ◆ **Waivers assumed acceptable.** Foundry waivers are granted case by case, and a design that assumes one will be granted may not tape out.
- ◆ **Verification on a database that then changes.** Any ECO after a clean run invalidates it, and re-running costs days.

WHERE THE EFFORT GOES	26 M/M	WHO IS ON IT ~12.6 FTE	CONNECTIONS
DRC execution and triage	8.0	Physical verification lead owns it	DEPENDS ON PDK-10 PD-08 PD-12 PD-13 PD-15 PD-16 S0-01
LVS and mismatch debug	7.0	Physical verification engineers	FEEDS INTO S0-09 S0-10 T0-01 T0-02 T0-09 TEST-08
Antenna verification and fixing	4.5	Layout engineers	TERMS HERE 3 ECO DRC LVS
Closure and waiver preparation	3.5	Macro owners	
Density and fill verification	3.0	Foundry liaison	

13 S0 · SIGNOFF

S0-04

EM and IR-drop signoff, static and dynamic

TAKES	COSTS	OWNER	CRITICAL PATH
8w w7–w15 of the stage	20.0 M/M ~10.9 people while it runs	EM/IR signoff lead SIPI liaison approves	yes slipping moves the program

WHY IT EXISTS

Sign off **electromigration and IR drop**—static and dynamic, with real switching—because a design that meets timing and fails EM will not survive its warranty.

EM and IR are reliability and performance at once. IR drop consumes timing margin the STA assumed; EM limits how long the part will work. Both are analyzed on the final database with real activity, and both can send routing back for changes that reopen timing.

WHAT IT DELIVERS

S0-D3

EM/IR and SI/PI signoff reports

The EM and IR half of the deliverable; S0-05 supplies the SI/PI half.

NEEDS FIRST

- Final database from PD-13
- PDN from PD-03
- Switching activity from DV-04 and SYN-06
- Chip power model and co-analysis from SIPI-04
- EM rules from PDK-02

DONE WHEN

- IR back-annotated into timing signoff
- Dynamic IR analyzed with realistic localized activity
- EM checked on signal nets as well as power

HOW IT GETS THERE

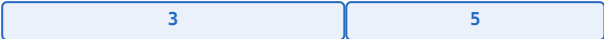
6 STEPS · 2 LANES

w7	w9	w11	w13	w15
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	EM/IR setup and activity scenario definition → EM/IR setup and activity scenarios	1.5w
2	Static IR drop analysis and closure → Static IR analysis and closure	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Dynamic IR analysis under realistic switching → Dynamic IR results under real switching runs in parallel	2w
4	Electromigration analysis on power and signal → EM analysis on power and signal nets	2w
5	IR-aware timing impact assessment → IR-aware timing impact assessment runs in parallel	1.5w
6	EM/IR closure and waiver disposition → EM/IR signoff reports	2.5w

Step 5 closes a loop that is often left open. IR drop lowers the local supply and slows the logic, and unless that is back-annotated into timing the STA has been run at a voltage the die does not actually see.

WATCH OUT FOR

- ◆ **IR not back-annotated into timing.** The design is then signed off at a supply voltage it never sees, and the margin was never real.
- ◆ **Dynamic IR analyzed with uniform activity.** Real switching is bursty and local, and uniform assumptions understate the worst case.
- ◆ **Package effects excluded.** Die-only IR is optimistic; the number that matters includes the package inductance SIPI-04 models.
- ◆ **Signal EM overlooked.** High-toggle nets on advanced nodes carry real EM exposure, and checking only power misses them.
- ◆ **EM fixes reopening timing.** Widening wires changes delay, and a fix applied after timing signoff invalidates it.

WHERE THE EFFORT GOES	20 M/M	WHO IS ON IT ~10.9 FTE	CONNECTIONS	TERMS HERE	7
Static and dynamic IR analysis	7.0	EM/IR signoff lead OWNS IT	DEPENDS ON SYN-06 PD-03 PD-10 PD-13 S0-01 SIPI-04	STA EM IR PDN SI/PI SIPI EM/IR	
EM analysis	5.0	Power integrity engineers			
Closure and waiver disposition	4.0	Reliability engineer	FEEDS INTO S0-02 S0-05 S0-06 S0-10 T0-02 MP-02		
IR-aware timing assessment	2.5	Timing engineer			
Setup and scenarios	1.5	SIPI liaison			

13 S0 · SIGNOFF

S0-05

Power and signal integrity signoff — crosstalk, noise

TAKES	COSTS	OWNER	CRITICAL PATH
6w w9–w15 of the stage	12.0 M/M ~8.7 people while it runs	SI signoff engineer Signoff lead approves	no has float

WHY IT EXISTS

Sign off **signal integrity**—crosstalk delay, noise, glitch—on the final database, and confirm the numbers the timing signoff assumed.

Crosstalk is timing. Coupling between adjacent nets changes delay in both directions and can inject glitches that propagate as functional failures. The analysis is separate from STA and its results feed straight back into it, which makes the two inseparable at signoff.

WHAT IT DELIVERS

S0-D3

EM/IR and SI/PI signoff reports

The signal integrity half of the deliverable.

NEEDS FIRST

- Final database and extraction from PD-13
- Crosstalk fixes from PD-10
- Timing analysis from SO-02
- Noise rules and thresholds from PDK-02
- PI results from SO-04

DONE WHEN

- Crosstalk delay included in the timing signoff numbers
- Glitch analysis run, not only delay analysis
- Noise thresholds set from library data, not defaults

HOW IT GETS THERE

6 STEPS · 2 LANES

w9	w11	w13	w15
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	SI signoff setup and coupling extraction → SI signoff setup and coupling extraction	1w
2	Crosstalk delay analysis and timing impact → Crosstalk delay analysis and timing impact	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Noise and glitch analysis → Noise and glitch analysis results runs in parallel	1.5w
4	Victim-aggressor analysis on critical nets → Victim-aggressor findings on critical nets	1.5w
5	Power integrity signoff reconciliation → Power integrity signoff reconciliation runs in parallel	1w
6	SI signoff closure and reporting → SI signoff report	2w

Step 4 is where the analysis becomes actionable. A global crosstalk number tells nobody what to fix; a victim-aggressor list on the critical paths tells routing exactly which nets to space or shield.

WATCH OUT FOR

- ◆ **Crosstalk reported and not reconciled with timing.** Two separate reports means the design is signed off against timing that excludes coupling.
- ◆ **Glitch analysis omitted.** Noise on a static net can propagate as a functional failure that no timing analysis would show.
- ◆ **Analysis on a subset of nets.** Coupling is a property of adjacency, and the critical net set is not the same as the coupled net set.
- ◆ **Fixes applied after timing signoff.** Spacing and shielding change delay, and a late fix invalidates the timing it was meant to protect.
- ◆ **Thresholds set by tool default.** Noise thresholds are a library and technology property, and defaults are either optimistic or unnecessarily punitive.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT	~8.7 FTE	CONNECTIONS	TERMS HERE	5
Crosstalk delay analysis	3.5	SI signoff engineer	OWNS IT	DEPENDS ON PD-10 PD-13 S0-01 S0-04 SIPI-06	STA SI PI SI/PI EM/IR	
Victim-aggressor analysis	2.5	STA engineer		FEEDS INTO S0-02 S0-10 T0-02		
Noise and glitch analysis	2.5	Routing engineers				
Closure and reporting	2.0	Library engineer				
Setup and extraction	1.5	Signoff lead				

13 S0 · SIGNOFF

S0-06

Chip-package-system co-analysis
signoff review against SIPI results

TAKES	COSTS	OWNER	CRITICAL PATH	
4w w11–w15 of the stage	5.0 M/M ~5.4 people while it runs	Signoff lead Timing engineer approves	no has float	
WHY IT EXISTS				
Review the chip-package-system co-verification result as a signoff item in its own right, rather than assuming the die-only analysis covers it. SIPI analyzed the die, package and board as one electrical system. This activity brings that result into the signoff record and reconciles it against the die-only numbers—because where they disagree, the system-level answer is the one silicon will produce.				
WHAT IT DELIVERS				
No key deliverable is owned here — this activity contributes to the ones below.				
NEEDS FIRST				
● Co-verification results from SIPI-11 ● Die-only IR from SO-04 ● Channel budgets from ARCH-04 and AMS-03 ● Chip power model from PD-11 ● Timing signoff from SO-02				
DONE WHEN				
● Die-only and system-level results reconciled, not merely compared ● Channel compliance reviewed against the specification mask ● Review completed before Design Freeze, not after				
HOW IT GETS THERE				
5 STEPS · 2 LANES				
w11	w12	w13	w14	w15
MAIN SEQUENCE				
1 2 5				
IN PARALLEL				
3 4				
#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE			TAT
1	SIPI result intake and scope confirmation → SIPI result intake record			0.75w
2	Die-only against system-level IR reconciliation → Die-only against system-level reconciliation			1.25w
3	Channel compliance review against the interface budgets → Channel compliance review findings			1w
			runs in parallel	

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Power-aware timing correlation review → Power-aware timing correlation review	1w runs in parallel
5	Co-verification signoff review and disposition → Co-verification signoff disposition	2w

Step 2 is the reconciliation that matters. Die-only IR analysis excludes package inductance, and the difference between the two numbers is margin the design either has or does not—a discrepancy that has to be resolved before mask release rather than explained afterwards.

WATCH OUT FOR

- ◆ **Co-verification treated as a package deliverable.** It constrains the die's signoff, and excluding it from the signoff record means freezing without it.
- ◆ **Discrepancies explained rather than resolved.** Where die-only and system-level disagree, one is wrong, and the difference is real margin.
- ◆ **Channel compliance reviewed against internal assumptions.** The interface specification's mask is what compliance means, and internal budgets can be more generous.
- ◆ **Review scheduled after Design Freeze.** The freeze is what gates mask release, and a co-verification finding afterwards has nowhere to go.
- ◆ **Power-aware timing not reviewed.** STA at nominal voltage and IR analysis showing droop are two views of the same design, and only the reconciliation is the truth.

WHERE THE EFFORT GOES	5 M/M	WHO IS ON IT	~5.4 FTE	CONNECTIONS	TERMS HERE	5
Co-verification review and disposition	2.0	Signoff lead owns it		DEPENDS ON PDK-02 PD-11 S0-02 S0-04 PKGD-05 PKGD-06 SIPI-03 SIPI-04 SIPI-07 SIPI-08 SIPI-09 SIPI-10 SIPI-11	STA IR SI/PI SIPI EM/IR	
IR reconciliation	1.5	SIPI lead				
Channel compliance review	0.8	Power integrity engineer				
Timing correlation review	0.8	Interface architect Timing engineer		FEEDS INTO S0-10 T0-04 T0-06		

13 S0 · SIGNOFF

S0-07

Reliability verification — ESD, latch-up, soft error / FIT

TAKES	COSTS	OWNER	CRITICAL PATH
6w w8–w14 of the stage	10.0 M/M ~7.2 people while it runs	Reliability signoff engineer Layout engineers approves	no has float

WHY IT EXISTS

Verify the **reliability requirements** the product will be qualified against—ESD paths, latch-up structures, soft error rate—on the final database rather than in the qualification lab.

Qualification failures are the most expensive kind: they arrive after silicon, after packaging and after months of stress testing. Every one of these checks has a design-time equivalent, and running them here converts a possible respin into a layout fix.

WHAT IT DELIVERS

S0-D4

Reliability reports — ESD, latch-up, FIT

This activity is the deliverable, and it is what MP-03 will be measured against.

NEEDS FIRST

- Final database from PD-13
- ESD strategy and clamps from PDK-07
- Latch-up rules from PDK-02
- Soft error requirements from the qualification plan
- AMS reliability results from AMS-14

DONE WHEN

- ESD paths traced through the full chip, not assumed from strategy
- Latch-up checked internally as well as at the ring
- FIT estimated against the product requirement

HOW IT GETS THERE

5 STEPS · 2 LANES

w8w10w12w14

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Reliability requirement collection and rule setup → Reliability requirements and rule setup	1w
2	ESD path verification through the full chip → Full-chip ESD path verification results	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Latch-up structure and guard ring verification → Latch-up and guard ring verification	1.5w runs in parallel
4	Soft error rate and FIT estimation → Soft error rate and FIT estimate	1.5w runs in parallel
5	Reliability closure and waiver disposition → Reliability signoff report and waivers	3.5w

Step 2 traces the discharge path through the actual layout rather than confirming the strategy. A clamp that exists and a path that reaches it are different claims, and only the full-chip trace establishes the second.

WATCH OUT FOR

- ◆ **ESD verified as a strategy rather than a path.** A clamp with no complete discharge path to it protects nothing, and only a full-chip trace finds the gap.
- ◆ **Latch-up checked only at the IO ring.** Internal structures near power domain boundaries can latch up too, and they are rarely checked.
- ◆ **Soft error rate not estimated.** FIT is a product requirement in many markets, and estimating it after qualification leaves no mitigation options.
- ◆ **Reliability waivers granted without lifetime analysis.** A waiver with no quantified impact cannot be defended when qualification fails.
- ◆ **Analysis on a database that then changes.** Late ECOs can break an ESD path, and nobody re-runs the check.

WHERE THE EFFORT GOES	10 M/M	WHO IS ON IT ~7.2 FTE	CONNECTIONS	TERMS HERE	4
ESD path verification	3.0	Reliability signoff engineer owns it	DEPENDS ON PDK-02 PDK-07 AMS-14 PD-13	AMS IO ESD FIT	
Latch-up verification	2.0	ESD engineer	FEEDS INTO S0-10 T0-02 MP-02 MP-03		
Closure and waiver disposition	2.0	Physical verification engineer			
Soft error and FIT estimation	1.5	Quality liaison			
Requirement setup	1.5	Layout engineers			

13 S0 · SIGNOFF

S0-08

DFM and lithography checks, CMP and hotspot analysis

TAKES	COSTS	OWNER	CRITICAL PATH
6w w8–w14 of the stage	10.0 M/M ~7.2 people while it runs	DFM engineer Yield engineer approves	no has float

WHY IT EXISTS

Check the design against **manufacturability**—lithography hotspots, CMP density, DFM recommended rules—so the foundry can print it at yield.

A DRC-clean design can still be difficult to manufacture. Litho hotspots print differently from what was drawn, and CMP density variation produces thickness variation that moves timing. Both are checked with foundry-supplied models and both are cheaper to fix here than to yield-learn in production.

WHAT IT DELIVERS

S0-D5

DFM and lithography hotspot report

This activity is the deliverable, and its compliance score is a yield predictor.

NEEDS FIRST

- Final database from PD-13
- DFM decks and litho models from the foundry
- Rule dispositions from PDK-02
- Fill and density from PD-15
- Defect density model from TECH-07

DONE WHEN

- Hotspots fixed or accepted with a yield impact estimate
- Compliance scored rather than assumed
- Fixes re-verified against DRC and timing

HOW IT GETS THERE

5 STEPS · 2 LANES

w8w10w12w14

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	DFM deck setup and model version confirmation → DFM deck setup and model versions	1w
2	Lithography hotspot detection and analysis → Litho hotspot analysis results	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	CMP and density simulation → CMP and density simulation results	1.5w runs in parallel
4	Recommended rule compliance scoring → Recommended rule compliance score	1.5w runs in parallel
5	Hotspot fixing and DFM closure → Hotspot fixes and DFM closure report	3.5w

Step 4 quantifies what `PDK-02` decided. Recommended-rule compliance is a score rather than a pass, and it correlates with yield—a design that scored 60% on the rules it chose to ignore is a design whose yield learning curve starts lower.

WATCH OUT FOR

- ◆ **DFM checked after routing is frozen.** Hotspot fixes are layout changes, and a frozen database has no room for them.
- ◆ **Recommended rules ignored without scoring.** The yield cost of ignoring them is real and unmeasured unless someone scores compliance.
- ◆ **Litho models outdated.** Foundry models improve through the node's life, and an old model finds the wrong hotspots.
- ◆ **CMP variation not linked to timing.** Thickness variation changes resistance and capacitance, and the timing impact is rarely fed back.
- ◆ **Fixes applied without re-verification.** A DFM fix is a layout change, and it can create DRC or timing problems of its own.

WHERE THE EFFORT GOES	10 M/M	WHO IS ON IT ~7.2 FTE	CONNECTIONS	TERMS HERE	4
Hotspot fixing and closure	3.5	DFM engineer OWNS IT	DEPENDS ON TECH-07 PDK-02 PD-08 PD-13 PD-15	CMP DRC LVS DFM	
Litho hotspot analysis	2.5	Physical verification engineer	FEEDS INTO S0-10 T0-02 FAB-06 MP-05		
CMP and density simulation	2.0	Layout engineers			
Rule compliance scoring	1.0	Foundry liaison			
Deck setup	1.0	Yield engineer			

13 S0 · SIGNOFF

S0-09

Final formal equivalence and LVS netlist consistency

TAKES	COSTS	OWNER	CRITICAL PATH
5w w10–w15 of the stage	8.0 M/M ~7.0 people while it runs	Equivalence engineer Signoff lead approves	yes slipping moves the program

WHY IT EXISTS

Prove for the last time that the **database being taped out is the design that was verified**—final formal equivalence and netlist-to-layout consistency.

Between the FFN and the tapeout database sit hundreds of ECOs, a scan reorder, fill and finishing. Each was individually correct; the question is whether the composition still implements the RTL that DV signed off. This is the last check that answers it.

WHAT IT DELIVERS

S0-D6

Final formal equivalence report

This activity is the deliverable, and it is the last evidence that verification's conclusions still apply.

NEEDS FIRST

- Final database and netlist from PD-13
- FFN from SYN-09
- ECO log from PD-14
- LVS results from SO-03
- Equivalence methodology from SYN-10

DONE WHEN

- Equivalence proved against the verified RTL, not only the FFN
- Every netlist difference traced to an approved ECO
- Run performed after the last database change

HOW IT GETS THERE

5 STEPS · 2 LANES

w10	w11	w12	w13	w14	w15
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Final equivalence setup against the FFN and the RTL → Final equivalence setup	1w
2	RTL to final netlist equivalence → RTL to final netlist equivalence results	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	ECO reconciliation against the change log → ECO reconciliation record	1w runs in parallel
4	Netlist to layout consistency confirmation → Netlist to layout consistency confirmation	1w runs in parallel
5	Non-equivalence resolution and final report → Final equivalence report	2.5w

Step 3 pairs the formal result with the ECO log from `PD-14`. Equivalence proves the netlists match; reconciliation proves that every difference between them was an ECO somebody approved rather than a change nobody noticed.

WATCH OUT FOR

- ◆ **Equivalence run against the FFN rather than the RTL.** That proves the ECOs preserved the netlist and not that the netlist implements the verified design.
- ◆ **ECO log not reconciled.** A difference that no ECO explains is an unapproved change, and it is exactly what this check exists to find.
- ◆ **Non-equivalences waived under tapeout pressure.** Each waiver is a region where the silicon may not do what verification proved.
- ◆ **Run before the last ECO.** Any change after the equivalence run invalidates it, and re-running takes days nobody has.
- ◆ **Layout consistency assumed from LVS.** LVS compares layout to netlist; that the netlist is the right one is a separate claim.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT ~7.0 FTE	CONNECTIONS	TERMS HERE	5
Non-equivalence resolution and report	3.0	Equivalence engineer owns it	DEPENDS ON RTL-06 DV-05 DV-06 DV-10 DV-11 DV-12 SYN-07 SYN-09 SYN-10 PD-13 PD-14 PD-16 S0-03	RTL ECO LVS FFN DV	
RTL to netlist equivalence	2.0	Physical design lead			
ECO reconciliation	1.5	Verification lead			
Setup	1.0	Physical verification engineer	FEEDS INTO S0-10 T0-02 T0-03		
Layout consistency	0.5	Signoff lead			

13 S0 · SIGNOFF

S0-10

Waiver review board and foundry waiver alignment

TAKES	COSTS	OWNER	CRITICAL PATH
6w w10–w16 of the stage	6.0 M/M ~4.3 people while it runs	Signoff lead Program manager approves	yes slipping moves the program

WHY IT EXISTS

Run the **waiver review board**—every violation that survives signoff, examined, owned and either fixed or accepted—and assemble the Design Freeze package.

Every large design tapes out with waivers. What distinguishes a controlled tapeout from a hopeful one is whether each waiver was understood, quantified and signed, or simply ran out of time. This board is where that distinction is made, and its package is what the Go / No-Go decision reads.

WHAT IT DELIVERS

S0-D7

Signoff summary and Design Freeze package

This activity is the deliverable, and it is what the tapeout decision is taken on.

NEEDS FIRST

- Waivers from every signoff activity
- Timing waivers from SO-02
- Physical verification waivers from SO-03
- Reliability and DFM waivers from SO-07 and SO-08
- Foundry waiver process from TECH-06

DONE WHEN

- Every waiver has a root cause, a quantified risk and a signature
- Foundry waivers submitted and answered before freeze
- Residual risk summarized in one statement for the gate

HOW IT GETS THERE

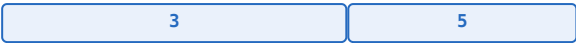
6 STEPS · 2 LANES

w10	w12	w14	w16
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Waiver intake and classification across all signoff domains → Classified waiver intake across signoff domains	1w
2	Root cause and risk quantification per waiver → Root cause and risk quantification per waiver	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Foundry waiver alignment and submission → Classified waiver register · Foundry waiver submissions and responses	1.5w runs in parallel
4	Board review and disposition → Board disposition records	1.5w
5	Residual risk statement → Residual risk statement	1w runs in parallel
6	Signoff summary and Design Freeze package assembly → Signoff summary and Design Freeze package	2w

Step 3 has a duration outside the program's control. Foundry waivers are reviewed on the foundry's schedule, and a design that assumes approval and does not receive it cannot release masks—which makes early submission worth more than a complete submission.

WATCH OUT FOR

- ◆ **Waivers accumulated rather than reviewed.** A register nobody dispositioned is a set of accepted risks nobody accepted.
- ◆ **Foundry waivers submitted late.** Their review is on the foundry's schedule, and an unapproved waiver blocks mask release.
- ◆ **Risk quantified qualitatively.** "Low risk" is not a basis for a tapeout decision; a stated failure mode and probability is.
- ◆ **Board without authority.** A review that cannot refuse a waiver is a review that approves everything.
- ◆ **Residual risk not summarized.** The Go / No-Go decision needs one statement of what is being accepted, not a register of two hundred entries.

WHERE THE EFFORT GOES	6 M/M	WHO IS ON IT ~4.3 FTE	CONNECTIONS	TERMS HERE	2
Package assembly	1.5	Signoff lead owns it	DEPENDS ON PDK-01 PDK-10 TC-07	STA DFM	
Root cause and quantification	1.5	Domain signoff owners	TC-08 S0-02 S0-03		
Board review and disposition	1.5	Foundry liaison	S0-04 S0-05 S0-06		
Foundry alignment	1.0	Chief architect	S0-07 S0-08 S0-09		
Residual risk statement	0.5	Program manager	S0-11 FEEDS INTO T0-03 T0-04 T0-05 MP-09		

13 S0 · SIGNOFF

S0-11

Gate-level simulation with final SDF

TAKES	COSTS	OWNER	CRITICAL PATH
5w w10–w15 of the stage	8.0 M/M ~7.0 people while it runs	Gate-level simulation engineer Signoff lead approves	no has float

WHY IT EXISTS

Run **gate-level simulation with the final SDF**—the last dynamic check that the design behaves correctly with real timing on the database being taped out.

Static timing analysis proves paths meet constraints; it does not prove the design works. Simulating with final timing catches what STA structurally cannot—race conditions, reset behavior under real delays, and any place where the constraints described a design different from the one built.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Final database and SDF from PD-13 and PD-09
- Gate-level environment from DV-11
- Test patterns and modes from DFT-08
- Timing signoff results from SO-02
- Power intent from SYN-07

DONE WHEN

- Simulated against the final SDF, not an earlier one
- Reset and initialization simulated with real delays
- X failures fixed rather than forced

HOW IT GETS THERE

5 STEPS · 2 LANES

w10	w11	w12	w13	w14	w15
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Final SDF generation and simulation setup → Final SDF and simulation setup	1w
2	Timing-annotated functional simulation → Timing-annotated functional simulation results	1.5w
3	Reset and initialization sequence simulation → Reset and initialization results	1w

runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Test mode simulation with final timing → Test mode simulation results	1w runs in parallel
5	Failure debug and signoff reporting → Gate-level signoff report	2.5w

Step 3 is the check most likely to find something. Reset behavior depends on relative delays across the die, and a sequence that works with estimated timing can fail with the real numbers—a failure that would otherwise appear at first power-on.

WATCH OUT FOR

- ◆ **Skipped because STA is clean.** They answer different questions, and the failures gate-level simulation finds are invisible to static analysis.
- ◆ **Run on an earlier database.** A simulation against pre-final timing says nothing about the database being taped out.
- ◆ **Reset simulation omitted.** Reset ordering under real delays is a common silicon failure and one of the cheapest to catch here.
- ◆ **X failures forced away.** Initializing signals to make the simulation run hides exactly the uninitialized state that will fail at power-on.
- ◆ **Runtime prohibiting meaningful scenarios.** Full-timing simulation is slow, and a scenario set trimmed to fit may exclude the case that matters.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT ~7.0 FTE	CONNECTIONS
Failure debug and reporting	3.0	Gate-level simulation engineer OWNS IT	DEPENDS ON DV-11 DFT-08 PD-09 PD-13 S0-02
Timing-annotated simulation	2.0	Timing engineer	FEEDS INTO S0-10 T0-02 BU-02
Reset and initialization simulation	1.5	Verification engineers	TERMS HERE 3
Test mode simulation	1.0	DFT engineer	STA DFT SDF
SDF and setup	0.5	Signoff lead	

13 S0 · SIGNOFF

S0-12

Signoff corner correlation against foundry decks

TAKES	COSTS	OWNER	CRITICAL PATH
5w w2-w7 of the stage	6.0 M/M ~5.2 people while it runs	Signoff methodology engineer Foundry liaison approves	no has float

WHY IT EXISTS

Correlate the **signoff flow against the foundry's own decks and the implementation tool**, so that a number produced here means the same thing everywhere it is read.

Three parties analyze this design: the implementation tool, the signoff tool and the foundry. If they disagree, the design can be closed in one and rejected by another. Establishing the correlation early is what lets physical design close with the right margin instead of re-closing after signoff reports.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Implementation timing results from PD-09
- Signoff flow from SO-01
- Foundry decks and derates from PDK-11
- Extraction decks from PDK-10
- Library models from PDK-03

DONE WHEN

- Correlation established on representative, difficult paths
- Margin guidance published and adopted by physical design
- Foundry deck differences understood, not assumed away

HOW IT GETS THERE

5 STEPS · 2 LANES

w2	w3	w4	w5	w6	w7
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Correlation methodology and reference case selection → Correlation methodology and reference cases	1w
2	Implementation to signoff tool correlation → Implementation to signoff correlation results	1.5w
3	Foundry deck and derate correlation → Foundry deck and derate correlation	1w

runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Extraction and parasitic correlation → Extraction and parasitic correlation	1w runs in parallel
5	Correlation findings and margin guidance → Margin guidance for physical design	2.5w

Step 5 is the output physical design actually uses. Knowing that signoff reads 40 ps more pessimistic than implementation on long paths lets `PD-09` close with that margin built in, rather than closing to zero and discovering the gap at `SO-02`.

WATCH OUT FOR

- ◆ **Correlation established after signoff starts.** The gap is then discovered on the final database, when closing it means re-opening closure.
- ◆ **Correlation on unrepresentative paths.** Tool differences concentrate on long, heavily coupled paths, and correlating on short clean ones shows agreement that does not generalize.
- ◆ **Extraction differences overlooked.** Parasitic differences between flows can exceed timing tool differences and are less often checked.
- ◆ **Margin guidance not adopted.** Correlation that physical design does not close against has produced information nobody used.
- ◆ **Foundry decks assumed identical to internal ones.** Derates and corner definitions can differ in detail, and the foundry's version is the one that decides acceptance.

WHERE THE EFFORT GOES	6 M/M	WHO IS ON IT ~5.2 FTE	CONNECTIONS	TERMS HERE	1
Correlation findings and guidance	2.0	Signoff methodology engineer OWNS IT	DEPENDS ON PDK-03 PDK-10 PDK-11 S0-01	STA	
Implementation to signoff correlation	1.5	STA engineers	RUNS WITH PD-09		
Extraction correlation	1.0	Extraction engineer	FEEDS INTO S0-02 T0-02		
Foundry deck correlation	1.0	Physical design liaison			
Methodology and reference cases	0.5	Foundry liaison			

MANUFACTURE

14

Tapeout & Mask Release (MTO)

T0

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w78 – w86	8 _w	11	47 _{M/M}	8 of 11

ENGINEERING ACTIVITIES					KEY DELIVERABLES	
T0-01	Final GDSII / OASIS assembly on the FFN closure database, with layer map verification		2w	6 _{M/M}	T0-D1	Released GDSII / OASIS database with checksum record
T0-02	Final full-chip verification re-run on the released database		2w	8 _{M/M}	T0-D2	Tapeout checklist with signoff matrix
T0-03	Tapeout checklist completion and owner signoff		3w	5 _{M/M}	T0-D3	Open issue and risk acceptance record
T0-04	Open-issue risk assessment and waiver acceptance		2w	3 _{M/M}	T0-D4	Go / No-Go decision minutes
T0-05	Go / No-Go decision meeting		0.5w	1 _{M/M}	T0-D5	FEOL MTO release package and mask order confirmation
T0-06	FEOL layer data preparation and MTO release to the mask shop		1.5w	3 _{M/M}	T0-D6	BEOL ECO log covering the fix window
T0-07	FEOL mask order confirmation and mask shop scheduling		1w	1 _{M/M}	T0-D7	BEOL MTO release package and mask order confirmation
T0-08	BEOL design fix window — metal-layer ECOs and late fixes while FEOL masks are being cut		4w	10 _{M/M}	T0-D8	Full mask set completion record
T0-09	BEOL layer re-verification — DRC, LVS, antenna and density on the fixed metal stack		2.5w	6 _{M/M}		
T0-10	BEOL layer data preparation and MTO release		1.5w	3 _{M/M}		
T0-11	BEOL mask order confirmation and full mask set completion tracking		1w	1 _{M/M}		

14 TO · TAPEOUT & MASK RELEASE (MT0)

T0-01

Final GDSII / OASIS assembly on the FFN closure database, with layer map verification

TAKES	COSTS	OWNER	CRITICAL PATH
2w w0-w2 of the stage	6.0 M/M ~13.0 people while it runs	Tapeout engineer Foundry liaison approves	yes slipping moves the program

WHY IT EXISTS

Assemble the **final GDSII or OASIS database** from the closed design—every block, every macro, every layer—and verify that the layer map the foundry will read matches the one the design was built with.

This is a mechanical activity with no tolerance for error. A layer misassigned in the stream-out produces masks that are wrong in a way no verification would have caught, because the design was correct and the translation was not.

WHAT IT DELIVERS

T0-D1

Released GDSII / OASIS database with checksum record

This activity is the deliverable, and the checksum is what makes every later reference unambiguous.

NEEDS FIRST

- Frozen signoff database from PD-13 and PD-16
- Hard macro GDS from AMS-15
- Foundry layer map and stream-out rules
- Chip finishing from PD-15
- Signoff clean status from SO-03

DONE WHEN

- Layer map verified against the foundry's definition, not the internal one
- Every macro GDS traced to its released version
- Database checksummed and recorded

HOW IT GETS THERE

4 STEPS · 2 LANES

w0	w1	w2
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Closure database intake and provenance check → Database checksum and provenance record	0.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	GDSII / OASIS stream out from the closed database → Streamed GDSII / OASIS database	0.75w
3	Macro and hard IP GDS merge → Merged macro and hard IP GDS	0.5w runs in parallel
4	Layer map verification against the foundry definition → Merged macro and IP layers · Layer map verification record	0.75w

Step 4 is checked against the foundry's layer map rather than the internal one. The two are usually identical and occasionally are not, and the difference is only visible at the mask shop—by which point the masks exist.

WATCH OUT FOR

- ◆ **Layer map mismatch.** A misassigned layer produces masks that are wrong in a way no design verification would have found.
- ◆ **Macro GDS from the wrong version.** An abstract that matched and a layout that did not is discovered at LVS, or worse, not at all.
- ◆ **Stream-out from an unfrozen database.** The design has to stop moving before it is streamed, or the streamed version is not the signed-off one.
- ◆ **No checksum.** Without one, the question "is this the database we signed off" has no answer.
- ◆ **Assembly compressed to hours.** Stream-out on a die this size takes real time, and rushing it is how a merge error is missed.

WHERE THE EFFORT GOES	6 M/M	WHO IS ON IT ~13.0 FTE	CONNECTIONS	TERMS HERE 6
Stream-out	2.5	Tapeout engineer owns it	DEPENDS ON AMS-15 SYN-09 PD-13 PD-15 PD-16 S0-03	IP LVS GDS OASIS CAD GDSII
Layer map verification	1.5	Physical design lead	FEEDS INTO T0-02 T0-03 T0-06 T0-08	
Macro merge	1.0	Macro owners		
Intake and provenance	1.0	CAD engineer		
		Foundry liaison		

14 TO · TAPEOUT & MASK RELEASE (MT0)

T0-02

Final full-chip verification re-run on the released database

TAKES	COSTS	OWNER	CRITICAL PATH
2w w0–w2 of the stage	8.0 M/M ~17.4 people while it runs	Physical verification engineer Foundry liaison approves	yes slipping moves the program

WHY IT EXISTS

Re-run **full-chip physical verification on the released database**—not on the one signoff checked, but on the exact bytes going to the mask shop.

Between the signoff run and the released database sit stream-out, macro merge and possibly a late fix. Each is a chance to introduce something. This re-run is short, expensive in compute and non-negotiable: it is the only check that the thing being released is the thing that was verified.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Released database from T0-01
- Verification decks from PDK-10
- Signoff results from SO-03
- Waiver list from SO-10
- Compute capacity for full-chip runs

DONE WHEN

- Verification run on the exact released database
- Checksums confirmed identical between verified and released
- No violations outstanding that were not already waived

HOW IT GETS THERE

5 STEPS · 2 LANES

w0

w1

w2

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Verification setup on the released database → Verification results on the released database	0.5w
2	Full-chip DRC re-run → Full-chip DRC re-run results	0.75w
3	Density and fill re-check → Density and fill confirmation	0.5w runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	LVS and antenna re-run → LVS and antenna re-run results	0.75w
5	Checksum confirmation against the released database → Checksum confirmation record	0.5w runs in parallel

Step 5 sounds pedantic and prevents the worst class of error. Verifying one database and releasing another has happened on real programs, and the only defense is confirming the checksum of what was verified against the checksum of what was sent.

WATCH OUT FOR

- ◆ **Skipped because signoff was clean.** Signoff checked a different database, and the difference between them is exactly what this run exists to catch.
- ◆ **Run on the signoff database rather than the released one.** That repeats the earlier check and proves nothing new.
- ◆ **Compute capacity unavailable.** A full-chip run takes days and needs to be scheduled, not requested.
- ◆ **New violations discovered with no time to fix.** Which is why S0-03 has to close cleanly rather than nearly.
- ◆ **Checksums not compared.** Verifying one file and releasing another is a documented failure mode with a trivial defense.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT ~17.4 FTE	CONNECTIONS	TERMS HERE 5
DRC re-run	3.0	Physical verification engineer OWNS IT	DEPENDS ON PDK-10 DV-11 SYN-10 S0-02 S0-03 S0-04 S0-05 S0-07 S0-08 S0-09 S0-11 S0-12 T0-01	DRC LVS OASIS CAD GDSII
LVS and antenna re-run	2.5	Tapeout engineer	FEEDS INTO T0-03 T0-05 T0-06	
Setup	1.0	CAD infrastructure		
Density re-check	1.0	Signoff lead		
Checksum confirmation	0.5	Foundry liaison		

14 TO · TAPEOUT & MASK RELEASE (MT0)

T0-03

Tapeout checklist completion and owner signoff

TAKES	COSTS	OWNER	CRITICAL PATH
3w w0-w3 of the stage	5.0 M/M ~7.2 people while it runs	Program manager Package lead approves	yes slipping moves the program

WHY IT EXISTS

Walk the **tapeout checklist**—every domain, every entry, evidence attached, owner signed—so the Go / No-Go decision is taken on facts rather than on confidence.

The checklist is the program's collected memory of what has to be true before masks are cut. Its value is entirely in the evidence: an entry ticked because the owner believes it is true is worth nothing, and the difference between a checklist and a ritual is whether anyone reads the attachments.

WHAT IT DELIVERS

T0-D2

Tapeout checklist with signoff matrix

This activity is the deliverable, and the signoff matrix is what makes accountability explicit.

NEEDS FIRST

- Signoff results from SO-02 through SO-11
- DFT signoff from DFT-08
- DV closure from DV-12
- Verification results from TO-02
- Waiver dispositions from SO-10

DONE WHEN

- Every entry carries evidence, not an assertion
- Signatures from accountable owners, not proxies
- Gaps escalated with time to close them

HOW IT GETS THERE

4 STEPS · 2 LANES

w0	w1	w2	w3
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Checklist assembly per domain → Assembled tapeout checklist	0.75w
2	Evidence collection per item → Evidence attached per item	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Gap chasing and escalation → Gap and escalation log	0.75w <i>runs in parallel</i>
4	Owner signoff collection → Owner signoff matrix	1.25w

Step 3 runs in parallel because it is where the time goes. Chasing an owner who has not signed, or an entry whose evidence does not exist, is the real work—the checklist itself takes an afternoon.

WATCH OUT FOR

- ◆ **Entries ticked without evidence.** A checklist of assertions is a ritual, and it produces confidence rather than information.
- ◆ **Signoff by proxy.** An entry signed by someone other than the accountable owner has moved the signature without moving the accountability.
- ◆ **Checklist assembled at tapeout.** It should be maintained across the program; assembling it in the last week means discovering gaps with no time to close them.
- ◆ **Domains missing from the checklist.** Test, package and reliability entries are easy to omit from a design-centred checklist and expensive to miss.
- ◆ **Gaps escalated too late.** An unsigned entry three days before the Go / No-Go is a decision the meeting cannot take.

WHERE THE EFFORT GOES	5 M/M	WHO IS ON IT ~7.2 FTE	CONNECTIONS	TERMS HERE 2
Evidence collection	2.0	Program manager OWNS IT	DEPENDS ON RTL-10 DV-10 DV-12 DFT-07 DFT-08 PD-14 S0-02 S0-09 S0-10 T0-01 T0-02	DV DFT
Signoff collection	1.5	Domain owners		
Checklist assembly	1.0	Signoff lead		
Gap escalation	0.5	DFT and test leads	FEEDS INTO T0-04 T0-05	
		Package lead		

14 T0 · TAPEOUT & MASK RELEASE (MT0)

T0-04

Open-issue risk assessment and waiver acceptance

TAKES	COSTS	OWNER	CRITICAL PATH
2w w1–w3 of the stage	3.0 M/M ~6.5 people while it runs	Program manager Chief architect approves	no has float

WHY IT EXISTS

Take every open issue the program is carrying—**bugs, waivers, unclosed items**—and turn it into an accepted risk with a name against it.

No large program tapes out with nothing open. The question is whether what remains was assessed and accepted, or simply ran out of time. This activity converts a list of loose ends into a set of decisions, each with an owner, a consequence and a mitigation.

WHAT IT DELIVERS

T0-D3

Open issue and risk acceptance record

This activity is the deliverable, and named ownership is what distinguishes acceptance from omission.

NEEDS FIRST

- Open bug list and risk statement from DV-12
- Waiver register from SO-10
- Test coverage gaps from DFT-07
- Package and assembly open items from PKGD-11
- Program risk register

DONE WHEN

- Every open item classified with a stated consequence
- Mitigation options identified where they exist
- Each accepted risk has a named owner

HOW IT GETS THERE

4 STEPS · 2 LANES

w1

w2

w3

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Open item consolidation across design, verification and test → Consolidated open item list	0.5w
2	Risk classification and consequence assessment → Risk classification with consequences	0.75w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Mitigation options per item → Mitigation options per item	0.5w runs in parallel
4	Risk acceptance and owner assignment → Risk acceptance record with named owners	0.75w

Step 3 is what gives the Go / No-Go meeting something to decide. An item with a mitigation—a test screen, a software workaround, a bin split—is a different decision from one without, and identifying which is which is this activity's contribution.

WATCH OUT FOR

- ◆ **Items carried without assessment.** An open issue nobody classified is a risk being accepted without anyone accepting it.
- ◆ **Risk stated qualitatively.** "Low risk" gives the decision room nothing; a failure mode, a probability and an affected population gives it a decision.
- ◆ **Mitigations not identified.** Many silicon risks can be screened at test or worked around in software, and the option is worth knowing before the gate.
- ◆ **Test and package items excluded.** Risk consolidation that covers only design misses the categories that most often cause a late problem.
- ◆ **Acceptance without an owner.** An accepted risk with no name against it has nobody watching for it at bring-up.

WHERE THE EFFORT GOES	3 M/M	WHO IS ON IT ~6.5 FTE	CONNECTIONS
Risk classification	1.3	Program manager OWNS IT	DEPENDS ON DV-12 DFT-07 S0-06 S0-10 T0-03 SIPI-11
Acceptance and ownership	0.8	Verification lead	FEEDS INTO T0-05 BU-10 MP-09
Consolidation	0.5	Signoff lead	
Mitigation options	0.5	Product engineering Chief architect	

14 TO · TAPEOUT & MASK RELEASE (MT0)

T0-05

Go / No-Go decision meeting

TAKES	COSTS	OWNER	CRITICAL PATH
0.5w w1.5–w2 of the stage	1.0 M/M ~8.7 people while it runs	Executive sponsor Product engineering approves	yes slipping moves the program

WHY IT EXISTS

Hold the **Go / No-Go decision**—the moment the program commits several million dollars of masks and a quarter of schedule to the design as it stands.

Half a week of activity and one man-month, preceded by three weeks of preparation. That ratio is correct: a gate argued in the room rather than before it is a gate whose outcome depends on who attended. The meeting exists to take a decision, record its conditions and name who owns them.

WHAT IT DELIVERS

T0-D4

Go / No-Go decision minutes

This activity is the deliverable, and the conditions attached are as important as the decision itself.

NEEDS FIRST

- Tapeout checklist from TO-03
- Risk acceptance record from TO-04
- Verification results from TO-02
- Foundry readiness from TECH-05
- Program schedule position from DEF-08

DONE WHEN

- Decision recorded with conditions and their owners
- Dissent captured in the minutes
- Conditions assigned before mask data is released

HOW IT GETS THERE

3 STEPS · 1 LANE

w1.5

w2

MAIN SEQUENCE

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Decision pack presentation — checklist, risks, readiness → Decision pack	0.2w
2	Decision, conditions and dissent capture → Go / No-Go decision with conditions · Dissent record	0.2w
3	Minutes, distribution and condition assignment → Minutes and condition assignments	0.1w

Step 2 captures dissent deliberately. The objection nobody minuted is the one that resurfaces at first silicon with nothing to point at, and recording it costs a line.

WATCH OUT FOR

- ◆ **Decision taken because the date arrived.** A gate that cannot say no is not a gate, and the organization learns that quickly.
- ◆ **Conditions accepted verbally.** A Go granted "subject to" something unrecorded is an unconditional Go.
- ◆ **Wrong people in the room.** The decision needs someone who can say no and someone who owns the consequence, and both have to attend.
- ◆ **Dissent unrecorded.** It is the cheapest insurance available and it is routinely omitted.
- ◆ **Material presented for the first time in the room.** A pack circulated on the day produces a decision based on whoever speaks most confidently.

WHERE THE EFFORT GOES	1 M/M	WHO IS ON IT ~8.7 FTE	CONNECTIONS	TERMS HERE	2
Decision pack presentation	0.4	Executive sponsor OWNS IT	DEPENDS ON DV-12 S0-10 T0-02 T0-03 T0-04	FEOL MT0	
Decision and conditions	0.4	Program manager	FEEDS INTO T0-06 T0-08 MP-09		
Minutes and distribution	0.2	Chief architect			
		Signoff lead			
		Product engineering			

14 TO · TAPEOUT & MASK RELEASE (MTO)

T0-06

FEOL layer data preparation and MTO release to the mask shop

TAKES	COSTS	OWNER	CRITICAL PATH
1.5w w2–w3.5 of the stage	3.0 M/M ~8.7 people while it runs	Tapeout engineer Procurement approves	yes slipping moves the program

WHY IT EXISTS

Prepare and release the **front-end layer data** to the mask shop—the first of the two mask tapeouts, and the one that starts the fab clock.

FEOL goes first because it is the longest lead item and the least likely to change. Releasing it now starts mask writing while the back-end layers are still absorbing late fixes, which is the entire reason the two releases are separated.

WHAT IT DELIVERS

T0-D5

FEOL MTO release package and mask order confirmation

This activity is the deliverable, and it is what starts the mask and fab clock.

NEEDS FIRST

- Released database from TO-01
- Go / No-Go decision from TO-05
- Foundry submission format and MDP requirements
- Layer split definition from TECH-02
- Mask order and slot from TECH-05

DONE WHEN

- Layer split matches the FEOL/BEOL plan exactly
- Submission acknowledged by the foundry in writing
- Released layers traceable to the database checksum

HOW IT GETS THERE

3 STEPS · 1 LANE

w2

w3

w3.5

MAIN SEQUENCE

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	FEOL layer selection and data preparation → FEOL layer data set	0.5w
2	Mask data preparation handoff — fracture, OPC scope → Mask data preparation handoff package	0.5w
3	Submission to the foundry and acceptance confirmation → Submission record and foundry acceptance	0.5w

Step 1's layer selection has to be exact. A layer released with FEOL that later needs a fix cannot be fixed in the BEOL window, and a layer held back that the mask shop needed delays the whole set.

WATCH OUT FOR

- ◆ **Layer split wrong.** A layer released as FEOL that later needs a fix cannot use the BEOL window, and the fix costs a new mask.
- ◆ **Released before the Go / No-Go.** Schedule pressure makes this tempting and it removes the gate's authority entirely.
- ◆ **MDP requirements misunderstood.** Fracture and OPC expectations differ per foundry, and a non-compliant submission is rejected.
- ◆ **Submission not acknowledged.** A release nobody confirmed received is a release that may not have started the clock.
- ◆ **Traceability lost between database and submission.** The submitted layers have to be provably from the checksummed database.

WHERE THE EFFORT GOES	3 M/M	WHO IS ON IT ~8.7 FTE	CONNECTIONS	TERMS HERE	7
		Tapeout engineer owns it	DEPENDS ON TECH-05 SO-06 T0-01 T0-02 T0-05	FEOL BEOL MTO OPC MDP OASIS GDSII	
Data preparation	1.5				
MDP handoff	1.0	Foundry liaison	FEEDS INTO T0-07 T0-08 T0-10 FAB-01		
Submission and acceptance	0.5	Physical verification engineer Program manager Procurement			

14 TO · TAPEOUT & MASK RELEASE (MT0)

T0-07

FEOL mask order confirmation and mask shop scheduling

TAKES	COSTS	OWNER	CRITICAL PATH
1w w3-w4 of the stage	1.0 M/M ~4.3 people while it runs	Procurement Finance partner approves	no has float

WHY IT EXISTS

Confirm the **FEOL mask order** and get a mask shop schedule the program can plan wafer start against.

One man-month of administration that fixes a date the whole downstream schedule hangs on. Mask writing takes weeks, and knowing when the FEOL set will be ready is what tells FAB-02 when wafers can start.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- FEOL submission record from TO-06
- Mask cost sheet from TECH-04
- Capacity and slot commitment from TECH-05
- Purchase authorization
- Mask shop schedule

DONE WHEN

- Order confirmed against the exact submitted layer set
- Mask completion date received and communicated
- Escalation contact identified

HOW IT GETS THERE

3 STEPS · 1 LANE

w3

w4

MAIN SEQUENCE

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Mask order confirmation and purchase authorization → Confirmed mask order	0.4w
2	Mask shop schedule confirmation → Mask shop schedule with completion date	0.3w
3	Delivery date communication to the program → Delivery date communication to the program	0.3w

Step 3 sounds trivial and is where schedules break. A mask completion date known only to the person who confirmed it is a date the rest of the program plans without.

WATCH OUT FOR

- ◆

Order placed without confirmed pricing. Mask cost is a large line item and an unconfirmed order produces an invoice nobody budgeted.
- ◆

Mask shop schedule assumed. Mask writing capacity is shared, and the completion date is the shop's to give rather than the program's to assume.
- ◆

Completion date not communicated. Wafer start planning depends on it, and a date held by one person is a date the program does not have.
- ◆

No escalation path. Mask shop delays happen, and knowing who to call is worth arranging before it is needed.
- ◆

Order and submission mismatched. The order has to name the exact layer set that was submitted.

WHERE THE EFFORT GOES	1 M/M	WHO IS ON IT ~4.3 FTE	CONNECTIONS
Order confirmation	0.5	Procurement owns it	DEPENDS ON TECH-04 TECH-05 T0-06
Schedule confirmation	0.3	Foundry liaison	FEEDS INTO T0-11 FAB-01 FAB-02
Communication	0.2	Program manager	TERMS HERE 2
		Tapeout engineer	FEOL MTO
		Finance partner	

14 TO · TAPEOUT & MASK RELEASE (MT0)

T0-08

BEOL design fix window — metal-layer ECOs and late fixes while FEOL masks are being cut

TAKES	COSTS	OWNER	CRITICAL PATH
4w w2–w6 of the stage	10.0 M/M ~10.9 people while it runs	Physical design lead Signoff lead approves	yes slipping moves the program

WHY IT EXISTS

Use the **four weeks between the two mask releases** to make metal-layer fixes—the last window in which a design change is possible without a new front-end mask set.

This window is the reason the releases are split. FEOL masks are being cut and cannot change; BEOL is still open, and any fix implementable in metal can still be made. It is a genuine and finite opportunity, and it closes on a date.

WHAT IT DELIVERS

T0-D6

BEOL ECO log covering the fix window

This activity is the deliverable, and the log is what makes the BEOL database traceable.

NEEDS FIRST

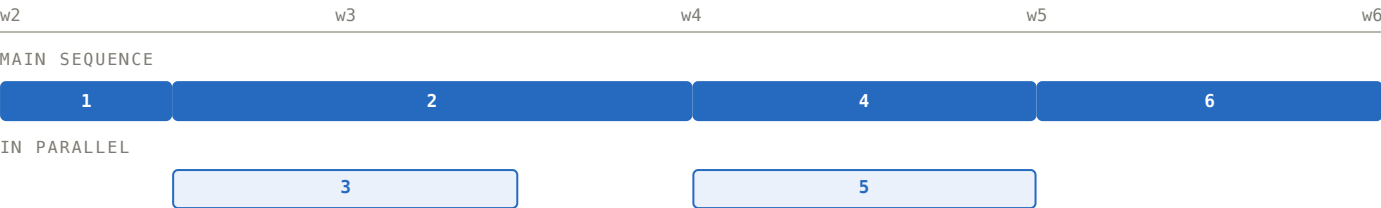
- Released database from TO-01
- Late findings from bring-up preparation, verification and signoff
- Spare cell and metal-only strategy from PD-14
- Go / No-Go conditions from TO-05
- Equivalence methodology from SO-09

DONE WHEN

- Every fix implementable in metal, none requiring base layers
- Combined database verified, not only individual fixes
- Window closed on its date, not extended

HOW IT GETS THERE

6 STEPS · 2 LANES



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Fix window scope and admission rules → Fix window scope and admission rules	0.5w
2	Metal-layer ECO implementation → Implemented metal-layer ECOs	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Fix prioritization and risk assessment → Fix prioritization and risk record	1w runs in parallel
4	ECO verification per fix → Per-fix verification results	1w
5	Equivalence and consistency check per fix → Per-fix equivalence and consistency results	1w runs in parallel
6	Window close and BEOL database freeze → Frozen BEOL database	1w

Step 1 has to be strict. Every late fix carries risk, and a window with no admission rules fills with changes that individually look safe and collectively make the BEOL database something nobody has verified as a whole.

WATCH OUT FOR

- ◆ **Window used as general contingency.** Fixes admitted without rules accumulate into a database that differs from the verified one in ways nobody assessed as a whole.
- ◆ **Fix requiring base layers.** If PD-14's spare cell strategy was not put in place, the fix cannot be made in metal and the window is useless.
- ◆ **Verification per fix but not in combination.** Individually correct changes can interact, and the combined database needs a check of its own.
- ◆ **Window extended.** BEOL mask release has a date; extending the window pushes the whole mask set and with it wafer start.
- ◆ **Equivalence skipped for speed.** A metal ECO can change function, and the only detector is an equivalence run.

WHERE THE EFFORT GOES	10 M/M	WHO IS ON IT ~10.9 FTE	CONNECTIONS	TERMS HERE	4
ECO implementation	4.0	Physical design lead owns it	DEPENDS ON PD-14 T0-01 T0-05 T0-06	FEOL BEOL MTO ECO	
ECO verification	2.5	Program manager	FEEDS INTO T0-09 T0-10 BU-11		
Prioritization and risk	1.5	Verification lead			
Equivalence checks	1.5	Equivalence engineer			
Window close	0.5	Signoff lead			

14 TO · TAPEOUT & MASK RELEASE (MT0)

T0-09

BEOL layer re-verification — DRC, LVS, antenna and density on the fixed metal stack

TAKES	COSTS	OWNER	CRITICAL PATH
2.5w w3.5–w6 of the stage	6.0 M/M ~10.4 people while it runs	Physical verification engineer Signoff lead approves	yes slipping moves the program

WHY IT EXISTS

Re-verify the **metal stack after the fix window**—DRC, LVS, antenna, density on the changed layers—before the back-end data is released.

The BEOL database is not the one that was signed off. It carries the fix window's changes, and every one of them touched routing. This re-verification is what makes the BEOL release as trustworthy as the FEOL one, and it has to fit inside the window's tail.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Frozen BEOL database from TO-08
- ECO log from TO-08
- Verification decks from PDK-10
- Timing analysis capability from SO-02
- Signoff baseline from SO-03

DONE WHEN

- Scope justified against what the fixes touched
- Timing re-confirmed for every applied fix
- Results complete before BEOL data is released

HOW IT GETS THERE

4 STEPS · 2 LANES

w3.5

w4.5

w5.5

w6

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Re-verification scope on the changed layers → Re-verification scope decision	0.5w
2	DRC and antenna verification on the metal stack → DRC and antenna results on the metal stack	1w
3	Timing impact confirmation for the applied fixes → Timing impact confirmation	0.75w <i>runs in parallel</i>

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	LVS and density re-check → LVS and density re-check results	1w

Step 1 decides between a targeted re-run and a full one. Targeted is faster and risks missing an interaction; full is safe and may not fit the window. The scope decision has to be made on what the fixes actually touched.

WATCH OUT FOR

- ◆ **Scope too narrow.** A targeted re-run that misses an interaction between two fixes releases a violation into the mask set.
- ◆ **Timing not re-confirmed.** Metal ECOs change delay, and a fix that breaks timing has replaced one problem with another.
- ◆ **Re-verification not fitting the window.** Full-chip runs take days, and a window planned without them cannot accommodate them.
- ◆ **LVS skipped because the change was metal-only.** Metal changes connectivity, which is exactly what LVS checks.
- ◆ **Results arriving after the release.** Verifying after submission is verifying something already committed to masks.

WHERE THE EFFORT GOES	6 M/M	WHO IS ON IT ~10.4 FTE	CONNECTIONS	TERMS HERE 6
DRC and antenna verification	2.5	Physical verification engineer OWNS IT	DEPENDS ON PDK-10 S0-02 S0-03 T0-08	FEOL BEOL MTO ECO DRC LVS
LVS and density re-check	2.0	Physical design lead	FEEDS INTO T0-10	
Scope definition	0.8	Timing engineer		
Timing confirmation	0.8	Tapeout engineer		
		Signoff lead		

14 TO · TAPEOUT & MASK RELEASE (MTO)

T0-10

BEOL layer data preparation and MTO release

TAKES	COSTS	OWNER	CRITICAL PATH
1.5w w6–w7.5 of the stage	3.0 M/M ~8.7 people while it runs	Tapeout engineer Configuration manager approves	yes slipping moves the program

WHY IT EXISTS

Prepare and release the **back-end layer data**—the second mask tapeout, and the moment the design is fully committed to silicon.

BEOL release is the real tapeout. After it nothing about the design can change without a new mask set, and the milestone the program has been working toward for eighty weeks is this submission being accepted.

WHAT IT DELIVERS

T0-D7

BEOL MTO release package and mask order confirmation

This activity is the deliverable, and the milestone the stage carries—Tapeout.

NEEDS FIRST

- Re-verified BEOL database from TO-09
- FEOL layer split from TO-06
- Foundry submission requirements
- Mask order framework from TO-07
- ECO log from TO-08

DONE WHEN

- Layer set exactly complementary to the FEOL release
- Submission acknowledged in writing
- Database difference from the original traced to the ECO log

HOW IT GETS THERE

3 STEPS · 1 LANE

w6

w7

w7.5

MAIN SEQUENCE

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	BEOL layer data preparation from the re-verified database → BEOL layer data set	0.5w
2	Mask data preparation handoff → Mask data preparation handoff package	0.5w
3	Submission to the foundry and acceptance confirmation → Submission record and foundry acceptance	0.5w

The layer set here has to be exactly the complement of what `TO-06` released. A layer released twice or not at all produces a mask set that is either wrong or incomplete, and the mask shop will not necessarily notice.

WATCH OUT FOR

- ◆ **Layer set not complementary to FEOL.** A layer released twice or omitted produces an incomplete or contradictory mask set.
- ◆ **Released before re-verification completes.** The fix window's changes are then committed unchecked.
- ◆ **Submission rejected on format.** BEOL layers have their own OPC and fracture requirements, and a rejection costs days against a running FEOL clock.
- ◆ **No acceptance confirmation.** An unconfirmed release may not have started the second mask clock.
- ◆ **Traceability broken.** The BEOL database differs from the checksummed original, and the difference has to be exactly the ECO log.

WHERE THE EFFORT GOES	3 M/M	WHO IS ON IT ~8.7 FTE	CONNECTIONS	TERMS HERE	6
		Tapeout engineer owns it	DEPENDS ON TO-06 TO-08 TO-09	FEOL BEOL MTO OPC	
Data preparation	1.5			MDP ECO	
MDP handoff	1.0	Foundry liaison	FEEDS INTO TO-11 FAB-03		
Submission and acceptance	0.5	Physical verification engineer			
		Program manager			
		Configuration manager			

14 TO · TAPEOUT & MASK RELEASE (MT0)

T0-11

BEOL mask order confirmation and full mask set completion tracking

TAKES	COSTS	OWNER	CRITICAL PATH
1w w7-w8 of the stage	1.0 M/M ~4.3 people while it runs	Procurement Operations planner approves	no has float

WHY IT EXISTS

Confirm the **BEOL order and track the mask set to completion**, so that FAB-05 knows when back-end layers will be available and the line does not stall mid-lot.

The BEOL masks are needed several weeks into wafer processing, not at wafer start. Tracking them to completion is what confirms that the parallel schedule the FEOL/BEOL split assumed actually holds.

WHAT IT DELIVERS

T0-D8

Full mask set completion record

This activity is the deliverable, and it is what confirms the split schedule held.

NEEDS FIRST

- BEOL submission record from TO-10
- Mask order framework from TO-07
- Mask shop schedules
- Wafer processing plan from FAB-04
- Cost sheet from TECH-04

DONE WHEN

- Order confirmed against the submitted layer set
- Set completion tracked to the last layer
- Availability communicated to fabrication against the line schedule

HOW IT GETS THERE

3 STEPS · 1 LANE

w7

w8

MAIN SEQUENCE

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	BEOL mask order confirmation → Confirmed BEOL mask order	0.4w
2	Full mask set completion tracking → Full mask set completion tracking	0.3w
3	Mask availability communication to fabrication → Mask availability communication to fabrication	0.3w

Step 2 tracks the whole set rather than the second half. A mask set is only usable when every layer is complete, and the last layer to finish is the one that decides.

WATCH OUT FOR

- ◆ **BEOL masks late against the wafer schedule.** Wafers reach the metallization steps and stop, which is worse than starting late.
- ◆ **Completion tracked per layer rather than per set.** The set is usable when the last layer is done, and per-layer optimism hides that.
- ◆ **Availability not communicated to fabrication.** The line plans against the date, and a date nobody passed on is a date the line does not have.
- ◆ **Order mismatched to submission.** The order has to name exactly the layers submitted, and BEOL sets are large.
- ◆ **No contingency for a mask defect.** Masks fail inspection sometimes, and a remake takes weeks.

WHERE THE EFFORT GOES	1 M/M	WHO IS ON IT ~4.3 FTE	CONNECTIONS
Order confirmation	0.4	Procurement owns it	DEPENDS ON T0-07 T0-10
Set completion tracking	0.4	Foundry liaison	FEEDS INTO FAB-03 FAB-05 FAB-09
Communication	0.2	Program manager Tapeout engineer Operations planner	TERMS HERE 2 BEOL MTO

MANUFACTURE

15

Fabrication

FAB

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w79 – w98	19 _w	10	30 _{M/M}	6 of 10

ENGINEERING ACTIVITIES				KEY DELIVERABLES		
FAB-01	FEOL mask set fabrication, inspection and qualification	7w	4	M/M	FAB-D1	FEOL and BEOL mask sets with qualification reports
FAB-02	Wafer start on FEOL mask availability	1w	1	M/M	FAB-D2	Processed engineering-lot wafers
FAB-03	BEOL mask set fabrication, inspection and qualification, in parallel with front-end processing	6w	4	M/M	FAB-D3	Inline, PCM and WAT data package
FAB-04	Front-end wafer processing	8w	3	M/M	FAB-D4	Wafer acceptance disposition record
FAB-05	Back-end-of-line processing	5w	2	M/M	FAB-D5	Wafer-out forecast versus actual log
FAB-06	Inline metrology and defect inspection monitoring	14w cont.	5	M/M	FAB-D6	First Silicon availability notice
FAB-07	E-test / PCM data review	3w	3	M/M		
FAB-08	Wafer acceptance test and lot disposition	3w	3	M/M		
FAB-09	Hot-lot management and WIP tracking	18w cont.	4	M/M		
FAB-10	Wafer shipment and logistics to sort and assembly	2w	1	M/M		

15 FAB · FABRICATION

FAB-01

FEOL mask set fabrication, inspection and qualification

TAKES	COSTS	OWNER	CRITICAL PATH
7w w0–w7 of the stage	4.0 M/M ~2.5 people while it runs	Foundry liaison Procurement approves	yes slipping moves the program

WHY IT EXISTS

Get the **front-end mask set written, inspected and qualified**—seven weeks of the mask shop's calendar that the program can watch but not accelerate.

EUV mask sets at this node are slow and expensive to produce, and a single mask failing inspection means a remake measured in weeks. The program's role is oversight rather than execution: confirm the data was accepted, review inspection results and escalate when a layer slips.

WHAT IT DELIVERS

FAB-D1

FEOL and BEOL mask sets with qualification reports

The front-end half of the deliverable.

NEEDS FIRST

- FEOL data submission from TO-06
- Mask order and schedule from TO-07
- Foundry mask inspection criteria
- Layer criticality assessment
- Mask cost and remake terms from TECH-04

DONE WHEN

- Every mask inspected and dispositioned, not only written
- Set qualified and released to the line
- Completion date communicated against the wafer start plan

HOW IT GETS THERE

6 STEPS · 2 LANES

w0	w2	w4	w6	w7
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Mask data acceptance and mask data preparation → Mask data preparation acceptance	1w
2	FEOL mask writing → Written FEOL mask set	3w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Pellicle and handling preparation → Pellicle and handling preparation record	1.5w runs in parallel
4	Mask inspection and defect repair → Inspection results per layer	2w
5	Mask defect disposition and remake decisions → Defect disposition decisions	1w runs in parallel
6	Mask qualification and release to the line → Qualified mask set released to the line	1w

Step 5 is where a schedule is saved or lost. A mask defect can sometimes be repaired, sometimes waived as non-printing, and sometimes requires a remake—and the three outcomes differ by weeks.

WATCH OUT FOR

- ◆ **Mask defect requiring a remake.** Weeks of schedule, and the program has no way to accelerate the replacement.
- ◆ **Inspection criteria not understood.** Whether a defect is printing depends on layer and position, and the disposition needs design input the foundry does not have.
- ◆ **Critical layers not prioritized.** Mask shops write in a sequence, and the layers on the critical path can sometimes be pulled forward if anyone asks.
- ◆ **Data preparation rejected late.** MDP problems surface days after submission and consume the schedule silently.
- ◆ **No visibility into progress.** A seven-week activity with no status reporting means the first news is either completion or a delay.

WHERE THE EFFORT GOES	4 M/M	WHO IS ON IT	~2.5 FTE	CONNECTIONS
Mask writing oversight	1.5	Foundry liaison	owns it	DEPENDS ON T0-06 T0-07
Inspection review	1.0	Tapeout engineer		FEEDS INTO FAB-02 FAB-04 FAB-09
Qualification	0.8	Program manager		TERMS HERE 4
MDP acceptance	0.8	Physical verification engineer		FEOL BEOL EUV MDP
		Procurement		

15 FAB · FABRICATION

FAB-02

Wafer start on FEOL mask availability

TAKES	COSTS	OWNER	CRITICAL PATH
1w w7–w8 of the stage	1.0 M/M ~4.3 people while it runs	Operations planner Program manager approves	yes slipping moves the program

WHY IT EXISTS

Start wafers **the moment the front-end masks are qualified**—lot definition, starting material, split strategy—because every day of delay here is a day at the far end of the program.

A single week of activity that gates the remaining eleven. The decisions it carries are small and consequential: how many wafers, what splits, which lots are hot, and whether any material is held back for a second attempt.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Qualified FEOL mask set from FAB-01
- Capacity and hot-lot terms from TECH-05
- Wafer quantity plan from the engineering lot requirement
- Split strategy from TECH-07 node risk
- Starting material specification

DONE WHEN

- Wafers started within days of mask qualification
- Quantity sufficient for bring-up, qual and samples
- Hot-lot priority invoked where it was negotiated

HOW IT GETS THERE

3 STEPS · 1 LANE

w7

w8

MAIN SEQUENCE

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Wafer start authorization and lot definition → Wafer start authorization	0.4w
2	Starting material and lot split decision → Lot definition with split strategy	0.3w
3	Start confirmation and schedule communication → Start confirmation and schedule communication	0.3w

Step 2 decides the split strategy—whether all wafers run the same process conditions or some are deliberately skewed. Splits cost wafers and buy information about process sensitivity that no simulation can provide.

WATCH OUT FOR

- ◆ **Start delayed after masks are ready.** Every day is a day at the far end of a schedule with no recovery mechanism.
- ◆ **Wafer quantity too small.** Bring-up, characterization, qualification and customer samples all draw from this lot, and running short means a second lot and a quarter.
- ◆ **No process splits.** A single condition gives one data point; splits reveal process sensitivity while there is still time to act on it.
- ◆ **Hot-lot status not invoked.** The priority negotiated in TECH-05 has to be requested at start, not assumed.
- ◆ **Starting material not specified.** Substrate type and specification affect device behavior, and defaulting is how an unexpected variable enters.

WHERE THE EFFORT GOES	1 M/M	WHO IS ON IT ~4.3 FTE	CONNECTIONS
Authorization	0.4	Operations planner owns it	DEPENDS ON TECH-05 T0-07 FAB-01
Lot split decision	0.4	Foundry liaison	FEEDS INTO FAB-04 FAB-06 FAB-09
Confirmation	0.2	Product engineering	TERMS HERE 1
		Device engineer	FEOL
		Program manager	

15 FAB · FABRICATION

FAB-03

BEOL mask set fabrication, inspection and qualification, in parallel with front-end processing

TAKES	COSTS	OWNER	CRITICAL PATH
6w w4–w10 of the stage	4.0 M/M ~2.9 people while it runs	Foundry liaison Physical verification engineer approves	no has float

WHY IT EXISTS

Write and qualify the **back-end mask set while front-end processing runs**—the parallelism the FEOL/BEOL split was designed to create.

This activity is the payoff of the two-stage release. BEOL masks are written during the weeks the wafers spend in front-end processing, so the fix window cost the program nothing in schedule as long as these masks are ready when the lots reach metallization.

WHAT IT DELIVERS

FAB-D1

FEOL and BEOL mask sets with qualification reports

The back-end half of the deliverable.

NEEDS FIRST

- BEOL data submission from TO-10
- Mask order from TO-11
- Front-end processing schedule from FAB-04
- Foundry inspection criteria
- Metallization start estimate

DONE WHEN

- Masks qualified before lots reach metallization
- Progress tracked against the front-end schedule, not in isolation
- First metal layers prioritized in the writing sequence

HOW IT GETS THERE

5 STEPS · 2 LANES

w4

w6

w8

w10

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	BEOL mask data acceptance and preparation → BEOL mask data acceptance	0.5w
2	BEOL mask writing → Written BEOL mask set · Qualified BEOL masks released to the line	2.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Defect disposition on inspected layers → Inspection and defect disposition results	1.5w runs in parallel
4	Inspection and qualification → BEOL mask inspection and qualification results	2w
5	Release to the line ahead of metallization → BEOL masks released to the line	1w

Step 5's date is the one that matters. The masks do not need to exist at wafer start; they need to exist before the lots reach the first metal layer, and that is several weeks of margin the split created deliberately.

WATCH OUT FOR

- ◆ **Masks late against metallization.** Lots stop mid-process, which wastes fab capacity and is harder to recover from than a late start.
- ◆ **Parallelism assumed rather than tracked.** The split only pays off if the BEOL schedule is actively compared against the front-end progress.
- ◆ **Defect disposition slower on BEOL layers.** There are more of them, and a per-layer process that worked for FEOL can bottleneck here.
- ◆ **Metallization start estimate stale.** The masks are needed against a moving date, and planning against the original estimate is planning against fiction.
- ◆ **No priority on the first metal layers.** They are needed first and can often be written first if anyone asks.

WHERE THE EFFORT GOES	4 M/M	WHO IS ON IT ~2.9 FTE	CONNECTIONS
Mask writing oversight	1.5	Foundry liaison owns it	DEPENDS ON T0-10 T0-11 FAB-04
Inspection and qualification	1.5	Tapeout engineer	FEEDS INTO FAB-05 FAB-09
Data acceptance	0.5	Operations planner	TERMS HERE 2
Release	0.5	Program manager	FEOL BEOL
		Physical verification engineer	

15 FAB · FABRICATION

FAB-04

Front-end wafer processing

TAKES	COSTS	OWNER	CRITICAL PATH
8w w7–w15 of the stage	3.0 M/M ~1.6 people while it runs	Foundry liaison Program manager approves	yes slipping moves the program

WHY IT EXISTS

Watch the wafers through **front-end processing**—transistor formation, contacts, local interconnect—and respond when the line reports something unexpected.

Eight weeks in which the program does almost nothing and cannot afford to do nothing. The lots are the foundry's to run; what belongs to the program is understanding the inline data, recognizing an excursion early and deciding what to do about it.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Started lots from FAB-02
- Inline metrology data from FAB-06
- Foundry process control data
- Excursion response framework
- Hot-lot priority from TECH-05

DONE WHEN

- Every excursion dispositioned against a framework, not improvised
- Inline data reviewed by the design team, not only the foundry
- Hot-lot priority maintained through the flow

HOW IT GETS THERE

6 STEPS · 2 LANES

w7	w9	w11	w13	w15
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Lot release and front-end entry confirmation → Lot release and front-end entry confirmation	1w
2	Transistor formation processing → Wafers through transistor formation	3w
3	Inline monitoring at front-end critical steps → Inline monitoring findings at critical steps	2w

runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Contact and local interconnect processing → Wafers through contact and local interconnect	3w
5	Excursion response and disposition → Excursion responses and lot dispositions	1.5w runs in parallel
6	Front-end exit and handoff to back end → Front-end processing progress · Front-end exit confirmation	1w

Step 5 needs a standing decision framework. When a lot goes out of control limits the choices are scrap, rework, continue at risk or split —and making that decision in the moment, without a framework, tends to produce whichever answer is least disruptive rather than best.

WATCH OUT FOR

- ◆ **Excursion response improvised.** Without a framework the decision defaults to whatever is least disruptive rather than what is best for the lot.
- ◆ **Inline data not reviewed by the program.** The foundry monitors its process; only the design team can say whether a deviation matters for this device.
- ◆ **Hot-lot priority lost mid-flow.** Priority granted at start can quietly lapse, and nobody notices until the cycle time slips.
- ◆ **Scrap decision taken too late.** A lot continuing through eight weeks of processing after a fatal excursion wastes capacity that a second lot needed.
- ◆ **No second lot in reserve.** If the first lot is lost, having no material started means restarting the whole cycle.

WHERE THE EFFORT GOES	3 M/M	WHO IS ON IT	~1.6 FTE	CONNECTIONS	TERMS HERE	2
Monitoring and data review	1.3	Foundry liaison OWNS IT		DEPENDS ON FAB-01 FAB-02 FAB-06	WAT PCM	
Excursion response	0.8	Device engineer		FEEDS INTO FAB-03 FAB-05 FAB-07		
Lot management	0.5	Yield engineer		FAB-09		
Handoff	0.5	Operations planner				
		Program manager				

15 FAB · FABRICATION

FAB-05

Back-end-of-line processing

TAKES	COSTS	OWNER	CRITICAL PATH
5w w14–w19 of the stage	2.0 M/M ~1.7 people while it runs	Foundry liaison Package liaison approves	yes slipping moves the program

WHY IT EXISTS

Complete the wafers through **back-end-of-line processing**—metallization, vias, top metal, passivation—and finish the lot.

The back end is where the design's routing becomes physical. It is also where the BEOL masks are consumed, which makes it the point at which the FEOL/BEOL split either paid off invisibly or produces a stall that everyone notices.

WHAT IT DELIVERS

FAB-D2

Processed engineering-lot wafers

This activity is where the wafers are finished—the deliverable the whole stage exists to produce.

NEEDS FIRST

- Front-end wafers from FAB-04
- Qualified BEOL masks from FAB-03
- Inline monitoring from FAB-06
- Process control data
- Metallization process specification

DONE WHEN

- Wafers completed without a mask-driven stall
- Split traceability maintained to lot exit
- Wafer count confirmed against the downstream requirement

HOW IT GETS THERE

5 STEPS · 2 LANES

w14	w15	w16	w17	w18	w19
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Back-end entry and mask availability confirmation → Back-end entry and mask availability confirmation	0.5w
2	Metallization and via formation → Wafers through metallization and via formation	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Back-end inline monitoring → Back-end inline monitoring findings	1.5w runs in parallel
4	Top metal and passivation → Wafers through top metal and passivation	1.5w
5	Wafer completion and exit → Metallized and via-formed wafers · Completed wafers with passivation · Lot exit confirmation	1w

Step 1 is a confirmation with real consequences. If `FAB-03` has not delivered qualified masks by this point the lots wait in the line, and waiting mid-process is more expensive than waiting at the start.

WATCH OUT FOR

- ◆ **BEOL masks not ready at back-end entry.** Lots stall mid-process, consuming fab capacity and adding cycle time nobody planned.
- ◆ **Metallization defects detected late.** Back-end defects affect yield directly, and inline detection is what allows a response.
- ◆ **Passivation and top metal treated as routine.** They affect bump attachment and package assembly, and problems here surface at ASSY-03.
- ◆ **Lot completion not confirmed against quantity.** Wafers lost during processing reduce the number available downstream, and the count has to be checked rather than assumed.
- ◆ **Process changes between split lots not tracked.** If splits were run, which wafer had which condition has to be traceable to the end.

WHERE THE EFFORT GOES	2 M/M	WHO IS ON IT ~1.7 FTE	CONNECTIONS	TERMS HERE 3
Monitoring	0.8	Foundry liaison owns it	DEPENDS ON T0-11 FAB-03 FAB-04 FAB-06	BEOL WAT PCM
Entry checks	0.5	Device engineer		
Completion	0.5	Yield engineer	FEEDS INTO FAB-07 FAB-08 FAB-10	
Handoff	0.3	Operations planner		
		Package liaison		

15 FAB · FABRICATION

FAB-06

Inline metrology and defect inspection monitoring

TAKES	COSTS	OWNER	CRITICAL PATH
14w cont. w5–w19 of the stage	5.0 M/M ~1.6 people while it runs	Yield engineer Product engineering approves	no has float

WHY IT EXISTS

Watch the **inline metrology and defect inspection data** across the whole run, so an excursion is caught while the lot can still be saved rather than at final test.

The foundry monitors its process against its own control limits; the program needs to know whether a deviation matters for this device. A defect density excursion on a layer the design uses heavily is a different event from one on a layer it barely touches, and only the design team can tell the difference.

WHAT IT DELIVERS

FAB-D3

Inline, PCM and WAT data package

The inline half of the deliverable, and the earliest signal the program gets.

NEEDS FIRST

- Foundry inline metrology data
- Defect inspection results
- Design sensitivity analysis from TECH-03 and SO-08
- Process control limits
- Yield model from TECH-07

DONE WHEN

- Monitoring focused on the design's actual sensitivities
- Excursions escalated while the lot can still be acted on
- Defects attributed to a source, not only counted

HOW IT GETS THERE

6 STEPS · 2 LANES

w5	w8	w11	w14	w17	w19
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Monitoring plan and critical step selection → Monitoring plan with critical step selection	1w
2	Inline metrology data review → Inline metrology review findings	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Excursion detection and escalation → Excursion detections and escalations runs in parallel	2w
4	Defect inspection review and classification → Reviewed and classified defect inspection results	2w
5	Defect Pareto and source attribution runs in parallel → Defect classification and Pareto · Source attribution findings	2w
6	Continuous monitoring across the lot → Inline monitoring log across the lot	9w

Step 1 chooses where to look. Monitoring every step equally wastes attention; selecting the layers and dimensions this design is most sensitive to—minimum-pitch metal, critical gate dimensions, via layers—is what makes the monitoring informative.

WATCH OUT FOR

- ◆ **Monitoring every step equally.** Attention spread evenly finds nothing; the design's sensitivities decide where to look.
- ◆ **Data reviewed only by the foundry.** Process control answers whether the line is in control, not whether this device is affected.
- ◆ **Excursions detected after the affected step.** Once the lot has moved on, the response options narrow sharply.
- ◆ **Defects classified but not attributed.** Knowing the count without the source gives no basis for a corrective action.
- ◆ **Monitoring stopping at front-end exit.** Back-end defects affect yield as much and are often less well monitored.

WHERE THE EFFORT GOES	5 M/M	WHO IS ON IT	~1.6 FTE	CONNECTIONS	TERMS HERE	3
Continuous monitoring	2.0	Yield engineer owns it		DEPENDS ON TECH-07 SO-08 FAB-02	WAT PCM DFM	
Defect classification	1.3	Device engineer		FEEDS INTO FAB-04 FAB-05 FAB-07		
Metrology review	1.0	Foundry liaison		FAB-08 MP-05 MP-08		
Escalation	0.8	DFM engineer				
		Product engineering				

15 FAB · FABRICATION

FAB-07

E-test / PCM data review

TAKES	COSTS	OWNER	CRITICAL PATH
3w w15–w18 of the stage	3.0 M/M ~4.3 people while it runs	Device engineer Product engineering approves	no has float

WHY IT EXISTS

Read the **e-test and PCM data**—device parameters measured on structures in the scribe line—and compare them against what the models predicted.

PCM data arrives before packaged parts and answers process questions weeks earlier than the lab can. Threshold voltages, resistances, capacitances and ring oscillator frequencies measured on real silicon are the first hard evidence of whether the design's assumptions about the process were right.

WHAT IT DELIVERS

FAB-D3

Inline, PCM and WAT data package

The PCM half of the deliverable, and the earliest device-level data available.

NEEDS FIRST

- E-test and PCM data from the foundry
- Model predictions from PDK-03 and TECH-03
- Test chip correlation from TC-07
- Process target specifications
- Split lot definitions from FAB-02

DONE WHEN

- Parameters compared against models, not only against targets
- Split lots analyzed separately
- Findings communicated before packaged parts arrive

HOW IT GETS THERE

4 STEPS · 2 LANES

w15	w16	w17	w18
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	PCM structure and parameter list confirmation → PCM parameter list and structures	0.5w
2	E-test data collection and review → E-test data review	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Across-wafer and lot-to-lot variability analysis → Across-wafer and lot variability analysis	1w runs in parallel
4	Parameter comparison against target and model → Parameter comparison against model and target	1.5w

Step 4 is where the value is. A threshold voltage 30 mV from target changes timing across the whole design, and knowing that before parts arrive lets bring-up be planned around it rather than surprised by it.

WATCH OUT FOR

- ◆ **PCM data not reviewed until parts arrive.** It answers process questions weeks earlier and costs nothing extra to read.
- ◆ **Comparison against target only.** The design was built against models; comparing against the model is what predicts silicon behavior.
- ◆ **Variability ignored.** Across-wafer parameter spread predicts the bin distribution, and the mean alone does not.
- ◆ **Split lots analyzed together.** If splits were run, aggregating them averages away exactly the information the splits were for.
- ◆ **No link to the test chip results.** TC-07 correlated models against silicon; this data either confirms that correlation or contradicts it.

WHERE THE EFFORT GOES	3 M/M	WHO IS ON IT ~4.3 FTE	CONNECTIONS	TERMS HERE	2
Parameter comparison	1.3	Device engineer OWNS IT	DEPENDS ON TC-07 FAB-04 FAB-05 FAB-06	WAT PCM	
Data review	1.0	Yield engineer	FEEDS INTO FAB-08 BU-07 MP-05 MP-08		
Variability analysis	0.8	Foundry liaison Library engineer Product engineering			

15 FAB · FABRICATION

FAB-08

Wafer acceptance test and lot disposition

TAKES	COSTS	OWNER	CRITICAL PATH
3w w16–w19 of the stage	3.0 M/M ~4.3 people while it runs	Product engineering Validation liaison approves	yes slipping moves the program

WHY IT EXISTS

Run **wafer acceptance testing and decide whether the lot is accepted**—the formal gate between the fab and everything downstream.

WAT is the contractual boundary. Parameters within specification mean the foundry delivered what was agreed; parameters outside mean a conversation about disposition, rework or credit. Either way the decision has to be taken deliberately rather than by default.

WHAT IT DELIVERS

FAB-D4

Wafer acceptance disposition record

This activity is the deliverable, and the record is what a later yield discussion refers back to.

NEEDS FIRST

- Completed wafers from FAB-05
- WAT specification and acceptance criteria
- PCM findings from FAB-07
- Inline findings from FAB-06
- Downstream schedule pressure from the program plan

DONE WHEN

- Disposition decided deliberately and recorded
- Deviations documented and communicated to bring-up
- Decision taken without holding the lot unnecessarily

HOW IT GETS THERE

4 STEPS · 2 LANES

w16	w17	w18	w19
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Acceptance criteria confirmation with the foundry → Confirmed acceptance criteria	0.5w
2	WAT execution and data review → WAT execution results	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Out-of-specification escalation with the foundry → Out-of-specification escalation record	0.75w runs in parallel
4	Lot disposition and release decision → Lot disposition and release decision	1.5w

Step 4 can accept a lot that is out of specification. On an engineering lot the priority is usually silicon in the lab rather than perfect silicon, and accepting a marginal lot with the deviation documented is often the right call—but it must be a call.

WATCH OUT FOR

- ◆ **Acceptance by default.** A lot released without a decision has been accepted by nobody, and the deviation is discovered later with no record.
- ◆ **Criteria not agreed in advance.** Negotiating acceptance criteria after the data exists favors whoever is more motivated.
- ◆ **Marginal lot rejected on principle.** On an engineering lot, silicon in the lab is usually worth more than a lot within every limit.
- ◆ **Deviations not documented.** A parameter out of specification affects bring-up interpretation, and the bring-up team has to know.
- ◆ **Disposition delayed.** Every day the lot sits undispositioned is a day of bring-up not happening.

WHERE THE EFFORT GOES	3 M/M	WHO IS ON IT ~4.3 FTE	CONNECTIONS	TERMS HERE	2
WAT review	1.3	Product engineering OWNS IT	DEPENDS ON FAB-05 FAB-06 FAB-07 TEST-03 TEST-05	WAT PCM	
Disposition	1.0	Device engineer			
Escalation	0.5	Foundry liaison	FEEDS INTO FAB-10 ASSY-01 BU-01		
Criteria confirmation	0.3	Program manager	MP-05		
		Validation liaison			

15 FAB · FABRICATION

FAB-09

Hot-lot management and WIP tracking

TAKES	COSTS	OWNER	CRITICAL PATH
18w cont. w1–w19 of the stage	4.0 M/M ~1.0 people while it runs	Program manager Downstream stage leads approves	no has float

WHY IT EXISTS

Track the lots and **maintain an honest wafer-out forecast**, because ten downstream activities are planning against a date this activity owns.

Assembly, test, bring-up and qualification all schedule against wafer-out. The forecast will move—fab schedules do—and the program's exposure is not the movement but the surprise. Four man-months across eighteen weeks buys a date everyone can plan against and early warning when it changes.

WHAT IT DELIVERS

FAB-D5

Wafer-out forecast versus actual log

This activity is the deliverable, and its value is in being current rather than in being right the first time.

NEEDS FIRST

- Foundry WIP reporting
- Lot status from FAB-04 and FAB-05
- Mask availability from FAB-01 and FAB-03
- Hot-lot terms from TECH-05
- Downstream schedule dependencies from DEF-08

DONE WHEN

- Forecast updated at every status change, not monthly
- Hot-lot priority actively maintained
- Downstream stages replanning against the current date

HOW IT GETS THERE

6 STEPS · 2 LANES

w1	w5	w9	w13	w17	w19
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	WIP tracking setup and reporting cadence → WIP tracking and reporting cadence	1w
2	Hot-lot priority management → Hot-lot priority management record	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Escalation on schedule deviation → Schedule deviation escalations runs in parallel	2w
4	Wafer-out forecast maintenance → Wafer-out forecast, maintained	2w
5	Forecast communication to downstream stages runs in parallel → Forecast communications to downstream stages	2w
6	Continuous tracking across the run → Yield tracking log across the run	13w

Step 5 is the reason this activity exists. A forecast held by the fab liaison and not distributed leaves `ASSY`, `TEST` and `BU` planning against the last number they heard, which is usually the original one.

WATCH OUT FOR

- ◆ **Forecast not updated.** Downstream stages plan against a stale date and discover the change when the parts do not arrive.
- ◆ **Hot-lot priority lapsing.** Priority is maintained by attention, and a lot that quietly loses it takes standard cycle time.
- ◆ **Deviations not escalated.** A slip identified and not raised becomes a slip everyone discovers at the same time, too late to replan.
- ◆ **Forecast presented without confidence.** A date with no stated uncertainty invites downstream teams to plan to the day.
- ◆ **Tracking limited to the fab.** Wafer-out is not the same as parts available; sort, logistics and assembly all sit between.

WHERE THE EFFORT GOES	4 M/M	WHO IS ON IT	~1.0 FTE	CONNECTIONS	TERMS HERE	1
		Program manager owns it		DEPENDS ON DEF-08 TECH-05 T0-11 FAB-01 FAB-02 FAB-03 FAB-04 TEST-11	WIP	
Continuous tracking	1.8					
Forecast maintenance	1.0	Foundry liaison				
Hot-lot management	0.8	Operations planner				
Escalation	0.5	Product engineering		FEEDS INTO PTV-12 ASSY-01 TEST-03 BU-01 MP-05		
		Downstream stage leads				

15 FAB · FABRICATION

FAB-10 Wafer shipment and logistics to sort and assembly			
TAKES	COSTS	OWNER	CRITICAL PATH
2w	1.0 M/M	Operations planner	yes
w17–w19 of the stage	~2.2 people while it runs	Program manager approves	slipping moves the program

WHY IT EXISTS

Move the wafers from the fab to **sort and assembly**—shipment, customs, documentation—and confirm First Silicon is actually available.

Wafers finished in a fab on another continent are not parts in a lab. Shipment, export documentation and customs sit between wafer-out and anyone touching silicon, and they are the part of the schedule most often left out of the plan entirely.

WHAT IT DELIVERS

FAB-D6

First Silicon availability notice

This activity is the deliverable, and it carries the stage's milestone.

NEEDS FIRST

- Accepted lots from FAB-08
- Export control classification from TECH-06
- Shipping and logistics arrangements from TECH-09
- Sort and assembly readiness from ASSY-01
- Delivery destination requirements

DONE WHEN

- Documentation prepared before wafer-out, not after
- Delivery confirmed at the destination
- First Silicon declared on delivery, not on wafer-out

HOW IT GETS THERE

3 STEPS · 1 LANE

w17

w18

w19

MAIN SEQUENCE

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Shipment planning and documentation → Shipment plan and documentation	0.5w
2	Customs and export control handling → Customs and export clearance	0.75w
3	Delivery to sort and assembly → Delivery confirmation to sort and assembly · First Silicon availability notice	0.75w

Step 2 is where a week can vanish. Export control classification and customs paperwork for engineering samples crossing borders is a real process, and starting it when the wafers are ready rather than while they are in the line is what makes it a delay.

WATCH OUT FOR

- ◆ **Customs paperwork started at wafer-out.** It can be prepared during processing and takes a week if it is not.
- ◆ **Export classification unresolved.** Advanced semiconductors carry export controls, and an unclassified shipment does not move.
- ◆ **Destination not ready.** Wafers arriving at a sort facility with no probe card or no capacity wait there instead of in the fab.
- ◆ **Shipment split without tracking.** Lots split across shipments for risk are useful only if which wafers went where is traceable.
- ◆ **Milestone declared at wafer-out rather than at delivery.** First Silicon means silicon someone can touch, and declaring it early hides a week of transit.

WHERE THE EFFORT GOES	1 M/M	WHO IS ON IT ~2.2 FTE	CONNECTIONS
Customs and documentation	0.5	Operations planner OWNS IT	DEPENDS ON TECH-06 TECH-09 FAB-05 FAB-08 PTV-12
Shipment planning	0.3	Compliance officer	FEEDS INTO ASSY-01 BU-01
Delivery tracking	0.3	Foundry liaison	
		Product engineering	
		Program manager	

INTEGRATE

16

Package & Substrate Design

PKGd

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w24 – w76	52 _w	11	110 _{M/M}	6 _{of 11}

ENGINEERING ACTIVITIES				
PKGD-01	Package architecture selection — 2.5D interposer type, organic substrate, layer count	8w	10	M/M
PKGD-02	Bump map, pitch and power-ground planning with physical design	10w	12	M/M
PKGD-03	Interposer / RDL routing design	14w	18	M/M
PKGD-04	Substrate stack-up, escape routing and package DRC	14w	18	M/M
PKGD-05	Package routing for signal integrity — length matching, reference plane and via design	10w	12	M/M
PKGD-06	Package PDN and decap footprint design (electrical closure owned by SIPI)	9w	10	M/M
PKGD-07	Thermal and mechanical simulation — warpage, co-planarity, TIM, lid	12w	12	M/M
PKGD-08	Test vehicle requirement definition and handoff to PTV	4w	3	M/M
PKGD-09	Substrate and interposer supplier selection and lead-time booking	10w	4	M/M
PKGD-10	OSAT selection and assembly process definition	9w	5	M/M
PKGD-11	Package design freeze, DRC and tooling release	6w	6	M/M

KEY DELIVERABLES	
PKGD-D1	Package architecture specification
PKGD-D2	Bump map and interposer / RDL database
PKGD-D3	Substrate design files (Gerber / ODB++) and stack-up
PKGD-D4	Package electrical design intent and model handoff to co-verification
PKGD-D5	Thermal and mechanical (warpage) simulation reports
PKGD-D6	Test vehicle requirement specification for PTV
PKGD-D7	Substrate and interposer PO with committed lead time
PKGD-D8	OSAT assembly process flow and agreement
PKGD-D9	Package Design Freeze package

16 PKGD · PACKAGE & SUBSTRATE DESIGN

PKGD-01

Package architecture selection — 2.5D interposer type, organic substrate, layer count

TAKES	COSTS	OWNER	CRITICAL PATH
8w w0–w8 of the stage	10.0 M/M ~5.4 people while it runs	Package architect Chief architect approves	yes slipping moves the program

WHY IT EXISTS

Choose the **package architecture**—interposer type, substrate construction, layer count, body size—and commit to it, because everything from bump pitch to assembly capacity follows.

This decision has the same character as the monolithic-versus-chiplet call in ARCH-02: it looks like an engineering choice and it is a supply chain commitment. A 2.5D silicon interposer is a different supplier, a different capacity queue and a different cost structure from an organic bridge, and switching later is not practical.

WHAT IT DELIVERS

PKGD-D1

Package architecture specification

This activity is the deliverable, and it is a supply chain commitment as much as a technical one.

NEEDS FIRST

- Package architecture assumption from DEF-01
- Partitioning and die count from ARCH-02
- Backend supply chain findings from TECH-09
- Bandwidth and HBM requirements from DEF-05
- Cost model from DEF-04

DONE WHEN

- Architecture selected with capacity indicated, not assumed
- At least two options evaluated and costed end to end
- Specification issued before bump planning starts

HOW IT GETS THERE

6 STEPS · 2 LANES

w0	w2	w4	w6	w8
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Package requirement consolidation — die size, power, interfaces, thermal → Package requirement consolidation	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Architecture option definition — interposer, bridge, organic → Defined package architecture options	2w
3	Capability and capacity screen per option → Architecture options with capability assessment · Capacity screen per option	1.5w runs in parallel
4	Substrate construction and layer count study → Substrate construction study	2w
5	Cost comparison per option with the DEF-04 model → Cost comparison	1.5w runs in parallel
6	Architecture selection and specification → Package architecture specification	2.5w

Step 3 can eliminate an option outright. Advanced interposer capacity is allocated quarters ahead to a small number of customers, and an architecture the program cannot get capacity for is not an option however good it looks technically.

WATCH OUT FOR

- ◆ **Architecture chosen without capacity.** Advanced packaging capacity is scarce and allocated far ahead; a technically ideal option with no allocation is not an option.
- ◆ **Decision taken after the bump map.** Package architecture constrains bump pitch and field, and deciding afterwards means renegotiating the die.
- ◆ **Cost compared on the package alone.** Interposer, substrate, assembly yield and test all differ per architecture, and comparing one line misleads.
- ◆ **Thermal requirement not in the selection.** Package thermal resistance is architecture-dependent and sets the achievable power envelope.
- ◆ **Single option evaluated.** Without an alternative the program has no negotiating position and no fallback if capacity is refused.

WHERE THE EFFORT GOES	10 M/M	WHO IS ON IT	~5.4 FTE	CONNECTIONS	TERMS HERE	3
Architecture option definition	2.5	Package architect OWNS IT		DEPENDS ON DEF-01 DEF-04 DEF-05 ARCH-02 TECH-09	P0 HBM (ESD) HBM	
Substrate construction study	2.0	Backend supply chain lead		FEEDS INTO PD-04 PKG-02 PKG-04 PKG-07 PKG-08 PKG-09 PKG-10 PTV-01 PTV-02 PTV-03		
Selection and specification	2.0	Thermal engineer				
Capacity screen	1.5	Product cost analyst				
Cost comparison	1.0	Chief architect				
Requirement consolidation	1.0					

16 PKGD · PACKAGE & SUBSTRATE DESIGN

PKGD-02

Bump map, pitch and power-ground planning with physical design

TAKES	COSTS	OWNER	CRITICAL PATH
10w w8–w18 of the stage	12.0 M/M ~5.2 people while it runs	Package designer Package architect approves	yes slipping moves the program

WHY IT EXISTS

Negotiate the **bump map with physical design**—pitch, count, power-to-signal ratio, position—until both the die and the package can build what the other needs.

The bump field is where two design teams share one constraint. The die wants bumps where its logic is; the package wants them where its escape routing can reach; and the power delivery requirement takes most of the field regardless. Convergence takes iteration, and the iteration has to finish before PD-04 freezes.

WHAT IT DELIVERS

PKGD-D2

Bump map and interposer / RDL database

The bump map half of the deliverable; PKGD-03 supplies the interposer routing.

NEEDS FIRST

- Package architecture from PKGD-01
- Bump budget and intent from ARCH-08
- Power envelope and delivery requirement from PD-03
- Interface lane counts from ARCH-04
- Substrate layer count from PKGD-04

DONE WHEN

- Escape feasibility confirmed on the substrate, not only on the die
- Power-to-signal ratio meets the delivery requirement
- Map converged before the PD-04 freeze date

HOW IT GETS THERE

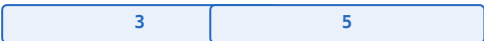
6 STEPS · 2 LANES

w8	w10	w12	w14	w16	w18
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Bump pitch and field definition against the architecture → Bump pitch and field definition	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Power-to-signal ratio derivation from the power envelope → Power-to-signal ratio	1.5w
3	Signal bump assignment per interface → Signal bump assignment per interface	2w runs in parallel
4	Escape feasibility check against the substrate → Escape feasibility findings	2w
5	Die-side placement iteration with physical design → Die-side placement iteration record	2w runs in parallel
6	Bump map convergence and handoff → Converged bump map	4.5w

Step 4 is the check that usually forces the iteration. A bump assignment that works on the die can require escape routing the substrate cannot provide at its layer count, and finding that early is what keeps the map from being redone after freeze.

WATCH OUT FOR

- ◆ **Bump map converged on the die side only.** A map that cannot escape on the substrate has to be redone, and by then the die may be placed around it.
- ◆ **Power bumps reduced to fit signals.** The interfaces then work and the die cannot be fed, which SIPI-04 finds much later.
- ◆ **Pitch chosen from the previous program.** Pitch depends on the interposer technology and the assembly process, both of which may have changed.
- ◆ **Signal assignment ignoring channel quality.** Which bump a high-speed signal uses affects its escape path and its crosstalk exposure.
- ◆ **Iteration continuing past the freeze date.** PD-04 and the substrate order both depend on this closing on time.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT	~5.2 FTE	CONNECTIONS	TERMS HERE	2
Bump pitch and field definition	3.0	Package designer OWNS IT		DEPENDS ON DEF-05 ARCH-04 ARCH-08 PKGD-01	RDL SI	
Signal assignment	2.5	Physical design liaison		FEEDS INTO PD-04 PKGD-03 PKGD-05 PKGD-06 PTV-04 SIPI-02 SIPI-06		
Escape feasibility	2.5	Power delivery engineer		LATER INPUT PD-03 +10w PD-04 +10w		
Convergence and handoff	2.0	SI engineer				
Power ratio derivation	1.0	Package architect				
Die-side iteration	1.0					

16 PKGD · PACKAGE & SUBSTRATE DESIGN

PKGD-03

Interposer / RDL routing design

TAKES	COSTS	OWNER	CRITICAL PATH
14w w16–w30 of the stage	18.0 M/M ~5.6 people while it runs	Interposer designer Package architect approves	yes slipping moves the program

WHY IT EXISTS

Route the **interposer and its redistribution layers**—the silicon layer that carries the die-to-HBM channels and everything else the substrate cannot.

On a 2.5D part the interposer carries the highest-density and highest-performance connections in the package. HBM channels alone run to thousands of nets at fine pitch, and their routing quality decides whether the memory interface trains at speed.

WHAT IT DELIVERS

PKGD-D2

Bump map and interposer / RDL database

The interposer half of the deliverable—the routed silicon layer itself.

NEEDS FIRST

- Bump map from PKGD-02
- HBM channel requirements from ARCH-03 and ARCH-04
- Interposer technology rules from the supplier
- Power delivery requirement from PD-03
- SI requirements from SIPI-07

DONE WHEN

- HBM channel lengths matched within the training budget
- Reference planes continuous under high-speed channels
- Interposer DRC clean against supplier rules

HOW IT GETS THERE

7 STEPS · 2 LANES

w16	w19	w22	w25	w28	w30
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Interposer layer stack and routing resource plan → Interposer layer stack and routing plan	2w
2	HBM channel routing and length management → Routed HBM channels with length management	4w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Die-to-die and remaining signal routing → Die-to-die and signal routing	3w runs in parallel
4	Power and ground distribution on the interposer → Interposer power distribution	3w
5	RDL routing and via planning → RDL routing and via plan	2.5w runs in parallel
6	Interposer DRC and manufacturability → Interposer DRC and manufacturability results	2.5w runs in parallel
7	Interposer database release → Released interposer database	5w

Step 2 dominates because HBM channels are numerous, fine-pitch and length-sensitive at once. Getting them routed with matched lengths and clean references is most of the interposer's difficulty and most of its schedule.

WATCH OUT FOR

- ◆ **HBM channel lengths unmatched.** Skew across a wide parallel interface eats the training margin, and the failure appears at bring-up as a memory that will not train at speed.
- ◆ **Reference plane discontinuities.** A channel crossing a plane split has an impedance discontinuity, and on HBM there are thousands of channels to get right.
- ◆ **Interposer power distribution underprovisioned.** Current for two HBM stacks and the die passes through the interposer, and its resistance is part of the IR budget.
- ◆ **Manufacturability checked late.** Interposer rules differ from die rules, and a design that violates them is a supplier rejection rather than a warning.
- ◆ **Routing started before the bump map converges.** Every bump change is an interposer rework, and the rework is expensive at this density.

WHERE THE EFFORT GOES	18 M/M	WHO IS ON IT	~5.6 FTE	CONNECTIONS	TERMS HERE	6
HBM channel routing	5.0	Interposer designer	owns it	DEPENDS ON ARCH-03 ARCH-04 PD-03 PKGD-02	RDL DRC IR SI HBM (ESD) HBM	
Power distribution	3.5	SI engineer		FEEDS INTO PKGD-04 PKGD-05 PKGD-11 PTV-04 PTV-05 SIPI-02 SIPI-07 TEST-03		
Signal and die-to-die routing	3.0	Power delivery engineer				
RDL and via planning	2.5	Supplier liaison				
Database release	2.0	Package architect				
DRC and manufacturability	2.0					

16 PKGD · PACKAGE & SUBSTRATE DESIGN

PKGD-04

Substrate stack-up, escape routing and package DRC

TAKES	COSTS	OWNER	CRITICAL PATH
14w w18–w32 of the stage	18.0 M/M ~5.6 people while it runs	Substrate designer Power integrity engineer approves	yes slipping moves the program

WHY IT EXISTS

Design the **substrate**—stack-up, escape routing, power planes, package DRC—the organic carrier that connects the interposer to the board.

The substrate is the part with the longest lead time and the least flexibility once ordered. Its layer count is fixed by escape routing needs and its cost scales with it, so the design converges on the smallest stack-up that can carry the escape—a trade made once and lived with.

WHAT IT DELIVERS

PKGD-D3

Substrate design files (Gerber / ODB++) and stack-up

This activity is the deliverable, and it is what the supplier manufactures from.

NEEDS FIRST

- Package architecture from PKGD-01
- Interposer footprint from PKGD-03
- Board requirements and ball map from EVB-02
- Supplier design rules from PKGD-09
- Power delivery requirement from SIPI-05

DONE WHEN

- Layer count justified by an attempted escape, not assumed
- Package DRC clean against the actual supplier’s rules
- Ball map agreed with the board design

HOW IT GETS THERE

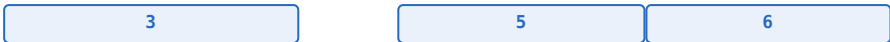
7 STEPS · 2 LANES

w18	w21	w24	w27	w30	w32
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Stack-up definition and layer count decision → Substrate stack-up and layer count	2w
2	Escape routing from the interposer footprint → Escape routing from the interposer	4w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Power and ground plane design → Power and ground plane design	3w runs in parallel
4	Board-side ball assignment and breakout → Board-side ball assignment	3w
5	Via structure and drill planning → Via and drill plan	2.5w runs in parallel
6	Package DRC against supplier rules → Package DRC results against supplier rules	2.5w runs in parallel
7	Substrate database release → Released substrate database	5w

Step 1 is the cost decision. Every substrate layer adds cost across the product's lifetime, and the layer count is set by whether the escape routing fits—which is not known until step 2 is attempted. The two steps iterate.

WATCH OUT FOR

- ◆ **Layer count set before escape is attempted.** A stack-up chosen for cost that cannot carry the escape has to be increased, and the cost model was built on the smaller number.
- ◆ **Supplier rules obtained late.** Substrate rules vary by supplier and technology, and designing to generic rules produces a design one supplier can build.
- ◆ **Power planes fragmented by escape routing.** Escape vias perforate the planes, and the resulting current path is what SIPI-03 will analyze.
- ◆ **Ball map not agreed with the board.** The substrate's board-side interface is the board's problem too, and changing it late affects both.
- ◆ **Design released before DRC closes.** A supplier rejection costs weeks against a twenty-week lead time.

WHERE THE EFFORT GOES	18 M/M	WHO IS ON IT	~5.6 FTE	CONNECTIONS	TERMS HERE	3
Escape routing	5.0	Substrate designer	OWNS IT	DEPENDS ON PKGD-01 PKGD-03 PKGD-09	PO DRC ODB	
Ball assignment and breakout	3.5	Package architect		FEEDS INTO PKGD-05 PKGD-06 PKGD-11		
Power and ground planes	3.0	Supplier liaison		PTV-05 SIPI-02 ASSY-08		
Stack-up definition	2.5	Board designer		EVB-01 EVB-02 EVB-06		
Via and drill planning	2.0	Power integrity engineer		TEST-01 TEST-02 TEST-04		
DRC and release	2.0					

16 PKGD · PACKAGE & SUBSTRATE DESIGN

PKGD-05

Package routing for signal integrity — length matching, reference plane and via design

TAKES	COSTS	OWNER	CRITICAL PATH
10w w26–w36 of the stage	12.0 M/M ~5.2 people while it runs	SI engineer Supplier liaison approves	no has float

WHY IT EXISTS

Route the package for **signal integrity**—length matching, reference continuity, via design, crosstalk separation—so the channels can carry what the interfaces need.

Package routing is where high-speed channels are made or ruined. A via with a long stub, a trace crossing a plane split, or a differential pair routed without its partner all cost margin that the PHY then has to make up. SIPI-07 will measure the result; this activity is what determines it.

WHAT IT DELIVERS

PKGD-D4

Package electrical design intent and model handoff to co-verification

The signal half of the deliverable—the design intent SIPI then verifies.

NEEDS FIRST

- Interposer and substrate routing from PKGD-03 and PKGD-04
- Channel budgets from AMS-03 and ARCH-04
- SI requirements and targets from SIPI-07
- Interface compliance requirements
- Supplier via and drill capability

DONE WHEN

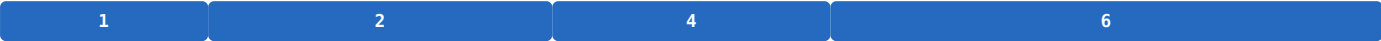
- Channel budget explicitly allocated between die, package and board
- Via stubs managed rather than accepted
- Routing reviewed against SI requirements before substrate release

HOW IT GETS THERE

6 STEPS · 2 LANES

w26	w28	w30	w32	w34	w36
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Channel budget allocation to the package → Channel budget allocation to the package	1.5w
2	Length matching and skew control per interface → Length-matched routing per interface	2.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Reference plane continuity and return path design → Reference plane continuity design	2w runs in parallel
4	Via structure design and stub management → Via structures with managed stubs	2w
5	Crosstalk separation and shielding → Crosstalk separation plan	2w runs in parallel
6	Routing review against the SI requirements → SI routing review record	4w

Step 4 is the one most often underestimated. Via stubs resonate at frequencies that matter for the fastest interfaces, and back-drilling or blind via structures are decisions with cost implications that belong to the substrate design rather than to routing.

WATCH OUT FOR

- ◆ **Via stubs unmanaged.** They resonate in band for the fastest interfaces, and the remedy—back-drilling or blind vias—is a substrate cost decision made too late.
- ◆ **Length matching without reference continuity.** Matched traces over a split plane have matched delay and mismatched impedance.
- ◆ **Channel budget not allocated.** Die, package and board each consume part of the loss budget, and without allocation each assumes the others left room.
- ◆ **Crosstalk managed by spacing alone.** On a dense substrate there is no spare space, and the answer is usually reference and via design instead.
- ◆ **Review after the substrate is released.** SI findings then have nowhere to go except into the PHY's margin.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT	~5.2 FTE	CONNECTIONS	TERMS HERE	3
Length matching and skew	3.0	SI engineer OWNS IT		DEPENDS ON ARCH-04 AMS-03 PKGD-02 PKGD-03 PKGD-04	PHY SI SIPI	
Via structure design	2.5	Package designer		RUNS WITH SIPI-07		
Reference and return path	2.0	Interposer designer		FEEDS INTO S0-06 PKGD-11 SIPI-08 EVB-05		
Crosstalk separation	2.0	PHY liaison				
Budget allocation	1.5	Supplier liaison				
Review	1.0					

16 PKGD · PACKAGE & SUBSTRATE DESIGN

PKGD-06

Package PDN and decap footprint design (electrical closure owned by SIPI)

TAKES	COSTS	OWNER	CRITICAL PATH
9w w28–w37 of the stage	10.0 M/M ~4.8 people while it runs	Package power integrity engineer Supplier liaison approves	no has float

WHY IT EXISTS

Design the **package power delivery**—plane structure, via arrays, decoupling footprints—knowing that its electrical closure belongs to SIPI rather than to this activity.

The package sits between the die's instantaneous current demand and the board's slower supply. Its planes, vias and decoupling determine the impedance in the mid-frequency band where neither the die's on-chip decap nor the board's bulk capacitors help. Designing it is here; proving it works is co-verification.

WHAT IT DELIVERS

PKGD-D4

Package electrical design intent and model handoff to co-verification

The power half of the deliverable, handed to SIPI for closure.

NEEDS FIRST

- Substrate stack-up from PKGD-04
- Power delivery requirement from PD-03
- Bump and ball power assignment from PKGD-02
- Impedance targets from SIPI-03
- Supplier capability for via arrays and mounting

DONE WHEN

- Decoupling footprints placed for loop inductance, not convenience
- Plane perforation analyzed rather than assumed away
- Design intent handed over in a form SIPI can analyze

HOW IT GETS THERE

6 STEPS · 2 LANES

w28	w30	w32	w34	w36	w37
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Package PDN topology and plane allocation → Package PDN topology and plane allocation	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Power via array design and current capability → Power via array design	2w
3	Decoupling capacitor footprint placement → Decoupling footprint placement	1.5w runs in parallel
4	Plane perforation and current path management → Plane perforation and current path analysis	2w
5	Interposer to substrate power transition → Interposer to substrate power transition design	1.5w runs in parallel
6	PDN design intent handoff to co-verification → PDN design intent handoff	3w

Step 3 places footprints rather than choosing values. Which capacitors go where is decided by `SIPI-05` against the impedance target; the package's job is to provide mounting positions close enough to the current draw for those choices to be available.

WATCH OUT FOR

- ◆ **Decoupling footprints too far from the current draw.** Mounting position determines the loop inductance, and a capacitor mounted far away is a capacitor that does not help.
- ◆ **Plane perforation ignored.** Escape vias perforate power planes, and the remaining current path can be far worse than the plane suggests.
- ◆ **Via array sized for DC only.** Transient current is what the package has to supply, and DC sizing understates the requirement.
- ◆ **Closure assumed to be someone else's problem.** If the design intent is not analysable, SIPI cannot close it and the iteration comes back here.
- ◆ **Interposer to substrate transition overlooked.** The power path crosses two interfaces, and the transition is where its resistance concentrates.

WHERE THE EFFORT GOES	10 M/M	WHO IS ON IT ~4.8 FTE	CONNECTIONS	TERMS HERE	3
PDN topology and planes	2.5	Package power integrity engineer OWNS IT	DEPENDS ON PD-03 PKGD-02 PKGD-04	DC PDN SIPI	
Power via arrays	2.0	Substrate designer	RUNS WITH SIPI-03		
Plane perforation management	2.0	SIPI liaison	FEEDS INTO S0-06 PKGD-11 SIPI-05		
Decoupling footprints	1.5	Die power delivery engineer	LATER INPUT SIPI-05 +3w		
Power transition design	1.0	Supplier liaison			
Handoff	1.0				

16 PKGD · PACKAGE & SUBSTRATE DESIGN

PKGD-07

Thermal and mechanical simulation
— warpage, co-planarity, TIM, lid

TAKES	COSTS	OWNER	CRITICAL PATH
12w w20–w32 of the stage	12.0 M/M ~4.3 people while it runs	Thermal engineer Die power liaison approves	no has float

WHY IT EXISTS

Simulate the **thermal and mechanical behavior** of the package—heat path, warpage, co-planarity, TIM and lid—before any of it is built.

A package at this power dissipates hundreds of watts through a stack of materials with different expansion coefficients. The thermal path decides the achievable power envelope; the mechanical behavior decides whether the assembly survives reflow. Both are simulated here and measured on real vehicles in PTV.

WHAT IT DELIVERS

PKGD-D5

Thermal and mechanical (warpage) simulation reports

This activity is the deliverable, and PTV is what validates it.

NEEDS FIRST

- Package architecture from PKGD-01
- Power map and hotspot data from PD-03 and SYN-06
- Material properties from the supplier
- Assembly reflow profile from PKGD-10
- Thermal requirements from DEF-03

DONE WHEN

- Thermal simulated against the real power map, not uniform power
- Warpage simulated across the full reflow profile
- Findings handed to PTV as vehicle requirements

HOW IT GETS THERE

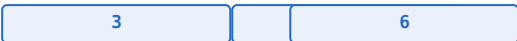
7 STEPS · 2 LANES

w20	w23	w26	w29	w32
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Thermal model construction — die, interposer, substrate, lid → Thermal model of the package stack	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Steady-state thermal simulation and Rjc estimation → Steady-state thermal results and Rjc estimate	2.5w
3	Hotspot and transient thermal analysis → Hotspot and transient analysis	2w runs in parallel
4	TIM selection and interface resistance study → TIM selection and interface study	2w runs in parallel
5	Warpage simulation across the reflow profile → Warpage simulation across reflow	3w
6	Mechanical stress and co-planarity analysis → Mechanical stress and co-planarity results	2w runs in parallel
7	Thermal and mechanical design recommendations → Thermal and mechanical design recommendations	4.5w

Step 5 predicts what `PTV-07` will measure. Warpage across the reflow profile is the mechanism behind most 2.5D assembly failures, and simulating it before the vehicles are built is what makes the vehicle results interpretable rather than merely surprising.

WATCH OUT FOR

- ◆ **Thermal simulated at uniform power.** An accelerator's power is concentrated in the compute clusters, and uniform assumptions understate the hotspot badly.
- ◆ **Warpage simulated at room temperature only.** The failure happens at reflow peak, and the room-temperature number says nothing about it.
- ◆ **TIM performance taken from the datasheet.** Interface resistance depends on bond line thickness and pressure, both of which are assembly properties.
- ◆ **Material properties assumed.** Substrate and underfill properties vary by supplier and lot, and simulation is only as good as the inputs.
- ◆ **Results not fed into the vehicle plan.** Simulation identifies which risks to measure, and a vehicle plan built independently measures the wrong things.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT	~4.3 FTE	CONNECTIONS	TERMS HERE	3
Warpage simulation	3.0	Thermal engineer	OWNS IT	DEPENDS ON DEF-03 PD-03 PKGD-01	TIM Rjc PTV	
Steady-state thermal	2.5	Mechanical engineer		RUNS WITH PKGD-10		
TIM and interface study	2.0	Package architect		FEEDS INTO PTV-02 PTV-03 PTV-07		
Hotspot and transient analysis	1.5	Supplier liaison		PTV-09 SIPI-10		
Mechanical stress analysis	1.5	Die power liaison		LATER INPUT PTV-07 +18w PTV-09 +18w		
Recommendations	1.5			FEEDS BACK INTO PKGD-08 PTV-01		

16 PKGD · PACKAGE & SUBSTRATE DESIGN

PKGD-08

Test vehicle requirement definition and handoff to PTV

TAKES	COSTS	OWNER	CRITICAL PATH
4w w10–w14 of the stage	3.0 M/M ~3.3 people while it runs	Package architect OSAT liaison approves	no has float

WHY IT EXISTS

Tell **PTV what the vehicles have to measure**—which risks, which structures, which conditions—early enough that the vehicles can be designed, built and stressed before wafer-out.

The test vehicle program has a fifty-two week window and a hard gate at product wafer-out. Its requirements have to arrive while there is time to act on them, which is why this activity sits at week ten of a fifty-two week stage rather than after the package design is finished.

WHAT IT DELIVERS

PKGD-D6

Test vehicle requirement specification for PTV

This activity is the deliverable, and its timing matters as much as its content.

NEEDS FIRST

- Package architecture from PKGD-01
- Thermal and warpage predictions from PKGD-07
- Assembly process direction from PKGD-10
- CPI and ULK risk from TECH-03
- Previous-program package failures

DONE WHEN

- Requirements issued with time for the vehicle program to finish before wafer-out
- Risks drawn from experience as well as from the design
- Feedback mechanism into package design agreed

HOW IT GETS THERE

4 STEPS · 2 LANES

w10	w11	w12	w13	w14
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Package risk identification for vehicle coverage → Package risk list for vehicle coverage	1w
2	Vehicle type requirements — mechanical, thermal, electrical → Vehicle type requirements	1.5w runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Measurement condition and structure specification → Measurement conditions and structure specification	1.5w
4	Requirement handoff and PTV plan review → Requirement handoff to PTV	1.5w

Step 1 has to be done on an incomplete package design, which is uncomfortable and necessary. The architecture is known, the materials are known and the assembly process is broadly known—enough to say what will be risky, which is all the vehicle plan needs.

WATCH OUT FOR

- ◆ **Requirements issued after the package design completes.** The vehicle program then cannot finish before wafer-out, and the gate is missed.
- ◆ **Risks identified from the design rather than from experience.** The failures that matter are the ones previous programs hit, and they are rarely visible in a design review.
- ◆ **Electrical vehicle omitted.** Daisy-chain continuity and assembly yield are as important as thermal and mechanical, and they need their own vehicle.
- ◆ **Conditions unspecified.** A vehicle measured at the wrong temperature or reflow profile answers a question nobody asked.
- ◆ **No feedback loop defined.** The vehicle results have to change something, and the mechanism for that has to be agreed up front.

WHERE THE EFFORT GOES	3 M/M	WHO IS ON IT ~3.3 FTE	CONNECTIONS	TERMS HERE	4
Risk identification	1.0	Package architect OWNS IT	DEPENDS ON TECH-03 PKG-01	ULK OSAT CPI PTV	
Measurement specification	0.8	Thermal and mechanical engineers	FEEDS INTO PTV-01 PTV-02 PTV-03		
Vehicle type requirements	0.8	Reliability engineer	PTV-04 EVB-08		
Handoff and review	0.5	PTV lead OSAT liaison	LATER INPUT PKG-07 +6w		

16 PKGD · PACKAGE & SUBSTRATE DESIGN

PKGD-09

Substrate and interposer supplier selection and lead-time booking

TAKES	COSTS	OWNER	CRITICAL PATH
10w	4.0 M/M	Backend supply chain lead	yes
w22–w32 of the stage	~1.7 people while it runs	Program manager approves	slipping moves the program

WHY IT EXISTS

Select the **substrate and interposer suppliers and book their lead time**—a twenty-week commitment that sets the date the package can be built.

Substrate lead time is the longest single external dependency in the back half of the program, and it is ordered against a design that is not finished. Selecting the supplier early, understanding their rules and booking capacity is what makes the order placeable the moment the design freezes.

WHAT IT DELIVERS

PKGD-D7

Substrate and interposer PO with committed lead time

This activity is the deliverable, and the committed date is what wafer-out has to be planned against.

NEEDS FIRST

- Package architecture from PKGD-01
- Substrate design requirements from PKGD-04
- Backend supply chain landscape from TECH-09
- Volume forecast and ramp from DEF-08
- Cost targets from DEF-04

DONE WHEN

- Lead time committed in writing, not indicative
- Supplier rules available before substrate design completes
- Order placeable the week the design freezes

HOW IT GETS THERE

6 STEPS · 2 LANES

w22	w24	w26	w28	w30	w32
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Supplier long-list and capability screen → Supplier capability screen	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Design rule and technology fit assessment → Design rule and technology fit assessment	2w
3	Capacity and lead time confirmation → Capacity and lead time confirmation	2w runs in parallel
4	Commercial negotiation and pricing → Commercial terms	2w runs in parallel
5	Supplier selection and qualification plan → Supplier selection and qualification plan	2.5w
6	Purchase order and lead time booking → Purchase order with committed lead time	4w

Step 6 takes four weeks because the order cannot be placed until the design is frozen, and the freeze is in `PKGD-11`. Everything before it exists so that the order can be placed the same week the design is released rather than a month later.

WATCH OUT FOR

- ◆ **Order placed after wafer-out is committed.** A twenty-week substrate against a fixed wafer-out date means wafers waiting for a package.
- ◆ **Supplier rules obtained after design starts.** A substrate designed to generic rules may be unbuildable by the chosen supplier.
- ◆ **Capacity assumed.** Advanced substrate capacity is constrained, and a supplier who can build it may not be able to build it when needed.
- ◆ **Single supplier with no alternative.** A supply disruption then has no mitigation, and substrates are not quickly re-sourced.
- ◆ **Lead time quoted without a commitment.** An indicative lead time is not a date, and the difference appears when the schedule depends on it.

WHERE THE EFFORT GOES	4 M/M	WHO IS ON IT ~1.7 FTE	CONNECTIONS
Supplier selection and qualification	1.0	Backend supply chain lead OWNS IT	DEPENDS ON DEF-08 TECH-09 PKGD-01
Design rule assessment	0.8	Substrate designer	FEEDS INTO PKGD-04 PKGD-11 PTV-05
Capacity and lead time	0.8	Procurement	ASSY-04 MP-10
Purchase order	0.8	Quality liaison	TERMS HERE 1
Commercial negotiation	0.5	Program manager	P0
Capability screen	0.3		

16 PKGD · PACKAGE & SUBSTRATE DESIGN

PKGD-10

OSAT selection and assembly process definition

TAKES	COSTS	OWNER	CRITICAL PATH
9w w24–w33 of the stage	5.0 M/M ~2.4 people while it runs	Backend supply chain lead PTV lead approves	no has float

WHY IT EXISTS

Select the **OSAT and define the assembly process**—bonding, underfill, molding, lid attach—because the process determines what the package design has to accommodate.

Assembly is not a service the package is handed to; it is a set of constraints the package is designed within. Bond force, reflow profile, underfill flow and lid attach all impose requirements, and choosing the OSAT late means designing a package that has to be adapted to whoever ends up building it.

WHAT IT DELIVERS

PKGD-D8

OSAT assembly process flow and agreement

This activity is the deliverable, and the agreement includes the responsibility model.

NEEDS FIRST

- Package architecture from PKGD-01
- Backend supply chain findings from TECH-09
- Thermal and mechanical requirements from PKGD-07
- Volume and ramp forecast from DEF-08
- Known-good-die strategy from TEST-01

DONE WHEN

- OSAT selected with committed capacity
- Process parameters known to the package design, not assumed
- Known-good-die responsibility agreed in writing

HOW IT GETS THERE

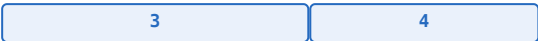
6 STEPS · 2 LANES

w24	w26	w28	w30	w32	w33
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	OSAT capability screen for the package architecture → OSAT capability screen	1.5w
2	Assembly process flow definition → Assembly process flow definition	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Bonding, underfill and molding process parameters → Process parameter specification	2w runs in parallel
4	Known-good-die handoff and responsibility model → Known-good-die responsibility model	1.5w runs in parallel
5	OSAT selection and capacity commitment → OSAT selection and capacity commitment	2w
6	Assembly agreement and process freeze plan → Assembly agreement	3.5w

Step 4 settles a question that becomes expensive later. When an assembled unit fails, whether the die or the assembly is responsible determines who bears the cost, and agreeing it while nobody has lost money is far easier than afterwards.

WATCH OUT FOR

- ◆ **OSAT selected after the package is designed.** The design then has to be adapted to the assembler's process rather than developed with it.
- ◆ **Advanced packaging capacity not committed.** The same constraint as the interposer, and it applies to assembly capacity independently.
- ◆ **Known-good-die responsibility undefined.** The first failed lot is the wrong moment to discover nobody agreed who owns the die.
- ◆ **Process parameters left to the OSAT.** Bond force and reflow profile affect warpage and CPI, and the package design has to know them.
- ◆ **Process frozen without vehicle data.** PTV-12 freezes the process window on evidence; freezing earlier means freezing on assumption.

WHERE THE EFFORT GOES	5 M/M	WHO IS ON IT ~2.4 FTE	CONNECTIONS	TERMS HERE	3
Assembly process definition	1.5	Backend supply chain lead owns IT	DEPENDS ON TECH-09 PKGD-01	OSAT CPI PTV	
OSAT selection	1.0	Package architect	RUNS WITH PKGD-07		
Process parameters	1.0	Quality liaison	FEEDS INTO PKGD-11 PTV-04 PTV-06		
Agreement and freeze plan	0.8	Procurement	ASSY-01 ASSY-02 ASSY-03		
Known-good-die model	0.8	PTV lead			

16 PKGD · PACKAGE & SUBSTRATE DESIGN

PKGD-11

Package design freeze, DRC and tooling release

TAKES	COSTS	OWNER	CRITICAL PATH
6w w46–w52 of the stage	6.0 M/M ~4.3 people while it runs	Package architect Procurement approves	yes slipping moves the program

WHY IT EXISTS

Freeze the **package design**, close its DRC, release the tooling—and start the twenty-week substrate lead time that everything downstream waits on.

The freeze is a schedule event more than a design one. The substrate order cannot be placed against a moving design, and the lead time that follows is what sets when packages can be built. Every week of delay here is a week at the far end, after wafer-out, where it is most visible and least recoverable.

WHAT IT DELIVERS

PKGD-D9

Package Design Freeze package

This activity is the deliverable, and it carries the stage's milestone.

NEEDS FIRST

- Interposer database from PKGD-03
- Substrate database from PKGD-04
- SI and PDN design from PKGD-05 and PKGD-06
- Co-verification findings from SIPI-11
- Supplier PO framework from PKGD-09

DONE WHEN

- Supplier has accepted the design, not only received it
- Order released the same week as the freeze
- Co-verification findings incorporated before freeze, not after

HOW IT GETS THERE

6 STEPS · 2 LANES

w46w48w50w52

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Design completeness review across interposer and substrate → Design completeness review record	1w
2	Final package DRC and supplier rule compliance → Final package DRC results	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Electrical intent confirmation against SIPI findings → Electrical intent confirmation	1w runs in parallel
4	Tooling and manufacturing data release → Released tooling and manufacturing data	1.5w
5	Supplier design review and acceptance → Supplier acceptance record	1.5w runs in parallel
6	Design freeze declaration and order release → Package Design Freeze declaration	2w

Step 5 runs in parallel and can send the design back. Suppliers review for manufacturability against their own process, and their acceptance is what makes the order placeable—a design frozen internally and rejected by the supplier has not frozen anything.

WATCH OUT FOR

- ◆ **Freeze slipping.** Twenty weeks of substrate lead time follow it, and the delay lands after wafer-out where nothing can absorb it.
- ◆ **Frozen before co-verification completes.** A SIPI finding after freeze is a package change against an order already placed.
- ◆ **Supplier acceptance assumed.** A design frozen internally and rejected on manufacturability has not frozen anything except the schedule.
- ◆ **Tooling data incomplete.** Manufacturing data is more than the design files, and a missing item stalls the order after it is placed.
- ◆ **Late bump map change from the die side.** PD-04 has to be frozen first, and a die-side change after this point is extremely expensive.

WHERE THE EFFORT GOES	6 M/M	WHO IS ON IT	~4.3 FTE	CONNECTIONS	TERMS HERE	5
Tooling and data release	1.5	Package architect OWNS IT		DEPENDS ON PD-04 PKGD-03 PKGD-04 PKGD-05 PKGD-06 PKGD-09 PKGD-10 PTV-08 SIPI-11	P0 DRC PDN SI SIPI	
Final DRC	1.5	Package and substrate designers				
Freeze and order release	1.0	SIPI liaison		FEEDS INTO PTV-12 ASSY-04 ASSY-05		
Completeness review	1.0					
Supplier review	0.5	Supplier liaison		MP-01 MP-03 MP-04		
Electrical confirmation	0.5	Procurement		MP-10 MP-11		

INTEGRATE

17

Package Test Vehicle (CPI / TTV)

PTV

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH		
w36 – w88	52 _w	12	96 _{M/M}	6 _{of 12}		
ENGINEERING ACTIVITIES				KEY DELIVERABLES		
PTV-01	Test vehicle strategy and risk item definition — CPI, warpage, thermal, ULK/BEOL stress		6w	5 M/M	PTV-D1	Test vehicle plan and risk coverage matrix
PTV-02	Mechanical test vehicle (MTV) design — bump array, dummy die stack, substrate coupon		9w	8 M/M	PTV-D2	MTV, TTV and daisy-chain vehicle designs
PTV-03	Thermal test vehicle (TTV) design — embedded heater array and RTD sensor placement		9w	8 M/M	PTV-D3	Built vehicle lots with assembly travelers
PTV-04	Daisy-chain electrical vehicle design — interconnect continuity and bump yield structures		8w	6 M/M	PTV-D4	Warpage and co-planarity data across the reflow profile
PTV-05	TV die fabrication and interposer / substrate vehicle build		16w	6 M/M	PTV-D5	CPI stress assessment report — ULK, bump and BEOL integrity
PTV-06	OSAT assembly of vehicles with process window DOE		11w	12 M/M	PTV-D6	Thermal characterization report with model correlation (Rjc, TIM, hotspot map)
PTV-07	Warpage and co-planarity measurement across the reflow profile (shadow moiré)		7w	8 M/M	PTV-D7	Board-level reliability data on vehicles
PTV-08	Chip-package interaction stress evaluation — low-k / ULK crack, bump and BEOL integrity		11w	14 M/M	PTV-D8	Frozen assembly process window definition
PTV-09	Thermal characterization — Rjc / Rja, TIM performance, hotspot mapping against the thermal model		9w	10 M/M	PTV-D9	Package validation complete record — gate for product wafer-out
PTV-10	Board-level reliability on vehicles — temperature cycle, drop, bend		14w	8 M/M		
PTV-11	Daisy-chain continuity test and assembly yield learning		9w	6 M/M		
PTV-12	Feedback into package design and assembly process window freeze		7w	5 M/M		

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-01

Test vehicle strategy and risk item definition
— CPI, warpage, thermal, ULK/BEOL stress

TAKES	COSTS	OWNER	CRITICAL PATH
6w w0–w6 of the stage	5.0 M/M ~3.6 people while it runs	PTV lead OSAT liaison approves	yes slipping moves the program

WHY IT EXISTS

Decide **which package risks the vehicles will actually retire**—chip-package interaction, warpage, thermal, assembly yield—and what evidence would count as retiring them.

The vehicle program has one shot and a hard gate at product wafer-out. Its value depends entirely on measuring the risks that would otherwise be discovered on real dies, and a vehicle plan built from what is easy to measure rather than from what is risky produces reassurance instead of information.

WHAT IT DELIVERS

PTV-D1

Test vehicle plan and risk coverage matrix

This activity is the deliverable, and the matrix is what every later result is reported against.

NEEDS FIRST

- Vehicle requirements from PKGD-08
- Thermal and warpage predictions from PKGD-07
- CPI and ULK risk from TECH-03
- Assembly process direction from PKGD-10
- Wafer-out forecast from FAB-09

DONE WHEN

- Every carried risk has defined retiring evidence
- Programme schedule closes before the wafer-out gate
- Plan reflects prioritized risk rather than measurement convenience

HOW IT GETS THERE

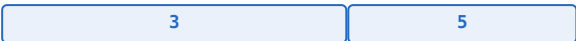
6 STEPS · 2 LANES

w0	w2	w4	w6
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Risk intake from package design and previous programs → Risk intake from package design and previous programs	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Risk prioritization against product impact → Package risk list with product impact	1.5w
3	Evidence definition — what would retire each risk → Evidence definition per risk	1.5w runs in parallel
4	Vehicle type allocation per risk → Vehicle type allocation	1.5w
5	Schedule feasibility against the wafer-out gate → Schedule feasibility finding	1w runs in parallel
6	Vehicle plan and risk coverage matrix → Test vehicle plan and risk coverage matrix	2w

Step 5 checks the whole program against one date. Fifty-two weeks of vehicle work has to finish before product wafers come out at week 98, and if the arithmetic does not close then the plan has to shrink rather than the gate move.

WATCH OUT FOR

- ◆ **Plan built from what is measurable.** Easy measurements produce data; the risk list produces information, and the two are not the same set.
- ◆ **Schedule not checked against the gate.** A fifty-two week program that finishes after wafer-out has validated nothing in time to matter.
- ◆ **Evidence undefined.** Without agreeing in advance what would retire a risk, the results become an argument rather than a conclusion.
- ◆ **Previous-program failures not consulted.** The package failure modes that recur are known, and they are the ones worth measuring.
- ◆ **Vehicle count driven by budget rather than by risk.** Fewer vehicles means fewer risks retired, and the trade should be explicit.

WHERE THE EFFORT GOES	5 M/M	WHO IS ON IT ~3.6 FTE	CONNECTIONS	TERMS HERE	4
Risk prioritization	1.5	PTV lead owns it	DEPENDS ON TECH-03 TECH-09 PKGD-01 PKGD-08	ULK OSAT CPI PTV	
Vehicle allocation	1.0	Package architect	FEEDS INTO PTV-02 PTV-03 PTV-04 PTV-12		
Plan and coverage matrix	1.0	Reliability engineer	LATER INPUT PKGD-07 +2w		
Evidence definition	0.8	Program manager			
Risk intake	0.8	OSAT liaison			

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-02

Mechanical test vehicle (MTV) design — bump array, dummy die stack, substrate coupon

TAKES	COSTS	OWNER	CRITICAL PATH
9w w5–w14 of the stage	8.0 M/M ~3.9 people while it runs	Mechanical engineer OSAT liaison approves	no has float

WHY IT EXISTS

Design the **mechanical test vehicle**—dummy die stack, representative bump array, substrate coupon—that carries the package's mechanical behavior without any of its electronics.

The MTV exists to be stressed. Its die are blank, its interposer is representative and its purpose is to reproduce the mechanical stack faithfully enough that warpage, co-planarity and CPI measured on it predict what the product will do.

WHAT IT DELIVERS

PTV-D2

MTV, TTV and daisy-chain vehicle designs

The mechanical vehicle of the three.

NEEDS FIRST

- Vehicle plan from PTV-01
- Package architecture from PKGD-01
- Mechanical predictions from PKGD-07
- Die thickness and bump specification from PD-04
- Material specifications from suppliers

DONE WHEN

- Mechanically representative in the dimensions that drive warpage
- Measurement features integrated, not added externally
- Released against a settled product stack

HOW IT GETS THERE

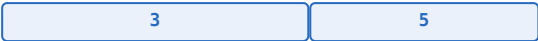
6 STEPS · 2 LANES

w5	w7	w9	w11	w13	w14
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Mechanical representativeness requirements → Mechanical representativeness requirements	1.5w
2	Dummy die and die stack design → Dummy die and stack design	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Bump array design representative of the product → Representative bump array	2w runs in parallel
4	Interposer and substrate coupon design → Interposer and substrate coupon design	2.5w
5	Measurement feature integration — strain, reference marks → Measurement feature integration	1.5w runs in parallel
6	MTV design release → Released MTV design	3w

Step 1 defines what "representative" means for mechanics, which is narrower than it sounds. Die thickness, bump metallurgy, underfill and substrate construction matter; the circuitry does not. Getting that distinction right is what makes the MTV cheap.

WATCH OUT FOR

- ◆ **Vehicle not mechanically representative.** Different die thickness or underfill produces warpage that does not predict the product's.
- ◆ **Measurement features omitted.** A vehicle with no reference marks or strain features can be measured externally and not internally.
- ◆ **Bump array simplified.** Bump density and distribution drive local stress, and a uniform array does not reproduce the product's concentrations.
- ◆ **Design released before the product's stack is settled.** A vehicle built on an assumption that changes measures the wrong package.
- ◆ **Coupon size unrepresentative.** Warpage scales with body size, and a smaller coupon understates it.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT ~3.9 FTE	CONNECTIONS	TERMS HERE	4
Coupon and substrate design	2.0	Mechanical engineer owns it	DEPENDS ON PKGD-01 PKGD-07 PKGD-08 PTV-01	OSAT MTV TTV CPI	
Dummy die and stack	1.8	Package designer	FEEDS INTO PTV-05 PTV-07 PTV-08		
Bump array design	1.5	Materials engineer			
Design release	1.5	Metrology engineer			
Measurement features	1.3	OSAT liaison			

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-03

Thermal test vehicle (TTV) design — embedded heater array and RTD sensor placement

TAKES	COSTS	OWNER	CRITICAL PATH
9w w5–w14 of the stage	8.0 M/M ~3.9 people while it runs	Thermal engineer OSAT liaison approves	no has float

WHY IT EXISTS

Design the **thermal test vehicle**—embedded heaters and temperature sensors laid out to match the product's power map—so the package's thermal path can be measured rather than simulated.

A TTV is a die that only makes heat, in the pattern the real die will make it. Uniform heaters measure an average the product will never experience; heaters matched to the compute cluster distribution measure the hotspot that decides the achievable power envelope.

WHAT IT DELIVERS

PTV-D2

MTV, TTV and daisy-chain vehicle designs

The thermal vehicle of the three.

NEEDS FIRST

- Vehicle plan from PTV-01
- Power map and hotspot distribution from PD-03 and SYN-06
- Thermal predictions from PKGD-07
- Package architecture from PKGD-01
- Sensor and heater process capability

DONE WHEN

- Heater zones match the product's real power distribution
- Sensors placed to see gradients, not only a peak
- Calibration approach defined before build

HOW IT GETS THERE

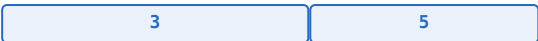
6 STEPS · 2 LANES

w5	w7	w9	w11	w13	w14
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Power map translation into heater zones → Heater zone map from the power distribution	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Heater array design and drive requirements → Heater array design and drive requirements	2w
3	RTD and temperature sensor placement → Sensor placement plan	2w runs in parallel
4	TTV die and interposer integration → TTV die and interposer integration	2.5w
5	Calibration approach for the sensors → Sensor calibration approach	1.5w runs in parallel
6	TTV design release → Released TTV design	3w

Step 3 decides what the vehicle can see. Sensors placed only at the die center measure a number the datasheet can quote; sensors distributed across the zones measure the gradient that determines whether the hotspot is survivable.

WATCH OUT FOR

- ◆ **Uniform heaters.** They measure an average power distribution the product will never have, and the hotspot goes unmeasured.
- ◆ **Sensors too few or badly placed.** Thermal gradients are what matter, and a single center sensor cannot see one.
- ◆ **Sensor calibration not planned.** An uncalibrated RTD gives a relative number, and thermal resistance needs an absolute one.
- ◆ **Heater drive capability underspecified.** Reproducing hundreds of watts through a test vehicle needs real power delivery, and the setup has to supply it.
- ◆ **Power map taken from an early estimate.** The distribution changes through implementation, and a TTV built on the architecture-era map measures the wrong hotspot.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT ~3.9 FTE	CONNECTIONS	TERMS HERE 4
TTV die integration	2.0	Thermal engineer OWNS IT	DEPENDS ON PKG01 PKG07 PKG08	OSAT MTV TTV RTD
Heater array design	1.8	TTV designer	PTV01	
Sensor placement	1.5	Metrology engineer	FEEDS INTO PTV05 PTV09	
Design release	1.5	Die power liaison		
Power map translation	1.3	OSAT liaison		

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-04

Daisy-chain electrical vehicle design —
interconnect continuity and bump yield structures

TAKES	COSTS	OWNER	CRITICAL PATH
8w w6–w14 of the stage	6.0 M/M ~3.3 people while it runs	Package test engineer Yield engineer approves	no has float

WHY IT EXISTS

Design the **daisy-chain electrical vehicle**—continuity structures through every interconnect level—so assembly yield and interconnect integrity can be measured electrically.

Daisy chains turn assembly quality into a resistance measurement. Every bump, every via and every ball becomes part of a chain that either conducts or does not, which makes an otherwise invisible assembly process observable and quantifiable.

WHAT IT DELIVERS

PTV-D2

MTV, TTV and daisy-chain vehicle designs

The electrical vehicle of the three.

NEEDS FIRST

- Vehicle plan from PTV-01
- Bump map and interconnect structure from PKGD-02 and PKGD-03
- Assembly process from PKGD-10
- Substrate ball map from PKGD-04
- Measurement capability at the OSAT

DONE WHEN

- Chains segmented finely enough to localize a failure
- Every interconnect level covered by at least one structure
- Measurement possible with equipment the OSAT actually has

HOW IT GETS THERE

6 STEPS · 2 LANES

w6	w8	w10	w12	w14
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Interconnect level inventory — bump, TSV, via, ball → Interconnect level inventory	1w
2	Daisy chain topology and segmentation design → Daisy chain topology with segmentation	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Kelvin structures for resistance measurement → Kelvin resistance structures	1.5w runs in parallel
4	Chain-to-location mapping for defect localization → Chain-to-location mapping	1.5w
5	Probe and measurement access design → Probe and measurement access design	1.5w runs in parallel
6	Daisy-chain vehicle design release → Released daisy-chain vehicle design	3.5w

Step 2's segmentation decides the diagnostic resolution. One long chain says whether the assembly worked; many short segments say where it failed, and the difference is what turns a yield number into a corrective action.

WATCH OUT FOR

- ◆ **Chains too long.** A single break makes the whole chain open, and the yield number carries no information about where or why.
- ◆ **No Kelvin structures.** Two-wire resistance includes the probe path, and marginal interconnects are exactly where the difference matters.
- ◆ **Chain-to-location mapping omitted.** A failing chain that cannot be located gives no basis for a process change.
- ◆ **Not all interconnect levels covered.** Bump, TSV, via and ball each fail differently, and a vehicle covering only bumps misses the rest.
- ◆ **Measurement access not planned.** A vehicle that can only be measured with equipment the OSAT does not have is a vehicle that is not measured.

WHERE THE EFFORT GOES	6 M/M	WHO IS ON IT ~3.3 FTE	CONNECTIONS
Chain topology and segmentation	1.5	Package test engineer OWNS IT	DEPENDS ON PKGD-02 PKGD-03 PKGD-08 PKGD-10 PTV-01
Design release	1.3	Package designer	FEEDS INTO PTV-05 PTV-11
Chain-to-location mapping	1.3	Failure analysis engineer	
Kelvin structures	1.0	OSAT liaison	TERMS HERE 4
Measurement access	1.0	Yield engineer	OSAT TSV MTV TTV

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-05

TV die fabrication and interposer / substrate vehicle build

TAKES	COSTS	OWNER	CRITICAL PATH
16w w14–w30 of the stage	6.0 M/M ~1.6 people while it runs	PTV lead Program manager approves	yes slipping moves the program

WHY IT EXISTS

Build the **vehicle hardware**—TV die, interposers, substrates—through the same suppliers and processes the product will use.

Sixteen weeks and six man-months: mostly waiting, on the same supplier queues the product will stand in. Using the real suppliers and the real processes is the point—a vehicle built through a different route measures a package the product will never have.

WHAT IT DELIVERS

PTV-D3

Built vehicle lots with assembly travelers

The material half of the deliverable; PTV-06 assembles them.

NEEDS FIRST

- Vehicle designs from PTV-02, PTV-03 and PTV-04
- Supplier capacity from PKGD-09
- Substrate and interposer processes from PKGD-03 and PKGD-04
- Build quantity from the DOE plan in PTV-06
- Material specifications

DONE WHEN

- Vehicles built through the product's own suppliers and processes
- Quantity sufficient for the assembly DOE
- Materials inspected before assembly, not after

HOW IT GETS THERE

6 STEPS · 2 LANES

w14	w18	w22	w26	w30
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Vehicle build planning and quantity definition → Vehicle build plan with quantities	1.5w
2	TV die fabrication — dummy, TTV and daisy-chain → Fabricated TV die	6w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Interposer vehicle build → Interposer vehicles	5w runs in parallel
4	Substrate vehicle build → Vehicle build plan and quantities · Substrate vehicles	5.5w
5	Incoming inspection and material qualification → Incoming inspection records	2.5w runs in parallel
6	Material readiness for assembly → Materials ready for assembly	3w

Step 4 sits on the same substrate queue as the product, which is why it needs planning rather than ordering. Vehicle substrates competing with product substrates for the same capacity is a scheduling problem the program creates for itself if nobody sequences them.

WATCH OUT FOR

- ◆ **Vehicles built through a different process.** A package assembled from materials the product will not use measures a package that will not exist.
- ◆ **Vehicle builds competing with product builds.** Same suppliers, same queue; without sequencing one delays the other.
- ◆ **Quantity too small for the DOE.** The process window study needs enough units to vary conditions, and a build sized for a single condition cannot support it.
- ◆ **Incoming inspection skipped.** A vehicle built from out-of-specification material produces results that describe the material rather than the package.
- ◆ **Build schedule not tracked against the gate.** Sixteen weeks of build inside a fifty-two week program leaves limited room for a slip.

WHERE THE EFFORT GOES	6 M/M	WHO IS ON IT	~1.6 FTE	CONNECTIONS	TERMS HERE	4
		PTV lead owns it		DEPENDS ON PKGD-03 PKGD-04 PKGD-09 PTV-02 PTV-03 PTV-04	DOE TV TTV PTV	
Substrate and interposer builds	2.0	Supply chain lead		FEEDS INTO PTV-06 PTV-12		
TV die fabrication	1.5	Package designer				
Incoming inspection	1.0	Quality engineer				
Build planning	0.8	Program manager				
Material readiness	0.8					

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-06

OSAT assembly of vehicles with process window DOE

TAKES	COSTS	OWNER	CRITICAL PATH
11w w28–w39 of the stage	12.0 M/M ~4.7 people while it runs	Assembly process engineer PTV lead approves	yes slipping moves the program

WHY IT EXISTS

Assemble the vehicles at the OSAT across a **designed experiment**—bond force, reflow profile, underfill, cure—so the assembly process window is measured rather than assumed.

The process window is what ASSY will run to. Establishing it needs deliberate variation: units assembled at the edges of each parameter, not just at nominal, so that the window's boundaries are known rather than the center being confirmed.

WHAT IT DELIVERS

PTV-D3

Built vehicle lots with assembly travelers

The assembly half of the deliverable, with the condition traceability that makes it useful.

NEEDS FIRST

- Built vehicle materials from PTV-05
- Assembly process definition from PKGD-10
- Process parameter ranges from the OSAT
- Thermal and mechanical predictions from PKGD-07
- Inspection capability at the OSAT

DONE WHEN

- Units assembled across the parameter ranges, not only at nominal
- Every unit traceable to its assembly conditions
- Inline inspection performed during assembly, not after

HOW IT GETS THERE

7 STEPS · 2 LANES

w28	w31	w34	w37	w39
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	DOE design across the assembly parameters → Assembly DOE design	1.5w
2	Bonding and die attach across conditions → Bonded vehicles across conditions	2.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Underfill and cure across conditions → Underfilled and cured vehicles	2w runs in parallel
4	Molding, lid and TIM attach → Molded and lidded vehicles	2w
5	Inline inspection during assembly — X-ray, CSAM → Inline inspection results	2w runs in parallel
6	Assembly traveler and condition traceability → Assembly travelers with condition traceability	2w runs in parallel
7	Assembled lot release to characterization → Assembled units across conditions · Assembled lots released to characterization	5w

Step 6 is what makes the DOE usable. Every unit has to be traceable to the exact conditions it was assembled under, and an assembly traveler that records the lot but not the condition turns a designed experiment back into a single build.

WATCH OUT FOR

- ◆ **Assembly at nominal only.** That confirms the center and says nothing about the boundaries, which is where production will occasionally operate.
- ◆ **Condition traceability lost.** Units that cannot be attributed to their assembly conditions turn the DOE into an expensive single build.
- ◆ **DOE too large for the unit count.** Varying every parameter needs units the build did not produce, and an under-powered DOE gives ambiguous results.
- ◆ **Inline inspection omitted.** Assembly defects found only at final characterization cannot be attributed to a process step.
- ◆ **OSAT running its standard process.** The point is deliberate variation, and an assembler defaulting to its recipe produces no window information.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT	~4.7 FTE	CONNECTIONS	TERMS HERE	5
Bonding and die attach	3.0	Assembly process engineer	owns it	DEPENDS ON PKGD-10 PTV-05	DOE OSAT TIM PTV CSAM	
Lot release and handling	2.5	OSAT process engineer		FEEDS INTO PTV-07 PTV-08 PTV-09		
Underfill and cure	2.0	Quality engineer		PTV-10 PTV-11 PTV-12		
Inline inspection	2.0	Metrology engineer				
Molding and lid attach	1.5	PTV lead				
DOE design and travelers	1.0					

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-07

Warpage and co-planarity measurement across the reflow profile (shadow moiré)

TAKES	COSTS	OWNER	CRITICAL PATH
7w w38–w45 of the stage	8.0 M/M ~5.0 people while it runs	Mechanical engineer Quality engineer approves	no has float

WHY IT EXISTS

Measure **warpage and co-planarity across the reflow profile**—not at room temperature, but at every point in the thermal cycle where the package deforms.

Warpage is the mechanism behind most 2.5D assembly failures. A package flat at room temperature can bow by more than a hundred microns at reflow peak, opening joints that then re-close as it cools—leaving a unit that looks fine and has intermittent connections.

WHAT IT DELIVERS

PTV-D4

Warpage and co-planarity data across the reflow profile

This activity is the deliverable, and the reflow-profile part is what distinguishes it from a room-temperature check.

NEEDS FIRST

- Assembled units with condition traceability from PTV-06
- Warpage simulation from PKGD-07
- Reflow profile from PKGD-10
- Measurement equipment and capability
- Co-planarity specification from the assembly requirements

DONE WHEN

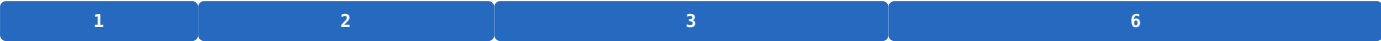
- Warpage measured across the full reflow profile
- Simulation correlated against measurement
- Results attributed to assembly conditions

HOW IT GETS THERE

6 STEPS · 2 LANES

w38w40w42w44w45

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Measurement setup — shadow moiré or equivalent → Measurement setup and method	1w
2	Room-temperature baseline across units → Room-temperature warpage baseline	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Warpage measurement across the reflow profile → Warpage across the reflow profile	2w
4	Co-planarity and ball flatness measurement → Co-planarity and ball flatness results	1.5w
5	Condition-to-warpage correlation from the DOE → Condition correlation from the DOE	1.5w
6	Comparison against the PKGD-07 simulation → Simulation comparison	2.5w

Step 6 is what makes the measurement reusable. A warpage number is one package; a validated simulation model predicts the next one, and correlating the two is what turns a measurement into a capability.

WATCH OUT FOR

- ◆ **Measured only at room temperature.** The failure occurs at reflow peak, and the room-temperature number is not predictive of it.
- ◆ **Simulation comparison skipped.** Without correlation the model stays unvalidated, and the next package has to be measured from scratch.
- ◆ **Co-planarity ignored.** Ball flatness determines whether the package mounts to the board reliably, and it is a separate property from warpage.
- ◆ **Too few units to see condition effects.** Warpage varies unit to unit, and distinguishing a condition effect from noise needs population.
- ◆ **Results not linked to assembly conditions.** Knowing the warpage without knowing which condition produced it gives no basis for a window.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT ~5.0 FTE	CONNECTIONS	TERMS HERE	1
Warpage across reflow	2.0	Mechanical engineer owns it	DEPENDS ON PKGD-07 PTV-02 PTV-06	DOE	
Simulation comparison	1.8	Metrology engineer	FEEDS INTO PTV-08 PTV-12 ASSY-06		
Room-temperature baseline	1.5	Simulation engineer	FEEDS BACK INTO PKGD-07		
Co-planarity measurement	1.5	Assembly process engineer			
Condition correlation	1.3	Quality engineer			

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-08

Chip-package interaction stress evaluation — low-k / ULK crack, bump and BEOL integrity

TAKES	COSTS	OWNER	CRITICAL PATH
11w w37–w48 of the stage	14.0 M/M ~5.5 people while it runs	Reliability engineer Die liaison approves	yes slipping moves the program

WHY IT EXISTS

Evaluate **chip-package interaction**—the stress the package puts into the die's low-k dielectric stack, and whether the bumps and back-end metal survive it.

CPI is the failure mode that makes 2.5D packaging risky at advanced nodes. Ultra-low-k dielectrics are mechanically weak, the package expands differently from the silicon, and the resulting stress concentrates exactly where the bumps meet the top metal. It cracks or it does not, and simulation alone is not enough to know.

WHAT IT DELIVERS

PTV-D5

CPI stress assessment report — ULK, bump and BEOL integrity

This activity is the deliverable, and it is the most consequential result the vehicle program produces.

NEEDS FIRST

- Assembled MTV units from PTV-06
- Die stack and ULK information from TECH-03
- Warpage results from PTV-07
- Thermal cycling capability
- Underfill and material properties

DONE WHEN

- Stress applied to a cycle count that would reveal accumulation
- Failures cross-sectioned and localized, not only detected
- Findings delivered before the package design freeze

HOW IT GETS THERE

6 STEPS · 2 LANES

w37	w40	w43	w46	w48
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	CPI stress evaluation plan and structure selection → CPI evaluation plan and structure selection	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Thermal cycling stress application → Thermal cycling stress results	3w
3	Bump and BEOL integrity inspection → Bump and BEOL integrity findings	2.5w runs in parallel
4	Low-k and ULK crack detection → ULK crack detection results	2.5w
5	Cross-section and failure analysis on affected units → Cross-section and failure analysis	2w runs in parallel
6	CPI assessment and design implications → CPI assessment with design implications	4w

Step 5 is destructive and necessary. Electrical and acoustic methods say something is wrong; only cross-sectioning shows what failed and where, and the design implication depends entirely on which layer cracked.

WATCH OUT FOR

- ◆ **CPI evaluated by simulation alone.** Material behavior at these dimensions is not predictable to the accuracy the decision needs.
- ◆ **Insufficient thermal cycles.** CPI failures accumulate; a short cycle count shows nothing and reassures wrongly.
- ◆ **Cross-sectioning skipped.** Detection without localization gives no design implication, and the implication is the output that matters.
- ◆ **Only nominal units stressed.** The DOE's edge conditions are where CPI is most likely, and testing only the center misses it.
- ◆ **Findings arriving after package design freezes.** A CPI problem needs an underfill, bump or design change, none of which are available after PKGD-11.

WHERE THE EFFORT GOES	14 M/M	WHO IS ON IT ~5.5 FTE	CONNECTIONS	TERMS HERE	6
CPI assessment and implications	3.5	Reliability engineer OWNS IT	DEPENDS ON TECH-03 PTV-02 PTV-06 PTV-07	DOE BEOL ULK MTV CPI FA	
Thermal cycling	3.0	Failure analysis engineer			
Crack detection	2.5	Materials engineer	FEEDS INTO PKGD-11 PTV-12 MP-04		
Cross-section and FA	2.5	Package architect			
Bump and BEOL inspection	2.5	Die liaison			

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-09

Thermal characterization — Rjc / Rja, TIM performance, hotspot mapping against the thermal model

TAKES	COSTS	OWNER	CRITICAL PATH
9w w38–w47 of the stage	10.0 M/M ~4.8 people while it runs	Thermal engineer Product engineering approves	no has float

WHY IT EXISTS

Characterize the package's **thermal performance**—junction-to-case and junction-to-ambient resistance, TIM behavior, hotspot mapping—and correlate it against the simulation.

The thermal path decides the achievable power envelope, and the product is sold on that envelope. Measuring it on a TTV that reproduces the real power distribution is what turns a simulated Rjc into a number the datasheet can carry.

WHAT IT DELIVERS

PTV-D6

Thermal characterization report with model correlation (Rjc, TIM, hotspot map)

This activity is the deliverable, and the correlation is what makes the model reusable.

NEEDS FIRST

- Assembled TTV units from PTV-06
- Thermal simulation from PKGD-07
- Sensor calibration approach from PTV-03
- Thermal measurement equipment
- TIM material options

DONE WHEN

- TIM performance measured as assembled, not as specified
- Hotspot mapped rather than averaged
- Thermal model correlated and updated

HOW IT GETS THERE

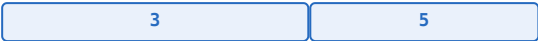
6 STEPS · 2 LANES

w38w40w42w44w46w47

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Thermal measurement setup and sensor calibration → Thermal measurement setup and calibration	1.5w
2	Steady-state Rjc and Rja measurement → Rjc and Rja measurements	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Hotspot mapping across the heater zones → Hotspot maps across zones	2w runs in parallel
4	TIM performance and bond line study → TIM performance and bond line results	2w
5	Transient thermal response measurement → Transient thermal response	1.5w runs in parallel
6	Correlation against the PKGD-07 thermal model → Model correlation results	3.5w

Step 4 measures the interface nobody can simulate reliably. TIM performance depends on bond line thickness, pressure and pump-out over time, and the datasheet value is a starting point rather than an answer.

WATCH OUT FOR

- ◆ **TIM performance taken from the datasheet.** Bond line thickness and pressure are assembly parameters, and the achieved performance differs from the specification.
- ◆ **Hotspot not measured.** The average junction temperature is not what limits the part; the hottest zone is.
- ◆ **Simulation not correlated.** The model then remains unvalidated, and every future thermal question needs another vehicle.
- ◆ **Transient response ignored.** An accelerator's power is bursty, and the transient behavior determines whether the package can absorb a workload spike.
- ◆ **Measured at a single ambient condition.** Rja depends on airflow and ambient, and one condition does not characterize the range the product will see.

WHERE THE EFFORT GOES	10 M/M	WHO IS ON IT	~4.8 FTE	CONNECTIONS	TERMS HERE	4
Model correlation	2.5	Thermal engineer	OWNS IT	DEPENDS ON PKGD-07 PTV-03 PTV-06	TTV TIM Rjc Rja	
Rjc and Rja measurement	2.0	Metrology engineer		FEEDS INTO PTV-12 ASSY-05 MP-02		
TIM study	2.0	Materials engineer		FEEDS BACK INTO PKGD-07 SIPI-10		
Hotspot mapping	1.8	Simulation engineer				
Transient measurement	1.8	Product engineering				

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-10

Board-level reliability on vehicles
— temperature cycle, drop, bend

TAKES	COSTS	OWNER	CRITICAL PATH
14w w36–w50 of the stage	8.0 M/M ~2.5 people while it runs	Reliability engineer Package architect approves	yes slipping moves the program

WHY IT EXISTS

Stress the vehicles through **board-level reliability**—temperature cycling, drop, bend—because the package has to survive being mounted in a system, not only being assembled.

Board-level reliability is where the package meets the customer's manufacturing process. Solder joints crack under thermal cycling, boards flex during handling, and a package that passes component qualification and fails board-level testing is a package the customer cannot use.

WHAT IT DELIVERS

PTV-D7

Board-level reliability data on vehicles

This activity is the deliverable, and characteristic life is what customers will ask for.

NEEDS FIRST

- Assembled units from PTV-06
- Board-level reliability standards (JEDEC)
- Customer board and mounting requirements
- Coupon board design capability
- Reliability chamber capacity

DONE WHEN

- Testing run to failure, with characteristic life established
- Coupon boards representative of customer mounting
- Failure modes identified, not only counted

HOW IT GETS THERE

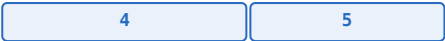
6 STEPS · 2 LANES

w36	w39	w42	w45	w48	w50
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Board-level test plan and coupon board design → Board-level test plan and coupon boards	2w
2	Coupon board fabrication and vehicle mounting → Mounted vehicle assemblies	2.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Temperature cycling to failure → Temperature cycling results to failure	4w
4	Drop and shock testing → Drop and shock results	2.5w
	runs in parallel	
5	Bend and flex testing → Bend and flex results	2w
	runs in parallel	
6	Failure mode analysis and characteristic life → Failure mode analysis and characteristic life	5.5w

Step 3 runs to failure rather than to a pass criterion. Characteristic life—the cycle count at which a defined fraction fails—is what a customer needs to assess the package in their system, and a pass at a fixed cycle count does not provide it.

WATCH OUT FOR

- ◆ **Testing to a pass criterion rather than to failure.** A pass tells the customer nothing about margin; characteristic life does.
- ◆ **Coupon board unrepresentative.** Board thickness and pad design affect solder joint stress, and a test board unlike the customer's predicts poorly.
- ◆ **Started too late to finish.** Temperature cycling to failure takes months, and it is the longest test in the vehicle program.
- ◆ **Failure modes not analyzed.** Knowing units failed at a cycle count without knowing how gives no route to improvement.
- ◆ **Only one mounting condition.** Different customers mount differently, and the package's sensitivity to that is worth knowing.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT ~2.5 FTE	CONNECTIONS
Failure analysis and characteristic life	2.5	Reliability engineer owns it	DEPENDS ON PTV-06
Temperature cycling	2.0	Board engineer	FEEDS INTO PTV-12 MP-04 MP-11
Coupon board and mounting	1.5	Failure analysis engineer	TERMS HERE 1
Drop and shock	1.0	Quality liaison	JEDEC
Bend and flex	1.0	Package architect	

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-11

Daisy-chain continuity test and assembly yield learning

TAKES	COSTS	OWNER	CRITICAL PATH
9w w39–w48 of the stage	6.0 M/M ~2.9 people while it runs	Package test engineer Quality engineer approves	no has float

WHY IT EXISTS

Measure **assembly yield electrically** through the daisy chains, localize what failed, and turn that into process learning before the product is assembled.

Assembly yield on a 2.5D package with hundreds of thousands of interconnects is not guaranteed. The daisy chains make it measurable, and the segmentation makes it attributable—which is what lets the OSAT fix a process problem rather than accept a yield number.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Assembled daisy-chain units from PTV-06
- Chain-to-location mapping from PTV-04
- Assembly conditions from the travelers
- Measurement capability at the OSAT
- Yield expectations from the cost model

DONE WHEN

- Failures localized to interconnect level and position
- Marginal joints identified by resistance, not only by continuity
- Feedback delivered before the process window freezes

HOW IT GETS THERE

6 STEPS · 2 LANES

w39	w41	w43	w45	w47	w48
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Continuity test setup and measurement plan → Continuity test setup and plan	1w
2	Chain continuity measurement across units → Chain continuity results across units	2w
3	Kelvin resistance measurement on interconnects → Kelvin resistance measurements	1.5w

runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Failure localization from chain segmentation → Failure localization findings	2w
5	Condition-to-yield correlation from the DOE → Condition-to-yield correlation	1.5w runs in parallel
6	Assembly yield learning and process feedback → Assembly yield learning and process feedback	4w

Step 5 is why the DOE was run. Yield varying with bond force or reflow profile identifies the parameter that matters, and that is the parameter the process window in `PTV-12` will be built around.

WATCH OUT FOR

- ◆ **Yield measured without localization.** A number with no location gives the OSAT nothing to act on.
- ◆ **Marginal interconnects passing continuity.** A high-resistance joint conducts and will fail in the field; Kelvin measurement is what distinguishes it.
- ◆ **Condition correlation not performed.** The DOE's whole purpose is to link yield to parameters, and skipping the analysis wastes the build.
- ◆ **Feedback arriving after the process freezes.** Learning that cannot change the process window is learning with no consequence.
- ◆ **Sample size too small for yield estimation.** Assembly yield is a rate, and estimating it needs enough units for the rate to be meaningful.

WHERE THE EFFORT GOES	6 M/M	WHO IS ON IT ~2.9 FTE	CONNECTIONS	TERMS HERE	2
Yield learning and feedback	1.5	Package test engineer owns it	DEPENDS ON PTV-04PTV-06	DOE OSAT	
Continuity measurement	1.3	Yield engineer	FEEDS INTO PTV-12ASSY-07ASSY-08		
Failure localization	1.3	OSAT process engineer	MP-05		
Kelvin measurement	1.0	Failure analysis engineer			
Condition correlation	1.0	Quality engineer			

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-12

Feedback into package design and assembly process window freeze

TAKES	COSTS	OWNER	CRITICAL PATH
7w w45–w52 of the stage	5.0 M/M ~3.1 people while it runs	PTV lead Program manager approves	yes slipping moves the program

WHY IT EXISTS

Freeze the **assembly process window**, feed the findings back into package design, and declare package validation complete—the gate product wafer-out has to clear.

This is the activity the whole stage exists for. It converts a large body of measurement into two things: a process window ASSY will run to, and a statement that the package has been validated. Product wafers must not come out before it is issued, because the alternative is learning about the package on real dies.

WHAT IT DELIVERS

- PTV-D8

Frozen assembly process window definition

The window ASSY will build the product to, derived from evidence rather than from the OSAT's standard recipe.
- PTV-D9

Package validation complete record — gate for product wafer-out

This activity carries the stage's milestone and the gate.

NEEDS FIRST

- Warpage results from PTV-07
- CPI assessment from PTV-08
- Thermal characterization from PTV-09
- Board-level reliability from PTV-10
- Assembly yield learning from PTV-11

DONE WHEN

- Every matrix risk marked retired or explicitly not retired
- Process window frozen with boundaries that exclude observed failures
- Window transferred to the OSAT as a specification, not a report

HOW IT GETS THERE

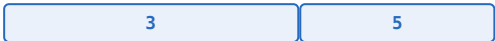
6 STEPS · 2 LANES

w45w47w49w51w52

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Result consolidation across all vehicle characterization → Consolidated vehicle results	1w
2	Risk retirement assessment against the coverage matrix → Risk retirement assessment against the matrix	1.5w
3	Package design feedback and change requests → Package design feedback and change requests	1.5w runs in parallel
4	Assembly process window definition and freeze → Frozen assembly process window	2w
5	Residual risk statement for unretired items → Residual risk statement	1w runs in parallel
6	Package validation complete declaration → Package validation complete declaration	2.5w

Step 2 is the honest accounting. Some risks in the matrix will not have been retired—a vehicle failed, a measurement was inconclusive—and the declaration has to say which, rather than declaring completion because the date arrived.

WATCH OUT FOR

- ◆ **Completion declared because the date arrived.** The gate exists to be capable of holding wafer-out, and a declaration that cannot say no is not a gate.
- ◆ **Unretired risks not stated.** A validation record that implies full coverage when some risks remain open misleads exactly the decision it informs.
- ◆ **Feedback arriving after PKGD-11.** Package design changes are not available after the freeze, so vehicle findings have to land before it.
- ◆ **Process window frozen too wide.** A window that accommodates every observed condition is not a window; the boundaries have to exclude what produced failures.
- ◆ **Window not transferred to the OSAT as a specification.** A window that exists in a report and not in the assembly instructions will not be followed.

WHERE THE EFFORT GOES	5 M/M	WHO IS ON IT ~3.1 FTE	CONNECTIONS	TERMS HERE 3
Validation declaration	1.3	PTV lead owns it	DEPENDS ON FAB-09 PKGD-11 PTV-01 PTV-05 PTV-06 PTV-07 PTV-08 PTV-09 PTV-10 PTV-11	OSAT CPI PTV
Process window definition	1.0	Package architect	FEEDS INTO FAB-10 ASSY-01 ASSY-03 ASSY-04 ASSY-07 MP-04	
Risk retirement assessment	1.0	Assembly process engineer	LATER INPUT ASSY-07 +13w MP-04 +26w	
Design feedback	1.0	Reliability engineer		
Residual risk statement	0.8	Program manager		

INTEGRATE

18

Chip-Package-System Co-Verification

SIPI

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w50 – w76	26 _w	11	85 _{M/M}	7 of 11

ENGINEERING ACTIVITIES					KEY DELIVERABLES	
SIPI-01	Chip power model (CPM/CPS) extraction per power domain and operating mode	7w	8	M/M	SIPI-D1	Chip power model (CPM/CPS) release per domain and mode
SIPI-02	Package and board model extraction — 3D EM, S-parameter and RLC network	8w	10	M/M	SIPI-D2	Extracted package and board electrical models
SIPI-03	Die-package-board PDN co-simulation — DC and AC impedance versus target	9w	12	M/M	SIPI-D3	PDN impedance and dynamic IR co-simulation report
SIPI-04	Dynamic voltage-drop analysis with package inductance and real switching profiles	8w	10	M/M	SIPI-D4	Decap budget and placement specification across die, package and board
SIPI-05	Decoupling capacitor budget and placement optimization across die, package and board	7w	7	M/M	SIPI-D5	Channel compliance report per interface, with margins
SIPI-06	SSN / SSO and simultaneous-switching noise analysis at the IO ring	5w	5	M/M	SIPI-D6	Eye, jitter and BER budget closure record
SIPI-07	High-speed channel simulation with extracted package models — insertion loss, crosstalk, return path	9w	11	M/M	SIPI-D7	Power-aware STA correlation report
SIPI-08	Eye diagram, jitter and BER budget closure per interface (PCIe/CXL, HBM, die-to-die)	8w	9	M/M	SIPI-D8	Chip-package-system co-verification signoff — tapeout gate
SIPI-09	Power-aware timing correlation — voltage drop back-annotated into STA	5w	5	M/M		
SIPI-10	Electro-thermal co-analysis coupled with the package thermal model	5w	5	M/M		
SIPI-11	Co-verification signoff review and criteria disposition with design and package teams	4w	3	M/M		

18 SIPI · CHIP-PACKAGE-SYSTEM CO-VERIFICATION

SIPI-01

Chip power model (CPM/CPS) extraction per power domain and operating mode

TAKES	COSTS	OWNER	CRITICAL PATH
7w w0–w7 of the stage	8.0 M/M ~5.0 people while it runs	Power modeling engineer Configuration manager approves	yes slipping moves the program

WHY IT EXISTS

Get a **chip power model** that represents the die's current demand honestly—per domain, per mode, under the switching the product actually produces.

Everything in this stage depends on this model. A CPM extracted under averaged activity understates the transient the package must supply, and every downstream conclusion inherits that error without any indication that it did.

WHAT IT DELIVERS

SIPI-D1

Chip power model (CPM/CPS) release per domain and mode

This activity is the deliverable, and everything else in the stage consumes it.

NEEDS FIRST

- Placed and routed database from PD-05 and PD-06
- Chip power model extraction from PD-11
- Switching activity from DV-04 and SYN-06
- Power domain definition from ARCH-05
- Model format requirements from the analysis tools

DONE WHEN

- Model built on the worst realistic transient, not an average
- Separate models per domain and mode
- Validated against die-level IR before release

HOW IT GETS THERE

6 STEPS · 2 LANES

w0	w2	w4	w6	w7
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Model requirement definition with physical design → Model requirements and format definition	1w
2	Switching scenario selection from real workloads → Switching scenario selection	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Domain and mode decomposition → Domain and mode decomposition	1.5w runs in parallel
4	CPM extraction and assembly → Assembled chip power model	2w
5	Model validation against die-level IR analysis → Validation against die-level IR	1.5w runs in parallel
6	Model release and version control → Extracted chip power model · Versioned model release	2.5w

Step 2 decides everything the stage can conclude. The worst realistic transient—a workload burst arriving after an idle period—is what the package has to absorb, and a model built on steady-state activity cannot represent it.

WATCH OUT FOR

- ◆ **Model built on averaged activity.** The transient is what the package must supply, and averaging removes exactly the event of interest.
- ◆ **Single model for all modes.** DVFS points and workloads draw very differently, and one model represents none of them well.
- ◆ **Not validated against die-level IR.** A CPM that disagrees with the die's own analysis is wrong somewhere, and nobody knows where.
- ◆ **Delivered late.** The stage cannot start without it, and the stage gates mask release.
- ◆ **Version drift.** Analysis run against a stale model produces conclusions about a die that has since changed.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT ~5.0 FTE	CONNECTIONS	TERMS HERE	5
CPM extraction	2.5	Power modeling engineer owns it	DEPENDS ON ARCH-05 DV-04 PD-05	IR PDN DVFS CPM	
Model validation	1.5	Physical design liaison	FEEDS INTO PD-11 SIPI-03 SIPI-04	SI/PI	
Scenario selection	1.5	Verification liaison	SIPI-09 SIPI-10		
Requirement definition	1.5	SI/PI lead	LATER INPUT		
Release and version control	1.0	Configuration manager	PD-11 +3w		

18 SIPI · CHIP-PACKAGE-SYSTEM CO-VERIFICATION

SIPI-02

Package and board model extraction —
3D EM, S-parameter and RLC network

TAKES	COSTS	OWNER	CRITICAL PATH
8w w2–w10 of the stage	10.0 M/M ~5.4 people while it runs	SI/PI extraction engineer CAD infrastructure approves	yes slipping moves the program

WHY IT EXISTS

Extract **electrical models of the package and the board**—3D field solution, S-parameters, RLC networks—so the system can be analyzed as one circuit rather than three assumptions.

Each of the three parties has been designing against a model of the other two. Extraction replaces those assumptions with a solved answer: what the interposer, substrate and board actually do electrically, at the frequencies the design cares about.

WHAT IT DELIVERS

SIPI-02

Extracted package and board electrical models

This activity is the deliverable, and it is what turns three design intents into one analysable circuit.

NEEDS FIRST

- Interposer database from PKGD-03
- Substrate database from PKGD-04
- Board design from EVB-03
- Bump and ball assignment from PKGD-02 and PD-04
- 3D field solver capability

DONE WHEN

- Frequency range covers both PDN and channel needs
- Board and socket included, not only die and package
- Models validated rather than assumed correct

HOW IT GETS THERE

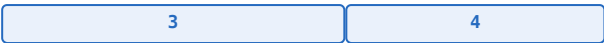
6 STEPS · 2 LANES

w2	w4	w6	w8	w10
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Extraction scope and frequency range definition → Extraction scope and frequency definition	1.5w
2	Interposer model extraction → Interposer electrical model	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Substrate model extraction → Substrate electrical model	2w runs in parallel
4	Board and socket model extraction → Board and socket models	1.5w runs in parallel
5	PDN network model assembly → Assembled PDN network model	2w
6	Model validation and correlation → Model validation results	2.5w

Step 1's frequency range decides the extraction cost and its usefulness. PDN analysis needs DC to the switching harmonics; channel analysis needs far higher. Extracting once for both is efficient and extracting for the wrong range wastes the effort entirely.

WATCH OUT FOR

- ◆ **Frequency range too narrow.** A model extracted for PDN analysis cannot support channel simulation, and re-extraction is expensive.
- ◆ **Extraction on an unfinished design.** Package routing changes invalidate the model, and re-extraction is measured in weeks.
- ◆ **Board model omitted.** The board is part of the PDN and part of the channel, and excluding it makes the analysis die-and-package only.
- ◆ **Models not validated.** A 3D extraction has convergence and meshing choices, and an unvalidated result can be confidently wrong.
- ◆ **Socket and connector effects ignored.** The bring-up platform includes them, and channel margin measured at bring-up will reflect them.

WHERE THE EFFORT GOES	10 M/M	WHO IS ON IT	~5.4 FTE	CONNECTIONS	TERMS HERE	6
Interposer extraction	2.5	SI/PI extraction engineer	OWNS IT	DEPENDS ON ARCH-08 PD-04 PKGD-02 PKGD-03 PKGD-04	DC PDN SI SI/PI RLC CAD	
Substrate extraction	2.0	Package designer		FEEDS INTO SIPI-03 SIPI-04 SIPI-06 SIPI-07 EVB-03		
PDN network assembly	2.0	Board designer		LATER INPUT EVB-03 +4w		
Model validation	1.5	SI lead				
Board and socket models	1.0	CAD infrastructure				
Scope definition	1.0					

18 SIPI · CHIP-PACKAGE-SYSTEM CO-VERIFICATION

SIPI-03

Die-package-board PDN co-simulation
— DC and AC impedance versus target

TAKES	COSTS	OWNER	CRITICAL PATH
9w w8–w17 of the stage	12.0 M/M ~5.8 people while it runs	Power integrity engineer SI/PI lead approves	yes slipping moves the program

WHY IT EXISTS

Simulate the **die, package and board as one power delivery network** and check its impedance against target across the frequency range that matters.

The three levels each cover a different band: on-die decap handles the highest frequencies, package decoupling the middle, board bulk capacitance the lowest. Analyzed separately each looks adequate; analyzed together the gaps between them appear, and those gaps are where the supply actually sags.

WHAT IT DELIVERS

SIPI-03

PDN impedance and dynamic IR co-simulation report

The impedance half of the deliverable; SIPI-04 supplies the dynamic IR half.

NEEDS FIRST

- Chip power model from SIPI-01
- Package and board models from SIPI-02
- Die PDN from PD-03
- Package PDN design from PKGD-06
- Board power design from EVB-05

DONE WHEN

- Impedance analyzed across the full frequency range as one network
- Anti-resonances identified and addressed
- Findings carry recommendations naming which level to change

HOW IT GETS THERE

6 STEPS · 2 LANES

w8	w10	w12	w14	w16	w17
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Impedance target derivation per domain → Impedance targets per domain	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Combined die-package-board PDN assembly → Combined PDN model	2w
3	DC resistance and static drop analysis → DC resistance and static drop results	1.5w runs in parallel
4	AC impedance analysis across frequency → AC impedance across frequency	2.5w
5	Anti-resonance identification and analysis → Anti-resonance analysis	1.5w runs in parallel
6	PDN findings and improvement recommendations → PDN findings and recommendations	3w

Step 5 finds what separate analyses cannot. Anti-resonances form between the package's inductance and the die's capacitance, at frequencies neither level would identify alone, and they are where impedance peaks far above target.

WATCH OUT FOR

- ◆ **Levels analyzed separately.** Each looks adequate in isolation and the transitions between them are where the impedance peaks.
- ◆ **Anti-resonances missed.** They occur at frequencies no single level would examine and can exceed target by a large factor.
- ◆ **Impedance target derived from a rule of thumb.** The target follows from the current step and the allowable ripple, and a generic value is either wasteful or inadequate.
- ◆ **DC analysis only.** Static drop is the easy half; the frequency-dependent behavior is where the problems are.
- ◆ **Findings without recommendations.** An impedance plot that exceeds target tells the team there is a problem, not which level to fix.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT	~5.8 FTE	CONNECTIONS	TERMS HERE	4
AC impedance analysis	3.0	Power integrity engineer owns it		DEPENDS ON PD-03 PD-11 SIPI-01 SIPI-02	DC IR PDN SI/PI	
Findings and recommendations	2.5	Die power delivery engineer		RUNS WITH PKGD-06		
PDN assembly	2.0	Package power engineer		FEEDS INTO S0-06 SIPI-04 SIPI-05 SIPI-11		
Anti-resonance analysis	2.0	Board designer				
DC analysis	1.5	SI/PI lead				
Target derivation	1.0					

18 SIPI · CHIP-PACKAGE-SYSTEM CO-VERIFICATION

SIPI-04

Dynamic voltage-drop analysis with package inductance and real switching profiles

TAKES	COSTS	OWNER	CRITICAL PATH
8w w12–w20 of the stage	10.0 M/M ~5.4 people while it runs	Power integrity engineer Package power engineer approves	yes slipping moves the program

WHY IT EXISTS

Analyze **dynamic voltage droop with the package's inductance included**, under the switching profiles the product actually produces—the number the die-only analysis is optimistic about.

Die-only dynamic IR excludes the package's series inductance, which is the dominant impedance at the frequencies a workload burst excites. The difference between the two numbers is real margin, and it is the reason this stage exists rather than trusting S0–04.

WHAT IT DELIVERS

SIPI-03

PDN impedance and dynamic IR co-simulation report

The dynamic IR half of the deliverable, and the number signoff should be using.

NEEDS FIRST

- Chip power model from SIPI-01
- Combined PDN model from SIPI-03
- Die-only IR results from PD-03 and SO-04
- Workload transient profiles from DV-04
- Decap configuration from SIPI-05

DONE WHEN

- Package inductance included in every result
- Die-only and co-analysis numbers reconciled
- Findings delivered before timing signoff closes

HOW IT GETS THERE

6 STEPS · 2 LANES

w12	w14	w16	w18	w20
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Transient scenario selection from the workload profiles → Transient scenario selection	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Combined transient simulation with package inductance → Combined transient simulation results	2.5w
3	Droop magnitude and duration analysis per domain → Droop magnitude and duration per domain	2w
4	Comparison against die-only IR results → Die-only comparison	1.5w runs in parallel
5	Worst-case scenario exploration → Worst-case scenario findings	1.5w runs in parallel
6	Dynamic IR findings and margin assessment → Dynamic IR margin assessment	2w

Step 4 is the comparison the program most needs. Where die-only analysis says 4.1% and the co-analysis says 5.0% against a 5% budget, the design has no margin—and the two numbers have to be reconciled before `SO-04` signs off on the optimistic one.

WATCH OUT FOR

- ◆ **Package inductance excluded.** It dominates at the frequencies a burst excites, and excluding it produces a comfortable and wrong answer.
- ◆ **Transient scenario too gentle.** The worst case is a burst after idle, and a scenario built from steady-state activity never reaches it.
- ◆ **Droop duration ignored.** A deep short droop and a shallow long one have different timing consequences, and only the magnitude is usually reported.
- ◆ **Die-only comparison not made.** Without it nobody knows whether S0-04's number is optimistic, and the two signoffs disagree silently.
- ◆ **Findings arriving after timing signoff.** Droop consumes timing margin, and a late finding invalidates closure.

WHERE THE EFFORT GOES	10 M/M	WHO IS ON IT	~5.4 FTE	CONNECTIONS	TERMS HERE	3
Transient simulation	3.0	Power integrity engineer OWNS IT		DEPENDS ON PD-04 PD-10 PD-11 SIPI-01 SIPI-02 SIPI-03	STA IR PDN	
Droop analysis	2.0	Die power delivery engineer				
Findings and margin assessment	2.0	Verification liaison		FEEDS INTO S0-04 S0-06 SIPI-05 SIPI-09 SIPI-11 EVB-03 BU-05		
Die-only comparison	1.5	Timing engineer				
Scenario selection and exploration	1.5	Package power engineer				

SIPI-05

Decoupling capacitor budget and placement optimization across die, package and board

TAKES	COSTS	OWNER	CRITICAL PATH
7w w14–w21 of the stage	7.0 M/M ~4.3 people while it runs	Power integrity engineer Program manager approves	no has float

WHY IT EXISTS

Allocate the **decoupling budget across die, package and board**—how much capacitance, of what type, mounted where—so the three levels cover the frequency range between them rather than overlapping.

Decoupling is a system-level allocation that is usually made three times independently. On-die capacitance is expensive in area, package capacitors are limited by mounting positions, and board bulk is cheap and slow. Optimizing them together is what covers the band; optimizing them separately leaves the middle uncovered.

WHAT IT DELIVERS

SIPI-D4

Decap budget and placement specification across die, package and board

This activity is the deliverable, and it is the only place the three levels are allocated together.

NEEDS FIRST

- PDN impedance results from SIPI-03
- Dynamic droop results from SIPI-04
- On-die decap capability and area cost from PD-03
- Package mounting positions from PKGD-06
- Board capacitance options from EVB-05

DONE WHEN

- Bands allocated so the three levels cover the range without gaps
- Package capacitors placed for loop inductance
- Allocation verified against the impedance target before handoff

HOW IT GETS THERE

6 STEPS · 2 LANES

w14

w16

w18

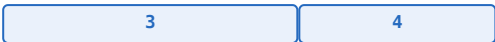
w20

w21

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Frequency band allocation across the three levels → Frequency band allocation	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	On-die decap requirement and area cost → On-die decap requirement with area cost	1.5w
3	Package capacitor selection and mounting position → Package capacitor selection and placement	1.5w runs in parallel
4	Board bulk capacitance requirement → Board bulk capacitance requirement	1w runs in parallel
5	Combined impedance verification with the allocation → Combined impedance verification	1.5w
6	Decap specification and handoff to each level → Decap specification per level	2.5w

Step 2 carries a real cost. On-die decoupling consumes die area that could have been logic, and the trade between area and package capacitors is a system decision rather than a physical design one.

WATCH OUT FOR

- ◆ **Each level decoupled independently.** The bands overlap in places and leave gaps in others, and the gaps are where the impedance peaks.
- ◆ **Package capacitors placed where there is room.** Mounting position determines loop inductance, and a capacitor far from the current draw contributes little.
- ◆ **On-die decap area not costed.** It is die area, and treating it as free produces a design that traded logic for decoupling without anyone deciding to.
- ◆ **Allocation not verified.** A budget that has not been re-simulated against the impedance target is a proposal rather than a solution.
- ◆ **Specification not handed to each level.** A system allocation that stays in a report is not implemented by anyone.

WHERE THE EFFORT GOES	7 M/M	WHO IS ON IT ~4.3 FTE	CONNECTIONS	TERMS HERE	2
Decap specification and handoff	1.5	Power integrity engineer OWNS IT	DEPENDS ON AMS-04 PD-03 PKGD-06 SIPI-03 SIPI-04	IR PDN	
Band allocation	1.3	Die power delivery engineer	FEEDS INTO SIPI-11 EVB-05 BU-05		
On-die requirement	1.3	Package power engineer	FEEDS BACK INTO PD-03 PKGD-06		
Package capacitor selection	1.3	Board designer			
Combined verification	1.0	Program manager			
Board requirement	0.8				

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SIPI-06

SSN / SSO and simultaneous-switching noise analysis at the IO ring

TAKES	COSTS	OWNER	CRITICAL PATH			
5w w15–w20 of the stage	5.0 M/M ~4.3 people while it runs	SI/PI engineer Interface architect approves	no has float			
WHY IT EXISTS						
Analyze simultaneous switching noise at the IO ring—the supply disturbance many outputs switching together inject into their own reference. SSN is a local problem with system consequences. When a wide interface switches together, the current through the shared return path moves the reference the receivers are measuring against, and the resulting noise appears as jitter and reduced eye margin on the very interface that caused it.						
WHAT IT DELIVERS						
No key deliverable is owned here — this activity contributes to the ones below.						
NEEDS FIRST						
● Package and board models from SIPI-02 ● IO library characteristics from PDK-07 ● Bump and ball assignment from PKGD-02 ● Interface switching patterns from ARCH-04 ● On-die IO power distribution from PD-03						
DONE WHEN						
● Worst-case simultaneous switching simulated per interface ● Reference shift included in the interface eye budget ● Mitigations identified before the bump map freezes						
HOW IT GETS THERE						
6 STEPS · 2 LANES						
w15	w16	w17	w18	w19	w20	
MAIN SEQUENCE						
1		2		4		
				6		
IN PARALLEL						
		3		5		
#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE					TAT
1	SSN analysis scope — which interfaces switch together → SSN analysis scope					1w
2	IO ring power and ground network modeling → IO ring power and ground models					1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Simultaneous switching simulation per interface → Simultaneous switching simulation results	1.5w runs in parallel
4	Ground bounce and reference shift analysis → Ground bounce and reference shift analysis	1.5w
5	Mitigation assessment — bump assignment, decoupling → Mitigation assessment	1w runs in parallel
6	SSN findings and interface impact → SSN findings per interface	1w

Step 5 usually points back at the bump map. The most effective SSN mitigation is more or better-placed power and ground bumps for that interface, which is a `PKGD-02` decision—and one that has to be made before the map freezes.

WATCH OUT FOR

- ◆ **SSN analyzed per pin rather than per interface.** The problem is simultaneity, and a single-pin analysis cannot see it.
- ◆ **Reference shift not included in the eye budget.** Ground bounce moves the receiver's threshold, and excluding it overstates the margin.
- ◆ **Mitigation identified after bump map freeze.** The most effective remedy is bump assignment, and it is unavailable after PKGD–02 closes.
- ◆ **Worst-case switching pattern not simulated.** All outputs switching the same direction simultaneously is the worst case and is rarely the default pattern.
- ◆ **IO ring modelled as ideal.** The ring's own power distribution is what carries the return current, and idealizing it removes the effect being analyzed.

WHERE THE EFFORT GOES	5 M/M	WHO IS ON IT ~4.3 FTE	CONNECTIONS	TERMS HERE	5
IO ring modeling	1.3	SI/PI engineer owns it	DEPENDS ON PDK-07 PD-03 PKGD-02 SIPI-02	IR PDN SSN IO SI/PI	
Switching simulation	1.0	IO library engineer	FEEDS INTO S0-05 SIPI-08 SIPI-11 EVB-03 EVB-05		
Ground bounce analysis	1.0	Package designer			
Mitigation assessment	1.0	Die power engineer			
Findings	0.8	Interface architect			

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SIPI-07

High-speed channel simulation with extracted package models — insertion loss, crosstalk, return path

TAKES	COSTS	OWNER	CRITICAL PATH
9w w8–w17 of the stage	11.0 M/M ~5.3 people while it runs	SI engineer Interface architect approves	yes slipping moves the program

WHY IT EXISTS

Simulate the **high-speed channels end to end with extracted package models**—insertion loss, crosstalk, return path, reflections—rather than against the idealized channel the PHY was designed to.

Every PHY closes its margin against an assumed channel. This activity produces the real one: interposer, substrate, board, socket, connectors and all their discontinuities. Where the real channel is worse than assumed, the margin the PHY reported never existed.

WHAT IT DELIVERS

SIPI-05

Channel compliance report per interface, with margins

This activity is the deliverable, and the margin is stated against the real channel.

NEEDS FIRST

- Package and board models from SIPI-02
- Channel budgets from AMS-03 and ARCH-04
- Package SI routing from PKGD-05
- Interface specifications and compliance masks
- PHY equalization capability from AMS-03

DONE WHEN

- Channels simulated end to end including socket and connectors
- Real channel reconciled against the PHY's assumed budget
- Findings delivered before the package design freeze

HOW IT GETS THERE

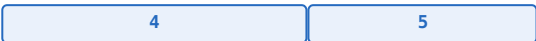
6 STEPS · 2 LANES

w8	w10	w12	w14	w16	w17
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Channel topology definition per interface → Channel topology per interface	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	End-to-end channel assembly from extracted models → End-to-end channel models	2w
3	Insertion loss and return loss analysis → Insertion and return loss results	2w
4	Crosstalk analysis — near and far end → Crosstalk analysis results	2w
5	Return path and discontinuity analysis → Return path and discontinuity findings	1.5w
6	Channel characterization against the PHY assumption → Channel characterization against PHY assumptions	3.5w

Step 6 is the reconciliation that matters. The PHY in `AMS-03` closed against a channel budget; this activity says what the channel actually is, and any gap between them is margin the interface does not have.

WATCH OUT FOR

- ◆ **Channel simulated on nominal models only.** Manufacturing variation moves impedance and loss, and the compliance margin has to survive it.
- ◆ **PHY assumption not reconciled.** The PHY closed against a budget; if the real channel is worse, the interface has no margin and nobody has noticed.
- ◆ **Crosstalk analyzed only at the near end.** Far-end crosstalk on a long dense channel can dominate, and it behaves differently.
- ◆ **Socket and connector excluded.** The bring-up platform includes them, and margin measured at bring-up will reflect them.
- ◆ **Findings after the package design freeze.** Channel improvements are routing changes, and they are unavailable after PKGD-11.

WHERE THE EFFORT GOES	11 M/M	WHO IS ON IT	~5.3 FTE	CONNECTIONS	TERMS HERE	2
Channel characterization	3.0	SI engineer	OWNS IT	DEPENDS ON ARCH-04 PDK-07 AMS-03 PKGD-03 SIPI-02	PHY SI	
Insertion and return loss	2.0	PHY designer		RUNS WITH PKGD-05		
Channel assembly	2.0	Package designer		FEEDS INTO S0-06 SIPI-08		
Crosstalk analysis	2.0	Board designer		FEEDS BACK INTO AMS-03		
Return path analysis	1.0	Interface architect				
Topology definition	1.0					

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SIPI-08

Eye diagram, jitter and BER budget closure per interface (PCIe/CXL, HBM, die-to-die)

TAKES	COSTS	OWNER	CRITICAL PATH
8w w15–w23 of the stage	9.0 M/M ~4.9 people while it runs	SI engineer SI/PI lead approves	yes slipping moves the program

WHY IT EXISTS

Close the **eye, jitter and BER budget per interface**—combining channel loss, crosstalk, SSN, PHY equalization and jitter into one number that either meets the compliance mask or does not.

This is where all the electrical work resolves into a pass or a fail. Every contribution—the channel from SIPI-07, the noise from SIPI-06, the PLL jitter from AMS-02 and the PHY’s own equalization—lands in a single budget, and the interface either has margin or it is shipping on hope.

WHAT IT DELIVERS

SIPI-D6

Eye, jitter and BER budget closure record

This activity is the deliverable—the statement that each interface will work.

NEEDS FIRST

- Channel characterization from SIPI-07
- SSN findings from SIPI-06
- PLL jitter from AMS-02
- PHY equalization and BER capability from AMS-03
- Interface compliance masks

DONE WHEN

- Every interface closed against its compliance mask with stated margin
- Jitter combined correctly rather than summed
- Budget holds across variation, not only at nominal

HOW IT GETS THERE

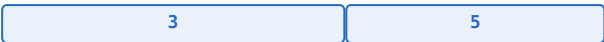
6 STEPS · 2 LANES

w15w17w19w21w23

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Budget structure and contribution allocation per interface → Budget structure and contribution allocation	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Jitter budget assembly — random, deterministic, correlated → Assembled jitter budget	2w
3	Statistical eye analysis with equalization → Statistical eye analysis results	2w <i>runs in parallel</i>
4	BER estimation and margin against the mask → BER estimation and mask margin	2w
5	Sensitivity to channel and process variation → Variation sensitivity analysis	1.5w <i>runs in parallel</i>
6	Budget closure and interface signoff → Assembled jitter budget per interface · Interface budget closure records	2.5w

Step 2 is where interfaces are usually lost. Jitter contributions from the PLL, the PHY, the channel and the SSN do not add linearly, and a budget that sums them naively either over-designs or, more often, over-promises.

WATCH OUT FOR

- ◆ **Jitter contributions summed naively.** Random and deterministic components combine differently, and getting it wrong produces a budget that is confidently wrong.
- ◆ **Equalization assumed at its maximum.** A PHY closing only with full equalization has no adaptation range left for a worse-than-nominal channel.
- ◆ **Variation sensitivity ignored.** The budget has to hold across manufacturing spread, not only at nominal.
- ◆ **SSN excluded from the budget.** It is a real contribution and it degrades the interface that generates it.
- ◆ **Interface failing the mask with no route to fix.** Late findings leave only channel changes, PHY changes or a rate reduction, and all three are expensive.

WHERE THE EFFORT GOES	9 M/M	WHO IS ON IT	~4.9 FTE	CONNECTIONS	TERMS HERE	6
Budget closure and signoff	2.0	SI engineer OWNS IT		DEPENDS ON AMS-02 AMS-03 PKG0-05 SIPI-06 SIPI-07	PLL PHY SSN SI SI/PI BER	
Jitter budget assembly	2.0	PHY designer				
BER estimation	1.8	Interface architect		FEEDS INTO S0-06 SIPI-11 BU-05 MP-11		
Statistical eye analysis	1.8	Clocking architect				
Variation sensitivity	1.0	SI/PI lead				
Budget structure	0.5					

18 SIPI · CHIP-PACKAGE-SYSTEM CO-VERIFICATION

SIPI-09

Power-aware timing correlation — voltage drop back-annotated into STA

TAKES	COSTS	OWNER	CRITICAL PATH
5w w18–w23 of the stage	5.0 M/M ~4.3 people while it runs	Power integrity engineer SI/PI lead approves	no has float

WHY IT EXISTS

Back-annotate the **voltage droop into static timing analysis**—because a design signed off at nominal voltage was signed off at a voltage the die does not see.

Timing and power are analyzed by different teams with different tools, and the connection between them is usually a margin someone assumed. This activity replaces the assumption: the droop map from SIPI-04 applied to the timing analysis, so that closure reflects the supply the logic actually has.

WHAT IT DELIVERS

SIPI-D7

Power-aware STA correlation report

This activity is the deliverable, and it connects two signoffs that otherwise run independently.

NEEDS FIRST

- Dynamic droop results from SIPI-04
- Timing analysis from PD-09 and SO-02
- Voltage-aware timing tool capability
- Region and instance mapping from PD-02
- Operating point definitions from ARCH-05

DONE WHEN

- Droop applied regionally rather than as a flat margin
- Reconciled with the SO-02 signoff assumption
- Recommendations targeted at the regions that need them

HOW IT GETS THERE

6 STEPS · 2 LANES

w18	w19	w20	w21	w22	w23
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Droop map generation per region and mode → Droop map per region and mode	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Voltage-aware timing analysis setup → Voltage-aware timing setup	1.5w
3	Timing re-analysis with back-annotated droop → Re-analyzed timing with droop	1.5w runs in parallel
4	Margin impact assessment per timing path group → Margin impact per path group	1.5w
5	Reconciliation with the SO-02 signoff margin → Reconciliation with signoff margin	1.5w runs in parallel
6	Power-aware timing findings and recommendations → Power-aware timing recommendations	1w

Step 5 is the reconciliation. If timing signoff already carries a flat voltage margin and this analysis shows the droop is worse in some regions and better in others, the flat margin was both too generous and insufficient—and the recommendation is regional rather than global.

WATCH OUT FOR

- ◆ **Flat voltage margin used instead.** Droop is regional, and a flat margin is simultaneously wasteful where droop is low and inadequate where it is high.
- ◆ **Analysis run after timing closure.** Findings then require reopening closure, which is expensive at this point in the schedule.
- ◆ **Droop map too coarse.** Region granularity determines whether the analysis can distinguish a hot spot from an average.
- ◆ **Reconciliation skipped.** Two signoffs with different voltage assumptions produce a design that satisfies neither cleanly.
- ◆ **Recommendations global rather than regional.** The remedy is usually local—more decoupling or a PDN change in one region—and a global response over-corrects.

WHERE THE EFFORT GOES	5 M/M	WHO IS ON IT ~4.3 FTE	CONNECTIONS	TERMS HERE 4
Timing re-analysis	1.3	Power integrity engineer owns it	DEPENDS ON PD-09 SIPI-01 SIPI-04	STA IR PDN SI/PI
Margin impact assessment	1.0	STA engineer	RUNS WITH S0-02	
Droop map generation	1.0	Signoff methodology engineer	FEEDS INTO S0-06 SIPI-11	
Reconciliation	1.0	Physical design liaison		
Findings	0.8	SI/PI lead		

18 SIPI · CHIP-PACKAGE-SYSTEM CO-VERIFICATION

SIPI-10

Electro-thermal co-analysis coupled with the package thermal model

TAKES

5w

w18–w23 of the stage

COSTS

5.0 M/M

~4.3 people while it runs

OWNER

Thermal engineer

SI/PI lead approves

CRITICAL PATH

no

has float

WHY IT EXISTS

Couple the **electrical and thermal analyses**—because leakage rises with temperature, temperature rises with power, and the two feed each other until they settle somewhere the isolated analyses did not predict.

Electro-thermal coupling matters at this power density. A hotspot raises local leakage, which raises local power, which raises the hotspot. Analyzing power at a fixed temperature and temperature at a fixed power finds an equilibrium that is optimistic in exactly the region that matters.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Package thermal model from PKGD-07
- Thermal characterization from PTV-09
- Chip power model from SIPI-01
- Leakage characterization from PDK-03
- Power map from PD-03

DONE WHEN

- Coupled analysis converged, or non-convergence investigated
- Compared against the isolated analyses to quantify the difference
- Gradients assessed alongside peak temperature

HOW IT GETS THERE

6 STEPS · 2 LANES

w18w19w20w21w22w23

1

2

4

6

3

5

#STEP, AND WHAT IT ADDS TO THE DELIVERABLETAT

1Coupling methodology and iteration scheme
→ Coupling methodology and iteration scheme1w

2Thermal model integration with the package model
→ Integrated electro-thermal model1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Leakage-temperature feedback modeling → Leakage-temperature feedback model	1w runs in parallel
4	Coupled electro-thermal iteration to convergence → Coupled analysis results	1.5w
5	Hotspot and gradient assessment → Hotspot and gradient assessment	1w runs in parallel
6	Findings against the isolated analyses → Comparison against isolated analyses	1w

Step 4 iterates to convergence, and whether it converges is itself information. A design where the leakage-temperature loop does not settle is a design with a thermal runaway risk, which is a different and more serious finding than a high temperature.

WATCH OUT FOR

- ◆ **Analyses run at fixed temperature and fixed power.** The equilibrium the coupled analysis finds is worse, and it is the one that occurs.
- ◆ **Leakage-temperature model too coarse.** The feedback is exponential, and a linear approximation understates the hotspot.
- ◆ **Non-convergence dismissed as a numerical issue.** It can indicate a real thermal runaway risk and deserves investigation rather than a damping factor.
- ◆ **Package thermal model unvalidated.** If PTV-09 has not correlated it, the coupled analysis inherits an unknown error.
- ◆ **Gradient ignored in favor of peak temperature.** Gradients drive mechanical stress and timing variation across the die, and the peak alone does not capture them.

WHERE THE EFFORT GOES	5 M/M	WHO IS ON IT ~4.3 FTE	CONNECTIONS	TERMS HERE 4
Coupled iteration	1.3	Thermal engineer owns IT	DEPENDS ON PD-03 PKGD-07 SIPI-01	STA IR PDN SI/PI
Thermal model integration	1.0	Power integrity engineer	FEEDS INTO S0-06 SIPI-11 MP-02	
Leakage feedback modeling	1.0	Device engineer	LATER INPUT PTV-09 +1w	
Hotspot assessment	1.0	Package thermal liaison		
Findings and comparison	0.8	SI/PI lead		

18 SIPI · CHIP-PACKAGE-SYSTEM CO-VERIFICATION

SIPI-11

Co-verification signoff review and criteria disposition with design and package teams

TAKES	COSTS	OWNER	CRITICAL PATH
4w w22–w26 of the stage	3.0 M/M ~3.3 people while it runs	SI/PI lead Signoff lead approves	yes slipping moves the program

WHY IT EXISTS

Hold the **co-verification signoff**—every result reviewed against named exit criteria, dispositioned, and signed—because mask release depends on it.

This is the gate the stage exists to reach. Without a signoff with criteria, co-verification is a collection of analyses that each conclude something and together conclude nothing. The disposition here is what 50–06 reviews and what makes the MTO gate at week 78 meaningful.

WHAT IT DELIVERS

SIPI-D8

Chip-package-system co-verification signoff — tapeout gate

This activity is the deliverable, and it carries the stage's milestone.

NEEDS FIRST

- Results from SIPI-03 through SIPI-10
- Exit criteria defined at stage start
- Package design freeze schedule from PKGD-11
- Signoff schedule from SO-06
- Program risk register

DONE WHEN

- Every analysis assessed against a named exit criterion
- Open items dispositioned by their owners, not by the analysis team
- Signoff complete before the package design freeze and the MTO gate

HOW IT GETS THERE

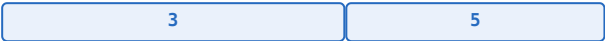
6 STEPS · 2 LANES

w22	w23	w24	w25	w26
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Exit criteria confirmation across all analyses → Confirmed exit criteria	0.75w
2	Result consolidation and margin summary → Consolidated results and margin summary	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Open item disposition with die, package and board owners → Open item dispositions with owners	1w runs in parallel
4	Signoff review with design and package teams → Signoff review record with design and package	1.25w
5	Residual risk statement → Residual risk statement	0.75w runs in parallel
6	Co-verification signoff declaration → Signoff review record · Co-verification signoff declaration	1w

Step 3 is a three-party disposition. Every open item belongs to the die, the package or the board, and closing it requires the owner in the room—which is why co-verification signoff is a joint review rather than a report circulated for comment.

WATCH OUT FOR

- ◆ **Signoff without exit criteria.** The analyses each conclude something and together conclude nothing that can be signed.
- ◆ **Open items dispositioned without their owners.** An item closed by the analysis team rather than by the die, package or board owner has not actually been closed.
- ◆ **Signoff after the package design freeze.** Findings then have nowhere to go, and the freeze was taken without them.
- ◆ **Residual risk not stated.** Some margin will be thin; saying which and how thin is what lets the tapeout decision be informed.
- ◆ **Declared to meet the date.** The gate exists to be able to hold mask release, and one that cannot is not a gate.

WHERE THE EFFORT GOES	3 M/M	WHO IS ON IT	~3.3 FTE	CONNECTIONS	TERMS HERE	3
		SI/PI lead OWNS IT		DEPENDS ON SIPI-03 SIPI-04 SIPI-05 SIPI-06 SIPI-08 SIPI-09 SIPI-10	MTO SI/PI BER	
Signoff review	1.0					
Result consolidation	0.8	Die power and timing owners				
Open item disposition	0.5					
Declaration	0.5	Package architect		FEEDS INTO S0-06 T0-04 PKG0-11		
Residual risk statement	0.3	Board designer		BU-05		
		Signoff lead				

INTEGRATE

19 Assembly & Package Build

ASSY

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH		
w98 – w106	8 _w	9	36 _{M/M}	5 _{of 9}		
ENGINEERING ACTIVITIES				KEY DELIVERABLES		
ASSY-01	Known-good-die sort and selection		3w	5 _{M/M}	ASSY-D1	Assembled units — bring-up, qualification and sample lots
ASSY-02	HBM stack procurement and incoming inspection		3w	2 _{M/M}	ASSY-D2	Assembly travelers and process data
ASSY-03	Die attach, micro-bump and thermo-compression bonding		3w	6 _{M/M}	ASSY-D3	Assembly yield report and failure pareto
ASSY-04	Interposer and substrate assembly, underfill, molding		3w	5 _{M/M}	ASSY-D4	Package-level inspection and test data
ASSY-05	Lid / TIM attach and ball attach		2w	3 _{M/M}	ASSY-D5	Unit allocation record across bring-up, qual and customers
ASSY-06	Assembly inline inspection — X-ray, CSAM, warpage metrology		2w	3 _{M/M}		
ASSY-07	Assembly yield analysis and process tuning		5w	6 _{M/M}		
ASSY-08	Package-level open / short and continuity test		2w	3 _{M/M}		
ASSY-09	Unit build for bring-up, qualification and customer samples		3w	3 _{M/M}		

19 ASSY · ASSEMBLY & PACKAGE BUILD

ASSY-01

Known-good-die sort and selection

TAKES	COSTS	OWNER	CRITICAL PATH
3w w0–w3 of the stage	5.0 M/M ~7.2 people while it runs	Product engineering Quality engineer approves	yes slipping moves the program

WHY IT EXISTS

Sort the wafers and pick the **known-good die** that will go into packages—because on a 2.5D part a bad die does not fail alone, it takes an interposer, two HBM stacks and an assembly with it.

Known-good-die screening is what makes multi-die packaging economically viable. The coverage achieved at sort determines how many assembled units fail for reasons that were detectable on the wafer, and each of those is a package's worth of material lost.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Accepted wafers from FAB-08 and FAB-10
- Wafer sort program from TEST-05
- Probe card from TEST-03
- KGD criteria from PKGD-10
- Assembly quantity requirement from ASSY-09

DONE WHEN

- Coverage sufficient for the assembled unit's economics
- Marginal criteria set against package cost, not die cost
- Every released die traceable to wafer and position

HOW IT GETS THERE

5 STEPS · 2 LANES

w0	w1	w2	w3
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Wafer receipt and sort setup on the probe station → Wafers received with sort setup complete	0.5w
2	Wafer sort execution against the KGD criteria → Sort execution results	1w
3	Marginal die disposition → Marginal die disposition record	0.75w runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Known-good-die selection and wafer mapping → Known-good-die map per wafer	1w
5	Die release to assembly with traceability → Die released to assembly with traceability	0.5w

Step 3 is where the economics live. A marginal die is cheap to scrap and expensive to assemble; the criteria for rejecting one have to be set against the assembled unit's cost rather than against the die's.

WATCH OUT FOR

- ◆ **KGD coverage below what the package economics need.** Every escaped die costs an assembled unit, and the coverage target follows from that cost rather than from a test convention.
- ◆ **Marginal criteria set against die cost.** On a 2.5D part the assembled unit is worth many times the die, and the reject threshold should reflect that.
- ◆ **Traceability lost between wafer and package.** An assembly failure that cannot be traced back to a die and its wafer position cannot be attributed.
- ◆ **Sort program not ready.** The probe card and program come from TEST, and a delay there stops assembly rather than delaying test.
- ◆ **Sorting only for function.** Parametric marginality—leakage, Vmin—predicts field failure and is worth screening even when the die functions.

WHERE THE EFFORT GOES	5 M/M	WHO IS ON IT ~7.2 FTE	CONNECTIONS	TERMS HERE 5
		Product engineering OWNS IT	DEPENDS ON TECH-09 FAB-08 FAB-09 FAB-10 PKG-10 PTV-12 TEST-03 TEST-05	OSAT KGD Vmin HBM (ESD) HBM
Sort execution	2.0	Test engineering		
KGD selection and mapping	1.3	Yield engineer		
Sort setup	0.8	Assembly liaison		
Marginal disposition	0.5	Quality engineer	FEEDS INTO ASSY-03 ASSY-07 ASSY-09	
Release and traceability	0.5			

19 ASSY · ASSEMBLY & PACKAGE BUILD

ASSY-02

HBM stack procurement and incoming inspection

TAKES	COSTS	OWNER	CRITICAL PATH
3w w0–w3 of the stage	2.0 M/M ~2.9 people while it runs	Procurement Supplier liaison approves	no has float

WHY IT EXISTS

Procure and inspect the **HBM stacks**—bought components that go into the package alongside the program's own die, and that fail in ways the program does not control.

HBM is a purchased part with its own supply constraints and its own quality distribution. Incoming inspection matters because an HBM stack that fails after assembly costs the whole unit, and the supplier's outgoing quality is not the program's to verify after the fact.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- HBM purchase orders from MP-10 and procurement
- Supplier test data and certificates
- HBM specification and requirements from ARCH-03
- Incoming inspection capability
- Assembly quantity requirement

DONE WHEN

- Every lot inspected and its supplier data reviewed
- Stacks confirmed against the package specification
- Lot traceability maintained into assembly

HOW IT GETS THERE

5 STEPS · 2 LANES

w0	w1	w2	w3
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	HBM procurement confirmation and delivery → Delivered HBM stacks	0.5w
2	Incoming inspection against specification → Incoming inspection results	1w
3	Supplier test data review → Supplier test data review	0.75w runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Stack qualification and release to assembly → Qualified stacks released to assembly	1w
5	Inventory and lot traceability → Inventory and lot traceability record	0.5w

Step 3 costs almost nothing and catches a class of problem inspection cannot. Supplier test data shows the distribution the delivered lot came from, and a lot at the edge of specification is worth knowing about before it is bonded in.

WATCH OUT FOR

- ◆ **HBM supply constrained.** It is a scarce component with long lead times, and a shortage stops assembly regardless of how many die are ready.
- ◆ **Incoming inspection skipped.** A marginal stack bonded into a unit costs the whole unit, and the fault is only visible afterwards.
- ◆ **Supplier data not reviewed.** The distribution the lot came from predicts its behavior, and the data is provided at no cost.
- ◆ **Lot traceability not maintained.** An HBM-related failure that cannot be traced to a supplier lot cannot be escalated usefully.
- ◆ **Specification mismatch discovered at assembly.** Stack height, bump pitch and thermal characteristics all have to match the package design.

WHERE THE EFFORT GOES	2 M/M	WHO IS ON IT	~2.9 FTE	CONNECTIONS	TERMS HERE	2
Incoming inspection	0.8	Procurement	OWNS IT	DEPENDS ON ARCH-03 TECH-09 PKGD-10	HBM (ESD) HBM	
Stack qualification	0.5	Quality engineer		FEEDS INTO ASSY-03 ASSY-07 MP-10		
Procurement coordination	0.4	Package engineer		MP-11		
Traceability	0.3	Operations planner				
		Supplier liaison				

19 ASSY · ASSEMBLY & PACKAGE BUILD

ASSY-03

Die attach, micro-bump and thermo-compression bonding

TAKES	COSTS	OWNER	CRITICAL PATH
3w w2–w5 of the stage	6.0 M/M ~8.7 people while it runs	OSAT process engineer Yield engineer approves	yes slipping moves the program

WHY IT EXISTS

Bond the die and HBM stacks to the interposer—**micro-bump preparation and thermo-compression bonding**—inside the process window the vehicles established.

This is the step the whole PTV program existed to de-risk. Bond force, temperature and time were characterized on vehicles; here they are applied to real die, and the process window is either followed or the vehicle work was wasted.

WHAT IT DELIVERS

ASSY-D2

Assembly travelers and process data

This activity is where the traveler starts and where the actual bond conditions are recorded.

NEEDS FIRST

- Known-good die from ASSY-01
- Qualified HBM stacks from ASSY-02
- Interposers from PKGD-09 supply
- Frozen process window from PTV-12
- Assembly process flow from PKGD-10

DONE WHEN

- Every unit bonded inside the frozen window
- Actual bond parameters recorded per unit
- Post-bond inspection coverage sufficient to catch localized defects

HOW IT GETS THERE

5 STEPS · 2 LANES

w2	w3	w4	w5
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Process setup to the frozen window → Process setup to the frozen window	0.5w
2	Micro-bump preparation and flux application → Prepared micro-bumps	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Bond parameter monitoring during the run → Bond parameter monitoring data runs in parallel	0.75w
4	Thermo-compression bonding of die and stacks → Bonded die and stack assemblies	1w
5	Post-bond inspection → Post-bond inspection results · Assembly travelers with actual conditions	0.5w

Step 3 records what actually happened. The frozen window says what the parameters should be; monitoring says what they were, and a unit that failed can only be attributed if its actual bond conditions were captured.

WATCH OUT FOR

- Running outside the frozen window. The PTV program established boundaries; operating outside them discards the evidence it produced.
- Bond parameters not recorded per unit. A failed unit cannot be attributed to its conditions, and the yield analysis loses its most useful variable.
- Bump preparation inconsistent. Flux and surface condition affect joint formation, and variability here shows up as scattered yield.
- Post-bond inspection sampled too thinly. Bonding defects are localized, and a sample that misses them lets them through to underfill where they are unfixable.
- Process drift across the run. Long runs drift, and without monitoring the drift is only visible in the yield trend afterwards.

WHERE THE EFFORT GOES	6 M/M	WHO IS ON IT ~8.7 FTE	CONNECTIONS	TERMS HERE	4
Thermo-compression bonding	2.5	OSAT process engineer owns it	DEPENDS ON PKGD-10 PTV-12 ASSY-01 ASSY-02	OSAT PTV HBM (ESD) HBM	
Bump preparation	1.5	Assembly process engineer	FEEDS INTO ASSY-04 ASSY-06 ASSY-07		
Process setup	1.0	Quality engineer			
Post-bond inspection	0.5	Package engineer			
Parameter monitoring	0.5	Yield engineer			

19 ASSY · ASSEMBLY & PACKAGE BUILD

ASSY-04

Interposer and substrate assembly, underfill, molding

TAKES	COSTS	OWNER	CRITICAL PATH
3w w3–w6 of the stage	5.0 M/M ~7.2 people while it runs	OSAT process engineer Reliability liaison approves	yes slipping moves the program

WHY IT EXISTS

Attach the interposer to the substrate and **underfill and mold** the assembly—the steps that make it mechanically robust and that lock in whatever went before.

Underfill is irreversible. Once it has flowed and cured, a bonding defect underneath is permanent and invisible to everything except acoustic imaging. The step's quality—void-free flow, complete fillet, correct cure—determines whether the mechanical stress the package will see is distributed or concentrated.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Bonded assemblies from ASSY-03
- Substrates from PKGD-09 supply
- Underfill and mold compound specifications
- Frozen process window from PTV-12
- Warpage data from PTV-07

DONE WHEN

- Void and fillet inspected before cure, not after
- Cure profile within the frozen window
- Warpage during the process inside the validated range

HOW IT GETS THERE

5 STEPS · 2 LANES

w3	w4	w5	w6
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Interposer to substrate attach → Interposer-to-substrate attached assemblies	0.5w
2	Underfill dispense and flow → Dispensed and flowed underfill	1w
3	Void and fillet inspection → Void and fillet inspection results	0.75w runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Underfill cure and molding → Underfilled assemblies	1w
5	Post-mold inspection → Cured and molded units · Post-mold inspection results	0.5w

Step 3 is the last chance to catch a bonding problem. Acoustic imaging before cure can still identify a void or an incomplete fill; after cure the unit is what it is, and the defect surfaces during thermal cycling months later.

WATCH OUT FOR

- ◆ **Underfill voids.** They concentrate stress and are the mechanism behind delayed CPI failures, and they are invisible after cure without acoustic imaging.
- ◆ **Incomplete fillet.** The fillet distributes stress at the die edge, and an incomplete one puts that stress into the die corner instead.
- ◆ **Cure profile deviation.** Under-cured underfill has different mechanical properties, and the difference appears as reliability rather than as a visible defect.
- ◆ **Warpage during cure exceeding the window.** The vehicle program measured it; running outside that window reintroduces the risk that was retired.
- ◆ **Inspection only after molding.** Once molded the assembly cannot be examined usefully, and the inspection has to happen before.

WHERE THE EFFORT GOES	5 M/M	WHO IS ON IT ~7.2 FTE	CONNECTIONS	TERMS HERE	2
Underfill dispense and cure	2.0	OSAT process engineer OWNS IT	DEPENDS ON PKG0-09 PKG0-11 PTV-12 ASSY-03	OSAT CPI	
Interposer attach	1.3	Assembly process engineer			
Molding	1.0	Quality engineer	FEEDS INTO ASSY-05 ASSY-06 ASSY-07		
Inspection	0.8	Materials engineer Reliability liaison			

19 ASSY · ASSEMBLY & PACKAGE BUILD

ASSY-05

Lid / TIM attach and ball attach

TAKES	COSTS	OWNER	CRITICAL PATH
2w w5–w7 of the stage	3.0 M/M ~6.5 people while it runs	OSAT process engineer Product engineering approves	yes slipping moves the program

WHY IT EXISTS

Finish the package—**TIM, lid and ball attach**—and produce units that can be mounted, cooled and tested.

These are the last steps and they determine two things the product is judged on: the thermal interface that decides the achievable power envelope, and the ball array that decides whether the package mounts reliably. Both were characterized on vehicles and both are executed here.

WHAT IT DELIVERS

ASSY-D1

Assembled units — bring-up, qualification and sample lots

This activity is where a unit becomes a finished package.

NEEDS FIRST

- Molded assemblies from ASSY-04
- TIM material and lid components
- Ball attach process from the frozen window
- TIM characterization from PTV-09
- Dimensional specification from PKGD-11

DONE WHEN

- TIM bond line within the characterized range
- Lid flatness and ball co-planarity inside specification
- Every unit inspected, not a sample

HOW IT GETS THERE

4 STEPS · 2 LANES

w5

w6

w7

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	TIM application and lid attach → Lidded units with applied TIM	0.5w
2	Ball attach and reflow → Ball-attached packages	0.75w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Lid flatness and co-planarity check → Lid flatness and co-planarity results	0.5w runs in parallel
4	Final visual and dimensional inspection → Final inspection results · Completed assembled units	0.75w

Step 1 sets the thermal performance the product will actually have. Bond line thickness under the lid is a process parameter, and the difference between the characterized value and the achieved one is watts of headroom.

WATCH OUT FOR

- ◆ **TIM bond line thicker than characterized.** Thermal resistance rises, and the power envelope the product was sold on shrinks.
- ◆ **Lid flatness out of specification.** The cooling solution then contacts unevenly, which produces hotspots the thermal design did not predict.
- ◆ **Ball co-planarity outside the mounting window.** The package does not mount reliably, and the customer discovers it during their assembly.
- ◆ **Dimensional drift from the specification.** The package has to fit sockets, heatsinks and board footprints that were designed to the drawing.
- ◆ **Final inspection sampled rather than complete.** On an engineering build every unit is precious, and a defect shipped to bring-up costs lab days.

WHERE THE EFFORT GOES	3 M/M	WHO IS ON IT ~6.5 FTE	CONNECTIONS
Lid and TIM attach	1.3	OSAT process engineer owns it	DEPENDS ON PKGD-11 PTV-09 ASSY-04
Ball attach	1.0	Thermal engineer	FEEDS INTO ASSY-08 ASSY-09
Inspection	0.8	Quality engineer	TERMS HERE 2
		Package engineer	OSAT TIM
		Product engineering	

19 ASSY · ASSEMBLY & PACKAGE BUILD

ASSY-06

Assembly inline inspection — X-ray, CSAM, warpage metrology

TAKES	COSTS	OWNER	CRITICAL PATH
2w w3–w5 of the stage	3.0 M/M ~6.5 people while it runs	Quality engineer Package engineer approves	no has float

WHY IT EXISTS

Inspect the assemblies **without taking them apart**—X-ray for joints, acoustic imaging for delamination and voids, metrology for warpage—because most assembly defects are invisible from outside.

A 2.5D package hides everything that matters. The joints are under the die, the underfill is under the interposer, and the only way to see any of it non-destructively is through imaging. What this activity catches is what does not reach bring-up as an unexplained failure.

WHAT IT DELIVERS

ASSY-D4

Package-level inspection and test data

This activity is the deliverable—the non-destructive record of what is inside each unit.

NEEDS FIRST

- Assemblies at each process stage from ASSY-03 to ASSY-05
- Inspection criteria from PTV vehicle results
- X-ray and CSAM capability at the OSAT
- Warpage limits from PTV-07
- Defect classification scheme

DONE WHEN

- Inspection at each process step, not only at the end
- Defects classified against a consistent scheme
- Findings fed back to the process that created them

HOW IT GETS THERE

4 STEPS · 2 LANES

w3	w4	w5
----	----	----

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Inspection plan and sampling strategy → Inspection plan and sampling strategy	0.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	X-ray inspection of joints and voids → X-ray inspection results	0.75w
3	Defect classification and disposition → Defect classification and disposition	0.5w runs in parallel
4	Acoustic imaging and warpage metrology → Acoustic imaging and warpage results	0.75w

Step 1's sampling strategy is a real decision on a small build. Full inspection is affordable at engineering-lot quantities and not at production volume, and starting with full coverage is what tells you what sampling would have missed.

WATCH OUT FOR

- ◆ **Sampling too thin on an engineering build.** Full inspection is affordable at these quantities, and a defect reaching bring-up costs far more than the inspection would have.
- ◆ **Defects classified inconsistently.** Without a scheme the same feature is a defect on Monday and acceptable on Tuesday, and the yield data becomes noise.
- ◆ **Inspection only at the end.** Defects are cheapest to catch at the step that created them, and end-of-line inspection cannot attribute them.
- ◆ **Warpage limits not applied.** The vehicle program set them; a unit outside them is a reliability risk regardless of whether it functions.
- ◆ **Findings not fed back to process.** Inspection that produces a pass/fail and no process signal wastes most of its value.

WHERE THE EFFORT GOES	3 M/M	WHO IS ON IT ~6.5 FTE	CONNECTIONS
X-ray and acoustic imaging	1.3	Quality engineer OWNS IT	DEPENDS ON PTV-07 ASSY-03 ASSY-04
Warpage metrology	0.8	OSAT metrology	FEEDS INTO ASSY-07 ASSY-08 MP-04
Defect classification	0.5	Assembly process engineer	TERMS HERE 3
Inspection plan	0.5	Failure analysis engineer	OSAT PTV CSAM
		Package engineer	

ASSY-07

Assembly yield analysis and process tuning

TAKES	COSTS	OWNER	CRITICAL PATH
5w w3–w8 of the stage	6.0 M/M ~5.2 people while it runs	Yield engineer Product cost analyst approves	no has float

WHY IT EXISTS

Analyze the **assembly yield**, build the failure pareto, find the root causes with the OSAT and tune the process inside the window the vehicles defined.

First-build assembly yield on a 2.5D part is rarely what the cost model assumed. This activity turns the gap into a set of attributed causes and process adjustments—and compares the result against what the vehicle program predicted, which is how the vehicle work gets validated in turn.

WHAT IT DELIVERS

ASSY-D3

Assembly yield report and failure pareto

This activity is the deliverable, and the pareto is what the OSAT acts on.

NEEDS FIRST

- Yield data from every assembly step
- Inspection results from ASSY-06
- Continuity results from ASSY-08
- PTV yield prediction from PTV-11
- Frozen process window from PTV-12

DONE WHEN

- Failures attributed to process steps, not only counted
- Tuning kept inside the validated window
- Yield compared against the vehicle prediction

HOW IT GETS THERE

6 STEPS · 2 LANES

w3	w4	w5	w6	w7	w8
----	----	----	----	----	----

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Yield data collection across process steps → Yield data by process step	0.75w
2	Failure pareto by mode and process step → Failure pareto by mode	1.25w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Comparison against the PTV yield prediction → Comparison against vehicle prediction	1w runs in parallel
4	Root cause analysis with the OSAT → Root cause findings	1.5w
5	Corrective action tracking → Corrective actions and their tracking	1w runs in parallel
6	Process tuning inside the frozen window → Process tuning record	1.5w

Step 6 stays inside the window deliberately. Tuning beyond the validated boundaries would abandon the evidence the vehicle program produced, and a yield improvement bought that way carries an unquantified reliability risk.

WATCH OUT FOR

- ◆ **Yield reported without attribution.** A number tells the program it has a problem; a pareto by process step tells the OSAT what to change.
- ◆ **Tuning outside the validated window.** It may improve yield and it discards the reliability evidence the vehicles produced.
- ◆ **Vehicle prediction not compared.** The comparison is what validates or invalidates the whole PTV program, and it is cheap to make.
- ◆ **Root cause pursued without the OSAT.** Process knowledge sits with the assembler, and analysis without them produces hypotheses rather than causes.
- ◆ **Corrective actions untracked.** An agreed action that nobody follows up is an action that did not happen.

WHERE THE EFFORT GOES	6 M/M	WHO IS ON IT	~5.2 FTE	CONNECTIONS	TERMS HERE	2
Root cause analysis	2.0	Yield engineer owns it		DEPENDS ON PTV-11 PTV-12 ASSY-01 ASSY-02 ASSY-03 ASSY-04 ASSY-06 ASSY-08	OSAT PTV	
Failure pareto	1.5	OSAT process engineer		FEEDS INTO MP-05 MP-10		
Process tuning	1.3	Quality engineer		FEEDS BACK INTO PTV-12		
Data collection	0.8	Package engineer				
Vehicle comparison	0.5	Product cost analyst				

19 ASSY · ASSEMBLY & PACKAGE BUILD

ASSY-08

Package-level open / short and continuity test

TAKES	COSTS	OWNER	CRITICAL PATH
2w w6–w8 of the stage	3.0 M/M ~6.5 people while it runs	Package test engineer Product engineering approves	no has float

WHY IT EXISTS

Test the assembled packages **electrically for opens, shorts and marginal joints** before they go anywhere—the cheapest possible screen on the most expensive units the program has.

A package-level continuity test costs minutes and catches assembly failures that would otherwise consume lab days at bring-up. On units this expensive, screening before shipping is not optional economics.

WHAT IT DELIVERS

ASSY-D4

Package-level inspection and test data

The electrical half of the deliverable, alongside the imaging from ASSY-06.

NEEDS FIRST

- Completed units from ASSY-05
- Continuity criteria from PTV-11
- Test hardware and fixtures
- Package pin map from PKGD-04
- Failure disposition process

DONE WHEN

- Every unit screened before release
- Marginal joints identified by resistance, not only continuity
- Failures fed into the yield pareto

HOW IT GETS THERE

4 STEPS · 2 LANES

w6

w7

w8

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Test setup and continuity criteria → Test setup and criteria	0.5w
2	Open and short test execution → Open and short results per unit	0.75w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Marginal resistance screening → Marginal resistance screening results	0.5w runs in parallel
4	Failure disposition and unit release → Failure disposition and released units	0.75w

Step 3 separates working from reliably working. A joint that conducts with elevated resistance passes a continuity test and fails in the field, and screening for it here is far cheaper than a field return.

WATCH OUT FOR

- ◆ **Screening skipped because units are scarce.** A failed unit consumed at bring-up costs more lab time than the whole screen would have.
- ◆ **Continuity only, no resistance.** Marginal joints conduct and fail later, and only a resistance measurement distinguishes them.
- ◆ **Criteria not derived from vehicle data.** PTV-11 measured what good and marginal look like, and the criteria should come from there.
- ◆ **Failures not fed to yield analysis.** An electrical failure is yield data and belongs in the pareto.
- ◆ **Test fixture damaging units.** On packages this expensive the fixture itself has to be qualified.

WHERE THE EFFORT GOES	3 M/M	WHO IS ON IT ~6.5 FTE	CONNECTIONS	TERMS HERE	1
Test execution	1.3	Package test engineer OWNS IT	DEPENDS ON PKG0-04 PTV-11 ASSY-05	OSAT	
Failure disposition	0.8	Quality engineer	ASSY-06		
Test setup	0.5	Yield engineer	FEEDS INTO ASSY-07 ASSY-09		
Marginal screening	0.5	OSAT test	FEEDS BACK INTO BU-01		
		Product engineering			

19 ASSY · ASSEMBLY & PACKAGE BUILD

ASSY-09

Unit build for bring-up, qualification and customer samples

TAKES	COSTS	OWNER	CRITICAL PATH
3w w5–w8 of the stage	3.0 M/M ~4.3 people while it runs	Program manager Operations planner approves	yes slipping moves the program

WHY IT EXISTS

Allocate the finished units—**bring-up, qualification, customer samples**—and build to that allocation, because the first build never has enough for everyone.

Three groups need units simultaneously and all of them are on the critical path of something. Bring-up needs them to start debugging, qualification needs them to start a months-long stress campaign, and customers were promised samples. The allocation is a program decision, not an assembly one.

WHAT IT DELIVERS

- ASSY-D5

Unit allocation record across bring-up, qual and customers

This activity is the deliverable, and it settles a contention that would otherwise be settled by whoever asks loudest.
- ASSY-D1

Assembled units — bring-up, qualification and sample lots

The units as allocated lots rather than as an undifferentiated build.

NEEDS FIRST

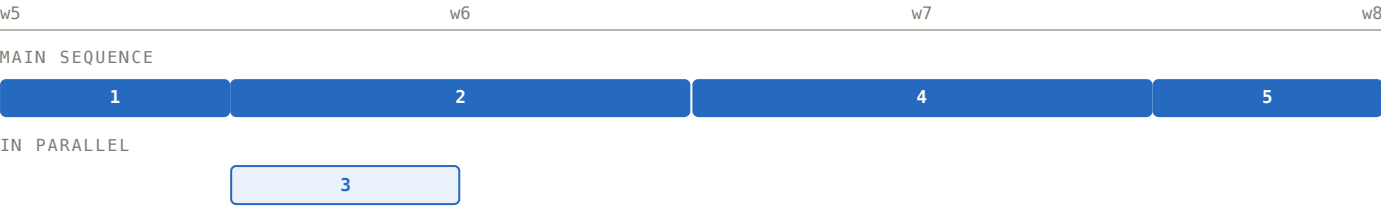
- Screened units from ASSY-08
- Bring-up requirement from BU-01
- Qualification vehicle requirement from MP-01
- Customer sample commitments from DEF-01
- Build quantity from the assembly run

DONE WHEN

- Allocation decided at program level, not by request order
- Qualification quantity sufficient to start its campaign
- Every allocated unit traceable to its lot and conditions

HOW IT GETS THERE

5 STEPS · 2 LANES



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Allocation requirement collection from each consumer → Allocation requirements per consumer	0.5w
2	Unit allocation and reservation → Unit allocation and reservation record	1w
3	Contention resolution and escalation runs in parallel → Contention resolution decisions	0.5w
4	Build to allocation across lots → Units built to allocation	1w
5	Release and traceability per allocation → Release and traceability per allocation	0.5w

Step 3 is a program decision escalated deliberately. Qualification starting late delays mass production; bring-up starting late delays everything; and customer samples are often contractual. Resolving that trade below the program level puts it in the wrong hands.

WATCH OUT FOR

- ◆ **Allocation decided by whoever asks first.** The three consumers have different downstream consequences, and the trade belongs to the program.
- ◆ **Qualification starved.** Reliability stress runs for months; starting it late moves mass production directly.
- ◆ **Customer samples cut.** They are frequently contractual, and missing them has commercial consequences the program may not see.
- ◆ **Build quantity below the sum of needs.** Which is normal, and it means the shortfall has to be decided rather than discovered.
- ◆ **Traceability lost per allocation.** A qualification failure has to be traceable to its lot and its assembly conditions.

WHERE THE EFFORT GOES	3 M/M	WHO IS ON IT	~4.3 FTE	CONNECTIONS
Allocation and reservation	1.0	Program manager OWNS IT		DEPENDS ON ASSY-01 ASSY-05 ASSY-08
Build to allocation	0.8	Validation lead		FEEDS INTO BU-01 BU-12 MP-01
Requirement collection	0.5	Quality lead		MP-02 MP-03 MP-04
Contention resolution	0.5	Product marketing		FEEDS BACK INTO
Release and traceability	0.3	Operations planner		TEST-11

VALIDATE & RAMP

20

Validation Hardware / EVB

EVB

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w52 – w90	38 _w	10	92 _{M/M}	5 of 10
ENGINEERING ACTIVITIES				KEY DELIVERABLES
EVB-01	Validation platform requirements and topology definition		5w 5 M/M	EVB-D1 Validation platform specification
EVB-02	EVB schematic design — power tree, clocking, host interface, connectors		10w 14 M/M	EVB-D2 EVB schematics, BOM and layout database
EVB-03	PCB layout with high-speed channel SI/PI simulation		12w 18 M/M	EVB-D3 Fabricated and assembled boards, rev A/B with quantity plan
EVB-04	PCB fabrication and assembly		8w 4 M/M	EVB-D4 Board bring-up report and known issues
EVB-05	Power delivery, VRM and telemetry design and bring-up		7w 8 M/M	EVB-D5 Debug and trace access documentation
EVB-06	Debug infrastructure — JTAG and trace pods, interposers, instrumentation headers		7w 8 M/M	EVB-D6 Lab setup and instrument reservation plan
EVB-07	Board bring-up with socketed or dummy parts		7w 10 M/M	
EVB-08	Thermal solution and cooling for the lab platform		7w 7 M/M	
EVB-09	Lab instrumentation reservation and test rack build		7w 8 M/M	
EVB-10	Minimum host-side enablement for power-on (BSP handoff boundary)		7w 10 M/M	

20 EVB · VALIDATION HARDWARE / EVB

EVB-01

Validation platform requirements and topology definition

TAKES	COSTS	OWNER	CRITICAL PATH
5w w0–w5 of the stage	5.0 M/M ~4.3 people while it runs	Validation lead Program manager approves	yes slipping moves the program

WHY IT EXISTS

Decide **what the silicon will be validated on**—how many platforms, in what configuration, at what quantity—before anyone draws a schematic.

The board is the only way to touch the silicon, and its capability bounds everything bring-up and characterization can do. A platform that cannot reach a rail, cannot socket a part, or cannot be temperature-soaked closes off measurements that the qualification plan assumes are available.

WHAT IT DELIVERS

EVB-D1

Validation platform specification

This activity is the deliverable, and everything the board team builds derives from it.

NEEDS FIRST

- Bring-up requirements from BU-02 through BU-08 planning
- Characterization plan from TEST-07
- Package pin map from PKGD-04
- Power envelope from ARCH-08
- Interface list from ARCH-03

DONE WHEN

- Requirements collected from all three consumers, not just bring-up
- Socket-versus-solder decided and its consequences accepted
- Quantity plan sized against parallel debug needs

HOW IT GETS THERE

6 STEPS · 2 LANES

w0	w1	w2	w3	w4	w5
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Validation requirement collection from bring-up, characterization and qual → Consolidated validation requirements	1w
2	Platform topology options — socketed, soldered, mezzanine → Platform topology options with trade-offs	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Reuse assessment against existing platforms → Reuse assessment	1w runs in parallel
4	Topology selection with its trade-offs recorded → Topology selection record	1.5w
5	Quantity and revision plan → Quantity and revision plan	1w runs in parallel
6	Platform specification → Validation platform specification	1.5w

Step 3 is where a schedule can be bought. An existing platform that carries 70% of the requirement, adapted, beats a new design by a quarter—and the assessment costs a week.

WATCH OUT FOR

- ◆ **Requirements collected from bring-up only.** Characterization and qualification also live on this platform, and their needs—soak capability, instrumented rails—differ.
- ◆ **Socket versus solder decided late.** It changes the board, the mechanical design and the unit consumption, and reversing it costs a board revision.
- ◆ **Quantity underestimated.** Boards are the constraint on parallel debug, and a second fabrication run costs two months.
- ◆ **Reuse not assessed.** An adaptable existing platform is the cheapest schedule available and it is often never looked for.
- ◆ **Specification written after design started.** The board then documents what was built rather than bounding what should be.

WHERE THE EFFORT GOES	5 M/M	WHO IS ON IT ~4.3 FTE	CONNECTIONS
Requirement collection	1.5	Validation lead owns it	DEPENDS ON ARCH-03 ARCH-08 TC-06 PKGD-04
Topology selection	1.3	Hardware architect	FEEDS INTO EVB-02 EVB-04 EVB-05 EVB-06 EVB-08 EVB-09 EVB-10
Specification	1.0	Characterization engineer	LATER INPUT TEST-07 +21w
Reuse assessment	0.8	Package engineer	
Quantity plan	0.5	Program manager	

20 EVB · VALIDATION HARDWARE / EVB

EVB-02

EVB schematic design — power tree, clocking, host interface, connectors

TAKES	COSTS	OWNER	CRITICAL PATH
10w w4–w14 of the stage	14.0 M/M ~6.1 people while it runs	Hardware engineer Hardware lead approves	yes slipping moves the program

WHY IT EXISTS

Design the **EVB schematic**—power tree, host interface, socket, clocking, debug headers—and its bill of materials.

The schematic is where the platform's capability is fixed. A rail without a sense point cannot be measured, a debug header omitted cannot be added, and a component with a 40-week lead time turns a board schedule into a procurement schedule.

WHAT IT DELIVERS

EVB-D2

EVB schematics, BOM and layout database

The schematic half of the deliverable.

NEEDS FIRST

- Platform specification from EVB-01
- Package pin map from PKGD-04
- Power rail requirements from PD-08
- Interface specifications from ARCH-03
- Debug requirements from EVB-06

DONE WHEN

- Every rail instrumented with a sense point
- BOM sourced with lead times confirmed against the schedule
- Schematic reviewed by validation, not only by hardware

HOW IT GETS THERE

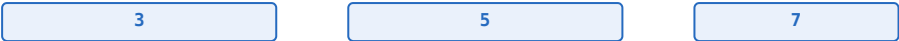
8 STEPS · 2 LANES

w4w6w8w10w12w14

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Block diagram and interface partition → Block diagram and interface partition	1w
2	Power tree and VRM schematic → Power tree and VRM schematic	2.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Clocking and reference schematic → Clocking and reference schematic	2w runs in parallel
4	Host interface and connector schematic → Host interface schematic	2.5w
5	Debug and instrumentation header design → Debug and instrumentation headers	2w runs in parallel
6	Socket and DUT interface schematic → Socket and DUT interface	2w
7	BOM construction and component sourcing check → Bill of materials with sourcing status	1.5w runs in parallel
8	Schematic review and release to layout → Reviewed and released schematic	2w

Step 7 runs alongside deliberately. Component lead times are discovered during sourcing, and a part found to be 40 weeks out while the schematic is still editable can be substituted; found after layout it cannot.

WATCH OUT FOR

- ◆ **Rails without sense points.** A rail that cannot be measured cannot be characterized, and the omission is only discovered during bring-up.
- ◆ **Long-lead components found after layout.** The substitution then costs a board revision instead of a schematic edit.
- ◆ **Debug headers designed for the debugger nobody uses.** The tooling has to be agreed with validation rather than assumed.
- ◆ **Pin map changing after schematic release.** The package pin map has to be frozen for the schematic to be worth reviewing.
- ◆ **Power tree sized without the real power envelope.** A VRM sized to an early estimate cannot supply the silicon's actual transients.

WHERE THE EFFORT GOES	14 M/M	WHO IS ON IT	~6.1 FTE	CONNECTIONS	TERMS HERE	4
Power tree design	3.5	Hardware engineer	owns it	DEPENDS ON ARCH-03 PD-08 PKGD-04 EVB-01 EVB-06	BOM DUT EVB VRM	
Host interface	3.0	Power engineer		FEEDS INTO EVB-03 EVB-04 EVB-05		
Socket interface	2.5	Validation engineer				
Clocking	2.0	Component engineer				
Review and release	1.5	Hardware lead				
BOM and sourcing	1.5					

20 EVB · VALIDATION HARDWARE / EVB

EVB-03

PCB layout with high-speed channel SI/PI simulation

TAKES	COSTS	OWNER	CRITICAL PATH
12w w12–w24 of the stage	18.0 M/M ~6.5 people while it runs	PCB layout engineer Fabrication liaison approves	yes slipping moves the program

WHY IT EXISTS

Lay out the PCB with **signal and power integrity simulated on the critical channels**, because a validation board carrying PCIe Gen6 or CXL is itself a high-speed design.

The board is in series with every interface measurement bring-up will make. A channel with a marginal board contribution produces link failures that look like silicon problems, and separating the two costs weeks of debug time that the simulation would have avoided.

WHAT IT DELIVERS

EVB-D2

EVB schematics, BOM and layout database

The layout half of the deliverable—what fabrication is built from.

NEEDS FIRST

- Released schematic and BOM from EVB-02
- Package model from SIPI-02
- Channel budget from SIPI-04
- Board PDN target from SIPI-06
- Mechanical constraints from EVB-08

DONE WHEN

- Board channel contribution inside its share of the SIPI budget
- SI and PI simulated during routing, not after
- Mechanical and thermal constraints met without a revision

HOW IT GETS THERE

8 STEPS · 2 LANES

w12w15w18w21w24

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Stack-up and impedance planning → Stack-up and impedance plan	1.5w
2	Placement and floorplan → Placement and floorplan	2.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	SI simulation on the critical channels → SI simulation results on critical channels	2w runs in parallel
4	High-speed channel routing → Routed high-speed channels	3w
5	PI simulation on the board PDN → PI simulation on the board PDN	2w runs in parallel
6	Power plane and PDN layout → Power plane layout	2.5w
7	Thermal and mechanical layout constraints → Thermal and mechanical constraint compliance	2w runs in parallel
8	DRC, review and release to fabrication → DRC-clean layout database released to fabrication	2.5w

Steps 3 and 5 run against the routing rather than after it. A channel simulated after the board is routed produces a report; simulated during, it produces a routing change.

WATCH OUT FOR

- ◆ **Board channel contribution not budgeted.** The SIPI channel budget covers package and board together, and a board that consumes more than its share leaves the silicon nothing.
- ◆ **SI simulated after routing.** The result is a report rather than a design change, and fixing it costs a revision.
- ◆ **Board PDN inadequate for the transient.** The silicon's di/dt is on the package model; a board PDN sized without it droops.
- ◆ **Stack-up chosen for cost before impedance.** Controlled impedance on the high-speed layers is not negotiable, and a cheaper stack-up that misses it is not cheaper.
- ◆ **Mechanical constraints applied last.** The cooling solution and socket need space, and finding that out after routing costs a revision.

WHERE THE EFFORT GOES	18 M/M	WHO IS ON IT	~6.5 FTE	CONNECTIONS	TERMS HERE	12
High-speed channel routing	5.0	PCB layout engineer owns IT		DEPENDS ON SIPI-02 SIPI-04 SIPI-06 EVB-02 EVB-08	BOM DRC DFM PDN SI PI SI/PI SIPI PCB EVB PCIe CXL	
SI simulation	3.5	SI/PI engineer		FEEDS INTO EVB-04 EVB-07		
Power plane layout	3.0	Hardware engineer		FEEDS BACK INTO SIPI-02		
Placement	2.5	Mechanical engineer				
PI simulation	2.0	Fabrication liaison				
DRC and release	2.0					

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EVB-04

PCB fabrication and assembly

TAKES	COSTS	OWNER	CRITICAL PATH
8w w23–w31 of the stage	4.0 M/M ~2.2 people while it runs	Hardware engineer Vendor liaison approves	yes slipping moves the program

WHY IT EXISTS

Get the boards **fabricated, assembled and inspected**—an eight-week vendor cycle that is mostly waiting and entirely on the critical path to bring-up.

This is procurement work with an engineering tail. The board has to arrive before silicon does, its quantity has to match the parallel-debug plan, and every board that arrives faulty is a debug station that does not exist.

WHAT IT DELIVERS

EVB-D3

Fabricated and assembled boards, rev A/B with quantity plan

This activity is the deliverable.

NEEDS FIRST

- Released layout database from EVB-03
- BOM with sourcing from EVB-02
- Quantity plan from EVB-01
- Fabrication and assembly vendor capacity
- Inspection criteria

DONE WHEN

- Boards delivered before silicon arrives
- First article inspected before the run completes
- Controlled impedance verified by coupon

HOW IT GETS THERE

6 STEPS · 2 LANES

w23	w25	w27	w29	w31
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Fabrication vendor selection and data release → Vendor selection and released fabrication data	1w
2	PCB fabrication → Fabricated bare boards	3w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Component procurement against lead times → Procured components	1.5w runs in parallel
4	Board assembly → Assembled boards	2.5w
5	First article inspection → First article inspection result	1w runs in parallel
6	Incoming inspection and acceptance → Incoming inspection and acceptance record	1.5w

Step 5 catches assembly problems on one board before they are replicated across the whole run. It costs a day and it is routinely skipped in the interest of speed, which is how twenty identically-wrong boards get built.

WATCH OUT FOR

- ◆ **Long-lead components blocking assembly.** The board is fabricated and waiting on one part, which is the most common way this activity slips.
- ◆ **First article skipped.** An assembly error is then replicated across the entire run and discovered at bring-up.
- ◆ **Quantity cut for cost.** Boards are the constraint on parallel debug, and the saving is repaid in serialized lab time.
- ◆ **Boards arriving after silicon.** Silicon waiting on a board is the worst schedule outcome the program can produce.
- ◆ **Controlled impedance not verified.** The stack-up was designed for it; the fabricated board has to be coupon-tested to confirm it.

WHERE THE EFFORT GOES	4 M/M	WHO IS ON IT ~2.2 FTE	CONNECTIONS
Assembly oversight	1.3	Hardware engineer owns it	DEPENDS ON EVB-01 EVB-02 EVB-03
Fabrication oversight	1.0	Component engineer	FEEDS INTO EVB-07 BU-01
Procurement	1.0	Quality engineer	TERMS HERE 4
Inspection	0.8	Program manager	BOM DFM PCB EVB
		Vendor liaison	

20 EVB · VALIDATION HARDWARE / EVB

EVB-05

Power delivery, VRM and telemetry design and bring-up

TAKES	COSTS	OWNER	CRITICAL PATH
7w w8–w15 of the stage	8.0 M/M ~5.0 people while it runs	Power engineer Test engineer approves	no has float

WHY IT EXISTS

Design and bring up the **power delivery, VRM and telemetry** on the platform—the rails the silicon runs on and the instrumentation that measures them.

Characterization is mostly a power measurement. Shmoo needs rails that can be swept, power validation needs rails that can be measured accurately, and both need protection that trips before an expensive part is destroyed rather than after.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Power rail requirements from PD-08
- Board PDN design from EVB-03
- Transient budget from SIPI-06
- Shmoo range from BU-07 planning
- Package power delivery from PKGD-05

DONE WHEN

- Rails sweepable across the full shmoo range
- Protection validated before any real part is inserted
- Telemetry accurate enough for characterization

HOW IT GETS THERE

6 STEPS · 2 LANES

w8	w10	w12	w14	w15
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Power tree requirement from the die and package → Power tree requirement	1w
2	VRM selection and design → VRM design	1.5w
3	Transient response design against the SIPI budget → Transient response design	1.5w runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Telemetry and monitoring design → Telemetry and monitoring capability	1.5w
5	Protection and sequencing design → Protection and sequencing design	1.5w <i>runs in parallel</i>
6	Power bring-up and validation on the board → Power bring-up and validation results	3w

Step 5 is what protects the units. A sequencing error or an overcurrent event on a first-silicon part costs a package worth many thousands, and the protection has to be validated before a real part is ever inserted.

WATCH OUT FOR

- ◆ **Rails not sweepable across the shmoo range.** Characterization then cannot reach the corners the qualification plan requires.
- ◆ **Protection untested before a real part goes in.** A sequencing fault destroys a first-silicon unit, and there are very few of them.
- ◆ **Telemetry accuracy insufficient.** Power measurements at the resolution characterization needs demand real instrumentation, not a rough monitor.
- ◆ **Transient response inadequate.** The die's di/dt is known from SIPI; a VRM that cannot follow it produces droop that looks like a silicon failure.
- ◆ **Sequencing not matching the silicon's requirement.** The power-up order comes from the design, and getting it wrong is a common first-board fault.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT ~5.0 FTE	CONNECTIONS	TERMS HERE 7
VRM design	2.5	Power engineer <i>owns it</i>	DEPENDS ON PD-08 PKGD-05 SIPI-05 SIPI-06 EVB-01 EVB-02	BOM PDN SI/PI SIPI EVB VRM Shmoo
Power bring-up	2.0	Hardware engineer		
Telemetry design	1.5	SI/PI engineer	FEEDS INTO EVB-07 BU-02 BU-07	
Transient response	1.3	Validation engineer		
Protection and sequencing	0.8	Test engineer		

20 EVB · VALIDATION HARDWARE / EVB

EVB-06

Debug infrastructure — JTAG and trace pods, interposers, instrumentation headers

TAKES	COSTS	OWNER	CRITICAL PATH
7w w10–w17 of the stage	8.0 M/M ~5.0 people while it runs	Validation engineer Mechanical engineer approves	no has float

WHY IT EXISTS

Build the **debug access**—JTAG and trace pods, interposers, probe points—because silicon debug is bounded by what can be observed.

When first silicon does something unexpected, the difference between a day and a month is whether the state that explains it can be read out. The DFT infrastructure provides the internal access; this activity provides the physical path to it.

WHAT IT DELIVERS

EVB-D5

Debug and trace access documentation

This activity is the deliverable, and it is what the bring-up team reads at 2am.

NEEDS FIRST

- Debug and trace architecture from DFT-08
- Silicon debug requirements from BU-09 planning
- Platform specification from EVB-01
- Package pin map from PKGD-04
- Debug tooling from the DFT flow

DONE WHEN

- Debug access matching what DFT actually exposes
- Trace bandwidth sufficient for the events being traced
- Debug path validated before silicon, not during a crisis

HOW IT GETS THERE

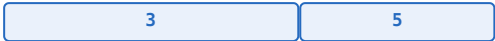
6 STEPS · 2 LANES

w10w12w14w16w17

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Debug requirements from validation and DFT → Debug requirements	1w
2	JTAG and trace pod interface design → JTAG and trace pod interface	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Logic analyser and scope access design → Logic analyser and scope access	1.5w runs in parallel
4	Interposer and probe access design → Interposer and probe access design	1.5w
5	Software debug tool integration → Software debug tool integration	1w runs in parallel
6	Debug infrastructure validation → Debug infrastructure validation results · Debug and trace access documentation	3w

Step 6 validates the debug path before it is needed. A trace pod that does not work is discovered during the crisis it was meant to resolve, which is the worst possible moment.

WATCH OUT FOR

- ◆ **Debug access designed without DFT.** The internal observability comes from DFT-08, and the board access has to match what it exposes.
- ◆ **Trace bandwidth insufficient.** Trace is only useful if it can keep up with the events being traced, and the bandwidth is a board design parameter.
- ◆ **Probe points unreachable once the cooling solution is mounted.** The thermal solution and the probe access compete for the same physical space.
- ◆ **Debug path unvalidated before silicon.** It is then debugged during the emergency it exists to resolve.
- ◆ **Software tooling not integrated.** A hardware debug path with no software to drive it is not a debug path.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT	~5.0 FTE	CONNECTIONS	TERMS HERE	4
Pod interface design	2.5	Validation engineer	OWNS IT	DEPENDS ON DFT-08 PKGD-04 EVB-01	BOM DFT JTAG EVB	
Infrastructure validation	2.0	DFT engineer		FEEDS INTO EVB-02 EVB-07 EVB-10		
Probe access design	1.5	Hardware engineer		BU-04 BU-05 BU-09		
Analyser access	1.3	Software engineer				
Tool integration	0.8	Mechanical engineer				

20 EVB · VALIDATION HARDWARE / EVB

EVB-07

Board bring-up with socketed or dummy parts

TAKES	COSTS	OWNER	CRITICAL PATH
7w w30–w37 of the stage	10.0 M/M ~6.2 people while it runs	Hardware engineer Mechanical engineer approves	yes slipping moves the program

WHY IT EXISTS

Bring the **board itself up**—with dummy parts or a socketed loopback—so that when silicon arrives, every failure is the silicon's.

This is the single highest-leverage activity in the stage. A board debugged before first silicon means bring-up starts on a known-good platform; a board debugged alongside first silicon means every anomaly has two possible causes and each takes days to separate.

WHAT IT DELIVERS

EVB-D4

Board bring-up report and known issues

This activity is the deliverable, and the known-issues list is what stops board faults being mistaken for silicon faults.

NEEDS FIRST

- Assembled boards from EVB-04
- Power subsystem from EVB-05
- Debug infrastructure from EVB-06
- Thermal solution from EVB-08
- Host enablement from EVB-10

DONE WHEN

- Board bring-up complete before silicon arrives
- Known issues documented for the bring-up team
- Every board in the fleet brought up, not just one

HOW IT GETS THERE

6 STEPS · 2 LANES

w30	w32	w34	w36	w37
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Bring-up plan and dummy part fitting → Bring-up plan and fitted dummy parts	1w
2	Power tree bring-up and rail verification → Power tree bring-up results · Board bring-up report and known issues	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Telemetry and protection validation → Telemetry and protection validation	1.5w runs in parallel
4	Clocking and reference validation → Clocking and reference validation	1.5w
5	Thermal solution fit check → Thermal fit check	1.5w runs in parallel
6	Interface loopback and signal validation → Interface loopback and signal validation results	3w

The whole activity is scheduled to complete before silicon arrives. Every hour spent here is an hour not spent, at ten times the cost, debugging a board while first silicon waits on the bench.

WATCH OUT FOR

- ◆ **Board bring-up deferred until silicon arrives.** Every anomaly then has two candidate causes, and separating them costs days each.
- ◆ **Known issues not documented.** The bring-up team then rediscovers each board quirk independently, once per engineer.
- ◆ **Dummy parts not representative.** A loopback that does not load the rails or exercise the channels validates less than it appears to.
- ◆ **Only one board brought up.** Board-to-board variation is real, and a fleet where only one is characterized is a fleet with unknown members.
- ◆ **Thermal fit checked after silicon insertion.** A cooling solution that does not fit is discovered with an irreplaceable part underneath it.

WHERE THE EFFORT GOES	10 M/M	WHO IS ON IT	~6.2 FTE	CONNECTIONS	TERMS HERE	1
Interface validation	3.0	Hardware engineer	owns it	DEPENDS ON EVB-03 EVB-04 EVB-05 EVB-06 EVB-08 EVB-10	SI	
Power bring-up	2.5	Power engineer		FEEDS INTO BU-01 BU-02		
Clocking validation	2.0	Validation engineer				
Telemetry validation	1.5	SI engineer				
Thermal fit	1.0	Mechanical engineer				

20 EVB · VALIDATION HARDWARE / EVB

EVB-08

Thermal solution and cooling for the lab platform

TAKES	COSTS	OWNER	CRITICAL PATH
7w w16–w23 of the stage	7.0 M/M ~4.3 people while it runs	Thermal engineer Hardware engineer approves	no has float

WHY IT EXISTS

Design and validate the **cooling for the lab platform**—which is not the product's thermal solution, but has to remove the same power.

A part that thermally throttles during characterization produces data about the cooling rather than about the silicon. The lab solution can be oversized and ugly in ways the product's cannot, and it should be, because measurement validity depends on it.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Package thermal envelope from PKGD-08
- Power envelope from ARCH-08
- Platform specification from EVB-01
- Characterization temperature range from TEST-07
- Mechanical constraints from EVB-03

DONE WHEN

- No throttling at maximum power during characterization
- Temperature held at set point, including the hot corner
- Mounting compatible with debug probe access

HOW IT GETS THERE

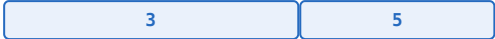
6 STEPS · 2 LANES

w16	w18	w20	w22	w23
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Thermal requirement from the package envelope → Thermal requirement	1w
2	Cooling solution selection → Cooling solution selection	1.5w
3	Airflow and enclosure design → Airflow and enclosure design	1.5w

runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Mounting and interface design → Mounting and interface design	1.5w
5	Temperature control and soak capability → Temperature control and soak capability	1w <i>runs in parallel</i>
6	Thermal validation on the platform → Thermal validation results	3w

Step 5 is what makes qualification-adjacent measurement possible. Characterization needs the part held at a temperature, not merely kept cool, and a solution that only cools cannot produce a hot-corner shmoo.

WATCH OUT FOR

- ❖ **Cooling sized to the product's solution rather than to the lab's need.** The lab platform can be oversized, and measurement validity is worth more than realism.
- ❖ **No temperature control, only cooling.** Characterization needs the part held at a set point, including a hot one.
- ❖ **Throttling during characterization.** The data then describes the cooling rather than the silicon, and it is often not noticed.
- ❖ **Mounting competing with probe access.** The debug points and the heatsink want the same space, and the conflict has to be resolved in layout.
- ❖ **Thermal validated only at nominal power.** The interesting measurements are at maximum power, which is where the solution is stressed.

WHERE THE EFFORT GOES	7 M/M	WHO IS ON IT ~4.3 FTE	CONNECTIONS
Cooling selection	2.0	Thermal engineer owns it	DEPENDS ON ARCH-08 PKG0-08 EVB-01
Thermal validation	2.0	Mechanical engineer	FEEDS INTO EVB-03 EVB-07 BU-06
Mounting design	1.5	Validation engineer	BU-07
Airflow design	1.0	Package engineer	LATER INPUT TEST-07 +3w
Temperature control	0.5	Hardware engineer	

20 EVB · VALIDATION HARDWARE / EVB

EVB-09 Lab instrumentation reservation and test rack build			
TAKES	COSTS	OWNER	CRITICAL PATH
7w	8.0 M/M	Lab manager	no
w24–w31 of the stage	~5.0 people while it runs	Facilities approves	has float

WHY IT EXISTS

Reserve the instruments and **build the lab racks**—because characterization needs equipment that is expensive, shared and booked months ahead.

A high-speed oscilloscope or a BERT is a capital item with a queue in front of it. Discovering during bring-up that the scope with the right bandwidth is committed elsewhere for six weeks does not delay the measurement by six weeks—it delays everything downstream of it.

WHAT IT DELIVERS

EVB-D6

Lab setup and instrument reservation plan

This activity is the deliverable, and it is what stops bring-up queueing for a scope.

NEEDS FIRST

- Characterization plan from TEST-07
- Bring-up plan from BU-02 through BU-08
- Platform specification from EVB-01
- Instrument pool and capital budget
- Lab space and infrastructure

DONE WHEN

- Instruments reserved against the bring-up window, not requested during it
- Automation working before bring-up starts
- Racks calibrated and their calibration current

HOW IT GETS THERE

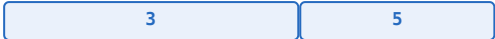
6 STEPS · 2 LANES

w24	w26	w28	w30	w31
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Instrument requirement from the characterization plan → Instrument requirement list	1w
2	Instrument reservation and procurement → Instrument reservations and procurement · Lab setup and instrument reservation plan	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Automation and control infrastructure → Automation and control infrastructure	1.5w runs in parallel
4	Test rack build and cabling → Built test racks with cabling	1.5w
5	Lab space and power provisioning → Lab space and power provisioning	1w runs in parallel
6	Rack integration and calibration → Rack integration and calibration record	3w

Step 3 decides whether characterization is manual or automatic. A shmoo run by hand takes days per corner; the same shmoo automated runs overnight, and the automation is written here rather than improvised during bring-up.

WATCH OUT FOR

- ◆ **Instruments booked too late.** High-bandwidth equipment is shared and queued, and a six-week wait delays everything downstream.
- ◆ **Automation deferred to bring-up.** It is then written under time pressure by people who should be debugging silicon.
- ◆ **Racks not calibrated.** Measurements from an uncalibrated setup are unusable for characterization and often not questioned.
- ◆ **Lab power or space insufficient.** A rack that cannot be powered is a rack that does not exist, and facilities lead times are long.
- ◆ **Instrument bandwidth below the signal rate.** A scope that cannot resolve the edge produces a plausible and wrong measurement.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT	~5.0 FTE	CONNECTIONS	TERMS HERE	1
Rack build and cabling	2.0	Lab manager	OWNS IT	DEPENDS ON EVB-01	BERT	
Integration and calibration	2.0	Characterization engineer		RUNS WITH TEST-07		
Instrument reservation	1.8	Automation engineer		FEEDS INTO BU-05 BU-07 BU-08		
Automation infrastructure	1.5	Validation lead				
Space provisioning	0.8	Facilities				

20 EVB · VALIDATION HARDWARE / EVB

EVB-10

Minimum host-side enablement for power-on (BSP handoff boundary)

TAKES	COSTS	OWNER	CRITICAL PATH
7w w26–w33 of the stage	10.0 M/M ~6.2 people while it runs	Validation software engineer Bring-up lead approves	no has float

WHY IT EXISTS

Provide the **minimum host-side software** needed to power on and talk to the part—boot, register access, enumeration, logging—without taking on the product's software stack.

SW/FW is outside this program's scope, but silicon cannot be brought up without something on the other end of the link. The boundary matters: this activity delivers exactly enough to reach first register read and no more, and where that line sits should be agreed rather than discovered.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Register map from RTL-06
- Boot sequence from ARCH-06
- Debug tooling from EVB-06
- Firmware team's scope and schedule
- Platform specification from EVB-01

DONE WHEN

- Boundary with firmware explicit and agreed
- Register access validated against the current map
- Enablement validated on the platform before silicon

HOW IT GETS THERE

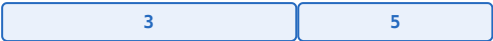
6 STEPS · 2 LANES

w26	w28	w30	w32	w33
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Handoff boundary definition with the firmware team → Handoff boundary definition	1w
2	Minimal boot and initialization software → Minimal boot and initialization software	1.5w
3	Host driver stub and enumeration → Host driver stub and enumeration	1.5w

runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Register access and diagnostic tooling → Register access and diagnostic tooling	1.5w
5	Logging and data capture → Logging and data capture	1w <i>runs in parallel</i>
6	Host enablement validation on the platform → Host enablement validation results	3w

Step 1 is the activity's most valuable hour. Without an explicit boundary, hardware validation writes software the firmware team is also writing, and each assumes the other is doing it.

WATCH OUT FOR

- ◆ **Boundary with firmware left implicit.** Either the work is duplicated or nobody does it, and both are discovered at power-on.
- ◆ **Register map drifting from the RTL.** A diagnostic tool built against a stale map reads the wrong addresses and reports plausible nonsense.
- ◆ **Scope creeping into the product stack.** Minimal enablement becomes a driver project, and the validation team stops validating.
- ◆ **Logging insufficient for post-mortem.** A failure that cannot be reconstructed from the log has to be reproduced, which on first silicon may not be possible.
- ◆ **Enablement unvalidated before silicon.** Software bugs then present as silicon bugs during the most expensive week of the program.

WHERE THE EFFORT GOES	10 M/M	WHO IS ON IT ~6.2 FTE	CONNECTIONS	TERMS HERE	2
Boot software	3.0	Validation software engineer owns it	DEPENDS ON ARCH-06 RTL-06 EVB-01 EVB-06	SW/FW RTL	
Enablement validation	2.5	Firmware liaison			
Diagnostic tooling	2.0	Design engineer	FEEDS INTO EVB-07 BU-02 BU-04		
Driver stub	1.5	Validation engineer			
Logging	1.0	Bring-up lead			

VALIDATE & RAMP

21

Test Development

TEST

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w54 – w96	42 _w	11	174 _{M/M}	4 _{of 11}
ENGINEERING ACTIVITIES				KEY DELIVERABLES
TEST-01	Test plan and coverage strategy — wafer sort, final test, system-level test		8w 12 _{M/M}	TEST-D1 Test plan and test coverage matrix
TEST-02	ATE platform selection and tester time booking		6w 4 _{M/M}	TEST-D2 Qualified probe card and load board
TEST-03	Probe card design, fabrication and qualification		20w 16 _{M/M}	TEST-D3 Wafer sort and final test programs, release-tagged
TEST-04	Load board / DUT board design, fabrication and bring-up		17w 16 _{M/M}	TEST-D4 Characterization test suite
TEST-05	Wafer sort test program development		16w 30 _{M/M}	TEST-D5 ATE-ready pattern set with debug log
TEST-06	Final / package test program development		16w 28 _{M/M}	TEST-D6 Test time and test cost model
TEST-07	Characterization test content — shmoo, Vmin/Fmax, PVT sweeps		12w 20 _{M/M}	TEST-D7 Test data infrastructure and yield database
TEST-08	DFT pattern porting to ATE format and pattern debug		10w 16 _{M/M}	
TEST-09	Test time and cost optimization — parallelism, site count		9w 12 _{M/M}	
TEST-10	Correlation between ATE, bench and system results		8w 10 _{M/M}	
TEST-11	Test data infrastructure — STDF collection and yield database		8w 10 _{M/M}	

TEST-01

Test plan and coverage strategy —
wafer sort, final test, system-level test

TAKES	COSTS	OWNER	CRITICAL PATH
8w w0–w8 of the stage	12.0 M/M ~6.5 people while it runs	Test architect Cost analyst approves	yes slipping moves the program

WHY IT EXISTS

Decide **what will be tested, where, and to what defect level**—the plan every other activity in the stage implements.

Test is where quality is bought and where cost per unit is set, and the two trade directly. The plan is the document where that trade is made explicitly: a DPPM target implies coverage, coverage implies content, content implies test time, and test time is money on every unit ever shipped.

WHAT IT DELIVERS

TEST-01

Test plan and test coverage matrix

This activity is the deliverable, and every other TEST activity implements part of it.

NEEDS FIRST

- Product specification and quality target from DEF-04
- DFT coverage from DFT-03 and DFT-05
- Package pin map from PKGD-04
- Quality and reliability requirement from MP-01
- Cost target from DEF-06

DONE WHEN

- DPPM target derived from actual coverage, not asserted
- Test time budgeted before content is written
- Sort and final coverage split deliberately

HOW IT GETS THERE

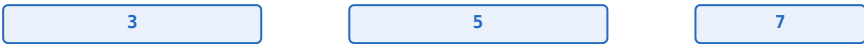
8 STEPS · 2 LANES

w0	w2	w4	w6	w8
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Coverage requirement from the specification and DFT → Coverage requirement	1w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Test flow architecture — sort, final, characterization → Test flow architecture across insertions	2w
3	Tester resource estimation → Tester resource estimate	1.5w runs in parallel
4	Coverage matrix construction → Test coverage matrix	2w
5	DPPM target derivation from the coverage → DPPM target derivation	1.5w runs in parallel
6	Test insertion and binning strategy → Test insertion and binning strategy · Reviewed test plan	1.5w
7	Burn-in and screening decision → Burn-in and screening decision	1w runs in parallel
8	Plan review and release → Reviewed and released test plan	1.5w

Step 5 is the one that closes the loop. Coverage without a DPPM target is an engineering preference; a DPPM target derived from the coverage is a commitment the customer can be given.

WATCH OUT FOR

- ◆ **Coverage chosen without a DPPM target.** There is then no way to say whether the test is sufficient, only whether it is thorough.
- ◆ **Test time not budgeted at plan time.** Content accumulates and the cost per unit is discovered at production readiness when it is expensive to change.
- ◆ **Burn-in decided late.** It is a capital and cycle-time commitment, and reversing it after qualification planning is expensive.
- ◆ **Sort and final coverage overlapping without intent.** Testing the same thing twice costs money on every unit and often happens by accretion.
- ◆ **DFT coverage assumed rather than measured.** The plan then promises a DPPM the actual pattern set cannot deliver.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT	~6.5 FTE	CONNECTIONS	TERMS HERE	2
Coverage matrix	3.0	Test architect owns it		DEPENDS ON DEF-04 DEF-06 ARCH-04 DFT-01 DFT-02 DFT-03 DFT-05 PKGD-04	DFT DPPM	
Test flow architecture	2.5	DFT engineer				
Binning strategy	2.0	Quality engineer				
DPPM derivation	1.8	Product engineering		FEEDS INTO TEST-02 TEST-03 TEST-04 TEST-05 TEST-06 TEST-07 TEST-09		
Review and release	1.5	Cost analyst				
Resource estimation	1.3					

TEST-02

ATE platform selection and tester time booking

TAKES	COSTS	OWNER	CRITICAL PATH
6w w4–w10 of the stage	4.0 M/M ~2.9 people while it runs	Test engineering manager ATE vendor liaison approves	no has float

WHY IT EXISTS

Choose the **ATE platform** and book the tester time—a capital and capacity decision that has to be made long before there is anything to test.

Tester capacity is a shared, scheduled resource. Program development time, characterization time and production capacity are all booked against the same pool, and a program that has not committed its slots competes with programs that have.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Test plan and flow architecture from TEST-01
- Pin count and interface rates from PKGD-04
- Available ATE platforms and their capability
- Test capacity and capital budget
- Production volume forecast from DEF-05

DONE WHEN

- Development, characterization and production slots all booked
- Instrument set covering the highest interface rate
- Pin count supporting the planned multi-site conversion

HOW IT GETS THERE

6 STEPS · 2 LANES

w4	w6	w8	w10
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Tester requirement from the test plan → Tester requirement specification	1w
2	ATE platform evaluation against the requirement → ATE platform evaluation	1.5w
3	Instrument option and pin-count analysis → Instrument option and pin-count analysis	1w

runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Platform selection and configuration → Platform selection and configuration	1.5w
5	Cost per tested unit model → Cost per tested unit model	1w <i>runs in parallel</i>
6	Tester time booking and capacity commitment → Tester time bookings across development, characterization and production	2w

Step 6 is why this runs early. Development, characterization and production all draw on the same testers, and slots that are not booked are taken.

WATCH OUT FOR

- ◆ **Tester time not booked ahead.** Capacity is shared, and a program without commitments waits behind programs with them.
- ◆ **Instrument options short of the interface rate.** High-speed content cannot be run, and the shortfall is discovered when the program is written.
- ◆ **Pin count insufficient for parallel sites.** Multi-site conversion later becomes impossible, and test cost stays where it started.
- ◆ **Platform chosen on capital cost alone.** Cost per tested unit depends on throughput as much as on the tester's price.
- ◆ **Development capacity not separated from production.** Program debug then competes with the ramp it is meant to enable.

WHERE THE EFFORT GOES	4 M/M	WHO IS ON IT ~2.9 FTE	CONNECTIONS	TERMS HERE 1
Platform evaluation	1.3	Test engineering manager owns it	DEPENDS ON DEF-05 DEF-08 PKGD-04 TEST-01	ATE
Tester time booking	1.0	Test architect	FEEDS INTO TEST-03 TEST-04 TEST-05 TEST-06 TEST-08 TEST-09 MP-07	
Selection and configuration	0.8	Operations planner		
Pin-count analysis	0.5	Cost analyst		
Cost model	0.5	ATE vendor liaison		

TEST-03

Probe card design, fabrication and qualification

TAKES	COSTS	OWNER	CRITICAL PATH
20w w10–w30 of the stage	16.0 M/M ~3.5 people while it runs	Probe engineer Operations planner approves	yes slipping moves the program

WHY IT EXISTS

Design, fabricate and qualify the **probe card**—the physical interface between the tester and the wafer, and the longest lead time in the stage.

Nothing can be sorted without it, and sort gates assembly. Twenty weeks of design and fabrication sit in front of the first wafer, which means the card has to be committed against a bump map that is frozen far earlier than most people expect.

WHAT IT DELIVERS

TEST-D2

Qualified probe card and load board

The probe card half of the deliverable.

NEEDS FIRST

- Bump map from PD-15 and PKGD-03
- Test plan and sort content from TEST-01
- ATE platform and interface from TEST-02
- Wafer sort touchdown plan
- Probe card vendor capability

DONE WHEN

- Card qualified on the tester before wafers arrive
- Contact resistance and planarity inside specification
- Spare card available against the sort volume

HOW IT GETS THERE

8 STEPS · 2 LANES

w10	w14	w18	w22	w26	w30
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Probe card requirement and bump map extraction → Probe card requirement and bump map	2w
2	Probe card design → Probe card design	3w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Contact resistance and planarity analysis → Contact resistance and planarity analysis	3w runs in parallel
4	Probe card fabrication → Fabricated probe card	8w
5	Touchdown life and cleaning strategy → Touchdown life and cleaning strategy	3w runs in parallel
6	Qualification on the tester → Tester qualification results	4w
7	Spare card and repair plan → Spare card and repair plan	2.5w runs in parallel
8	Release to wafer sort → Probe card released to sort	3w

Step 4's eight weeks are pure vendor lead time and cannot be compressed. Step 1's bump map therefore has to be frozen roughly five months before sort, which is well before most people believe the design is stable.

WATCH OUT FOR

- ◆ **Bump map changing after card release.** Eight weeks of fabrication are lost, and sort—and therefore assembly—moves by that much.
- ◆ **Contact resistance out of specification.** Marginal contact produces failures that look like silicon defects and corrupts the whole sort yield.
- ◆ **Planarity insufficient across the touchdown area.** Some probes contact and some do not, and the pattern is subtle enough to be missed.
- ◆ **No spare card.** A damaged card with a ten-week replacement stops sort entirely, and cards are damaged routinely.
- ◆ **Touchdown life below the wafer volume.** Card degradation then appears as a yield trend that gets chased as a process problem.

WHERE THE EFFORT GOES	16 M/M	WHO IS ON IT	~3.5 FTE	CONNECTIONS	TERMS HERE	1
Probe card design	4.0	Probe engineer	owns it	DEPENDS ON PD-15 FAB-09 PKGD-03 TEST-01 TEST-02	ATE	
Qualification	3.5	Test engineer				
Requirement and bump map	2.5	Physical design		FEEDS INTO FAB-08 ASSY-01 TEST-05		
Contact analysis	2.5	Probe card vendor		MP-07		
Touchdown strategy	2.0	Operations planner				
Spare plan	1.5					

TEST-04 Load board / DUT board design, fabrication and bring-up			
TAKES	COSTS	OWNER	CRITICAL PATH
17w	16.0 M/M	Test hardware engineer	no
w12–w29 of the stage	~4.1 people while it runs	Socket vendor approves	has float

WHY IT EXISTS

Design, build and bring up the **load board and socket**—the package-level equivalent of the probe card, and what every final-test and characterization measurement passes through.

The load board carries the tester's signals to the package at full rate and delivers its power at full current. It is a high-speed board design in its own right, and its own contribution to a measurement has to be small enough that what is measured is the silicon.

WHAT IT DELIVERS

TEST-D2

Qualified probe card and load board

The load board half of the deliverable.

NEEDS FIRST

- Package pin map and ball map from PKGD-04
- Test plan and final content from TEST-01
- ATE platform interface from TEST-02
- Power envelope from ARCH-08
- Socket vendor capability

DONE WHEN

- Power delivery adequate at maximum test current
- Socket insertion life above the planned test volume
- Board channel contribution small against the measurement margin

HOW IT GETS THERE

8 STEPS · 2 LANES

w12

w16

w20

w24

w28

w29

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Load board requirement and socket selection → Load board requirement and socket selection	2w
2	Load board design with signal integrity → Load board design with SI results	3.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Socket contact and insertion life analysis → Socket contact and life analysis	2.5w runs in parallel
4	Fabrication and assembly → Fabricated and assembled load board	5w
5	Power delivery design on the load board → Load board design · Load board power delivery design	2.5w runs in parallel
6	Bring-up on the tester → Tester bring-up results	3.5w
7	Correlation fixture design → Correlation fixture	2w runs in parallel
8	Release to final test → Load board released to final test	3w

Step 5 is routinely underestimated. A high-power part draws current through the socket, and a load board whose power delivery droops produces failures at speed that are indistinguishable from silicon marginality.

WATCH OUT FOR

- ◆ **Load board power delivery droop.** It produces at-speed failures that look exactly like silicon marginality and consume weeks of debug.
- ◆ **Socket insertion life below the test volume.** Contact degrades, yield trends downward, and the cause is looked for in the process.
- ◆ **Board channel loss consuming the measurement margin.** The interface test then fails on the board rather than on the part.
- ◆ **Socket selected before the package is dimensioned.** A socket that does not fit the finished package is discovered very late.
- ◆ **No correlation fixture.** ATE and bench results then cannot be reconciled, and both are distrusted.

WHERE THE EFFORT GOES	16 M/M	WHO IS ON IT	~4.1 FTE	CONNECTIONS	TERMS HERE	3
Load board design	4.0	Test hardware engineer owns it		DEPENDS ON ARCH-08 PKGD-04 TEST-01 TEST-02	SI SI/PI ATE	
Tester bring-up	3.0	SI/PI engineer		FEEDS INTO TEST-06 TEST-07 TEST-10 MP-07		
Fabrication and assembly	2.5	Test engineer				
Power delivery design	2.5	Package engineer				
Socket analysis	2.0	Socket vendor				
Release	2.0					

TEST-05

Wafer sort test program development

TAKES	COSTS	OWNER	CRITICAL PATH
16w w16–w32 of the stage	30.0 M/M ~8.1 people while it runs	Test engineer Assembly liaison approves	yes slipping moves the program

WHY IT EXISTS

Write the **wafer sort test program**—the screen that decides which die are worth assembling into a package.

On a 2.5D part, sort is the most economically consequential test in the flow. Everything it lets through carries an interposer, two HBM stacks and an assembly with it, so its coverage is measured against package cost rather than against die cost.

WHAT IT DELIVERS

TEST-D3

Wafer sort and final test programs, release-tagged

The sort half of the deliverable.

NEEDS FIRST

- Test plan and coverage matrix from TEST-01
- ATE patterns from TEST-08
- Probe card from TEST-03
- Memory BIST from DFT-06
- Scan architecture from DFT-03

DONE WHEN

- Guard bands derived from package cost and measurement uncertainty
- Coverage sufficient for the assembly economics
- Wafer map format agreed with assembly

HOW IT GETS THERE

8 STEPS · 2 LANES

w16	w20	w24	w28	w32
-----	-----	-----	-----	-----

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Sort flow and test list definition → Sort flow and test list	2w
2	DC and continuity test development → DC and continuity tests	3.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Parametric and process monitor test development → Parametric and monitor tests	2.5w runs in parallel
4	Scan and structural test integration → Integrated scan and structural content	3.5w
5	Binning and wafer map output → Binning and wafer map output	2.5w runs in parallel
6	Functional and memory BIST content → Functional and BIST content	3.5w
7	Test limits and guard-band setting → Test limits and guard bands	2.5w runs in parallel
8	Program integration and debug on the tester → Debugged sort program	3.5w

Step 7 is where the program's economics are set. Limits too tight throw away good die; too loose and packages are built on bad ones. The guard bands should come from measurement uncertainty and package cost, not from convention.

WATCH OUT FOR

- ◆ **Guard bands set by convention.** On a part where an escape costs an assembled package, the limits should be derived from that cost.
- ◆ **Coverage below what the package economics need.** Assembly yield then absorbs the difference, at many times the price.
- ◆ **Program debugged only in simulation.** Tester behavior differs from the model, and the difference is found on the first wafer.
- ◆ **Wafer map output not matching the assembly requirement.** KGD selection depends on the map format, and a mismatch stops assembly.
- ◆ **Test time growing without check.** Sort time multiplies across every wafer ever run, and content accretes silently.

WHERE THE EFFORT GOES	30 M/M	WHO IS ON IT	~8.1 FTE	CONNECTIONS	TERMS HERE	8
Scan and structural integration	6.5	Test engineer OWNS IT		DEPENDS ON AMS-04 AMS-09 DFT-03 DFT-06 DFT-10 TEST-01 TEST-02 TEST-03 TEST-08	KGD DC DFT BIST ATE DPPM HBM (ESD) HBM	
Functional and BIST	6.0	DFT engineer		FEEDS INTO FAB-08 ASSY-01 TEST-09 TEST-11 MP-06		
DC and continuity	5.5	Product engineering				
Integration and debug	5.0	Yield engineer				
Parametric development	4.0	Assembly liaison				
Binning and limits	3.0					

21 TEST · TEST DEVELOPMENT

TEST-06

Final / package test program development

TAKES	COSTS	OWNER	CRITICAL PATH
16w w20–w36 of the stage	28.0 M/M ~7.6 people while it runs	Test engineer Yield engineer approves	no has float

WHY IT EXISTS

Write the **final test program** that runs on the assembled package—at speed, at temperature, across every interface the product ships with.

Final test is the last screen before a customer sees the part. It exercises what sort could not: the assembled interfaces, the HBM stacks, the full power envelope, and the at-speed behavior of a part that is now mechanically complete.

WHAT IT DELIVERS

TEST-D3

Wafer sort and final test programs, release-tagged

The final-test half of the deliverable.

NEEDS FIRST

- Test plan from TEST-01
- Load board from TEST-04
- ATE patterns from TEST-08
- HBM interface specification from ARCH-03
- Binning strategy from TEST-01

DONE WHEN

- HBM interfaces exercised at their operating rate
- At-speed content run across the temperature range
- Test time inside the cost model's allowance

HOW IT GETS THERE

8 STEPS · 2 LANES

w20	w24	w28	w32	w36
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Final test flow definition → Final test flow	2w
2	Package-level DC and continuity content → Package-level DC and continuity content	3.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Power and thermal test content → Power and thermal test content	2.5w runs in parallel
4	At-speed functional content → At-speed functional content	3.5w
5	Binning and grading content → Binning and grading content	2.5w runs in parallel
6	Interface and HBM test content → Interface and HBM test content	3.5w
7	Final test limits and guard bands → Final test limits and guard bands	2.5w runs in parallel
8	Program integration and debug → Debugged final test program	3.5w

Step 6 is where the 2.5D part differs most from a monolithic one. The HBM interfaces exist only after assembly, they cannot be tested at sort, and they are the most likely thing to be wrong about an assembled unit.

WATCH OUT FOR

- ◆ **HBM interface content insufficient.** The interfaces only exist after assembly, and a marginal one shipped is a field failure.
- ◆ **At-speed content not run at temperature.** Timing marginality appears at the hot corner, and testing only at room temperature misses it.
- ◆ **Binning content added late.** Grading is how the product line is built, and retrofitting it changes the flow and its cost.
- ◆ **Test time not tracked against the cost model.** Final test dominates per-unit test cost and grows quietly.
- ◆ **Program debugged without real assembled units.** The first units are precious, and the program has to be ready when they exist.

WHERE THE EFFORT GOES	28 M/M	WHO IS ON IT	~7.6 FTE	CONNECTIONS	TERMS HERE	5
At-speed functional	6.0	Test engineer owns it		DEPENDS ON ARCH-03 TEST-01 TEST-02 TEST-04 TEST-08	DC ATE DPPM HBM (ESD) HBM	
Interface and HBM content	6.0	Product engineering		FEEDS INTO TEST-10 TEST-11 BU-05		
DC and continuity	5.0	Interface engineer		MP-06		
Integration and debug	4.5	Thermal engineer		FEEDS BACK INTO TEST-09		
Power and thermal	3.5	Yield engineer				
Binning and limits	3.0					

TEST-07

Characterization test content —
shmoo, Vmin/Fmax, PVT sweeps

TAKES		COSTS	OWNER	CRITICAL PATH
12w		20.0 M/M	Characterization engineer	no
w24–w36 of the stage		~7.2 people while it runs	Quality engineer approves	has float
WHY IT EXISTS				
Build the characterization content—shmoos, corners, parametric sweeps, datasheet parameter measurement—that turns silicon into a specified product. Production test asks whether a part passes. Characterization asks what the part actually does, across voltage, frequency and temperature, over a statistically meaningful sample. Every number in the datasheet comes from here, and so do the guard bands production test uses.				
WHAT IT DELIVERS				
TEST-D4 Characterization test suite This activity is the deliverable, and every datasheet number traces to it.				
NEEDS FIRST				
- Datasheet parameter list from DEF-04 - Qualification requirements from MP-01 - Test plan from TEST-01 - Load board and correlation fixture from TEST-04 - Lab instrumentation from EVB-09				
DONE WHEN				
- Sample size supporting the datasheet's confidence claim - Corners spanning the qualification range - Every datasheet number traceable to a measurement				
HOW IT GETS THERE				8 STEPS · 2 LANES
w24	w27	w30	w33	w36
MAIN SEQUENCE				
1	2	4	6	8
IN PARALLEL				
	3	5	7	
#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE			TAT
1	Characterization requirement from qualification and marketing → Characterization requirement			1.5w
2	Shmoo and corner content → Shmoo and corner content			2.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Temperature and voltage sweep automation → Sweep automation	2w runs in parallel
4	Parametric characterization content → Parametric characterization content	2.5w
5	Statistical sample plan → Statistical sample plan	2w runs in parallel
6	Datasheet parameter test content → Datasheet parameter content	2.5w
7	Data analysis and reporting pipeline → Data analysis and reporting pipeline	1.5w runs in parallel
8	Content integration and correlation setup → Integrated characterization suite	3w

Step 5 decides whether the results mean anything. A datasheet number from three units is an anecdote; the sample size follows from the distribution being characterized and the confidence the specification claims.

WATCH OUT FOR

- ◆ **Sample size too small for the claim.** A datasheet parameter from a handful of units is a number without confidence behind it.
- ◆ **Corners not spanning the qualification range.** Qualification then stresses conditions characterization never measured.
- ◆ **Sweeps not automated.** Manual shmoo consumes engineer-weeks per corner, and the corners get quietly reduced.
- ◆ **Datasheet parameters not traceable to a measurement.** A customer question about a number then has no answer.
- ◆ **Analysis pipeline built after the data.** Data is collected in a format nobody can analyze, and the campaign is re-run.

WHERE THE EFFORT GOES	20 M/M	WHO IS ON IT	~7.2 FTE	CONNECTIONS	TERMS HERE	1
Shmoo and corner content	4.5	Characterization engineer	OWNS IT	DEPENDS ON DEF-04 TEST-01 TEST-04	Shmoo	
Datasheet parameters	4.0	Product engineering		RUNS WITH EVB-09		
Parametric content	3.5	Statistician or yield engineer		FEEDS INTO TEST-10 TEST-11 BU-07		
Integration and correlation	3.0	Automation engineer		BU-08 MP-02 MP-08		
Sweep automation	2.5	Quality engineer		MP-12		
Sample plan and reporting	2.5			LATER INPUT MP-01 +16w		
				FEEDS BACK INTO EVB-01 EVB-08		

TEST-08

DFT pattern porting to ATE format and pattern debug

TAKES	COSTS	OWNER	CRITICAL PATH
10w w22–w32 of the stage	16.0 M/M ~7.0 people while it runs	Test engineer Product engineering approves	yes slipping moves the program

WHY IT EXISTS

Port the **DFT patterns to the tester's format** and debug them there—the step between a pattern that simulates and a pattern that runs.

ATPG produces patterns against an idealized model. The tester has finite memory, real timing resolution, protocol constraints and a specific pin electronics behavior. Every one of those turns a pattern that passes in simulation into one that fails on hardware, and finding out which is the work.

WHAT IT DELIVERS

TEST-05

ATE-ready pattern set with debug log

This activity is the deliverable.

NEEDS FIRST

- ATPG patterns from DFT-03
- Compression architecture from DFT-04
- ATE platform and memory depth from TEST-02
- Timing constraints from SO-03
- Tester timing model

DONE WHEN

- Pattern set fitting tester memory without coverage loss
- Timing mapped within the tester's resolution
- Patterns simulated against the tester model before silicon

HOW IT GETS THERE

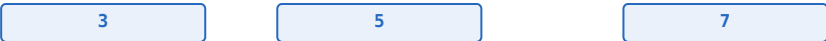
8 STEPS · 2 LANES

w22	w24	w26	w28	w30	w32
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	ATPG pattern handoff from DFT → Pattern handoff record	1.5w
2	Pattern translation to the tester format → Translated tester-format patterns	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Pattern volume and memory-depth reduction → Volume and memory reduction runs in parallel	1.5w
4	Timing and protocol mapping → Timing and protocol mapping	2.5w
5	Compression and streaming setup → Compression and streaming setup runs in parallel	1.5w
6	Pattern simulation against the tester model → Tester-model simulation results	2w
7	Pattern regression and version control → Pattern regression under version control runs in parallel	1.5w
8	Pattern debug on silicon or emulation → Debugged ATE-ready pattern set	2w

Step 3 is a hard constraint, not an optimization. Tester pattern memory is finite, and a pattern set that exceeds it simply cannot be loaded—which is discovered at translation, not at ATPG.

WATCH OUT FOR

- ◆ **Pattern volume exceeding tester memory.** The set cannot be loaded at all, and reduction after ATPG loses coverage.
- ◆ **Timing not mappable to the tester's resolution.** At-speed patterns then run at the wrong speed, silently.
- ◆ **Patterns not simulated against the tester model.** Failures are then debugged on silicon at tester rates rather than in simulation.
- ◆ **Pattern set not versioned.** A yield excursion cannot be correlated to a pattern change nobody recorded.
- ◆ **Debug deferred until first silicon.** Pattern problems and silicon problems then arrive together and cannot be separated.

WHERE THE EFFORT GOES	16 M/M	WHO IS ON IT ~7.0 FTE	CONNECTIONS
Timing and protocol mapping	4.0	Test engineer owns it	DEPENDS ON DFT-03DFT-04DFT-07 DFT-08SO-03TEST-02
Pattern debug	3.5	DFT engineer	FEEDS INTO TEST-05TEST-06MP-06
Translation	3.0	ATE applications engineer	TERMS HERE 4 DFTATPGATECAD
Tester-model simulation	2.5	CAD engineer	
Volume reduction	1.5	Product engineering	
Regression and version control	1.5		

TEST-09

Test time and cost optimization — parallelism, site count

TAKES	COSTS	OWNER	CRITICAL PATH
9w w8–w17 of the stage	12.0 M/M ~5.8 people while it runs	Test engineer Operations planner approves	no has float

WHY IT EXISTS

Reduce **test time and test cost per unit**—an activity that starts early, because content is far cheaper to not write than to remove.

Test time is paid on every unit for the product's whole life. A second saved at final test on a million units is real money, and the savings available at plan time are much larger than the ones available after the program is written and qualified.

WHAT IT DELIVERS

TEST-D6

Test time and test cost model

This activity is the deliverable.

NEEDS FIRST

- Test plan and flow from TEST-01
- Sort content from TEST-05
- Final content from TEST-06
- Tester cost per hour from TEST-02
- Expected yield and failure distribution

DONE WHEN

- Optimization done before qualification freezes the content
- Coverage trades made explicitly against the DPPM target
- Cost model including a yield assumption

HOW IT GETS THERE

8 STEPS · 2 LANES

w8	w10	w12	w14	w16	w17
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Test time baseline measurement → Test time baseline	1.5w
2	Test time contributor analysis → Contributor analysis by test block	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Redundant coverage identification → Redundant coverage findings	1.5w runs in parallel
4	Content reduction and reordering → Content reduction and reordering	2w
5	Adaptive and skip-on-fail strategy → Adaptive and skip-on-fail strategy	1.5w runs in parallel
6	Parallelism and multi-site assessment → Parallelism and multi-site assessment	1.75w
7	Tester resource cost per unit → Tester resource cost per unit	1.25w runs in parallel
8	Cost model update and release → Cost per unit model	1.75w

Step 4's reordering is often the largest single win and costs nothing. Running the tests that fail most first means failing parts leave the tester early, and the average test time drops without any coverage change.

WATCH OUT FOR

- ◆ **Optimization started after qualification.** Any content change then requires re-qualification, which is why this starts early.
- ◆ **Coverage traded away silently.** Time reduction that loses coverage has to be an explicit decision against the DPPM target.
- ◆ **Reordering not done.** It is free and it is the largest single lever, and it is routinely overlooked.
- ◆ **Multi-site assumed without pin count.** The tester configuration from TEST-02 has to support it, and retrofitting it does not work.
- ◆ **Cost model without a yield assumption.** Cost per good unit depends on yield, and a model without it understates the cost.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT	~5.8 FTE	CONNECTIONS	TERMS HERE	1
Contributor analysis	3.0	Test engineer	OWNS IT	DEPENDS ON DFT-02 TEST-01 TEST-02 TEST-05	DPPM	
Content reduction	2.5	Product engineering		FEEDS INTO MP-06 MP-07		
Parallelism assessment	2.0	Cost analyst		LATER INPUT TEST-06 +3w		
Cost model	1.8	Yield engineer		FEEDS BACK INTO DEF-06 DFT-07		
Redundancy identification	1.5	Operations planner				
Adaptive strategy	1.3					

21 TEST · TEST DEVELOPMENT

TEST-10

Correlation between ATE, bench and system results

TAKES	COSTS	OWNER	CRITICAL PATH
8w w32–w40 of the stage	10.0 M/M ~5.4 people while it runs	Product engineering Quality engineer approves	no has float

WHY IT EXISTS

Correlate **ATE, bench and system results** so that a part passing on the tester behaves the same way in a customer's system.

Three measurement environments produce three numbers for the same parameter, and they never agree exactly. What matters is whether the disagreement is understood: an unexplained gap between tester and system is either an escape path or an over-screen, and both are expensive.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Final test results from TEST-06
- Bench measurements from BU-08
- System-level results from BU-05
- Measurement uncertainty from TEST-07
- Load board and correlation fixture from TEST-04

DONE WHEN

- Every material discrepancy root-caused, not accepted
- Golden units retained and re-measurable
- Guard bands updated from what correlation found

HOW IT GETS THERE

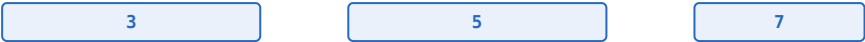
8 STEPS · 2 LANES

w32	w34	w36	w38	w40
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Correlation plan and unit selection → Correlation plan and unit selection	1w
2	ATE-to-bench correlation → ATE-to-bench correlation results	2w
3	Measurement uncertainty budget → Measurement uncertainty budget	1.5w

runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	ATE-to-system correlation → ATE-to-system correlation results	2w
5	Golden unit designation and retention → Designated golden units	1.5w runs in parallel
6	Discrepancy analysis and root cause → Discrepancy analysis and root causes	1.5w
7	Correlation regression setup → Correlation regression	1w runs in parallel
8	Guard-band adjustment and release → Adjusted guard bands	1.5w

Step 5 is what makes correlation repeatable. A retained set of golden units re-measured after any change is how tester drift, load board degradation and program regressions are caught before they become yield excursions.

WATCH OUT FOR

- ◆ **Discrepancy accepted without root cause.** The gap is either an escape or an over-screen, and 'they just differ' resolves neither.
- ◆ **Correlation on too few units.** Two units that agree prove very little about the population.
- ◆ **No golden units retained.** Drift in the tester, the load board or the program then has nothing to be measured against.
- ◆ **System correlation skipped.** ATE-to-bench is the easy half; ATE-to-system is the one the customer experiences.
- ◆ **Guard bands not updated.** The correlation produced the information and nothing acted on it.

WHERE THE EFFORT GOES	10 M/M	WHO IS ON IT	~5.4 FTE	CONNECTIONS	TERMS HERE	1
ATE-to-bench correlation	2.5	Product engineering owns it		DEPENDS ON DFT-11 TEST-04 TEST-06 TEST-07	ATE	
ATE-to-system correlation	2.3	Test engineer		FEEDS INTO MP-06 MP-08 MP-12		
Discrepancy analysis	2.0	Characterization engineer		LATER INPUT BU-05 +14w BU-07 +17w BU-08 +19w		
Uncertainty budget	1.5	Validation engineer				
Guard-band adjustment	1.3	Quality engineer				
Golden unit retention	0.5					

TEST-11

Test data infrastructure — STDF collection and yield database

TAKES	COSTS	OWNER	CRITICAL PATH
8w w26–w34 of the stage	10.0 M/M ~5.4 people while it runs	Test data engineer IT infrastructure approves	no has float

WHY IT EXISTS

Build the **test data infrastructure**—STDF collection, parsing, a yield database and the analysis on top of it—so that test results become information rather than files.

Every yield decision the program makes for the rest of its life is made from this data. Without a pipeline, the data exists as per-lot files nobody can query, and questions that should take an hour take a week or go unasked.

WHAT IT DELIVERS

TEST-D7

Test data infrastructure and yield database

This activity is the deliverable.

NEEDS FIRST

- Sort and final test output formats from TEST-05 and TEST-06
- Yield analysis requirements from MP-05
- Characterization data needs from TEST-07
- Traceability scheme from ASSY-01 and ASSY-09
- Data infrastructure and storage

DONE WHEN

- Traceability keys carried from the first lot
- Pipeline running before first silicon data arrives
- Retention policy covering the field-return horizon

HOW IT GETS THERE

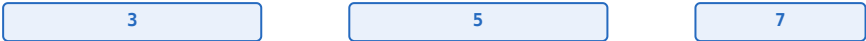
8 STEPS · 2 LANES

w26	w28	w30	w32	w34
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Data requirement from yield, quality and characterization → Data requirements	1w
2	STDF collection and parsing pipeline → STDF collection and parsing pipeline	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Traceability keys — lot, wafer, unit, assembly → Traceability key scheme	1.5w runs in parallel
4	Yield database and schema → Yield database and schema	2w
5	Retention and archival policy → Retention and archival policy	1.5w runs in parallel
6	Analysis and dashboard layer → Analysis and dashboard layer	1.5w
7	Alerting on yield excursions → Excursion alerting	1w runs in parallel
8	Deployment and access provisioning → Deployed and accessible infrastructure	1.5w

Step 3 is the piece that is hardest to add later. A unit's test result is only useful if it can be joined to its wafer position, its assembly lot and its bring-up history, and those keys have to be carried from the beginning.

WATCH OUT FOR

- ◆ **Traceability keys added retroactively.** Historical data then cannot be joined, and the most valuable early lots become unanalysable.
- ◆ **Pipeline built after the first lots run.** The first silicon's data is the most informative the program will ever have, and it is the data most often lost.
- ◆ **Retention policy absent.** Field returns arrive years later, and the test record for that unit has to still exist.
- ◆ **Dashboards without an alerting layer.** A yield excursion is noticed at the weekly review rather than the same day.
- ◆ **Schema not covering characterization.** Characterization data has a different shape, and a sort-only schema forces it into a separate silo.

WHERE THE EFFORT GOES	10 M/M	WHO IS ON IT ~5.4 FTE	CONNECTIONS	TERMS HERE	1
Collection pipeline	2.5	Test data engineer owns IT	DEPENDS ON TEST-05 TEST-06 TEST-07	STDF	
Yield database	2.3	Yield engineer	FEEDS INTO FAB-09 MP-05 MP-09		
Dashboard layer	2.0	Quality engineer	LATER INPUT ASSY-09 +15w		
Traceability keys	1.5	Test engineer			
Deployment	1.0	IT infrastructure			
Retention policy	0.8				

VALIDATE & RAMP

22

Silicon Bring-up

BU

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w102 – w120	18 _w	12	153 _{M/M}	6 of 12
ENGINEERING ACTIVITIES				KEY DELIVERABLES
BU-01	Sample receipt, incoming inspection and board mounting		1.5w 2 M/M	BU-D1 Bring-up report with per-milestone health status
BU-02	Power-on, power sequencing and basic health check		3w 8 M/M	BU-D2 Characterization data set — V/F/T shmoo and power measurements
BU-03	Reset, clocking and PLL lock validation		3w 8 M/M	BU-D3 Interface compliance results with training margins
BU-04	Boot, firmware load and functional smoke test		4w 12 M/M	BU-D4 Errata list with workarounds
BU-05	Interface bring-up — PCIe/CXL link training, HBM training, die-to-die		8w 26 M/M	BU-D5 Failure analysis reports
BU-06	Memory subsystem and bandwidth validation		6w 16 M/M	BU-D6 Respin versus metal-fix decision record
BU-07	Shmoo across voltage, frequency and temperature		7w 16 M/M	BU-D7 Customer sample release package
BU-08	Performance validation against the architecture model		6w 16 M/M	
BU-09	Silicon debug of anomalies and failure analysis requests		10w 28 M/M	
BU-10	Errata capture, workaround definition and documentation		7w 10 M/M	
BU-11	Respin versus metal-fix decision analysis — ECO scope, cost, schedule		4w 5 M/M	
BU-12	Customer sample readiness and release package		5w 6 M/M	

22 BU · SILICON BRING-UP

BU-01

Sample receipt, incoming inspection and board mounting

TAKES	COSTS	OWNER	CRITICAL PATH
1.5w w0–w1.5 of the stage	2.0 M/M ~5.8 people while it runs	Bring-up lead Program manager approves	yes slipping moves the program

WHY IT EXISTS

Receive the first units, **check them and mount them**—a short activity whose only job is to make sure the most valuable objects the program has ever produced are not damaged before they are powered.

There are very few first-silicon units and each represents the whole program's schedule. A unit destroyed by a mounting error or an unnoticed shipping defect is not replaceable inside any useful timeframe.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Allocated units from ASSY-09
- Screened unit record from ASSY-08
- Brought-up boards from EVB-07
- Board known-issue list from EVB-07
- ESD and handling procedures

DONE WHEN

- Every unit inspected before mounting
- Units mounted only on validated boards
- Unit register started and maintained

HOW IT GETS THERE

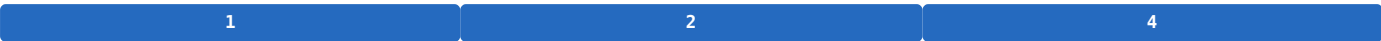
4 STEPS · 2 LANES

w0

w1

w1.5

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Sample receipt and incoming inspection → Incoming inspection record	0.5w
2	Board mounting and mechanical fit → Units mounted on boards	0.5w
3	Unit tracking register and allocation record → Unit tracking and allocation register	0.5w

runs in parallel

#

STEP, AND WHAT IT ADDS TO THE DELIVERABLE

TAT

4

Pre-power checklist against the board's known issues
→ [Pre-power checklist completion](#)

0.5w

Step 4 is the reason board bring-up happened first. The checklist is only meaningful because `EVB-07` already established which board behaviors are normal and which are faults.

WATCH OUT FOR

- ◆ **Handling damage on an irreplaceable unit.** ESD or mechanical damage at mounting costs a unit the program cannot replace.
- ◆ **Mounting onto an unvalidated board.** A board fault then destroys the part, and EVB-07 exists to prevent exactly this.
- ◆ **No unit tracking.** Every subsequent result has to be attributable to a specific unit and its history.
- ◆ **Incoming inspection skipped in the rush.** Shipping damage is real and is much cheaper to find before power than after.
- ◆ **All units mounted at once.** A systematic board or handling fault then takes the whole fleet rather than one unit.

WHERE THE EFFORT GOES	2 M/M	WHO IS ON IT	~5.8 FTE	CONNECTIONS	TERMS HERE	2
Board mounting	0.8	Bring-up lead owns it		DEPENDS ON FAB-08 FAB-09 FAB-10 ASSY-09 EVB-04 EVB-07	ESD FA	
Incoming inspection	0.5	Hardware engineer				
Pre-power checklist	0.5	Quality engineer		FEEDS INTO BU-02		
Unit tracking	0.3	Lab technician		LATER INPUT ASSY-08 +0.5w		
		Program manager				

22 BU · SILICON BRING-UP

BU-02

Power-on, power sequencing and basic health check

TAKES	COSTS	OWNER	CRITICAL PATH
3w w1–w4 of the stage	8.0 M/M ~11.6 people while it runs	Bring-up lead Program manager approves	yes slipping moves the program

WHY IT EXISTS

Apply power for the first time—**sequenced, current-limited and instrumented**—and establish that the part is alive.

First power-on is the moment the whole design becomes physical. It is also the moment a shorted rail or a sequencing error can destroy a unit in milliseconds, which is why it is done with limits, with measurement, and with the ability to stop.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Mounted units from BU-01
- Power subsystem and protection from EVB-05
- Power sequencing specification from ARCH-08
- Expected rail currents from PD-08
- Telemetry from EVB-05

DONE WHEN

- Current limits in place before first application
- Rail currents profiled against expectation
- No unexplained anomaly carried into the next milestone

HOW IT GETS THERE

5 STEPS · 2 LANES

w1	w2	w3	w4
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MAIN SEQUENCE

1	2	4	5
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IN PARALLEL

3

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Power sequencing verification against the specification → Power sequencing verification	0.5w
2	First power application with current limits → First power application record	1w
3	Anomaly triage on the first application → Anomaly triage findings	0.75w runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Rail voltage and current profiling → Rail voltage and current profiles	1w
5	Health check and go/no-go for the next step → Health check and go/no-go decision	0.5w

Step 2's current limits are what turn a fatal event into a diagnostic one. A rail that draws far more than expected trips the limit and the part survives to be investigated; without the limit it is destroyed and the information goes with it.

WATCH OUT FOR

- ◆ **Power applied without current limits.** A short then destroys the unit instead of revealing itself.
- ◆ **Sequencing not verified against the specification.** A wrong power-up order damages the part and the fault is the platform's, not the silicon's.
- ◆ **Rail currents not profiled.** The first real power data is available at exactly this moment and is easy to not capture.
- ◆ **Multiple units powered before the first is understood.** A systematic fault then takes several units instead of one.
- ◆ **Anomalies pushed past to reach the next milestone.** Something unexplained at power-on rarely stays harmless.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT ~11.6 FTE	CONNECTIONS
First power application	2.5	Bring-up lead OWNS IT	DEPENDS ON ARCH-08 DV-06 DFT-08 DFT-10 PD-08 S0-11 EVB-05 EVB-07 EVB-10 BU-01
Rail profiling	2.0	Power engineer	
Sequencing verification	1.5	Design engineer	
Anomaly triage	1.3	Validation engineer	FEEDS INTO BU-03 BU-09 MP-05
Health check	0.8	Program manager	

22 BU · SILICON BRING-UP

BU-03

Reset, clocking and PLL lock validation

TAKES	COSTS	OWNER	CRITICAL PATH
3w w2–w5 of the stage	8.0 M/M ~11.6 people while it runs	Bring-up engineer Validation engineer approves	no has float

WHY IT EXISTS

Validate **reset, clocking and PLL lock**—the infrastructure everything else depends on, and the layer where a fault masquerades as a fault somewhere else.

A PLL that locks unreliably or a reset domain released in the wrong order produces symptoms all over the chip: boot failures, interface training failures, sporadic functional errors. Clearing this layer first means the failures found later are real.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Powered units from BU-02
- PLL specifications from AMS-04
- Clock architecture from ARCH-05
- Reset architecture from RTL-04
- Clock instrumentation from EVB-09

DONE WHEN

- PLL lock verified across the full operating range
- Reset domain release order confirmed against the architecture
- Jitter measured with the board's contribution separated

HOW IT GETS THERE

5 STEPS · 2 LANES

w2	w3	w4	w5
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MAIN SEQUENCE

1	2	4	5
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IN PARALLEL

3

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Reference clock and input validation → Reference clock validation	0.5w
2	PLL lock across the frequency range → PLL lock results across the range	1w
3	Reset sequence and domain release verification → Reset sequence and domain release verification	0.75w runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Clock tree and domain verification → Clock tree and domain verification	1w
5	Jitter and stability measurement → Jitter and stability measurements	0.5w

Step 5's jitter measurement is easy to skip because the part appears to work without it. It matters because interface training margin depends on clock quality, and a jitter problem found here explains a class of failures that would otherwise be chased in the SerDes.

WATCH OUT FOR

- ◆ **PLL lock validated only at nominal.** The lock range matters because shmoo and characterization will operate outside nominal.
- ◆ **Reset domain release order not verified.** The resulting failures appear anywhere on the chip and are attributed to the wrong block.
- ◆ **Jitter unmeasured.** Interface training margin depends on it, and a jitter problem is then debugged inside the SerDes.
- ◆ **Clock tree assumed correct because the part boots.** A misconfigured divider is compatible with booting and incompatible with performance.
- ◆ **Clock quality measured through the board without accounting for it.** The board contributes jitter and its contribution has to be separated.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT ~11.6 FTE	CONNECTIONS	TERMS HERE 3
PLL lock validation	2.5	Bring-up engineer OWNS IT	DEPENDS ON ARCH-05 AMS-02 AMS-04 AMS-16 RTL-04 DV-07 DFT-04 BU-02	PLL SerDes AMS
Clock tree verification	2.0	AMS engineer		
Reset sequence verification	1.5	Design engineer	FEEDS INTO BU-04 BU-05 BU-09	
Jitter measurement	1.3	Characterization engineer		
Reference validation	0.8	Validation engineer		

22 BU · SILICON BRING-UP

BU-04

Boot, firmware load and functional smoke test

TAKES	COSTS	OWNER	CRITICAL PATH
4w w4–w8 of the stage	12.0 M/M ~13.0 people while it runs	Bring-up lead Program manager approves	yes slipping moves the program

WHY IT EXISTS

Boot the part—ROM, firmware load, register access, first functional response—and confirm that the design does what it was written to do.

First boot is the milestone that turns a powered die into a working processor. It is also the point at which the design, the firmware, the board and the host software are all exercised together for the first time, which is why it is the milestone most likely to expose an integration assumption nobody wrote down.

WHAT IT DELIVERS

BU-D1

Bring-up report with per-milestone health status

First boot is the report's central milestone and where it starts being published widely.

NEEDS FIRST

- Clocking-validated units from BU-03
- Host enablement from EVB-10
- Boot ROM and firmware from the firmware team
- Register map from RTL-06
- Debug access from EVB-06

DONE WHEN

- Successful register read and device identity confirmed
- Boot-time deviations captured as errata candidates
- Status broadcast to the program

HOW IT GETS THERE

6 STEPS · 2 LANES

w4	w5	w6	w7	w8
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Boot ROM and initialization sequence execution → Boot ROM execution record	0.75w
2	Firmware load and handoff → Firmware load and handoff results	1.25w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Boot failure triage → Boot failure triage findings runs in parallel	1w
4	Register access and device identity read → Register access and identity read	1.25w
5	First bring-up report and status broadcast → First bring-up report runs in parallel	1w
6	Functional smoke test → Functional smoke test results	0.75w

Step 4's first successful register read is the moment the program knows the silicon is fundamentally sound. Everything before it is infrastructure and everything after it is degrees of correctness.

WATCH OUT FOR

- ◆ **Boot failure with no state visible.** Without debug access the failure is a symptom with no information, and debug access comes from EVB-06.
- ◆ **Firmware and hardware assumptions differing.** The handoff is where two teams' expectations meet, and nobody wrote them down.
- ◆ **Register map drift.** The tool reads addresses the silicon does not implement and reports plausible garbage.
- ◆ **Smoke test too shallow.** A part that boots and does nothing else has passed a milestone without demonstrating much.
- ◆ **Status not broadcast.** First boot is the news the whole program is waiting for, and silence is read as failure.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT	~13.0 FTE	CONNECTIONS	TERMS HERE	1
Firmware load and handoff	3.5	Bring-up lead owns it		DEPENDS ON RTL-06 RTL-07 DV-04 DV-08 EVB-06 EVB-10 BU-03	ROM	
Register access	3.0	Firmware engineer				
Boot sequence execution	2.5	Design engineer		FEEDS INTO BU-05 BU-06 BU-07 BU-09 BU-10		
Failure triage	1.8	Validation software engineer				
Smoke test	1.3	Program manager				

22 BU · SILICON BRING-UP

BU-05

Interface bring-up — PCIe/CXL link training, HBM training, die-to-die

TAKES	COSTS	OWNER	CRITICAL PATH
8w w6–w14 of the stage	26.0 M/M ~14.1 people while it runs	Interface engineer Bring-up lead approves	yes slipping moves the program

WHY IT EXISTS

Bring up the **interfaces**—PCIe/CXL link training, HBM training, die-to-die—which on this part is the largest single body of bring-up work and the most likely to produce a respin decision.

Interfaces are where design, package, board and protocol all meet. Training failures are notoriously hard to attribute because every layer can cause them, and the margin measurements taken here are what tell the program whether the part is comfortable or on the edge.

WHAT IT DELIVERS

BU-D3

Interface compliance results with training margins

This activity is the deliverable, and the margins are what say whether the part is comfortable.

NEEDS FIRST

- Booted units from BU-04
- SIPI channel budget and predictions from SIPI-04 and SIPI-05
- Debug and trace access from EVB-06
- Lab instrumentation from EVB-09
- Interface specifications from ARCH-03

DONE WHEN

- Every interface's margin measured, not only its training result
- Failures attributed to a layer against the SIPI prediction
- Interoperability confirmed on more than the lab's host

HOW IT GETS THERE

8 STEPS · 2 LANES

w6	w8	w10	w12	w14
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Interface bring-up order and plan → Interface bring-up order and plan	1w
2	PCIe/CXL link training → PCIe/CXL link training results	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Link margin and eye measurement → Link margin and eye measurements	2w runs in parallel
4	HBM interface training → HBM training results	2.5w
5	Training failure debug → Training failure debug findings	2w runs in parallel
6	Die-to-die interface bring-up → Interface bring-up plan · Die-to-die interface results	1.5w
7	Interoperability against host platforms → Interoperability results	1.5w runs in parallel
8	Interface stability and compliance run → Interface stability and compliance data	1w

Step 3 is not optional even when training succeeds. A link that trains with no margin trains today and fails at temperature or on a different host, and only the eye measurement distinguishes the two.

WATCH OUT FOR

- ◆ **Training success accepted without margin measurement.** A zero-margin link works on the bench and fails in the field.
- ◆ **Failures not attributable across layers.** Design, package, board and protocol can all cause the same symptom, and the SIPI predictions are what separate them.
- ◆ **HBM training treated as a vendor problem.** The interface is the program's, and its training parameters have to be understood locally.
- ◆ **Interoperability tested against one host.** Customers use many, and a link that only works with the lab's host is not compliant.
- ◆ **Compliance run deferred.** Formal compliance is a gate for certification and takes lab time that has to be booked.

WHERE THE EFFORT GOES	26 M/M	WHO IS ON IT	~14.1 FTE	CONNECTIONS	TERMS HERE	5
HBM training	7.0	Interface engineer	OWNS IT	DEPENDS ON AMS-03 AMS-16 DV-07 DFT-11 SIPI-04 SIPI-05 SIPI-08 SIPI-11 EVB-06 EVB-09 TEST-06 BU-03 BU-04	SerDes SI/PI SIPI HBM (ESD) HBM	
PCIe/CXL training	6.0	SI/PI engineer		FEEDS INTO BU-06 BU-07 BU-08 BU-09 BU-10 MP-11		
Margin and eye measurement	4.5	Design engineer		FEEDS BACK INTO TEST-10		
Training failure debug	4.0	Validation engineer				
Die-to-die bring-up	2.5	Bring-up lead				
Compliance and interoperability	2.0					

22 BU · SILICON BRING-UP

BU-06

Memory subsystem and bandwidth validation

TAKES	COSTS	OWNER	CRITICAL PATH
6w w8–w14 of the stage	16.0 M/M ~11.6 people while it runs	Memory engineer Thermal engineer approves	no has float

WHY IT EXISTS

Validate the **memory subsystem**—controller, address map, bandwidth, latency, ECC—because on an AI accelerator the memory system is most of the performance.

HBM training establishes that the links work; this establishes that the memory system delivers. Bandwidth and latency against the architecture model is the measurement the product's performance claims rest on, and it is where architectural assumptions meet a real memory controller.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Trained HBM interfaces from BU-05
- Memory subsystem model from ARCH-04
- Address map from RTL-05
- Memory test content from TEST-06
- Thermal capability from EVB-08

DONE WHEN

- ECC validated by injected errors, not by absence of failures
- Bandwidth and latency compared against the architecture model
- Efficiency characterized, not only peak

HOW IT GETS THERE

6 STEPS · 2 LANES

w8

w10

w12

w14

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Memory controller initialization and configuration → Controller initialization and configuration	1w
2	Address map and pattern validation → Address map and pattern validation	1.5w
3	ECC and error handling validation → ECC and error handling validation	1.5w runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Bandwidth measurement against the model → Bandwidth measurements against the model	1.5w
5	Refresh and thermal behavior → Refresh and thermal behavior data	1.5w <i>runs in parallel</i>
6	Latency and efficiency characterization → Latency and efficiency characterization	2w

Step 3 has to be done deliberately by injecting errors. ECC that has never corrected anything is ECC that has never been shown to work, and a broken correction path is silent until it matters.

WATCH OUT FOR

- ❖ **ECC never exercised.** A correction path that has never corrected an injected error has not been validated.
- ❖ **Bandwidth measured without the model to compare against.** A number without a target says nothing about whether the design met its intent.
- ❖ **Refresh behavior at temperature not checked.** HBM refresh scales with temperature and the effect on bandwidth is real.
- ❖ **Address map validated only at the boundaries.** Aliasing errors sit in the middle of the map and pass boundary tests.
- ❖ **Efficiency ignored in favor of peak.** Peak bandwidth is a datasheet number; achieved efficiency is what workloads see.

WHERE THE EFFORT GOES	16 M/M	WHO IS ON IT ~11.6 FTE	CONNECTIONS	TERMS HERE 4
Bandwidth measurement	4.5	Memory engineer owns it	DEPENDS ON ARCH-04 RTL-05 EVB-08 BU-04 BU-05	HBM (ESD) HBM ECC AI
Latency characterization	3.5	Architect	FEEDS INTO BU-08 BU-10 MP-12	
Address map validation	3.0	Design engineer		
ECC validation	2.5	Validation engineer		
Controller initialization	1.5	Thermal engineer		
Refresh behavior	1.0			

22 BU · SILICON BRING-UP

BU-07

Shmoo across voltage, frequency and temperature

TAKES	COSTS	OWNER	CRITICAL PATH
7w w9–w16 of the stage	16.0 M/M ~9.9 people while it runs	Characterization engineer Design engineer approves	no has float

WHY IT EXISTS

Shmoo the part across voltage, frequency and temperature—the measurement that establishes how much margin the design actually has.

Every operating condition the product will be specified at comes out of this. The shmoo shows where the part stops working, and the distance between that boundary and the nominal operating point is the margin the datasheet, the guard bands and the qualification plan all depend on.

WHAT IT DELIVERS

BU-D2

Characterization data set — V/F/T shmoo and power measurements

This activity is the deliverable.

NEEDS FIRST

- Booted and interface-trained units from BU-04 and BU-05
- Sweep automation from EVB-09
- Thermal control from EVB-08
- Rail sweep capability from EVB-05
- Characterization content from TEST-07

DONE WHEN

- Shmoo across a fleet, not a single unit
- Temperature corners included, not only room
- No throttling inside the measured region

HOW IT GETS THERE

6 STEPS · 2 LANES

w9	w11	w13	w15	w16
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Shmoo plan and automation setup → Shmoo plan and automation	1w
2	Voltage-frequency shmoo at room temperature → Voltage-frequency shmoo at room	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Power measurement across the sweep → Power measurements across the sweep	1.5w runs in parallel
4	Temperature-extended shmoo → Temperature-extended shmoo	1.5w
5	Unit-to-unit variation across the fleet → Unit-to-unit variation data	1.5w runs in parallel
6	Shmoo analysis and margin extraction → Margin extraction and analysis	3w

Step 5 is what separates a characteristic from an anecdote. One unit's shmoo is that unit; the fleet's spread is what the population looks like, and the specification has to be set against the population.

WATCH OUT FOR

- ◆ **Shmoo on a single unit.** The result describes that unit and is used as if it described the population.
- ◆ **Room temperature only.** The hot and cold corners are where the margin is smallest and where qualification will operate.
- ◆ **Manual sweeps.** The corner count then gets reduced to fit the available engineer time, silently.
- ◆ **Thermal throttling inside the sweep.** The shmoo then maps the cooling solution rather than the silicon.
- ◆ **Margin measured but not fed into guard bands.** The data exists and production test still uses convention.

WHERE THE EFFORT GOES	16 M/M	WHO IS ON IT	~9.9 FTE	CONNECTIONS	TERMS HERE	1
		Characterization engineer owns it		DEPENDS ON AMS-13 FAB-07 EVB-05 EVB-08 EVB-09 TEST-07 BU-04 BU-05	Shmoo	
Temperature-extended shmoo	4.0	Validation engineer				
Voltage-frequency shmoo	3.5	Product engineering				
Margin extraction	3.0	Thermal engineer		FEEDS INTO BU-08 BU-11 MP-06 MP-08 MP-12		
Power measurement	2.5	Design engineer		FEEDS BACK INTO TEST-10		
Unit-to-unit variation	2.0					
Automation setup	1.0					

22 BU · SILICON BRING-UP

BU-08

Performance validation against the architecture model

TAKES	COSTS	OWNER	CRITICAL PATH
6w w11–w17 of the stage	16.0 M/M ~11.6 people while it runs	Performance engineer Product marketing approves	no has float

WHY IT EXISTS

Measure **real performance against the architecture model**—the check on whether the product does what it was sold as doing.

The whole program was justified by a performance projection. This is where that projection meets a physical part running real workloads, and any gap has to be attributed: to the model, to the design, to the memory system, or to the software driving it.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Functional units from BU-04 through BU-06
- Architecture performance model from ARCH-02
- Workload definitions from DEF-03
- Power measurement from BU-07
- Lab instrumentation from EVB-09

DONE WHEN

- Every material gap attributed to a cause
- Efficiency measured alongside throughput
- Software's contribution to the measurement established

HOW IT GETS THERE

6 STEPS · 2 LANES

w11w13w15w17

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Benchmark and workload setup → Benchmark and workload setup	1w
2	Performance measurement across workloads → Performance measurements across workloads	1.5w
3	Power efficiency measurement → Power efficiency measurements	1.5w runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Comparison against the architecture model → Comparison against the architecture model	1.5w
5	Bottleneck profiling → Bottleneck profiling results	1.5w <i>runs in parallel</i>
6	Gap analysis and attribution → Gap analysis and attribution	2w

Step 6 is where the program learns something durable. A gap attributed to the model improves the next program's projections; a gap attributed to the design becomes an errata or a respin input; an unattributed gap teaches nothing.

WATCH OUT FOR

- ◆ **Performance measured on unrepresentative workloads.** A benchmark that is not what customers run produces a number that does not predict their experience.
- ◆ **Gap reported without attribution.** Nobody can act on it, and the next program repeats the modeling error.
- ◆ **Software limiting the measurement.** An immature stack understates the silicon, and the difference has to be established rather than assumed.
- ◆ **Efficiency ignored.** Performance per watt is the metric this product class is bought on, and raw throughput alone misleads.
- ◆ **Measured at a single operating point.** Real deployment spans conditions, and performance at nominal is not the whole picture.

WHERE THE EFFORT GOES	16 M/M	WHO IS ON IT ~11.6 FTE	CONNECTIONS
Performance measurement	4.0	Performance engineer owns it	DEPENDS ON DEF-03 ARCH-02 DV-09 EVB-09 TEST-07 BU-05 BU-06 BU-07
Model comparison	3.5	Architect	
Gap analysis	3.0	Validation software engineer	FEEDS INTO BU-11 MP-12
Bottleneck profiling	2.5	Design engineer	
Efficiency measurement	2.0	Product marketing	FEEDS BACK INTO TEST-10
Benchmark setup	1.0		

22 BU · SILICON BRING-UP

BU-09

Silicon debug of anomalies and failure analysis requests

TAKES	COSTS	OWNER	CRITICAL PATH
10w w4–w14 of the stage	28.0 M/M ~12.2 people while it runs	Silicon debug engineer Failure analysis engineer approves	yes slipping moves the program

WHY IT EXISTS

Debug what is wrong—reproduce it, isolate it, extract the internal state, find the root cause, and get failure analysis where physical evidence is needed.

This is the activity the whole DFT and debug infrastructure investment exists to serve. It runs alongside every other bring-up activity because anomalies arrive from all of them, and its throughput largely determines how long bring-up takes.

WHAT IT DELIVERS

BU-D5

Failure analysis reports

This activity is the deliverable, and physical FA is its most conclusive form.

NEEDS FIRST

- Anomalies from every bring-up activity
- DFT and trace access from DFT-08 and EVB-06
- Simulation and emulation environments from DV-08 and DV-11
- Design database and RTL from RTL-09
- FA laboratory capability

DONE WHEN

- Root cause established rather than symptom suppressed
- Anomalies reproduced in simulation where possible
- Findings captured so they are debugged once

HOW IT GETS THERE

8 STEPS · 2 LANES

w4	w6	w8	w10	w12	w14
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Anomaly intake and triage process → Anomaly intake and triage records	1w
2	Reproduction and isolation → Reproduction and isolation results	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Simulation and emulation cross-check → Simulation and emulation cross-check findings	2w runs in parallel
4	Internal state extraction via DFT and trace → Extracted internal state	2.5w
5	Design-team escalation and analysis → Design-team analysis	2w runs in parallel
6	Hypothesis test and root cause → Root cause determinations	2.5w
7	Debug knowledge capture → Debug knowledge base	1.5w runs in parallel
8	Failure analysis request and disposition → Failure analysis reports and dispositions	2w

Step 3 is the highest-leverage step in the activity. An anomaly reproduced in simulation can be examined with full visibility and unlimited retries, which is an environment no amount of silicon debug can match.

WATCH OUT FOR

- ◆ **Symptoms treated instead of causes.** A workaround without a root cause hides a mechanism that resurfaces in the field.
- ◆ **No path back to simulation.** Debug then happens only on silicon, with limited visibility and long turnaround per hypothesis.
- ◆ **Design team engaged late.** The people who wrote the logic can often recognize a symptom in minutes, and validation debugs alone for days first.
- ◆ **Units consumed by destructive FA too readily.** Physical analysis destroys the unit, and there are very few.
- ◆ **Findings not captured.** The same anomaly is then debugged twice by different engineers.

WHERE THE EFFORT GOES	28 M/M	WHO IS ON IT	~12.2 FTE	CONNECTIONS
Root cause determination	7.0	Silicon debug engineer	owns it	DEPENDS ON DV-08 DV-11 DFT-08 DFT-11 EVB-06 BU-02 BU-03 BU-04 BU-05
Internal state extraction	6.0	Design engineer		FEEDS INTO BU-10 BU-11 MP-05
Reproduction and isolation	5.0	DFT engineer		TERMS HERE 4 RTL DV DFT FA
Simulation cross-check	4.0	DV engineer		
FA and disposition	3.5	Failure analysis engineer		
Knowledge capture	2.5			

22 BU · SILICON BRING-UP

BU-10 Errata capture, workaround definition and documentation			
TAKES	COSTS	OWNER	CRITICAL PATH
7w	10.0 M/M	Product engineering	no
w10–w17 of the stage	~6.2 people while it runs	Technical writer approves	has float

WHY IT EXISTS

Turn what is wrong into a **documented errata list with workarounds**—the honest account of the silicon's behavior that customers and the next revision both need.

Every part ships with errata. What distinguishes a well-run program is that its errata are known, characterized, worked around and disclosed, rather than discovered by a customer. This activity is where debug findings become a product document.

WHAT IT DELIVERS

BU-D4

Errata list with workarounds

This activity is the deliverable, and it is what ships alongside the product.

NEEDS FIRST

- Root causes from BU-09
- Failure analysis results from BU-09
- Design impact assessment from RTL-09
- Customer use models from DEF-03
- Documentation standards from MP-12

DONE WHEN

- Every workaround validated on silicon, not only described
- Severity assessed with customer use models in view
- Errata versioned against silicon revision

HOW IT GETS THERE

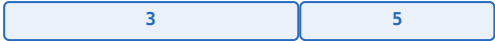
6 STEPS · 2 LANES

w10	w12	w14	w16	w17
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Errata intake and classification → Errata intake and classification	1w
2	Impact and severity assessment → Impact and severity assessment	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Customer-visibility determination → Customer-visibility determination	1.5w runs in parallel
4	Workaround definition and validation → Validated workarounds	1.5w
5	Fix-versus-document decision input → Fix-versus-document input	1w runs in parallel
6	Errata documentation and release → Released errata documentation	3w

Step 4's validation is the part that gets skipped. A workaround that has been described but never run is a hypothesis, and shipping it as a mitigation is how a known issue becomes a field failure anyway.

WATCH OUT FOR

- ◆ **Workarounds documented but never validated.** The mitigation then fails in the field and the errata gave false comfort.
- ◆ **Severity assessed by engineering alone.** Customer impact depends on use models, and the assessment needs that perspective.
- ◆ **Errata suppressed for commercial reasons.** Undisclosed known issues discovered by a customer damage far more than the disclosure would have.
- ◆ **Errata list not versioned against silicon revision.** Which errata apply to which stepping becomes ambiguous the moment a second stepping exists.
- ◆ **Documentation deferred to the end.** The context is freshest at root cause, and reconstructing it months later loses detail.

WHERE THE EFFORT GOES	10 M/M	WHO IS ON IT ~6.2 FTE	CONNECTIONS
Workaround definition and validation	3.0	Product engineering owns it	DEPENDS ON DV-12 T0-04 BU-04 BU-05 BU-06 BU-09
Errata documentation	2.5	Design engineer	FEEDS INTO BU-11 BU-12 MP-12
Impact assessment	2.0	Validation engineer	
Visibility determination	1.5	Product marketing	
Fix decision input	1.0	Technical writer	

22 BU · SILICON BRING-UP

BU-11

Respin versus metal-fix decision analysis — ECO scope, cost, schedule

TAKES	COSTS	OWNER	CRITICAL PATH
4w w13–w17 of the stage	5.0 M/M ~5.4 people while it runs	Program manager Executive sponsor approves	yes slipping moves the program

WHY IT EXISTS

Decide whether the silicon needs a **respin, a metal fix, or nothing**—the single most consequential decision the program makes after tapeout.

A full-mask respin costs a mask set and roughly a quarter of schedule. A metal fix costs far less but can only reach defects in the routing layers. Shipping as-is with workarounds costs nothing now and possibly a great deal later. The decision has to weigh all three against evidence rather than optimism.

WHAT IT DELIVERS

BU-D6

Respin versus metal-fix decision record

This activity is the deliverable, and the record is what the program is held to.

NEEDS FIRST

- Errata list and severity from BU-10
- Root causes from BU-09
- Margin data from BU-07
- Performance gaps from BU-08
- Mask and fab cost from TECH-08

DONE WHEN

- Metal-fix feasibility established physically, not assumed
- Margin evidence weighed alongside functional defects
- Decision recorded with what was known at the time

HOW IT GETS THERE

6 STEPS · 2 LANES

w13	w14	w15	w16	w17
-----	-----	-----	-----	-----

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Defect list and fix scoping → Defect list and fix scope	0.75w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Metal-fix feasibility assessment → Metal-fix feasibility assessment	1w
3	Full-mask respin scoping → Full-mask respin scope	0.75w <i>runs in parallel</i>
4	Cost, schedule and risk comparison → Cost, schedule and risk comparison	1.25w
5	Customer and market impact input → Customer and market impact assessment	0.75w <i>runs in parallel</i>
6	Decision and its record → Recorded decision with its rationale	1w

Step 2 is bounded by physics rather than by will. A metal fix can only change routing layers, and whether the required change fits inside them is a question with a factual answer that has to be established before the decision is framed.

WATCH OUT FOR

- ◆ **Decision made on schedule pressure alone.** Shipping a part with a real defect to protect a date routinely costs more than the delay would have.
- ◆ **Metal-fix feasibility assumed.** It is a physical question about which layers the fix touches, and it is often answered optimistically.
- ◆ **Margin data not weighed.** A part with adequate function and no margin fails in the field, and the shmoo already said so.
- ◆ **Decision unrecorded.** Six months later nobody can reconstruct what was known and what was accepted.
- ◆ **Customer input excluded.** Whether an errata is tolerable depends on what customers do with the part.

WHERE THE EFFORT GOES	5 M/M	WHO IS ON IT	~5.4 FTE	CONNECTIONS
Cost, schedule and risk comparison	1.5	Program manager owns it		DEPENDS ON TECH-08 TO-08 BU-07 BU-08 BU-09 BU-10
Metal-fix feasibility	1.3	Design lead		
Respin scoping	1.0	Product engineering		FEEDS INTO BU-12 MP-09 MP-10
Decision record	0.8	Product marketing		
Market impact input	0.5	Executive sponsor		

22 BU · SILICON BRING-UP

BU-12

Customer sample readiness and release package

TAKES	COSTS	OWNER	CRITICAL PATH
5w w13–w18 of the stage	6.0 M/M ~5.2 people while it runs	Product marketing Program manager approves	no has float

WHY IT EXISTS

Get **samples to customers** with everything they need to use them—screened units, documentation, errata, and a channel to report back.

Customer samples start the evaluation cycle that decides design wins, and they start it months before mass production. What ships with the unit determines whether the customer's first week is productive or spent rediscovering issues the program already knows about.

WHAT IT DELIVERS

BU-D7

Customer sample release package

This activity is the deliverable—units plus everything needed to use them.

NEEDS FIRST

- Allocated sample units from ASSY-09
- Errata list from BU-10
- Decision record from BU-11
- Preliminary datasheet from MP-12
- Customer commitments from DEF-01

DONE WHEN

- Errata pack shipping with every sample
- Units screened before shipment
- Feedback channel established and staffed

HOW IT GETS THERE

6 STEPS · 2 LANES

w13	w14	w15	w16	w17	w18
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Sample readiness criteria → Sample readiness criteria	0.75w
2	Unit selection and screening for samples → Screened and selected sample units	1.25w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Customer documentation and errata pack → Customer documentation and errata pack	1w runs in parallel
4	Release package assembly → Assembled release package	1.5w
5	Feedback channel establishment → Feedback channel	1w runs in parallel
6	Sample shipment and support setup → Shipped samples with support in place	1.5w

Step 5 is the one that pays back most. A customer who finds something and has an obvious route to report it gives the program early field data; one who does not gives it a design loss with no explanation.

WATCH OUT FOR

- ◆ **Samples shipped without errata.** The customer spends their first week rediscovering known issues and loses confidence in the part.
- ◆ **Units not screened before shipping.** A marginal sample creates a customer-visible failure that the program cannot investigate remotely.
- ◆ **Documentation lagging the silicon.** A datasheet describing behavior the sample does not have generates support load and doubt.
- ◆ **No feedback channel.** Customer findings then arrive as rumor or not at all, and the program loses its earliest field data.
- ◆ **Samples shipped before the respin decision.** If a respin follows, customers are evaluating a stepping that will not ship.

WHERE THE EFFORT GOES	6 M/M	WHO IS ON IT	~5.2 FTE	CONNECTIONS
Release package assembly	1.8	Product marketing	OWNS IT	DEPENDS ON ASSY-09 BU-10 BU-11
Unit selection and screening	1.3	Product engineering		FEEDS INTO MP-10 MP-12
Documentation pack	1.3	Applications engineer		
Shipment and support	1.0	Technical writer		
Feedback channel	0.8	Program manager		

VALIDATE & RAMP

23

Qualification & Production

MP

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w106 – w132	26 _w	12	208 _{M/M}	6 of 12
ENGINEERING ACTIVITIES				KEY DELIVERABLES
MP-01	Qualification plan definition against JEDEC / AEC standards		6w 8 M/M	MP-D1 Qualification plan and JEDEC-compliant qualification report
MP-02	Reliability stress — HTOL, HTS, temperature cycle, uHAST, THB		18w 30 M/M	MP-D2 Reliability and package qualification data packages
MP-03	ESD (HBM, CDM) and latch-up qualification		6w 8 M/M	MP-D3 Production test program release
MP-04	Package qualification — MSL, drop, bend, board-level reliability		12w 18 M/M	MP-D4 Yield model versus cost target report
MP-05	Yield learning, failure pareto and defect analysis		20w cont. 40 M/M	MP-D5 Production readiness review signoff
MP-06	Production test program release and guard-band validation		9w 20 M/M	MP-D6 Ramp plan and supply commitment
MP-07	Test time reduction and multi-site conversion		9w 16 M/M	MP-D7 Compliance certificates
MP-08	Process corner and split-lot validation		9w 14 M/M	MP-D8 Datasheet and product documentation set
MP-09	Production readiness review and change control (PCN) setup		6w 8 M/M	MP-D9 Mass Production release record
MP-10	Capacity, supply chain and ramp commitment — wafer starts, substrate, HBM		12w 12 M/M	
MP-11	Compliance and certification — PCIe/CXL compliance, RoHS/REACH, safety		12w 18 M/M	
MP-12	Documentation release — datasheet, user guide, errata, application notes		10w 16 M/M	

23 MP · QUALIFICATION & PRODUCTION

MP-01

Qualification plan definition against JEDEC / AEC standards

TAKES	COSTS	OWNER	CRITICAL PATH
6w w0–w6 of the stage	8.0 M/M ~5.8 people while it runs	Reliability engineer Operations planner approves	yes slipping moves the program

WHY IT EXISTS

Define **what qualification means for this product**—which standards apply, which stresses run, on how many units, and against what acceptance criteria.

Qualification is the evidence that the part survives its intended life. The plan is where that evidence is designed: too little and a customer's own qualification finds what yours did not, too much and months of chamber time are spent on stresses nobody required.

WHAT IT DELIVERS

MP-D1

Qualification plan and JEDEC-compliant qualification report

The plan half of the deliverable; the report follows from executing it.

NEEDS FIRST

- Product requirements and quality target from DEF-04
- Application and use conditions from DEF-03
- Customer qualification requirements
- Package construction from PKGD-11
- Unit availability from ASSY-09

DONE WHEN

- Chamber capacity booked against the full stress duration
- Sample sizes supporting the reliability claim
- Customer-specific requirements included before release

HOW IT GETS THERE

6 STEPS · 2 LANES

w0w2w4w6

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Standard selection — JEDEC, AEC and customer applicability → Applicable standard selection	1w
2	Stress matrix and sample size definition → Stress matrix with sample sizes	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Qualification vehicle and lot definition → Qualification vehicle and lot definition	1w runs in parallel
4	Lab and chamber capacity booking → Chamber and lab bookings	1.5w
5	Customer qualification requirement collection → Customer requirement record · Released qualification plan	1w runs in parallel
6	Plan review and release → Reviewed and released qualification plan	2w

Step 4 is why this activity runs first. HTOL occupies a chamber for a thousand hours, and chambers are shared; a booking made late pushes the whole qualification campaign and therefore mass production.

WATCH OUT FOR

- ◆ **Chamber capacity booked late.** Stress duration is fixed and capacity is shared, so a late booking moves mass production directly.
- ◆ **Sample sizes below the confidence the claim needs.** A reliability claim from too few units does not support the specification.
- ◆ **Customer requirements collected after the plan.** A customer-specific stress discovered later restarts a months-long campaign.
- ◆ **Three lots not planned.** Standards generally require multiple production lots, and a single-lot qualification is not accepted.
- ◆ **Use conditions assumed rather than specified.** The acceleration factors depend on them, and wrong conditions make the whole campaign non-representative.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT	~5.8 FTE	CONNECTIONS	TERMS HERE	3
		Reliability engineer owns it		DEPENDS ON DEF-03 DEF-04 PKGD-11 ASSY-09	HTOL JEDEC AEC	
Stress matrix and sample sizes	2.5	Quality manager		FEEDS INTO MP-02 MP-03 MP-04		
Plan review and release	1.8	Package engineer		MP-09		
Standard selection	1.5	Product marketing		FEEDS BACK INTO TEST-07		
Lab and chamber booking	1.0	Operations planner				
Vehicle definition	0.8					
Customer requirements	0.5					

23 MP · QUALIFICATION & PRODUCTION

MP-02

Reliability stress — HTOL, HTS, temperature cycle, uHAST, THB

TAKES	COSTS	OWNER	CRITICAL PATH
18w w5–w23 of the stage	30.0 M/M ~7.2 people while it runs	Reliability engineer Lab technician approves	yes slipping moves the program

WHY IT EXISTS

Run the **reliability stresses**—HTOL, high-temperature storage, temperature cycling, uHAST, THB—that demonstrate the silicon survives its specified life.

This is eighteen weeks of mostly waiting, and it is the longest fixed-duration item between first silicon and mass production. HTOL alone is a thousand hours; it cannot be shortened, only started earlier or run in parallel.

WHAT IT DELIVERS

MP-D2

Reliability and package qualification data packages

The silicon-reliability half of the deliverable.

NEEDS FIRST

- Qualification plan from MP-01
- Qualification units from ASSY-09
- Pre-stress characterization from TEST-07
- Chamber capacity from MP-01
- Failure analysis capability

DONE WHEN

- Pre-stress baseline recorded for every unit
- Interim readouts taken at the planned intervals
- Every stress failure analyzed to a mechanism

HOW IT GETS THERE

8 STEPS · 2 LANES

w5w9w13w17w21w23

MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Sample preparation and pre-stress characterization → Pre-stress characterization baseline	1.5w
2	Stress chamber loading and campaign start → Loaded stress campaign	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	uHAST and THB execution → uHAST and THB results	3w runs in parallel
4	HTOL execution → HTOL results	6w
5	Interim readouts at defined intervals → Interim readout data	3w runs in parallel
6	Temperature cycle and HTS execution → Temperature cycle and HTS results	4.5w
7	Failure analysis on stress failures → Stress failure analyses	2.5w runs in parallel
8	Final readout, analysis and report → Reliability data package	4w

Step 5's interim readouts are what turn a pass/fail campaign into a learning one. A parameter drifting at 500 hours predicts a failure at 1000 that would otherwise arrive as a surprise at the end of a four-month run.

WATCH OUT FOR

- ◆ **Campaign started late.** The duration is fixed, so every week of delay is a week of mass production delay, one for one.
- ◆ **No pre-stress characterization.** Drift cannot be measured without a baseline, and a failure cannot be distinguished from a part that was always marginal.
- ◆ **Interim readouts skipped.** Failures then arrive only at the end, with no time left to respond before the ramp.
- ◆ **Stress failures not analyzed.** A failure without a mechanism cannot be judged as systematic or as a one-off, and the distinction decides the ramp.
- ◆ **Units insufficient for the full matrix.** A stress dropped for lack of units is a gap in the qualification the customer will find.

WHERE THE EFFORT GOES	30 M/M	WHO IS ON IT	~7.2 FTE	CONNECTIONS	TERMS HERE	4
HTOL execution	8.0	Reliability engineer owns it		DEPENDS ON AMS-14 S0-04 S0-07 PTV-09 SIPI-10 ASSY-09 TEST-07 MP-01	HTOL HTS THB uHAST	
Temperature cycle and HTS	6.0	Test engineer				
Readout and analysis	5.0	Failure analysis engineer		FEEDS INTO MP-05 MP-09 MP-12		
uHAST and THB	4.5	Quality engineer				
Interim readouts	3.5	Lab technician				
Failure analysis	3.0					

23 MP · QUALIFICATION & PRODUCTION

MP-03

ESD (HBM, CDM) and latch-up qualification

TAKES	COSTS	OWNER	CRITICAL PATH
6w w6–w12 of the stage	8.0 M/M ~5.8 people while it runs	ESD engineer Failure analysis engineer approves	no has float

WHY IT EXISTS

Qualify the part's **ESD and latch-up robustness**—HBM and CDM discharge, and latch-up immunity—the handling-survival properties every downstream user depends on.

ESD failures happen in assembly lines, in labs and in the field, and they are almost never traced back to the part that could not take them. The classification levels are what tell a customer's manufacturing engineer how to handle the device.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- ESD protection design from AMS-09 and PD-11
- Package construction from PKGD-11
- Qualification plan from MP-01
- Qualification units from ASSY-09
- ESD test laboratory capability

DONE WHEN

- CDM evaluated against the package's actual capacitance
- Latch-up tested at the hot corner
- Classification meeting customer assembly requirements

HOW IT GETS THERE

6 STEPS · 2 LANES

w6	w8	w10	w12
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	ESD and latch-up test plan and pin classification → ESD and latch-up test plan	1w
2	HBM ESD characterization → HBM ESD classification	1.5w
3	Pin classification and protection review → Pin classification and protection review	1w

runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	CDM ESD characterization → CDM ESD classification	1.5w
5	Failure analysis on ESD failures → ESD failure analyses	1w <i>runs in parallel</i>
6	Latch-up test and report → Latch-up test results and report	2w

Step 4 matters more than step 2 on a package this size. CDM stress scales with package capacitance, and a large 2.5D package charges to a far higher energy than a small one—the CDM level is usually the binding constraint.

WATCH OUT FOR

- ◆ **CDM underestimated for the package size.** A large package stores far more charge, and the CDM level is usually the binding one.
- ◆ **ESD failures not analyzed.** Which pin and which protection element failed is what makes the result actionable for a fix.
- ◆ **Latch-up tested only at room temperature.** Latch-up susceptibility increases with temperature, and the hot corner is the real test.
- ◆ **Classification below what customer assembly requires.** A part that cannot survive a customer's line is a part they cannot buy.
- ◆ **Pin classification incomplete.** Untested pin combinations are unqualified pins, and there is no assumption to fall back on.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT	~5.8 FTE	CONNECTIONS	TERMS HERE	4
HBM ESD characterization	2.0	ESD engineer <i>OWNS IT</i>		DEPENDS ON PDK-07 AMS-09 AMS-14 PD-11 S0-07 PKG0-11 ASSY-09 MP-01	ESD HBM (ESD) CDM HBM	
CDM ESD characterization	2.0	Circuit designer				
Latch-up test	1.8	Package engineer		FEEDS INTO MP-09 MP-12		
Pin classification	1.0	Quality engineer				
Test plan	0.8	Failure analysis engineer				
Failure analysis	0.5					

23 MP · QUALIFICATION & PRODUCTION

MP-04

Package qualification — MSL, drop, bend, board-level reliability

TAKES	COSTS	OWNER	CRITICAL PATH
12w w8–w20 of the stage	18.0 M/M ~6.5 people while it runs	Package reliability engineer Failure analysis engineer approves	no has float

WHY IT EXISTS

Qualify the **package**—moisture sensitivity, board-level reliability, drop and bend—which on a 2.5D structure is where the reliability risk actually concentrates.

The silicon's reliability is a well-understood function of process and use conditions. The package's is not: a large interposer on an organic substrate with two HBM stacks has thermal expansion mismatches on every interface, and this is where they are proven or found out.

WHAT IT DELIVERS

MP-D2

Reliability and package qualification data packages

The package half of the deliverable.

NEEDS FIRST

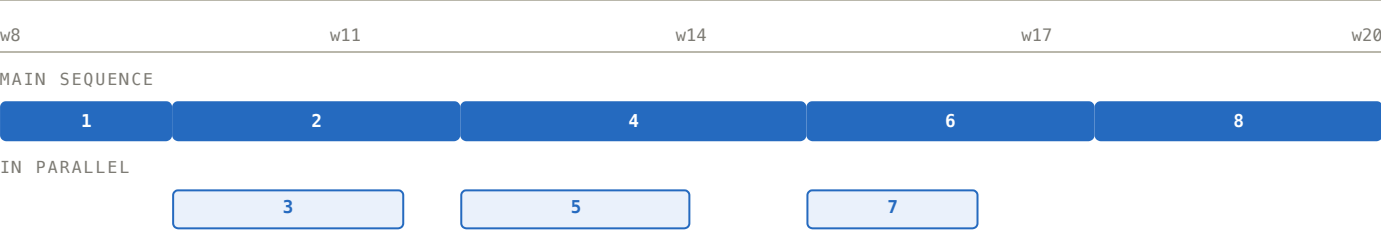
- Qualification plan from MP-01
- Package units from ASSY-09
- PTV reliability results from PTV-10
- Package construction from PKGD-11
- Board-level test capability

DONE WHEN

- MSL classification compatible with customer assembly
- Board-level life meeting the application requirement
- Results compared against the PTV prediction

HOW IT GETS THERE

8 STEPS · 2 LANES



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	MSL preconditioning and moisture classification → MSL classification	1.5w
2	Board-level reliability — temperature cycling on board → Board-level reliability results	2.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Solder joint reliability modeling correlation → Solder joint reliability model correlation	2w runs in parallel
4	Drop and bend testing → Drop and bend results	3w
5	Warpage and co-planarity tracked through stress → Warpage and co-planarity through stress	2w runs in parallel
6	Package-level readout and acoustic imaging → Package readout and acoustic imaging	2.5w
7	Failure analysis on package failures → Package failure analyses	1.5w runs in parallel
8	Analysis and qualification report → Package qualification report	2.5w

Step 3 is what turns this campaign into a predictive model rather than a single data point. Correlating measured joint life against the model means the next package variant can be assessed without repeating twelve weeks of testing.

WATCH OUT FOR

- ◆ **MSL classification worse than the assembly process needs.** A high moisture sensitivity forces bake-and-handle procedures on every customer line.
- ◆ **Board-level reliability short of the application requirement.** The joints, not the die, then set the product's life.
- ◆ **PTV results not compared.** The vehicles predicted this behavior, and the comparison validates or invalidates the whole vehicle program.
- ◆ **Warpage measured only before stress.** The interesting change is what stress does to it, and that requires measurement through the campaign.
- ◆ **Failures attributed to the package without analysis.** A CPI failure and a solder joint failure need different fixes and look similar from outside.

WHERE THE EFFORT GOES	18 M/M	WHO IS ON IT	~6.5 FTE	CONNECTIONS	TERMS HERE	5
Board-level reliability	4.5	Package reliability engineer	owns it	DEPENDS ON PKGD-11 PTV-08 PTV-10 PTV-12 ASSY-06 ASSY-09 MP-01	CPI PTV MSL HBM (ESD) HBM	
Drop and bend testing	3.5	Package engineer				
MSL preconditioning	3.0	Materials engineer				
Readout and acoustic imaging	2.5	Quality engineer		FEEDS INTO MP-09 MP-12		
Joint reliability correlation	2.5	Failure analysis engineer		FEEDS BACK INTO PTV-12		
Warpage tracking	2.0					

23 MP · QUALIFICATION & PRODUCTION

MP-05

Yield learning, failure pareto and defect analysis

TAKES	COSTS	OWNER	CRITICAL PATH
20w cont. w2–w22 of the stage	40.0 M/M ~8.7 people while it runs	Yield engineer Cost analyst approves	yes slipping moves the program

WHY IT EXISTS

Run **yield learning** continuously through the ramp—pareto, defect analysis, corrective action, and a yield model checked against the cost target.

Yield is the difference between a product that makes money and one that does not, and first-ramp yield is never where the cost model assumed. This activity runs for the whole stage because yield learning is a rate, not an event: what matters is how fast the pareto's top item is removed and replaced by the next.

WHAT IT DELIVERS

MP-D4

Yield model versus cost target report

This activity is the deliverable, and the model is what the business plan is checked against.

NEEDS FIRST

- Test data infrastructure from TEST-11
- Fab yield data from FAB-09
- Assembly yield from ASSY-07
- Bring-up failure analysis from BU-09
- Cost model from DEF-06

DONE WHEN

- Yield split by die, assembly and test
- Corrective actions verified against the pareto
- Yield model reconciled with the cost target

HOW IT GETS THERE

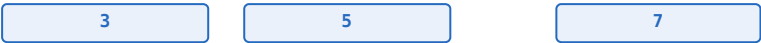
8 STEPS · 2 LANES

w2	w6	w10	w14	w18	w22
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Yield data pipeline and baseline establishment → Yield data pipeline and baseline	2w
2	Failure pareto by bin and failure mode → Failure pareto by bin and mode	3.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Wafer-level spatial signature analysis → Spatial signature analysis	3w runs in parallel
4	Defect analysis and root cause → Defect root causes	4.5w
5	Assembly and test yield split → Assembly and test yield split	3w runs in parallel
6	Corrective action and yield improvement → Corrective actions and improvement record	5w
7	Excursion detection and response → Excursion detection and responses	3w runs in parallel
8	Yield model against the cost target → Yield model against cost target	5w

Step 5 is the split that decides who acts. Die yield, assembly yield and test yield have different owners and different fixes, and a combined number tells none of them what to do.

WATCH OUT FOR

- ◆ **Yield reported as one number.** Die, assembly and test yield have different owners, and a combined figure directs nobody.
- ◆ **Spatial signatures not analyzed.** Edge, center and radial patterns each point at a different process cause and are visible only on the wafer map.
- ◆ **Corrective actions not verified.** An action taken without confirming the pareto moved is an action that may have done nothing.
- ◆ **Excursions detected at the weekly review.** By then a week of material is affected, and detection has to be automatic.
- ◆ **Yield model not checked against the cost target.** The gap between achieved and assumed yield is the product's margin, and it can go unnoticed for months.

WHERE THE EFFORT GOES	40 M/M	WHO IS ON IT	~8.7 FTE	CONNECTIONS
Defect analysis and root cause	10.0	Yield engineer owns it		DEPENDS ON DEF-04 DEF-06 TECH-07 AMS-08 DFT-03 DFT-11 S0-08 FAB-06 FAB-07 FAB-08 FAB-09 PTV-11 ASSY-07 TEST-11 BU-02 BU-09 MP-02 MP-08
Corrective action and improvement	9.0	Process engineer		
Failure pareto	7.0	Assembly engineer		
Yield model	6.0	Test engineer		
Spatial signature analysis	4.5	Cost analyst		FEEDS INTO MP-06 MP-09 MP-10
Excursion response	3.5			

23 MP · QUALIFICATION & PRODUCTION

MP-06 Production test program release and guard-band validation			
TAKES	COSTS	OWNER	CRITICAL PATH
9w	20.0 M/M	Product engineering	yes
w10–w19 of the stage	~9.7 people while it runs	Operations approves	slipping moves the program

WHY IT EXISTS

Release the **production test program**—frozen, guard-band-validated, qualified and version-controlled—for every unit the product will ever ship.

After this release, a change to test content requires re-qualification. That makes the release a genuine freeze, and it makes the guard-band validation that precedes it the last chance to be sure the limits neither throw away good parts nor let bad ones through.

WHAT IT DELIVERS

MP-D3

Production test program release

This activity is the deliverable, and after it the content is frozen.

NEEDS FIRST

- Sort and final programs from TEST-05 and TEST-06
- Characterization data from TEST-07 and BU-07
- Correlation results from TEST-10
- Yield data from MP-05
- Production tester fleet from TEST-02

DONE WHEN

- Guard bands derived from characterization, not convention
- Both escape and over-kill quantified
- Flow validated across multiple lots and testers

HOW IT GETS THERE

8 STEPS · 2 LANES

w10	w12	w14	w16	w18	w19
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Release candidate program freeze → Frozen release candidate	1w
2	Guard-band validation against characterization → Guard-band validation results	2w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Test escape and over-kill analysis → Escape and over-kill analysis	1.5w runs in parallel
4	Production flow validation on volume lots → Production flow validation on volume lots	2w
5	Limit adjustment and re-validation → Limit adjustments	1.5w runs in parallel
6	Program qualification and signoff → Program qualification and signoff	2w
7	Production tester porting and correlation → Tester porting and correlation	1.25w runs in parallel
8	Release and version control → Released, version-controlled production programs	2w

Step 3 quantifies both directions of error. Escapes cost the customer's trust and over-kill costs yield, and a limit set without measuring both is optimizing one at the other's expense without knowing it.

WATCH OUT FOR

- ◆ **Guard bands not validated against characterization.** Limits then rest on convention rather than on what the silicon actually does.
- ◆ **Over-kill unmeasured.** Good parts are discarded on every lot and the loss is attributed to yield rather than to the limits.
- ◆ **Flow validated on one lot.** Lot-to-lot variation is the thing the flow has to tolerate, and one lot cannot show it.
- ◆ **Porting across the tester fleet without correlation.** Tester-to-tester differences then present as yield differences by machine.
- ◆ **Released without version control.** A yield excursion cannot be correlated with a program change nobody can identify.

WHERE THE EFFORT GOES	20 M/M	WHO IS ON IT ~9.7 FTE	CONNECTIONS
Guard-band validation	5.0	Product engineering owns it	DEPENDS ON DFT-07 DFT-08 DFT-10 TEST-05 TEST-06 TEST-08 TEST-09 TEST-10 BU-07 MP-05 MP-08
Production flow validation	4.0	Test engineer	FEEDS INTO MP-07 MP-09 MP-10
Program qualification	3.5	Quality engineer	TERMS HERE 1 DPPM
Escape and over-kill analysis	3.0	Yield engineer	
Release and version control	2.5	Operations	
Tester porting	2.0		

23 MP · QUALIFICATION & PRODUCTION

MP-07

Test time reduction and multi-site conversion

TAKES	COSTS	OWNER	CRITICAL PATH
9w w14–w23 of the stage	16.0 M/M ~7.7 people while it runs	Test engineer Yield engineer approves	no has float

WHY IT EXISTS

Convert production test to **multi-site** and cut test time—the change that makes the cost per unit sustainable at volume.

Single-site testing is fine for a hundred units and ruinous for a million. Multi-site conversion multiplies throughput per tester and divides the capital needed for the ramp, and it is a real engineering exercise because parallel sites interfere with each other in ways single-site testing never reveals.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Released production programs from MP-06
- Tester pin count and configuration from TEST-02
- Test time analysis from TEST-09
- Load board and probe card designs from TEST-03 and TEST-04
- Volume forecast from MP-10

DONE WHEN

- Site-to-site bias measured and within tolerance
- Parallel power delivery adequate at full current
- Conversion complete before capital is committed to the ramp

HOW IT GETS THERE

8 STEPS · 2 LANES

w14	w16	w18	w20	w22	w23
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Multi-site feasibility on the production hardware → Multi-site feasibility assessment · Converted multi-site hardware	1w
2	Multi-site load board and probe card conversion → Converted multi-site load board and probe card	2w
3	Test time reduction on the released content → Test time reduction on released content	1.5w runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Parallel program adaptation → Adapted parallel programs	2w
5	Site-to-site variation analysis → Site-to-site variation analysis	1.5w <i>runs in parallel</i>
6	Multi-site correlation and validation → Multi-site correlation and validation	2w
7	Throughput and capacity model update → Updated throughput and capacity model	1.25w <i>runs in parallel</i>
8	Release of the converted flow → Released converted flow	2w

Step 5 is what makes the conversion trustworthy. Sites differ—in contact, in power delivery, in noise coupling—and a systematic site bias shows up as a yield difference that gets chased as a process problem for weeks.

WATCH OUT FOR

- ◆ **Site-to-site bias undetected.** It appears as a yield difference and is investigated as a process problem for weeks.
- ◆ **Pin count insufficient for the site count.** The tester configuration from TEST-02 bounds this, and it cannot be changed now.
- ◆ **Parallel power delivery inadequate.** Four parts drawing full current through one load board droop in ways one part does not.
- ◆ **Conversion done after the ramp starts.** Capital is then committed at single-site throughput and cannot be recovered.
- ◆ **Time reduction changing coverage.** Any content change after MP-06 requires re-qualification, which is expensive at this point.

WHERE THE EFFORT GOES	16 M/M	WHO IS ON IT ~7.7 FTE	CONNECTIONS
Multi-site hardware conversion	4.0	Test engineer OWNS IT	DEPENDS ON TEST-02 TEST-03 TEST-04 TEST-09 MP-06
Parallel program adaptation	3.5	Test hardware engineer	FEEDS INTO MP-09 MP-10
Correlation and validation	3.0	Product engineering	
Test time reduction	2.5	Operations planner	
Site variation analysis	1.8	Yield engineer	
Capacity model	1.3		

23 MP · QUALIFICATION & PRODUCTION

MP-08

Process corner and split-lot validation

TAKES	COSTS	OWNER	CRITICAL PATH
9w w8–w17 of the stage	14.0 M/M ~6.8 people while it runs	Product engineering Yield engineer approves	no has float

WHY IT EXISTS

Validate the design across **process corners** using split lots—material deliberately fabricated at the edges of the process window.

Production silicon spans the process distribution, and the units bring-up characterized came from one lot near its center. Split lots produce the fast and slow corners on purpose, which is the only way to know whether the design's margin survives the population it will actually be built from.

WHAT IT DELIVERS

No key deliverable is owned here — this activity contributes to the ones below.

NEEDS FIRST

- Split lot material from FAB-06
- Corner models from PDK-05
- Characterization content from TEST-07
- Shmoo baseline from BU-07
- Process window from TECH-05

DONE WHEN

- Split material verified to be at the intended corners
- Cpk computed against the specification
- Model-to-silicon correlation established at both corners

HOW IT GETS THERE

8 STEPS · 2 LANES

w8	w10	w12	w14	w16	w17
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Split-lot design across process corners → Split-lot design	1w
2	Split lot fabrication and tracking → Fabricated and tracked split lots	2w
3	Parametric spread and Cpk analysis → Parametric spread and Cpk analysis	1.5w

runs in parallel

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
4	Corner characterization on split material → Corner characterization results	2w
5	Design margin at the corners → Design margin at the corners	1.5w runs in parallel
6	Corner-to-model correlation → Corner-to-model correlation	2w
7	Process window recommendation → Process window recommendation	1.25w runs in parallel
8	Corner signoff and report → Corner signoff report	2w

Step 6 closes a loop opened right back at signoff. The design was timed against corner models; measuring real corner silicon says whether those models were pessimistic, accurate, or—the dangerous case—optimistic.

WATCH OUT FOR

- ◆ **Corner models optimistic against real silicon.** The design was signed off against those models, and a gap here means the margin does not exist.
- ◆ **Split lots not actually at the corners.** Material that drifted back toward center proves nothing about the edges.
- ◆ **Cpk not computed.** Capability against the specification is what predicts production yield, and a pass/fail result does not.
- ◆ **Corner characterization skipped for schedule.** The population's edges are then only seen in production, as an unexplained yield tail.
- ◆ **Process window recommendation not fed back to the fab.** The data exists and the process keeps running at its original center point.

WHERE THE EFFORT GOES	14 M/M	WHO IS ON IT	~6.8 FTE	CONNECTIONS
Corner characterization	3.5	Product engineering owns it		DEPENDS ON TECH-05 PDK-05 FAB-06 FAB-07 TEST-07 TEST-10 BU-07
Split lot execution	3.0	Process engineer		FEEDS INTO MP-05 MP-06 MP-09
Corner-to-model correlation	2.5	Characterization engineer		TERMS HERE 1 Shmoo
Cpk analysis	2.0	Design engineer		
Design margin assessment	1.8	Yield engineer		
Window recommendation	1.3			

23 MP · QUALIFICATION & PRODUCTION

MP-09

Production readiness review and change control (PCN) setup

TAKES	COSTS	OWNER	CRITICAL PATH
6w w18–w24 of the stage	8.0 M/M ~5.8 people while it runs	Program manager Executive sponsor approves	yes slipping moves the program

WHY IT EXISTS

Hold the **production readiness review**—the cross-functional gate where every function states whether it is ready—and stand up change control for everything that follows.

This is the program's last formal decision point. After it, the product is in production and every change goes through customer notification. The review's value is that it forces each function to say ready or not ready on the record, which is a different act from being asked in a meeting whether there are any concerns.

WHAT IT DELIVERS

- MP-D5

Production readiness review signoff

This activity is the deliverable—the record of who declared ready and on what evidence.
- MP-D9

Mass Production release record

The formal statement that the product is in production, with its conditions.

NEEDS FIRST

- Qualification results from MP-02, MP-03 and MP-04
- Test program release from MP-06
- Yield model from MP-05
- Supply commitment from MP-10
- Compliance certificates from MP-11

DONE WHEN

- Every conditional item with an owner, a date and a consequence
- Change control and PCN process established before release
- Dissenting positions recorded, not resolved off the record

HOW IT GETS THERE

6 STEPS · 2 LANES

w18	w20	w22	w24
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Readiness criteria and evidence collection → Readiness criteria and collected evidence	1w
2	Cross-functional readiness review → Cross-functional review record	1.5w
3	Change control and document baseline → Change control and document baseline	1w <i>runs in parallel</i>
4	Gap closure and conditional item tracking → Gap closure and conditional item tracking	1.5w
5	Escalation of unresolved items → Escalation record	1w <i>runs in parallel</i>
6	Signoff and PCN process establishment → Signed-off readiness and established PCN process · Mass production release record	2w

Step 4's conditional items are where a review either works or becomes theatre. A conditional pass with an owner, a date and a consequence is a real mechanism; a conditional pass without them is an unconditional pass with a note attached.

WATCH OUT FOR

- ◆ **Review held without evidence.** Function-by-function assurances are not the same as qualification data, and a review that accepts them is not a gate.
- ◆ **Conditional items without owners or dates.** The condition is then never closed and the pass was unconditional in practice.
- ◆ **Change control not established.** Post-production changes then reach customers without notification, which is a contractual failure.
- ◆ **Dissent not recorded.** A function that said not-ready and was overruled is the program's most valuable record if something goes wrong.
- ◆ **Document baseline not frozen.** Which datasheet, which errata, which test program was released becomes ambiguous immediately.

WHERE THE EFFORT GOES	8 M/M	WHO IS ON IT ~5.8 FTE	CONNECTIONS	TERMS HERE	1
Cross-functional review	2.3	Program manager OWNS IT	DEPENDS ON DFT-10 PD-15 S0-10	PCN	
Evidence collection	1.8	Quality manager	T0-04 T0-05 TEST-11		
Gap closure and tracking	1.5	Function leads	BU-11 MP-01 MP-02		
Signoff and PCN setup	1.3	Document control	MP-03 MP-04 MP-05		
Change control baseline	0.8	Executive sponsor	MP-06 MP-07 MP-08		
Escalation	0.5		MP-10 MP-11 MP-12		

23 MP · QUALIFICATION & PRODUCTION

MP-10

Capacity, supply chain and ramp commitment — wafer starts, substrate, HBM

TAKES	COSTS	OWNER	CRITICAL PATH
12w w10–w22 of the stage	12.0 M/M ~4.3 people while it runs	Operations manager Program manager approves	yes slipping moves the program

WHY IT EXISTS

Commit the **supply chain and the ramp**—wafers, substrates, HBM, assembly and test capacity—against a demand forecast, with the lead times and second sources that make the commitment real.

A ramp is a chain of commitments made months before the volume they serve. Substrates have long lead times, HBM is allocated across the industry, and assembly capacity is booked. Any one of them short stops the ramp, and none of them can be fixed quickly.

WHAT IT DELIVERS

MP-D6

Ramp plan and supply commitment

This activity is the deliverable.

NEEDS FIRST

- Demand forecast from DEF-05 and product marketing
- Yield model from MP-05
- Test throughput from MP-07
- HBM supply position from ASSY-02
- Substrate and interposer supply from PKGD-09

DONE WHEN

- Every long-lead item committed with buffer
- Ramp built on the achieved yield, not the assumed one
- Single-source nodes identified with a contingency

HOW IT GETS THERE

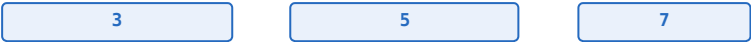
8 STEPS · 2 LANES

w10	w13	w16	w19	w22
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Demand forecast and volume commitment → Demand forecast and volume commitment	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Wafer, substrate and HBM supply commitment → Wafer, substrate and HBM supply commitments	2.5w
3	Lead time and buffer analysis → Lead time and buffer analysis	2w runs in parallel
4	Assembly and test capacity commitment → Assembly and test capacity commitments	2.5w
5	Cost per unit at volume → Cost per unit at volume	2w runs in parallel
6	Ramp plan and build schedule → Ramp plan and build schedule	2.5w
7	Logistics and distribution setup → Logistics and distribution setup	1.5w runs in parallel
8	Supply chain risk and second-source plan → Supply chain risk and second-source plan	3w

Step 8 runs last but covers everything before it. Each commitment made in steps 2 and 4 is a single point of failure, and the plan has to say what happens if any one of them fails—before it does.

WATCH OUT FOR

- ◆ **HBM allocation insufficient.** It is allocated across the industry and a shortfall stops the ramp regardless of everything else.
- ◆ **Substrate lead time underestimated.** Advanced substrates are long-lead and capacity-constrained, and the buffer has to reflect that.
- ◆ **Ramp planned against an optimistic yield.** Volume commitments then require more wafer starts than were booked.
- ◆ **No second source anywhere in the chain.** Every single-source node is a single point of failure for the whole product.
- ◆ **Capacity committed before the readiness review.** A conditional readiness outcome then meets an unconditional supply commitment.

WHERE THE EFFORT GOES	12 M/M	WHO IS ON IT	~4.3 FTE	CONNECTIONS	TERMS HERE	2
Supply commitment	3.0	Operations manager owns it		DEPENDS ON DEF-05 TECH-04 TECH-05 PKGD-09 PKGD-11 ASSY-02 ASSY-07 BU-11 BU-12 MP-05 MP-06 MP-07	HBM (ESD) HBM	
Capacity commitment	2.5	Procurement				
Ramp plan	2.0	Product marketing				
Risk and second-source plan	1.8	Cost analyst		FEEDS INTO MP-09		
Lead time analysis	1.5	Program manager				
Cost at volume	1.3					

23 MP · QUALIFICATION & PRODUCTION

MP-11

Compliance and certification — PCIe/CXL compliance, RoHS/REACH, safety

TAKES	COSTS	OWNER	CRITICAL PATH
12w w10–w22 of the stage	18.0 M/M ~6.5 people while it runs	Compliance engineer Product marketing approves	no has float

WHY IT EXISTS

Obtain the **compliance and certification** the product cannot ship without—interface compliance, EMC and safety, and materials declarations.

These are gates rather than engineering activities: without a PCIe or CXL compliance listing, without EMC certification, without RoHS and REACH declarations, the product cannot be sold in most markets regardless of how well it works.

WHAT IT DELIVERS

MP-D7

Compliance certificates

This activity is the deliverable, and without it the product cannot ship.

NEEDS FIRST

- Interface compliance results from BU-05
- Materials declarations from the supply chain
- Package construction from PKGD-11
- Applicable regulatory requirements
- Certification body and plugfest schedules

DONE WHEN

- Compliance listing obtained, not just tested
- Materials declarations collected for every component
- Export classification completed before shipment

HOW IT GETS THERE

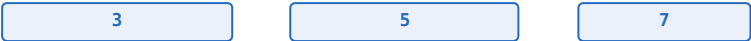
8 STEPS · 2 LANES

w10	w13	w16	w19	w22
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Applicable standards and certification scope → Applicable standards and scope	1.5w
2	PCIe/CXL compliance testing at a plugfest → PCIe/CXL compliance test results	2.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
3	Compliance failure remediation → Remediation record	2w runs in parallel
4	EMC and safety testing → EMC and safety test results	2.5w
5	Documentation and declaration preparation → Compliance documentation and declarations	2w runs in parallel
6	Materials compliance — RoHS, REACH, conflict minerals → Materials compliance declarations	2.5w
7	Export classification → Export classification	1.5w runs in parallel
8	Certificate issue and registration → Issued certificates and registrations	3w

Step 2 depends on an event calendar the program does not control. Plugfests happen on fixed dates a few times a year, and missing one costs a quarter rather than a week.

WATCH OUT FOR

- ◆ **Plugfest date missed.** They occur a few times a year on fixed dates, and missing one delays the listing by a quarter.
- ◆ **Materials declarations not collected from suppliers.** Every component in the package needs one, and collecting them takes months.
- ◆ **Compliance failure discovered at the plugfest.** Pre-compliance testing exists precisely so that formal testing is a confirmation.
- ◆ **Export classification left late.** A high-performance accelerator may be controlled, and the classification determines which markets exist.
- ◆ **Certification scope assessed for one region.** Requirements differ by market, and a missing certification closes a region entirely.

WHERE THE EFFORT GOES	18 M/M	WHO IS ON IT	~6.5 FTE	CONNECTIONS	TERMS HERE	5
PCIe/CXL compliance	4.5	Compliance engineer	OWNS IT	DEPENDS ON DEF-05 ARCH-07 PKGD-11 PTV-10 SIPI-08 ASSY-02 BU-05	EMC RoHS REACH PCIe CXL	
EMC and safety testing	4.0	Interface engineer				
Materials compliance	3.5	Regulatory affairs				
Certificate issue and registration	2.5	Supply chain		FEEDS INTO MP-09 MP-12		
Failure remediation	2.0	Product marketing				
Documentation and export	1.5					

23 MP · QUALIFICATION & PRODUCTION

MP-12

Documentation release — datasheet, user guide, errata, application notes

TAKES	COSTS	OWNER	CRITICAL PATH
10w w14–w24 of the stage	16.0 M/M ~7.0 people while it runs	Technical writer Product marketing approves	no has float

WHY IT EXISTS

Release the **product documentation**—datasheet, user guide, errata, application notes—the artefacts a customer actually receives alongside the silicon.

The documentation is the product as far as most customers experience it. A datasheet parameter that cannot be traced to a measurement, a programming model that omits a required sequence, or an errata list that lags the silicon each cost far more support effort than the writing would have.

WHAT IT DELIVERS

MP-D8

Datasheet and product documentation set

This activity is the deliverable.

NEEDS FIRST

- Characterization data from TEST-07 and BU-07
- Errata list from BU-10
- Performance results from BU-08
- Compliance classifications from MP-11
- Programming model from RTL-06 and ARCH-06

DONE WHEN

- Every datasheet parameter traceable to a measurement
- Errata current with the shipping silicon
- Documentation under revision and distribution control

HOW IT GETS THERE

8 STEPS · 2 LANES

w14	w16	w18	w20	w22	w24
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MAIN SEQUENCE



IN PARALLEL



#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
1	Documentation scope and ownership → Documentation scope and ownership · Reviewed and released documentation set	1.5w

#	STEP, AND WHAT IT ADDS TO THE DELIVERABLE	TAT
2	Datasheet from characterization data → Datasheet	2w
3	Parameter traceability to measurement → Parameter traceability record	1.5w runs in parallel
4	User guide and programming model → User guide and programming model documentation	2w
5	Localization and format production → Localized and formatted documentation outputs	1.5w runs in parallel
6	Errata and application notes → Errata and application notes	2w
7	Revision and distribution control → Revision and distribution control	1.5w runs in parallel
8	Technical review, approval and release → Localized and formatted outputs	2.5w

Step 3 is what makes the datasheet defensible. Every number should point at a measurement, its conditions and its sample size, so that a customer's question about a parameter has an answer rather than a search.

WATCH OUT FOR

- ◆ **Datasheet parameters without traceable measurements.** A customer question then has no answer, and the number cannot be defended.
- ◆ **Errata lagging the silicon.** The most useful document ships out of date, which is worse than shipping it late.
- ◆ **Programming model incomplete.** Required initialization sequences omitted from the guide become support cases, one per customer.
- ◆ **Documentation written from specifications rather than measurements.** It then describes the intended product rather than the one shipping.
- ◆ **No revision control on released documents.** Which datasheet revision a customer holds becomes unknowable.

WHERE THE EFFORT GOES	16 M/M	WHO IS ON IT	~7.0 FTE	CONNECTIONS
Datasheet	4.5	Technical writer	owns it	DEPENDS ON TEST-07 TEST-10 BU-06 BU-07 BU-08 BU-10 BU-12 MP-02 MP-03 MP-04 MP-11
User guide	3.5	Product engineering		FEEDS INTO MP-09
Errata and application notes	2.8	Applications engineer		
Review and release	2.5	Design engineer		
Parameter traceability	1.8	Product marketing		
Revision control	1.0			

Glossary

TERMS

Every abbreviation used anywhere in this document, expanded, with what it is and why a program manager would care. 184 entries. Each activity page lists the terms it uses.

PROGRAM AND COMMERCIAL

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ASP	Average Selling Price What the part is expected to sell for. With yield and cost it decides whether the program has a margin.
BOM	Bill of Materials The itemized list of what goes into the product. An IP BOM lists every IP block the die needs and where it comes from.
BSP	Board Support Package The minimum software that lets a host boot and talk to the board. Its boundary with the firmware team has to be agreed, not assumed.
CI	Continuous Integration Automated build-and-test on every change. In hardware it usually means nightly regressions rather than per-commit.
DOE	Design Of Experiments A structured set of experimental splits chosen so each variable's effect can be separated. How process windows get characterized.
FTE	Full-Time Equivalent One person working full time. Derived here as $M/M \div TAT$ in months — the team size implied while the activity runs.
KPI	Key Performance Indicator A measurable target derived from a requirement. A requirement without a KPI cannot be tested at the end.
M/M	Man-Month One person working for one month. The unit of effort in this template; summing it across a stage gives the headcount the stage consumes.
NDA	Non-Disclosure Agreement The contract that has to exist before a foundry, IP vendor or customer will share anything useful. Routinely on the critical path early.
NRE	Non-Recurring Engineering One-off cost — masks, tooling, IP licences — as opposed to per-unit cost. Dominates the business case for an advanced node.
PCN	Product Change Notification The formal notice to customers that something about a shipping product has changed. Once in production, changes go through it.
PO	Purchase Order The binding commitment to buy. Long-lead items (masks, substrates, HBM) need one far earlier than intuition suggests.

PPA	Power, Performance, Area The three quantities every chip design trades against each other. A "PPA target" is the contract the architecture must meet.
PRD	Product Requirements Document The baselined statement of what the product must do. Everything downstream is an answer to it.
RFI	Request For Information An early, non-binding enquiry to a supplier — used to scope options before quoting.
RFQ	Request For Quotation A formal request for price and terms. Precedes a purchase order.
SW/FW	Software / Firmware Out of scope for this template, which covers design and silicon enablement only — but the handoff boundary still has to be agreed.
TAT	Turn-Around Time Elapsed calendar time an activity occupies, in weeks. Not the same as effort — a 20-week activity may take one person or ten.
WIP	Work In Progress Material already started but not finished. In a fab it is weeks of committed capacity that cannot be redirected.

PROCESS, FOUNDRY AND MANUFACTURING

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BEOL	Back End Of Line The metal interconnect layers built above the transistors. Taped out about a month after FEOL to buy design-fix time.
CMP	Chemical Mechanical Planarization The polishing step that keeps each layer flat. Drives metal-density fill rules in layout.
CPI	Chip-Package Interaction Mechanical stress where die and package meet, driven by their different thermal expansion. The main reliability risk in 2.5D construction.
CoWoS	Chip-on-Wafer-on-Substrate A 2.5D packaging technology: die placed on a silicon interposer, which sits on the substrate. The construction assumed here.
DK	Design Kit Used interchangeably with PDK; sometimes the assembly or package equivalent supplied by the OSAT.
DRM	Design Rule Manual The foundry document defining what geometry is legal. Its version has to match the DRC deck actually being run.
DTCO	Design-Technology Co-Optimization Tuning design style and process options together rather than treating the process as fixed. Where standard-cell track height gets decided.
EUV	Extreme Ultraviolet lithography The lithography used at advanced nodes. Its layer count drives mask cost and cycle time.
FEOL	Front End Of Line The transistor-forming steps of wafer fabrication. Taped out first in a split MTO, because it has the longest cycle time.

HD	High Density A standard-cell flavor optimized for area rather than speed; HPC is its high-performance counterpart. Process option menus offer both.
KGD	Known Good Die A die screened at wafer sort as worth packaging. On a 2.5D part an escaped bad die costs an entire assembled module.
MDP	Mask Data Preparation Turning the design database into mask-writer input — fracturing, OPC, job deck. Weeks of work between tapeout and first wafer.
MPW	Multi-Project Wafer A shared mask set carrying several designs. The cheap way to get a test chip fabricated.
MTO	Mask Tape-Out The release of the design database to mask making. The irreversible commitment — masks cost millions and take weeks.
MTV	Mechanical Test Vehicle A test vehicle for warpage, stress and assembly mechanics rather than electrical behavior.
OPC	Optical Proximity Correction Pre-distorting mask shapes so the printed result matches intent. Part of mask data prep, after tapeout.
OSAT	Outsourced Semiconductor Assembly and Test The subcontractor that packages and tests the die. Owns the assembly process knowledge the program depends on.
PCM	Process Control Monitor The test structures WAT measures. Their readings are the earliest signal that a lot drifted.
PDK	Process Design Kit The foundry's package of models, rules and libraries that makes a process usable. Its version and maturity gate everything downstream.
RDL	Redistribution Layer Fine wiring on the die or interposer surface that moves connections to where the bumps are.
TSV	Through-Silicon Via A vertical connection through a die or interposer. What makes stacked and 2.5D construction possible.
TTV	Thermal Test Vehicle A test vehicle carrying heaters and sensors, used to characterize the package thermal path before real silicon exists.
TV	Test Vehicle A structure built purely to learn about a process, not to sell. Retires risk before the real product commits.
ULK	Ultra Low-K dielectric The fragile low-capacitance insulator in BEOL. Mechanically weak, which is why chip-package interaction stress matters.
WAT	Wafer Acceptance Test Parametric measurement on scribe-line structures. Tells you whether the process was in spec independently of whether the chip works.

IP AND LIBRARIES

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AMS	Analog / Mixed-Signal Circuitry that is not purely digital. Characterized by simulation rather than by static timing, and on a different schedule from the digital flow.
HV	High Voltage Devices rated above the core supply — used in IO and power management.
IP	Intellectual Property block A pre-designed functional block — a PHY, a memory compiler, a controller — bought or reused rather than written.
LDO	Low-Dropout regulator An on-die linear regulator giving a quiet local supply to sensitive analog blocks.
NoC	Network on Chip The on-die interconnect fabric carrying traffic between blocks.
OTP	One-Time Programmable memory Fuse-based memory written once after manufacture — used for trim values, keys and repair settings.
PHY	Physical layer The analog and mixed-signal circuitry that drives an interface's wires — as opposed to the digital controller above it.
PLL	Phase-Locked Loop The circuit that multiplies a reference clock up to the operating frequency. Its lock range bounds the shmoo.
RF	Radio Frequency High-frequency analog circuitry.
ROM	Read-Only Memory Fixed on-die memory, typically holding boot code.
SRAM	Static Random-Access Memory On-die memory. Usually generated by a compiler; when the compiler misses the PPA target, a custom instance has to be developed.
SerDes	Serialiser / Deserialiser The circuit that turns parallel data into a high-speed serial stream and back. The heart of every high-speed interface.
VCO	Voltage-Controlled Oscillator The tunable oscillator inside a PLL.

DESIGN AND IMPLEMENTATION

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AOCV	Advanced On-Chip Variation Depth- and distance-dependent OCV derating — less pessimistic than a flat margin.
CDC	Clock Domain Crossing A signal passing between unrelated clocks. Unsynchronized crossings fail intermittently and are invisible to simulation.
CPM	Chip Power Model A compact model of the die's current demand, handed to package and board teams so they can size the PDN.
CPS	Chip Power/Signal model CPM extended with signal behavior, for combined power and signal co-analysis.

CTS	Clock Tree Synthesis Building the buffer network that delivers the clock. Sets skew, and consumes a large share of dynamic power.
DC	Direct current Zero frequency. A regulator specified only at DC says nothing about how it behaves where the digital load actually switches.
DFM	Design For Manufacturability Changes that improve yield without changing function — via doubling, spacing, fill.
DRC	Design Rule Check Verifying that layout geometry obeys the foundry rules. A tapeout gate.
DVFS	Dynamic Voltage and Frequency Scaling Changing supply and clock at runtime to trade power against performance.
ECO	Engineering Change Order A late, targeted change to a design that is otherwise frozen. A metal-only ECO touches routing layers alone.
EM	Electromigration Metal atoms drifting under sustained current, eventually opening a wire. Bounds how much current a given wire width may carry. In “3D EM” it means electromagnetic instead — field solving for package and board models.
FFN	Final Full Netlist The netlist released for the last physical-design turn — RTL frozen, ECOs closed, no functional change beyond it.
GLS	Gate-Level Simulation Simulating the actual netlist with real delays. Slow, but catches what RTL simulation and STA both miss.
I/O	Input / Output The circuitry and pads connecting the die to the outside world.
IPC	Instructions Per Cycle How much work the design retires per clock. Performance is frequency × IPC, so a max frequency target only means something alongside the IPC assumption behind it.
IR	IR drop Supply voltage lost across the resistance of the power grid ($I \times R$). Too much and the logic misses timing that STA said was fine.
LEC	Logical Equivalence Checking Proving two netlists implement the same function — used after every synthesis or ECO step.
LVS	Layout Versus Schematic Verifying that the layout implements the intended netlist. The other tapeout gate.
MCMM	Multi-Corner Multi-Mode The same thing as MMMC; vendors differ on word order.
MMMC	Multi-Mode Multi-Corner Analyzing every operating mode against every process/voltage/temperature corner at once. The standard implementation setup.
N0/N1/N2	Netlist drop 0 / 1 / 2 The successive netlist releases from synthesis to physical design. N0 is flow-flush — structurally representative but functionally incomplete — and each drop raises quality.
OCV	On-Chip Variation Allowance for variation between devices on the same die. AOCV and POCV are its advanced and parametric refinements.

PDN	Power Delivery Network Everything carrying current from the regulator to the transistors — board, package and on-die grid together.
POCV	Parametric On-Chip Variation Statistical OCV using per-cell sigma rather than a derate table.
PVT	Process, Voltage, Temperature The three axes of operating variation a design must survive. "Corners" are their extremes.
QoR	Quality of Results The bundle of timing, area, power and congestion numbers a tool run produces. What "the netlist got better" actually means.
RDC	Reset Domain Crossing The same hazard for signals passing between independently reset regions.
RTL	Register Transfer Level The synthesisable hardware description — Verilog or SystemVerilog. "RTL freeze" means no further functional change.
SDC	Synopsys Design Constraints The file describing clocks, exceptions and IO timing. Wrong constraints make timing closure meaningless in both directions.
SSN	Simultaneous Switching Noise Supply disturbance when many outputs switch together. Couples into interfaces as jitter.
SSO	Simultaneous Switching Output Many outputs switching at once; the current step they draw is what produces SSN.
STA	Static Timing Analysis Exhaustive timing checking without simulation. The basis of timing signoff.
UPF	Unified Power Format The machine-readable description of power domains, isolation and retention. Carries power intent through the whole flow.
VDD	Supply voltage The positive rail powering a block. "VDD collapse" is a deliberate momentary droop used as an SRAM write assist.

VERIFICATION AND DFT

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ATPG	Automatic Test Pattern Generation Computing the input patterns that expose manufacturing defects, and the coverage they reach.
BIRA	Built-In Redundancy Analysis On-die logic that works out which spare rows or columns would repair a failing memory.
BISR	Built-In Self-Repair Applying that repair automatically. Turns memory defects from scrap into yield.
BIST	Built-In Self-Test On-die logic that tests a block without external equipment.
BSDL	Boundary Scan Description Language The machine-readable description of a part's boundary-scan behavior, delivered to board makers.
DFT	Design For Test Structure added to the design purely so it can be tested and debugged — scan, BIST, compression, trace.

DUT	Device Under Test Whatever is currently being tested — a block in simulation or a packaged part on a tester.
DV	Design Verification Proving the RTL does what the specification says, before any silicon exists.
FPGA	Field-Programmable Gate Array Reconfigurable hardware used to prototype the design far faster than simulation.
ICL	Instrument Connectivity Language Describes how on-die instruments are wired to the IJTAG network.
IJTAG	Internal JTAG (IEEE 1687) A standard for reaching instruments inside the chip through the JTAG port.
JTAG	Joint Test Action Group (IEEE 1149.1) The standard serial access port used for boundary scan, debug and configuration.
MBIST	Memory Built-In Self-Test BIST for embedded memories — the only practical way to test thousands of SRAM instances.
OCC	On-Chip Clock Controller Generates the fast clock pulses needed for at-speed scan testing.
PDL	Procedural Description Language Describes how to operate those instruments, alongside ICL.
SDF	Standard Delay Format Back-annotated delays from the physical design, used to make gate-level simulation realistic.
STIL	Standard Test Interface Language A vendor-neutral test pattern format. Translated to tester-native format before use.
TAP	Test Access Port The JTAG interface pins and controller.
UVM	Universal Verification Methodology The standard SystemVerilog framework for building constrained-random testbenches.
VIP	Verification IP A bought testbench component that models a protocol — PCIe, CXL, HBM — so the design can be checked against it.
WGL	Waveform Generation Language An older pattern interchange format, still used by some testers.

SIGNAL, POWER AND PACKAGE

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BERT	Bit Error Rate Tester Instrument that drives a high-speed link and counts errors — how link margin is actually measured.
CSAM	C-mode Scanning Acoustic Microscopy Ultrasonic imaging that finds voids and delamination inside a sealed package without destroying it.
EM/IR	Electromigration and IR drop The two current-related checks on the on-die power grid, usually run as one analysis.
EVB	Evaluation Board The lab platform built to bring up and characterize the silicon. Its capability bounds what can be measured.

PCB	Printed Circuit Board The board the packaged part is mounted on.
PI	Power Integrity Whether the supply stays inside its window under real switching current.
PTV	Package Test Vehicle The vehicle program that de-risks assembly and package construction before product silicon commits.
RTD	Resistance Temperature Detector A resistive on-die or in-package temperature sensor. Needs calibration before its reading is an absolute temperature rather than a relative one.
Rja	Thermal resistance, junction to ambient The same, measured to the surrounding air — so it includes the cooling solution.
Rjc	Thermal resistance, junction to case How many degrees the die rises per watt, measured to the package case. The package's share of the thermal path.
SI	Signal Integrity Whether a high-speed signal still resembles itself at the far end — loss, reflection, crosstalk, jitter.
SI/PI	Signal and Power Integrity The two treated together, because on a high-speed interface they interact.
SIPI	Signal Integrity / Power Integrity The stage that owns electrical closure across die, package and board together.
TIM	Thermal Interface Material The compound between die and lid or heatsink. Its bond line thickness directly sets how much power the part can dissipate.
VRM	Voltage Regulator Module The board-level regulator supplying the part. Its transient response has to follow the die's current steps.
TEST	12
ATE	Automatic Test Equipment The production tester. Its capability, pin count and cost per hour shape the whole test strategy.
DPPM	Defective Parts Per Million The escape rate the test flow is designed to achieve. Coverage without a DPPM target is an engineering preference, not a commitment.
FP8	8-bit floating point A low-precision format used for training and inference.
Fmax	Maximum operating frequency The highest clock a part sustains at a given voltage and temperature.
INT8	8-bit integer A low-precision numeric format used for inference. Throughput is quoted per format.
MLPerf	MLPerf benchmark suite The industry-standard machine-learning benchmark set customers compare parts with.
PPM	Parts Per Million Rate unit; DPPM is its defective-parts form.

STDF	Standard Test Data Format The industry format for per-unit test results. The raw material of all yield analysis.
Shmoo	Shmoo plot A pass/fail map across two swept variables, usually voltage against frequency. Shows where the part stops working, and so how much margin it has.
TOPS	Tera-Operations Per Second Throughput measure for an accelerator — 10^{12} operations a second.
TOPS/W	Tera-Operations Per Second per Watt Efficiency measure. The metric this product class is actually bought on.
V_{min}	Minimum operating voltage The lowest supply at which a part still works. Its distribution across units is a core characterization result.

RELIABILITY AND COMPLIANCE

15

AEC	Automotive Electronics Council The body defining automotive-grade qualification, stricter than commercial JEDEC.
CDM	Charged Device Model The ESD model where the package itself is charged and discharges through a pin. Scales with package size, so it usually binds on a large 2.5D part.
EMC	Electromagnetic Compatibility Regulatory limits on emitted and tolerated interference. A market-access gate.
ESD	Electrostatic Discharge A static discharge event the part must survive. Classified by model.
FA	Failure Analysis Physically finding out why a part failed. Usually destroys the unit, which matters when there are only a handful.
FIT	Failures In Time Failures per billion device-hours. The unit reliability commitments are written in.
HBM (ESD)	Human Body Model The ESD model representing a person touching a pin. Distinct from HBM the memory — context decides which is meant.
HTOL	High Temperature Operating Life A thousand hours of powered operation at elevated temperature. The core reliability stress, and the longest fixed item before mass production.
HTS	High Temperature Storage Unpowered bake, which exercises diffusion and interface degradation.
JEDEC	Joint Electron Device Engineering Council The body whose standards define memory interfaces and reliability qualification methods.
MSL	Moisture Sensitivity Level How long a package may sit in ambient air before it must be baked prior to soldering. A poor level forces handling procedures on every customer line.
REACH	Registration, Evaluation, Authorization and Restriction of Chemicals The broader EU chemicals regulation, with the same supplier-declaration burden.
RoHS	Restriction of Hazardous Substances EU restriction on certain materials. Needs a declaration from every component supplier.

THB	Temperature Humidity Bias Powered stress in heat and humidity — finds corrosion and moisture-driven failure.
uHAST	unbiased Highly Accelerated Stress Test Pressurized heat and humidity without bias. Accelerates moisture ingress into the package.

INTERFACES AND MEMORY

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BER	Bit Error Rate How often a link corrupts a bit. The headline quality number for a high-speed interface.
CXL	Compute Express Link A coherent protocol layered on the PCIe physical layer, used for memory and accelerator attachment.
D2D	Die-to-Die The link between two dies inside one package.
ECC	Error Correcting Code Redundancy that detects and repairs memory errors. Untested correction logic is indistinguishable from working correction logic.
GB/s	Gigabytes per second Bandwidth unit used for memory and interface targets.
HBM	High Bandwidth Memory Stacked DRAM placed beside the die on an interposer. Two stacks are assumed here. Also stands for Human Body Model in ESD contexts.
HBM3	High Bandwidth Memory generation 3 The HBM generation assumed in this template; 3E is its extended-speed revision.
HBM3E	High Bandwidth Memory 3 Extended The extended-speed revision of HBM3.
PCIe	Peripheral Component Interconnect Express The host interface standard. Compliance is obtained at scheduled plugfests, not on demand.
PSRR	Power Supply Rejection Ratio How well an analog block ignores noise on its supply. Sets how much PDN noise a PHY tolerates.
RX	Receiver The receiving end of a serial link, including its equalization and clock recovery.
TX	Transmitter The driving end of a serial link.
TX/RX	Transmit / Receive The two directions of a serial link.
UCIe	Universal Chiplet Interconnect Express The standard die-to-die interface for chiplet construction.

DATA FORMATS AND TOOLING

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.lib	Liberty timing library Timing, power and noise characterization of a cell across corners. Missing corners silently limit signoff.
AI	Artificial Intelligence The product's application domain in this template — a training and inference accelerator.

CAD	Computer-Aided Design (methodology team) The group that owns tool flows, scripts and infrastructure rather than the design itself.
CDL	Circuit Description Language The transistor-level netlist used as the reference for LVS.
EDA	Electronic Design Automation The design-tool industry and its licences — a real capacity constraint during peak implementation.
GDS	Graphic Data System (GDSII) The layout database format handed to mask making. OASIS is its modern, smaller replacement.
GDSII	Graphic Data System II The full name of the GDS layout format handed to mask making.
HPC	High Performance Computing The other workload family such parts serve, with different bandwidth and precision needs.
ID	Identifier The stable reference used to trace a requirement, activity or deliverable through the program.
LEF	Library Exchange Format The abstract physical view of a cell or macro — outline, pins, blockages — used for placement and routing.
LLM	Large Language Model The workload class driving the memory bandwidth and capacity targets assumed here.
ML	Machine Learning The broader workload family the product serves.
OASIS	Open Artwork System Interchange Standard A compact layout format used instead of GDSII for large databases.
ODB	ODB++ board database A PCB manufacturing data format.
QRC	Parasitic extraction (Quantus RC) Extracting resistance and capacitance from layout so timing reflects real wires.
RLC	Resistance, Inductance, Capacitance The passive network form used for package and board models.

Schedule tensions

SCHED

Every link in this template was authored as an engineering relationship, then checked against the calendar. These 49 pass the first test and fail the second: the source activity starts **after** the activity that needs it has already closed.

Some are expected — TC-08 and PTV-12 exist precisely to feed results backwards, and chip-package co-design iterates by construction. Others are real: an input the template assumes is available is not, and the receiving activity has to proceed on a provisional version. Both are listed, largest gap first, because only a program manager can tell them apart.

NEEDS IT	WINDOW	SOURCE	WINDOW	GAP	WHAT THE SOURCE IS
DEF-06	w0-w4	TEST-09	w62-w71	+58w	Test time and cost optimization — parallelism, site count
DEF-04	w3-w7	TC-08	w56-w62	+49w	Feedback into the production design — model correction and margin decisions
PDK-03	w12-w20	TC-08	w56-w62	+36w	Feedback into the production design — model correction and margin decisions
DEF-04	w3-w7	AMS-08	w38-w48	+31w	Statistical margin and yield analysis — importance sampling, sigma-Vmin, bitcell variation
PDK-03	w12-w20	TC-07	w50-w62	+30w	Silicon characterization and correlation to simulation
ARCH-01	w6-w20	DV-09	w48-w60	+28w	Performance and bandwidth validation against the architecture model
PTV-12	w81-w88	MP-04	w114-w126	+26w	Package qualification — MSL, drop, bend, board-level reliability
ARCH-06	w18-w22	SYN-11	w44-w66	+22w	Timing, area and power reporting against budget, per drop
EVB-01	w52-w57	TEST-07	w78-w90	+21w	Characterization test content — shmoo, Vmin/Fmax, PVT sweeps
TEST-10	w86-w94	BU-08	w113-w119	+19w	Performance validation against the architecture model
PKGD-07	w44-w56	PTV-07	w74-w81	+18w	Warpage and co-planarity measurement across the reflow profile (shadow moiré)
PKGD-07	w44-w56	PTV-09	w74-w83	+18w	Thermal characterization — Rjc / Rja, TIM performance, hotspot mapping against the thermal model
TEST-10	w86-w94	BU-07	w111-w118	+17w	Shmoo across voltage, frequency and temperature
PDK-12	w24-w30	PD-01	w46-w53	+16w	Flow setup and MMC environment build on the N0 flow-flush netlist — scripts, decks, runtimes, capacity
TEST-07	w78-w90	MP-01	w106-w112	+16w	Qualification plan definition against JEDEC / AEC standards
TEST-11	w80-w88	ASSY-09	w103-w106	+15w	Unit build for bring-up, qualification and customer samples
DEF-09	w5-w8	IPR-10	w22-w26	+14w	IP maturity risk rating and second-source contingency
TEST-10	w86-w94	BU-05	w108-w116	+14w	Interface bring-up — PCIe/CXL link training, HBM training, die-to-die
PTV-12	w81-w88	ASSY-07	w101-w106	+13w	Assembly yield analysis and process tuning

NEEDS IT	WINDOW	SOURCE	WINDOW	GAP	WHAT THE SOURCE IS
PKGD-02	w32–w42	PD-04	w52–w60	+10w	Bump and RDL planning with chip-package co-design feedback
PKGD-02	w32–w42	PD-03	w52–w60	+10w	Power delivery network design and early IR analysis
DFT-09	w34–w44	PD-02	w51–w60	+7w	Floorplan, macro placement and partition definition
RTL-03	w28–w44	AMS-15	w50–w58	+6w	Hard macro abstraction and view generation (LEF, .lib, CDL, GDS, behavioural model)
DFT-07	w40–w56	TEST-09	w62–w71	+6w	Test time and cost optimization — parallelism, site count
DFT-07	w40–w56	PD-12	w62–w68	+6w	Scan chain reordering and DFT-aware routing
PKGD-08	w34–w38	PKGD-07	w44–w56	+6w	Thermal and mechanical simulation — warpage, co-planarity, TIM, lid
ARCH-06	w18–w22	PDK-06	w27–w31	+5w	Custom / pushed-rule memory instance decision, scope and schedule impact assessment
IPR-06	w15–w22	DV-01	w26–w34	+4w	Verification plan and coverage model definition
SIPI-02	w52–w60	EVB-03	w64–w76	+4w	PCB layout with high-speed channel SI/PI simulation
PD-03	w52–w60	SIPI-05	w64–w71	+4w	Decoupling capacitor budget and placement optimization across die, package and board
AMS-01	w18–w24	PDK-06	w27–w31	+3w	Custom / pushed-rule memory instance decision, scope and schedule impact assessment
SIPI-01	w50–w57	PD-11	w60–w65	+3w	Chip power model (CPM/CPS) extraction and handoff to package co-verification
PKGD-06	w52–w61	SIPI-05	w64–w71	+3w	Decoupling capacitor budget and placement optimization across die, package and board
EVB-08	w68–w75	TEST-07	w78–w90	+3w	Characterization test content — shmoo, Vmin/Fmax, PVT sweeps
TEST-09	w62–w71	TEST-06	w74–w90	+3w	Final / package test program development
DFT-09	w34–w44	PD-01	w46–w53	+2w	Flow setup and MMMC environment build on the N0 flow-flush netlist — scripts, decks, runtimes, capacity
PTV-01	w36–w42	PKGD-07	w44–w56	+2w	Thermal and mechanical simulation — warpage, co-planarity, TIM, lid
AMS-03	w22–w56	SIPI-07	w58–w67	+2w	High-speed channel simulation with extracted package models — insertion loss, crosstalk, return path
DEF-09	w5–w8	TECH-08	w9–w13	+1w	Second-source and node-migration strategy assessment
SYN-03	w45–w49	AMS-15	w50–w58	+1w	Hard macro abstraction and view generation (LEF, .lib, CDL, GDS, behavioural model)
SIPI-10	w68–w73	PTV-09	w74–w83	+1w	Thermal characterization — Rjc / Rja, TIM performance, hotspot mapping against the thermal model
BU-01	w102–w103.5	ASSY-08	w104–w106	+0.5w	Package-level open / short and continuity test
IPR-01	w4–w9	ARCH-04	w9–w15	+0w	Interface and protocol selection (PCIe gen, CXL, UCIe, HBM3/3E)
ARCH-06	w18–w22	PDK-05	w22–w28	+0w	Compiler instance PPA characterization against block budgets — density, Vmin, access time, leakage
DFT-03	w24–w34	AMS-09	w34–w40	+0w	Redundancy and repair scheme integration with the BISR architecture
RTL-03	w28–w44	AMS-16	w44–w60	+0w	AMS-digital co-simulation and integration support

NEEDS IT	WINDOW	SOURCE	WINDOW	GAP	WHAT THE SOURCE IS
TC-01	w22-w27	PDK-06	w27-w31	+0w	Custom / pushed-rule memory instance decision, scope and schedule impact assessment
AMS-13	w40-w56	TC-08	w56-w62	+0w	Feedback into the production design — model correction and margin decisions
DFT-06	w32-w42	SYN-01	w42-w50	+0w	SDC constraint development and validation